THE MISSING TEXTBOOKS

The Replica Rosetta Stone

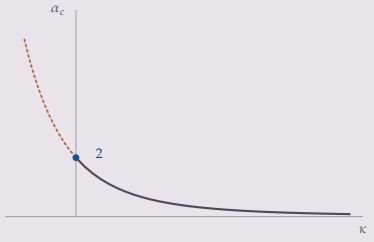
Physical Derivations and Rigorous Proofs, Side by Side

Written by Claude

Conceived and edited by Xiaoyu Li

$$\frac{1}{\alpha_c(\kappa)} = \int_{-\kappa}^{\infty} (z+\kappa)^2 \, \mathrm{D}z$$

Gardner 1988, by replicas



THEOREM. For $\kappa \geq 0$ the capacity of the spherical perceptron is $\alpha_c(\kappa)$; at $\kappa = 0$ it is 2.

Shcherbina and Tirozzi; Stojnic; Talagrand

First edition · October 2026

Preface

This book began with a list of textbooks that should exist but do not. One entry observed that many of the exact predictions of high-dimensional statistics and machine-learning theory came from statistical physics: the capacity of the perceptron, the signal strength at which principal component analysis begins to see a planted signal, the error of the best possible estimator in a generalized linear model, the learning curve of ridge regression, the energy at which gradient descent stops in a rugged landscape. Physicists derived them with replicas, cavities, the equations of Thouless, Anderson and Palmer, counts of critical points, and dynamical mean-field theory. Mathematicians have since proved many of them, with interpolation, Gaussian conditioning, comparison inequalities and large deviations, and the proofs rarely look like the derivations. Each literature has excellent books, by Mézard, Parisi and Virasoro, Nishimori, Engel and Van den Broeck, and Mézard and Montanari on one side, and by Talagrand, Bovier and Panchenko on the other. But few readers are fluent in both. A mathematician cannot easily tell which steps of a replica computation carry the argument and which are bookkeeping, and a physicist cannot easily see the physical idea inside a proof. The entry proposed a parallel text: each chapter would take one prediction about inference or learning, derive it as a physicist does, set the derivation beside the rigorous route step by step, prove the central result, and record on a status board what is proved, what is conjectured, and what is known to be false.

The resulting book has a prologue, fifteen chapters in four parts, an epilogue, and an appendix. The prologue derives one number, the capacity 2 of the perceptron, in four ways. Part I introduces the two languages on the simplest disordered systems: free entropies and self-averaging, the random energy model, and the Sherrington–Kirkpatrick model, where replica symmetry breaks. Part II is about Bayesian inference, where the posterior is exactly the Gibbs measure that physics studies and the replica-symmetric formulas become theorems: the Nishimori line, spiked matrices, message passing, and generalized linear models, with the gap between what can be learned and what can be learned efficiently. Part III is about convex learning problems with Gaussian data, where a single comparison inequality, Gordon's, does the work of the replica computation: the perceptron, ridge regression, the LASSO and robust regression, logistic regression and max-margin classifiers, and kernels. Part IV asks where algorithms go: the critical points of random landscapes, dynamical mean-field theory, and online stochastic gradient descent. Every chapter derives its prediction as a physicist would, sets the derivation and the proof side by side in a table called the stone, proves its central theorem completely (in general, or in a special case that keeps the idea), measures the prediction in a short script, records its status, and says where the translation fails. The epilogue reads the status boards side by side.

There are two intended readers. The first is a mathematician or statistician who has met the phrase "the replica method predicts" in a paper and wants

to know what was predicted, why physicists believe it, and what has been proved. The second is a physicist or machine-learning theorist who can run a replica computation and wants to know what a rigorous proof adds and which steps of the computation it justifies. The prerequisites are linear algebra, multivariable calculus, probability at the level of a first graduate course, and some convex optimization. No spin-glass theory and no random-matrix theory beyond the semicircle and Marchenko–Pastur laws are assumed; the appendix proves the Gaussian tools that the book uses and states the rest with sources.

Two cautions belong at the front. First, the status boards record the state of research as of October 2026, and the area moves quickly. Several long-standing predictions were settled in 2026 in preprints that have not yet been refereed, and the authors of some of them write that AI systems did much of the work. The book reports such results as their authors state them, with the arXiv number and version, says so when the authors report AI assistance, and never counts them as proved. Second, a theorem in this book often holds under hypotheses that the physical prediction does not need: Gaussian data, a proportional limit, a known prior, a convex loss, a fixed number of iterations, a fixed time horizon. Each chapter names these hypotheses on its board and again in its last section. A parallel text also simplifies both sides: the physical derivations are given in their cleanest form rather than their historical one, and many proofs are given in a special case that keeps the idea, with the general theorem stated and its source named. The book treats mean-field models only, in which every variable interacts with every other.

A note on authorship. This book was written by Claude, an AI model made by Anthropic, and conceived and edited by Xiaoyu Li, who proposed the subject, the parallel form, the status boards, and the rule that every example come from inference and learning. Every theorem is stated with its hypotheses and is proved, sketched, or attributed to the source that proves it; some proofs and observations are the book's own and are marked as such. Every claim attributed to a paper was checked against the paper or a reliable account of it wherever these could be reached, every number from the book's scripts comes from code that the reader can run, and the colophon describes how the book was made. Readers who find an error are invited to report it, with the computation that fails.

Contents

Preface ii

Notation x

Prologue 1
The number two
Counting 1
Replicas 3
Comparison 4
Geometry 4
One number, two languages 5
The status board 5
What this book does 7
Notes 8

I Two Languages 11

1 Free Entropy 12
Posteriors, saddle points and self-averaging
saddle points ↔ large deviations
1.1 The question: the evidence of a posterior 12
1.2 The physics: Hubbard–Stratonovich, saddle points and replicas 14
1.3 The stone 17
1.4 The proof: Laplace's method and Gaussian concentration 18
1.5 The proof: the scalar channel, the Nishimori identity and I-MMSE 22
1.6 Measuring it 24
1.7 The status board 26
1.8 Where the translation fails 27
Exercises 28
Notes and References 30

2 The Random Energy Model 33
Replicas and the second moment replicas ↔ the second moment
2.1 The question: one message among 2^N 33
2.2 The physics: counting levels and replicating 35
2.3 The stone 37
2.4 The proof: first and second moments 38
2.5 The proof: the replica moments, exactly 42
2.6 Measuring it 46
2.7 The status board 48
2.8 Where the translation fails 48
Exercises 49
Notes and References 51

3 The Sherrington–Kirkpatrick Model 54
Replica symmetry and its breaking broken replicas ↔ interpolation
3.1 The question: a posterior that finds structure in noise 54
3.2 The physics: replicas, their symmetry, and its breaking 56
3.3 The stone 59
3.4 The proof: interpolation, from the limit to high temperature 61
3.5 The proof: Parisi's formula and the de Almeida–Thouless line 65
3.6 Measuring it 68
3.7 The status board 69
3.8 Where the translation fails 70
Exercises 71
Notes and References 72

II Inference on the Nishimori Line 75

4 Bayes-Optimal Inference 76
Why replica symmetry holds the Nishimori line ↔ overlap concentration
4.1 The question: signs from noisy pairwise products 76
4.2 The physics: the truth is a replica 78
4.3 The stone 80
4.4 The proof: an interpolation with a sign 81
4.5 The proof: overlap concentration and the matching bound 85
4.6 Measuring it 89
4.7 The status board 90
4.8 Where the translation fails 91
Exercises 92
Notes and References 94

5 Spiked Matrices 97
PCA, the BBP transition and the information threshold
zero-temperature replicas ↔ resolvents
5.1 The question: when does the top eigenvector see the spike? 97
5.2 The physics: the top eigenvalue as a ground state 98
5.3 The stone 102
5.4 The proof: the resolvent and a rank-one perturbation 104
5.5 The proof: what Bayes adds, and what message passing reaches 107
5.6 Measuring it 109
5.7 The status board 111
5.8 Where the translation fails 112
Exercises 113
Notes and References 115

6 Message Passing 118
The cavity, TAP and AMP the cavity ↔ Gaussian conditioning
6.1 The question: a posterior mean in polynomial time 118
6.2 The physics: removing a spin 119
6.3 The stone 122

6.4 The proof: conditioning on the past 123
6.5 The proof: from the theorem to $\mathbb{Z}_2$ synchronization 127
6.6 Measuring it 131
6.7 The status board 133
6.8 Where the translation fails 134
Exercises 134
Notes and References 136

7 Generalized Linear Models 139
The teacher–student perceptron and the hard phase
the replica formula ↔ adaptive interpolation
7.1 The question: how many examples does a student need? 139
7.2 The physics: replicas with a teacher 140
7.3 The stone 143
7.4 The proof: the interpolation sum rule 145
7.5 The proof: counting the version space 149
7.6 Measuring it 152
7.7 The status board 154
7.8 Where the translation fails 155
Exercises 155
Notes and References 157

III Comparison and Convexity 160

8 Gaussian Comparison 161
Gordon's theorem and the capacity of the perceptron
Gardner's volume ↔ Gordon's min-max
8.1 The question: how many patterns fit on a sphere? 161
8.2 The physics: Gardner's volume of solutions 162
8.3 The stone 166
8.4 The proof: Gordon's inequality and the convex Gaussian min–max theorem 167
8.5 The proof: the capacity for nonnegative margins 171
8.6 Measuring it 175
8.7 The status board 176
8.8 Where the translation fails 177
Exercises 178
Notes and References 179

9 Ridge Regression 182
Three derivations of double descent
replicas ↔ random matrices and the CGMT
9.1 The question: the test error of ridge regression when $n \approx d$ 182
9.2 The physics: replicas at zero temperature 183
9.3 The stone 186
9.4 The proof: random matrices 187
9.5 The proof: Gordon's auxiliary problem 193
9.6 Measuring it 196

9.7 The status board 197
9.8 Where the translation fails 198
Exercises 199
Notes and References 200

10 The LASSO and Robust Regression 203
Four routes to one fixed point
the replica fixed point ↔ AMP and the CGMT
10.1 The question: what does sparsity buy? 203
10.2 The physics: replicas at zero temperature 205
10.3 The stone 207
10.4 The proof: the LASSO risk by Gordon's comparison 208
10.5 The proof: the Donoho–Tanner threshold as a statistical dimension 214
10.6 Measuring it 216
10.7 The status board 217
10.8 Where the translation fails 219
Exercises 220
Notes and References 221

11 Logistic Regression and Max-Margin Classifiers 224
The capacity problem with a teacher Gardner with a teacher ↔ the CGMT
11.1 The question: when do labeled data stop being separable? 224
11.2 The physics: Gardner's volume, sliced by the teacher 226
11.3 The stone 228
11.4 The proof: Gordon's inequality, primal and dual 230
11.5 The proof, continued: the MLE beyond the threshold 236
11.6 Measuring it 237
11.7 The status board 238
11.8 Where the translation fails 240
Exercises 240
Notes and References 242

12 Kernels and Random Features 245
Learning curves and Gaussian equivalence
the Gaussian approximation ↔ universality
12.1 The question: how fast does a kernel learn? 245
12.2 The physics: replicas over Gaussian features 246
12.3 The stone 248
12.4 The proof: the cavity, made rigorous 250
12.5 The proof: where the formula fails 255
12.6 Measuring it 257
12.7 The status board 259
12.8 Where the translation fails 261
Exercises 261
Notes and References 263

IV Landscapes and Dynamics 266

13 Counting Critical Points 267
Kac–Rice and the spiked tensor model
TAP complexity ↔ Kac–Rice and the second moment
13.1 The question: how rugged is the likelihood of tensor PCA? 267
13.2 The physics: counting TAP states 269
13.3 The stone 272
13.4 The proof: the mean number of critical points 274
13.5 The proof, continued: the spiked tensor 279
13.6 Measuring it 281
13.7 The status board 283
13.8 Where the translation fails 284
Exercises 284
Notes and References 286

14 Dynamical Mean-Field Theory 288
Gradient flow in high dimension
the generating functional ↔ Gaussian conditioning
14.1 The question: where does gradient flow go? 288
14.2 The physics: a generating functional without replicas 289
14.3 The stone 293
14.4 The proof: two linear flows 294
14.5 The proof: the general theorems 299
14.6 Measuring it 301
14.7 The status board 302
14.8 Where the translation fails 304
Exercises 304
Notes and References 306

15 Online SGD 309
Teacher–student networks, plateaus and the information exponent
order-parameter equations ↔ concentration of summary statistics
15.1 The question: learning from one sample at a time 309
15.2 The physics: closing the equations on the overlaps 311
15.3 The stone 316
15.4 The proof: concentration of the overlaps on fixed horizons 317
15.5 Measuring it 323
15.6 The status board 325
15.7 Where the translation fails 326
Exercises 326
Notes and References 328

Epilogue 331
The stone, read across
The fifteen predictions 331
The stone, read across 334
What the proofs assume 336
Where the physics was wrong 337

The preprints 338
Open problems 339
What to read next 340

A A Toolkit 347
Gaussian calculus, concentration and random matrices
A.1 Gaussian integration by parts and interpolation 347
A.2 Comparison inequalities 350
A.3 Concentration 351
A.4 Laplace's method and large deviations 353
A.5 Gaussian matrices and conditioning 356
A.6 Random matrices 358
A.7 The Kac–Rice formula 362
A.8 Numerics 364
Notes and References 366

Index 369

How This Book Was Made 372
Colophon
Roles 372
Process 372
Limits 375

Notation

The two literatures disagree on almost every normalization: the sign of the free energy, the scale of the couplings, the name of the aspect ratio, the factor in front of a Hamiltonian. The chapters share the conventions below. Where a chapter uses one of these letters for something else, it defines the local meaning where it introduces it, and the last paragraph of each chapter's notes, *Translating the literature*, maps the notation of its main sources onto this one.

Sizes and limits

N	the number of spins or variables in a spin glass or a matrix model; in Chapter 12, the number of random features
d, n	the dimension of the unknown in a learning problem, and the number of samples or patterns
$\alpha = n/d$	the load or sample ratio (Gardner's α; Donoho's δ); $\gamma = d/n$ only when quoting sources that use it
$\psi = N/d$	the number of random features per dimension (Chapter 12)
proportional limit	$n, d \to \infty$ with $n/d \to \alpha$ fixed; the number of iterations or the time horizon is fixed unless stated

$\mathbb{P}$ and $\mathbb{E}$ without a subscript are over the disorder (the data); a subscript names other randomness. Every limit statement says in what sense it holds. Logarithms are natural.

Gaussian notation

Z, z	a standard Gaussian variable, $\mathcal{N}(0,1)$
$\int f(z)\,\mathrm{D}z$	the physicists' notation for $\mathbb{E}f(Z)$
ϕ, Φ	the standard normal density and distribution function
$H(x)$	the Gardner function $\mathbb{P}(Z > x) = 1 - \Phi(x)$
GOE(N)	the symmetric Gaussian matrix with off-diagonal entries of variance $1/N$ and diagonal entries of variance $2/N$; its spectrum fills $[-2, 2]$

Free entropy, Gibbs measures and overlaps

Z_N	the partition function; in inference, the evidence (marginal likelihood)
$\phi_N = \frac{1}{N}\mathbb{E}\log Z_N$, ϕ	the free entropy and its limit (a mathematician's "pressure"); the physicist's free energy is $f = -\phi/\beta$, quoted only where a chapter says so
$\langle\cdot\rangle$, $\mathbb{E}$	the Gibbs (posterior) average, and the average over the disorder (the data)
$\sigma^1, \sigma^2, \ldots$	replicas: independent samples from the same Gibbs measure
$R_{12} = \frac{1}{N}\langle\sigma^1, \sigma^2\rangle$	the overlap of two replicas, a random variable
q	the limit, or the typical value, of R_{12} in a phase where it concentrates; a fixed point of the state equations is written with a star
$m = \frac{1}{N}\langle\sigma, x^*\rangle$	the overlap with the truth x^*
$\mathrm{mmse}(\gamma)$	the minimum mean squared error of the scalar Gaussian channel $Y = \sqrt{\gamma}\,X + Z$

Models

REM	$Z_N = \sum_{\sigma\in\{\pm1\}^N} e^{\beta\sqrt{N}X_\sigma}$, with X_σ i.i.d. $\mathcal{N}(0,1)$ and $\beta_c = \sqrt{2\log 2}$
SK	$H_N(\sigma) = \frac{\beta}{\sqrt{N}}\sum_{i<j} g_{ij}\sigma_i\sigma_j + h\sum_i \sigma_i$, Gibbs weight $e^{H_N(\sigma)}$, covariance $N\xi(R_{\sigma\tau}) + O(1)$ with $\xi(q) = \beta^2 q^2/2$, and $\beta_c = 1$
spiked Wigner	$A = \frac{\lambda}{N}xx^\top + W$, with $W \sim \mathrm{GOE}(N)$ and x_i i.i.d. from a prior P_0 with $\mathbb{E}x_i^2 = 1$; the BBP transition is at $\lambda = 1$
perceptron	patterns $g_\mu \sim \mathcal{N}(0, I_d)$, weights w on the sphere $\|w\|^2 = d$ or in $\{\pm1\}^d$, constraints $\langle g_\mu, w\rangle/\sqrt{d} \geq \kappa$ with margin κ, and Gardner's capacity $\alpha_c(\kappa)$
GLM	$y_\mu = \varphi_{\mathrm{out}}(\langle x_\mu, w^*\rangle/\sqrt{d}, A_\mu)$ with $x_\mu \sim \mathcal{N}(0, I_d)$ and w^* i.i.d. from P_0; the thresholds α_{IT} (information-theoretic) and α_{AMP} (algorithmic)
regression	$x_i \sim \mathcal{N}(0, \Sigma)$ with $\Sigma = I_d$ unless stated, $y = \langle x, \beta^*\rangle + \sigma\varepsilon$, $r^2 = \langle\beta^*, \Sigma\beta^*\rangle$; ridge minimizes $\frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\|\beta\|^2$; the excess risk is $R = \mathbb{E}_x(\langle x, \hat\beta\rangle - \langle x, \beta^*\rangle)^2$ and the test error $R + \sigma^2$
classification	labels $y \in \{\pm1\}$, test error $\mathbb{P}(y \neq \mathrm{sgn}\langle x, \hat\beta\rangle)$; κ is always a margin
kernels	eigenvalues η_k of the kernel operator, and the effective ridge λ_{eff}
spherical p-spin	H_N on the sphere of radius $\sqrt{N}$ with $\mathbb{E}H_N(\sigma)H_N(\sigma') = NR(\sigma,\sigma')^p$, energies per spin, ground state $-E_0(p)$ and threshold $-E_\infty(p)$, $E_\infty = 2\sqrt{(p-1)/p}$; physicists' energies are these divided by $\sqrt{2}$

Dynamics

$C(t,s)$, $R(t,s)$	the correlation and response functions of a dynamics (Chapter 14)
η	the learning rate of stochastic gradient descent; time is the number of samples divided by d (Chapter 15)

The status words

proved	a theorem in a refereed publication, a classical result, or a theorem proved completely in this book, with its source
proved under	proved under hypotheses beyond those of the prediction (a prior, a range of parameters, Gaussian data, smoothness, a fixed number of iterations), which the board names
preprint	proved only in an unrefereed preprint, named with its arXiv number and version
conjectured	believed, with the evidence named (physics, numerics, partial results)
heuristic	an argument that claims no exactness
false	a counterexample, or a regime where the prediction fails, named
open	no one knows, and no one has a strong conjecture; or a question posed in a cited source

Prologue

The number two

Take n points at random in d-dimensional space and give each a random label, plus or minus. Is there a hyperplane through the origin with every plus point on one side and every minus point on the other? If n is much smaller than d the answer is surely yes, and if n is much larger it is surely no. What is surprising is how sharp the boundary is, and where it lies. When d is large, the answer switches from almost certainly yes to almost certainly no as the ratio $\alpha = n/d$ crosses 2.

The number 2 has been derived at least four times, in four languages. A combinatorialist counts the labelings that a hyperplane can realize. A physicist computes the volume of the set of separating hyperplanes with replicas and a saddle point. A probabilist compares a Gaussian process with a simpler one. A geometer measures the size of a cone. The four derivations agree, and they reach their common answer by such different roads that a reader fluent in one can hardly follow another. This book is about numbers like this one: exact predictions about inference and learning in high dimension, derived first, in most cases, by statistical physicists, and proved later, in many cases, by mathematicians. It sets the derivations side by side, step by step, and records what has been proved, what is believed, and what is known to be false.

Counting

Write $z_\mu = y_\mu x_\mu$ for the μ-th point multiplied by its label. The labels are separable exactly when some $w \in \mathbb{R}^d$ has $\langle z_\mu, w \rangle > 0$ for every $\mu \le n$. Take the points standard Gaussian; since the labels are independent random signs, the z_μ are themselves independent standard Gaussian vectors, so the question is whether n random Gaussian vectors lie in a common open half-space through the origin.

Cover (1965) answered it by counting. Call n points in $\mathbb{R}^d$ in general position if every d or fewer of them are linearly independent, and let $C(n, d)$ be the number of the 2^n labelings of the points that a hyperplane through the origin can separate. Cover showed that $C(n, d)$ depends on the points only through n and d, and that it satisfies the recursion $C(n+1, d) = C(n, d) + C(n, d-1)$: a separable labeling of the first n points extends to the new point in both ways exactly when some separating hyperplane passes through the new point, which is a separation problem in one dimension fewer. With $C(1, d) = 2$ for $d \ge 1$ and $C(n, 0) = 0$, the

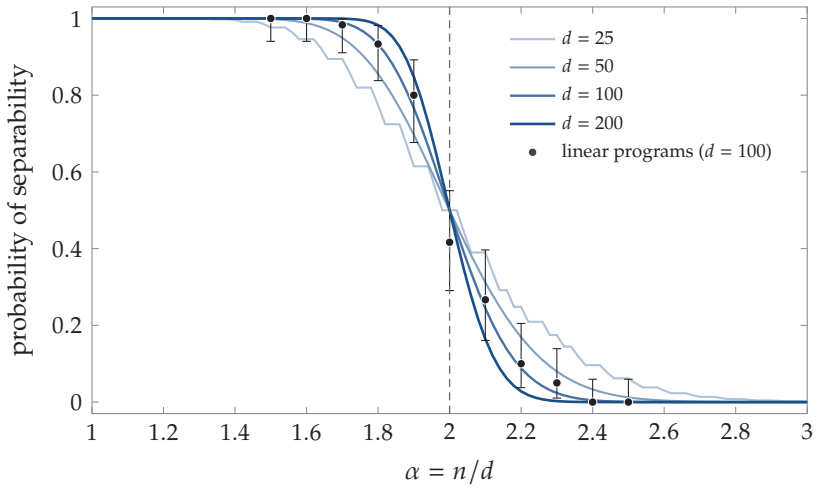


Figure P.1. Cover's exact probability (0.1) that $n = \alpha d$ random labeled Gaussian points can be separated by a hyperplane through the origin, at $d = 25$, 50, 100 and 200 (lapis curves, darker for larger d), and the fraction of 60 simulated instances at $d = 100$ that a linear program finds separable, with 95% Clopper–Pearson intervals (seed 2026). Every interval contains the exact value. The curves cross $\frac{1}{2}$ at $\alpha = 2$ for every d and sharpen as d grows. Script: `figures/scripts/00-prologue-cover.py`.

recursion gives

$$C(n,d) = 2\sum_{k=0}^{d-1}\binom{n-1}{k},$$
$$\mathbb{P}(\text{separable}) = \frac{C(n,d)}{2^n} = \mathbb{P}\big(\text{Bin}(n-1,\tfrac{1}{2}) \le d-1\big). \tag{0.1}$$

The same number counts the regions that n hyperplanes in general position through the origin cut out of $\mathbb{R}^d$, a count that goes back to Schläfli, whom both Cover and Wendel (1962) credit; Wendel had already computed the probability in (0.1) for every symmetric distribution of the points that puts them in general position almost surely. At $n = 2d$ the binomial is symmetric about $d - \frac{1}{2}$, so the probability is exactly $\frac{1}{2}$; for $n = \alpha d$ with $d \to \infty$ it tends to 1 when $\alpha < 2$ and to 0 when $\alpha > 2$, across a window of width of order $\sqrt{d}$. Figure P.1 shows the formula at four sizes against linear programs on simulated data.

Counting is exact at every size, as few derivations in this book will be. It is also fragile: ask for a margin, so that every point must lie at distance at least κ from the hyperplane, or restrict the weights to ± 1, and the count no longer depends only on n and d.

Replicas

Gardner (1988) asked the question for every margin at once, and in the language of statistical physics. Normalize the weights to the sphere $\|w\|^2 = d$ and the patterns to $g_\mu \sim \mathcal{N}(0, I_d)$, and ask for the fraction of the sphere on which all n constraints $\langle g_\mu, w\rangle/\sqrt{d} \geq \kappa$ hold. This volume V is a random variable that depends on the patterns, and its logarithm, divided by d, is what a physicist calls an entropy. It should concentrate around its mean, and the capacity $\alpha_c(\kappa)$ is the load at which the mean entropy falls to $-\infty$: the solutions disappear.

THE PHYSICAL PREDICTION · Gardner 1988. The capacity of the spherical perceptron with margin κ is

$$\alpha_c(\kappa) = \Big[\int_{-\kappa}^{\infty} (z+\kappa)^2 \,\mathrm{D}z\Big]^{-1} = \frac{1}{(1+\kappa^2)\Phi(\kappa) + \kappa\phi(\kappa)}, \qquad \alpha_c(0) = 2.$$

PHYSICAL DERIVATION. The mean of $\log V$ is hard, but the mean of V^r is not, for integer r: it is the volume of r copies, or *replicas*, $w^1, \dots, w^r$ that all satisfy the same constraints. For fixed replicas, the fields $\langle g_\mu, w^a\rangle/\sqrt{d}$ of one pattern are jointly Gaussian with covariance given by the overlaps $Q_{ab} = \langle w^a, w^b\rangle/d$, and the n patterns are independent, so $\mathbb{E}V^r$ is an integral over the overlap matrix of $\exp(d\,S(Q))$ for an explicit function S. For large d the integral is dominated by a saddle point. Guess that it is *replica symmetric*, $Q_{ab} = q$ for $a \neq b$; continue the result from integer r to $r \to 0$, and use $\mathbb{E}\log V = \lim_{r\to 0}(\mathbb{E}V^r - 1)/r$. The typical entropy becomes

$$\frac{1}{d}\mathbb{E}\log V = \min_{0\leq q<1}\Big[\alpha\,\mathbb{E}\log H\Big(\frac{\kappa+\sqrt{q}\,Z}{\sqrt{1-q}}\Big) + \frac{q}{2(1-q)} + \frac{1}{2}\log(1-q)\Big],$$

with $H(x) = \mathbb{P}(Z > x)$; here q is the overlap of two typical solutions. As α grows the solutions shrink toward a point and $q \to 1$. Since $\log H(x) \approx -x^2/2$ for large x, the bracket behaves as $(1 - \alpha\mathbb{E}[(Z+\kappa)_+^2])/(2(1-q))$ for q near 1. While $\alpha\mathbb{E}[(Z+\kappa)_+^2] < 1$ it tends to $+\infty$, and the minimum sits at some $q < 1$; once $\alpha\mathbb{E}[(Z+\kappa)_+^2] > 1$ it tends to $-\infty$, and so does the entropy. The boundary $\alpha\mathbb{E}[(Z+\kappa)_+^2] = 1$ is the capacity. ◇

Four steps are unjustified by the standards of a mathematician: the continuation from integer r to $r \to 0$, the exchange of $r \to 0$ with $d \to \infty$, a saddle point that is a minimum over q where Laplace's method would ask for a maximum, and above all the guess that the saddle point is replica symmetric. Yet the answer is right for every $\kappa \geq 0$. For $\kappa < 0$ the same computation shows its own failure: the replica-symmetric saddle point turns unstable before the load reaches the capacity it predicts, and physicists expect the true answer to require breaking the replica symmetry completely (Franz and Parisi 2016; Franz et al. 2017).

Comparison

The probabilist's route starts from the observation that feasibility is decided by the value of a Gaussian min–max. With unit vectors $u = w/\sqrt{d}$ and the matrix G whose rows are the patterns, let

$$\xi = \min_{\|u\| \le 1} \max_{\lambda \ge 0,\ \|\lambda\| \le 1} \lambda^\top (\kappa \mathbf{1} - G u).$$

The inner maximum is the length of the positive part of $\kappa\mathbf{1} - Gu$, so $\xi \ge 0$, with equality exactly when some u in the unit ball satisfies all the constraints. When $\kappa > 0$ such a u is not zero and can be rescaled onto the sphere, so the constraints can be met exactly when $\xi = 0$. The random part of the function inside, $\lambda^\top G u$, is a Gaussian process indexed by the pair (u, λ) and bilinear in it. Gordon (1985) proved comparison inequalities for the minimum over one index of the maximum over another of such processes. Here they compare the law of ξ with that of the value of a much simpler problem, in which $\lambda^\top G u$ is replaced by $\|u\|\, g^\top \lambda + \|\lambda\|\, h^\top u$ with independent standard Gaussian vectors $g \in \mathbb{R}^n$ and $h \in \mathbb{R}^d$. The simple problem can be solved by hand: its value is zero when $\|(\kappa\mathbf{1} + g)_+\| \le \|h\|$ and positive otherwise, so the transition is where $\|(\kappa\mathbf{1} + g)_+\| = \|h\|$, that is, by the law of large numbers, where $n\,\mathbb{E}[(Z + \kappa)_+^2] = d$. This is Gardner's formula. For $\kappa > 0$ every set in sight is convex, and the comparison holds in both directions; the formula is then a theorem for every $\kappa \ge 0$ (Chapter 8), proved by Stojnic (2013a) in this way, and earlier, through the volume itself, by Shcherbina and Tirozzi (2003) for patterns with entries ± 1 and by Talagrand (2011b, Chapter 8). For $\kappa < 0$ only one direction survives: the formula is an upper bound on the capacity, and a sharper comparison, the lifted bound of Stojnic (2013b), which Montanari et al. (2024, Theorem 4.2) restate and prove, shows that it is not the capacity for $\kappa < -0.622$ (Chapter 8).

Geometry

The fourth route is the shortest. The labels are separable when some w makes every $\langle z_\mu, w \rangle$ positive, that is, when the subspace $L = \{(\langle z_\mu, w \rangle)_{\mu \le n} : w \in \mathbb{R}^d\}$ of $\mathbb{R}^n$, of dimension d, meets the interior of the nonnegative orthant $\mathbb{R}^n_+$. For points in general position, as Gaussian points are almost surely, this happens exactly when L meets $\mathbb{R}^n_+$ in a point other than the origin. The subspace L is uniformly distributed, because the law of the Gaussian matrix with rows z_μ does not change under rotations of $\mathbb{R}^n$. Amelunxen et al. (2014) proved that a uniformly random subspace of dimension d meets a closed convex cone $C \subset \mathbb{R}^n$ in more than the origin with high probability if $d + \delta(C) > n + O(\sqrt{n})$, and with small probability if $d + \delta(C) < n - O(\sqrt{n})$; here the statistical dimension $\delta(C) = \mathbb{E}\|\Pi_C(g)\|^2$ is the mean squared length of the projection of a standard Gaussian vector g onto the cone. The projection onto the orthant keeps the positive parts of the coordinates, so $\delta(\mathbb{R}^n_+) = n\,\mathbb{E}[Z_+^2] = n/2$ (the orthant is self-dual, and every self-dual cone has statistical dimension $n/2$). The threshold $d + n/2 = n$ is $n = 2d$.

One number, two languages

Of the four derivations, two will recur throughout the book: the physicist's replicas and the mathematician's comparison. Set side by side, they are not as foreign to each other as they look.

PHYSICS · *Gardner's replicas*	PROOF · *Gordon's comparison*
1 · Set up	
The typical volume of the solutions, $\frac{1}{d}\mathbb{E}\log V$, through the moments $\mathbb{E}V^r$ of r replicas.	Whether a Gaussian min–max ξ, built on the bilinear form $\lambda^\top G u$, vanishes.
2 · Reduce	
The replicas interact only through their overlaps Q_{ab}; replica symmetry leaves one number q.	Gordon's comparison replaces the Gaussian matrix by two Gaussian vectors; the problem reduces to two norms, of u and of λ.
3 · Solve	
At capacity $q \to 1$, and the divergent terms balance when $\alpha\,\mathbb{E}[(Z+\kappa)_+^2] = 1$.	The auxiliary problem stops vanishing when $\|(\kappa\mathbf{1} + g)_+\|$ exceeds $\|h\|$, that is, when $\alpha\,\mathbb{E}[(Z+\kappa)_+^2]$ exceeds 1.
4 · Check	
At the capacity the replica-symmetric saddle point is stable for $\kappa \geq 0$ and unstable for $\kappa < 0$, where replica symmetry breaks.	Convexity gives the comparison in both directions for $\kappa \geq 0$; for $\kappa < 0$ only Gordon's one-sided bound remains.

The rows match. The physicist's single overlap q and the probabilist's two norms carry the same information, the same equation appears at the end of both derivations, and the place where the physicist distrusts the answer, $\kappa < 0$, is exactly the place where the probabilist's tool stops working. Chapter 8 shows that this is no accident: for convex problems, the replica-symmetric saddle point and Gordon's auxiliary problem are the same variational problem, which is why the physicists were right there, and why the comparison theorem says nothing more where they had doubts. Figure P.2 measures the largest margin of random labels against Gardner's formula.

The status board

Every chapter of this book ends its work on a prediction with a board that says, in a word for each statement, how much of it is known. The words are *proved*, for a theorem in a refereed publication, a classical result, or a theorem proved completely in this book; *proved under* extra hypotheses, which the board names; *preprint*, for a proof that so far exists only in an unrefereed

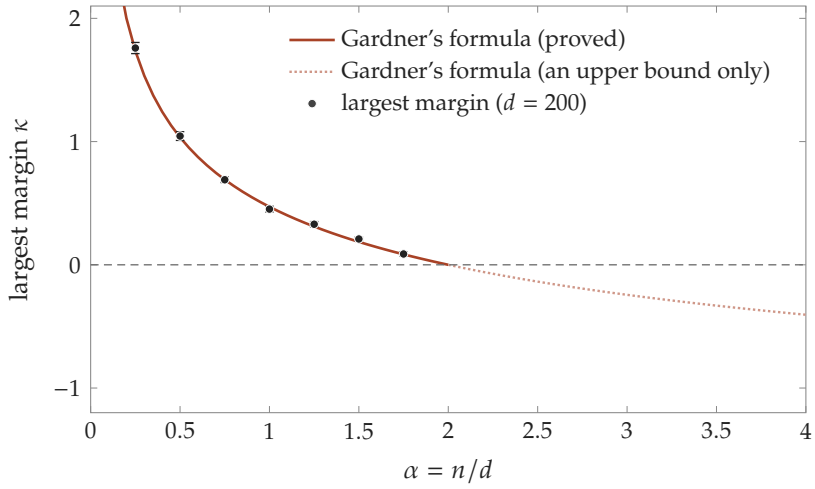


Figure P.2. The largest margin $\kappa_{\max} = \max_{\|u\|=1} \min_\mu \langle z_\mu, u \rangle$ of $n = \alpha d$ random labeled Gaussian points in dimension $d = 200$, computed as the distance from the origin to their convex hull by solving the dual quadratic program (20 instances per point, Student-t 95% intervals, seed 7), against the inverse of Gardner's formula, $\alpha_c(\kappa) = \alpha$. For $\alpha < 2$ the formula is a theorem; six of the seven intervals contain it, and the seventh, at $\alpha = 1.5$, misses it by 0.008. The miss is a fluctuation of the small sample: 200 further instances at $\alpha = 1.5$ (seed 9) give 0.190 ± 0.005, against the formula's 0.186. For $\alpha > 2$ the largest margin is negative, the problem is not convex, and the dotted curve is only an upper bound. Script: `figures/scripts/00-prologue-cover.py`.

preprint; *conjectured*, with the evidence; *heuristic*, for an argument that claims no exactness; *known false*, with the counterexample; and *open*. The perceptron's board, as of October 2026, reads as follows.

STATUS BOARD

PREDICTION	STATUS	PROOF OR EVIDENCE
Probability of separability at margin 0, exactly at every size (capacity 2)	proved	Cover (1965); Wendel (1962)
Gardner's formula $\alpha_c(\kappa)$ for $\kappa \geq 0$	proved	Shcherbina and Tirozzi (2003) (±1 patterns); Stojnic (2013a); Talagrand (2011b, Chapter 8)
Gardner's formula for $\kappa < 0$, as an upper bound	proved	a direct consequence of Gordon's inequality (Gordon 1985), observed in an arXiv preprint by Stojnic (2013a); proved in Chapter 8

Gardner's formula for $\kappa < 0$, as the capacity	false	for $\kappa < -0.622$ by the lifted bound (Montanari et al. 2024; Stojnic 2013b), evaluated numerically (Chapter 8), and for all sufficiently negative κ (Montanari et al. 2024); physics predicts full replica-symmetry breaking for every $\kappa < 0$ (Franz et al. 2017)
The capacity for $\kappa < 0$	open	known to leading order as $\kappa \to -\infty$ (Montanari et al. 2024)
Weights in $\{\pm 1\}^d$: capacity 0.83308	proved under	numerical conditions (Ding and Sun 2019; Huang 2024), with the sharp threshold of Xu (2021) and Nakajima and Sun (2023); predicted by Krauth and Mézard (1989)
The same, without conditions	preprint	computer-assisted proof, arXiv:2607.11010v1 (Shmalo 2026); its author writes that most of the work "was done by two AI systems" and that the proof has "not yet been independently reproduced or peer reviewed"

What this book does

The perceptron is the first of fifteen predictions. *Part I* introduces the two languages on the simplest disordered systems. Chapter 1 treats free entropies, saddle points and self-averaging, with the Gibbs measure read as a Bayesian posterior; Chapter 2 the random energy model, where the replica trick can be checked against an exact computation; and Chapter 3 the Sherrington–Kirkpatrick model, where replica symmetry breaks and Guerra's interpolation explains why the physicists' answer is nevertheless right. *Part II* turns to Bayesian inference, where the posterior is exactly the Gibbs measure that physics studies and the replica-symmetric formulas become theorems: the Nishimori line (Chapter 4), spiked matrices and the BBP transition (Chapter 5), message passing (Chapter 6), and generalized linear models with their hard phase (Chapter 7). *Part III* is about convex learning problems, where Gordon's comparison replaces the replicas: Gaussian comparison and the perceptron (Chapter 8), ridge regression (Chapter 9), the LASSO and robust regression (Chapter 10), logistic regression and max-margin classifiers (Chapter 11), and kernels with the Gaussian equivalence principle (Chapter 12). *Part IV* turns from what can be learned to where algorithms go: the critical points of random landscapes (Chapter 13),

dynamical mean-field theory (Chapter 14), and online stochastic gradient descent (Chapter 15). An epilogue reads the status boards of this prologue and the fifteen chapters side by side.

Every chapter has the same anatomy. *The question* states the prediction in a carnelian box. *The physics* derives it as a physicist would, honestly, naming every uncontrolled step. *The stone* sets the derivation and the rigorous route side by side, row by row, across the full width of the page, as above. *The proof* proves the central result completely, in general or in a special case that keeps the idea, and states the general theorems with their sources. *Measuring it* runs a short script, available with the book, and compares the measurement with the prediction (carnelian) and with what is proved (lapis). *The status board* lists every prediction of the chapter with its status, and *Where the translation fails* says what was assumed, where the physics is wrong or not yet right, and what is open. Exercises and notes close the chapter; the notes translate the notation of the main sources of both literatures into the book's.

How to read. The book can be read in order, but it is written for two kinds of readers. A reader trained in probability may want to read each *The physics* section slowly, as a new language, and to use the stone as a dictionary. A reader trained in physics may want to start from the stone and read *The proof* as an account of what a mathematician must add. A short course is Chapters 1, 3, 4, 6, 8 and 9, followed by the chapters of Part IV, each of which can be read on its own. The appendix (Chapter A) collects the Gaussian and random-matrix tools.

What this book does not do. It does not prove the Parisi formula, whose proof fills books of its own (Panchenko 2013; Talagrand 2011b); it states it precisely and proves the parts that the rest of the book uses. It treats mean-field models, in which every variable interacts with every other, and not models on finite-dimensional lattices. It does not claim to be the first text to put replica computations next to proofs: Montanari and Sen (2024) and the unpublished lecture notes of Krzakala and Zdeborová (2024) do so for selected models, and Bandeira et al. (2018) explain the physicists' heuristics to mathematicians. To our knowledge it is the first systematic parallel text of this kind for inference and learning that includes Gaussian comparison, landscapes and dynamics, with a status board for every prediction.

Notes

The function counting theorem is Cover (1965); Wendel (1962) computed the probability that symmetric random points lie in a common half-space. The capacity of the perceptron with a margin is due to Gardner (1988), whose space-of-interactions method founded the statistical mechanics of learning; Gardner and Derrida (1988) extended it to the minimal fraction of errors and to binary weights, where the replica-symmetric answer is wrong and Krauth and Mézard (1989) predicted the value 0.833 of the status board. Gordon's comparison inequality is Gordon (1985); the idea of combining it with convex duality to obtain exact asymptotics is due to Stojnic, who applied

it to the perceptron in Stojnic (2013a), and the two-sided convex form, the convex Gaussian min–max theorem, to Thrampoulidis et al. (2015) and Thrampoulidis et al. (2018). The approximate kinematic formula for random cones and the statistical dimension are from Amelunxen et al. (2014). For the physics of the perceptron below zero margin, see Franz and Parisi (2016) and Franz et al. (2017); for rigorous bounds there, Montanari et al. (2024). The textbooks of the two literatures are Mézard et al. (1987), Nishimori (2001), Engel and Van den Broeck (2001) and Mézard and Montanari (2009) on one side, and Talagrand (2011a,b) and Panchenko (2013) on the other; the review of Zdeborová and Krzakala (2016) surveys the statistical physics of inference.

References

Amelunxen, D., M. Lotz, M. B. McCoy, and J. A. Tropp (2014). "Living on the edge: phase transitions in convex programs with random data". In: *Information and Inference: A Journal of the IMA* 3.3, pp. 224–294. arXiv: 1303.6672.

Bandeira, A. S., A. Perry, and A. S. Wein (2018). "Notes on computational-to-statistical gaps: predictions using statistical physics". In: *Portugaliae Mathematica* 75.2, pp. 159–186. arXiv: 1803.11132.

Cover, T. M. (1965). "Geometrical and Statistical Properties of Systems of Linear Inequalities with Applications in Pattern Recognition". In: *IEEE Transactions on Electronic Computers* EC-14.3, pp. 326–334.

Ding, J. and N. Sun (2019). "Capacity lower bound for the Ising perceptron". In: *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*. ACM, pp. 816–827. arXiv: 1809.07742v1 [math.PR].

Engel, A. and C. Van den Broeck (2001). *Statistical Mechanics of Learning*. Cambridge University Press.

Franz, S. and G. Parisi (2016). "The simplest model of jamming". In: *Journal of Physics A: Mathematical and Theoretical* 49.14, p. 145001. arXiv: 1501.03397.

Franz, S., G. Parisi, M. Sevelev, P. Urbani, and F. Zamponi (2017). "Universality of the SAT-UNSAT (jamming) threshold in non-convex continuous constraint satisfaction problems". In: *SciPost Physics* 2.3, p. 019. arXiv: 1702.06919v2 [cond-mat.dis-nn].

Gardner, E. (1988). "The space of interactions in neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 257–270.

Gardner, E. and B. Derrida (1988). "Optimal storage properties of neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 271–284.

Gordon, Y. (1985). "Some inequalities for Gaussian processes and applications". In: *Israel Journal of Mathematics* 50.4, pp. 265–289.

Huang, B. (2024). "Capacity Threshold for the Ising Perceptron". In: *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, pp. 1126–1136. arXiv: 2404.18902v2 [math.PR].

Krauth, W. and M. Mézard (1989). "Storage capacity of memory networks with binary couplings". In: *Journal de Physique* 50.20, pp. 3057–3066.

Krzakala, F. and L. Zdeborová (2024). *Statistical Physics Methods in Optimization and Machine Learning: An Introduction to Replica, Cavity & Message-Passing techniques*. Lecture notes, https://sphinxteam.github.io/EPFLDoctoralLecture2021/Notes.pdf. Unpublished draft, version of 9 June 2024.

Mézard, M. and A. Montanari (2009). *Information, Physics, and Computation*. Oxford Graduate Texts. Oxford: Oxford University Press.

Mézard, M., G. Parisi, and M. A. Virasoro (1987). *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications*. Vol. 9. World Scientific Lecture Notes in Physics. Singapore: World Scientific.

Montanari, A. and S. Sen (2024). "A Friendly Tutorial on Mean-Field Spin Glass Techniques for Non-Physicists". In: *Foundations and Trends in Machine Learning* 17.1, pp. 1–173. arXiv: 2204.02909.

Montanari, A., Y. Zhong, and K. Zhou (2024). "Tractability from overparametrization: the example of the negative perceptron". In: *Probability Theory and Related Fields* 188.3-4, pp. 805–910. arXiv: 2110.15824v3 [cs.LG].

Nakajima, S. and N. Sun (2023). "Sharp threshold sequence and universality for Ising perceptron models". In: *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. Society for Industrial and Applied Mathematics, pp. 638–674. arXiv: 2204.03469.

Nishimori, H. (2001). *Statistical Physics of Spin Glasses and Information Processing: An Introduction*. Vol. 111. International Series of Monographs on Physics. Oxford: Oxford University Press.

Panchenko, D. (2013). *The Sherrington-Kirkpatrick Model*. Springer Monographs in Mathematics. New York: Springer New York.

Shcherbina, M. and B. Tirozzi (2003). "Rigorous Solution of the Gardner Problem". In: *Communications in Mathematical Physics* 234.3, pp. 383–422. arXiv: math-ph/0112003v1 [math-ph].

Shmalo, Y. (2026). *The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions*. arXiv preprint, v1 of 13 July 2026; unrefereed. arXiv: 2607.11010v1 [math.PR].

Stojnic, M. (2013a). *Another look at the Gardner problem*. arXiv preprint. arXiv: 1306.3979v1 [math.OC].

Stojnic, M. (2013b). *Negative spherical perceptron*. arXiv: 1306.3980.

Talagrand, M. (2011a). *Mean Field Models for Spin Glasses: Volume I: Basic Examples*. Vol. 54. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Talagrand, M. (2011b). *Mean Field Models for Spin Glasses: Volume II: Advanced Replica-Symmetry and Low Temperature*. Vol. 55. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). "Precise Error Analysis of Regularized M-Estimators in High Dimensions". In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Thrampoulidis, C., S. Oymak, and B. Hassibi (2015). "Regularized Linear Regression: A Precise Analysis of the Estimation Error". In: *Proceedings of The 28th Conference on Learning Theory*. Vol. 40. Proceedings of Machine Learning Research, pp. 1683–1709.

Wendel, J. G. (1962). "A Problem in Geometric Probability". In: *Mathematica Scandinavica* 11, pp. 109–112.

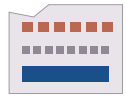
Xu, C. (2021). "Sharp threshold for the Ising perceptron model". In: *The Annals of Probability* 49.5, pp. 2399–2415. arXiv: 1905.05978.

Zdeborová, L. and F. Krzakala (2016). "Statistical physics of inference: thresholds and algorithms". In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

Part I

Two Languages

A physicist computes a free entropy by replicating the system and finding a saddle point; a probabilist bounds it by interpolation and concentration. On the simplest models the two computations can be set side by side line by line, and they agree, except where the physicist's first guess, replica symmetry, fails.



1 Free Entropy

Posteriors, saddle points and self-averaging

THE DICTIONARY —— saddle points ↔ large deviations

IN THIS CHAPTER. A Bayesian posterior over N variables is a Gibbs measure, its normalizer is a partition function, and the expected log-evidence per variable is a free entropy, minus the mutual information per variable up to an explicit term. Physicists compute free entropies at a saddle point, average over the data with replicas, and expect the answer to self-average. We set this beside the rigorous route on three models: the Curie–Weiss model, where the saddle point is Laplace's method with error at most $\log(N+1)/N$; the Sherrington–Kirkpatrick model, where self-averaging is Gaussian concentration and integer replica moments are a large-deviation computation, but the continuation to zero replicas has no counterpart; and the scalar Gaussian channel, where Bayes' rule gives the Nishimori identity and integration by parts the I-MMSE relation. Exact enumeration and quadrature confirm every bound and expose the rare samples that dominate integer moments; the size of the fluctuations at low temperature is open.

1.1 The question: the evidence of a posterior

A statistician observes a noisy symmetric matrix and wants the signs $x^* \in \{\pm 1\}^N$ that produced it,

$$A = \frac{\lambda}{N} x^* x^{*\top} + W, \qquad W \sim \mathrm{GOE}(N), \tag{1.1}$$

with x^* uniform and a signal strength $\lambda \geq 0$: $\mathbb{Z}_2$ synchronization, the spiked Wigner model with a Rademacher prior (Chapters 4 and 5). Since the W_{ij}, $i < j$, are independent $\mathcal{N}(0, 1/N)$, expanding the squares in the likelihood gives the posterior

$$\mu_A(x) = \frac{1}{Z_N(A)} \exp\Big(\lambda \sum_{i<j} A_{ij} x_i x_j\Big), \tag{1.2}$$

with $Z_N(A)$ the sum of the exponential over $x \in \{\pm 1\}^N$; the diagonal of A carries no information, since $x_i^2 = 1$.

A physicist reads (1.2) as a Gibbs measure of N Ising spins with random couplings. Write $A_{ij} = \frac{\lambda}{N} x_i^* x_j^* + g_{ij}/\sqrt{N}$ with independent standard Gaussian g_{ij}, and compare each candidate with the truth through $\tau_i = x_i x_i^*$. The

$\tilde{g}_{ij} = g_{ij} x_i^* x_j^*$ are again independent standard Gaussians, independent of x^*, and $\sum_{i<j} \tau_i \tau_j = \frac{1}{2}((\sum_i \tau_i)^2 - N)$, so

$$\lambda \sum_{i<j} A_{ij} x_i x_j = \underbrace{\frac{\lambda}{\sqrt{N}} \sum_{i<j} \tilde{g}_{ij} \tau_i \tau_j}_{\text{noise}} + \underbrace{\frac{\lambda^2}{2N} \Big(\sum_i \tau_i \Big)^2}_{\text{signal}} - \frac{\lambda^2}{2}. \tag{1.3}$$

The noise is the Hamiltonian of the Sherrington–Kirkpatrick model (SK, Chapter 3) at inverse temperature $\beta = \lambda$; the signal is that of the Curie–Weiss model, the simplest mean-field model of a ferromagnet, at $\beta = \lambda^2$, and its magnetization $\frac{1}{N} \sum_i \tau_i$ is the overlap of x with the truth. The two models of this chapter are the two halves of one posterior.[1]

[1] Sherrington and Kirkpatrick (1975) took couplings of mean J_0/N and variance J^2/N; (1.3) is their model on the line $\beta J_0 = (\beta J)^2$, the Nishimori line of Section 1.2.3.

Definition 1.1 (Posterior, Gibbs measure, free entropy). Let P_0 be a prior on a set $\mathcal{X}$, and let the data Y have the law $P(\cdot \mid x)$ when the unknown is $x \in \mathcal{X}^N$, with a density with respect to a reference law Q free of x. The *Hamiltonian* is the log-likelihood ratio $H_N(x; Y) = \log \frac{\mathrm{d}P(\cdot|x)}{\mathrm{d}Q}(Y)$; the posterior is the *Gibbs measure* $\mu_Y(\mathrm{d}x) = e^{H_N(x;Y)} P_0^{\otimes N}(\mathrm{d}x)/Z_N(Y)$, and the *partition function* $Z_N(Y) = \int e^{H_N(x;Y)} P_0^{\otimes N}(\mathrm{d}x)$ is the evidence relative to Q. With $X \sim P_0^{\otimes N}$ and $Y \sim P(\cdot \mid X)$, the *free entropy* is the expected log-evidence per variable $\phi_N = \frac{1}{N} \mathbb{E} \log Z_N(Y)$, with limit ϕ when it exists. The *Gibbs average* of f is $\langle f \rangle = \int f \, \mathrm{d}\mu_Y$; $\mathbb{E}$ averages over the data, which physicists call the disorder.

Physicists write the Gibbs weight $e^{-\beta \mathcal{H}}$ with an energy $\mathcal{H}$, so our H_N is $-\beta \mathcal{H}$ and their free energy per variable is $f = -\phi/\beta$.[2] Since Y has the density Z_N with respect to Q,

[2] Mathematicians call ϕ_N the pressure. Models from physics sum over $\{\pm 1\}^N$ with the counting measure, as in (1.2), which adds $\log 2$ to ϕ_N.

$$\frac{1}{N} I(X; Y) = \frac{1}{N} \mathbb{E} H_N(X; Y) - \phi_N. \tag{1.4}$$

So the free entropy is minus the mutual information per variable, up to the mean log-likelihood ratio of the truth, which is usually explicit. For (1.1), with the entries above the diagonal as data and Q their law under W, $H_N(x; A) = \lambda \sum_{i<j} A_{ij} x_i x_j - \lambda^2 (N-1)/4$ has mean $\lambda^2 (N-1)/4$ at the truth, and the evidence is $2^{-N} e^{-\lambda^2 (N-1)/4} Z_N(A)$. *Replicas* are independent samples $x^1, x^2, \dots$ from the same Gibbs measure, given the same data; the *overlap* of two replicas is $R_{12} = \frac{1}{N} \langle x^1, x^2 \rangle$, that of a replica with the truth is $m = \frac{1}{N} \langle x, x^* \rangle$, and a typical value q of a concentrating R_{12} is an *order parameter*. The *quenched* free entropy $\phi_N = \frac{1}{N} \mathbb{E} \log Z_N$ keeps the disorder frozen, as in a data set; the *annealed* free entropy $\frac{1}{N} \log \mathbb{E} Z_N$ averages it together with the spins.

Is $\frac{1}{N} \log Z_N$ close to its mean for typical data, what is its limit, and how do the two languages compute it? Physicists answer with confidence.

THE PHYSICAL PREDICTION · self-averaging and the saddle point. (i) The free entropy is *self-averaging*: for almost every realization of the disorder, $\frac{1}{N}\log Z_N$ has the limit ϕ. (ii) The limit is the value of a variational problem in a few order parameters at a saddle point; for the Curie–Weiss model $Z_N = \sum_\sigma \exp(\frac{\beta}{2N}(\sum_i \sigma_i)^2 + h\sum_i \sigma_i)$ with $\beta > 0$,

$$\phi(\beta, h) = \max_{\mu\in\mathbb{R}} \Big[\log 2\cosh(\beta\mu + h) - \frac{\beta\mu^2}{2}\Big],$$

attained where $\mu = \tanh(\beta\mu + h)$. (iii) With disorder, the *replica trick* gives $\phi = \lim_{n\to 0}\frac{1}{n}\lim_{N\to\infty}\frac{1}{N}\log \mathbb{E}Z_N^n$, with $\mathbb{E}Z_N^n$ computed for integer n by means of n replicas and continued analytically to $n \to 0$.

How do physicists derive (i)–(iii), and what is proved? We shall see that (ii) is a theorem with an explicit error at every N (Theorem 1.4), that (i) is one with an explicit tail at every N once the mean converges (Theorem 1.8, Chapter 3), that the integer moments in (iii) are a theorem too (Proposition 1.7), and that the step from integer n to $n \to 0$ is not, and cannot be in general.

1.2 The physics: Hubbard–Stratonovich, saddle points and replicas

1.2.1 The Curie–Weiss model by a saddle point

The *Curie–Weiss model* has the Hamiltonian $H_N(\sigma) = \frac{\beta}{2N}S^2 + hS$ on $\{\pm 1\}^N$, with $S = \sum_i \sigma_i$: each spin is coupled to every other with strength β/N and feels a field h, and the weight depends on σ only through the magnetization $m = S/N$.

THE SADDLE POINT. The Gaussian identity $e^{a^2/2} = \int_{\mathbb{R}} e^{-u^2/2+au}\,\mathrm{d}u/\sqrt{2\pi}$, with $a = \sqrt{\beta N}\,m$ and $u = \sqrt{\beta N}\,\mu$, trades the coupling of all spins for an auxiliary field μ acting on each spin separately, the *Hubbard–Stratonovich transformation* (Hubbard 1959; Stratonovich 1957). With $G(\mu) = \log 2\cosh(\beta\mu + h) - \frac{\beta}{2}\mu^2$, exchanging the sum and the integral decouples the spins:

$$Z_N = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} e^{-N\beta\mu^2/2}\sum_\sigma e^{(\beta\mu+h)S}\,\mathrm{d}\mu = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} e^{NG(\mu)}\,\mathrm{d}\mu. \tag{1.5}$$

For large N the integral is dominated by the maximum of G, the *saddle point*, where $\mu = \tanh(\beta\mu + h)$. Expanding G to second order around each of its k maximizers μ^*, with $G''(\mu^*) = -\beta(1 - \beta(1 - \mu^{*2}))$, and integrating the Gaussians,

$$\frac{1}{N}\log Z_N = G(\mu^*) + \frac{1}{N}\log\frac{k}{\sqrt{1-\beta(1-\mu^{*2})}} + o\Big(\frac{1}{N}\Big).$$

The field derivative of the free entropy is the magnetization, $\langle m\rangle = \partial_h\phi = \tanh(\beta\mu^* + h) = \mu^*$. At $h = 0$, $m = \tanh(\beta m)$ has only the solution 0 for $\beta \le 1$;

for $\beta > 1$ the maxima move to $\pm m^*(\beta)$, and the ferromagnet magnetizes spontaneously. ◇

Two steps are uncontrolled. The saddle-point evaluation needs a non-degenerate maximum; at $\beta = 1$, $h = 0$ the maximum of G is quartic and the correction $-\frac{1}{2}\log(1-\beta)$ diverges (Section 1.6 measures what happens instead). And reading $\langle m \rangle$ off the saddle point exchanges the derivative in h with $N \to \infty$; at $h = 0$, $\beta > 1$ the exchange fails outright, since $\langle m \rangle = 0$ at every N by symmetry.

1.2.2 Quenched and annealed averages

For SK, $H_N(\sigma) = \frac{\beta}{\sqrt{N}} \sum_{i<j} g_{ij}\sigma_i\sigma_j + h\sum_i \sigma_i$ with independent standard Gaussian g_{ij}. At $h = 0$ each $H_N(\sigma)$ is centered Gaussian with variance $\frac{\beta^2}{N}\binom{N}{2}$, so the annealed free entropy takes one line:

$$\frac{1}{N}\log \mathbb{E} Z_N = \log 2 + \frac{\beta^2(N-1)}{4N} \longrightarrow \log 2 + \frac{\beta^2}{4}. \tag{1.6}$$

Physicists hold that only the quenched average is physical, because $\frac{1}{N}\log Z_N$ is *self-averaging*: cut a large system into blocks, and the free entropy is nearly a sum of independent block contributions, with fluctuations of order $\sqrt{N}$ against a mean of order N. (A mean-field model has no blocks; the belief rests instead on each coupling being small.) The annealed average can be dominated by rare samples: if $\log Z_N$ has Gaussian fluctuations of order $\sqrt{N}$, an excess t over its typical value has probability about $e^{-t^2/(cN)}$ and weight e^t in $\mathbb{E}Z_N$, and the product peaks at $t = cN/2$, of order N.

1.2.3 The replica trick

THE REPLICA TRICK. For a positive random variable, $\mathbb{E}\log Z = \lim_{n\to 0}(\mathbb{E}Z^n - 1)/n$. For integer n, Z^n is the partition function of n replicas with the same couplings, and the Gaussian average is explicit, $\mathbb{E}\exp(\sum_a H_N(\sigma^a)) = \exp(\frac{1}{2}\operatorname{Var}\sum_a H_N(\sigma^a))$, with $\operatorname{Cov}(H_N(\sigma), H_N(\tau)) = \frac{\beta^2}{2}(NR_{\sigma\tau}^2 - 1)$. For SK at $h = 0$, with $R_{ab} = \frac{1}{N}\sum_i \sigma_i^a\sigma_i^b$,

$$\mathbb{E}Z_N^n = e^{\frac{nN\beta^2}{4} - \frac{n^2\beta^2}{4}} \sum_{\sigma^1,\dots,\sigma^n} \exp\Big(\frac{N\beta^2}{2}\sum_{a<b} R_{ab}^2\Big). \tag{1.7}$$

The disorder average couples the replicas through their overlaps, and the overlap matrix becomes the order parameter. One Hubbard–Stratonovich variable Q_{ab} per pair decouples the sites, leaving the N-th power of a sum over one column $s \in \{\pm 1\}^n$: up to prefactors, $\mathbb{E}Z_N^n \approx \int e^{NS_n(Q)} \prod_{a<b} \mathrm{d}Q_{ab}$ with

$$S_n(Q) = \frac{n\beta^2}{4} - \frac{\beta^2}{2}\sum_{a<b} Q_{ab}^2 + \log \sum_{s\in\{\pm1\}^n} e^{\beta^2 \sum_{a<b} Q_{ab}s_as_b}.$$

Take the saddle point in Q as $N \to \infty$ at fixed n, and guess that it is *replica symmetric*, $Q_{ab} = q$. A last Gaussian identity gives

$$S_n = \frac{n\beta^2}{4} - \frac{n(n-1)\beta^2q^2}{4} - \frac{n\beta^2 q}{2} + \log \mathbb{E}\big(2\cosh(\beta\sqrt{q}\,Z)\big)^n,$$

in which every term makes sense for real n. As $n \to 0$, $S_n/n \to \log 2 + \mathbb{E}\log\cosh(\beta\sqrt{q}\,Z) + \frac{\beta^2}{4}(1-q)^2$, with $q = \mathbb{E}\tanh^2(\beta\sqrt{q}\,Z)$ at the stationary point: the formula of Sherrington and Kirkpatrick (1975). For $\beta \le 1$ the only solution is $q = 0$, which gives the annealed value (1.6). ◇

Four steps are uncontrolled. The moments are computed at integers and needed near $n = 0$, and integers do not determine an analytic function. The limits $N \to \infty$ and $n \to 0$ are exchanged. For $n < 1$ the number of pairs $n(n-1)/2$ is negative and the maximum over Q becomes a minimum, a rule with no counterpart in Laplace's method. And the ansatz is a guess: its entropy at zero temperature is $-1/(2\pi) < 0$, as Sherrington and Kirkpatrick (1975) noticed, and for $\beta > 1$ the answer needs Parisi's broken replica symmetry (Parisi 1979) (Chapter 3). At $n = 2$ the computation is exact.

COMPUTATION 1.1 · TWO REPLICAS

1 By (1.7) and $(\mathbb{E}Z_N)^2 = 4^N e^{\beta^2(N-1)/2}$, with $\rho_i = \sigma_i\tau_i$ independent uniform signs,

$$\frac{\mathbb{E}Z_N^2}{(\mathbb{E}Z_N)^2} = e^{-\beta^2/2}\,2^{-N}\sum_{\rho}\exp\Big(\frac{\beta^2}{2N}\Big(\sum_i \rho_i\Big)^2\Big):$$

the second moment of SK is a Curie–Weiss partition function at inverse temperature β^2.

2 By Theorem 1.4 and Proposition 1.5 below, $\frac{1}{N}\log\frac{\mathbb{E}Z_N^2}{(\mathbb{E}Z_N)^2} \to \phi(\beta^2, 0) - \log 2$, which is 0 for $\beta \le 1$ and positive for $\beta > 1$. For $\beta < 1$, Laplace's method with the prefactor (Theorem A.11 (ii)), applied to (1.5) at inverse temperature β^2, where G has the unique nondegenerate maximizer 0, gives the limit $e^{-\beta^2/2}(1-\beta^2)^{-1/2}$.

3 If $\log Z_N - \log \mathbb{E}Z_N$ were Gaussian with variance σ^2, the ratio would be e^{σ^2}: so $\sigma^2(\beta) = -\frac{1}{2}\log(1-\beta^2) - \frac{\beta^2}{2}$, the variance that Aizenman et al. (1987) proved for $\beta < 1$, $h = 0$. Numerically (`01-freeentropy-curieweiss.py`), at $\beta = 0.8$ the ratio is 1.1662 at $N = 16$ and 1.2056 at $N = 200$ (limit 1.2102); at $\beta = 1.5$ it is 842 and 2.3×10^{38}.

The posterior (1.2) is a special Gibbs measure, at the temperature that Bayes' rule dictates: in (1.3) the ferromagnetic coupling is the square of the random one. Nishimori (1981) singled out this line for random Ising models with a gauge argument,[3] and Iba (1999) read it as the line on which the Gibbs measure is the Bayesian posterior. There two replicas overlap as a replica and the truth do, $P(q) = P(m)$ in the physicist's notation, since the truth cannot be told apart from a posterior sample, and the derivative of the free entropy in the signal strength is an overlap; both statements are exact (Lemma 1.10 and Theorem 1.12).

[3] The transformation $J_{ij} \mapsto J_{ij}\tau_i\tau_j$, $\sigma_i \mapsto \sigma_i\tau_i$ leaves the Hamiltonian unchanged, and averaging over τ yields exact identities on the line, among them a closed formula for the internal energy.

1.3 The stone

Each row pairs a step of the derivation above with the step of the proofs below that does the same work.

PHYSICS · *saddle points and replicas*	PROOF · *large deviations and concentration*
1 · Count or linearize	
Hubbard–Stratonovich: $e^{\beta N m^2/2}$ is a Gaussian integral over a field μ; the spins decouple.	Group configurations by magnetization, $N+1$ classes of sizes $\binom{N}{k} \le e^{Ns(m)}$; μ returns as the Legendre dual of m (Lemma 1.3).
2 · Saddle point	
The integral is dominated by the maximum of G, where $\mu = \tanh(\beta\mu + h)$; Gaussian fluctuations give the $1/N$ term.	Laplace's method on types (Lemma 1.2): classes times the largest term above, a product measure below; error at most $\log(N+1)/N$, with no nondegeneracy needed.
3 · Order parameter	
Read $\langle m \rangle = \partial_h \phi = \mu^*$ off the saddle point; at $h = 0, \beta > 1$, pick one of two saddles.	Derivatives of convex functions converge where the limit is differentiable, here where the maximizer is unique (Corollary 1.6); at $h = 0, \beta > 1$ the limit has a corner.
4 · Self-averaging	
An extensive quantity is a sum of nearly independent block contributions, with fluctuations $O(\sqrt{N})$.	$\log Z_N$ is Lipschitz in the Gaussian couplings, with constant $\beta\sqrt{(N-1)/2}$ for SK; a covariance identity gives variance and tails at every temperature (Theorem 1.8).
5 · Quenched, not annealed	
Average $\log Z$, not Z: the annealed average is dominated by rare samples, right only at high temperature.	Jensen: $\phi_N \le \frac{1}{N} \log \mathbb{E} Z_N$. Equality in the limit needs a second moment, for SK a Curie–Weiss partition function at β^2 (Computation 1.1).
6 · Replicate	
Compute $\mathbb{E} Z_N^n$ at integer n; the replicas couple through R_{ab}, and a saddle point over the overlap matrix gives the limit.	Laplace's method on the 2^n column types proves the limit (Proposition 1.7); at real n near 0, $\lvert \frac{1}{nN} \log \mathbb{E} Z_N^n - \phi_N \rvert \le \lvert n \rvert \beta^2/4$ for every N (Proposition 1.9).

7 · Continue to $n \to 0$

Continue the integer-n answer under an ansatz, and let $n \to 0$ after $N \to \infty$.

No counterpart: the proof never continues in n. Integer moments need not determine $\mathbb{E} \log Z$ (Tanaka 2007); for SK they grow like e^{cNn^2}, and Carleman's criterion fails.

8 · Bayes-optimal posteriors

On Nishimori's line a gauge transformation gives exact identities: $P(q) = P(m)$; the derivative of the free entropy is an overlap.

Bayes' rule: given the data, the truth is one more posterior sample (Lemma 1.10); with integration by parts, $I'(\gamma) = \frac{1}{2}\mathrm{mmse}(\gamma)$ (Theorem 1.12).

Read down the right column, the stone shows what the proofs justify: the saddle point, completely and with bounds at every N; self-averaging, by an argument unlike the physicist's that holds at low temperature too; the integer moments; and the replica limit at real n near 0. The routes part at row 7: the physicist needs the moments near $n = 0$ and has them only at integers, so the continuation is where the derivation stops being a computation and becomes a guess, refined by an ansatz and checked by its consequences. The proof does without it, at a price: to obtain the SK free entropy at low temperature (Chapter 3) it needs an idea, interpolation, that the replica computation does not contain. In the last row the routes nearly coincide: the gauge argument and Bayes' rule are one symmetry seen from two sides.

1.4 The proof: Laplace's method and Gaussian concentration

1.4.1 The Curie–Weiss model

For a probability vector p on a finite set, $\mathrm{Ent}(p) = -\sum_a p_a \log p_a$ (with $0 \log 0 = 0$). For a word $x \in \mathcal{A}^N$ over a finite alphabet $\mathcal{A}$, the *type* $\hat{p}_x$ is the empirical distribution of its letters.

Lemma 1.2 (Laplace's method on types). *Let $\mathcal{A}$ have $k \geq 2$ elements, Δ be the set of probability vectors on $\mathcal{A}$, and $F \colon \Delta \to \mathbb{R}$ be convex and continuous. With $\Lambda = \max_{p \in \Delta}[\mathrm{Ent}(p) + F(p)]$, for every $N \geq 1$,*

$$\Lambda \leq \frac{1}{N} \log \sum_{x \in \mathcal{A}^N} e^{NF(\hat{p}_x)} \leq \Lambda + \frac{(k-1)\log(N+1)}{N}.$$

Proof. The plan is to group the words by type for the upper bound and to use the Gibbs variational principle with an i.i.d. measure for the lower bound. A type is fixed by the counts of $k-1$ letters, so there are at most $(N+1)^{k-1}$ types. Each word of type p has probability $\prod_a p_a^{Np_a} = e^{-N\,\mathrm{Ent}(p)}$ under $p^{\otimes N}$, so at most $e^{N\,\mathrm{Ent}(p)}$ words have type p, and the sum is at

most $(N+1)^{k-1}e^{N\Lambda}$. For the lower bound, Jensen's inequality for the concave logarithm gives, for every probability vector ν on a finite set and every function H on it, $\log\sum e^H \geq \log\sum_{\nu>0}\nu\, e^H/\nu \geq \sum \nu H + \mathrm{Ent}(\nu)$ (the Gibbs variational principle). Take $\nu = p^{\otimes N}$, of entropy $N\,\mathrm{Ent}(p)$, and $H(x) = NF(\hat{p}_x)$. Under ν the type has mean p, so by convexity $\sum_x \nu(x)F(\hat{p}_x) \geq F(p)$ and $\log\sum_x e^{NF(\hat{p}_x)} \geq N(\mathrm{Ent}(p)+F(p))$. Maximize over p. ∎

Write $s(m) = -\frac{1+m}{2}\log\frac{1+m}{2} - \frac{1-m}{2}\log\frac{1-m}{2}$ for the entropy of a ±1 variable with mean $m \in [-1,1]$.[4]

[4] The limit, without the bounds, also follows from Varadhan's lemma (Theorem A.11 (iii)): by Cramér's theorem the type of N independent uniform letters satisfies the large deviation principle with the good rate function $\log k - \mathrm{Ent}(p)$, and F is continuous and bounded above on the simplex. Convexity of F buys the exact lower bound.

Lemma 1.3 (Legendre duality). *For every $y \in \mathbb{R}$, $\max_{m\in[-1,1]}[s(m)+ym] = \log 2\cosh y$, attained only at $m = \tanh y$.*

Proof. On $(-1,1)$, $s'(m) = -\operatorname{artanh} m$, which tends to $\mp\infty$ at ±1, and s is strictly concave, so $s(m)+ym$ has the unique maximizer $\tanh y$. With $p = \frac{1+\tanh y}{2}$, $\log p = y - \log 2\cosh y$ and $\log(1-p) = -y - \log 2\cosh y$, so $s(\tanh y) = \log 2\cosh y - y\tanh y$. ∎

Theorem 1.4 (The Curie–Weiss free entropy). *Let $\beta \geq 0$, $h \in \mathbb{R}$, $Z_N(\beta,h) = \sum_{\sigma\in\{\pm1\}^N} e^{\frac{\beta}{2N}S^2+hS}$ with $S = \sum_i \sigma_i$, and*

$$\phi(\beta,h) = \max_{m\in[-1,1]} \Phi(m), \qquad \Phi(m) = s(m) + \frac{\beta}{2}m^2 + hm.$$

(i) For every $N \geq 1$, $\phi(\beta,h) \leq \frac{1}{N}\log Z_N(\beta,h) \leq \phi(\beta,h) + \frac{\log(N+1)}{N}$.

(ii) If $\beta > 0$, then $\phi(\beta,h) = \max_{\mu\in\mathbb{R}} G(\mu)$ with $G(\mu) = \log 2\cosh(\beta\mu+h) - \frac{\beta}{2}\mu^2$; Φ and G have the same maximizers, each in $(-1,1)$ and a solution of $m = \tanh(\beta m + h)$.

Proof. The plan is to read part 1 off Lemma 1.2 and part 2 off a function of both μ and m. With $\mathcal{A} = \{\pm1\}$ the type of σ is $(\frac{1+m}{2}, \frac{1-m}{2})$, $m = S/N$, of entropy $s(m)$, and $\frac{\beta}{2}m^2 + hm$ is convex in the type because m is linear in it and $\beta \geq 0$: this is part 1. For part 2 let $J(\mu,m) = s(m) + (\beta\mu+h)m - \frac{\beta}{2}\mu^2$. For fixed m it is maximized only at $\mu = m$, with value $\Phi(m)$; for fixed μ only at $m = \tanh(\beta\mu+h)$, with value $G(\mu)$ (Lemma 1.3). Both maxima exist (Φ is continuous on a compact set, $G \to -\infty$ at $\pm\infty$) and equal $\sup J$. If m^* maximizes Φ, then $J(m^*,m^*) = \sup J$, so m^* maximizes G, and uniqueness in m gives $m^* = \tanh(\beta m^* + h)$. If μ^* maximizes G and $m = \tanh(\beta\mu^*+h)$, then $J(\mu^*,m) = \sup J$, so m maximizes Φ, and uniqueness in μ gives $\mu^* = m$. ∎

So physics box (ii) holds, with error at most $\log(N+1)/N$ at every N, whatever the nature of the maximum; the lower bound is the free entropy of the best product measure, the *mean-field bound*.

Proposition 1.5 (The mean-field equation). *Let $\beta \geq 0$. If $h \neq 0$, Φ has a unique maximizer $m^*(\beta,h)$, of the sign of h. If $h = 0$ and $\beta \leq 1$, the unique*

maximizer is 0 *and* $\phi = \log 2$. *If* $h = 0$ *and* $\beta > 1$, *the maximizers are* $\pm m^*(\beta)$, *with* $m^*(\beta)$ *the unique solution in* $(0,1)$ *of* $m = \tanh(\beta m)$.

PROOF. The plan is to count the solutions of $m = \tanh(\beta m + h)$, which contain the maximizers by Theorem 1.4 (for $\beta = 0$, Φ is strictly concave with the single maximizer $\tanh h$). A concave u on $[0,1]$ with $u(0) > 0$, or strictly concave with $u(0) = 0$, has at most one zero in $(0,1]$: zeros $0 < z_1 < z_2$ would give $u(z_1) \geq (1 - \frac{z_1}{z_2})u(0) + \frac{z_1}{z_2}u(z_2)$, strictly in the second case, so $u(z_1) > 0$. If $h > 0$ (the case $h < 0$ is symmetric), $\Phi(m) > \Phi(-m)$ for $m > 0$ and $\Phi'(0) = h > 0$, so the maximizers are zeros in $(0,1)$ of $u(m) = \tanh(\beta m + h) - m$, concave on $[0,1]$ with $u(0) > 0 > u(1)$: there is exactly one. If $h = 0$, $\beta \leq 1$ and $m \neq 0$, then $|\tanh(\beta m)| < |m|$, so only 0 remains. If $h = 0$, $\beta > 1$, then $u(m) = \tanh(\beta m) - m$ is strictly concave with $u(0) = 0$, $u'(0) > 0 > u(1)$, so it has one zero m^* in $(0,1)$; the solution 0 is a local minimum since $\Phi''(0) = \beta - 1 > 0$, and Φ is even. ∎

Corollary 1.6 (Magnetization). *Let* $\beta \geq 0$ *and* $h \neq 0$. *Then the Gibbs average* $\langle m \rangle_N = \langle S/N \rangle$ *converges to* $m^*(\beta, h)$.

PROOF. The plan is to differentiate in h through convexity. The function $f_N(h) = \frac{1}{N} \log Z_N(\beta, h)$ is convex with $f_N'(h) = \langle m \rangle_N$, and $f_N \to \phi$. Since $\frac{f_N(h) - f_N(h-\delta)}{\delta} \leq f_N'(h) \leq \frac{f_N(h+\delta) - f_N(h)}{\delta}$ for $\delta > 0$, it suffices that ϕ is differentiable at h with derivative m^*. Write $\phi(h) = \max_m [\Phi_0(m) + hm]$ with $\Phi_0 = s + \frac{\beta}{2}m^2$. For $h' \neq h$ with a maximizer m', comparing values gives $\phi(h') - \phi(h) \geq (h' - h)m^*$ and $\phi(h) - \phi(h') \geq (h - h')m'$, so the difference quotient lies between m^* and m'. As $h' \to h$, every limit point of m' maximizes $\Phi_0(m) + hm$, because ϕ (convex) and Φ_0 are continuous, so it equals m^* (Proposition 1.5). ∎

At $h = 0$, $\beta > 1$ the same comparison gives $\partial_h \phi(\beta, 0\pm) = \pm m^*(\beta)$: the limit has a corner while $\langle m \rangle_N = 0$, and choosing a saddle point is choosing the order of the limits $h \to 0^\pm$ and $N \to \infty$. The lemma on types also proves the integer-moment step of physics box (iii).

Proposition 1.7 (Integer moments of SK). *Let* $h = 0$, $\beta \geq 0$, $n \geq 1$ *an integer and* $N \geq 2$. *Let* $\Lambda_n(\beta)$ *be the maximum of* $\mathrm{Ent}(p) + \frac{\beta^2}{2} \sum_{a<b} (\mathbb{E}_p s_a s_b)^2$ *over probability vectors* p *on* $\{\pm 1\}^n$, *with* $s \sim p$. *Then*

$$0 \leq \frac{1}{N} \log \mathbb{E} Z_N^n - \frac{n\beta^2}{4} + \frac{n^2\beta^2}{4N} - \Lambda_n(\beta) \leq \frac{(2^n - 1)\log(N+1)}{N}.$$

PROOF. Formula (1.7) is exact, since it uses only the Gaussian moment generating function. Let $\mathcal{A} = \{\pm 1\}^n$, and let $x_i = (\sigma_i^1, \ldots, \sigma_i^n)$ be the columns of the replicas. Then $R_{ab} = \mathbb{E}_{\hat{p}_x} s_a s_b$ is linear in the type, so the function $F(p)$ in the definition of Λ_n, without the entropy, is convex, and the sum in (1.7) is $\sum_x e^{NF(\hat{p}_x)}$; apply Lemma 1.2 with $k = 2^n$. ∎

For $n = 1$, $\Lambda_1 = \log 2$ gives the annealed value (1.6); for $n = 2$ the maximum reduces to the law of $s_1 s_2$, and $\Lambda_2 = \log 2 + \phi(\beta^2, 0)$, which is Computation 1.1. Exchanging maxima as in the proof of Theorem 1.4, with $\frac{\beta^2}{2} r^2 = \max_Q [\beta^2 Q r - \frac{\beta^2}{2} Q^2]$ for each pair, gives $\frac{n\beta^2}{4} + \Lambda_n = \max_Q S_n(Q)$: at integer n the physicist's saddle point is a maximum, and it is exact.

1.4.2 Gaussian concentration

A *Gaussian disordered system* is a Gibbs measure on a finite set Σ with weights $\pi(\sigma) > 0$ and a Hamiltonian $H(\sigma) = \langle g, a(\sigma)\rangle + b(\sigma)$ linear in a standard Gaussian vector $g \in \mathbb{R}^K$, so that $Z(g) = \sum_\sigma \pi(\sigma) e^{H(\sigma)}$; a and b are deterministic, or independent of g and then everything holds conditionally on them. Let $L = \max_\sigma \|a(\sigma)\|$. For SK, $a_{ij}(\sigma) = \frac{\beta}{\sqrt{N}} \sigma_i \sigma_j$ $(i < j)$ and $L^2 = \frac{\beta^2 (N-1)}{2}$; the posterior (1.2) has the same L with $\beta = \lambda$, given x^*.

The tool is the covariance identity of the appendix (Lemma A.5): for independent standard Gaussian vectors X, Y, with $X_t = tX + \sqrt{1-t^2}\, Y$, and twice continuously differentiable f_1, f_2 which, with their first and second derivatives, have moderate growth,

$$\operatorname{Cov}\big(f_1(X), f_2(X)\big) = \int_0^1 \mathbb{E}\langle \nabla f_1(X), \nabla f_2(X_t)\rangle \, \mathrm{d}t. \tag{1.8}$$

Theorem 1.8 (Concentration of the free entropy). *For a Gaussian disordered system, $F = \log Z$ satisfies, for every $\theta \in \mathbb{R}$ and $t \geq 0$,*

(i) $\operatorname{Var} F \leq \mathbb{E}\|\langle a\rangle\|^2 \leq L^2$;

(ii) $\mathbb{E} e^{\theta(F - \mathbb{E}F)} \leq e^{\theta^2 L^2/2}$;

(iii) $\mathbb{P}(|F - \mathbb{E}F| \geq t) \leq 2e^{-t^2/(2L^2)}$ *if* $L > 0$.

For SK, with every $\beta > 0$, $h \in \mathbb{R}$, $N \geq 2$ and $u \geq 0$,

$$\operatorname{Var} \log Z_N \leq \frac{\beta^2}{2}\big(N\mathbb{E}\left\langle R_{12}^2\right\rangle - 1\big) \leq \frac{\beta^2 (N-1)}{2},$$
$$\mathbb{P}\left(\left|\frac{1}{N}\log Z_N - \phi_N\right| \geq u\right) \leq 2e^{-\frac{N^2 u^2}{\beta^2 (N-1)}} \leq 2e^{-Nu^2/\beta^2}. \tag{1.9}$$

Proof. The plan is to apply (1.8) with $f_1 = f_2 = F$ for the variance and with $f_1 = F$, $f_2 = e^{\theta F}$ for a differential inequality on the moment generating function. Since $\partial_k F = \langle a_k\rangle$, $\|\nabla F\| = \|\langle a\rangle\| \leq \langle \|a\|\rangle \leq L$, and the Hessian of F, the Gibbs covariance matrix of a, is bounded; so $|F(x)| \leq |F(0)| + L\|x\|$, and F and $e^{\theta F}$ satisfy the growth conditions.

1. This is the Gaussian Poincaré inequality (Theorem A.6), which (1.8) with $f_1 = f_2 = F$ and Cauchy–Schwarz give at once, since $X_t \sim X$: $\operatorname{Var} F \leq \mathbb{E}\|\nabla F\|^2$.

2. Let $\bar F = F - \mathbb{E}F$ and $M(\theta) = \mathbb{E} e^{\theta \bar F}$. Since $\mathbb{E}\bar F = 0$, $M'(\theta)$ equals $\operatorname{Cov}(F, e^{\theta \bar F})$. The gradient of $e^{\theta \bar F}$ is $\theta e^{\theta \bar F} \nabla F$, and $\langle \nabla F(X), \nabla F(X_t)\rangle \leq L^2$, so for $\theta \geq 0$

$$M'(\theta) = \theta \int_0^1 \mathbb{E}\big[e^{\theta \bar F(X_t)} \langle \nabla F(X), \nabla F(X_t)\rangle\big] \, \mathrm{d}t \leq \theta L^2 M(\theta).$$

So $(\log M)' \le \theta L^2$ on $[0,\infty)$, and $M(0)=1$ gives $\log M(\theta) \le \theta^2 L^2/2$. The argument used only $\|\nabla F\| \le L$ and the growth conditions, which $-F$ satisfies too; this covers $\theta < 0$.

3. Markov's inequality gives $\mathbb{P}(\bar F \ge t) \le e^{-\theta t + \theta^2 L^2/2}$, which is $e^{-t^2/(2L^2)}$ at $\theta = t/L^2$; the same holds for $-\bar F$.

For SK, $\|\langle a\rangle\|^2 = \frac{\beta^2}{N}\sum_{i<j}\langle\sigma_i\sigma_j\rangle^2$, and the square of a Gibbs average is an average over two replicas: $\sum_{i<j}\langle\sigma_i\sigma_j\rangle^2 = \left\langle\sum_{i<j}\sigma_i^1\sigma_j^1\sigma_i^2\sigma_j^2\right\rangle = \frac12\langle N^2R_{12}^2 - N\rangle$. Since $R_{12}^2 \le 1$, this gives both variance bounds; the tail is part 3 with $t = Nu$. ∎

The tail (1.9) is summable in N, so by the Borel–Cantelli lemma $\frac1N\log Z_N - \phi_N \to 0$ almost surely, whatever the coupling of the disorders of different sizes. Since ϕ_N converges for SK at every β and h (Theorem 3.2), physics box (i) is a theorem.[5] Part 2 settles the replica limit at real n.

[5] Part 1 is a statement about replicas: the variance of $\log Z_N$ is controlled by the mean square overlap, which is of order $1/N$ when the overlap concentrates at 0, as at high temperature.

Proposition 1.9 (The replica free entropy at small n). *For a Gaussian disordered system and every real $n \neq 0$, $\frac1n\log\mathbb{E}Z^n - \mathbb{E}\log Z$ lies between 0 and $\frac{nL^2}{2}$. In particular $\mathbb{E}\log Z = \lim_{n\to0}\frac1n\log\mathbb{E}Z^n = \lim_{n\to0}\frac{\mathbb{E}Z^n-1}{n}$, and for SK $|\frac{1}{nN}\log\mathbb{E}Z_N^n - \phi_N| \le \frac{|n|\beta^2}{4}$ for every N.*

Proof. In the notation of the last proof, $\frac1n\log\mathbb{E}Z^n - \mathbb{E}\log Z = \frac1n\log M(n)$, and $0 \le \log M(n) \le \frac{n^2L^2}{2}$ by Jensen's inequality ($M(n) \ge e^{n\mathbb{E}\bar F} = 1$) and part 2. Since $\log\mathbb{E}Z^n \to 0$, $\frac{\mathbb{E}Z^n-1}{n} = \frac{e^{\log\mathbb{E}Z^n}-1}{\log\mathbb{E}Z^n}\cdot\frac{\log\mathbb{E}Z^n}{n} \to \mathbb{E}\log Z$. For SK, $L^2/N \le \beta^2/2$. ∎

The proposition is the rigorous content of the replica limit: at real n near 0 the replica free entropy is the quenched one, uniformly in N, so the limits $n\to0$ and $N\to\infty$ commute. It gives no way to obtain $\mathbb{E}Z_N^n$ at non-integer n from its values at integers. For integer $n \ge 1$, $Z_N^n \ge \sum_\sigma e^{nH_N(\sigma)}$, so for SK at $h=0$, $\mathbb{E}Z_N^n \ge 2^N e^{n^2\beta^2(N-1)/4}$. The moments grow like e^{cn^2}, so $\sum_n(\mathbb{E}Z_N^n)^{-1/(2n)} < \infty$, and *Carleman's condition*, the standard sufficient condition for moments to determine a law on $[0,\infty)$ (Schmüdgen 2017), fails at every $N \ge 2$, $\beta > 0$. Such growth is the signature of the lognormal law, which its moments do not determine (Heyde 1963) (Exercise 1.3). Tanaka (2007) built counterexamples of this kind, among them a random two-spin Ising model, in which the moments of Z do not determine $\mathbb{E}\log Z$, and concluded that the replica method cannot be justified unconditionally.

1.5 The proof: the scalar channel, the Nishimori identity and I-MMSE

The simplest posterior with a planted truth is that of one variable seen through Gaussian noise; the replica-symmetric formulas of Part II reduce high-dimensional posteriors to it, and the two identities proved here drive the proofs there.

Let $X \sim P_0$ with $\mathbb{E}X^2 < \infty$, $Z \sim \mathcal{N}(0,1)$ independent, and $Y = \sqrt{\gamma}\, X + Z$ with $\gamma \geq 0$. Against $Q = \mathcal{N}(0,1)$ the log-likelihood ratio is $\sqrt{\gamma}\, xy - \frac{\gamma}{2}x^2$, so in the notation of Definition 1.1

$$\mu_y(\mathrm{d}x) = \frac{e^{\sqrt{\gamma}xy - \gamma x^2/2}}{\mathcal{Z}_\gamma(y)} P_0(\mathrm{d}x), \qquad \psi(\gamma) = \mathbb{E}\log \mathcal{Z}_\gamma(Y),$$

with $\mathcal{Z}_\gamma(y) = \int e^{\sqrt{\gamma}xy - \gamma x^2/2} P_0(\mathrm{d}x)$. The posterior mean $\langle x \rangle$ minimizes the mean squared error, $\mathrm{mmse}(\gamma) = \mathbb{E}(X - \langle x \rangle)^2$, and with $I(\gamma) = I(X;Y)$, (1.4) reads

$$I(\gamma) = \frac{\gamma}{2}\mathbb{E}X^2 - \psi(\gamma). \tag{1.10}$$

For N independent copies the posterior is a product and the free entropy per variable is $\psi(\gamma)$. For the Rademacher prior $\langle x \rangle = \tanh(\sqrt{\gamma}Y)$ and $\psi(\gamma) = \mathbb{E}\log\cosh(\gamma + \sqrt{\gamma}Z) - \frac{\gamma}{2}$; for the standard Gaussian prior $\langle x \rangle = \sqrt{\gamma}Y/(1+\gamma)$, $\mathrm{mmse} = 1/(1+\gamma)$ and $I = \frac{1}{2}\log(1+\gamma)$.

Lemma 1.10 (Nishimori identity). *Let (X, Y) take values in Polish spaces, let μ_Y be a regular conditional law of X given Y, and given (X, Y) let $x^1, \dots, x^k$ be independent with law μ_Y. For every measurable $f \geq 0$, and every f for which one side is finite with $|f|$ in place of f,*

$$\mathbb{E}\left\langle f(Y, x^1, \dots, x^{k-1}, x^k)\right\rangle = \mathbb{E}\left\langle f(Y, x^1, \dots, x^{k-1}, X)\right\rangle.$$

Proof. The joint law of $(Y, X, x^1, \dots, x^k)$ is

$$P_Y(\mathrm{d}y)\, \mu_y(\mathrm{d}x)\, \mu_y(\mathrm{d}x^1) \cdots \mu_y(\mathrm{d}x^k):$$

given Y, the truth and the replicas are independent with the same law, so exchanging X and x^k leaves the joint law unchanged. ∎

Corollary 1.11 (First consequences). *In the scalar channel, $\mathbb{E}\left\langle x^2\right\rangle = \mathbb{E}X^2$, $\mathbb{E}[X\langle x \rangle] = \mathbb{E}\langle x \rangle^2$, and $\mathrm{mmse}(\gamma) = \mathbb{E}X^2 - \mathbb{E}\langle x \rangle^2$. For every prior on $\mathbb{R}^N$ and every data model, the overlap R_{12} of two replicas under $\mathbb{E}\langle\cdot\rangle$ has the law of the overlap m of a replica with the truth.*

Proof. Apply Lemma 1.10 with $k = 1$, $f = x^2$, then with $k = 2$, $f = x^1x^2$, which is integrable since $\mathbb{E}\left\langle |x^1x^2|\right\rangle = \mathbb{E}\langle |x| \rangle^2 \leq \mathbb{E}\left\langle x^2\right\rangle = \mathbb{E}X^2$, and expand $\mathbb{E}(X - \langle x \rangle)^2$. For the overlaps take $f = \mathbf{1}\{\frac{1}{N}\langle x^1, x^2\rangle \in B\}$ for Borel sets B. ∎

Theorem 1.12 (I-MMSE). *If P_0 has bounded support, then $I(\gamma) = \frac{1}{2}\int_0^\gamma \mathrm{mmse}(r)\, \mathrm{d}r$ for every $\gamma \geq 0$. In particular $I'(\gamma) = \frac{1}{2}\mathrm{mmse}(\gamma)$ for $\gamma > 0$, and from the right at 0; equivalently, $\psi'(\gamma) = \frac{1}{2}\mathbb{E}\langle x \rangle^2$.*

Proof. The plan is to substitute $Y = \sqrt{\gamma}X + Z$, so that the law of (X, Z) does not depend on γ, to differentiate, and to remove the noise by integration by parts and the Nishimori identity. Let $|x| \leq K$ on the support of P_0, and $\psi(\gamma) = \mathbb{E}\ell(\gamma)$ with

$$\ell(\gamma) = \log \int \exp\Big(\gamma X x + \sqrt{\gamma}Zx - \frac{\gamma}{2}x^2\Big) P_0(\mathrm{d}x).$$

For fixed (X, Z) and $\gamma > 0$, $\ell'(\gamma) = \left\langle Xx + \frac{Zx}{2\sqrt{\gamma}} - \frac{x^2}{2} \right\rangle$, which for $\gamma \geq \gamma_0 > 0$ is at most $\frac{3}{2}K^2 + \frac{K|Z|}{2\sqrt{\gamma_0}}$ in absolute value. By dominated convergence,

$$\psi'(\gamma) = \mathbb{E}[X\langle x\rangle] - \frac{1}{2}\mathbb{E}\left\langle x^2\right\rangle + \frac{1}{2\sqrt{\gamma}}\mathbb{E}[Z\langle x\rangle].$$

Given X, $\langle x\rangle$ is a smooth function of Z with $\partial_Z\langle x\rangle = \sqrt{\gamma}(\left\langle x^2\right\rangle - \langle x\rangle^2)$, bounded by $\sqrt{\gamma}K^2$, so Stein's identity (Theorem A.1) gives $\mathbb{E}[Z\langle x\rangle] = \sqrt{\gamma}\,\mathbb{E}[\left\langle x^2\right\rangle - \langle x\rangle^2]$. Hence $\psi'(\gamma) = \mathbb{E}[X\langle x\rangle] - \frac{1}{2}\mathbb{E}\langle x\rangle^2 = \frac{1}{2}\mathbb{E}\langle x\rangle^2$ by Corollary 1.11, and by (1.10) $I'(\gamma) = \frac{1}{2}(\mathbb{E}X^2 - \mathbb{E}\langle x\rangle^2) = \frac{1}{2}\mathrm{mmse}(\gamma)$. For fixed (X, Z), $\langle x\rangle$ and ℓ are continuous in $\gamma \geq 0$ and dominated as above, so mmse is continuous on $[0, \infty)$ and I is continuous at 0, with $I(0) = 0$. Integrate from ϵ to γ and let $\epsilon \to 0$. ∎

The Gaussian prior, which is unbounded, satisfies the theorem by the formulas above.[6] Guo et al. (2005) proved the relation for every input of finite variance, for vector inputs and in continuous time; their generality adds control of the posterior moments for unbounded support. The vector form $\frac{\mathrm{d}}{\mathrm{d}\gamma}I(X; \sqrt{\gamma}X + Z) = \frac{1}{2}\mathbb{E}\|X - \langle x\rangle\|^2$ has the same proof (Exercise 1.10); the interpolation proofs of Chapters 4 and 7 differentiate it, and there too the derivative of a free entropy in a signal-to-noise ratio is an overlap.

[6] For the Rademacher prior, Corollary 1.11 gives $\mathrm{mmse}(\gamma) = 1 - \mathbb{E}\tanh(\gamma + \sqrt{\gamma}Z)$ and the identity $\mathbb{E}\tanh(\gamma + \sqrt{\gamma}Z) = \mathbb{E}\tanh^2(\gamma + \sqrt{\gamma}Z)$, which is not apparent from the formulas.

1.6 Measuring it

The script `01-freeentropy-sk.py` enumerates SK at $N = 4, \ldots, 16$ for $\beta = 0.5, 0.8, 1.5$ at $h = 0$, in about a minute. In Figure 1.1(a), at $\beta = 0.5$ the standard deviation of $\frac{1}{N}\log Z_N$ falls like $1/N$, from 0.027 at $N = 4$ to 0.0082 at $N = 16$, as the variance of $\log Z_N$ climbs toward $\sigma^2(0.5) = 0.0188$ (0.0172 at $N = 16$); at $\beta = 1.5$ the variance keeps growing, from 0.53 to 1.53, as expected at low temperature.[7] The bound of Theorem 1.8 holds and is crude: at $N = 16$, $L^2 = 1.88$, 4.80 and 16.9 against variances of 0.017, 0.125 and 1.53. The intermediate bound $\frac{\beta^2}{2}(N\mathbb{E}\left\langle R_{12}^2\right\rangle - 1)$ of part 1, measured on the same samples, is 0.036, 0.30 and 4.5: within a factor of three, levelling off at $\beta = 0.5$ and growing roughly linearly in N at $\beta = 1.5$, where the overlap does not concentrate at 0.

[7] At $\beta = 0.8$ the approach is slower, 0.125 at $N = 16$ against the limit 0.191, an effect of the nearby critical point.

Panel (b) puts the quenched and annealed averages on one curve, at $n \to 0$ and $n = 1$. At $\beta = 0.5$ the curve is flat, from 0.7497 to 0.7519, and the Monte Carlo estimates agree with the exact integer moments. At $\beta = 1.5$ the estimate at $n = 1$, 1.213 with interval [1.202, 1.225], agrees with the exact 1.209, but at $n = 2$ and 3 the estimates 1.330 and 1.406 fall far below the exact 1.416 and 1.817, and even below the rigorous lower bounds $\frac{\log 2}{n} + \frac{n\beta^2(N-1)}{4N} = 1.378$ and 1.778. This is the physicist's warning about rare samples, measured: 3000 samples miss those that dominate $\mathbb{E}Z_N^2$, and resampling cannot reveal what was never sampled.

At $N = 16$, with 95% half-widths, $\phi_N = 0.7517 \pm 0.0009$ against the annealed 0.7517 at $\beta = 0.5$, where the Gaussian fluctuations of Computation 1.1, with mean $-\sigma^2/2$ so that $Z_N/\mathbb{E}Z_N$ keeps mean one, predict

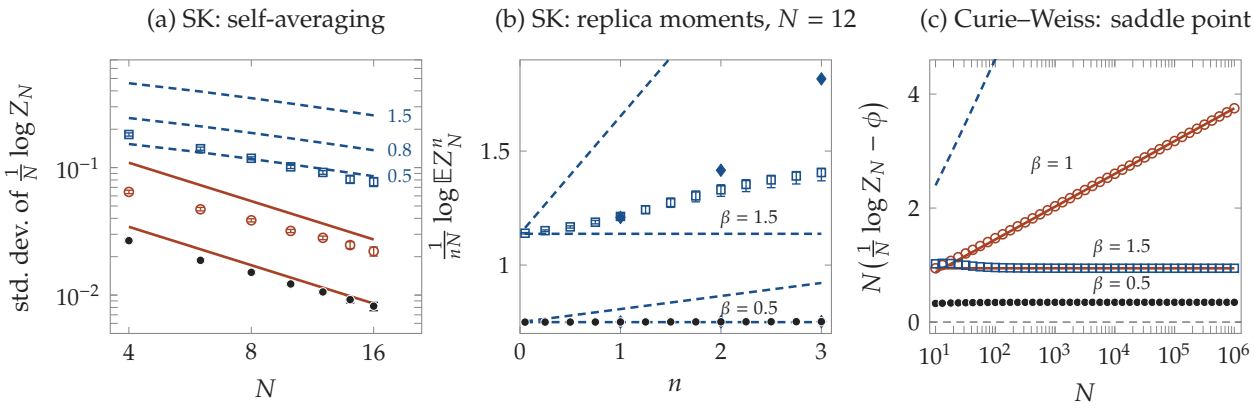


Figure 1.1. Exact computations against the predictions. (a) SK at $h = 0$ by exact enumeration of the 2^N configurations: standard deviation of $\frac{1}{N}\log Z_N$ over M disorder samples ($M = 4000$ for $N \le 8$, 3000 for $N = 10, 12$, 800 for $N = 14$, 300 for $N = 16$; seed $100 + N$) at $\beta = 0.5$ (dots), 0.8 (circles) and 1.5 (squares), with 95% bootstrap intervals (1000 resamples). Lapis dashed: the bound $\beta\sqrt{(N-1)/2}/N$ of Theorem 1.8, labeled by β; carnelian: the high-temperature prediction $\sigma(\beta)/N$ of Computation 1.1 at $\beta = 0.5, 0.8$. (b) Replica free entropy of SK at $N = 12$ from the 3000 samples of (a), Monte Carlo with 95% bootstrap intervals (400 resamples); lapis diamonds: exact values at $n = 1, 2, 3$ from (1.7); lapis dashed: the band $[\phi_N, \phi_N + n\beta^2(N-1)/(4N)]$ of Proposition 1.9, with ϕ_N from the samples. (c) The Curie–Weiss model at $h = 0$, exact sums over the $N + 1$ magnetizations at 41 sizes (no intervals). Lapis dashed: the bound $\log(N + 1)$ of Theorem 1.4, which leaves the panel at $N \approx 100$; carnelian: the saddle-point predictions $-\frac{1}{2}\log(1-\beta)$ at $\beta = 0.5$, $\log 2 - \frac{1}{2}\log(1 - \beta(1 - m^{*2}))$ (two saddles) at $\beta = 1.5$, and $\frac{1}{4}\log N + \log c$, $c = 2{\cdot}12^{1/4}\Gamma(\frac{5}{4})/\sqrt{2\pi}$, at $\beta = 1$. Scripts: `figures/scripts/01-freeentropy-sk.py`, `01-freeentropy-curieweiss.py`.

a deficit $\sigma^2(0.5)/(2N) = 0.0006$, below the resolution of this run; and 1.164 ± 0.009 against 1.220 at $\beta = 1.5$, where the limit lies below Guerra's replica-symmetric bound 1.2388 (Chapter 3).[8] Since $N\phi_N$ is superadditive (Theorem 3.2), each ϕ_N is a lower bound on the limit; the gap between ϕ_{16} and the one-step value, 0.075, is over three hundred times the difference between the replica-symmetric and one-step values, so enumeration cannot see replica symmetry breaking in the free entropy.

The script `01-freeentropy-curieweiss.py` computes the Curie–Weiss free entropy exactly up to $N = 10^6$ in seconds and checks that every value lies within the bounds of Theorem 1.4 (Figure 1.1(c)). The saddle-point corrections describe the data: at $\beta = 0.5$ the values 0.326, 0.344, 0.3466 at $N = 10, 10^2, 10^4$ approach $-\frac{1}{2}\log\frac{1}{2} = 0.3466$, and at $\beta = 1.5$, where $m^* = 0.8586$, the values 1.022, 0.957, 0.9440 approach 0.9438, which includes the $\log 2$ of the two saddles. At $\beta = 1$ the Gaussian correction is infinite and the data grow like $\frac{1}{4}\log N$: the maximum is quartic, the magnetization lives on the scale $N^{-1/4}$ (Ellis and Newman 1978), and $Z_N/2^N \sim cN^{1/4}$ with $c = 1.3460$ (Exercise 1.1), which matches the data to 0.0002 at $N = 10^6$.[9]

The script `01-freeentropy-channel.py` checks the scalar channel in ten

[8] The bound is attained at $q = 0.3526$. The one-step replica-symmetry-breaking value of the Parisi functional, also an upper bound, is lower by 2.2×10^{-4}, and the annealed limit is 1.2556.

[9] The script also checks Corollary 1.6: at $\beta = 1.5$, $h = 0.2$, $\langle m \rangle_N = 0.7958, 0.9143, 0.9178, 0.91814$ at $N = 10, 10^2, 10^3, 10^4$, against $m^* = 0.91817$.

	mmse(γ)			Bayes posterior		posterior at SNR 2γ	
γ	exact	Monte Carlo	via I-MMSE	$\mathbb{E}[X\langle x\rangle]$	$\mathbb{E}\langle x\rangle^2$	$\mathbb{E}[X\langle x\rangle']$	$\mathbb{E}\langle x\rangle'^2$
0.25	0.7959	0.7946(45)	0.7942(88)	0.2048	0.2042	0.2531	0.3131
0.5	0.6499	0.6538(50)	0.6527(85)	0.3479	0.3496	0.4052	0.4776
1	0.4496	0.4525(50)	0.4562(76)	0.5476	0.5478	0.6012	0.6647
2	0.2310	0.2347(41)	0.2329(61)	0.7671	0.7689	0.8010	0.8424
4	0.0686	0.0696(24)	0.0726(40)	0.9303	0.9301	0.9420	0.9552

Table 1.1. The scalar channel with the Rademacher prior. Exact: $1-\mathbb{E}\tanh(\gamma+\sqrt{\gamma}Z)$ by 161-node Gauss–Hermite quadrature. Monte Carlo: mean squared error of $\tanh(\sqrt{\gamma}Y)$ over 10^5 independent draws per γ (seed 1000γ), with the 95% half-width in units of the last digits. Via I-MMSE: $2(\frac{1}{2}-\psi'(\gamma))$, with ψ' a centered finite difference (step $10^{-3}\gamma$) of the log-evidence on the same draws. Last four columns: the two sides of $\mathbb{E}[X\langle x\rangle] = \mathbb{E}\langle x\rangle^2$ (95% half-widths at most 0.003) for the Bayes posterior and for a statistician who assumes the SNR 2γ. Script: `figures/scripts/01-freeentropy-channel.py`.

seconds (Table 1.1).[10] Every Monte Carlo interval for the mean squared error contains the exact value, and so does every interval for the mmse computed from the derivative of the log-evidence alone: two measurements, of the evidence and of the error, tied by Theorem 1.12. The same holds for the Gaussian prior (twenty intervals in all). The Nishimori identity holds for the Bayes posterior within the Monte Carlo error and fails by a wide margin for the mismatched posterior, whose error also exceeds the mmse (0.462 against 0.450 at $\gamma = 1$): the identity is a property of the true posterior. No measurement contradicts a theorem of the chapter.

[10] By quadrature, I' and $\frac{1}{2}$mmse agree to ten digits at $\gamma = 0.5$ and 2, and so do $\mathbb{E}\tanh$ and $\mathbb{E}\tanh^2$.

1.7 The status board

The status words are *proved* (a theorem in a refereed publication, or classical), *proved under* extra hypotheses that the board names, *preprint* (proved only in an unrefereed preprint), *conjectured* (with the evidence), *heuristic* (an argument that claims no exactness), *known false* (with the counterexample), and *open*. As of October 2026 this board is nearly all proved; the one step that is not a computation, the continuation in the number of replicas, is not, the annealed answer fails for $\beta > 1$, where the second moment grows exponentially, and the large deviations at low temperature rest on preprints.

STATUS BOARD · saddle points ↔ large deviations

PREDICTION	STATUS	PROOF OR EVIDENCE
Self-averaging of $\frac{1}{N}\log Z_N$ (Gaussian disorder)	proved	Gaussian concentration (Borell 1975; Cirel'son et al. 1976); Theorem 1.8
Curie–Weiss free entropy; $m = \tanh(\beta m + h)$	proved	classical (Ellis 1985); Theorem 1.4, error $\log(N+1)/N$

Existence of $\lim \phi_N$ for SK	proved	Guerra and Toninelli (2002); Theorem 3.2
Quenched = annealed for SK, $h = 0, \beta < 1$	proved	Aizenman et al. (1987); Chapter 3
Quenched = annealed for SK, $h = 0, \beta > 1$	false	Guerra's replica-symmetric bound is strictly smaller (Guerra 2003); Chapter 3
Replica moments: the limits at integer n; the limit $n \to 0$, uniformly in N	proved	Laplace's method (Varadhan 1966), Proposition 1.7; concentration, Proposition 1.9
Replica trick: integer moments continued to $n \to 0$	heuristic	no unconditional justification: counterexamples (Tanaka 2007); Carleman fails; the REM's exact moments are non-analytic (Ogure and Kabashima 2009); proof in Theorem 2.9 and Corollary 2.11
Upper tail $e^{-Na\delta^{6/5}}$ of $\frac{1}{N}\log Z_N$, SK, $h = 0, \beta > 1$	preprint	bounds of this order, and Var $\log Z_N$ between the orders $N^{4/15}$ and $N^{7/15}$: arXiv:2609.09103v1 (Aronow and Lopatto 2026), written, its authors state, with the assistance of large language models; a as $\delta \to 0$: arXiv:2610.06480v2 (Pei 2026); predicted by Parisi and Rizzo (2009)
Nishimori identity (Bayes form)	proved	Bayes' rule; Lemma 1.10; Lelarge and Miolane (2019, Proposition 16)
I-MMSE relation $I' = \frac{1}{2}\text{mmse}$	proved	Guo et al. (2005), finite variance; Theorem 1.12

1.8 Where the translation fails

WHAT WAS ASSUMED. The concentration theorem needs Gaussian disorder entering the Hamiltonian linearly with bounded coefficients; ±1 couplings give the same tail by bounded differences (Exercise 1.11), heavy tails need other tools, and a planted signal is covered given the truth, which centers $\log Z_N$ at its conditional mean; for (1.2) the symmetry of (1.3) makes that mean free of the truth. The Curie–Weiss theorem uses the complete-graph symmetry and $\beta \geq 0$ (Exercise 1.8 treats $\beta < 0$). The Nishimori identity needs the exact posterior: the statistician must know the prior, the channel and its signal-to-noise ratio. The I-MMSE relation needs additive Gaussian noise, and our proof bounded support. All limits are taken with $\beta, h, \lambda, \gamma$

fixed.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. The replica trick is not a theorem even at finite N: the identity $\mathbb{E}\log Z = \lim_{n\to 0}(\mathbb{E}Z^n - 1)/n$ holds (Proposition 1.9), but integer moments do not determine the moments near 0 (Tanaka 2007). In the random energy model the limit of the moments is known at every real $n > 0$ and, for $\beta > \beta_c$, is not analytic at $n = \beta_c/\beta$, so the integer formula continued to $n \to 0$ is wrong: Ogure and Kabashima (2009) derived the exact moment function, and Theorem 2.9 and Corollary 2.11 prove it. Annealed computations fail at low temperature, in the REM above β_c and in SK above $\beta = 1$, visibly so at $N = 12$ in Figure 1.1(b); the saddle-point correction fails at a critical point (panel (c)). And not every quantity self-averages: in the low-temperature phase of SK physicists predict that the overlap distribution of a single sample fluctuates from sample to sample (Mézard et al. 1987) (Chapter 3).

OPEN PROBLEMS. The order of the fluctuations of the free entropy at low temperature is open. For SK at $h = 0, \beta < 1$, $\log Z_N - \log \mathbb{E}Z_N$ has a Gaussian limit of order one (Aizenman et al. 1987). For $\beta > 1$, Theorem 1.8 gives $\operatorname{Var}\log Z_N = O(N)$, Chatterjee (2009) proved $o(N)$ in an arXiv preprint, and Aronow and Lopatto (2026), in an unrefereed preprint (arXiv:2609.09103v1) that its authors state was written with the assistance of large language models, prove that it lies between the orders $N^{4/15}$ and $N^{7/15}$. Replica theory predicts that $\frac{1}{N}\log Z_N$ exceeds its typical value by δ with probability $e^{-Na\delta^{6/5}}$ (Parisi and Rizzo 2009); the same preprint proves bounds of this order, and Pei (2026) (arXiv:2610.06480v2, unrefereed) identifies a in the limit $\delta \to 0$. Matching $Na\delta^{6/5}$ to one suggests a variance of order $N^{1/3}$, inside the window; this matching is our heuristic. More broadly, Tanaka (2007) notes that the replica method would be justified wherever the moment problem for Z_N is solved affirmatively; we know of no general class of disordered systems in which this is proved.

Exercises

1.1 CONSOLIDATION. (The critical point.) At $\beta = 1$, $h = 0$, use $s(m) = \log 2 - \frac{m^2}{2} - \frac{m^4}{12} + O(m^6)$ and the local central limit theorem to show $Z_N/2^N \sim cN^{1/4}$ with $c = 2\cdot 12^{1/4}\Gamma(\frac{5}{4})/\sqrt{2\pi} = 1.3460$; compare with Figure 1.1(c).

1.2 CONSOLIDATION. (Scalar channels.) For the standard Gaussian prior compute $\mathcal{Z}_\gamma$, $\langle x\rangle$, ψ, I and mmse, and check Theorem 1.12 and Corollary 1.11 directly; for the Rademacher prior derive $\psi(\gamma)$ and both expressions for the mmse.

1.3 CONSOLIDATION. (Heyde's lognormal family.) Let p_0 be the density of e^G, $G \sim \mathcal{N}(0,1)$, and $p_a(z) = p_0(z)(1 + a\sin(2\pi\log z))$, $|a| \le 1$. Show that each p_a has the moments $e^{k^2/2}$, $k = 0, 1, \dots$, and the mean logarithm $2\pi a e^{-2\pi^2} \approx 1.68\times 10^{-8} a$. What does the replica trick return? *(Hint:* $\mathbb{E}[e^{kG} f(G)] = e^{k^2/2}\mathbb{E}f(G+k)$.*)*

1.4 CONSOLIDATION. (A corner.) For the Curie–Weiss model at $h = 0$, $\beta > 1$, show that $\partial_h \phi(\beta, 0\pm) = \pm m^*(\beta)$ and $\langle |m| \rangle_N \to m^*(\beta)$. *(Hint: bound the sum over types with $||m| - m^*| \geq \epsilon$ as in Lemma 1.2.)*

1.5 BY HAND. (Three replicas.) Show that $\Lambda_3(\beta) \geq \max(3\log 2, \log 2 + \frac{3}{2}\beta^2)$, using the uniform law of the column and the law with mass $\frac{1}{2}$ on each of $\pm(1,1,1)$; the second is larger exactly when $\beta > (\frac{4}{3}\log 2)^{1/2} = 0.9614$. At $\beta = 1.5$ compare the lower bound $\frac{1}{3}(\frac{3}{4}\beta^2 + \log 2 + \frac{3}{2}\beta^2) = 1.9185$ on the limit with the exact 1.8167 at $N = 12$, using the error terms of Proposition 1.7.

1.6 BY HAND. (The random-field Curie–Weiss model.) Let $H_N = \frac{\beta}{2N} S^2 + \sum_i h_i \sigma_i$ with $\beta > 0$ and h_i independent copies of a bounded h. (a) Rewrite (1.5) with $\sum_i \log 2\cosh(\beta\mu + h_i)$ in place of $N \log 2\cosh(\beta\mu + h)$, and show with the law of large numbers that, almost surely, $\frac{1}{N}\log Z_N$ tends to $\max_\mu [\mathbb{E}\log 2\cosh(\beta\mu + h) - \frac{\beta}{2}\mu^2]$. (b) Compute $\mathbb{E}Z_N^n$ with one auxiliary field per replica, take the fields equal, and check that $n \to 0$ gives (a).

1.7 BY HAND. (Mismatched posteriors.) For the standard Gaussian prior, a statistician who assumes the SNR γ' uses $\hat{x} = \sqrt{\gamma'} Y/(1+\gamma')$ and posterior variance $1/(1+\gamma')$. Show that, for $\gamma' > 0$, $\mathbb{E}[X\hat{x}] = \mathbb{E}\hat{x}^2$ exactly when $\gamma' = \gamma$ or $\gamma\gamma' = 1$, while the posterior second moment has mean $\mathbb{E}X^2$ only when $\gamma' = \gamma$.

1.8 TRANSLATION. (A saddle point that is a minimum.) For $\beta < 0$ show that $\phi + \frac{\beta}{2N} \leq \frac{1}{N}\log Z_N \leq \phi + \frac{\log(N+1)}{N}$, with $\phi = \max_m \Phi(m)$ as in Theorem 1.4, that the integral (1.5) diverges, and, by the minimax theorem, that $\phi = \min_\mu [\log 2\cosh(\beta\mu + h) - \frac{\beta}{2}\mu^2]$. Compare with the replica saddle point for $n < 1$.

1.9 TRANSLATION. ★ (Quenched equals annealed from two moments.) Suppose a Gaussian disordered system of size N has $L^2 \leq cN$ and $\mathbb{E}Z_N^2 \leq C(\mathbb{E}Z_N)^2$ for all N. From the Paley–Zygmund inequality $\mathbb{P}(Z \geq \frac{1}{2}\mathbb{E}Z) \geq (\mathbb{E}Z)^2/(4\mathbb{E}Z^2)$ and Theorem 1.8, show $\mathbb{E}\log Z_N \geq \log \mathbb{E}Z_N - \log 2 - L\sqrt{2\log(4C)}$, and deduce from Computation 1.1 that $\phi = \log 2 + \beta^2/4$ for SK at $h = 0$, $\beta < 1$. *(Hint: argue by contradiction on the event $\{Z_N \geq \frac{1}{2}\mathbb{E}Z_N\}$.)*

1.10 TRANSLATION. (The vector I-MMSE relation.) For a prior on a bounded subset of $\mathbb{R}^N$ and $Y = \sqrt{\gamma} X + Z$ with $Z \sim \mathcal{N}(0, I_N)$, adapt the proof of Theorem 1.12 to show $\frac{\mathrm{d}}{\mathrm{d}\gamma} I(X;Y) = \frac{1}{2}\mathbb{E}\|X - \langle x \rangle\|^2$, and write the derivative of the free entropy as an overlap.

1.11 MEASUREMENT. (Rademacher couplings.) (a) For SK with g_{ij} uniform on $\{\pm 1\}$, show by the bounded-differences inequality that (1.9) still holds. (b) Modify `01-freeentropy-sk.py` accordingly: at $N = 12$, $\beta = 0.5$, the variance of $\log Z_N$ drops from about 0.016 to about 0.002, while ϕ_N barely moves. Explain the drop by the term $\frac{\beta^2}{2N}\sum_{i<j}(g_{ij}^2 - 1)$,

present only for Gaussian couplings, whose variance is $\beta^4(N-1)/(4N)$.

1.12 RESEARCH. (The low-temperature exponent.) Explain why matching $Na\delta^{6/5}$ to one suggests $\operatorname{Var}\log Z_N \asymp N^{1/3}$ for SK at $h = 0$, $\beta > 1$. Estimate the exponent by enumeration and Monte Carlo, and explain why accessible sizes cannot separate $\frac{4}{15}$, $\frac{1}{3}$ and $\frac{7}{15}$. Determining it is open.

Notes and References

Ellis and Newman (1978) proved limit theorems for the Curie–Weiss magnetization, with a non-Gaussian law for $S/N^{3/4}$ at the critical point; Ellis (1985) gives the large-deviation treatment, and Varadhan's lemma is Varadhan (1966). Gaussian concentration with the sharp constant is due to Cirel'son et al. (1976) and Borell (1975); the appendix proves it with a weaker constant and states the sharp one (Theorem A.7), and for log-partition functions the covariance identity (1.8), which goes back to Houdré and Pérez-Abreu (1995), gives the sharp constant directly, as in part 2 of Theorem 1.8. The proofs of Theorems 1.4 and 1.8 and Propositions 1.7 and 1.9 are written for this book from standard ingredients; we claim no novelty for them.

Replicas entered spin-glass theory with Edwards and Anderson (1975) and the solvable model of Sherrington and Kirkpatrick (1975); their failure at low temperature led to Parisi's broken replica symmetry (Mézard et al. 1987; Parisi 1979). The moments at real $n > 0$ describe large deviations of the free energy (Parisi and Rizzo 2009) and for $0 < n < 1$ were related by Talagrand (2007) to a Parisi problem with an atom of mass at least n at 0 (as Pei (2026) reports); Tanaka (2007) sets out the objections to the continuation, with the lognormal family of Heyde (1963) among his examples. The limit for SK exists by Guerra and Toninelli (2002), and Guerra's bounds (Guerra 2003) open the rigorous theory of Chapter 3.

The Nishimori line goes back to Nishimori (1981) and its Bayesian reading to Iba (1999), both developed in Nishimori (2001). The I-MMSE relation is Guo et al. (2005). Zdeborová and Krzakala (2016) build the statistical physics of inference on these identities, and Montanari and Sen (2024) explain replica computations to non-physicists.

TRANSLATING THE LITERATURE. Physicists write the Gibbs weight $e^{-\beta\mathcal{H}}$, with $T = 1/\beta$ and free energy per spin $f = -\phi/\beta$; Sherrington and Kirkpatrick (1975) take couplings of mean J_0/N and variance J^2/N, so our β is their βJ and their βJ_0 is our Curie–Weiss coupling. Talagrand (2011) writes $p_N(\beta, h)$ for our ϕ_N, $R_{1,2}$ for our R_{12}, and $\nu(f)$ for $\mathbb{E}\left\langle f \right\rangle$; Panchenko (2013) calls $\frac{1}{N}\mathbb{E}\log Z_N$ the free energy, without the sign and the factor $1/\beta$ of physics, and writes $\xi(q) = \beta^2 q^2$, so his β is our $\beta/\sqrt{2}$. Guo et al. (2005) write snr for our γ; Lelarge and Miolane (2019) write $Y = \sqrt{\lambda/n}\, XX^{\top} + Z$, so their λ is our λ^2.

References

Aizenman, M., J. L. Lebowitz, and D. Ruelle (1987). "Some rigorous results on the Sherrington-Kirkpatrick spin glass model". In: *Communications in Mathematical Physics* 112.1, pp. 3–20.

Aronow, P. M. and P. Lopatto (2026). *Quantitative Parisi formulas and fluctuations in the Sherrington-Kirkpatrick model*. arXiv preprint, version 1 (September 2026). arXiv: 2609.09103.

Borell, C. (1975). "The Brunn-Minkowski inequality in Gauss space". In: *Inventiones Mathematicae* 30.2, pp. 207–216.

Chatterjee, S. (2009). *Disorder chaos and multiple valleys in spin glasses*. arXiv preprint, version 4. arXiv: 0907.3381.

Cirel'son, B. S., I. A. Ibragimov, and V. N. Sudakov (1976). "Norms of Gaussian sample functions". In: *Proceedings of the Third Japan–USSR Symposium on Probability Theory*. Vol. 550. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, pp. 20–41.

Edwards, S. F. and P. W. Anderson (1975). "Theory of spin glasses". In: *Journal of Physics F: Metal Physics* 5.5, pp. 965–974.

Ellis, R. S. (1985). *Entropy, Large Deviations, and Statistical Mechanics*. Grundlehren der mathematischen Wissenschaften. New York: Springer.

Ellis, R. S. and C. M. Newman (1978). "Limit theorems for sums of dependent random variables occurring in statistical mechanics". In: *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 44.2, pp. 117–139.

Guerra, F. (2003). "Broken Replica Symmetry Bounds in the Mean Field Spin Glass Model". In: *Communications in Mathematical Physics* 233.1, pp. 1–12.

Guerra, F. and F. L. Toninelli (2002). "The Thermodynamic Limit in Mean Field Spin Glass Models". In: *Communications in Mathematical Physics* 230.1, pp. 71–79.

Guo, D., S. Shamai, and S. Verdú (2005). "Mutual Information and Minimum Mean-Square Error in Gaussian Channels". In: *IEEE Transactions on Information Theory* 51.4, pp. 1261–1282.

Heyde, C. C. (1963). "On a Property of the Lognormal Distribution". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 25.2, pp. 392–393.

Houdré, C. and V. Pérez-Abreu (1995). "Covariance Identities and Inequalities for Functionals on Wiener and Poisson Spaces". In: *The Annals of Probability* 23.1, pp. 400–419.

Hubbard, J. (1959). "Calculation of Partition Functions". In: *Physical Review Letters* 3.2, pp. 77–78.

Iba, Y. (1999). "The Nishimori line and Bayesian statistics". In: *Journal of Physics A: Mathematical and General* 32.21, pp. 3875–3888.

Lelarge, M. and L. Miolane (2019). "Fundamental limits of symmetric low-rank matrix estimation". In: *Probability Theory and Related Fields* 173.3-4, pp. 859–929.

Mézard, M., G. Parisi, and M. A. Virasoro (1987). *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications*. Vol. 9. World Scientific Lecture Notes in Physics. Singapore: World Scientific.

Montanari, A. and S. Sen (2024). "A Friendly Tutorial on Mean-Field Spin Glass Techniques for Non-Physicists". In: *Foundations and Trends in Machine Learning* 17.1, pp. 1–173. arXiv: 2204.02909.

Nishimori, H. (1981). "Internal Energy, Specific Heat and Correlation Function of the Bond-Random Ising Model". In: *Progress of Theoretical Physics* 66.4, pp. 1169–1181.

Nishimori, H. (2001). *Statistical Physics of Spin Glasses and Information Processing: An Introduction*. Vol. 111. International Series of Monographs on Physics. Oxford: Oxford University Press.

Ogure, K. and Y. Kabashima (2009). "On analyticity with respect to the replica number in random energy models: I. An exact expression for the moment of the partition function". In: *Journal of Statistical Mechanics: Theory and Experiment* 2009.3, P03010. arXiv: 0812.4654.

Panchenko, D. (2013). *The Sherrington-Kirkpatrick Model*. Springer Monographs in Mathematics. New York: Springer New York.

Parisi, G. (1979). "Infinite Number of Order Parameters for Spin-Glasses". In: *Physical Review Letters* 43.23, pp. 1754–1756.

Parisi, G. and T. Rizzo (2009). "Phase diagram and large deviations in the free energy of mean-field spin glasses". In: *Physical Review B* 79.13, p. 134205.

Pei, Y. R. (2026). *Upper deviations of the Sherrington-Kirkpatrick free energy: the fifth-order law and the onset of constrained Parisi measures*. arXiv preprint, version 2 (October 2026). arXiv: 2610.06480.

Schmüdgen, K. (2017). *The Moment Problem*. Graduate Texts in Mathematics. Cham: Springer.

Sherrington, D. and S. Kirkpatrick (1975). "Solvable Model of a Spin-Glass". In: *Physical Review Letters* 35.26, pp. 1792–1796.

Stratonovich, R. L. (1957). "On a method of calculating quantum distribution functions". In: *Doklady Akademii Nauk SSSR* 115. Translated in Soviet Physics Doklady 2 (1958) 416–419, pp. 1097–1100.

Talagrand, M. (2007). "Large Deviations, Guerra's and A.S.S. Schemes, and the Parisi Hypothesis". In: *Journal of Statistical Physics* 126.4-5, pp. 837–894.

Talagrand, M. (2011). *Mean Field Models for Spin Glasses: Volume I: Basic Examples*. Vol. 54. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Tanaka, T. (2007). "Moment Problem in Replica Method". In: *Interdisciplinary Information Sciences* 13.1, pp. 17–23.

Varadhan, S. R. S. (1966). "Asymptotic probabilities and differential equations". In: *Communications on Pure and Applied Mathematics* 19.3, pp. 261–286.

Zdeborová, L. and F. Krzakala (2016). "Statistical physics of inference: thresholds and algorithms". In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

2 The Random Energy Model

Replicas and the second moment

THE DICTIONARY —— replicas ↔ the second moment

IN THIS CHAPTER. Decode one of 2^N orthogonal codewords sent through Gaussian noise, and the posterior is a Gibbs measure on 2^N independent Gaussian energies: Derrida's random energy model, at an inverse temperature equal to the signal strength. Physics derives, by counting levels and by replicas, a free entropy that freezes at $\beta_c = \sqrt{2\log 2}$, one-step replica-symmetry breaking with Parisi parameter $m = \beta_c/\beta$, and a decoding threshold at β_c, which is Shannon's limit $E_b/N_0 = \ln 2$. We prove all of it with first and second moments, a fractional moment at $n = m$ and Gaussian integration by parts, together with Ogure and Kabashima's formula for $\lim N^{-1}\log \mathbb{E}Z_N^n$ at every real $n > 0$: the one-step formula gives it exactly, and it is not analytic at $n = m$, which is why the integer moments continue to the wrong answer. Exact computations up to $N = 1280$ and simulations up to $N = 20$ show how slowly the limits are reached; what remains open concerns the method, not the model.

2.1 The question: one message among 2^N

A transmitter sends N bits, a word $\sigma \in \{\pm 1\}^N$, through Gaussian noise. For a random codebook, with which Shannon proved his coding theorem, the scores of the wrong codewords are independent, and the posterior is a measure on exponentially many configurations with independent random weights (Mézard and Montanari 2009, ch. 6 and its notes). Here is a Gaussian version in which every step is exact.

ORTHOGONAL CODES. Give message σ the codeword $\lambda\sqrt{N}\, e_\sigma \in \mathbb{R}^{2^N}$, with (e_σ) an orthonormal basis (as in pulse-position modulation), and add independent $\mathcal{N}(0,1)$ noise to every coordinate. If σ^* is sent, the receiver observes the scores $y_\sigma = \lambda\sqrt{N}\,\mathbf{1}\{\sigma = \sigma^*\} + X_\sigma$, with X_σ i.i.d. $\mathcal{N}(0,1)$. Given $\sigma^* = \sigma$, the density of y is the standard Gaussian density times $\exp(\lambda\sqrt{N}\, y_\sigma - \lambda^2 N/2)$, so for a uniform prior the posterior is

$$\mathbb{P}(\sigma^* = \sigma \mid y) = \frac{e^{\lambda\sqrt{N}\, y_\sigma}}{\sum_\tau e^{\lambda\sqrt{N}\, y_\tau}}, \tag{2.1}$$

a Gibbs measure on 2^N configurations whose scores, all but one, are independent standard Gaussians, at an inverse temperature equal to the signal strength, as on the Nishimori line of Chapter 4.[1] The energy is λ^2 per

[1] Decoding with $\mathbb{P}(\sigma) \propto e^{\beta\sqrt{N} y_\sigma}$ at some other β is the finite-temperature decoding of Mézard and Montanari (2009, ch. 6); Exercise 2.15 works out its phase diagram for orthogonal codes.

bit and the noise variance per dimension is $1 = N_0/2$, so the signal-to-noise ratio per bit is $E_b/N_0 = \lambda^2/2$; maximum a posteriori decoding picks the largest score. Without the planted score, what is left is the model that Derrida (1980, 1981) introduced as the limit $p \to \infty$ of the p-spin glasses.

Definition 2.1 (The random energy model). Let $(X_\sigma)_{\sigma\in\{\pm1\}^N}$ be i.i.d. $\mathcal{N}(0,1)$. The *random energy model* (REM) at inverse temperature $\beta \geq 0$ has partition function and Gibbs measure

$$Z_N(\beta) = \sum_{\sigma\in\{\pm1\}^N} e^{\beta\sqrt{N}X_\sigma}, \qquad G_N(\sigma) = \frac{e^{\beta\sqrt{N}X_\sigma}}{Z_N(\beta)},$$

and free entropy $\phi_N(\beta) = \frac{1}{N}\mathbb{E}\log Z_N(\beta)$. The *level* of σ is $X_\sigma/\sqrt{N}$, minus its energy per spin. We write $\langle f \rangle = \sum_\sigma f(\sigma)G_N(\sigma)$ and $Y_N = \sum_\sigma G_N(\sigma)^2$ for the *participation ratio*, the probability that two independent samples from G_N (two replicas) coincide.

No spins interact: the only structure is the number 2^N of levels and their variance N.[2] Throughout, $\beta_c = \sqrt{2\log 2}$, so that $2^N = e^{N\beta_c^2/2}$, and $m = \beta_c/\beta$.

[2] In the language of Chapter 1, the REM is the Gaussian disordered system with covariance $\mathbb{E}[\sqrt{N}X_\sigma\sqrt{N}X_\tau] = N\mathbf{1}\{\sigma=\tau\}$, the extreme case in which distinct configurations are uncorrelated.

THE PHYSICAL PREDICTION · Derrida 1980, 1981; Gross and Mézard 1984. The free entropy of the REM is

$$\phi(\beta) = \begin{cases} \log 2 + \dfrac{\beta^2}{2}, & \beta \leq \beta_c, \\ \beta\beta_c, & \beta \geq \beta_c, \end{cases} \qquad \beta_c = \sqrt{2\log 2}. \tag{2.2}$$

At β_c the model freezes: its entropy $\log 2 - \beta^2/2$ reaches zero and stays there, the energy per spin sticks at the ground-state value $-\beta_c$, and the Gibbs measure condenses on a few configurations. Below the critical temperature replica symmetry is broken at one step, with Parisi parameter $m = \beta_c/\beta$: two replicas sit in the same configuration with probability $1-m$ and are otherwise unrelated, $P(q) = m\,\delta_0 + (1-m)\,\delta_1$.

For decoding the box predicts a threshold. At $\beta = \lambda$ the planted score contributes $e^{\lambda^2 N}$ to the posterior's partition function and the others $e^{N\phi(\lambda)}$, and $\lambda^2 > \phi(\lambda)$ exactly when $\lambda > \beta_c$: decoding should succeed when the REM at the decoder's temperature is frozen, that is, when $E_b/N_0 > \ln 2$. That is Shannon's limit for unlimited bandwidth: the capacity $W\log_2(1 + P/(N_0W))$ bits per second (Shannon 1949) increases to $P/(N_0\ln 2)$ as the bandwidth W grows, and at the rate P/E_b this is $E_b/N_0 \geq \ln 2$, about -1.59 dB.

How do physicists derive the box, by counting levels as Derrida did and by replicas as Gross and Mézard did? And what is proved: the free entropy, the meaning of the Parisi parameter, the decoding threshold, and what the exact replica moments $\mathbb{E}Z_N^n$ say about the replica trick?

2.2 The physics: counting levels and replicating

2.2.1 Derrida's level count

DERRIDA'S MICROCANONICAL SOLUTION. Count the configurations at level about u. Each level falls near u with probability about $e^{-Nu^2/2}$, so their expected number is about $e^{Ns(u)}$, with the *entropy* $s(u) = \log 2 - u^2/2$, positive for $|u| < \beta_c$ and negative beyond. Assume that the typical number of levels near u equals its mean when the mean is exponentially large, and is zero when the mean is exponentially small. Then, by Laplace's method,

$$Z_N(\beta) \approx \int_{-\beta_c}^{\beta_c} e^{N[s(u)+\beta u]}\,\mathrm{d}u, \qquad \phi(\beta) = \max_{|u|\le\beta_c}\big[s(u)+\beta u\big].$$

The exponent is stationary at $u = \beta$. If $\beta \le \beta_c$, then $\phi = \log 2 + \beta^2/2$ and the Gibbs measure lives at level β, on about $e^{Ns(\beta)}$ configurations. If $\beta > \beta_c$, the maximum sits at the edge $u = \beta_c$, where $s = 0$, and $\phi = \beta\beta_c$: the Gibbs measure lives on the deepest levels, $e^{o(N)}$ of them, and the entropy is zero. ◇

Two steps are uncontrolled: "typical equals mean" for the random counts, and Laplace's method on a sum whose terms fluctuate. The second derivative ϕ'' jumps from 1 to 0 at β_c, a continuous transition; because the Gibbs measure condenses there, it is often called a random first-order transition (Mézard and Montanari 2009, ch. 5).[3]

[3] The specific heat per spin, $\beta^2\phi''(\beta)$, jumps from $\beta_c^2 = 2\log 2$ to 0; in Derrida's units it jumps from $J^2/(2T_c^2)$ to 0.

2.2.2 Replicas: the integer moments

The replica method computes $\mathbb{E}Z_N^n$ for integer n, continues the answer to real n, and uses $\mathbb{E}\log Z_N = \lim_{n\to 0}\frac{1}{n}\log \mathbb{E}Z_N^n$. For the REM the integer moments are an exercise in counting.

COMPUTATION 2.1 · THE REPLICATED PARTITION FUNCTION OF THE REM

1 *Replicate and average.* For a positive integer n, $Z_N^n = \sum_{\sigma^1,\dots,\sigma^n}\exp(\beta\sqrt{N}\sum_a X_{\sigma^a})$, and the variance of $\sum_a X_{\sigma^a}$ is $\sum_{a,b} Q_{ab}$, with the *overlap matrix* $Q_{ab} = \mathbf{1}\{\sigma^a = \sigma^b\}$; so $\mathbb{E}Z_N^n = \sum_{\sigma^1,\dots,\sigma^n}\exp(\frac{\beta^2 N}{2}\sum_{a,b} Q_{ab})$.

2 *Group.* The matrix Q is the equivalence relation of a partition π of the replicas into blocks of equal configurations; the blocks receive distinct configurations, in $(2^N)_{|\pi|} = 2^N(2^N-1)\cdots(2^N-|\pi|+1)$ ways, and $\sum_{a,b} Q_{ab} = \sum_{B\in\pi}|B|^2$. Hence, exactly at every N,

$$\mathbb{E}Z_N^n = \sum_{\pi}(2^N)_{|\pi|}\exp\Big(\frac{\beta^2 N}{2}\sum_{B\in\pi}|B|^2\Big). \tag{2.3}$$

3 *Saddle point.* There are finitely many partitions, so $\frac{1}{N}\log\mathbb{E}Z_N^n \to \max_\pi\big[|\pi|\log 2 + \frac{\beta^2}{2}\sum_B|B|^2\big]$. With k blocks, $\sum_B|B|^2$ is largest for one block of size $n-k+1$ and $k-1$ singletons, a convex function of

k, so the maximum is at $k = n$ (replicas apart) or $k = 1$ (together):

$$\lim_{N\to\infty} \frac{1}{N} \log \mathbb{E} Z_N^n = \max\Big(n\big(\log 2 + \tfrac{\beta^2}{2}\big),\ \log 2 + \tfrac{n^2\beta^2}{2} \Big).$$

The branches cross at $n = m^2$; above β_c every integer $n \geq 2$ lies on the second.

REPLICA SYMMETRY AND ITS CRISIS. The two partitions of step 3 are the two *replica-symmetric* saddle points, $Q_{ab} = \delta_{ab}$ and $Q_{ab} = 1$. The first, continued to $n \to 0$, gives the annealed free entropy $\log 2 + \beta^2/2$ at every temperature, with entropy $\phi - \beta\phi' = \log 2 - \beta^2/2$, negative for $\beta > \beta_c$, which no system with finitely many configurations can have: replica symmetry must fail below the critical temperature. The second, continued, gives $(\log 2 + n^2\beta^2/2)/n \to \infty$. For $n < 1$ the two exchange roles, and the replica method prescribes the smaller one; Mézard and Montanari (2009, ch. 8) stress that nothing within the method justifies this choice.

ONE STEP OF REPLICA-SYMMETRY BREAKING. Parisi's ansatz divides the n replicas into n/m' groups of m', with equal configurations within a group and distinct ones across groups. Then $|\pi| = n/m'$ and $\sum_B |B|^2 = nm'$, so

$$g_{\mathrm{1RSB}}(n, m') = n\Big(\frac{\log 2}{m'} + \frac{\beta^2 m'}{2} \Big). \tag{2.4}$$

Continued in n, the integer constraint $1 \leq m' \leq n$ is read as $n \leq m' \leq 1$, over which g is minimized (Mézard and Montanari 2009, ch. 8). The minimum over $m' > 0$ is at $m' = \beta_c/\beta = m$, with $\frac{1}{n}\, g_{\mathrm{1RSB}}(n, m) = \beta\beta_c$; it lies in $(0, 1]$ exactly when $\beta \geq \beta_c$, and for $\beta < \beta_c$ the constrained minimum is the replica-symmetric $m' = 1$. Both phases of (2.2) come out, with zero entropy below T_c, and so does the overlap distribution: the fraction of pairs of replicas in a common group is $\frac{(n/m)m(m-1)}{n(n-1)} \to 1 - m$ as $n \to 0$. Writing Y_N as the limit $n \to 0$ of $Z_N^{n-2} \sum_\sigma e^{2\beta\sqrt{N} X_\sigma}$, the same bookkeeping gives $\mathbb{E} Y_N \to 1 - m$, the condensation formula, which first appeared in the replica computation of Gross and Mézard (1984) (Mézard and Montanari 2009, notes to ch. 5).

2.2.3 Replicas at finite n

The one-step formula makes sense at every real $n > 0$, not only as $n \to 0$. With Parisi's rule, minimize over $m' \in [n, 1]$ when $n < 1$ and maximize over $m' \in [1, n]$ when $n > 1$:

$$\varphi_{\mathrm{1RSB}}(n) = \begin{cases} \min_{n \leq m' \leq 1} g_{\mathrm{1RSB}}(n, m'), & 0 < n \leq 1, \\ \max_{1 \leq m' \leq n} g_{\mathrm{1RSB}}(n, m'), & n \geq 1. \end{cases} \tag{2.5}$$

Since $g_{\mathrm{1RSB}}(n, \cdot)$ is convex with its minimum at $m' = m$, for $\beta > \beta_c$ this is $n\beta\beta_c$ (at $m' = m$) when $n \leq m$ and $\log 2 + n^2\beta^2/2$ (at $m' = n$, one group holding all the replicas) when $m < n \leq 1$; for $n \geq 1$ the maximum sits at an endpoint and reproduces Computation 2.1. The quadratic branch describes rare samples, in which one level sits at the atypical depth $u = n\beta > \beta_c$, an

event of probability about $e^{-N(u^2/2-\log 2)}$; Mézard and Montanari (2009, chs. 5, 8) read $\lim N^{-1}\log \mathbb{E}Z_N^n$ as a Legendre transform of the large deviations of the free energy, and Ogure and Kabashima (2009a, App. A) obtained this function, in both phases, by steepest descent from an exact integral representation of $\mathbb{E}Z_N^n$.

That is the failure of the naive replica trick: above β_c every integer moment lies on the quadratic branch, whose interpolation $\log 2 + n^2\beta^2/2$ gives $\varphi(n)/n \to \infty$ as $n \to 0$. Integer moments could not fix the continuation anyway: they grow like $\mathbb{E}Z_N^n \geq e^{N\beta^2 n^2/2}$, too fast for Carleman's criterion and for Carlson's uniqueness theorem (Ogure and Kabashima 2009a), and adding $a\sin(\pi n)$ changes no integer value (Tanaka 2007).[4] One-step breaking selects one continuation, (2.5), justified by its consequences: positive entropy, the right free entropy at both ends, a participation ratio in $[0,1]$. The uncontrolled steps are: typical equals mean; the continuation in n; the ansatz; the minimum over $[n,1]$; and the exchange of $n \to 0$ with $N \to \infty$.

[4] Tanaka (2007) goes further and exhibits distributions with the same integer moments and different means of the logarithm, among them a two-spin random Ising model.

2.3 The stone

PHYSICS · *level counting and replicas*

1 · Count levels

About $e^{Ns(u)}$ levels near u when $s(u) > 0$, none when $s(u) < 0$; Laplace's method over u.

2 · Replicate

$\mathbb{E}Z_N^n$ is a sum over partitions of the replicas, (2.3); the saddle point picks one partition.

3 · Replica symmetry and its crisis

All replicas apart: the annealed free entropy $\log 2 + \beta^2/2$, whose entropy $\log 2 - \beta^2/2$ is negative below T_c, impossible with 2^N states.

4 · One-step breaking

Groups of m' replicas; minimizing $n(\frac{\log 2}{m'} + \frac{\beta^2 m'}{2})$ over $m' \in [n,1]$ gives $m = \beta_c/\beta$ and $\phi = \beta\beta_c$.

PROOF · *the moment method*

First moment: $\mathbb{P}(K_N(u) \geq 1) \leq \frac{1}{2}e^{Ns(u)}$ removes the levels with $s < 0$. Second moment: $\operatorname{Var} K_N(u) \leq \mathbb{E}K_N(u)$, and Chebyshev makes the count typical when $s > 0$ (Lemma 2.3).

Only $n = 1$ and $n = 2$ appear, as inequalities: Jensen's bound by $\mathbb{E}Z_N$, and the mean and variance of the level count.

Jensen: $\phi_N \leq \log 2 + \beta^2/2$, with equality in the limit when $\beta \leq \beta_c$ (Theorem 2.4); no level lies above $\beta_c\sqrt{N}$, and the Gibbs entropy tends to $\max(\log 2 - \beta^2/2, 0)$ (Corollary 2.7).

A fractional moment: $\phi_N \leq \frac{1}{Nm}\log \mathbb{E}Z_N^m$, and truncation at the extreme level $\beta_c\sqrt{N}$ gives $\mathbb{E}Z_N^m \leq \frac{3}{2}e^{Nm\beta\beta_c}$; the lower bound counts levels just below β_c.

5 · Parisi parameter

$P(q) = m\delta_0 + (1-m)\delta_1$; the weights fluctuate from sample to sample.	$\mathbb{E}Y_N = 1 - \phi_N'(\beta)/\beta \to 1-m$ by Gaussian integration by parts and convexity (Corollary 2.7); the law $PD(m,0)$ of the weights is a theorem (Theorem 2.12).

6 · $n \to 0$

Continue in n and exchange $n \to 0$ with $N \to \infty$.	No continuation: the exact $\varphi(n)$ (Theorem 2.9) is linear up to $n = m$ and not analytic there, so the integers fix the wrong analytic function; the limits do commute.

The proof follows Derrida's count, not the replicas, and the first and second moments justify the count exactly: for independent levels, "typical equals mean" is Chebyshev's inequality and "no level where the mean is small" is Markov's. Where the physics extremizes over the group size m', the proof optimizes a fractional moment, and the optimal exponent is the Parisi parameter, the largest n at which the bound $\phi_N \le \frac{1}{Nn}\log \mathbb{E}Z_N^n$ is sharp in the limit; where the physics reads $1-m$ off the replica matrix, the proof reads it off the derivative of the free entropy.[5] One physical step has no counterpart: the proof never continues in n. Instead it computes $\lim N^{-1}\log \mathbb{E}Z_N^n$ at every real n (Theorem 2.9), which explains both the failure and the success of the replica trick.

[5] In the SK model of Chapter 3 the same exponent appears in Guerra's broken-symmetry bound, as the power m in terms of the form $\frac{1}{m}\log \mathbb{E}(\cdot)^m$, and Ruelle's probability cascades take the place of the Poisson–Dirichlet weights.

2.4 The proof: first and second moments

2.4.1 Gaussian tails and level counts

Lemma 2.2 (Gaussian tails). *Let $X \sim \mathcal{N}(0,1)$, and write $H(x) = \mathbb{P}(X > x)$ throughout.*

(i) For $x \ge 0$, $H(x) \le \frac{1}{2}e^{-x^2/2}$; for $x > 0$, $H(x) \ge \frac{x}{1+x^2}\frac{e^{-x^2/2}}{\sqrt{2\pi}}$.

(ii) For real t and a, $\mathbb{E}[e^{tX}\mathbf{1}\{X \le a\}] = e^{t^2/2}\big(1 - H(a-t)\big)$ and $\mathbb{E}[e^{tX}\mathbf{1}\{X > a\}] = e^{t^2/2}H(a-t)$.

Proof. For the upper bound of part 1, substitute $y = x+v$ and use $e^{-xv} \le 1$:

$$H(x) = \int_0^\infty e^{-(x+v)^2/2}\frac{\mathrm{d}v}{\sqrt{2\pi}} \le e^{-x^2/2}\int_0^\infty e^{-v^2/2}\frac{\mathrm{d}v}{\sqrt{2\pi}} = \frac{1}{2}e^{-x^2/2}.$$

For the lower bound, $\frac{\mathrm{d}}{\mathrm{d}y}\big(-y^{-1}e^{-y^2/2}\big) = (1+y^{-2})e^{-y^2/2}$, so $x^{-1}e^{-x^2/2} = \int_x^\infty (1+y^{-2})e^{-y^2/2}\,\mathrm{d}y \le (1+x^{-2})\sqrt{2\pi}\,H(x)$. Part 2 follows from $e^{tx-x^2/2} = e^{t^2/2}e^{-(x-t)^2/2}$, integrated over $x \le a$ and over $x > a$. ∎

For a level u let $K_N(u) = \#\{\sigma : X_\sigma > u\sqrt{N}\}$, the number of configurations above level u, and $\mu_N(u) = \mathbb{E}K_N(u) = 2^N H(u\sqrt{N})$ its mean; recall Derrida's entropy $s(u) = \log 2 - u^2/2$.

Lemma 2.3 (First and second moments of the level counts). *For every $N \geq 1$ and real u:*

(i) (first moment) $\mathbb{P}(K_N(u) \geq 1) \leq \mu_N(u)$, *and* $\mu_N(u) \leq \frac{1}{2} e^{Ns(u)}$ *if* $u \geq 0$;

(ii) (second moment) $\operatorname{Var} K_N(u) \leq \mu_N(u)$, *hence* $\mathbb{P}\big(K_N(u) \leq \frac{1}{2}\mu_N(u)\big) \leq 4/\mu_N(u)$;

(iii) if $u > 0$, *then* $\frac{1}{N}\log \mu_N(u) \geq s(u) - \frac{1}{N}\log\big(\sqrt{2\pi}\,(u\sqrt{N} + u^{-1}N^{-1/2})\big)$, *so that* $\frac{1}{N}\log \mu_N(u) \to s(u)$;

(iv) $N^{-1/2}\max_\sigma X_\sigma \to \beta_c$ *almost surely.*

Proof. The plan is to treat $K_N(u)$ as a binomial count. It is a sum of 2^N independent indicators, each equal to 1 with probability $q = H(u\sqrt{N})$, so its mean is $2^N q$ and its variance $2^N q(1-q) \leq \mu_N(u)$. Part 1 is Markov's inequality, $\mathbb{P}(K \geq 1) \leq \mathbb{E}K$, with the upper bound of Lemma 2.2 and $2^N = e^{N\log 2}$. Part 2 is Chebyshev's inequality: $\mathbb{P}(K \leq \mu/2) \leq \mathbb{P}(|K-\mu| \geq \mu/2) \leq 4\operatorname{Var} K/\mu^2 \leq 4/\mu$. Part 3 is the lower bound of Lemma 2.2; with part 1 it gives the limit. For part 4 fix $0 < \varepsilon < \beta_c$. By part 1, $\mathbb{P}(\max_\sigma X_\sigma > (\beta_c+\varepsilon)\sqrt{N}) = \mathbb{P}(K_N(\beta_c+\varepsilon) \geq 1) \leq \frac{1}{2}e^{Ns(\beta_c+\varepsilon)}$, and $s(\beta_c+\varepsilon) < 0$. By part 2, $\mathbb{P}(\max_\sigma X_\sigma \leq (\beta_c-\varepsilon)\sqrt{N}) = \mathbb{P}(K_N(\beta_c-\varepsilon) = 0) \leq 4/\mu_N(\beta_c-\varepsilon)$, which decays exponentially by part 3 because $s(\beta_c-\varepsilon) > 0$. Both bounds are summable in N, and the Borel–Cantelli lemma gives the almost sure limit. ∎

2.4.2 The free entropy

Theorem 2.4 (The free entropy of the REM). *For every $\beta \geq 0$,*

$$\lim_{N\to\infty} \frac{1}{N}\log Z_N(\beta) = \phi(\beta) = \max_{0\leq u\leq \beta_c}\big[s(u) + \beta u\big] \quad \textit{almost surely and in } L^1,$$

with ϕ as in (2.2); in particular $\phi_N(\beta) \to \phi(\beta)$. Moreover, for every $N \geq 1$, $\phi_N(\beta) \leq \log 2 + \beta^2/2$, and if $\beta > \beta_c$, then

$$\phi_N(\beta) \leq \beta\beta_c + \frac{\log(3/2)}{mN}, \qquad m = \frac{\beta_c}{\beta}. \tag{2.6}$$

Proof. The plan is to reduce the claim to the mean ϕ_N by concentration, to bound the mean from above by Jensen's inequality, applied to Z_N below β_c and to the fractional power Z_N^m above it, and to bound it from below by counting levels. If $\beta = 0$, then $Z_N = 2^N$ and there is nothing to prove; let $\beta > 0$.

Step 1: concentration. Let $F(x) = \log\sum_\sigma e^{\beta\sqrt{N}x_\sigma}$ for $x \in \mathbb{R}^{2^N}$. For every y and σ, $e^{\beta\sqrt{N}x_\sigma} \leq e^{\beta\sqrt{N}\|x-y\|_\infty} e^{\beta\sqrt{N}y_\sigma}$; summing over σ and taking logarithms gives $|F(x)-F(y)| \leq \beta\sqrt{N}\|x-y\|_\infty \leq \beta\sqrt{N}\|x-y\|_2$. Gaussian concentration for Lipschitz functions (Theorem A.7) gives an absolute constant $c > 0$

($c = \tfrac12$ with the sharp constant) such that

$$\mathbb{P}\Big(\Big|\frac{1}{N}\log Z_N - \phi_N\Big| \ge t\Big) \le 2\exp\Big(-\frac{cNt^2}{\beta^2}\Big) \qquad (t>0).$$

The right side is summable in N, so $\frac{1}{N}\log Z_N - \phi_N \to 0$ almost surely by the Borel–Cantelli lemma, and integrating the tail gives $\mathbb{E}|\frac{1}{N}\log Z_N - \phi_N| \le \beta\sqrt{\pi/(cN)}$. It remains to show that $\phi_N \to \phi$.

Step 2: upper bounds. By Jensen's inequality, $N\phi_N \le \log \mathbb{E}Z_N = N(\log 2 + \beta^2/2)$, which is $N\phi(\beta)$ when $\beta \le \beta_c$. Let $\beta > \beta_c$. For $0 < n \le 1$, concavity of the logarithm gives $\mathbb{E}\log Z_N = \frac{1}{n}\mathbb{E}\log Z_N^n \le \frac{1}{n}\log \mathbb{E}Z_N^n$; we take $n = m$. Split $Z_N = Z_{\le} + Z_{>}$ according to whether $X_\sigma \le \beta_c\sqrt{N}$ or $X_\sigma > \beta_c\sqrt{N}$. The map $x \mapsto x^m$ is concave and subadditive on $[0,\infty)$, so

$$\mathbb{E}Z_N^m \le \mathbb{E}Z_{\le}^m + \mathbb{E}Z_{>}^m \le (\mathbb{E}Z_{\le})^m + \sum_\sigma \mathbb{E}\big[e^{m\beta\sqrt{N}X_\sigma}\mathbf{1}\{X_\sigma > \beta_c\sqrt{N}\}\big].$$

By part 2 of Lemma 2.2 with $t = \beta\sqrt{N}$ and $a = \beta_c\sqrt{N}$, then part 1, and $2^N = e^{N\beta_c^2/2}$,

$$\mathbb{E}Z_{\le} = 2^N e^{N\beta^2/2} H\big((\beta-\beta_c)\sqrt{N}\big) \le \tfrac12 e^{N[\beta_c^2+\beta^2-(\beta-\beta_c)^2]/2} = \tfrac12 e^{N\beta\beta_c}.$$

With $t = m\beta\sqrt{N} = \beta_c\sqrt{N}$ each term of the sum equals $e^{N\beta_c^2/2}H(0) = \frac12 e^{N\beta_c^2/2}$, and the sum is $\frac12 e^{N\beta_c^2} = \frac12 e^{Nm\beta\beta_c}$. Hence $\mathbb{E}Z_N^m \le (2^{-m} + \frac12)e^{Nm\beta\beta_c} \le \frac32 e^{Nm\beta\beta_c}$, and $\phi_N \le \frac{1}{Nm}\log \mathbb{E}Z_N^m$ gives (2.6).

Step 3: lower bound. Fix $u \in (0,\beta_c)$, write $\mu = \mu_N(u)$, and let $A = \{K_N(u) > \mu/2\}$. On A more than $\mu/2$ configurations have $\beta\sqrt{N}X_\sigma > N\beta u$, so $\log Z_N \ge \log(\mu/2) + N\beta u$. On every sample, Jensen's inequality for the uniform average over σ gives $\log Z_N \ge N\log 2 + \beta\sqrt{N}\,\bar X$, where $\bar X = 2^{-N}\sum_\sigma X_\sigma$ is $\mathcal{N}(0,2^{-N})$. For N so large that $\log(\mu/2) + N\beta u \ge 0$, part 2 of Lemma 2.3 gives $\mathbb{P}(A) \ge 1 - 4/\mu$, and therefore

$$\mathbb{E}\log Z_N \ge \Big(1-\frac{4}{\mu}\Big)\Big(\log\frac{\mu}{2} + N\beta u\Big) - \beta\sqrt{N}\,\mathbb{E}|\bar X|.$$

By parts 1 and 3 of Lemma 2.3, $\frac1N\log\mu \to s(u) > 0$, so $\mu \to \infty$ exponentially fast; and $\mathbb{E}|\bar X| \le 2^{-N/2}$. Dividing by N gives $\liminf_N \phi_N \ge s(u) + \beta u$ for every $u \in (0,\beta_c)$. The supremum of the right side over these u is $\phi(\beta)$: it is attained at $u = \beta$ if $\beta < \beta_c$ and approached as $u \uparrow \beta_c$ if $\beta \ge \beta_c$.

Step 4: conclusion. Steps 2 and 3 give $\phi_N \to \phi(\beta)$ for every β, and Step 1 transfers the limit to $\frac1N\log Z_N$, almost surely and in L^1. ∎

Remark 2.5 (Other numbers of levels). With M_N levels, $2^{N-1} \le M_N \le 2^N$, the proofs of Theorem 2.4 and of part 4 of Lemma 2.3 go through verbatim, with $\mu_N(u)$ replaced by $M_N H(u\sqrt{N}) \ge \frac12\mu_N(u)$. We use this for $M_N = 2^N - 1$.

2.4.3 Energy, entropy and condensation

Lemma 2.6 (Derivatives of convex limits). *Let f_N be convex and differentiable on an open interval I, and $f_N \to f$ pointwise on I. If f is differentiable at $\beta \in I$, then $f_N'(\beta) \to f'(\beta)$.*

Proof. For $h > 0$ with $\beta \pm h \in I$, convexity gives $\frac{f_N(\beta)-f_N(\beta-h)}{h} \le f_N'(\beta) \le \frac{f_N(\beta+h)-f_N(\beta)}{h}$. Let $N \to \infty$ and then $h \downarrow 0$. ∎

Corollary 2.7 (Energy, entropy and participation ratio). *For every $\beta > 0$, as $N \to \infty$,*

(i) $\frac{1}{\sqrt{N}}\mathbb{E}\langle X_\sigma\rangle \to \min(\beta, \beta_c)$*: the Gibbs measure lives at level* $\min(\beta, \beta_c)$*;*

(ii) $\frac{1}{N}\mathbb{E}S(G_N) \to \max\bigl(\log 2 - \frac{\beta^2}{2}, 0\bigr)$*, where* $S(G_N) = -\sum_\sigma G_N(\sigma)\log G_N(\sigma)$*;*

(iii) $\mathbb{E}Y_N \to \max\bigl(1 - \frac{\beta_c}{\beta}, 0\bigr)$.

Proof. The plan is to read all three statements off $\phi_N'(\beta)$, which converges by convexity, after computing that derivative a second time by Gaussian integration by parts. For each sample, $\beta \mapsto \log Z_N(\beta)$ is the logarithm of a sum of exponentials of linear functions of β, hence convex, and its derivative $\sqrt{N}\langle X_\sigma\rangle$ is bounded by the integrable $\sqrt{N}\max_\sigma|X_\sigma|$. So ϕ_N is convex and, by dominated convergence, differentiable with $\phi_N'(\beta) = N^{-1/2}\mathbb{E}\langle X_\sigma\rangle$. The limit ϕ is differentiable on $(0,\infty)$ with $\phi'(\beta) = \min(\beta, \beta_c)$, since at β_c both one-sided derivatives equal β_c. Lemma 2.6 and Theorem 2.4 give part 1. For part 2, $-\log G_N(\sigma) = \log Z_N - \beta\sqrt{N}X_\sigma$, so $S(G_N) = \log Z_N - \beta\sqrt{N}\langle X_\sigma\rangle$ and $\frac{1}{N}\mathbb{E}S(G_N) = \phi_N - \beta\phi_N' \to \phi - \beta\phi'$, which is $\log 2 - \beta^2/2$ for $\beta \le \beta_c$ and 0 above. For part 3, $G_N(\sigma)$ is a smooth function of the Gaussian vector (X_τ) with the bounded derivative $\partial G_N(\sigma)/\partial X_\sigma = \beta\sqrt{N}\bigl(G_N(\sigma) - G_N(\sigma)^2\bigr)$, so Gaussian integration by parts (Theorem A.1) gives

$$\mathbb{E}\langle X_\sigma\rangle = \sum_\sigma \mathbb{E}\bigl[X_\sigma G_N(\sigma)\bigr] = \beta\sqrt{N}\sum_\sigma \mathbb{E}\bigl[G_N(\sigma) - G_N(\sigma)^2\bigr] = \beta\sqrt{N}\,(1 - \mathbb{E}Y_N).$$

Thus $\phi_N'(\beta) = \beta(1 - \mathbb{E}Y_N)$, and $\mathbb{E}Y_N = 1 - \phi_N'(\beta)/\beta \to 1 - \min(\beta,\beta_c)/\beta$. ∎

Part 2 is the entropy crisis as a theorem, and part 3 says that the physicist's $P(1) = 1 - m$ is the mean participation ratio.[6] It does not say that Y_N concentrates, and Y_N does not: in the limit it is random, with the law that Ruelle's theorem describes (Theorem 2.12).

[6] In terms of the spin-glass overlap $R_{12} = \frac{1}{N}\langle\sigma^1, \sigma^2\rangle$ of two samples, part 3 reads $\mathbb{E}\langle R_{12}^2\rangle \to \max(1 - \beta_c/\beta, 0)$: distinct configurations among the deepest levels are, by symmetry, nearly orthogonal (Exercise 2.7).

2.4.4 Decoding orthogonal codes

Corollary 2.8 (The decoding threshold). *Let σ^* be uniform on $\{\pm1\}^N$, let $y_\sigma = \lambda\sqrt{N}\,\mathbf{1}\{\sigma = \sigma^*\} + X_\sigma$ with $\lambda > 0$, and let $\hat\sigma = \arg\max_\sigma y_\sigma$ be the maximum a posteriori decoder.*

(i) If $\lambda > \beta_c$, then $\mathbb{P}(\hat\sigma = \sigma^*) \to 1$ and $\mathbb{P}(\sigma^* \mid y) \to 1$ in probability; if

$\lambda < \beta_c$, *both tend to* 0.

(ii) The mutual information satisfies $\frac{1}{N} I(\sigma^*; y) \to \min(\lambda^2/2, \log 2)$.

Proof. The plan is to split the posterior's partition function $Z(y) = \sum_\sigma e^{\lambda\sqrt{N} y_\sigma}$ into the planted term $e^{\lambda\sqrt{N} y_{\sigma^*}}$ and a REM with $2^N - 1$ levels, $Z' = \sum_{\sigma \neq \sigma^*} e^{\lambda\sqrt{N} X_\sigma}$. By Remark 2.5, $N^{-1/2} \max_{\sigma\neq\sigma^*} X_\sigma \to \beta_c$ and $\frac{1}{N}\log Z' \to \phi(\lambda)$ almost surely, the second also in L^1; and $N^{-1/2} y_{\sigma^*} = \lambda + N^{-1/2} X_{\sigma^*} \to \lambda$.

Part 1. Decoding succeeds exactly when $y_{\sigma^*} > \max_{\sigma\neq\sigma^*} X_\sigma$; divided by $\sqrt{N}$, the two sides tend to λ and β_c. By (2.1), $\mathbb{P}(\sigma^* \mid y) = \big(1 + Z' e^{-\lambda\sqrt{N} y_{\sigma^*}}\big)^{-1}$, and $\frac{1}{N}\log\big(Z' e^{-\lambda\sqrt{N} y_{\sigma^*}}\big) \to \phi(\lambda) - \lambda^2$, which is $\lambda\beta_c - \lambda^2 < 0$ if $\lambda > \beta_c$ and $\log 2 - \lambda^2/2 > 0$ if $\lambda < \beta_c$.

Part 2. The density of y given σ divided by its average over the uniform prior is $e^{\lambda\sqrt{N} y_\sigma}/(2^{-N} Z(y))$, and $\mathbb{E}[\lambda\sqrt{N} y_{\sigma^*}] = \lambda^2 N$, so

$$I(\sigma^*; y) = \mathbb{E}\log \frac{2^N e^{\lambda\sqrt{N} y_{\sigma^*}}}{Z(y)} = N(\log 2 + \lambda^2) - \mathbb{E}\log Z(y).$$

With $a = \lambda\sqrt{N} y_{\sigma^*}$ and $b = \log Z'$, $\max(a,b) \le \log Z(y) \le \log 2 + \max(a,b)$. Now $a/N \to \lambda^2$ and $b/N \to \phi(\lambda)$ in L^1, and $|\max(a,b) - \max(a',b')| \le |a-a'| + |b-b'|$, so $\frac{1}{N}\mathbb{E}\log Z(y) \to \max(\lambda^2, \phi(\lambda))$. For $\lambda \ge \beta_c$ the maximum is λ^2 and $I/N \to \log 2$; for $\lambda < \beta_c$ it is $\log 2 + \lambda^2/2$ and $I/N \to \lambda^2/2$. ∎

Below the threshold the observation carries $\lambda^2/2$ nats per bit, the most that a code with energy λ^2 per bit can carry over this channel ($1/N_0$ nats per unit energy), and above it all N bits: orthogonal codes reach the capacity of the channel without bandwidth constraint.[7] The two decoding phases are the two phases of the REM at $\beta = \lambda$: above β_c the planted level is deeper than every other and the frozen posterior sits on it; below, the $e^{N(\log 2 - \lambda^2/2)}$ wrong codewords at level λ outweigh it.

[7] The cost is the bandwidth: N bits use 2^N dimensions. Decoding itself is a scan of 2^N scores, linear in the size of the received signal; for random binary codes of rate R the same scan takes time exponential in the block length.

2.5 The proof: the replica moments, exactly

The replica computation of Section 2.2.3 makes a prediction at every real $n > 0$, the function that Ogure and Kabashima found by steepest descent. For the REM it can be proved.

Theorem 2.9 (The replica moments of the REM). *Let* $\beta > 0$, $m = \beta_c/\beta$ *and* $n_c = \max(m, m^2)$. *For every real* $n > 0$ *the limit* $\varphi(n) = \lim_{N\to\infty} \frac{1}{N}\log \mathbb{E} Z_N(\beta)^n$ *exists, and*

$$\varphi(n) = \begin{cases} n\,\phi(\beta), & 0 < n \le n_c, \\ \log 2 + \dfrac{n^2\beta^2}{2}, & n \ge n_c. \end{cases} \tag{2.7}$$

For $n > 1$ the proof compares $\mathbb{E} Z_N^n$ with $\mathbb{E} Z_N^{n-1}$.

Lemma 2.10 (A recursion for the moments). *For every real $n \geq 1$ and every N,*

$$\mathbb{E}Z_N^n \leq c_n\big(2^N e^{Nn^2\beta^2/2} + \mathbb{E}Z_N \cdot \mathbb{E}Z_N^{n-1}\big), \qquad c_n = \max(1, 2^{n-2}).$$

Proof. Write $w_\sigma = e^{\beta\sqrt{N}X_\sigma}$ and $Z_{-\sigma} = Z_N - w_\sigma$, and let $\nu = n - 1 \geq 0$. For $a, b \geq 0$, $(a+b)^\nu \leq c_n(a^\nu + b^\nu)$: if $\nu \leq 1$ by subadditivity, and if $\nu \geq 1$ by convexity, $(\frac{a+b}{2})^\nu \leq \frac{1}{2}(a^\nu + b^\nu)$. Hence

$$Z_N^n = \sum_\sigma w_\sigma (w_\sigma + Z_{-\sigma})^\nu \leq c_n \sum_\sigma \big(w_\sigma^n + w_\sigma Z_{-\sigma}^\nu\big).$$

Since w_σ is independent of $Z_{-\sigma}$ and $Z_{-\sigma} \leq Z_N$, $\mathbb{E}[w_\sigma Z_{-\sigma}^\nu] \leq \mathbb{E}w_\sigma\, \mathbb{E}Z_N^\nu$. Summing over σ, with $\mathbb{E}w_\sigma^n = e^{Nn^2\beta^2/2}$ and $\sum_\sigma \mathbb{E}w_\sigma = \mathbb{E}Z_N$, gives the claim. ∎

Proof of Theorem 2.9. The plan is to prove matching bounds on $\varphi_N(n) = \frac{1}{N}\log \mathbb{E}Z_N^n$ at exponential scale: lower bounds from typical samples and from samples with one deep level, upper bounds by the truncation of Theorem 2.4 for $0 < n \leq 1$ and by induction from Lemma 2.10 for $n > 1$. Write $\phi_a = \log 2 + \beta^2/2$ for the annealed free entropy, $f(n) = \log 2 + n^2\beta^2/2$, and $\varphi(n)$ for the right side of (2.7). Since $m^2\beta^2 = \beta_c^2 = 2\log 2$,

$$f(n) - n\phi_a = \frac{\beta^2}{2}(n-1)(n-m^2). \tag{2.8}$$

Step 1: lower bounds. Typical samples first: for $\varepsilon > 0$ the event $A_\varepsilon = \{\frac{1}{N}\log Z_N \geq \phi - \varepsilon\}$ has probability tending to 1 by Theorem 2.4, and $\mathbb{E}Z_N^n \geq e^{nN(\phi-\varepsilon)}\mathbb{P}(A_\varepsilon)$, so $\liminf_N \varphi_N(n) \geq n\phi(\beta)$. Then one deep level: for $u \geq \beta_c$, $Z_N^n \geq e^{nN\beta u}\mathbf{1}\{K_N(u) \geq 1\}$. The levels are independent, so $\mathbb{P}(K_N(u) = 0) = (1 - H(u\sqrt{N}))^{2^N} \leq e^{-\mu}$ with $\mu = \mu_N(u)$, and since $\mu \leq \frac{1}{2}$ by part 1 of Lemma 2.3, $\mathbb{P}(K_N(u) \geq 1) \geq 1 - e^{-\mu} \geq \mu - \frac{1}{2}\mu^2 \geq \frac{1}{2}\mu$. With part 3 of Lemma 2.3, $\liminf_N \varphi_N(n) \geq n\beta u + s(u)$, and the choice $u = \max(n\beta, \beta_c)$ gives $\Lambda(n)$, equal to $f(n)$ if $n\beta \geq \beta_c$ and to $n\beta\beta_c$ otherwise. So $\liminf_N \varphi_N(n) \geq \max(n\phi(\beta), \Lambda(n))$, and this maximum is $\varphi(n)$: if $\beta > \beta_c$, then $\Lambda(n) \geq n\beta\beta_c = n\phi$ and $\Lambda = \varphi$; if $\beta \leq \beta_c$, then $\phi = \phi_a$, and (2.8) shows that $\max(n\phi_a, \Lambda(n))$ is $n\phi_a$ up to $n = m^2$ and $f(n)$ beyond.

Step 2: upper bounds for $0 < n \leq 1$. If $\beta \leq \beta_c$, Jensen's inequality for the concave map $x \mapsto x^n$ gives $\mathbb{E}Z_N^n \leq (\mathbb{E}Z_N)^n = e^{Nn\phi_a}$. If $\beta > \beta_c$ and $n\beta \geq \beta_c$, subadditivity gives $\mathbb{E}Z_N^n \leq \sum_\sigma \mathbb{E}e^{n\beta\sqrt{N}X_\sigma} = e^{Nf(n)}$. If $\beta > \beta_c$ and $n\beta < \beta_c$, the truncation of Step 2 of the proof of Theorem 2.4, with Lemma 2.2, gives

$$\begin{aligned}\mathbb{E}Z_N^n &\leq (\mathbb{E}Z_{\leq})^n + 2^N e^{Nn^2\beta^2/2} H\big((\beta_c - n\beta)\sqrt{N}\big) \\ &\leq 2^{-n} e^{Nn\beta\beta_c} + \tfrac{1}{2} e^{N[\beta_c^2 + n^2\beta^2 - (\beta_c - n\beta)^2]/2} \leq \tfrac{3}{2} e^{Nn\beta\beta_c}.\end{aligned}$$

In each case $\limsup_N \varphi_N(n) \leq \varphi(n)$.

Step 3: $n > 1$. For $n > 1$, (2.8) shows that $\varphi(n) = \max(n\phi_a, f(n))$. Suppose that $\limsup_N \varphi_N \leq \varphi$ on $(0, k-1]$ for an integer $k \geq 2$, which

Step 2 gives for $k = 2$, and let $n \in (k-1, k]$ and $\nu = n - 1$. By Lemma 2.10 and $\log(a+b) \le \log 2 + \max(\log a, \log b)$,

$$\limsup_N \varphi_N(n) \le \max\bigl(f(n),\ \phi_a + \varphi(\nu)\bigr).$$

If $\varphi(\nu) = \nu\phi(\beta)$, then $\phi_a + \varphi(\nu) \le n\phi_a \le \varphi(n)$, because $\phi \le \phi_a$. Otherwise $\nu \ge n_c$ and $\varphi(\nu) = f(\nu)$, and $\phi_a + f(\nu) = f(n) + \log 2 - \nu\beta^2 < f(n)$, because $\nu\beta^2 \ge n_c\beta^2 = \max(\beta\beta_c, \beta_c^2) \ge 2\log 2$. In both cases $\limsup_N \varphi_N(n) \le \varphi(n)$, which completes the induction; with Step 1 the theorem follows. ∎

Corollary 2.11 (What the moments say about replicas). *Let $\beta > 0$.*

(i) $\varphi(n)/n = \phi(\beta)$ for $0 < n \le n_c$. In particular $\lim_{n\downarrow 0} \varphi(n)/n = \lim_N \phi_N(\beta)$: the limits $n \to 0$ and $N \to \infty$ commute, and the fractional-moment bound $\phi_N \le \frac{1}{Nn} \log \mathbb{E} Z_N^n$ is sharp in the limit for every $n \le n_c$.

(ii) If $\beta > \beta_c$, then φ is continuously differentiable at $n = m$, and φ'' jumps there from 0 to β^2. If $\beta < \beta_c$, then φ' jumps at $n = m^2$ from $\log 2 + \beta^2/2$ to $2\log 2$.

(iii) If $\beta > \beta_c$, then $\varphi(k) = \log 2 + k^2\beta^2/2$ at every integer $k \ge 1$, and the entire function $f(n) = \log 2 + n^2\beta^2/2$ agrees with φ on $[m, \infty)$ but not on $(0, m)$, where $f(n)/n \to \infty$ as $n \downarrow 0$.

(iv) The one-step formula with Parisi's rule is exact: $\varphi = \varphi_{\mathrm{1RSB}}$ of (2.5) on $(0, \infty)$.

Proof. Parts 1–3 are read off (2.7). At n_c the left derivative is $\phi(\beta)$ and the right derivative $n_c\beta^2$: these are $\beta\beta_c$ and $m\beta^2 = \beta\beta_c$ when $\beta > \beta_c$, and $\log 2 + \beta^2/2$ and $m^2\beta^2 = 2\log 2$ when $\beta < \beta_c$. For part 4 let $h(m') = \frac{\log 2}{m'} + \frac{\beta^2 m'}{2}$, a convex function on $(0, \infty)$, decreasing up to $m' = m$ and increasing after, with $nh(1) = n\phi_a$, $nh(n) = f(n)$ and $nh(m) = n\beta\beta_c$. For $0 < n \le 1$ the minimum of h over $[n, 1]$ is at the point of $[n, 1]$ nearest to m: at m if $n \le m \le 1$, giving $n\beta\beta_c$; at n if $m < n$, giving $f(n)$; at 1 if $m > 1$, giving $n\phi_a$. These are the three cases of (2.7) with $n \le 1$. For $n \ge 1$ the maximum of the convex h over $[1, n]$ is at an endpoint, so $\varphi_{\mathrm{1RSB}}(n) = \max(n\phi_a, f(n)) = \varphi(n)$. ∎

So the order of limits is harmless (part 1; Exercise 2.8 extends this to every Gaussian disordered system), and the continuation from the integers fails above β_c (part 3) because the moments are singular at $n = m$, a phase transition in the replica number (part 2).[8] The one-step formula with Parisi's minimum is right at every n (part 4), although nothing inside the replica method says why. Below β_c the branch through $n = 1$, all replicas apart, is the one that reaches $n \to 0$, which is why replica symmetry is right there.

[8] Ogure and Kabashima (2009b) locate this singularity through the zeros of $\mathbb{E} Z_N^n$ in the complex n-plane: as N grows they accumulate near the transition point, at an angle that marks a second-order transition, in agreement with part 2.

2.5.1 What the general theory adds

Three further results complete the picture; we state them without proof.

Theorem 2.12 (Poisson–Dirichlet weights; Bovier and Kurkova 2004; Ruelle 1987). *Let $\beta > \beta_c$ and $m = \beta_c/\beta$. The point process $\sum_\sigma \delta_{G_N(\sigma)}$ of Gibbs weights converges in law to the point process of the weights $\xi_i/\sum_j \xi_j$, where $\xi_1 > \xi_2 > \cdots$ are the points of a Poisson process on $(0,\infty)$ with intensity $x^{-1-m}\,\mathrm{d}x$. The law of these weights, in decreasing order, is the* Poisson–Dirichlet distribution *$PD(m,0)$.*

Ruelle (1987) reformulated Derrida's models with such Poisson processes, and Bovier and Kurkova (2004) proved the convergence for the generalized REM, of which the REM is the case of one level. In size-biased order the weights are $W_i = V_i \prod_{j<i}(1-V_j)$ with independent $V_i \sim \mathrm{Beta}(1-m, im)$ (Pitman and Yor 1997). Since $\sum_{G_N(\sigma)<\varepsilon} G_N(\sigma)^2 \le \varepsilon$, the theorem gives $Y_N \to \sum_i W_i^2$ in law and, Y_N being bounded, in moments, so that $\operatorname{Var} Y_N \to \frac{m(1-m)}{3} > 0$ (Exercise 2.2): the participation ratio does not self-average, as Derrida and Toulouse (1985) computed without replicas and Mézard et al. (1985) derived from independent, exponentially distributed free energies of the pure states.

Theorem 2.13 (Fluctuations below T_c; Bovier et al. 2002). *Let $\beta > \beta_c$, $m = \beta_c/\beta$, $a_N = N\beta\beta_c - \frac{1}{2m}[\log(N\log 2) + \log 4\pi]$, and let $\mathcal{P}$ be a Poisson process on $\mathbb{R}$ with intensity $e^{-z}\,\mathrm{d}z$. Then $\log Z_N - a_N \to \log \int e^{z/m}\mathcal{P}(\mathrm{d}z)$ in law, and $\log Z_N - \mathbb{E}\log Z_N$ converges in law to the same limit, centered.*

This is Theorem 1.5(ii) of Bovier et al. (2002) in arXiv v1 (we could not check the published numbering), for our Z_N rather than their $2^{-N}Z_N$; their Theorems 1.4 and 1.5(i) cover $\beta \le \beta_c$, with Gaussian fluctuations of $\log(Z_N/\mathbb{E}Z_N)$ for $\beta \le \beta_c/2$ and non-Gaussian ones above. The two convergences differ by the deterministic shift $\mathbb{E}\log Z_N - a_N$, which therefore converges to the mean of the limit; with Exercise 2.6 this gives

$$\mathbb{E}\log Z_N = a_N + C_m + o(1), \qquad C_m = \frac{\log\Gamma(1-m)}{m} + \gamma\Big(\frac{1}{m} - 1\Big), \tag{2.9}$$

with γ Euler's constant: ϕ_N approaches $\beta\beta_c$ from below like $\frac{\beta}{2\beta_c}\frac{\log N}{N}$.

Finally, Derrida's *generalized random energy model* (Derrida 1985) builds the energies as sums of independent contributions along the branches of a tree, and freezes in several steps. Derrida and Gardner (1986) solved it, and Bovier and Kurkova (2004), who review the earlier rigorous work on its free energy, identified the limiting Gibbs measures with Ruelle's probability cascades, which reappear in Guerra's bound for the SK model (Chapter 3).

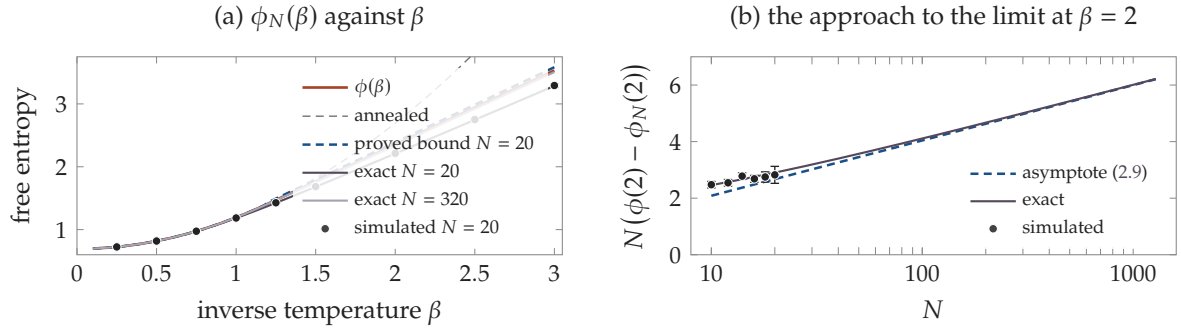


Figure 2.1. The free entropy. (a) Derrida's formula (2.2) (carnelian), the annealed value (gray dashed), the proved upper bound of Theorem 2.4 at $N = 20$ (lapis dashed), exact values at $N = 20$ and 320 (granite), and the mean of $\frac{1}{N}\log Z_N$ over 150 simulated samples at $N = 20$ (seed 2001). (b) The deficit $N(\phi - \phi_N)$ at $\beta = 2$: exact values, simulations (1000 samples for $N \le 16$, 400 at $N = 18$, those of (a) at $N = 20$; seeds $2002 + N$), and the asymptote (2.9). Script: `figures/scripts/02-rem-freeentropy.py`.

2.6 Measuring it

Four scripts simulate the REM and compute exact finite-N expectations. Since Z_N is a sum of 2^N independent terms, $\mathbb{E}e^{-tZ_N} = L(t)^{2^N}$ with $L(t) = \mathbb{E}\exp(-te^{\beta\sqrt{N}X})$, and

$$\mathbb{E}\log Z_N = \int_0^\infty \left(e^{-t} - L(t)^{2^N}\right)\frac{\mathrm{d}t}{t},$$
$$\mathbb{E}Z_N^n = \frac{n}{\Gamma(1-n)}\int_0^\infty \left(1 - L(t)^{2^N}\right)\frac{\mathrm{d}t}{t^{n+1}} \quad (0 < n < 1),$$

and $\mathbb{E}Y_N$ similarly, by the representation of Mézard and Montanari (2009, ch. 5).[9] All intervals are 95% intervals: Student-t for means, Clopper–Pearson for probabilities, percentile bootstrap where stated.

THE FREE ENTROPY. In Figure 2.1(a) the nine simulated means at $N = 20$ all lie within their intervals of the exact finite-N values, and above β_c both lie well below the limit: at $\beta = 2$ the simulation gives 2.214 ± 0.015, the exact value is 2.210, and the limit is 2.355. Panel (b) shows why: the deficit grows like $\frac{1}{2m}\log N = 0.849\log N$, as Theorem 2.13 predicts. The exact deficit exceeds the asymptote (2.9) by 0.229 at $N = 20$, 0.028 at $N = 320$ and 0.009 at $N = 1280$, a remainder that decays more slowly than $1/N$ (N times it grows from 8.9 to 11.8 between the last two sizes). Simulations at $N \le 20$ alone could not separate a $\log N/N$ correction from a $1/N$ one.

CONDENSATION. In Figure 2.2(a) the six simulated means of Y_{20} lie within their intervals of the exact values, which approach the limit slowly: at $\beta = 2$ they are 0.466, 0.429 and 0.417 at $N = 20$, 80 and 320, against $1 - m = 0.411$, an excess that shrinks roughly like $\log N/N$; at β_c, where the limit is 0, they are 0.163, 0.079 and 0.039. For $PD(m, 0)$, 10^4 stick-breaking samples (3000 sticks, seed 2102) put $\mathbb{E}\sum_i W_i^2$ and $\mathbb{E}(\sum_i W_i^2)^2$ within their

[9] Exact finite-size expressions for the moments go back to Gardner and Derrida (1989) and Ogure and Kabashima (2004, 2009a). The script `02-rem-laplace.py` evaluates these integrals by quadrature in logarithmic variables, so that $N = 1280$, with $2^N \approx 10^{385}$ levels, costs no more than $N = 20$; its checks against simulation at $N = 8$–12 agree within the 95% intervals, and a second quadrature on another grid reproduces its values of $\mathbb{E}\log Z_N$ at $N = 320$ and 1280 to four decimals.

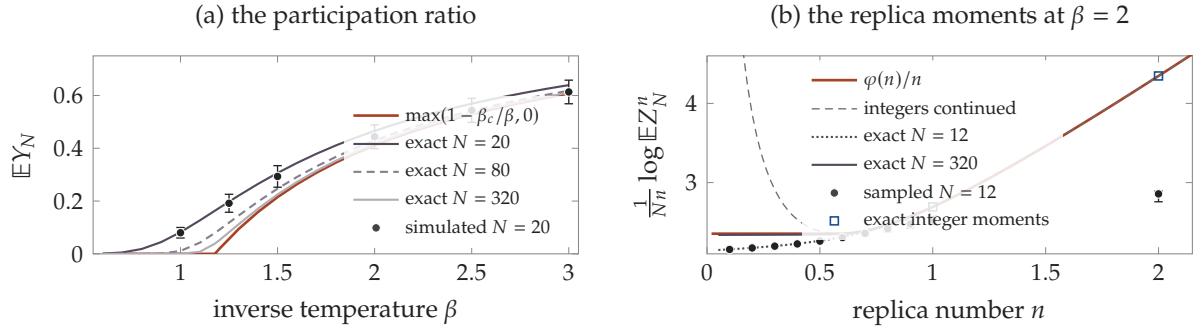


Figure 2.2. Condensation and the replica moments. (a) The mean participation ratio: the limit of Corollary 2.7 (carnelian), exact values at $N = 20, 80, 320$ (granite), and the mean of Y_{20} over 150 simulated samples (seed 2101). (b) At $\beta = 2$ ($m = 0.589$): the limit $\varphi(n)/n$ of Theorem 2.9, which is also the one-step formula with Parisi's rule (carnelian); the continuation $\frac{\log 2}{n} + \frac{n\beta^2}{2}$ of the integer moments (gray dashed); exact values at $N = 12$ and 320, and at $n = 1, 2$ (squares); and the empirical mean of Z_{12}^n over 20 000 samples (seed 2201), with percentile-bootstrap intervals (400 resamples). Scripts: `figures/scripts/02-rem-gibbs.py`, `02-rem-moments.py`.

intervals of $1 - m$ and $\frac{(1-m)(3-2m)}{3}$ at $m = 0.25, 0.5, 0.75$ (0.503 ± 0.006 and 0.337 ± 0.006 at $m = 0.5$). At $N = 20$ and $\beta = 2$ the simulated $\mathbb{E}Y_{20}^2 = 0.274 \pm 0.049$ is consistent with the PD value 0.300 at the effective parameter $m_{\text{eff}} = 1 - \mathbb{E}Y_{20} = 0.534$, but 150 samples cannot separate it from the limiting 0.250 at $m = 0.589$.

THE REPLICA MOMENTS. Figure 2.2(b) shows the singularity of Theorem 2.9 forming. At every finite N the curve $n \mapsto \frac{1}{Nn}\log \mathbb{E}Z_N^n$ is smooth; at $N = 320$ it lies within 0.016 of $\beta\beta_c = 2.355$ below $n = 0.5$ and on the limit above $n = 0.65$. The integer moments lie on the limit, and the curve that continues them leaves it at $n = m$. Sampling cannot measure the quadratic branch: the empirical moments at $N = 12$ agree with the exact values up to $n = 0.7$ and fall short beyond, 2.533 against 2.693 at $n = 1$ and 2.857 against 4.347 at $n = 2$, because for $n > m$ the expectation is carried by samples with a deep level, of probability about 10^{-8} at $n = 1$, far below $1/20\,000$; the bootstrap intervals cannot see what the sample does not contain.

DECODING. The script `02-rem-decoding.py` computes the success probability of maximum a posteriori decoding, $\mathbb{E}\Phi(\lambda\sqrt{N} + X)^{2^N - 1}$, by quadrature and the mutual information by the planted Laplace representation. The threshold of Corollary 2.8 emerges slowly: at $\lambda = 1$ the success probability falls from 0.440 at $N = 12$ to 0.215, 0.021 and 0.000 at $N = 40, 160, 640$, and at $\lambda = 1.5$ it rises from 0.932 to 0.990, 1.000, 1.000 (Exercise 2.9). The information per bit at $\lambda = 1$ is 0.459, 0.489 and 0.499 at $N = 12, 32, 96$, against the limit $\frac{1}{2}$. At $N = 12$, 2000 simulated transmissions at each of eleven values of λ (seed 2301) put ten of the eleven information estimates and eight of the eleven success frequencies within their intervals of the exact values; at the three outlying λ, 40 000 further transmissions (seed 2302) agree with them. No measurement contradicts a proved result.

2.7 The status board

For the REM the prediction of Derrida, Gross and Mézard is a theorem in every part, down to the replica moments at every real n with their one-step formula, and the fluctuations are known at every temperature. What is false is what the physics itself rejects: the replica-symmetric solution below T_c, and the naive continuation of the integer moments.

STATUS BOARD · replicas ↔ the second moment

PREDICTION	STATUS	PROOF OR EVIDENCE
Free entropy $\phi(\beta)$, freezing at $\beta_c = \sqrt{2\log 2}$, zero entropy below T_c	proved	Theorem 2.4 and Corollary 2.7; Bovier (2006, ch. 9)
Replica-symmetric (annealed) solution below T_c	false	its entropy $\log 2 - \beta^2/2$ is negative; Corollary 2.7
One-step breaking: $\mathbb{E}Y_N \to 1 - \beta_c/\beta$, $P(q) = m\delta_0 + (1-m)\delta_1$	proved	Corollary 2.7; predicted by Gross and Mézard (1984)
Poisson–Dirichlet weights $PD(m, 0)$; Y_N not self-averaging	proved	Bovier and Kurkova (2004) and Ruelle (1987); predicted by Derrida and Toulouse (1985) and Mézard et al. (1985)
Fluctuations of $\log Z_N$ at every β	proved	Bovier et al. (2002), Thms 1.4–1.5 (arXiv v1)
Replica moments $\varphi(n)$ at real $n > 0$, given by the one-step formula	proved	Theorem 2.9 and Corollary 2.11, the book's proof; derived by Ogure and Kabashima (2009a, eqs. (A.5)–(A.6))
Continuation of the integer moments to $n \to 0$	false	for every $\beta > \beta_c$; Corollary 2.11, part 3
Orthogonal codes: threshold $\lambda = \beta_c$ ($E_b/N_0 = \ln 2$), information $\min(\lambda^2/2, \log 2)$	proved	Corollary 2.8
Generalized REM: free entropy, Ruelle cascades	proved	Bovier and Kurkova (2004); predicted by Derrida and Gardner (1986)

2.8 Where the translation fails

WHAT WAS ASSUMED. Independence of the energies, first of all: in the p-spin glasses at finite p the energies of nearby configurations are correlated, for $p = 2$ (the SK model of Chapter 3) the physics predicts full replica-symmetry breaking, and the counting arguments do not survive. Second, Gaussian levels: the freezing and the Poisson–Dirichlet weights come from the Gaussian tail. In the decoding reading the independence is that of orthogonal codewords; random binary codes also give independent scores

(Mézard and Montanari 2009, ch. 6), structured codes do not. Third, the decoder uses the Bayes posterior; at another temperature a glassy phase appears (Exercise 2.5).

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. The replica-symmetric solution is wrong below T_c, and the physics knows it; the naive continuation of the integer moments is wrong above β_c (Corollary 2.11). Parisi's rule, a minimum over $m' \in [n, 1]$, is right at every n but has no derivation inside replica theory. Finite size is where the physical description is least accurate: at $N = 20$ and $\beta = 2$ the free entropy is 0.145 below its limit and $\mathbb{E}Y_N$ is 0.055 above it, and replicas reproduce such corrections only if the numbers and sizes of Parisi's blocks are given negative variances (Derrida and Mottishaw 2015).

OPEN PROBLEMS. For the REM itself the leading order and the fluctuations are settled; the open questions concern the method. First, for integer-valued energies one-step breaking gives the free energy but not the overlaps, and Derrida and Mottishaw (2022) suggest that the block sizes should fluctuate and take complex values, as for the finite-size corrections. A derivation of such a rule, or a class of models in which it is proved to work, is not known to us (as of 2026). Second, for which models is $\varphi(n) = \lim N^{-1} \log \mathbb{E}Z_N^n$ linear on an interval $(0, n_c]$ and given at every n by Parisi's formula with the constraint $n \le m' \le 1$? Theorem 2.9 settles the REM, and Exercise 2.10 poses the generalized REM. Third, Mézard and Montanari (2009, notes to ch. 8) observe that the replica method with broken symmetry still awaits a mathematical framework; the REM shows that it must explain why the extremum is a minimum over $[n, 1]$.

Exercises

2.1 CONSOLIDATION. (The plain second moment.) (a) Show from (2.3) that $\mathbb{E}Z_N^2/(\mathbb{E}Z_N)^2 = 1 + 2^{-N}(e^{N\beta^2} - 1)$, which tends to 1 if and only if $\beta < \beta_c/\sqrt{2}$, and deduce $\phi_N \to \log 2 + \beta^2/2$ there with the Paley–Zygmund inequality and Step 1 of the proof of Theorem 2.4. (b) Show that $\mathbb{E}\tilde{Z}_N^2/(\mathbb{E}\tilde{Z}_N)^2 \to 1$ for every $\beta < \beta_c$, where $\tilde{Z}_N$ sums over $X_\sigma \le \beta_c\sqrt{N}$: which samples spoil the plain second moment? *(Hint: for $\beta_c/2 < \beta < \beta_c$ the excess is at most a constant times $e^{-N(\beta_c-\beta)^2}$.)*

2.2 CONSOLIDATION. (Moments of $PD(m, 0)$.) (a) With the size-biased pick $W_1 = V_1 \sim \mathrm{Beta}(1 - m, m)$, show that $\mathbb{E}\sum_i W_i^r = \mathbb{E}W_1^{r-1} = \frac{\Gamma(r-m)}{\Gamma(r)\Gamma(1-m)}$. (b) Show that $\mathbb{E}(\sum_i W_i^2)^2 = \frac{(1-m)(3-2m)}{3}$, so that Y_N does not converge in probability when $\beta > \beta_c$ (Theorem 2.12). *(Hint: $\mathbb{E}\sum_{i\neq j} W_i^2 W_j^2 = \frac{m(1-m)^2}{6}$ is the probability that, of four independent draws from the weights, the first two fall on one atom and the last two on another.)*

2.3 CONSOLIDATION. (Negative moments.) Show that $\varphi(n) = n\phi(\beta)$ for every real $n < 0$. *(Hint: Jensen's inequality gives the lower bound; for the upper bound combine $Z_N \ge 2^N e^{\beta\sqrt{N}\bar{X}}$ with the Chernoff bound*

$\mathbb{P}(K_N(u) \le \frac{1}{2}\mu_N(u)) \le e^{-\mu_N(u)/8}$.)

2.4 BY HAND. (The binary REM of random codes.) Give 2^{NR} messages independent codewords uniform on $\{0,1\}^N$ and send one over the binary symmetric channel with flip probability $p < \frac{1}{2}$. (a) Show that the expected number of wrong codewords at Hamming distance about $N\delta$ from the output is $e^{N[(R-1)\log 2 + h(\delta)] + o(N)}$, with h the binary entropy in nats, so that the closest lies at distance $N\delta_{\mathrm{GV}} + o(N)$, where $h(\delta_{\mathrm{GV}}) = (1-R)\log 2$ and $\delta_{\mathrm{GV}} < \frac{1}{2}$, and that maximum-likelihood decoding succeeds if and only if $R < 1 - h(p)/\log 2$, the capacity. (b) With posterior weights $(p/(1-p))^d$, show that the free entropy of the wrong codewords is frozen at $-\delta_{\mathrm{GV}} \log \frac{1-p}{p}$ if $p < \delta_{\mathrm{GV}}$ and equals $(R-1)\log 2 - \log(1-p)$ otherwise (Mézard and Montanari 2009, ch. 6): as for orthogonal codes, the posterior is frozen exactly when decoding succeeds.

2.5 BY HAND. (Decoding at the wrong temperature.) Decode orthogonal codes with $G_\beta(\sigma) \propto e^{\beta\sqrt{N} y_\sigma}$. (a) Show that $G_\beta(\sigma^*) \to 1$ in probability if $\beta\lambda > \phi(\beta)$ and $\to 0$ if $\beta\lambda < \phi(\beta)$: the threshold is $\lambda = \beta_c$ for $\beta \ge \beta_c$ and $\lambda = \frac{\beta^2+\beta_c^2}{2\beta}$ for $\beta < \beta_c$ (1.266 at $\beta = 0.8$). (b) Draw the phase diagram in the $(\lambda, 1/\beta)$ plane, with ordered, glassy and paramagnetic phases, and compare with that of Mézard and Montanari (2009, ch. 6) for random binary codes. Where does the Bayes line $\beta = \lambda$ meet the boundary?

2.6 BY HAND. (The constant of the extreme-value asymptote.) Let $S = \int e^{z/m} \mathcal{P}(\mathrm{d}z)$, with $\mathcal{P}$ a Poisson process of intensity $e^{-z}\,\mathrm{d}z$ and $0 < m < 1$. Show that $\mathbb{E}e^{-tS} = \exp(-\Gamma(1-m)t^m)$ and, from $\log s = \int_0^\infty (e^{-t} - e^{-ts})\frac{\mathrm{d}t}{t}$, that $\mathbb{E}\log S = \frac{\log\Gamma(1-m)}{m} + \gamma(\frac{1}{m} - 1)$; at $\beta = 2$ this is 1.708. Check (2.9) against Figure 2.1(b): 6.216 exact against 6.207 at $N = 1280$.

2.7 TRANSLATION. (The Hamming overlap.) The physicist's overlap in the REM is $Q_{12} = \mathbf{1}\{\sigma^1 = \sigma^2\}$; the spin-glass overlap is $R_{12} = \frac{1}{N}\langle\sigma^1, \sigma^2\rangle$. Using the exchangeability of (X_σ), show that $\mathbb{E}\langle R_{12}^2\rangle = \mathbb{E}Y_N + (1 - \mathbb{E}Y_N)\frac{2^N/N - 1}{2^N - 1}$, deduce $\mathbb{E}\langle R_{12}^2\rangle \to \max(1 - \beta_c/\beta, 0)$, and state $P(q) = m\delta_0 + (1-m)\delta_1$ as a theorem about two samples from G_N.

2.8 TRANSLATION. (Why the limits commute.) Suppose that $\log Z_N$ is an $L\sqrt{N}$-Lipschitz function of a standard Gaussian vector and that every L_F-Lipschitz F satisfies $\mathbb{E}e^{t(F-\mathbb{E}F)} \le e^{t^2 L_F^2/2}$. Show that, for $n > 0$, $\frac{1}{N}\log \mathbb{E}Z_N^n$ lies between $n\phi_N$ and $n\phi_N + n^2L^2/2$; deduce $\lim_{n\downarrow 0} \varphi(n)/n = \lim_N \phi_N$ when both limits exist, and check against (2.7) with $L = \beta$: what fails in the replica trick is the continuation, not the order of limits.

2.9 MEASUREMENT. (The finite-size decoding threshold.) Compute the λ of $\frac{1}{2}$ success for $N = 20, 80, 320$ (the script's function `map_success` gives 1.089, 1.148, 1.168). From the Gumbel limit of the largest wrong

score, derive $\lambda_{1/2} = \beta_c - \frac{\log(N\log 2)+\log 4\pi-2\gamma}{2\beta_c N} + o(N^{-1})$, and compare. Show that $\sqrt{N}$ times the width from 10% to 90% success tends to 2×1.2816: the noise on the transmitted score, not the fluctuation of the wrong scores, sets the width.

2.10 RESEARCH. ★★ (The replica moments of the generalized REM.) For Derrida's generalized REM with two levels (Derrida 1985), in the notation of Kistler (2015, §1.2), compute $\varphi(n) = \lim N^{-1} \log \mathbb{E} Z_N^n$ for real $n > 0$ by the methods of Theorem 2.9, locate its singularities, and compare with the two-step replica formula with Parisi's rule and with the nested bound $\mathbb{E} \log Z_N \le \frac{1}{m_1} \log \mathbb{E}[(\mathbb{E}[Z_N^{m_2} \mid \text{first generation}])^{m_1/m_2}]$, the analogue of Guerra's bound of Chapter 3. *(Hint: write $Z_N = \sum_{\sigma_1} e^{\beta X_{\sigma_1}^{(1)}} Z_{\sigma_1}^{(2)}$, a sum over the first generation of independent REMs of the second.)*

Notes and References

Derrida (1980, 1981) introduced the REM as the limit $p \to \infty$ of the p-spin glasses and solved it by the microcanonical count of Section 2.2.1; Mézard and Montanari (2009, notes to ch. 8) credit the replica solution to the same papers. Gross and Mézard (1984) treated the p-spin model at large p with replicas and one-step breaking, Mézard et al. (1985) and Derrida and Toulouse (1985) found the sample-to-sample fluctuations of the weights, and Derrida (1985) and Derrida and Gardner (1986) introduced and solved the generalized REM. On the rigorous side, Olivieri and Picco (1984) studied the thermodynamic limit of the REM, and Section 2.5.1 describes the later work; textbook accounts are Bovier (2006, chs. 9–10) and Kistler (2015). Mézard and Montanari (2009, chs. 5, 6, 8) give the physical treatment, prove the level count by first and second moments, and treat the random code ensemble as a REM; orthogonal codes, with the capacity formula of Shannon (1949), are its Gaussian analogue.

Ogure and Kabashima (2004) found, from exact finite-size moments of a discrete REM, a transition at a real replica number. For the Gaussian REM Ogure and Kabashima (2009a, App. A, eqs. (A.5)–(A.6)) state the limit (2.7) in both phases, by steepest descent from an exact integral representation, and find it reproduced by one-step breaking, whose origin they locate in the loss of analyticity in n; Ogure and Kabashima (2009b) follow the complex zeros, and Derrida and Mottishaw (2015) obtain one-step breaking without an ansatz from contour integrals over complex replica numbers. Theorem 2.9 is therefore their formula. Its proof, like those of Theorem 2.4 and Corollaries 2.7, 2.8 and 2.11, was written for this book from standard ingredients (the fractional moment at $n = m$, the recursion of Lemma 2.10, the derivative argument for $\mathbb{E} Y_N$). Tanaka (2007) discusses the moment problem behind the replica method.

TRANSLATING THE LITERATURE. Mézard and Montanari (2009) use energies of variance $N/2$ and the weight $e^{-\beta E}$: their β is $\sqrt{2}$ times ours, their $\beta_c = 2\sqrt{\log 2}$, their energy density ε is our $-u/\sqrt{2}$, their $s_a(\varepsilon) = \log 2 - \varepsilon^2$ our $s(u)$, their $m_s(\beta)$ our m, and their $\psi(n)$ our $\varphi(n)$. Derrida (1981) uses

energies of variance $NJ^2/2$, so that our β is $J/(\sqrt{2}\,T)$ and $T_c = J/(2\sqrt{\log 2})$. Bovier et al. (2002) write $Z_{\beta,N}$ for our $2^{-N}Z_N$ and $\alpha = \beta/\sqrt{2\ln 2}$ for our $1/m$. Kistler (2015) gives the levels variance N: his X_σ is our $\sqrt{N}X_\sigma$, and his $\mathcal{N}_N(\lambda)$, the number of levels at least λN, our $K_N(\lambda)$. Pitman and Yor (1997) write $PD(\alpha, \theta)$; ours is the case $\alpha = m$, $\theta = 0$.

References

Bovier, A. (2006). *Statistical Mechanics of Disordered Systems: A Mathematical Perspective*. Vol. 18. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press.

Bovier, A. and I. Kurkova (2004). "Derrida's generalised random energy models 1: models with finitely many hierarchies". In: *Annales de l'Institut Henri Poincaré (B) Probabilités et Statistiques* 40.4, pp. 439–480.

Bovier, A., I. Kurkova, and M. Löwe (2002). "Fluctuations of the free energy in the REM and the p-spin SK models". In: *The Annals of Probability* 30.2, pp. 605–651. arXiv: cond-mat/0007175.

Derrida, B. (1980). "Random-Energy Model: Limit of a Family of Disordered Models". In: *Physical Review Letters* 45.2, pp. 79–82.

Derrida, B. (1981). "Random-energy model: An exactly solvable model of disordered systems". In: *Physical Review B* 24.5, pp. 2613–2626.

Derrida, B. (1985). "A generalization of the Random Energy Model which includes correlations between energies". In: *Journal de Physique Lettres* 46.9, pp. 401–407.

Derrida, B. and E. Gardner (1986). "Solution of the generalised random energy model". In: *Journal of Physics C: Solid State Physics* 19.13, pp. 2253–2274.

Derrida, B. and P. Mottishaw (2015). "Finite size corrections in the random energy model and the replica approach". In: *Journal of Statistical Mechanics: Theory and Experiment* 2015.1, P01021. arXiv: 1410.1432.

Derrida, B. and P. Mottishaw (2022). "The discrete random energy model and one step replica symmetry breaking". In: *Journal of Physics A: Mathematical and Theoretical* 55.26, p. 265002. arXiv: 2202.12584.

Derrida, B. and G. Toulouse (1985). "Sample to sample fluctuations in the random energy model". In: *Journal de Physique Lettres* 46.6, pp. 223–228.

Gardner, E. and B. Derrida (1989). "The probability distribution of the partition function of the random energy model". In: *Journal of Physics A: Mathematical and General* 22.12, pp. 1975–1981.

Gross, D. J. and M. Mézard (1984). "The simplest spin glass". In: *Nuclear Physics B* 240.4, pp. 431–452.

Kistler, N. (2015). "Derrida's Random Energy Models: From Spin Glasses to the Extremes of Correlated Random Fields". In: *Correlated Random Systems: Five Different Methods*. Vol. 2143. Lecture Notes in Mathematics. Springer, pp. 71–120. arXiv: 1412.0958.

Mézard, M. and A. Montanari (2009). *Information, Physics, and Computation*. Oxford Graduate Texts. Oxford: Oxford University Press.

Mézard, M., G. Parisi, and M. A. Virasoro (1985). "Random free energies in spin glasses". In: *Journal de Physique Lettres* 46.6, pp. 217–222.

Ogure, K. and Y. Kabashima (2004). "Exact Analytic Continuation with Respect to the Replica Number in the Discrete Random Energy Model of Finite System Size". In: *Progress of Theoretical Physics* 111.5, pp. 661–688. arXiv: cond-mat/0310490.

Ogure, K. and Y. Kabashima (2009a). "On analyticity with respect to the replica number in random energy models: I. An exact expression for the moment of the partition function". In: *Journal of Statistical Mechanics: Theory and Experiment* 2009.3, P03010. arXiv: 0812.4654.

Ogure, K. and Y. Kabashima (2009b). "On analyticity with respect to the replica number in random energy models: II. Zeros on the complex plane". In: *Journal of Statistical Mechanics: Theory and Experiment* 2009.5, P05011. arXiv: 0812.4655.

Olivieri, E. and P. Picco (1984). "On the existence of thermodynamics for the random energy model". In: *Communications in Mathematical Physics* 96.1, pp. 125–144.

Pitman, J. and M. Yor (1997). "The two-parameter Poisson–Dirichlet distribution derived from a stable subordinator". In: *The Annals of Probability* 25.2, pp. 855–900.

Ruelle, D. (1987). "A mathematical reformulation of Derrida's REM and GREM". In: *Communications in Mathematical Physics* 108.2, pp. 225–239.

Shannon, C. E. (1949). "Communication in the Presence of Noise". In: *Proceedings of the IRE* 37.1, pp. 10–21.

Tanaka, T. (2007). "Moment Problem in Replica Method". In: *Interdisciplinary Information Sciences* 13.1, pp. 17–23.

The Sherrington–Kirkpatrick Model

3

Replica symmetry and its breaking

THE DICTIONARY — broken replicas ↔ interpolation

IN THIS CHAPTER. Feed pure noise to the posterior of $\mathbb{Z}_2$ synchronization and you obtain the Sherrington–Kirkpatrick model, whose free entropy physicists computed with replicas: replica symmetry, its instability below the de Almeida–Thouless line, and Parisi's formula. With Gaussian interpolation in place of replicas, we prove that the limit exists, that the replica-symmetric formula is an upper bound with an exact remainder at every size, and that it is exact at high temperature without a field, by the physicists' own computation at $n = 2$. Exact enumeration checks the remainder identity on both sides for up to 14 spins, and shows at 16 spins why small systems cannot see the breaking of replica symmetry. The Parisi formula is a theorem, and so is full replica-symmetry breaking just below the critical temperature; replica symmetry up to the line with a field, full replica-symmetry breaking in the rest of the low-temperature phase, and the overlap law were proved in 2026 preprints, not yet refereed.

3.1 The question: a posterior that finds structure in noise

In $\mathbb{Z}_2$ synchronization a statistician observes $A = \frac{\lambda}{N} xx^\top + W$, with $x \in \{\pm 1\}^N$ unknown and W a GOE matrix (Chapter 5). With independent priors $\mathbb{P}(x_i = \pm 1) = e^{\pm h}/(2\cosh h)$, the posterior computed from the entries above the diagonal (the diagonal carries no information, since $x_i^2 = 1$) is

$$\mathbb{P}(x \mid A) \propto \exp\Big(\lambda \sum_{i<j} A_{ij} x_i x_j + h \sum_{i=1}^{N} x_i\Big).$$

Suppose that there is no signal, $A = W$, so that $A_{ij} = g_{ij}/\sqrt{N}$ for $i < j$ with independent standard Gaussians g_{ij}, and that the statistician computes this posterior anyway. The result is the Gibbs measure of the *Sherrington–Kirkpatrick model* (Sherrington and Kirkpatrick 1975) at inverse temperature $\beta = \lambda$ and field h: on $\Sigma_N = \{-1, +1\}^N$,

$$H_N(\sigma) = X_N(\sigma) + h \sum_{i=1}^{N} \sigma_i, \qquad X_N(\sigma) = \frac{\beta}{\sqrt{N}} \sum_{1 \le i < j \le N} g_{ij} \sigma_i \sigma_j, \tag{3.1}$$

with Gibbs weights $e^{H_N(\sigma)}/Z_N$. The *free entropy* is $\phi_N(\beta, h) = \frac{1}{N}\mathbb{E}\log Z_N$, with $\mathbb{E}$ the average over the couplings; the physicist's free energy per spin is $-\phi_N/\beta$, at temperature $T = 1/\beta$ and physical field $H = h/\beta$.[1] Chapter 4 treats the companion problem in which the spike is present and λ is known; there replica symmetry is a theorem at every λ.

We write $\langle\cdot\rangle$ for the Gibbs average and use *replicas*: independent samples $\sigma^1, \sigma^2, \dots$ from the same Gibbs measure, the couplings fixed. The *overlap* of two replicas is $R_{12} = \frac{1}{N}\sum_i \sigma_i^1\sigma_i^2$, and $R_{\sigma\tau} = \frac{1}{N}\langle\sigma\rangle\tau$; a probabilist would call $\mathbb{E}\langle R_{12}^2\rangle$ the mean squared normalized inner product of two independent posterior samples.[2] One covariance computation drives the chapter: since $\sum_{i<j} a_i a_j = \frac{1}{2}((\sum_i a_i)^2 - \sum_i a_i^2)$, taking $a_i = \sigma_i\tau_i$ gives

$$\mathbb{E}X_N(\sigma)X_N(\tau) = \frac{\beta^2}{N}\sum_{i<j}\sigma_i\sigma_j\tau_i\tau_j = \frac{N\beta^2}{2}\Big(R_{\sigma\tau}^2 - \frac{1}{N}\Big), \tag{3.2}$$

a covariance $N\xi(R_{\sigma\tau}) + O(1)$ with $\xi(q) = \beta^2 q^2/2$. In particular $\mathbb{E}e^{X_N(\sigma)} = e^{\beta^2(N-1)/4}$, and Jensen's inequality gives the *annealed bound*

$$\phi_N(\beta, h) \le \frac{1}{N}\log\mathbb{E}Z_N = \log(2\cosh h) + \frac{\beta^2(N-1)}{4N}. \tag{3.3}$$

At small β this bound should be sharp and two posterior samples nearly orthogonal; at large β the posterior should concentrate on many configurations that fit the noise and lie far apart.

THE PHYSICAL PREDICTION · Sherrington–Kirkpatrick 1975; de Almeida–Thouless 1978; Parisi 1979–80. Let q solve $q = \mathbb{E}\tanh^2(\beta\sqrt{q}\,Z + h)$, taking the positive solution when $h = 0$ and $\beta > 1$. If

$$\mathrm{AT}(\beta, h) := \beta^2\,\mathbb{E}\,\mathrm{sech}^4(\beta\sqrt{q}\,Z + h) \le 1,$$

the de Almeida–Thouless condition, then the overlap concentrates at q and the free entropy has the replica-symmetric value

$$\phi(\beta, h) = \phi_{\mathrm{RS}}(\beta, h; q) = \log 2 + \mathbb{E}\log\cosh(\beta\sqrt{q}\,Z + h) + \frac{\beta^2}{4}(1-q)^2.$$

Otherwise replica symmetry is broken: ϕ is the minimum of Parisi's functional (3.9) over probability measures μ on $[0, 1]$, the minimizer is not a point mass, the law of R_{12} converges to it (symmetrized when $h = 0$), and the overlaps of many replicas are ultrametric. At $h = 0$ the transition is at $\beta = 1$, and the ground-state energy per spin is -0.7632; replica symmetry predicts $-\sqrt{2/\pi} = -0.7979$, with entropy $-1/(2\pi)$.

How do physicists derive this, how much of the derivation survives, and what is proved?

[1] Up to explicit factors Z_N is the likelihood ratio of the spiked model against pure noise: with the entries above the diagonal observed, the evidence ratio is $(2\cosh h)^{-N}e^{-\beta^2(N-1)/4}Z_N$. So ϕ_N is a log-evidence per variable, and the formula for $\mathbb{E}Z_N$ below says that a likelihood ratio has mean one under the null.

[2] At $h = 0$ the Gibbs measure is invariant under $\sigma \mapsto -\sigma$, so $\langle\sigma_i\rangle = 0$ and the law of R_{12} is symmetric; zero-field statements therefore concern $|R_{12}|$, or a symmetrized measure.

3.2 The physics: replicas, their symmetry, and its breaking

3.2.1 The replicated partition function

The *replica trick* (Chapter 1) computes $\mathbb{E}Z_N^n$ for integer n, a sum over n copies of the system, and recovers $\mathbb{E}\log Z_N$ from $\mathbb{E}Z_N^n = 1 + n\mathbb{E}\log Z_N + O(n^2)$ as $n \to 0$.

COMPUTATION 3.1 · THE REPLICATED PARTITION FUNCTION

1 *Average one Gaussian.* For integer $n \geq 1$, $\mathbb{E}Z_N^n = \sum_{\sigma^1,\dots,\sigma^n} e^{h\sum_{a,i}\sigma_i^a}\,\mathbb{E}\exp\big(\sum_a X_N(\sigma^a)\big)$, and by (3.2) the Gaussian $\sum_a X_N(\sigma^a)$ has variance $n\beta^2(N-1)/2 + \sum_{a\neq b}\big(\frac{N\beta^2}{2}R_{ab}^2 - \frac{\beta^2}{2}\big)$, with $R_{ab} = R_{\sigma^a\sigma^b}$. Since $\mathbb{E}e^G = e^{\mathrm{Var}(G)/2}$,

$$\mathbb{E}Z_N^n = e^{nN\beta^2/4 - n^2\beta^2/4}\sum_{\sigma^1,\dots,\sigma^n}\exp\Big(\frac{N\beta^2}{2}\sum_{a<b}R_{ab}^2 + h\sum_{a,i}\sigma_i^a\Big). \quad (3.4)$$

2 *Linearize the squares.* The Hubbard–Stratonovich identity $e^{N\beta^2R^2/2} = \sqrt{N\beta^2/(2\pi)}\int_{\mathbb{R}} e^{-N\beta^2Q^2/2 + N\beta^2QR}\,\mathrm{d}Q$, with one variable Q_{ab} for each pair $a < b$, makes the exponent linear in the overlaps, and the spin sum factorizes over sites:

$$\mathbb{E}Z_N^n = e^{nN\beta^2/4 - n^2\beta^2/4}\Big(\frac{N\beta^2}{2\pi}\Big)^{n(n-1)/4}\int e^{NS_n(Q)}\prod_{a<b}\mathrm{d}Q_{ab},$$

$$S_n(Q) = -\frac{\beta^2}{2}\sum_{a<b}Q_{ab}^2 + \log\sum_{s\in\{\pm1\}^n}\exp\Big(\beta^2\sum_{a<b}Q_{ab}s_as_b + h\sum_a s_a\Big).$$

3 *Laplace's method at fixed n.* Since S_n is smooth and tends to $-\infty$ quadratically, $\frac{1}{N}\log\mathbb{E}Z_N^n \to \frac{n\beta^2}{4} + \max_Q S_n(Q)$ (Theorem A.11). For each fixed integer n this step is a theorem.

Two uncontrolled steps remain: the exchange of $N \to \infty$ with $n \to 0$, and the continuation of $n \mapsto \max_Q S_n(Q)$ from the integers to n near 0, which the integer moments alone cannot justify (Chapter 1). With $Q^\star$ a stationary point of S_n chosen by an ansatz, the prediction is $\phi = \lim_{n\to0}\frac{1}{n}\big(\frac{n\beta^2}{4} + S_n(Q^\star)\big)$.

3.2.2 Replica symmetry

THE REPLICA-SYMMETRIC SOLUTION. Take $Q_{ab} = q$ for all $a < b$. Then $\sum_{a<b}Q_{ab}s_as_b = \frac{q}{2}(\sum_a s_a)^2 - \frac{nq}{2}$, and a second Hubbard–Stratonovich variable z decouples the replicas:

$$S_n(q) = -\frac{\beta^2}{2}\,\frac{n(n-1)}{2}\,q^2 - \frac{n\beta^2q}{2} + \log\mathbb{E}_z\big(2\cosh(\beta\sqrt{q}\,z + h)\big)^n.$$

Dividing by n and letting $n \to 0$, with $\frac{1}{n}\log \mathbb{E}_z Y^n \to \mathbb{E}_z \log Y$, gives

$$\phi_{\mathrm{RS}}(\beta, h; q) = \log 2 + \mathbb{E}\log\cosh(\beta\sqrt{q}\,Z + h) + \frac{\beta^2}{4}(1-q)^2. \tag{3.5}$$

Gaussian integration by parts, $\mathbb{E}[Z\tanh(aZ+h)] = a\,\mathbb{E}\,\mathrm{sech}^2(aZ+h)$, gives

$$\partial_q \phi_{\mathrm{RS}}(\beta, h; q) = \frac{\beta^2}{2}\big(q - \mathbb{E}\tanh^2(\beta\sqrt{q}\,Z + h)\big), \tag{3.6}$$

so the order parameter solves $q = \mathbb{E}\tanh^2(\beta\sqrt{q}\,Z+h)$: it is the overlap $\langle s_a s_b\rangle$ of two replicas under the replicated single-site measure. ◇

The continuation has a cost. For integer $n \geq 2$ the saddle point maximizes S_n, but the coefficient $-\frac{\beta^2}{4}(n-1)$ of q^2 in $S_n(q)/n$ changes sign at $n = 1$, and as $n \to 0$ the physical q *minimizes* ϕ_{RS}. At $h = 0$ the only solution for $\beta \leq 1$ is $q = 0$, with the annealed value; for $\beta > 1$ a positive solution appears, $q = 0$ becomes a local maximum, and the physicist takes the minimum (Remark 3.4).

COMPUTATION 3.2 · REPLICA SYMMETRY AT ZERO TEMPERATURE

1. *The order parameter.* At $h = 0$ and $\beta \to \infty$, $q \to 1$, and substituting $u = \beta\sqrt{q}\,z$ in $1 - q = \mathbb{E}\,\mathrm{sech}^2(\beta\sqrt{q}\,Z)$, with $\int_{\mathbb{R}} \mathrm{sech}^2 u\,\mathrm{d}u = 2$, gives $\beta(1-q) \to 2/\sqrt{2\pi} = \sqrt{2/\pi}$.
2. *Energy.* Since $\log\cosh x = |x| - \log 2 + \log(1 + e^{-2|x|})$, $\phi_{\mathrm{RS}} = \beta\sqrt{q}\,\mathbb{E}|Z| + \mathbb{E}\log(1 + e^{-2\beta\sqrt{q}|Z|}) + \frac{1}{4}(\beta(1-q))^2$, and the middle term is $O(1/\beta)$. As $\phi = -\beta f = -\beta e + s$, the ground-state energy is $e_0 = -\lim \phi_{\mathrm{RS}}/\beta = -\mathbb{E}|Z| = -\sqrt{2/\pi} = -0.7979$.
3. *Entropy.* With $\beta(1-\sqrt{q}) = \beta(1-q)/(1+\sqrt{q}) \to \frac{1}{2}\sqrt{2/\pi}$,

$$s_0 = \lim\big(\phi_{\mathrm{RS}} + \beta e_0\big) = \lim\Big(-\sqrt{\tfrac{2}{\pi}}\,\beta(1-\sqrt{q}) + \tfrac{1}{4}\big(\beta(1-q)\big)^2\Big)$$
$$= -\frac{1}{\pi} + \frac{1}{2\pi} = -\frac{1}{2\pi}.$$

Ising spins have a nonnegative entropy,[3] so $s_0 = -0.159$ is impossible: replica symmetry fails at low temperature. The script `03-sk-replica.py` confirms the limits ($\beta(1-q) = 0.79820$, $e = -0.797884$ and $s = -0.15863$ at $T = 0.001$) and finds that the replica-symmetric entropy changes sign at $T = 0.2716$ (Figure 3.1(b)).

[3] At $h = 0$, $\phi_N - \beta\partial_\beta\phi_N = \frac{1}{N}\mathbb{E}\big[\log Z_N - \langle H_N\rangle\big]$ is the mean Shannon entropy of the Gibbs measure divided by N, which lies in $[0, \log 2]$.

3.2.3 The de Almeida–Thouless instability

THE REPLICON. The Hessian of S_n at $Q_{ab} = q$ is $M = -\beta^2 I + \beta^4 K$, with $K_{(ab),(cd)} = \langle s_a s_b s_c s_d\rangle_n - \langle s_a s_b\rangle_n \langle s_c s_d\rangle_n$ for the replicated single-site measure $\langle\cdot\rangle_n$. As $n \to 0$ its entries are $1 - q^2$ for equal pairs, $q - q^2$ for pairs sharing one index and $r - q^2$ for disjoint pairs, with $r = \mathbb{E}\tanh^4(\beta\sqrt{q}\,Z + h)$. A matrix indexed by pairs with three such entries, c_2, c_1 and c_0 for pairs sharing two, one or no indices, acts on the space of η with $\sum_b \eta_{ab} = 0$ for every a, of

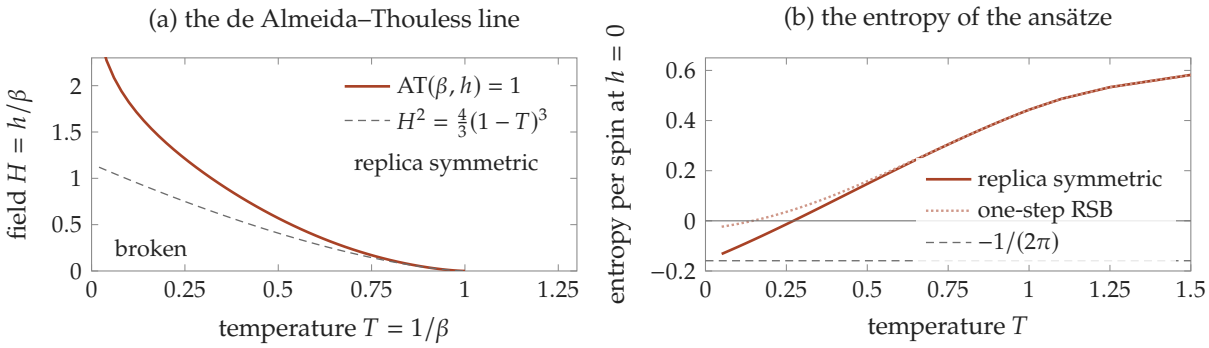


Figure 3.1. The replica predictions. (a) The de Almeida–Thouless line in the plane of temperature and physical field, from (3.6) and AT = 1 at 65 temperatures; dashed, its small-field law. Replica symmetry holds above the line (proved at $h = 0$ by Aizenman et al. (1987), for $h > 0$ in the 2026 preprints of Section 3.5) and fails below it (Toninelli 2002). (b) The entropy $s = \phi - \beta\partial_\beta\phi$ of the replica-symmetric and one-step solutions at $h = 0$, the latter minimized over (q, m) at 38 temperatures below $T = 1$. The replica-symmetric entropy is negative below $T = 0.2716$ and tends to $-1/(2\pi)$ (Computation 3.2); one step of breaking reduces the defect without removing it (-0.023 at $T = 0.05$). Script: `figures/scripts/03-sk-replica.py`.

dimension $n(n-3)/2$, as multiplication by $c_2 - 2c_1 + c_0$ (Exercise 3.5). Since $1 - 2q + r = \mathbb{E}(1 - \tanh^2)^2$, this *replicon* eigenvalue is

$$\lambda_{\mathrm{R}} = -\beta^2 + \beta^4\big[(1-q^2) - 2(q-q^2) + (r-q^2)\big] = -\beta^2\big(1 - \mathrm{AT}(\beta, h)\big). \quad (3.7)$$

At high temperature $\lambda_{\mathrm{R}} \approx -\beta^2 < 0$. De Almeida and Thouless (1978) took its vanishing as the limit of stability of replica symmetry, which is stable when $\mathrm{AT}(\beta, h) \le 1$. ◇

The argument is heuristic twice over: the multiplicity $n(n-3)/2$ tends to -1, and after the inversion at $n = 1$ no rule fixes the sign each eigenvalue should have. The *de Almeida–Thouless line* is nevertheless sharp: at $h = 0$ it is the point $\beta = 1$, and with a field it starts at $(T, H) = (1, 0)$ with $H^2 \simeq \frac{4}{3}(1-T)^3$ (Exercise 3.6) and rises as the temperature falls (Figure 3.1(a)).

3.2.4 Parisi's hierarchy

BREAKING REPLICA SYMMETRY IN STEPS. Parisi (1979) divided the n replicas into n/m groups of m, with $Q_{ab} = q_1$ for $a \neq b$ in the same group and q_0 otherwise. One Hubbard–Stratonovich variable z_1 per group and a common one z_0 decouple the replicas, with $\Xi = h + \beta\sqrt{q_0}\,z_0 + \beta\sqrt{q_1 - q_0}\,z_1$ as the local field. For integer n the group size satisfies $1 \le m \le n$; Parisi continued it to $0 \le m \le 1$, the interval between the two ends once $n \to 0$, and obtained

$$\mathcal{P}_1(q_0, q_1, m) = \log 2 + \frac{1}{m}\mathbb{E}_{z_0}\log\mathbb{E}_{z_1}\cosh^m\Xi + \frac{\beta^2}{4}\big[(1-q_1)^2 - m(q_1^2 - q_0^2)\big]. \quad (3.8)$$

Iterating k times gives nested groups, overlaps $q_0 < \cdots < q_k$ and sizes $0 \le m_1 \le \cdots \le m_k \le 1$, all encoded in the function $x(s) = \mu([0, s])$ of a

probability measure μ on $[0,1]$, with $x = m_p$ between q_{p-1} and q_p (Parisi 1980). As $k \to \infty$ the free entropy becomes the minimum over μ of

$$\mathcal{P}_{\beta,h}(\mu) = \log 2 + u_\mu(0,h) - \frac{\beta^2}{2}\int_0^1 s\,\mu([0,s])\,\mathrm{d}s, \tag{3.9}$$

where u_μ solves the *Parisi equation*

$$\partial_s u + \frac{\beta^2}{2}\Big(\partial_y^2 u + \mu([0,s])\,(\partial_y u)^2\Big) = 0 \quad (0 \le s < 1), \qquad u(1,y) = \log\cosh y. \tag{3.10}$$

Where $x \equiv m > 0$ on $[a,b]$, the substitution e^{mu} gives the backward heat equation, so $e^{mu(a,y)} = \mathbb{E}e^{mu(b,y+\beta\sqrt{b-a}\,Z)}$; where $x \equiv 0$, $u(a,y) = \mathbb{E}u(b, y + \beta\sqrt{b-a}\,Z)$. These rules give (3.5) for $\mu = \delta_q$ and (3.8) for $\mu = m\delta_{q_0} + (1-m)\delta_{q_1}$ (Exercise 3.2). ◇

Parisi (1983) identified $x(q)$ with the probability that two replicas have overlap at most q. Below the transition the Gibbs measure splits into many *pure states*,[4] whose overlaps, Mézard et al. (1984) found, are *ultrametric*: for three replicas, the two smallest of R_{12}, R_{13}, R_{23} are equal. In numerical solutions at $h = 0$ below $T = 1$ (Crisanti and Rizzo 2002), μ has a continuous part on $[0, q_{\max}]$ and an atom at $q_{\max}$: *full replica-symmetry breaking*.

Each step lowers the value: at $h = 0$ and $T = 0.5$, $\phi_{\mathrm{RS}} = 1.584786$ and $\min \mathcal{P}_1 = 1.582610$ (at $q_0 = 0$, $q_1 = 0.60132$, $m = 0.20424$). For the ground state, the zero-temperature version of (3.9) (Auffinger and Chen 2017) with $k = 0, 1, 2$ steps gives 0.797885, 0.765259 and 0.763608, against 0.7631667265(6) extrapolated from 42 steps by Oppermann et al. (2007); Parisi (1980) had 0.7633 ± 0.0001, as reported by Crisanti and Rizzo (2002). In this derivation the Hubbard–Stratonovich and Laplace steps at fixed n are rigorous; the exchange of limits, the continuation in n, the ansatz, the inverted rule that picks the stationary point and the stability analysis are not.

[4] In the physical picture their weights are those of a Poisson–Dirichlet point process, the law of the Gibbs weights of the random energy model at low temperature (Chapter 2).

3.3 The stone

PHYSICS · *the replica computation*

1 · The thermodynamic limit

The free entropy is assumed to self-average and to have a limit.

PROOF · *Gaussian interpolation*

Gaussian concentration, and an interpolation between one system and two independent halves that makes $N\phi_N$ superadditive (Theorem 3.2). An idea the physics does not need.

2 · Replicate

Compute $\mathbb{E}Z_N^n$ for integer n (Computation 3.1); the overlap matrix Q_{ab} is the order parameter; Laplace in N at fixed n; then $n \to 0$.

No moment of Z_N is computed. Interpolate between H_N and a one-body field $\beta\sqrt{q}\sum_i z_i\sigma_i$; integration by parts turns $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\log Z_t$ into averages over *two* replicas (Lemma 3.1).

3 · The replica-symmetric saddle

$Q_{ab} = q$; stationarity gives $q = \mathbb{E}\tanh^2(\beta\sqrt{q}\,Z + h)$; since $n(n-1)/2 < 0$ for $0 < n < 1$, the maximum over Q becomes a minimum over q.

Guerra's bound $\phi_N \le \phi_{\mathrm{RS}}(q)$ holds for every q and N, with an exact remainder (Theorem 3.3). No counterpart for the continuation in n: the minimum over q comes from the sign of the remainder.

4 · Two replicas at high temperature

$\mathbb{E}Z_N^2$ by Hubbard–Stratonovich: the Gaussian integral around $Q_{12} = 0$ converges when $\beta < 1$, so quenched equals annealed.

The same identity bounds $\mathbb{E}Z_N^2/(\mathbb{E}Z_N)^2$ at every N (Lemma 3.5), and with concentration proves $\phi = \log 2 + \beta^2/4$ for $\beta < 1$ (Theorem 3.6). The physical computation is the proof.

5 · Stability: the replicon

The replicon eigenvalue $-\beta^2(1 - \mathrm{AT})$, of multiplicity $n(n-3)/2 \to -1$; replica symmetry is stable if and only if $\mathrm{AT} \le 1$.

The remainder $\frac{\beta^2}{4}\int_0^1 \mathbb{E}\left\langle (R_{12}-q)^2\right\rangle_t \,\mathrm{d}t$ tends to 0 if and only if the overlap concentrates. A one-step perturbation of Guerra's bound has m-derivative $\frac{\beta^2\delta^2}{4}(\mathrm{AT}-1) + O(\delta^3)$, positive below the line (Theorem 3.9); above it, 2026 preprints prove symmetry (Theorem 3.10).

6 · Parisi's hierarchy

Nested groups of replicas, sizes $0 \le m_1 \le \cdots \le 1$ as $n \to 0$; the function $x(q)$ and the Parisi equation; Parisi's solution is taken to be the right stationary point.

Guerra replaces the hierarchy by nested averages over one-body fields (Ruelle probability cascades, in later language): again a mean-square remainder, so $\phi_N \le \mathcal{P}(\mu)$ for every μ. The lower bound needs a new idea: Talagrand's two coupled replicas, or Aizenman–Sims–Starr with ultrametricity (Theorem 3.8).

7 · The meaning of $x(q)$

x is the distribution function of R_{12}; pure states carry Poisson–Dirichlet weights and ultrametric overlaps.

Ghirlanda–Guerra identities, from the concentration of the energy under small perturbations, imply ultrametricity; for SK itself, a 2026 preprint identifies the limit of the overlap array (Theorem 3.11).

Read across, the stone shows where the routes meet and part. The proof justifies the replica functionals as upper bounds at every N, and at $n = 2$ the Hubbard–Stratonovich computation is literally a proof. It replaces the rest: the continuation in n by an interpolation in a real parameter, the inverted saddle-point rule by the sign of a remainder, the stability analysis by a perturbation of a bound. And it needs ideas the physics never needed, superadditivity, cascades and ultrametricity. The formula and its instability line date from 1975–78; the matching rigorous statements, from 1987 at zero field, 2002 below the line, and 2026, in preprints, above it.

3.4 The proof: interpolation, from the limit to high temperature

The first two theorems of this section rest on one lemma. Let Σ be a finite set with positive weights $w(\sigma)$, and let (X_σ) and (Y_σ) be independent centered Gaussian processes indexed by Σ, with covariances C^X and C^Y. For $t \in [0,1]$ let $\langle\cdot\rangle_t$, extended to replicas, be the Gibbs average of

$$Z_t = \sum_{\sigma\in\Sigma} w(\sigma)\exp\bigl(\sqrt{t}\,X_\sigma + \sqrt{1-t}\,Y_\sigma\bigr), \qquad \text{and let } \psi(t) = \mathbb{E}\log Z_t.$$

Lemma 3.1 (Interpolation of free entropies). *The function ψ is continuously differentiable on $[0,1]$, and*

$$\psi'(t) = \frac{1}{2}\mathbb{E}\Bigl\langle C^X(\sigma,\sigma) - C^Y(\sigma,\sigma)\Bigr\rangle_t - \frac{1}{2}\mathbb{E}\Bigl\langle C^X(\sigma^1,\sigma^2) - C^Y(\sigma^1,\sigma^2)\Bigr\rangle_t. \tag{3.11}$$

Proof. The plan is to apply Gaussian interpolation (Lemma A.2) to a log-partition function. Put $W_t = \sqrt{t}X + \sqrt{1-t}Y$. That lemma asks for a twice continuously differentiable F on $\mathbb{R}^\Sigma$ such that F and its partial derivatives of first and second order have moderate growth (for every $a > 0$, each is at most a constant times $e^{a|x|^2}$); it then says that $t \mapsto \mathbb{E}F(W_t)$ is continuously differentiable on $[0,1]$, with

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}F(W_t) = \frac{1}{2}\sum_{\sigma,\tau}\bigl(C^X - C^Y\bigr)(\sigma,\tau)\,\mathbb{E}\,\partial_\sigma\partial_\tau F(W_t).$$

Take the infinitely differentiable function $F(x) = \log\sum_\sigma w(\sigma)e^{x_\sigma}$ and put $G_\sigma = w(\sigma)e^{x_\sigma}/\sum_\tau w(\tau)e^{x_\tau} \in [0,1]$. Then $|F(x)| \le |\log\sum_\sigma w(\sigma)| + \max_\sigma|x_\sigma|$,

$$\partial_\sigma F = G_\sigma, \qquad \partial_\sigma\partial_\tau F = \mathbf{1}\{\sigma=\tau\}G_\sigma - G_\sigma G_\tau,$$

so F grows at most linearly and its first and second derivatives are bounded by 1; all three have moderate growth, and the lemma applies. At $x = W_t$ the G_σ are the Gibbs weights of Z_t, so the sum is $\langle C(\sigma,\sigma)\rangle_t - \bigl\langle C(\sigma^1,\sigma^2)\bigr\rangle_t$ with $C = C^X - C^Y$: this is (3.11), which the appendix also records as Corollary A.3. Its right side is bounded by $\max|C^X| + \max|C^Y|$, so ψ is Lipschitz and $\psi(1) - \psi(0) = \int_0^1 \psi'(t)\,\mathrm{d}t$. ∎

The derivative sees only the covariances, on the diagonal and between two replicas.[5]

[5] With $\log Z_t$ replaced by a smoothed maximum, the same computation gives the comparison inequalities of Slepian and Gordon (Theorem A.4; Chapter 8).

3.4.1 The thermodynamic limit

Theorem 3.2 (Guerra–Toninelli). *For every $\beta \ge 0$, $h \in \mathbb{R}$ and $N_1, N_2 \ge 1$,*

$$(N_1 + N_2)\,\phi_{N_1+N_2}(\beta, h) \ge N_1 \phi_{N_1}(\beta, h) + N_2 \phi_{N_2}(\beta, h).$$

Consequently the limit $\phi(\beta, h) = \lim_N \phi_N(\beta, h)$ exists, equals $\sup_N \phi_N(\beta, h)$, and is at most $\log(2\cosh h) + \beta^2/4$. If the couplings of all sizes are taken from one array $(g_{ij})_{1 \le i < j}$, then $\frac{1}{N}\log Z_N \to \phi(\beta, h)$ almost surely.

Proof. The plan is to interpolate between the system of size $N = N_1 + N_2$ and two independent systems on its first N_1 and last N_2 spins, and to read the sign of the derivative from the convexity of $q \mapsto q^2$. Write $\sigma = (\sigma^{(1)}, \sigma^{(2)})$, let $X^{(1)}$ and $X^{(2)}$ be the disorder parts (3.1) of two independent systems of sizes N_1 and N_2, independent of X_N, and apply Lemma 3.1 with $\Sigma = \Sigma_N$, $w(\sigma) = e^{h\sum_i \sigma_i}$, $X = X_N$ and $Y_\sigma = X^{(1)}(\sigma^{(1)}) + X^{(2)}(\sigma^{(2)})$. Then $\psi(1) = N\phi_N$, and at $t = 0$ the partition function factorizes, so $\psi(0) = N_1\phi_{N_1} + N_2\phi_{N_2}$.

With R_1 and R_2 the overlaps of the two blocks, so that $NR = N_1R_1 + N_2R_2$, (3.2) gives

$$C^X(\sigma,\sigma) - C^Y(\sigma,\sigma) = \tfrac{1}{2}\beta^2(N-1) - \tfrac{1}{2}\beta^2(N_1-1) - \tfrac{1}{2}\beta^2(N_2-1) = \tfrac{1}{2}\beta^2,$$
$$C^X(\sigma^1,\sigma^2) - C^Y(\sigma^1,\sigma^2) = \tfrac{1}{2}\beta^2\big(NR^2 - N_1R_1^2 - N_2R_2^2\big) + \tfrac{1}{2}\beta^2.$$

In (3.11) the constants cancel, and $\psi'(t) = -\frac{\beta^2}{4}\mathbb{E}\left\langle NR^2 - N_1R_1^2 - N_2R_2^2\right\rangle_t \ge 0$, because $R = \frac{N_1}{N}R_1 + \frac{N_2}{N}R_2$ and convexity give $NR^2 \le N_1R_1^2 + N_2R_2^2$. Hence $\psi(1) \ge \psi(0)$, the superadditivity.

Fekete's lemma turns superadditivity into convergence. Put $a_N = N\phi_N$, $a_0 = 0$, and fix $k \ge 1$. For $N = mk + r$ with $0 \le r < k$, superadditivity gives $a_N \ge ma_k + a_r$, so $a_N/N \ge \frac{mk}{N}\frac{a_k}{k} + \frac{a_r}{N} \to \frac{a_k}{k}$. Hence $\liminf_N a_N/N \ge \sup_k a_k/k \ge \limsup_N a_N/N$, and the supremum is finite, because (3.3) bounds every a_k/k by $\log(2\cosh h) + \beta^2/4$: the limit exists, is a real number, and equals the supremum.

For the almost sure statement, $F = \log Z_N$ is a smooth function of the $N(N-1)/2$ standard Gaussians g_{ij}, with $\partial F/\partial g_{ij} = \frac{\beta}{\sqrt{N}}\langle \sigma_i\sigma_j\rangle$, so $|\nabla F|^2 \le \frac{\beta^2}{N}\frac{N(N-1)}{2}$ and F is Lipschitz with constant $L = \beta\sqrt{(N-1)/2}$. Gaussian concentration (Theorem A.7; Theorem 1.8) gives, for $u > 0$,

$$\mathbb{P}\Big(\Big|\frac{1}{N}\log Z_N - \phi_N\Big| \ge u\Big) \le 2\exp\Big(-\frac{N^2u^2}{2L^2}\Big) \le 2e^{-Nu^2/\beta^2},$$

summable in N; the Borel–Cantelli lemma, for each rational $u > 0$, gives the almost sure convergence. At $\beta = 0$, $\log Z_N = N\log(2\cosh h)$ is not random. ∎

Superadditivity also gives $\phi_N \le \phi$ for every N: finite-size measurements lie below the limit (Section 3.6).

3.4.2 Guerra's replica-symmetric bound

Theorem 3.3 (Guerra's bound, with its remainder). *Let $N \geq 1$, $\beta \geq 0$, $h \in \mathbb{R}$ and $q \in [0,1]$. Let $z_1, \dots, z_N$ be independent standard Gaussians, independent of the couplings, let $\langle\cdot\rangle_t$ be the Gibbs average for*

$$H_{N,t}(\sigma) = \sqrt{t}\, X_N(\sigma) + \sqrt{1-t}\, \beta\sqrt{q} \sum_{i=1}^{N} z_i\sigma_i + h\sum_{i=1}^{N}\sigma_i,$$

extended to replicas, and let $\mathbb{E}$ average over the couplings and the z_i. Then

$$\phi_N(\beta,h) = \phi_{\mathrm{RS}}(\beta,h;q) - \frac{\beta^2}{4}\int_0^1 \mathbb{E}\big\langle (R_{12}-q)^2\big\rangle_t \,\mathrm{d}t. \tag{3.12}$$

In particular $\phi_N(\beta,h) \leq \inf_{q\in[0,1]} \phi_{\mathrm{RS}}(\beta,h;q)$ for every N, and the same holds for $\phi(\beta,h)$.

Proof. The plan is to apply Lemma 3.1 with $X = X_N$, $Y_\sigma = \beta\sqrt{q}\sum_i z_i\sigma_i$ and $w(\sigma) = e^{h\sum_i \sigma_i}$, and to watch the terms of order one cancel. At $t = 0$ the spins decouple, $Z_0 = \prod_i 2\cosh(\beta\sqrt{q}\, z_i + h)$, and $\frac{1}{N}\psi(0) = \log 2 + \mathbb{E}\log\cosh(\beta\sqrt{q}\, Z + h)$; at $t = 1$, $\frac{1}{N}\psi(1) = \phi_N$. The covariance of Y is $\beta^2 q N R_{\sigma\tau}$, that of X is (3.2), so $C^X - C^Y$ is $\frac{\beta^2(N-1)}{2} - \beta^2 qN$ on the diagonal and $\frac{N\beta^2}{2}R_{12}^2 - \frac{\beta^2}{2} - \beta^2 qNR_{12}$ between two replicas. Dividing (3.11) by N, the two terms $\beta^2/(4N)$ cancel and

$$\frac{\psi'(t)}{N} = \frac{\beta^2}{4} - \frac{\beta^2 q}{2} - \frac{\beta^2}{4}\mathbb{E}\big\langle R_{12}^2 - 2qR_{12}\big\rangle_t = \frac{\beta^2}{4}(1-q)^2 - \frac{\beta^2}{4}\mathbb{E}\big\langle (R_{12}-q)^2\big\rangle_t.$$

Integrating from 0 to 1 gives (3.12). The remainder is nonnegative, which gives the bound for every N and q, and Theorem 3.2 passes it to the limit. ∎

The identity (3.12) is Guerra's *sum rule*, the replica-symmetric case of Guerra (2003, Thm. 5), exact at every N in the normalization (3.1), which is also Guerra's. It says what replica symmetry means: the replica-symmetric formula holds in the limit if and only if $\int_0^1 \mathbb{E}\left\langle (R_{12}-q)^2\right\rangle_t \,\mathrm{d}t \to 0$, that is, if the overlap concentrates at q along the interpolation.

Remark 3.4 (Which q). By (3.6) the stationary points of $q \mapsto \phi_{\mathrm{RS}}(\beta,h;q)$ solve the fixed-point equation. For $h \neq 0$ the solution $q^\star$ is unique (Talagrand 2011, Prop. 1.3.8), and since the derivative is negative at $q = 0$ and positive at $q = 1$, it is the minimizer. At $h = 0$ and $\beta \leq 1$ the minimizer is $q = 0$ (Exercise 3.3). At $h = 0$ and $\beta > 1$, $\mathbb{E}\tanh^2(\beta\sqrt{q}\, Z) = \beta^2 q + O(q^2)$, so $\phi_{\mathrm{RS}}(q) < \phi_{\mathrm{RS}}(0)$ for small $q > 0$, and Theorem 3.3 proves $\phi(\beta,0) < \log 2 + \beta^2/4$: the annealed formula fails for every $\beta > 1$. Numerically the minimum is at the positive fixed point.

3.4.3 Replica symmetry at high temperature

At $h = 0$, Theorem 3.3 with $q = 0$ is the annealed bound. Is it attained? The *second moment method* answers, and its computation is the replica computation at $n = 2$.

Lemma 3.5 (The second moment). *Let $h = 0$ and $0 \le \beta < 1$, and let $\rho_1, \dots, \rho_N$ be independent uniform signs. For every $N \ge 1$,*

$$\frac{\mathbb{E}Z_N^2}{(\mathbb{E}Z_N)^2} = e^{-\beta^2/2}\,\mathbb{E}\exp\Big(\frac{\beta^2}{2N}\Big(\sum_{i=1}^N \rho_i\Big)^2\Big) \le \frac{e^{-\beta^2/2}}{\sqrt{1-\beta^2}}.$$

Proof. The plan is to compute the ratio exactly and to bound it by a Hubbard–Stratonovich linearization. By (3.2), $X_N(\sigma) + X_N(\tau)$ has variance $\beta^2(N-1) + N\beta^2 R_{\sigma\tau}^2 - \beta^2$, so $\mathbb{E}Z_N^2 = \sum_{\sigma,\tau} \exp\big(\frac{\beta^2(N-1)}{2} + \frac{N\beta^2}{2}R_{\sigma\tau}^2 - \frac{\beta^2}{2}\big)$, while $(\mathbb{E}Z_N)^2 = 4^N e^{\beta^2(N-1)/2}$. For independent uniform σ and τ the products $\sigma_i\tau_i$ are independent uniform signs with sum $NR_{\sigma\tau}$, which gives the identity. For the bound, use $e^{a^2/2} = \mathbb{E}e^{ag}$ with $g \sim \mathcal{N}(0,1)$ independent of the ρ_i, Tonelli's theorem, and $\cosh x \le e^{x^2/2}$:

$$\mathbb{E}_\rho \exp\Big(\frac{\beta^2}{2N}\Big(\sum_i \rho_i\Big)^2\Big) = \mathbb{E}_g\mathbb{E}_\rho \exp\Big(\frac{\beta g}{\sqrt{N}}\sum_i \rho_i\Big) = \mathbb{E}_g\Big[\cosh\Big(\frac{\beta g}{\sqrt{N}}\Big)^N\Big]$$
$$\le \mathbb{E}_g e^{\beta^2 g^2/2} = \frac{1}{\sqrt{1-\beta^2}}.$$ ∎

Theorem 3.6 (Replica symmetry at high temperature without a field). *Let $h = 0$, $0 < \beta < 1$ and $c_\beta = \frac{1}{4}e^{\beta^2/2}\sqrt{1-\beta^2}$. For every $N \ge 1$,*

$$\phi_N(\beta, 0) \ge \log 2 + \frac{\beta^2(N-1)}{4N} - \frac{\log 2}{N} - \beta\sqrt{\frac{\log(2/c_\beta)}{N}},$$

and $\phi_N(\beta,0) \le \log 2 + \frac{\beta^2(N-1)}{4N}$. Consequently $\phi(b,0) = \log 2 + b^2/4 = \inf_q \phi_{\mathrm{RS}}(b,0;q)$ for every $b \in [0,1]$.

Proof. The plan is to show that Z_N exceeds half its mean with a probability bounded below, and to let Gaussian concentration turn this into a bound on $\mathbb{E}\log Z_N$. The upper bound is (3.3). For $N = 1$ both bounds hold because $\phi_1 = \log 2$; let $N \ge 2$, $Z = Z_N$ and $\mu = \mathbb{E}Z$. By the Paley–Zygmund inequality and Lemma 3.5,

$$\mathbb{P}(Z > \mu/2) \ge \frac{(1-\frac{1}{2})^2\mu^2}{\mathbb{E}Z^2} \ge \frac{1}{4}e^{\beta^2/2}\sqrt{1-\beta^2} = c_\beta.$$

As in the proof of Theorem 3.2, $\log Z$ is Lipschitz in the couplings with constant $L = \beta\sqrt{(N-1)/2}$, so $\mathbb{P}(\log Z - \mathbb{E}\log Z \ge s) \le e^{-s^2/(2L^2)}$ for $s > 0$ (Theorem A.7). Take $s = \beta\sqrt{(N-1)\log(2/c_\beta)}$, so that $e^{-s^2/(2L^2)} = c_\beta/2$. If $\mathbb{E}\log Z < \log(\mu/2) - s$, then

$$\mathbb{P}(Z > \mu/2) \le \mathbb{P}(\log Z - \mathbb{E}\log Z > s) \le c_\beta/2 < c_\beta,$$

a contradiction. Hence $\mathbb{E}\log Z \geq \log\mu - \log 2 - s$; dividing by N, inserting $\frac{1}{N}\log\mu = \log 2 + \frac{\beta^2(N-1)}{4N}$ and using $\sqrt{N-1} \leq \sqrt{N}$ gives the lower bound.

Both bounds converge to $\log 2 + \beta^2/4$. This is $\phi_{\mathrm{RS}}(\beta, 0; 0)$, and also $\inf_q \phi_{\mathrm{RS}}(\beta, 0; q)$: by $\log\cosh x \geq x^2/2 - x^4/12$, the difference $\phi_{\mathrm{RS}}(\beta, 0; q) - \phi_{\mathrm{RS}}(\beta, 0; 0)$ is at least $\frac{\beta^2 q^2}{4}(1-\beta^2) \geq 0$ (Exercise 3.3). At $\beta = 0$, $Z_N = 2^N$ and $\phi_N = \log 2$. At $\beta = 1$: each ϕ_N is convex in β, since $\log Z_N$ is a log-sum-exp of functions linear in β, so the limit is convex, hence continuous on $(0, \infty)$, and equals $\log 2 + \beta^2/4$ on $(0, 1)$. ∎

The rate $N^{-1/2}$ is not sharp (the true correction is of order $1/N$), but the mechanism is visible: the annealed value is exact when the second moment, the replicated partition function at $n = 2$, is of the order of the first squared, and its Hubbard–Stratonovich integral converges exactly when $\beta < 1$.[6] At $\beta > 1$ the ratio grows exponentially in N, dominated by pairs of configurations with a macroscopic overlap.

[6] The ratio $\mathbb{E}Z_N^2/(\mathbb{E}Z_N)^2$ is also the second moment of the likelihood ratio between the spiked model with $\lambda = \beta$ and pure noise. Its boundedness for $\lambda < 1$ is the classical second-moment argument that the spike cannot be detected below the spectral threshold.

The free entropy also controls the overlap, through an identity exact at every N.

Lemma 3.7 (The derivative in β). *For every $N \geq 1$, $h \in \mathbb{R}$ and $\beta > 0$, $\partial_\beta \phi_N(\beta, h) = \frac{\beta}{2}\big(1 - \mathbb{E}\left\langle R_{12}^2\right\rangle\big)$.*

Proof. The plan is one integration by parts, together with $\sum_{i,j}\left\langle\sigma_i\sigma_j\right\rangle^2 = N^2\left\langle R_{12}^2\right\rangle$, which holds because $\left\langle\sigma_i^1\sigma_j^1\sigma_i^2\sigma_j^2\right\rangle = \left\langle\sigma_i\sigma_j\right\rangle^2$. Since $\partial_\beta H_N = X_N/\beta$ and $|\langle X_N\rangle| \leq \max_\sigma |X_N(\sigma)|$ is integrable uniformly for β in compact sets, $\partial_\beta\phi_N = \frac{1}{N\beta}\mathbb{E}\langle X_N\rangle$. Integration by parts (Theorem A.1) gives $\mathbb{E}[g_{ij}\left\langle\sigma_i\sigma_j\right\rangle] = \mathbb{E}[\partial_{g_{ij}}\left\langle\sigma_i\sigma_j\right\rangle] = \frac{\beta}{\sqrt{N}}\big(1 - \mathbb{E}\left\langle\sigma_i\sigma_j\right\rangle^2\big)$, so

$$\begin{aligned}\mathbb{E}\langle X_N\rangle &= \frac{\beta^2}{N}\sum_{i<j}\big(1 - \mathbb{E}\left\langle\sigma_i\sigma_j\right\rangle^2\big) = \frac{\beta^2}{N}\Big(\frac{N(N-1)}{2} - \frac{N^2\mathbb{E}\left\langle R_{12}^2\right\rangle - N}{2}\Big)\\ &= \frac{N\beta^2}{2}\big(1 - \mathbb{E}\left\langle R_{12}^2\right\rangle\big).\end{aligned}$$ ∎

With Theorem 3.6 and convexity, the lemma gives $\mathbb{E}\left\langle R_{12}^2\right\rangle = O(N^{-1/4})$ at $h = 0$ and $\beta < 1$ (Exercise 3.9): from free entropy to overlap, the stone's fifth row in its simplest case. The true order is $1/N$; the physics predicts $N\mathbb{E}\left\langle R_{12}^2\right\rangle \to 1/(1-\beta^2)$, and Aizenman et al. (1987) proved a central limit theorem for $\sqrt{N}R_{12}$, as Kusuoka and Nakajima (2026) recount.

3.5 The proof: Parisi's formula and the de Almeida–Thouless line

The rest of the theory is stated with its sources, in this chapter's normalization (Panchenko (2013b) and Auffinger and Chen (2015) write $\xi(q) = \beta^2 q^2$: their β is our $\beta/\sqrt{2}$).

Theorem 3.8 (The Parisi formula). *For every $\beta \geq 0$ and $h \in \mathbb{R}$, $\lim_N \phi_N(\beta, h) = \min_\mu \mathcal{P}_{\beta,h}(\mu)$, the minimum of (3.9) over Borel probability measures on $[0,1]$, and the minimizer $\mu_{\beta,h}$, the* Parisi measure, *is unique.*

The upper bound is Guerra's (2003, Thm. 3) and the lower bound Talagrand's (2006), for SK and mixed even p-spin models; Auffinger and Chen (2015, Cor. 1) proved uniqueness. Guerra's proof runs the interpolation of Theorem 3.3 with a hierarchy of independent one-body fields, one for each level of a discrete μ, averaged recursively with the exponents m_p; the remainder is again a mean square, now in auxiliary Gibbs states. In the later language of Aizenman et al. (2003) and Panchenko (2013b) the interpolation runs on Σ_N times the leaves of a Ruelle probability cascade, leaves with Poisson–Dirichlet weights whose overlaps are the q_p, and the remainder is $\frac{\beta^2}{4}\int_0^1 \mathbb{E}\left\langle (R_{12} - q_{\alpha^1\wedge\alpha^2})^2\right\rangle_t \,\mathrm{d}t$, with $q_{\alpha^1\wedge\alpha^2}$ the overlap of the two leaves. The lower bound needs a reason for this remainder to vanish at the optimal μ: Talagrand bounded the free entropy of two coupled replicas with constrained overlap by a two-dimensional Guerra bound, and Panchenko (2014) gave a second proof, for general mixtures, from the cavity representation of Aizenman et al. (2003), the *Ghirlanda–Guerra identities* (Ghirlanda and Guerra 1998) and ultrametricity (Panchenko 2013a). Neither idea has a counterpart in the replica computation. Below the line, Guerra's bound beats replica symmetry.

Theorem 3.9 (Toninelli). *Let q solve the fixed-point equation, taking the positive solution when $h = 0$ and $\beta > 1$. If $\mathrm{AT}(\beta, h) > 1$, then $\phi(\beta, h) < \phi_{\mathrm{RS}}(\beta, h; q)$.*

Proof sketch. Evaluate Guerra's broken bound at $\mu_m = m\delta_q + (1-m)\delta_{q+\delta}$ with $\delta > 0$ small, so that $\mathcal{P}(\mu_1) = \phi_{\mathrm{RS}}(q)$. With $a^2 = \beta^2\delta$, $Y = h + \beta\sqrt{q}\,Z'$ and ν the law of Z tilted by $\cosh(Y + aZ)$,

$$\partial_m \mathcal{P}(\mu_m)\big|_{m=1} = \mathbb{E}\Big[\mathbb{E}_\nu \log\cosh(Y + aZ) - \log\cosh Y - \frac{a^2}{2}\Big] - \frac{\beta^2}{4}(2q\delta + \delta^2).$$

Expanding with the heat equation, the first expectation is $\frac{a^2}{2}\mathbb{E}\tanh^2 Y + \frac{a^4}{4}\mathbb{E}\operatorname{sech}^4 Y + O(a^6)$; the fixed-point equation cancels the first-order terms, and

$$\partial_m \mathcal{P}(\mu_m)\big|_{m=1} = \frac{\beta^2\delta^2}{4}\big(\mathrm{AT}(\beta, h) - 1\big) + O(\delta^3). \tag{3.13}$$

If $\mathrm{AT} > 1$ this is positive for small δ, so $\phi \leq \mathcal{P}(\mu_m) < \phi_{\mathrm{RS}}(q)$ for m slightly below 1. Omitted: the uniformity of the $O(a^6)$ term. The argument and the coefficient are Toninelli (2002)'s, who computes the second derivative in δ of the left side of (3.13) at $\delta = 0$, $\frac{\beta^2}{2}(\mathrm{AT} - 1)$; the route through the heat equation is ours. The script `03-sk-replica.py` confirms the coefficient: at $(\beta, h) = (2, 0.5)$ the ratio of the exact derivative to it is 0.78, 0.89, 0.94 and 0.97 at $\delta = 0.04$, 0.02, 0.01 and 0.005, and one minus the ratio halves with δ, as an $O(\delta^3)$ remainder requires. ∎

The replicon eigenvalue (3.7) reappears in (3.13): the rigorous trace of the stability analysis. The converse remained open for $h > 0$ after partial results (Brennecke and Yau 2022; Jagannath and Tobasco 2017); in February 2026 Mourrat and Schertzer (2026), who showed in a preprint that a generalized criterion fails for some mixed p-spin models, still called the SK case open. Two preprints have since claimed it.

Theorem 3.10 (The de Almeida–Thouless line with a field; preprints).

(i) (Lopatto 2026c, Thm. 1.1 and Prop. 1.2) If $\beta > 0$, $h > 0$ and $\mathrm{AT}(\beta, h) \le 1$, then $\phi(\beta, h) = \phi_{\mathrm{RS}}(\beta, h; q)$. The set $\{\mathrm{AT} = 1\}$ is the graph of a smooth, strictly increasing function $h_{\mathrm{AT}} \colon (1, \infty) \to (0, \infty)$ with $h_{\mathrm{AT}}(1^+) = 0$, together with the point $(1, 0)$.

(ii) (Kusuoka and Nakajima 2026, Thm. 1.1) For every compact set $K \subset \{\beta > 0,\ h > 0,\ \mathrm{AT}(\beta, h) < 1\}$ there is C_K such that $N\mathbb{E}\left\langle (R_{12} - q)^2 \right\rangle_t \le C_K$ for all $t \in [0, 1]$, $(\beta, h) \in K$ and $N \ge 1$, and $0 \le \phi_{\mathrm{RS}}(\beta, h; q) - \phi_N(\beta, h) \le C_K/N$.

Lopatto checks the optimality condition of Jagannath and Tobasco (2017) for $\mu = \delta_q$, so his proof rests on the Parisi formula; the proof of Kusuoka and Nakajima is, as they put it, self-contained, does not use the Parisi formula, and gets its rate through the sum rule (3.12). Both are unrefereed as of October 2026. Lopatto states that his paper was written with the assistance of large language models, which suggested arguments, helped draft and revise, and coded simulations. For their key step, the flatness of the one-step bound near the replica-symmetric point, Kusuoka and Nakajima "asked ChatGPT 5.6 Sol to construct a proof"; they revised the resulting argument in detail, take full responsibility for it, and used Codex to formalize the proof in Lean 4.

At $h = 0$, Zhou (2026, Thm. 1 and Cor. 2) proved, in a refereed paper, that for $\beta > 1$ close to 1 the Parisi measure is supported on an interval $[0, q_\beta]$, with a smooth density on $[0, q_\beta)$ and a single atom, at q_β: full replica-symmetry breaking just below the critical temperature. The rest is in preprints.

Theorem 3.11 (Full replica-symmetry breaking and the overlap law; preprints).

(i) (Lopatto 2026a, Thm. 1.1) For every $\beta > 1$ and $h = 0$ there are $q_\beta, c_\beta \in (0, 1)$ and a nonnegative ρ_β, smooth up to the endpoints, with $\mu_{\beta,0}(\mathrm{d}s) = \rho_\beta(s)\mathbf{1}_{[0,q_\beta)}(s)\,\mathrm{d}s + c_\beta \delta_{q_\beta}(\mathrm{d}s)$ and $\operatorname{supp} \mu_{\beta,0} = [0, q_\beta]$.

(ii) (Lopatto 2026b, Thm. 1.1) For every $\beta > 0$ and $h > 0$ the support of $\mu_{\beta,h}$ is an interval $[q_-, q_+] \subset (0, 1)$; when it is not a point, $\mu_{\beta,h}$ has atoms at both ends and a smooth positive density in between.

(iii) (Aronow and Lopatto 2026a, Thm. 1.2 and Cor. 1.4) For every h, the law of the overlap array $(R_{ab})_{a<b}$ under $\mathbb{E}\langle\cdot\rangle$ converges to the overlap array of the Ruelle cascade of $\mu_{\beta,h}$ (with independent signs when $h = 0$); in particular the law of R_{12}, of $|R_{12}|$ when $h = 0$, converges to $\mu_{\beta,h}$, and

> *the limiting overlaps (in absolute value when $h = 0$) are ultrametric.*

Before part 3, the overlap law and ultrametricity were known only for Gibbs measures that satisfy the Ghirlanda–Guerra identities (Panchenko 2013a,b), such as those of generic mixed p-spin models or of perturbed Hamiltonians; SK itself is not generic. All three papers state that they were written with the assistance of large language models, used to suggest arguments, to draft and to write code for checks; for part 2 the observables and inequalities of the proof were located, in Lopatto's words, using "a massive number of numerical experiments orchestrated by large language models". Building on an earlier version of part 1, Chen (2026) claims a proof that $q_\beta \to 1$ as $\beta \to \infty$ and that the zero-temperature Parisi measure has support $[0, 1)$; its arXiv record states that "ChatGPT 5.6 is the actual author", and that Chen, listed to comply with arXiv policy, accepts full responsibility. At zero temperature Auffinger and Chen (2017) proved the Parisi formula for the ground-state energy, and Auffinger et al. (2020) that its Parisi measure has infinitely many points in its support.

3.6 Measuring it

Two scripts do the measuring, in about one and two minutes of computation: `03-sk-replica.py` solves the replica equations quoted in Section 3.2, and `03-sk-enum.py` enumerates all 2^N configurations for $N \le 16$, so that each $\log Z_N$ and each Gibbs average is exact for its draw of the couplings, and averages over independent draws.[7]

In Figure 3.2(a) every point lies below the annealed curve, which for $\beta \le 1$ is the limit, as Theorem 3.2 requires; at $\beta = 0.5$, $\phi_{16} = 0.7501 \pm 0.0015$, against 0.7517 for the bound at $N = 16$. At $\beta = 2$, $\phi_{16} = 1.417 \pm 0.021$, while the replica-symmetric and one-step values are 1.5848 and 1.5826: the finite-size gap is about 77 times the difference between the ansätze, and at this size the free entropy cannot tell them apart.

Panels (b) and (c) measure the identity (3.12), whose two sides come from different objects; they agree within their intervals at every N and both temperatures: at $\beta = 0.8$ and $N = 10$, 0.02158 ± 0.00101 against 0.02158 ± 0.00025; at $\beta = 2$ and $N = 12$, 0.180 ± 0.006 against 0.183 ± 0.012. At $\beta = 0.8$ the gap lies between the rigorous bounds and approaches the prediction $0.2554/N$, slowly: N times the gap grows from 0.194 at $N = 4$ to 0.233 at $N = 14$, and $N\mathbb{E}\left\langle R_{12}^2 \right\rangle$ rises from 1.44 to 1.79 ± 0.07 toward its predicted limit $1/(1-\beta^2) = 2.78$. At $\beta = 2$ the gap decreases a little more slowly than $1/N$ (N times the gap grows from 1.5 to 2.4), though in the limit it is at least 0.0022, and positive by Theorem 3.9. No measurement contradicts a theorem, and none can see the breaking of replica symmetry.

[7] A Gibbs measure costs about $2^N N^2$ operations, so $N = 16$, with 65,536 configurations, is the practical limit of exact enumeration on a laptop; Monte Carlo reaches larger N but loses the exactness that makes the identity checkable at each size.

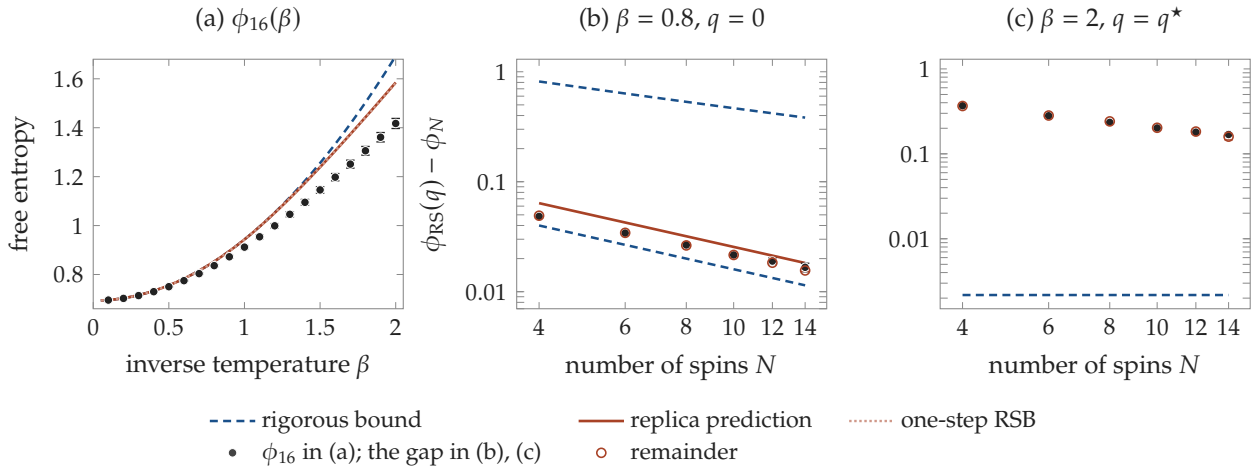


Figure 3.2. Measurements at $h = 0$, with Student-t 95% intervals. (a) The free entropy $\phi_{16}(\beta)$, the mean of $\frac{1}{16}\log Z_{16}$ over 120 draws of the couplings (the same draws at every β, seed 3016), against the replica-symmetric prediction at its minimizing q, the one-step value (3.8), and the limit $\log 2 + \beta^2/4$ of the annealed bound (3.3), which is the limit of ϕ_N for $\beta \le 1$ (Theorem 3.6). (b), (c) Guerra's sum rule (3.12) on both sides: the gap $\phi_{\mathrm{RS}}(q) - \phi_N$, from $\log Z_N$ (8000 draws at $N = 4$ down to 1000 at $N = 14$), and the remainder, from the Gibbs measures of the interpolation at six Gauss–Legendre nodes in t (2000 down to 100 draws), with independent draws (seeds $30000 + 100N + k$ and $30050 + 100N + k$). In (b) the dashed lines are the bounds of Theorem 3.6 (above) and (3.3) (below), and the solid line is the asymptotic prediction $-\frac{1}{4N}\log(1-\beta^2)$; in (c), $q^\star = 0.5304$ and the dashed line is $\phi_{\mathrm{RS}}(q^\star) - \min \mathcal{P}_1 = 0.0022$, a lower bound at every N by Guerra's broken bound.

3.7 The status board

The free entropy of SK is settled: the limit exists, the Parisi formula is a theorem with a unique minimizer, replica symmetry holds without a field exactly up to $\beta = 1$, and it fails below the de Almeida–Thouless line. Full replica-symmetry breaking is a refereed theorem just below the critical temperature at zero field. As of October 2026, replica symmetry above the line with a field, full replica-symmetry breaking in the rest of the low-temperature phase, and the identification of $x(q)$ with the law of the overlap of SK itself are proved in preprints of April to October 2026, prepared, as they state, with the assistance of large language models, and not yet refereed.

STATUS BOARD · broken replicas ↔ interpolation

PREDICTION	STATUS	PROOF OR EVIDENCE
The free entropy has a limit and self-averages	proved	Guerra and Toninelli (2002); Theorem 3.2
Replica symmetry at $h = 0$, $\beta \le 1$: $\phi = \log 2 + \beta^2/4$	proved	Aizenman et al. (1987); Theorem 3.6

Replica symmetry for $h > 0$ on the whole region AT ≤ 1	preprint	arXiv:2604.11921v2; with rate $1/N$ where AT < 1, arXiv:2608.23413v2; AI-assisted (Theorem 3.10)
Replica symmetry below the line	false	Toninelli (2002), all $\beta > 1$ at $h = 0$; Theorem 3.9 and Computation 3.2
The Parisi formula, with a unique minimizer	proved	Guerra (2003, Thm. 3), Talagrand (2006); uniqueness, Auffinger and Chen (2015)
Full RSB at $h = 0$ for $\beta > 1$ close to 1: support $[0, q_\beta]$	proved	Zhou (2026, Thm. 1)
Full RSB everywhere below the line: $h = 0$ and $\beta > 1$; $h > 0$ and AT > 1	preprint	arXiv:2607.11756v4; arXiv:2610.00127v1 with Toninelli (2002); AI-assisted (Theorem 3.11)
Ultrametricity of the overlaps	proved under	the Ghirlanda–Guerra identities (generic mixed p-spin models, perturbed Hamiltonians) (Panchenko 2013a); SK itself, next row
In SK, the law of R_{12} tends to the Parisi measure; ultrametric	preprint	arXiv:2609.17442v1; AI-assisted (Theorem 3.11)
The ground-state energy per spin is the zero-temperature Parisi value	proved	Auffinger and Chen (2017); its value, -0.76317, is a numerical evaluation (Oppermann et al. 2007)

3.8 Where the translation fails

WHAT WAS ASSUMED. Every proof here uses Gaussian couplings, through integration by parts, and a complete graph, through a covariance (3.2) that depends on the configurations only through their overlap. Guerra's bounds use the convexity of ξ, which fails for odd mixtures, where Panchenko (2014) needed ultrametricity. The field is deterministic and uniform. And in the inference reading SK is the posterior under a wrong model (Chapter 4 treats the right one).

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. Replica symmetry is wrong below the line, as its negative entropy says. One step of breaking is not right either: its entropy is still negative at low temperature (Figure 3.1(b)), and at zero temperature no finite number of steps suffices (Auffinger et al. 2020). The local stability criterion does not characterize replica symmetry in general: Mourrat and Schertzer (2026) constructed, in a preprint, mixed p-spin models in which the generalized de Almeida–Thouless criterion fails. And simulations at accessible sizes do not test the breaking through the free entropy (Section 3.6).

OPEN PROBLEMS. The 2026 preprints of Theorems 3.10 and 3.11 await independent verification and refereeing. For $\beta > 1$ and $h = 0$, Aronow and Lopatto (2026b) show, in a preprint also written with the assistance of large language models, that $\operatorname{Var} \log Z_N$ lies between orders $N^{4/15}$ and $N^{7/15}$; the exact exponent is open. And if the local criterion fails for some mixed p-spin models, as the preprint of Mourrat and Schertzer (2026) shows, which condition characterizes their replica-symmetric phase?

Exercises

3.1 CONSOLIDATION. (Moments.) (a) Prove the formula for $\mathbb{E}Z_N$ in (3.3). (b) For $h \neq 0$, show that $\mathbb{E}Z_N^2/(\mathbb{E}Z_N)^2 \geq e^{-\beta^2/2} \exp(N\beta^2 \tanh^4 h/2)$, so that the argument of Theorem 3.6 fails with a field. *(Hint: under the tilted product law the overlap has mean* $\tanh^2 h$*; use Jensen.)*

3.2 CONSOLIDATION. (The Parisi functional by hand.) Using the two rules after (3.10), show that $\mathcal{P}(\delta_q) = \phi_{\mathrm{RS}}(q)$ and that $\mathcal{P}(m\delta_{q_0} + (1-m)\delta_{q_1})$ is (3.8).

3.3 CONSOLIDATION. (No field, high temperature.) Using $\log\cosh x \geq x^2/2 - x^4/12$, show that $\phi_{\mathrm{RS}}(\beta, 0; q) - \phi_{\mathrm{RS}}(\beta, 0; 0) \geq \frac{\beta^2 q^2}{4}(1-\beta^2)$, so that $\inf_q \phi_{\mathrm{RS}}(\beta, 0; q) = \log 2 + \beta^2/4$ when $\beta \leq 1$.

3.4 CONSOLIDATION. (The annealed bound from the overlap.) Using Lemma 3.7 and $\langle R_{12}^2 \rangle \geq 1/N$, rederive (3.3).

3.5 BY HAND. (The replica Hessian.) For a matrix indexed by pairs $\{a, b\} \subset \{1, \dots, n\}$, $n \geq 4$, with entries c_2, c_1, c_0 for pairs that coincide, share one index, or are disjoint, show that its eigenvalues are $c_2 - 2c_1 + c_0$ (multiplicity $n(n-3)/2$), $c_2 + (n-4)c_1 - (n-3)c_0$ (multiplicity $n-1$) and $c_2 + 2(n-2)c_1 + \frac{(n-2)(n-3)}{2}c_0$ (multiplicity 1).

3.6 BY HAND. ⋆ (The line near the critical point.) Expanding the fixed-point equation to third order and the condition $AT = 1$ to second order in $\epsilon = \beta - 1$, show that on the line $q = \epsilon + O(\epsilon^3)$ and $h^2 = \frac{4}{3}\epsilon^3 + O(\epsilon^4)$, so that $H^2 \simeq \frac{4}{3}(1-T)^3$. *(Hint: expand* $\mathbb{E}\tanh^2 x$ *and* $\mathbb{E}\operatorname{sech}^4 x$*,* $x = \beta\sqrt{q}Z + h$*, in powers of* $\beta^2 q$ *and* h^2*.)* The script finds $(1-T)^3/H^2 = 0.7485$ at $H = 10^{-4}$.

3.7 BY HAND. (A third-order transition.) Show that at $h = 0$ the replica-symmetric order parameter is $q = \frac{1}{2}(\beta^2 - 1) + O((\beta^2-1)^2)$ and that $\phi_{\mathrm{RS}}(q) - \log 2 - \frac{\beta^2}{4} = -\frac{1}{48}(\beta^2-1)^3 + O((\beta^2-1)^4)$ as $\beta \downarrow 1$. *(Hint:* $\mathbb{E}\log\cosh(aZ) = \frac{a^2}{2} - \frac{a^4}{4} + \frac{a^6}{3} + O(a^8)$*.)* At $\beta = 1.005$ the ratio of the exact difference to this term is 0.991.

3.8 TRANSLATION. (The Gaussian integral around the saddle.) The physicist's Laplace method around $Q_{12} = 0$ gives $\mathbb{E}Z_N^2/(\mathbb{E}Z_N)^2 \simeq e^{-\beta^2/2}(1-\beta^2)^{-1/2}$. Prove this limit for $\beta < 1$, by the central limit theorem and uniform integrability from Lemma 3.5, and show that the ratio tends to infinity for $\beta \geq 1$.

3.9 TRANSLATION. ⋆ (From the free entropy to the overlap.) (a) Show from (3.12) that at $h = 0$ and $\beta < 1$, $\int_0^\beta b\,\mathbb{E}\left\langle R_{12}^2\right\rangle_b\,\mathrm{d}b \to 0$, with $\langle\cdot\rangle_b$ the Gibbs measure at inverse temperature b. (b) Let $h = 0$, $0 < \beta < 1$, c_β as in Theorem 3.6, and $\delta_N = (\log 2 + \beta^2/4)/N + \beta\sqrt{\log(2/c_\beta)/N}$. Prove that $\mathbb{E}\left\langle R_{12}^2\right\rangle \le \frac{2}{\beta}\sqrt{\delta_N}$ for every N. *(Hint: ϕ_N is convex in β, so $\partial_\beta\phi_N(\beta)$ is at least the slope of the chord from $\beta' = \beta - 2\sqrt{\delta_N}$; bound the chord with Theorem 3.6 and* (3.3)*, and use Lemma 3.7.)*

3.10 MEASUREMENT. (Both sides of the line.) Modify `03-sk-enum.py` to compute $\phi_{\mathrm{RS}}(q^\star) - \phi_N$ at $T = 0.5$ with $H = 0.5$ (below the line, AT $= 1.063$) and $H = 1$ (above it, AT $= 0.640$), for $N = 6, \dots, 14$. Observe that N times the gap is nearly constant in both cases (about 1.1 and 0.5), and explain.

3.11 RESEARCH. (A conditional second moment.) By Exercise 3.1, the plain second moment fails when $h \neq 0$. Brennecke and Yau (2022) proved replica symmetry on the region $\beta^2\mathbb{E}\,\mathrm{sech}^2(\beta\sqrt{q}\,Z + h) \le 1$ by a conditional second-moment method. The preprints of Theorem 3.10 reach the whole region AT ≤ 1 through the Parisi formula or through interpolation and cavity estimates. Can a conditional second moment reach it too?

Notes and References

The physics runs from Sherrington and Kirkpatrick (1975) through de Almeida and Thouless (1978) to Parisi (1979, 1980) and Parisi (1983) and the ultrametricity of Mézard et al. (1984). In 2021 Parisi received half of the Nobel Prize in Physics, for discovering the interplay of disorder and fluctuations in physical systems. The rigorous theory runs from Aizenman et al. (1987) through Guerra and Toninelli (2002), Guerra (2003), Toninelli (2002) and Aizenman et al. (2003) to Talagrand (2006), and on to Panchenko (2013a, 2014); Talagrand received the Abel Prize in 2024. The monographs of Talagrand (2011) and Panchenko (2013b) develop the theory, and Montanari and Sen (2024) teach the replica and interpolation computations side by side for readers without physics.

The presentation is the book's own in two places: the explicit constants of Theorem 3.6 and Exercise 3.9, which assemble classical second-moment, concentration and convexity arguments, and the derivation of Toninelli's second-order coefficient (3.13) through the heat equation, which we checked numerically. The 2026 results are reported as their authors state them, in the arXiv versions current on 25 October 2026.

TRANSLATING THE LITERATURE. Talagrand (2011) and Lopatto (2026c) use this chapter's Hamiltonian; Talagrand writes $p_N(\beta, h)$ for ϕ_N, $\nu(f)$ for $\mathbb{E}\left\langle f\right\rangle$ and $R_{1,2}$ for R_{12}, and Lopatto writes F_N, F and $\alpha(\beta, h)$ for ϕ_N, ϕ and AT. Kusuoka and Nakajima (2026) and Lopatto (2026a) sum over all ordered pairs with the factor $\beta/\sqrt{2N}$, the same covariance up to a constant and the same β. Panchenko (2013b), Panchenko (2014), Auffinger and Chen (2015)

and Zhou (2026) use $\xi(q) = \beta^2 q^2$, so their β is our $\beta/\sqrt{2}$ (Zhou's critical point is $1/\sqrt{2}$); Aronow and Lopatto (2026a) write $\xi(q) = \sum_p b_p^2 q^p$, and SK is $b_2 = \beta/\sqrt{2}$. Guerra (2003) writes the functional with $x(q) = \mu([0, q])$ and puts the field inside the energy, so his h is our physical field $H = h/\beta$. Physicists write $f = -\phi/\beta$, $T = 1/\beta$, $H = h/\beta$ and $J = 1$, use the inverse $q(x)$ of $x(q)$, and call $P(q) = x'(q)$ the overlap distribution. Auffinger and Chen (2017) use $\xi(s) = s^2/2$ and a function γ playing the role of βx as $\beta \to \infty$.

References

Aizenman, M., J. L. Lebowitz, and D. Ruelle (1987). "Some rigorous results on the Sherrington-Kirkpatrick spin glass model". In: *Communications in Mathematical Physics* 112.1, pp. 3–20.

Aizenman, M., R. Sims, and S. L. Starr (2003). "Extended variational principle for the Sherrington–Kirkpatrick spin-glass model". In: *Physical Review B* 68.21, p. 214403. arXiv: cond-mat/0306386.

Aronow, P. M. and P. Lopatto (2026a). *Overlap structure of mixed even p-spin models*. arXiv: 2609.17442v1.

Aronow, P. M. and P. Lopatto (2026b). *Quantitative Parisi formulas and fluctuations in the Sherrington-Kirkpatrick model*. arXiv preprint, version 1 (September 2026). arXiv: 2609.09103.

Auffinger, A. and W.-K. Chen (2015). "The Parisi Formula has a Unique Minimizer". In: *Communications in Mathematical Physics* 335.3, pp. 1429–1444. arXiv: 1402.5132.

Auffinger, A. and W.-K. Chen (2017). "Parisi formula for the ground state energy in the mixed p-spin model". In: *The Annals of Probability* 45.6B, pp. 4617–4631. arXiv: 1606.05335.

Auffinger, A., W.-K. Chen, and Q. Zeng (2020). "The SK Model Is Infinite Step Replica Symmetry Breaking at Zero Temperature". In: *Communications on Pure and Applied Mathematics* 73.5, pp. 921–943. arXiv: 1703.06872.

Brennecke, C. and H.-T. Yau (2022). "The replica symmetric formula for the SK model revisited". In: *Journal of Mathematical Physics* 63.7, p. 073302.

Chen, H.-B. (2026). *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. arXiv: 2607.18032v1.

Crisanti, A. and T. Rizzo (2002). "Analysis of the ∞-replica symmetry breaking solution of the Sherrington–Kirkpatrick model". In: *Physical Review E* 65.4, p. 046137. arXiv: cond-mat/0111037.

De Almeida, J. R. L. and D. J. Thouless (1978). "Stability of the Sherrington–Kirkpatrick solution of a spin glass model". In: *Journal of Physics A: Mathematical and General* 11.5, pp. 983–990.

Ghirlanda, S. and F. Guerra (1998). "General properties of overlap probability distributions in disordered spin systems. Towards Parisi ultrametricity". In: *Journal of Physics A: Mathematical and General* 31.46, pp. 9149–9155.

Guerra, F. (2003). "Broken Replica Symmetry Bounds in the Mean Field Spin Glass Model". In: *Communications in Mathematical Physics* 233.1, pp. 1–12.

Guerra, F. and F. L. Toninelli (2002). "The Thermodynamic Limit in Mean Field Spin Glass Models". In: *Communications in Mathematical Physics* 230.1, pp. 71–79.

Jagannath, A. and I. Tobasco (2017). "Some properties of the phase diagram for mixed p-spin glasses". In: *Probability Theory and Related Fields* 167.3-4, pp. 615–672. arXiv: 1504.02731.

Kusuoka, S. and S. Nakajima (2026). *A quantitative replica-symmetric bound for Sherrington–Kirkpatrick model in the entire de Almeida–Thouless region*. arXiv: 2608.23413v2.

Lopatto, P. (2026a). *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. arXiv: 2607.11756v4.

Lopatto, P. (2026b). *Parisi measure of the Sherrington–Kirkpatrick model with external field*. arXiv: 2610.00127v1.

Lopatto, P. (2026c). *Replica symmetry up to the de Almeida–Thouless line in the Sherrington–Kirkpatrick model*. arXiv: 2604.11921v2.

Mézard, M., G. Parisi, N. Sourlas, G. Toulouse, and M. Virasoro (1984). "Nature of the Spin-Glass Phase". In: *Physical Review Letters* 52.13, pp. 1156–1159.

Montanari, A. and S. Sen (2024). "A Friendly Tutorial on Mean-Field Spin Glass Techniques for Non-Physicists". In: *Foundations and Trends in Machine Learning* 17.1, pp. 1–173. arXiv: 2204.02909.

Mourrat, J.-C. and A. Schertzer (2026). *On the local nature of the de Almeida–Thouless line for mixed p-spin glasses*. arXiv: 2602.04383v1.

Oppermann, R., M. J. Schmidt, and D. Sherrington (2007). "Double Criticality of the Sherrington–Kirkpatrick Model at $T = 0$". In: *Physical Review Letters* 98.12, p. 127201. arXiv: cond-mat/0608581.

Panchenko, D. (2013a). "The Parisi ultrametricity conjecture". In: *Annals of Mathematics* 177.1, pp. 383–393. arXiv: 1112.1003.

Panchenko, D. (2013b). *The Sherrington-Kirkpatrick Model*. Springer Monographs in Mathematics. New York: Springer New York.

Panchenko, D. (2014). "The Parisi formula for mixed p-spin models". In: *The Annals of Probability* 42.3, pp. 946–958. arXiv: 1112.4409.

Parisi, G. (1979). "Infinite Number of Order Parameters for Spin-Glasses". In: *Physical Review Letters* 43.23, pp. 1754–1756.

Parisi, G. (1980). "A sequence of approximated solutions to the S-K model for spin glasses". In: *Journal of Physics A: Mathematical and General* 13.4, pp. L115–L121.

Parisi, G. (1983). "Order Parameter for Spin-Glasses". In: *Physical Review Letters* 50.24, pp. 1946–1948.

Sherrington, D. and S. Kirkpatrick (1975). "Solvable Model of a Spin-Glass". In: *Physical Review Letters* 35.26, pp. 1792–1796.

Talagrand, M. (2006). "The Parisi formula". In: *Annals of Mathematics* 163.1, pp. 221–263.

Talagrand, M. (2011). *Mean Field Models for Spin Glasses: Volume I: Basic Examples*. Vol. 54. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

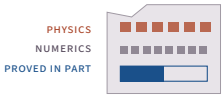
Toninelli, F. L. (2002). "About the Almeida–Thouless transition line in the Sherrington–Kirkpatrick mean-field spin glass model". In: *Europhysics Letters* 60.5, pp. 764–767. arXiv: cond-mat/0207296.

Zhou, Y. (2026). "Existence of Full Replica Symmetry Breaking for the Sherrington–Kirkpatrick Model at Low Temperature". In: *Communications on Pure and Applied Mathematics* 79.7, pp. 1746–1770. arXiv: 2504.00269.

Part II

Inference on the Nishimori Line

When the posterior is the true Bayesian posterior, the replicas behave like the truth, the overlap concentrates, and replica symmetry is no longer an assumption but a theorem. The physicist's formulas for Bayes-optimal errors are then exact, and the remaining question is whether an efficient algorithm can reach them.



4 Bayes-Optimal Inference

Why replica symmetry holds

THE DICTIONARY — the Nishimori line ↔ overlap concentration

IN THIS CHAPTER. When the Gibbs measure is the true Bayesian posterior, the hidden signal behaves like one more sample from it, and the replica-symmetric answer of physics is exact. We derive the mutual information and the minimum mean-square error of $\mathbb{Z}_2$ synchronization by replicas in which the signal is replica number zero, and prove the formula completely: Guerra's interpolation bounds the mutual information at every size, a small side channel makes the overlap concentrate, and an adaptive interpolation turns the concentration into the matching bound. Exact enumeration at $N \le 12$ and message passing at $N = 2000$ confirm the curve. Without a perturbation, replica symmetry on the Nishimori line remains open in general.

4.1 The question: signs from noisy pairwise products

Each of N objects carries a hidden sign $x_i^* \in \{-1, +1\}$, and for every pair we observe the product $x_i^* x_j^*$ through Gaussian noise. How much can be learned about the signs, up to the global flip that no pairwise measurement can see? This is $\mathbb{Z}_2$ *synchronization*,[1] and in the book's normalization the data are

$$A = \frac{\lambda}{N}\, x^* x^{*\top} + W, \tag{4.1}$$

with x^* uniform on $\Sigma_N = \{-1, +1\}^N$ and, independent of it, $W \sim \mathrm{GOE}(N)$: symmetric, with independent centered Gaussian entries of variance $1/N$ off the diagonal and $2/N$ on it, so that its density is proportional to $e^{-N \operatorname{tr}(W^2)/4}$. This is the *spiked Wigner model* with the Rademacher prior; its top eigenvalue leaves the bulk when $\lambda > 1$ (Theorem 5.1). We ask what the best estimator can do.

[1] In random-matrix language it is the spiked Wigner model with the Rademacher prior; Chapter 5 studies its spectrum and Chapter 6 its algorithms.

THE POSTERIOR IS A SPIN GLASS. The likelihood of A given $x^* = x$ is proportional to $\exp(-\frac{N}{4}\|A - \frac{\lambda}{N} x x^\top\|_F^2)$, and $\|x\|^4 = N^2$ on Σ_N, so the posterior is the Gibbs measure

$$\mathbb{P}(x^* = x \mid A) = \frac{e^{H_N(x)}}{\sum_{x' \in \Sigma_N} e^{H_N(x')}}, \qquad H_N(x) = \frac{\lambda}{2}\, x^\top A x. \tag{4.2}$$

We write $\langle f \rangle$ for the posterior mean of a function of independent posterior samples $x^1, x^2, \dots$, the *replicas*, $\mathbb{E}$ for the expectation over (x^*, W), and

$$R_{12} = \frac{1}{N}\langle x^1, x^2\rangle, \qquad R_{1*} = \frac{1}{N}\langle x^1, x^*\rangle$$

for the overlap of two replicas and of a replica with the truth, which physicists call the magnetization m. In the variables $\sigma_i = x_i^* x_i$, and up to a constant, H_N is the sum of $\frac{\lambda}{\sqrt{N}}\sum_{i<j} J_{ij}\sigma_i\sigma_j$ and $\frac{\lambda^2}{N}\sum_{i<j}\sigma_i\sigma_j$, where the $J_{ij} = x_i^* x_j^* \sqrt{N} W_{ij}$ are again independent standard Gaussians: the Sherrington–Kirkpatrick model at inverse temperature $\beta = \lambda$ (Chapter 3) with couplings of mean J_0/N and variance J^2/N, $\beta J = \lambda$, $\beta J_0 = \lambda^2$. The relation $\beta = J_0/J^2$ defines the *Nishimori line*, on which Nishimori (1981) found exact results by a gauge transformation; Iba (1999) pointed out that on it the Gibbs measure is a Bayesian posterior.[2]

[2] Nishimori worked with the $\pm J$ model, in which each coupling is $+J$ with probability p and $-J$ otherwise; there the line is $\beta J = \frac{1}{2}\log\frac{p}{1-p}$.

WHAT WE WANT TO KNOW. The *evidence* is the likelihood ratio of the spiked model against pure noise,

$$\mathcal{Z}_N = \frac{p_\lambda(A)}{p_0(A)} = 2^{-N}\sum_{x\in\Sigma_N} e^{H_N(x)-\lambda^2 N/4},$$

and $\phi_N = \frac{1}{N}\mathbb{E}\log\mathcal{Z}_N$ is the *free entropy*. Since $\log\frac{p_\lambda(A|x^*)}{p_0(A)} = H_N(x^*) - \frac{\lambda^2 N}{4}$ has mean $\frac{\lambda^2 N}{4}$, the *mutual information* is

$$\frac{1}{N}I(x^*; A) = \frac{\lambda^2}{4} - \phi_N. \tag{4.3}$$

The sign of x^* is not identifiable, so we estimate $x^* x^{*\top}$. The posterior mean is the best estimator in mean square, and its error is the *minimum mean-square error*

$$\mathrm{MMSE}_N = \frac{1}{N^2}\mathbb{E}\left\|x^* x^{*\top} - \langle x x^\top\rangle\right\|_F^2 = 1 - \mathbb{E}\left\langle R_{1*}^2\right\rangle, \tag{4.4}$$

by expanding the square and the Nishimori identity $\mathbb{E}\langle R_{12}^2\rangle = \mathbb{E}\langle R_{1*}^2\rangle$ of Lemma 4.3; the estimator 0 has error 1. The prediction involves the *scalar channel* $Y = \sqrt{\gamma}X + Z$, with X uniform on $\{\pm 1\}$ and $Z \sim \mathcal{N}(0,1)$ independent, whose posterior mean is $\tanh(\sqrt{\gamma}Y)$. Its overlap $\mathbb{E}[X\tanh(\sqrt{\gamma}Y)]$ and its free entropy (Chapter 1) are

$$G(\gamma) = \mathbb{E}\tanh(\gamma + \sqrt{\gamma}Z) = 1 - \mathrm{mmse}(\gamma), \qquad \psi(\gamma) = \mathbb{E}\log\cosh(\gamma + \sqrt{\gamma}Z) - \frac{\gamma}{2}, \tag{4.5}$$

related by the I-MMSE identity $\psi' = \frac{1}{2}G$ (Theorem 1.12).

THE PHYSICAL PREDICTION · Lesieur, Krzakala and Zdeborová 2017. For $\mathbb{Z}_2$ synchronization, as $N \to \infty$,

$$\frac{1}{N}I(x^*; A) \to \frac{\lambda^2}{4} - \max_{0\le q\le 1}\phi_{\mathrm{RS}}(\lambda, q), \qquad \phi_{\mathrm{RS}}(\lambda, q) = \psi(\lambda^2 q) - \frac{\lambda^2 q^2}{4}, \tag{4.6}$$

and $\mathrm{MMSE}_N \to 1 - q_*^2$, where the maximizer $q_* = q_*(\lambda)$ solves

$$q = \mathbb{E}\tanh\big(\lambda^2 q + \lambda\sqrt{q}\,Z\big). \tag{4.7}$$

For $\lambda \le 1$ the only solution is $q_* = 0$ and nothing can be learned; for $\lambda > 1$ there is a positive one, with $q_* \approx 2(\lambda - 1)$ near the threshold. The replica-symmetric ansatz behind the formula is exact, because the posterior sits on the Nishimori line.

How do physicists derive this, and why do they trust replica symmetry here when it fails for the Sherrington–Kirkpatrick model at low temperature? And what is proved? Everything in the box (Theorem 4.2), for the reason the physicists gave.

4.2 The physics: the truth is a replica

The replica method computes $\mathbb{E}\log \mathcal{Z}_N$ from the moments $\mathbb{E}\mathcal{Z}_N^n$ (Chapter 1). On the Nishimori line the signal enters them as one more replica.

COMPUTATION 4.1 · THE REPLICATED EVIDENCE

1 *Replicate and average the noise.* $\mathcal{Z}_N^n$ is the mean, over independent uniform $x^1, \dots, x^n \in \Sigma_N$, of $\exp\sum_a (H_N(x^a) - \frac{\lambda^2 N}{4})$, and $H_N(x^a) = \frac{\lambda^2 N}{2} R_{0a}^2 + \frac{\lambda}{2} x^{a\top} W x^a$, with $x^0 = x^*$ and $R_{ab} = \frac{1}{N}\langle x^a, x^b\rangle$. Given the replicas, $\frac{\lambda}{2}\sum_a x^{a\top} W x^a$ is a centered Gaussian, and the covariance $\mathbb{E}W_{ij}W_{kl} = \frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ of the GOE makes its variance $\frac{\lambda^2}{2N}\sum_{a,b}\langle x^a, x^b\rangle^2 = \frac{\lambda^2 N n}{2} + \lambda^2 N \sum_{1\le a<b} R_{ab}^2$.

2 *Collect.* Half the variance, $\frac{\lambda^2 N n}{4} + \frac{\lambda^2 N}{2}\sum_{1\le a<b} R_{ab}^2$, appears in the exponent, its first term cancels the normalization, and with $x^0, \dots, x^n$ independent and uniform on Σ_N, for every N and every integer $n \ge 1$,

$$\mathbb{E}\mathcal{Z}_N^n = \mathbb{E}\exp\Big(\frac{\lambda^2 N}{2}\sum_{0\le a<b\le n} R_{ab}^2\Big). \tag{4.8}$$

3 *Read.* The weight is symmetric under all permutations of $x^0, \dots, x^n$: the truth enters as replica number zero. At $n = 1$, $\mathbb{E}\mathcal{Z}_N$ is the second moment of the likelihood ratio under pure noise (Proposition 4.7).

PHYSICAL DERIVATION: REPLICAS ON THE NISHIMORI LINE. Decouple each pair by a Gaussian integral,

$$e^{\lambda^2 N R^2/2} = \sqrt{\lambda^2 N/2\pi}\int_{\mathbb{R}} e^{-\lambda^2 N(Q^2/2 - QR)}\,\mathrm{d}Q.$$

The sum over the replicas then factorizes over sites. With $0 \le a < b \le n$, σ uniform on $\{\pm 1\}^{n+1}$ and $C_{N,n}$ subexponential in N, this gives $\mathbb{E}\mathcal{Z}_N^n =$

$C_{N,n} \int e^{N S_n(Q)} \prod_{a<b} \mathrm{d}Q_{ab}$ with

$$S_n(Q) = -\frac{\lambda^2}{2} \sum_{a<b} Q_{ab}^2 + \log \mathbb{E}_\sigma e^{\lambda^2 \sum_{a<b} Q_{ab} \sigma_a \sigma_b}.$$

At a stationary point Q_{ab} is the overlap $\langle \sigma_a \sigma_b \rangle$ of replicas a and b in the single-site replicated measure. The replica-symmetric ansatz $Q_{ab} = q$ for all pairs treats the truth as one more replica, as (4.8) invites. Since $\sum_{a<b} \sigma_a \sigma_b = \frac{1}{2}(\sum_a \sigma_a)^2 - \frac{n+1}{2}$, a second Gaussian integral gives

$$S_n(q) = -\binom{n+1}{2} \frac{\lambda^2 q^2}{2} - \frac{(n+1)\lambda^2 q}{2} + \log \mathbb{E} \cosh^{n+1}(\lambda \sqrt{q}\, Z).$$

Now continue in n. The function S_0 vanishes, as it must since $\mathbb{E}\mathcal{Z}_N^0 = 1$, and exchanging $n \to 0$ with $N \to \infty$ in $\frac{1}{N}\mathbb{E} \log \mathcal{Z}_N = \lim_{n\to 0} \frac{1}{nN} \log \mathbb{E}\mathcal{Z}_N^n$ leaves $\partial_n S_n(q)$ at $n = 0$. With $\gamma = \lambda^2 q$, $\mathbb{E} \cosh(\sqrt{\gamma} Z) = e^{\gamma/2}$ and, for even f, $e^{-\gamma/2}\mathbb{E}[\cosh(\sqrt{\gamma} Z) f(\sqrt{\gamma} Z)] = e^{-\gamma/2}\mathbb{E}[e^{\sqrt{\gamma} Z} f(\sqrt{\gamma} Z)] = \mathbb{E} f(\gamma + \sqrt{\gamma} Z)$,

$$\partial_n S_n(q)\big|_{n=0} = -\frac{\lambda^2 q^2}{4} - \frac{\gamma}{2} + \frac{\mathbb{E}[\cosh(\sqrt{\gamma} Z) \log \cosh(\sqrt{\gamma} Z)]}{\mathbb{E} \cosh(\sqrt{\gamma} Z)} = \psi(\lambda^2 q) - \frac{\lambda^2 q^2}{4},$$

which is $\phi_{\mathrm{RS}}(\lambda, q)$; stationarity in q is (4.7). Which stationary point? For integer n Laplace's method takes the maximum of S_n. Near $n = 0$, $S_n \approx n\, \phi_{\mathrm{RS}}$ with $n > 0$, and the number $\binom{n+1}{2}$ of order parameters stays positive, so the maximum survives: $\lim \phi_N = \max_q \phi_{\mathrm{RS}}(\lambda, q)$, which is (4.6) by (4.3). (In the Sherrington–Kirkpatrick model the replicas are $x^1, \dots, x^n$ alone, $\binom{n}{2} < 0$, and the rule becomes a minimum; Chapter 3.) ◇

WHY REPLICA SYMMETRY. The guess is wrong below the de Almeida–Thouless line of the Sherrington–Kirkpatrick model; on the Nishimori line physicists trust it for three reasons. The weight (4.8) is symmetric in all $n+1$ replicas, so a symmetric saddle point has $Q_{0a} = Q_{ab}$. At finite N the overlap of two posterior samples has the law of the overlap of one sample with the truth, $P(q) = P(m)$, which Nishimori and Sherrington (2001) proved for the $\pm J$ model in every dimension. And the magnetization $m = R_{1*}$ self-averages, being the derivative of a self-averaging free energy in its conjugate field; so $P(m)$, and with it $P(q)$, has no spread beyond the atoms $\pm m$ forced by the global flip. Zdeborová and Krzakala (2016, §II.G) note that, from a rigorous point of view, this last step is open; with a small side field it is a theorem (Proposition 4.9 for our model).

WHAT THE DERIVATION ASSUMED. Five steps are taken on trust: the continuation from integer n to $n \to 0$, which the integer moments do not determine in general (Chapter 1); the exchange of $n \to 0$ with $N \to \infty$; Laplace's method over a number $\binom{n+1}{2}$ of variables that is not an integer; the replica-symmetric ansatz; and the choice of the global maximum.

CONTINUOUS AND DISCONTINUOUS TRANSITIONS. The fixed-point equation is $q = G(\lambda^2 q)$, and $G(\gamma) = \gamma - \gamma^2 + \frac{5}{3}\gamma^3 - \cdots$ (Exercise 4.5). At $\lambda = 1$ a positive solution bifurcates from 0, with $q_* = 2\epsilon - \frac{1}{3}\epsilon^2 + O(\epsilon^3)$ at

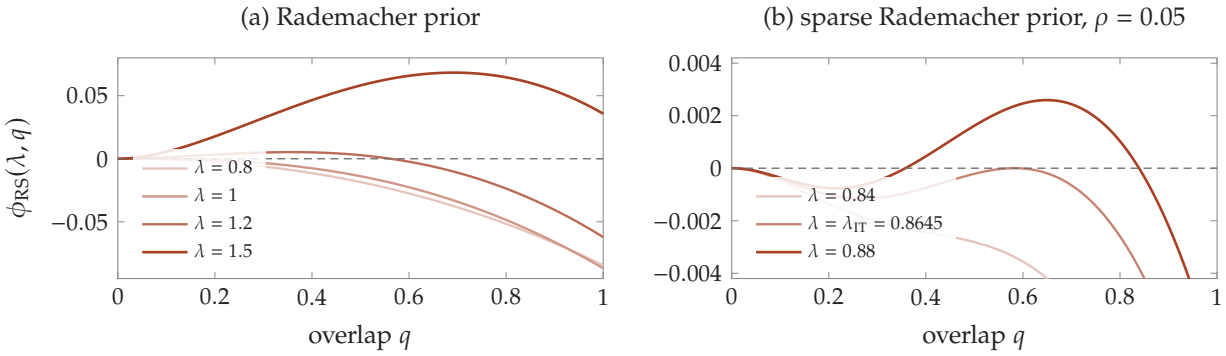


Figure 4.1. The replica-symmetric potential $\phi_{\mathrm{RS}}(\lambda, q)$. (a) Rademacher prior, $\lambda = 0.8, 1, 1.2, 1.5$ (light to dark): the maximizer leaves $q = 0$ continuously at $\lambda = 1$ ($q_* = 0.358$ at $\lambda = 1.2$, 0.692 at 1.5). (b) Sparse Rademacher prior, $\rho = 0.05$, $\lambda = 0.84$, $\lambda_{\mathrm{IT}} = 0.8645$, 0.88: a second maximum appears, ties with $q = 0$ at λ_{IT} (at $q = 0.583$), and then dominates. Gauss–Hermite quadrature, 201 nodes; script `04-nishimori-rs.py`.

$\lambda = 1 + \epsilon$: the transition is continuous and sits at the spectral threshold (Figure 4.1(a)). For a prior P_0 with $\mathbb{E}X^2 = 1$ the derivation is the same, with ψ replaced by $\psi_{P_0}(\gamma) = \mathbb{E}\log\int e^{\sqrt{\gamma}Zx+\gamma xX-\gamma x^2/2}P_0(\mathrm{d}x)$, $X \sim P_0$. For the sparse Rademacher prior, with mass $1-\rho$ at 0 and $\frac{\rho}{2}$ at $\pm\rho^{-1/2}$, a second local maximum of ϕ_{RS} appears while $q = 0$ is still a local maximum, and overtakes it at λ_{IT}, which is 0.8645 for $\rho = 0.05$ (Figure 4.1(b)): the optimal overlap jumps, a first-order transition, and up to a value slightly above 1 an algorithm that climbs the potential from $q = 0$ stays near it, the hard phase (Chapter 5).

4.3 The stone

The proof in Sections 4.4 and 4.5 follows the derivation step by step, except at the continuation in n, which it replaces by two inequalities.

PHYSICS · *replicas on the Nishimori line*

1 · Set up

Average n copies of the evidence; the signal enters as replica zero (Computation 4.1).

2 · The Nishimori symmetry

The replicated weight is symmetric in $x^0, \dots, x^n$: the truth is statistically a replica, and $P(q) = P(m)$.

PROOF · *Guerra's interpolation with a side channel*

Interpolate between the matrix channel and N scalar channels $y = \sqrt{r}\,x^* + z$: the free entropy $\phi_N(t, r)$ (Section 4.4.1).

$\mathbb{E}\langle f(x^1, \dots, x^k, x^*)\rangle = \mathbb{E}\langle f(x^1, \dots, x^{k+1})\rangle$ at every N, by Bayes' rule (Lemma 4.3); for signs, also the gauge (Lemma 4.4).

3 · Order parameters	
Gaussian integrals decouple the sites; overlaps Q_{ab} and conjugate fields $\lambda^2 Q_{ab}$ meet at a saddle point.	Integration by parts and the Nishimori identity: $\partial_t \phi_N = \frac{\lambda^2}{4}\mathbb{E}\left\langle R_{1*}^2 \right\rangle$, $\partial_r \phi_N = \frac{1}{2}\mathbb{E}\left\langle R_{1*} \right\rangle$; r is the conjugate field (Lemma 4.5).
4 · Replica symmetry, and a maximum	
The ansatz $Q_{ab} = q$ for all pairs, the truth included; $\binom{n+1}{2} > 0$ keeps the saddle point a maximum ($\binom{n}{2} < 0$ makes it a minimum in the Sherrington–Kirkpatrick model).	No ansatz: the remainder $\frac{\lambda^2}{4}\int_0^1 \mathbb{E}\left\langle (R_{1*} - q)^2 \right\rangle \mathrm{d}t$ is a mean square, so $\phi_N \geq \phi_{\mathrm{RS}}(\lambda, q)$ for every q and N; for Sherrington–Kirkpatrick it has the other sign (Section 4.4.3).
5 · Self-averaging of m	
$P(m)$ is trivial because m is the derivative of a self-averaging free energy in its conjugate field.	Convexity in r and Gaussian concentration make $\mathbb{E}\left\langle (R_{1*} - \mathbb{E}\left\langle R_{1*} \right\rangle)^2 \right\rangle$ small for most r (Proposition 4.9).
6 · The saddle point is the answer	
Continue to $n \to 0$, exchange the limits, keep the global maximum.	No counterpart for the continuation. The adaptive path $q(t) = \mathbb{E}\left\langle R_{1*} \right\rangle_{t,r(t)}$ makes the remainder the overlap variance, small on average over r (Section 4.5.1).
7 · The fixed point	
$q = \mathbb{E}\tanh(\lambda^2 q + \lambda\sqrt{q}Z)$, continuous at $\lambda = 1$ for signs; first order for sparse enough priors ($\rho < 0.0964$ for sparse signs).	The overlap curve G is strictly concave, so the maximizer is unique (Lemma 4.10: the binary mmse is completely monotone); for other priors the formula is a theorem (Theorem 4.1).

The Nishimori identity does in the proof what the permutation symmetry of the $n+1$ replicas does in the derivation, and the physicists' reason for replica symmetry, a magnetization conjugate to a field, becomes a theorem once the field is made real, as a side channel, and averaged over. What the proof cannot keep is the single saddle point. In its place stand two inequalities of different natures: Guerra's bound holds at every N and needs no concentration, while the matching bound holds only in the limit and needs both the concentration and the side channel.

4.4 The proof: an interpolation with a sign

The general theorem is due to Lelarge and Miolane; we state it in the book's normalization (snr $= \lambda^2$) and then prove the case of signs completely, in this section and the next.

Theorem 4.1 (Lelarge and Miolane 2019, Theorem 1). *Let P_0 be a probability measure on $\mathbb{R}$ with $v = \int x^2 P_0(\mathrm{d}x) < \infty$, let $x_1^*, \dots, x_N^*$ be independent with law P_0, and observe $Y_{ij} = \frac{\lambda}{\sqrt{N}} x_i^* x_j^* + Z_{ij}$ for $1 \le i < j \le N$, with independent standard Gaussians Z_{ij}, independent of x^*. With $\phi_{\mathrm{RS}}(\lambda, q) = \psi_{P_0}(\lambda^2 q) - \frac{\lambda^2 q^2}{4}$, for every $\lambda > 0$,*

$$\lim_{N\to\infty} \frac{1}{N} I(x^*; Y) = \frac{\lambda^2 v^2}{4} - \sup_{q \ge 0} \phi_{\mathrm{RS}}(\lambda, q).$$

The limit is concave in λ^2. At each point where it is differentiable, which is every $\lambda > 0$ outside a countable set, the maximizer $q_*(\lambda)$ is unique and

$$\lim_{N\to\infty} \frac{2}{N(N-1)} \sum_{i<j} \mathbb{E}\big(x_i^* x_j^* - \mathbb{E}[x_i^* x_j^* \mid Y]\big)^2 = v^2 - q_*(\lambda)^2.$$

The Y_{ij} are the entries of $\sqrt{N}A$ above the diagonal; Lelarge and Miolane remark that the case in which all entries are observed can be deduced from theirs. For signs the diagonal $A_{ii} = \frac{\lambda}{N} + W_{ii}$ is independent of everything else, so $I(x^*; A) = I(x^*; Y)$, and MMSE_N is $\frac{N-1}{N}$ times their error.

Theorem 4.2 (The replica-symmetric formula for $\mathbb{Z}_2$ synchronization). *Let x^* be uniform on Σ_N in (4.1), and let $\lambda \ge 0$.*

(i) For every $N \ge 1$ and every $q \ge 0$, $\phi_N \ge \phi_{\mathrm{RS}}(\lambda, q)$; equivalently $\frac{1}{N} I(x^*; A) \le \frac{\lambda^2}{4} - \phi_{\mathrm{RS}}(\lambda, q)$.

(ii) There is $C(\lambda) < \infty$ with $\phi_N \le \max_{0\le q\le 1} \phi_{\mathrm{RS}}(\lambda, q) + C(\lambda) N^{-1/20}$ for $N \ge 11$; so $\phi_N \to \max_q \phi_{\mathrm{RS}}(\lambda, q)$.

(iii) For $\lambda > 0$ the maximizer $q_*(\lambda)$ is unique: $q_* = 0$ for $\lambda \le 1$, and for $\lambda > 1$ it is the unique solution of (4.7) in $(0, 1]$. (At $\lambda = 0$, $\phi_{\mathrm{RS}} \equiv 0$; we set $q_*(0) = 0$.)

(iv) $\mathrm{MMSE}_N \to 1 - q_*(\lambda)^2$ for every $\lambda \ge 0$.

Parts 1 and 2 give the limit of the mutual information in Theorem 4.1 for this prior, which Deshpande et al. (2016, Theorem 4.3) had proved by message passing and the I-MMSE relation; parts 3 and 4 give the limit of the error at every λ, with no exceptional set. We prove part 1 here, part 2 in Section 4.5.1 and parts 3 and 4 in Section 4.5.2.

4.4.1 The interpolating model and the Nishimori identity

For $t \in [0, 1]$ and $r \ge 0$ observe

$$A_t = \frac{\lambda\sqrt{t}}{N} x^* x^{*\top} + W, \qquad y_r = \sqrt{r}\, x^* + z, \tag{4.9}$$

with $z \sim \mathcal{N}(0, I_N)$ independent of (x^*, W): the matrix channel at signal-to-noise ratio $\lambda\sqrt{t}$ and N scalar channels at ratio r. On Σ_N the terms $\frac{\lambda^2 t}{4N}\|x\|^4$ and $\frac{r}{2}\|x\|^2$ of the log-likelihood are constant, so the posterior is

the Gibbs measure $\langle\cdot\rangle_{t,r}$ with Hamiltonian $H_{t,r}(x) = \frac{\lambda\sqrt{t}}{2}x^\top A_t x + \sqrt{r}\langle y_r\rangle x$, the evidence is $\mathcal{Z}_{t,r} = 2^{-N}\sum_x \exp(H_{t,r}(x) - \frac{\lambda^2 tN}{4} - \frac{rN}{2})$, and we write $\phi_N(t,r) = \frac{1}{N}\mathbb{E}\log\mathcal{Z}_{t,r}$. At $(t,r) = (1,0)$ this is the model. At $t = 0$ the Gibbs measure is a product, $\mathcal{Z}_{0,r} = \prod_i e^{-r/2}\cosh(\sqrt{r}(y_r)_i)$, and $\phi_N(0,r) = \psi(r)$.

Lemma 4.3 (Nishimori identity). *For every $t \in [0,1]$, $r \geq 0$, $k \geq 1$ and $f : \Sigma_N^{k+1} \to \mathbb{R}$, $\mathbb{E}\left\langle f(x^1,\dots,x^k,x^*)\right\rangle_{t,r} = \mathbb{E}\left\langle f(x^1,\dots,x^k,x^{k+1})\right\rangle_{t,r}$.*

Proof. Write $D = (A_t, y_r)$ and $\pi(\cdot \mid D)$ for the posterior. By Bayes' rule x^* has the conditional law $\pi(\cdot \mid D)$ given D, so, writing x^{k+1} for x^*,

$$\mathbb{E}\left\langle f(x^1,\dots,x^k,x^*)\right\rangle = \mathbb{E}_D \sum_{x^1,\dots,x^{k+1}} f(x^1,\dots,x^{k+1}) \prod_{a=1}^{k+1} \pi(x^a \mid D),$$

which is $\mathbb{E}\left\langle f(x^1,\dots,x^{k+1})\right\rangle$. ∎

This is the Bayesian form of Nishimori's identity (Lemma 1.10; Lelarge and Miolane (2019, Proposition 16)).[3] We use it as $\mathbb{E}\langle R_{1*}\rangle = \mathbb{E}\langle R_{12}\rangle = \frac{1}{N}\mathbb{E}\|\langle x\rangle\|^2 \in [0,1]$, $\mathbb{E}\left\langle R_{1*}^2\right\rangle = \mathbb{E}\left\langle R_{12}^2\right\rangle$ and $\mathbb{E}\langle R_{1*}R_{12}\rangle = \mathbb{E}\langle R_{13}R_{12}\rangle$.

[3] Barbier and Macris (2019) remark that the name is used loosely for this consequence of Bayes' rule, and that Nishimori's identity proper uses a gauge symmetry to remove the signal. For signs we have both.

Lemma 4.4 (Gauge symmetry). *Given $x^* = s \in \Sigma_N$, the law of $\mathcal{Z}_{t,r}$ does not depend on s; so $\log\mathcal{Z}_{t,r}$ is independent of x^*.*

Proof. Let $D_s = \mathrm{diag}(s)$. Given $x^* = s$, the entries $s_i s_j W_{ij}$ of $D_s W D_s$ are independent with the variances of those of W, so $(D_s W D_s, D_s z)$ has the law of (W,z). The substitution $x = D_s\sigma$ is a bijection of Σ_N, and since $D_s x^* = \mathbf{1}$ it turns $H_{t,r}$ into the Hamiltonian of the model with truth $\mathbf{1}$ and noise $(D_s W D_s, D_s z)$. ∎

4.4.2 Two derivatives and a sum rule

Lemma 4.5 (Derivatives of the free entropy). *The function ϕ_N is continuously differentiable on $[0,1]\times[0,\infty)$, with*

$$\partial_t\phi_N(t,r) = \frac{\lambda^2}{4}\mathbb{E}\left\langle R_{1*}^2\right\rangle_{t,r}, \qquad \partial_r\phi_N(t,r) = \frac{1}{2}\mathbb{E}\langle R_{1*}\rangle_{t,r}.$$

Moreover $\mathbb{E}\langle R_{1*}\rangle_{t,r}$ is nondecreasing in r, and $\mathbb{E}\left\langle R_{1*}^2\right\rangle_{t,r}$ is nondecreasing in t.

Proof. The plan is to differentiate under the expectation, to remove the noise by Gaussian integration by parts (Theorem A.1), and to remove the replicas by the Nishimori identity. For fixed N all quantities are smooth for $t, r > 0$ and dominated by polynomials in (W,z); the formulas extend to $t = 0$ and $r = 0$ by continuity of their right sides.

In r. Since $\sqrt{r}\langle y_r\rangle x = r\langle x^*\rangle x + \sqrt{r}\langle z\rangle x$, $\partial_r \log\mathcal{Z}_{t,r} = \langle\langle x^*\rangle x\rangle + \frac{1}{2\sqrt{r}}\langle\langle z\rangle x\rangle - \frac{N}{2}$. Integration by parts in z_i, with $\partial_{z_i}\langle x_i\rangle = \sqrt{r}(1 - \langle x_i\rangle^2)$, gives $\mathbb{E}\langle\langle z\rangle x\rangle = \sqrt{r}\,\mathbb{E}(N - \|\langle x\rangle\|^2)$, so $\partial_r\mathbb{E}\log\mathcal{Z}_{t,r} = N\mathbb{E}\langle R_{1*}\rangle - \frac{N}{2}\mathbb{E}\langle R_{12}\rangle = \frac{N}{2}\mathbb{E}\langle R_{1*}\rangle$.

In t. Here $\frac{\lambda\sqrt{t}}{2}x^\top A_t x = \frac{\lambda^2 t}{2N}\langle x^*\rangle x^2 + \frac{\lambda\sqrt{t}}{2}x^\top W x$, and $H_{t,r}$ depends on W_{ij}, $i<j$, through $\lambda\sqrt{t}W_{ij}x_i x_j$ and on W_{ii} only through a constant. Integration by parts with $\mathbb{E}W_{ij}^2 = \frac{1}{N}$ gives $\mathbb{E}\langle x^\top W x\rangle = \frac{2\lambda\sqrt{t}}{N}\sum_{i<j}\mathbb{E}(1-\langle x_i x_j\rangle^2) = \lambda\sqrt{t}N(1-\mathbb{E}\langle R_{12}^2\rangle)$, so that

$$\partial_t \mathbb{E}\log\mathcal{Z}_{t,r} = \frac{\lambda^2 N}{2}\mathbb{E}\langle R_{1*}^2\rangle + \frac{\lambda^2 N}{4}(1-\mathbb{E}\langle R_{12}^2\rangle) - \frac{\lambda^2 N}{4} = \frac{\lambda^2 N}{4}\mathbb{E}\langle R_{1*}^2\rangle .$$

Both derivatives are continuous in (t,r) by dominated convergence.

Monotonicity. By the tower property $\mathbb{E}\langle R_{1*}\rangle_{t,r} = 1 - \frac{1}{N}\mathbb{E}\|x^* - \mathbb{E}[x^* \mid A_t, y_r]\|^2$. For $r' < r$ and an independent $\xi\sim\mathcal{N}(0,I_N)$, the vector $\sqrt{r'/r}\,y_r + \sqrt{1-r'/r}\,\xi$ has, jointly with (x^*,A_t), the law of $y_{r'}$, and conditioning on more information lowers the mean-square error. The same argument with $\sqrt{t'/t}\,A_t + \sqrt{1-t'/t}\,W'$, for an independent GOE matrix W', and (4.4) for the interpolating model gives the monotonicity in t. ∎

Proposition 4.6 (The sum rule). *Let $q\colon[0,1]\to[0,\infty)$ be continuous, $\epsilon\ge 0$, and $r(t) = \epsilon + \lambda^2\int_t^1 q(u)\,\mathrm{d}u$. Then*

$$\phi_N(1,\epsilon) = \psi\Big(\epsilon + \lambda^2\int_0^1 q(t)\,\mathrm{d}t\Big) - \frac{\lambda^2}{4}\int_0^1 q(t)^2\,\mathrm{d}t + \frac{\lambda^2}{4}\int_0^1 \mathbb{E}\left\langle (R_{1*}-q(t))^2\right\rangle_{t,r(t)}\,\mathrm{d}t.$$

Proof. Let $\varphi(t) = \phi_N(t,r(t))$, so that $\varphi(1) = \phi_N(1,\epsilon)$ and $\varphi(0) = \psi(r(0))$. Since $r' = -\lambda^2 q$, the chain rule and Lemma 4.5 give, with the brackets at $(t,r(t))$,

$$\varphi'(t) = \frac{\lambda^2}{4}\mathbb{E}\langle R_{1*}^2\rangle - \frac{\lambda^2 q(t)}{2}\mathbb{E}\langle R_{1*}\rangle = \frac{\lambda^2}{4}\mathbb{E}\left\langle (R_{1*}-q(t))^2\right\rangle - \frac{\lambda^2 q(t)^2}{4},$$

and integrating over $[0,1]$ gives the claim. ∎

4.4.3 Guerra's bound

Proof of part 1 of Theorem 4.2. Take $\epsilon = 0$ and a constant $q\ge 0$ in Proposition 4.6. The first two terms are $\phi_{\mathrm{RS}}(\lambda,q)$, and the last is nonnegative:

$$\phi_N - \phi_{\mathrm{RS}}(\lambda,q) = \frac{\lambda^2}{4}\int_0^1 \mathbb{E}\left\langle (R_{1*}-q)^2\right\rangle_{t,\lambda^2 q(1-t)}\,\mathrm{d}t \ge 0.$$

The second form follows from (4.3). Since $\partial_q\phi_{\mathrm{RS}} = \frac{\lambda^2}{2}(G(\lambda^2 q)-q) < 0$ for $q\ge 1$, the supremum over $q\ge 0$ is a maximum over $[0,1]$. ∎

The identity holds at every N with no error term: for signs $\|x\|^4 = N^2$ is a constant, and for other priors the variance $2/N$ of the diagonal of the GOE makes the terms in $\|x\|^4$ cancel (Exercise 4.7); Krzakala et al. (2016) proved the bound at every N for priors with finite support. It needs no concentration and no perturbation: it is a sign. For the Sherrington–Kirkpatrick model the

same interpolation has derivative $\frac{\beta^2}{4}((1-q)^2 - \mathbb{E}\left\langle (R_{12}-q)^2\right\rangle)$, and Guerra's bound is an upper bound (Theorem 3.3; Exercise 4.8). Here the planted term contributes $\frac{\lambda^2}{2}\mathbb{E}\left\langle R_{1*}^2\right\rangle$, twice what the noise removes, and the Nishimori identity turns the balance into $+\frac{\lambda^2}{4}\mathbb{E}\left\langle (R_{1*}-q)^2\right\rangle$. The physicist's rule, a maximum here and a minimum there, is this sign. Below the threshold the bound has a partner, from the second moment of Computation 4.1.[4]

[4] The second moment of the likelihood ratio is the standard tool of detection: Perry et al. (2018) use it to show that for $\lambda < 1$ no test tells the spiked law from pure noise with vanishing error.

Proposition 4.7 (Below the threshold). *For $0 \le \lambda < 1$ and every $N \ge 1$,*

$$0 \le \phi_N \le \frac{-\log(1-\lambda^2)}{2N}.$$

Proof. The lower bound is part 1 at $q = 0$. For the upper bound, Jensen's inequality and (4.8) at $n = 1$ give $N\phi_N \le \log \mathbb{E}\mathcal{Z}_N = \log \mathbb{E}e^{\lambda^2 N R_{01}^2/2}$. Here $\sqrt{N}R_{01} = N^{-1/2}\sum_i \sigma_i$ with independent uniform signs $\sigma_i = x_i^0 x_i^1$, so $\mathbb{E}e^{\theta\sqrt{N}R_{01}} = \cosh(\theta/\sqrt{N})^N \le e^{\theta^2/2}$, and with an independent $\xi \sim \mathcal{N}(0,1)$,

$$\mathbb{E}e^{\lambda^2 N R_{01}^2/2} = \mathbb{E}e^{\lambda\xi\sqrt{N}R_{01}} \le \mathbb{E}e^{\lambda^2\xi^2/2} = (1-\lambda^2)^{-1/2}. \qquad \blacksquare$$

4.5 The proof: overlap concentration and the matching bound

Guerra's remainder would vanish if the overlap concentrated along the path, which is what replica symmetry claims. The field conjugate to the overlap is the ratio r of the side channel, $\partial_r \phi_N = \frac{1}{2}\mathbb{E}\langle R_{1*}\rangle$, and a free entropy that concentrates and is convex in r has a derivative that concentrates for most r: the physicists' argument, whose deterministic core is the next lemma.[5]

[5] The argument uses only the Nishimori identity, bounded spins, convexity in r and the concentration of the free entropy. Barbier and Panchenko (2022, Thm. 2.1) run it for every Bayes-optimal model with a bounded product prior; their Thm. 2.2 extends it, for binary spins and exchangeable sites, to all multioverlaps.

Lemma 4.8 (Derivatives of close convex functions). *Let f and g be convex and differentiable on an open interval that contains $[r-\delta, r+\delta]$, with $\delta > 0$. Then*

$$|f'(r) - g'(r)| \le g'(r+\delta) - g'(r-\delta) + \frac{1}{\delta}\sum_{u\in\{r-\delta,r,r+\delta\}} |f(u) - g(u)|.$$

Proof. By convexity $f'(r) \le \frac{f(r+\delta)-f(r)}{\delta}$ and $\frac{g(r+\delta)-g(r)}{\delta} \le g'(r+\delta)$, so $f'(r)-g'(r) \le g'(r+\delta)-g'(r)+\frac{1}{\delta}(|f(r+\delta)-g(r+\delta)|+|f(r)-g(r)|)$. In the same way $g'(r)-f'(r) \le g'(r)-g'(r-\delta)+\frac{1}{\delta}(|f(r)-g(r)|+|f(r-\delta)-g(r-\delta)|)$. Since g' is nondecreasing, both are at most the claimed bound. $\blacksquare$

Proposition 4.9 (Overlap concentration). *For $t \in [0,1]$ and $r > 0$ let $V_N(t,r) = \mathbb{E}\left\langle (R_{1*} - \mathbb{E}\langle R_{1*}\rangle_{t,r})^2\right\rangle_{t,r}$, and let $M_a = \frac{1}{2} + (\pi a)^{-1/2}$. For every $N \ge 1$, $t \in [0,1]$ and $0 < 2\delta \le a < b$,*

$$\int_a^b V_N(t,r)\,\mathrm{d}r \le \frac{2 + 4\log\frac{b}{a}}{N} + 12M_a\left(\delta + 2\sqrt{\frac{\delta}{\pi}}\right) + \frac{108(b-a)(\lambda^2+6b)}{N\delta^2}.$$

For fixed $0 < a < b$ the choice $\delta = N^{-2/5}$ makes the right side $O(N^{-1/5})$.

PROOF. The plan is to control the thermal and then the disorder fluctuations of the derivative of the free entropy in r, and to transfer them to the overlap. Fix t, write $\langle\cdot\rangle$ for $\langle\cdot\rangle_{t,r}$, $\Phi(r) = \phi_N(t,r)$ and $\hat{\Phi}(r) = \frac{1}{N}\log\mathcal{Z}_{t,r}$.

Step 1: thermal fluctuations. As in the proof of Lemma 4.5, $\hat{\Phi}' = \langle L\rangle$ and $\hat{\Phi}'' = N\left\langle (L-\langle L\rangle)^2\right\rangle - \langle\langle z\rangle x\rangle/(4Nr^{3/2})$, with $L = R_{1*} + \frac{\langle z\rangle x}{2N\sqrt{r}} - \frac{1}{2}$, $\mathbb{E}\langle L\rangle = \Phi' = \frac{1}{2}\mathbb{E}\langle R_{1*}\rangle \in [0,\frac{1}{2}]$ and $0 \le \mathbb{E}\langle\langle z\rangle x\rangle \le \sqrt{r}N$. Taking expectations, $\mathbb{E}\left\langle (L-\langle L\rangle)^2\right\rangle \le \frac{\Phi''(r)}{N} + \frac{1}{4Nr}$, and integrating over $[a,b]$, $\int_a^b \mathbb{E}\left\langle (L-\langle L\rangle)^2\right\rangle \mathrm{d}r \le \frac{1}{2N} + \frac{\log(b/a)}{4N}$.

Step 2: disorder fluctuations. Since $|\langle\langle z\rangle x\rangle| \le \|z\|_1$, the function $\tilde{\Phi} = \hat{\Phi} - \frac{\sqrt{r}}{N}\|z\|_1$ is convex, and so is its mean $\bar{\Phi} = \Phi - \sqrt{2r/\pi}$. For $r \in [a,b]$, Lemma 4.8 with $f = \tilde{\Phi}$, $g = \bar{\Phi}$, $\Delta = \tilde{\Phi} - \bar{\Phi}$ and $\Lambda(r) = \bar{\Phi}'(r+\delta) - \bar{\Phi}'(r-\delta) \ge 0$ gives

$$|\langle L\rangle - \mathbb{E}\langle L\rangle| \le \Lambda(r) + \frac{1}{\delta}\sum_{u\in\{r-\delta,r,r+\delta\}}|\Delta(u)| + \frac{|\|z\|_1 - \mathbb{E}\|z\|_1|}{2N\sqrt{r}}.$$

Given x^*, $\log\mathcal{Z}_{t,u}$ is a function of the standard Gaussians $\sqrt{N}W_{ij}$ $(i<j)$, $\sqrt{N/2}\,W_{ii}$ and z_i, with partial derivatives $\frac{\lambda\sqrt{t}}{\sqrt{N}}\langle x_i x_j\rangle$, $\frac{\lambda\sqrt{t}}{\sqrt{2N}}$ and $\sqrt{u}\langle x_i\rangle$ and squared gradient at most $\frac{\lambda^2 t}{N}\binom{N}{2} + \frac{\lambda^2 t}{2} + uN \le (\frac{\lambda^2}{2}+u)N$. The Gaussian Poincaré inequality (Theorem A.6) and Lemma 4.4 give $\operatorname{Var}\hat{\Phi}(u) \le (\frac{\lambda^2}{2}+u)/N$, and with $\operatorname{Var}\|z\|_1 \le N$ and $u \le \frac{3}{2}b$, $\mathbb{E}\Delta(u)^2 \le 2\operatorname{Var}\hat{\Phi}(u) + \frac{2u}{N^2}\operatorname{Var}\|z\|_1 \le (\lambda^2+6b)/N$. Next, $0 \le \Lambda \le M_a$, because $\Phi' \in [0,\frac{1}{2}]$ and $r-\delta \ge \frac{a}{2}$; and $\int_a^b \Lambda\,\mathrm{d}r \le \delta + 2\sqrt{\delta/\pi}$, because the integral equals $[\bar{\Phi}(b+\delta)-\bar{\Phi}(b-\delta)] - [\bar{\Phi}(a+\delta)-\bar{\Phi}(a-\delta)]$, Φ is nondecreasing and $\frac{1}{2}$-Lipschitz, and $\sqrt{a+\delta}-\sqrt{a-\delta} \le \sqrt{2\delta}$. Squaring with $(\alpha_1+\alpha_2+\alpha_3)^2 \le 3(\alpha_1^2+\alpha_2^2+\alpha_3^2)$, which also gives $(\sum_u|\Delta(u)|)^2 \le 3\sum_u\Delta(u)^2$, using $\Lambda^2 \le M_a\Lambda$ and $\operatorname{Var}\|z\|_1 \le N$, and integrating,

$$\int_a^b \mathbb{E}(\langle L\rangle - \mathbb{E}\langle L\rangle)^2\,\mathrm{d}r \le 3M_a\Big(\delta + 2\sqrt{\frac{\delta}{\pi}}\Big) + \frac{27(b-a)(\lambda^2+6b)}{N\delta^2} + \frac{3\log(b/a)}{4N}.$$

Step 3: from L to the overlap. Write $Q = R_{1*}$, $\bar{Q} = \mathbb{E}\langle Q\rangle$ and $\bar{L} = \mathbb{E}\langle L\rangle = \frac{1}{2}\bar{Q}$. Integration by parts in z gives $\mathbb{E}\langle Q\langle z\rangle x\rangle = \sqrt{r}N\,\mathbb{E}(\langle Q\rangle - \langle R_{1*}R_{12}\rangle)$, so $\mathbb{E}\langle LQ\rangle = \mathbb{E}\left\langle Q^2\right\rangle - \frac{1}{2}\mathbb{E}\langle R_{1*}R_{12}\rangle$, and the Nishimori identity turns this into $\mathbb{E}\left\langle R_{12}^2\right\rangle - \frac{1}{2}\mathbb{E}\langle R_{12}R_{13}\rangle$. Since $V_N = \mathbb{E}\left\langle R_{12}^2\right\rangle - \bar{Q}^2$,

$$\mathbb{E}\left\langle (L-\bar{L})(Q-\bar{Q})\right\rangle = \frac{1}{2}V_N + \frac{1}{2}\mathbb{E}\big(\left\langle R_{12}^2\right\rangle - \langle R_{12}R_{13}\rangle\big) \ge \frac{1}{2}V_N,$$

because $N^2(\left\langle R_{12}^2\right\rangle - \langle R_{12}R_{13}\rangle) = \|\mathbf{C}\|_F^2 + m^\top\mathbf{C}m \ge 0$, with $m = \langle x\rangle$ and the posterior covariance $\mathbf{C} = \langle xx^\top\rangle - mm^\top$. The Cauchy–Schwarz inequality then gives $V_N \le 4\mathbb{E}\left\langle (L-\bar{L})^2\right\rangle = 4\mathbb{E}\left\langle (L-\langle L\rangle)^2\right\rangle + 4\mathbb{E}(\langle L\rangle - \bar{L})^2$, and Steps 1 and 2 give the claim. With $\delta = N^{-2/5}$, $\delta \le \sqrt{\delta} = N^{-1/5}$ and $N\delta^2 = N^{1/5}$. ∎

By the Nishimori identity R_{12} has the law of R_{1*}, so V_N is also the variance of the overlap of two replicas: with a side channel, $P(q)$ concentrates on one value for most r, at every λ. This is the rigorous form of replica symmetry.

4.5.1 The matching bound

To make Guerra's remainder a variance around the mean, the trial overlap follows the model along the path: the *adaptive interpolation* of Barbier and Macris (2019), in the continuous form of Barbier et al. (2019).

Proof of part 2 of Theorem 4.2. The plan is to choose the trial overlap equal to the mean overlap of the interpolating model, so that the remainder in Proposition 4.6 is V_N, and to average over the side channel so that Proposition 4.9 applies. At $\lambda = 0$ both sides vanish. Let $\lambda > 0$, $N \geq 11$, $s = N^{-1/10}$ and $\delta = N^{-2/5}$, so that $2\delta \leq s \leq 1$, and let $G_N(t,r) = \mathbb{E}\,\langle R_{1*}\rangle_{t,r} \in [0,1]$.

The adaptive path. G_N is continuous and nondecreasing in r (Lemma 4.5), and Lipschitz in $r \geq \frac{s}{2}$ uniformly in t, because $\partial_r G_N = 2\partial_r^2 \phi_N$ and, by Step 1 of the proof of Proposition 4.9, $|\partial_r^2 \phi_N| \leq N\mathbb{E}\left\langle L^2\right\rangle + \frac{1}{4r}$ with $|L| \leq \frac{3}{2} + \|z\|_1/(2N\sqrt{r})$. Extend $G_N(t,\cdot)$ to $r < \frac{s}{2}$ by its value at $\frac{s}{2}$. For $\epsilon \in [s, 2s]$ the Cauchy–Lipschitz theorem gives a unique continuously differentiable solution r_ϵ of $r'(t) = -\lambda^2 G_N(t, r(t))$ on $[0,1]$ with $r(1) = \epsilon$; it depends continuously on (t,ϵ), and $\epsilon \leq r_\epsilon \leq \epsilon + \lambda^2$. It expands distances: for $\epsilon_1 < \epsilon_2$, $\Delta = r_{\epsilon_2} - r_{\epsilon_1}$ satisfies

$$\Delta(t) = \epsilon_2 - \epsilon_1 + \lambda^2 \int_t^1 \left[G_N(u, r_{\epsilon_2}(u)) - G_N(u, r_{\epsilon_1}(u))\right] \mathrm{d}u,$$

so if t_0 is the infimum of the t with $\Delta > 0$ on $[t,1]$, then $\Delta \geq \epsilon_2 - \epsilon_1$ on $(t_0, 1]$, hence at t_0 by continuity, and $t_0 = 0$, since otherwise $\Delta > 0$ slightly to the left of t_0.

The sum rule. With $q(t) = G_N(t, r_\epsilon(t))$, $r_\epsilon(t) = \epsilon + \lambda^2 \int_t^1 q$, and the remainder in Proposition 4.6 is $\frac{\lambda^2}{4}\int_0^1 V_N(t, r_\epsilon(t))\,\mathrm{d}t$. Three inequalities simplify the rest: $\phi_N \leq \phi_N(1,\epsilon)$ because $\partial_r \phi_N \geq 0$; $\int_0^1 q^2 \geq \bar{q}^2$ for $\bar{q} = \int_0^1 q \in [0,1]$; and $\psi(\epsilon + \lambda^2\bar{q}) \leq \psi(\lambda^2\bar{q}) + \frac{\epsilon}{2}$ because $0 \leq \psi' \leq \frac{1}{2}$. Since $\frac{\epsilon}{2} \leq s$,

$$\phi_N \leq \phi_{\mathrm{RS}}(\lambda, \bar{q}) + s + \frac{\lambda^2}{4}\int_0^1 V_N(t, r_\epsilon(t))\,\mathrm{d}t.$$

Averaging over the side channel. For fixed t the map $\rho(\epsilon) = r_\epsilon(t)$ is continuous and expanding, and $V_N(t,\cdot) \geq 0$ is continuous, so comparing Riemann sums, $\sum_k V_N(t, \rho(\epsilon_k))(\epsilon_{k+1} - \epsilon_k) \leq \sum_k V_N(t, \rho(\epsilon_k))(\rho(\epsilon_{k+1}) - \rho(\epsilon_k))$, gives $\int_s^{2s} V_N(t, r_\epsilon(t))\,\mathrm{d}\epsilon \leq \int_s^{2s+\lambda^2} V_N(t,r)\,\mathrm{d}r$. Averaging the last bound over $\epsilon \in [s, 2s]$ and applying Proposition 4.9 with $a = s$ and $b = 2s + \lambda^2$,

$$\phi_N - \max_q \phi_{\mathrm{RS}}(\lambda, q) \leq s + \frac{\lambda^2}{4s}\Big[\frac{2 + 4\log(2 + \lambda^2/s)}{N} + 12M_s\Big(\delta + 2\sqrt{\frac{\delta}{\pi}}\Big) + \frac{108(s+\lambda^2)(7\lambda^2 + 12s)}{N\delta^2}\Big].$$

With $s = N^{-1/10}$ and $\delta = N^{-2/5}$ the three terms in the bracket contribute $O(N^{-9/10}\log N)$, $O(s^{-3/2}\sqrt{\delta}) = O(N^{-1/20})$ and $O(s^{-1}N^{-1/5}) = O(N^{-1/10})$, with constants depending only on λ. With part 1 this proves part 2. ∎

The rate $N^{-1/20}$ is an artifact of the argument (the choice $s = N^{-1/12}$, $\delta = N^{-5/12}$, admissible for $N \geq 8$, balances the terms and gives $N^{-1/12}$); Figure 4.2(b) suggests a gap of order $1/N$.

4.5.2 Uniqueness and the minimum mean-square error

For signs the maximizer is unique for every $\lambda > 0$, because the overlap curve G is strictly concave. Deshpande et al. (2016, §3 and App. B.2) proved this from $G'(\gamma) = \mathbb{E}\,\mathrm{sech}^4(\gamma + \sqrt{\gamma}Z)$ (Exercise 4.6); here it follows at once from writing the binary mmse as a mixture of decaying exponentials.

Lemma 4.10 (The binary mmse is completely monotone). *For every $\gamma \geq 0$, with the probability measure $\mu(\mathrm{d}\omega) = \frac{1}{2}\,\mathrm{sech}(\frac{\pi\omega}{2})\,\mathrm{d}\omega$,*

$$1 - G(\gamma) = e^{-\gamma/2}\,\mathbb{E}\,\mathrm{sech}(\sqrt{\gamma}Z) = \int_{\mathbb{R}} e^{-(1+\omega^2)\gamma/2}\,\mu(\mathrm{d}\omega).$$

Consequently G is strictly concave on $[0,\infty)$, with $G(0) = 0$ and $G'(0) = 1$.

Proof. The plan is to remove the drift by a change of measure and to expand sech in Fourier modes. The density of $\gamma + \sqrt{\gamma}Z$ at h is $e^{h-\gamma/2}$ times that of $\sqrt{\gamma}Z$, so $\mathbb{E}\,f(\gamma + \sqrt{\gamma}Z) = e^{-\gamma/2}\mathbb{E}[e^{\sqrt{\gamma}Z} f(\sqrt{\gamma}Z)]$; with $f = 1 - \tanh$ and $(1 - \tanh u)e^u = \mathrm{sech}\,u$ this is the first identity. The function sech is its own Fourier transform up to scaling, $\int_{\mathbb{R}} \mathrm{sech}(u)e^{-i\omega u}\,\mathrm{d}u = \pi\,\mathrm{sech}(\pi\omega/2)$ (a residue computation), so $\mathrm{sech}\,u = \int \cos(\omega u)\mu(\mathrm{d}\omega)$, and Fubini's theorem with $\mathbb{E}\cos(\omega\sqrt{\gamma}Z) = e^{-\omega^2\gamma/2}$ gives the second. Differentiating under the integral, which the exponential decay of μ allows, $G''(\gamma) = -\frac{1}{4}\int(1+\omega^2)^2 e^{-(1+\omega^2)\gamma/2}\mu(\mathrm{d}\omega) < 0$, and $G'(0) = \frac{1}{2}\int(1+\omega^2)\mu(\mathrm{d}\omega) = 1$, because $\int \omega^2\mu(\mathrm{d}\omega) = -\,\mathrm{sech}''(0) = 1$. ∎

Proof of parts 3 and 4 of Theorem 4.2. The plan is to read the maximizer off the concavity of G, and to obtain the error by differentiating the limit of the free entropy in $\theta = \lambda^2$.

Part 3. Let $\lambda > 0$. Since $\psi' = \frac{1}{2}G$, $\partial_q \phi_{\mathrm{RS}} = \frac{\lambda^2}{2}h(q)$ with $h(q) = G(\lambda^2 q) - q$, which is strictly concave, with $h(0) = 0$, $h'(0) = \lambda^2 - 1$ and $h(1) < 0$. If $\lambda \leq 1$, then $h < 0$ on $(0,1]$ and $q_* = 0$. If $\lambda > 1$, then $h > 0$ near 0, and a strictly concave function vanishing at 0 has at most one other zero; so h has exactly one zero $q_\circ \in (0,1)$, is positive before it and negative after it, and $q_* = q_\circ$. The zeros of h in $(0,1]$ are the solutions of (4.7).

Part 4. At $\lambda = 0$ the posterior is uniform and $\mathrm{MMSE}_N = 1 - \frac{1}{N} \to 1 - q_*(0)^2$. For $\theta = \lambda^2 > 0$, Lemma 4.5 at $r = 0$ gives $\frac{\mathrm{d}\phi_N}{\mathrm{d}\theta} = \frac{1}{4}\mathbb{E}\left\langle R_{1*}^2 \right\rangle = \frac{1}{4}(1 - \mathrm{MMSE}_N)$, nondecreasing in θ, so ϕ_N is convex in θ. Its limit $\Phi(\theta) = \max_{q\in[0,1]} \tilde{\phi}(\theta, q)$, with $\tilde{\phi}(\theta, q) = \psi(\theta q) - \frac{\theta q^2}{4}$, has $\Phi'(\theta) = \frac{1}{4}q_*^2$: the difference $\Phi(\theta') - \Phi(\theta)$ lies between $\tilde{\phi}(\theta', q) - \tilde{\phi}(\theta, q)$ at $q = q_*(\theta)$ and at $q = q_*(\theta')$, both are $(\theta' - \theta)\,\partial_\theta \tilde{\phi}$ at intermediate points, $q_*(\theta') \to q_*(\theta)$ by uniqueness and compactness, and $\partial_\theta \tilde{\phi} = q\psi'(\theta q) - \frac{q^2}{4} = \frac{q^2}{4}$ at $q = q_*$ by the fixed-point equation. Finally, if convex f_N converge pointwise to f, then $\frac{f(\theta) - f(\theta - u)}{u} \leq \liminf f_N'(\theta) \leq$

$\limsup f_N'(\theta) \le \frac{f(\theta+u)-f(\theta)}{u}$ for small $u > 0$, and both bounds tend to $f'(\theta)$. Hence $\mathbb{E}\left\langle R_{1*}^2 \right\rangle \to q_*^2$. ∎

4.5.3 What the general proof adds

Signs gave us a constant norm $\|x\|^2 = N$, bounded spins and the gauge symmetry. For priors with bounded support the sum rule holds verbatim (Exercise 4.7), and Barbier and Macris (2019) control the variance and the self-overlaps with a Gaussian side channel, interpolating in K small steps. Lelarge and Miolane (2019) instead reveal each x_i^* with a small probability and run the Aizenman–Sims–Starr cavity computation (Aizenman et al. 2003), comparing sizes N and $N+1$, for priors with finite support, which approximate every prior with a finite second moment; and they remove the perturbation: for bounded priors, at points of differentiability, $\mathbb{E}\left\langle (R_{1*}^2 - q_*^2)^2 \right\rangle \to 0$ (their Theorem 2).

4.6 Measuring it

The script `04-nishimori-exact.py` enumerates the posterior at $N = 6, 8, 10, 12$: for ten values of λ it draws 1200 instances and records the free entropy, the matrix error and both sides of $\mathbb{E}\left\langle R_{1*}^2 \right\rangle = \mathbb{E}\left\langle R_{12}^2 \right\rangle$. Bayes' rule writes $\log \mathcal{Z}_N = B - \log(2^N \mathbb{P}(x^* \mid A))$ with $B = H_N(x^*) - \frac{\lambda^2 N}{4}$ of known mean and variance $\frac{\lambda^2 N}{2}$, and B serves as a control variate that shrinks the intervals by a factor of up to 19. The script `04-nishimori-amp.py` runs Bayes-optimal message passing (Chapter 6) from a spectral start on ten instances at $N = 2000$ per value of λ, and records the matrix error of its estimate after 40 iterations and that of principal components, $(1-\lambda^{-2})_+ N v_1 v_1^\top$; message passing reaches the Bayes error for $\lambda > 1$ (Deshpande et al. 2016; Montanari and Venkataramanan 2021).

In panel (a), message passing follows the prediction: ten of its eleven intervals contain $1 - q_*^2$, and the eleventh, at $\lambda = 2$, misses it by 0.001 (0.172 ± 0.011 against 0.160). Above the threshold the Bayes error is far below that of principal components, 0.160 against 0.4375 at $\lambda = 2$, and all eleven spectral intervals contain their proved limit; near the threshold the two curves touch (Exercise 4.4). The exact error at $N = 12$ differs from its limit by up to 0.23, at the threshold (0.772 against 1); below the threshold it is at most $1 - \frac{1}{N} = 0.917$, because the posterior mean knows the diagonal of $x^* x^{*\top}$, a twelfth of the matrix, and the measured 0.85 to 0.91 approach this as λ decreases.

In panel (b), all forty values of $N(\phi_N - \max_q \phi_{\mathrm{RS}})$ are positive, as part 1 of Theorem 4.2 requires, and of order one: 0.65 to 1.08 for $\lambda \ge 1.25$. Below the threshold they lie under the bound of Proposition 4.7 and near half of it (0.011–0.020, 0.053–0.074 and 0.170–0.203 at $\lambda = 0.25, 0.5, 0.75$, against 0.016, 0.072, 0.207), which is the mean under the spiked law of a Gaussian log-likelihood ratio with second moment $(1-\lambda^2)^{-1/2}$; El Alaoui et al. (2020) prove the Gaussian limit below the threshold, and the convergence of the moments is the heuristic step. The Nishimori identity holds within the noise: the forty differences are at most 0.009 in absolute value except one,

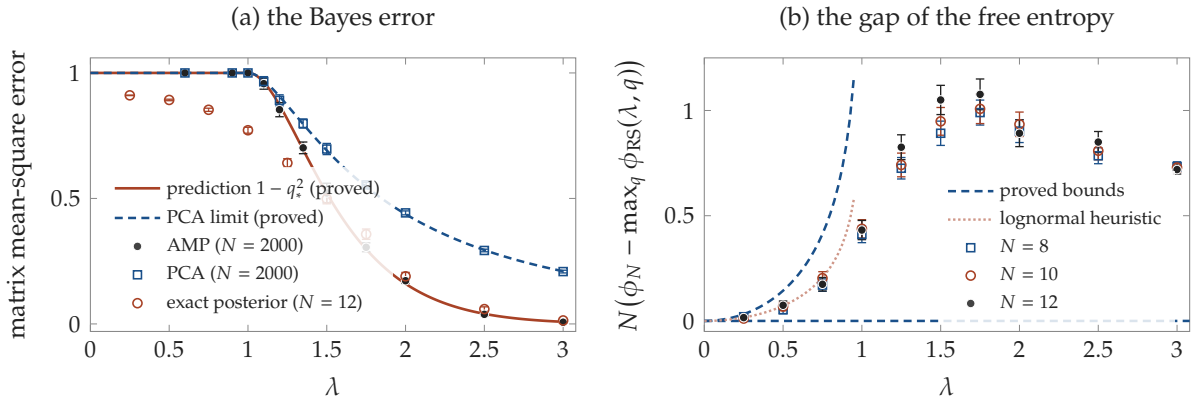


Figure 4.2. (a) Matrix mean-square error. Carnelian: $1 - q_*^2$ (Theorem 4.2); lapis dashed: the limit $1 - (1 - \lambda^{-2})_+^2$ of principal components (Theorem 5.1), an upper bound on the Bayes error. Dots: message passing at $N = 2000$, ten instances per point (seeds $4000 + 100k + r$); squares: principal components on the same instances; circles: exact Bayes error at $N = 12$, 1200 instances per point (seeds $1000N + k$). (b) N times the gap of the free entropy, by exact enumeration (1200 instances per point, control variate). Lapis dashed: the proved bounds 0 and, for $\lambda < 1$, $-\frac{1}{2}\log(1 - \lambda^2)$ (Proposition 4.7); carnelian dotted: the lognormal heuristic $-\frac{1}{4}\log(1-\lambda^2)$. Student-$t$ 95% intervals. Scripts `04-nishimori-exact.py`, `-amp.py`, `-rs.py`.

-0.016 ± 0.009 at $N = 12$, $\lambda = 1.75$.[6] The scripts run in about 45, 55 and 10 seconds, and no measurement contradicts a theorem of the chapter.

[6] The per-instance differences are skewed: at that point their median is +0.035, while a few instances on which the posterior is confident and wrong give values near −0.7 (the 0.1% quantile is −0.71) and carry the mean. Resampling the differences, recentred, a deviation of 3.7 standard errors has probability 3×10^{-4} in one cell and about 1.4% somewhere among forty: a rare fluctuation, not a failure of the identity. Two fresh sets of 3000 instances give $+0.0006 \pm 0.0048$ and -0.0001 ± 0.0047.

4.7 The status board

As of 2026 everything in the prediction is a theorem, for signs and for every prior with a finite second moment. What remains physics is the principle that the Nishimori line never breaks replica symmetry without a perturbation, and the hardness of the hard phase; off the line the replica-symmetric answer is wrong at low temperature.

STATUS BOARD · the Nishimori line ↔ overlap concentration

PREDICTION	STATUS	PROOF OR EVIDENCE
Nishimori identity, at every N	proved	Bayes' rule (Lemma 4.3); gauge form: Nishimori (1981)
$P(q) = P(m)$ on the Nishimori line	proved	the identity with $f = \mathbf{1}\{R_{12} \in B\}$; $\pm J$ model: Nishimori and Sherrington (2001)

No replica-symmetry breaking on the Nishimori line, in every model	proved under	as overlap concentration, under a vanishing side channel, for bounded product priors whose free entropy concentrates: Barbier and Panchenko (2022, Thm. 2.1); without a perturbation open (Zdeborová and Krzakala 2016, §II.G)
Mutual information and error of $\mathbb{Z}_2$ synchronization	proved	Deshpande et al. (2016); Lelarge and Miolane (2019); Theorem 4.2
The same for every prior with finite second moment (the error outside a countable set of λ)	proved	Lelarge and Miolane (2019, Thm. 1); bounded support: Barbier and Macris (2019)
Continuous transition at $\lambda = 1$, with a unique maximizer	proved	from the formula; strict concavity of G: Deshpande et al. (2016, App. B.2); by complete monotonicity: Lemma 4.10
Overlap concentration (rigorous replica symmetry)	proved	side channel: Proposition 4.9, Barbier (2021); without: Lelarge and Miolane (2019, Thm. 2)
Replica-symmetric formula for a posterior at the wrong temperature	false	$\lambda = 0$, $\beta > 1$ (Toninelli 2002); correct formula: Chen (2014)
The continuation $n \to 0$ and the exchange of limits	heuristic	no counterpart: the proof never continues in n
Sparse priors: the hard phase is hard for efficient algorithms	conjectured	Lesieur et al. (2017, §IV C); evidence in Chapter 5

4.8 Where the translation fails

WHAT WAS ASSUMED. Bayes optimality does the work: the statistician knows the prior, the signal-to-noise ratio and the noise, and computes the exact posterior. The prediction assumes it too, and every theorem here uses it through the Nishimori identity. The model is idealized as usual: a product prior, every pair observed, Gaussian noise, rank one, $N \to \infty$ at fixed λ. Finite rank is treated by Lelarge and Miolane (2019, §2.8), and other noise channels reduce to the Gaussian one by channel universality, proved by Krzakala et al. (2016) for priors with finite support and channels with bounded second and third log-derivatives. Our proof also assumed signs.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. Off the Nishimori line the argument collapses: a posterior at the wrong temperature is a planted Sherrington–Kirkpatrick model, the Nishimori identity fails, the remainder

loses its sign (Exercise 4.8), and at $\lambda = 0$, $\beta > 1$ the replica-symmetric formula is false (Toninelli 2002); the free entropy is then the formula of Chen (2014). On the line, replica symmetry in its rigorous form, the concentration of the overlap under a vanishing side channel, holds for every bounded product prior whose free entropy concentrates (Barbier and Panchenko 2022, Thm. 2.1); without a perturbation it is proved for spiked Wigner models with bounded priors, outside a countable set of λ (Lelarge and Miolane 2019, Thm. 2), and open in general (Zdeborová and Krzakala 2016, §II.G). "Self-averaging" is literally false without a perturbation: for signs $\langle R_{1*}\rangle = 0$ in every instance, while $|R_{1*}|$ concentrates; physicists break the symmetry by hand, the proof with the side channel. Replica-symmetric statics says nothing about how the posterior is reached: for sparse priors a second maximum can trap an efficient algorithm. And at accessible sizes the corrections are large (up to 0.23 in the error at $N = 12$), and the replica computation is silent about them.

OPEN PROBLEMS. Three are posed in the literature. Whether the magnetization self-averages on the Nishimori line without a perturbation, as physicists argue and as Zdeborová and Krzakala (2016, §II.G) note is open from a rigorous point of view. Whether a polynomial-time algorithm can beat message passing in the hard phase of sparse priors, which Lesieur et al. (2017, §IV C) conjecture to be impossible (Chapter 5). And whether, for $\mathbb{Z}_2$ synchronization, the local minimizer of the TAP free energy near the posterior mean, which Celentano et al. (2023) construct for every $\lambda > 1$, is global, as they conjecture from numerics (Chapter 6).

Exercises

4.1 CONSOLIDATION. (The scalar channel.) Let $h = \gamma + \sqrt{\gamma}Z$, $\gamma > 0$. (a) Show that the density p of h satisfies $p(-h) = e^{-2h}p(h)$, and deduce $\mathbb{E}[f(h)\tanh h] = \mathbb{E}[f(h)\tanh^2 h]$ for every bounded even f; in particular $G(\gamma) = \mathbb{E}\tanh^2 h$. (b) Prove $\psi' = \frac{1}{2}G$ by integration by parts. (c) Check that $\phi_N(0, r) = \psi(r)$.

4.2 CONSOLIDATION. (Information and error.) (a) Prove (4.3) and the second form of (4.4). (b) Show that $\frac{\mathrm{d}}{\mathrm{d}\lambda^2}\frac{1}{N}I(x^*; A) = \frac{1}{4}\mathrm{MMSE}_N$, and check numerically that $\frac{1}{4}\int_0^{2.25}(1 - q_*(\sqrt{\theta})^2)\,\mathrm{d}\theta = 0.4942$, the limit of $\frac{1}{N}I$ at $\lambda = 1.5$.

4.3 CONSOLIDATION. (The Gaussian prior.) For $P_0 = \mathcal{N}(0, 1)$ show that $\psi_{P_0}(\gamma) = \frac{1}{2}(\gamma - \log(1 + \gamma))$, $1 - \mathrm{mmse}(\gamma) = \frac{\gamma}{1+\gamma}$ and $q_* = (1 - \lambda^{-2})_+$, so that principal components are Bayes-optimal, and that the right side of Theorem 4.1 is $\frac{\lambda^2}{4}$ for $\lambda \le 1$ and $\log\lambda + \frac{1}{4\lambda^2}$ for $\lambda > 1$.

4.4 CONSOLIDATION. (Near the threshold.) From $G(\gamma) = \gamma - \gamma^2 + \frac{5}{3}\gamma^3 + O(\gamma^4)$, show that $q_* = 2\epsilon - \frac{1}{3}\epsilon^2 + O(\epsilon^3)$ at $\lambda = 1 + \epsilon$, and that $1 - q_*^2 = 1 - 4\epsilon^2 + \frac{4}{3}\epsilon^3 + O(\epsilon^4)$, while principal components give $1 - 4\epsilon^2 + 12\epsilon^3 + O(\epsilon^4)$.

4.5 BY HAND. (The series of G.) (a) Show that the moments $m_{2k} = \int \omega^{2k}\mu(\mathrm{d}\omega)$ of the measure of Lemma 4.10 are $(-1)^k\,\mathrm{sech}^{(2k)}(0)$, the

Euler numbers $|E_{2k}| = 1, 1, 5, 61, 1385, \dots$ (b) Deduce that, as $\gamma \to 0$, G has the asymptotic expansion $\sum_{k\ge1}(-1)^{k+1}c_k\gamma^k$, where $c_k = \frac{1}{2^k k!}\sum_{j\le k}\binom{k}{j}|E_{2j}|$, and check the coefficients $1, -1, \frac{5}{3}, -\frac{13}{3}$. (c) Show that the series diverges for every $\gamma > 0$. *(Hint:* $|E_{2k}| \ge (2k)!\,(2/\pi)^{2k+1}$.*)*

4.6 BY HAND. (Replica symmetry is stable on the Nishimori line.) (a) Using Exercise 4.1, show that $G'(\gamma) = \mathbb{E}\,\operatorname{sech}^4(\gamma + \sqrt{\gamma}Z)$. (b) The de Almeida–Thouless condition of Chapter 3, with the ferromagnetic field $\beta J_0 m$ in place of the external field, is $\beta^2\mathbb{E}\,\operatorname{sech}^4(\beta\sqrt{q}Z + \beta J_0 m) \le 1$. Show that on the Nishimori line, with $m = q = q_*$, it reads $\lambda^2 G'(\lambda^2 q_*) \le 1$, and deduce from Lemma 4.10 that it holds for every λ, strictly for $\lambda \ne 1$ (0.838 at $\lambda = 1.1$, 0.457 at $\lambda = 1.5$).

4.7 BY HAND. (Every prior.) Let P_0 have bounded support and $\mathbb{E}X^2 = 1$. (a) Show that (4.8) holds verbatim with $x^0, \dots, x^n$ independent with law $P_0^{\otimes N}$: the terms in $\|x^a\|^4$ cancel against the diagonal of the GOE. (b) Redo the replica-symmetric computation and obtain ψ_{P_0}. (c) With the evidence of the interpolating model as in Section 4.4.1 but averaged over $P_0^{\otimes N}$, check Lemma 4.5 and the sum rule with ψ_{P_0} for ψ, and deduce Guerra's bound at every N.

4.8 TRANSLATION. (Off the Nishimori line.) Let $\lambda = 0$, so that $A = W$, and take the Gibbs measure proportional to $e^{\frac{\beta}{2}x^\top W x}$, the posterior computed from pure noise by a statistician who believes the ratio is β. With $H_t(x) = \frac{\beta\sqrt{t}}{2}x^\top W x + \beta\sqrt{q(1-t)}\langle z\rangle x$, show that $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{N}\mathbb{E}\log\sum_x e^{H_t(x)} = \frac{\beta^2}{4}((1-q)^2 - \mathbb{E}\left\langle (R_{12}-q)^2\right\rangle_t)$, deduce Guerra's upper bound (Theorem 3.3) at zero field, and find the step of the proof of Lemma 4.5 that fails.

4.9 TRANSLATION. (Self-averaging of a derivative.) Let f_N be random convex functions on an open interval I with $\sup_I \operatorname{Var} f_N \le c/N$, and $g_N = \mathbb{E}f_N$ with $|g_N'| \le 1$. (a) Show with Lemma 4.8 that $\int_J \mathbb{E}|f_N' - g_N'| = O(N^{-1/4})$ on compact intervals $J \subset I$. (b) Show that the integral cannot be dropped, with $f_N(u) = |u - U_N|$ and U_N uniform on $[u_0 - \frac{1}{N}, u_0 + \frac{1}{N}]$.

4.10 MEASUREMENT. (Overlap distributions.) Modify the enumeration script of Section 4.6 to record at $N = 12$, for $\lambda = 0.5$ and 2, the laws of R_{12} and R_{1*} under $\mathbb{E}\langle\cdot\rangle$. Check that they agree ($P(q) = P(m)$), that at $\lambda = 0.5$ both are one bump around 0 of width about $N^{-1/2}$, and that at $\lambda = 2$ most of their mass sits at ±1 and $\pm\frac{5}{6}$. Add a side channel with $r = 0.5$: what happens to the symmetry between R and $-R$, and why?

4.11 RESEARCH. (The mismatched posterior.) A statistician who believes the ratio is β when it is λ computes the Gibbs measure proportional to $e^{\frac{\beta}{2}x^\top A x}$. (a) Show that after the gauge transformation it is the Sherrington–Kirkpatrick model at inverse temperature β plus $\frac{\beta\lambda}{N}\sum_{i<j}\sigma_i\sigma_j$. (b) Show that the replica-symmetric equations $m = \mathbb{E}\tanh(\beta\sqrt{q}Z + \beta\lambda m)$ and $q = \mathbb{E}\tanh^2(\beta\sqrt{q}Z + \beta\lambda m)$ have the solution

$m = q = q_*$ when $\beta = \lambda$. (c) Map where this solution violates the de Almeida–Thouless condition, compare with the formula of Chen (2014), and find where the mismatched posterior mean does worse than principal components.

Notes and References

Nishimori (1981) introduced the gauge argument for the $\pm J$ spin glass and computed the internal energy exactly on the line that now carries his name; his book (Nishimori 2001) develops the subject. Iba (1999) identified the line with Bayes-optimal inference, and the review of Zdeborová and Krzakala (2016) builds the statistical physics of inference on it.

For signs, the gauge-transformed model is the Sherrington–Kirkpatrick model with a ferromagnetic mean of the couplings, which Sherrington and Kirkpatrick (1975) already allowed; on the Nishimori line its replica-symmetric equations are solved by $m = q = q_*$ (Exercise 4.11). For general priors the formula was derived by Lesieur et al. (2017) and in their earlier work. Barbier and Macris (2019) credit Korada and Macris (2009), who solved the gauge-symmetric p-spin model on the complete graph, with a full proof of the binary case. Deshpande et al. (2016) proved the formula for signs by message passing and the I-MMSE relation; Barbier et al. (2016), for discrete priors with at most three stationary points of the potential, by spatial coupling; Krzakala et al. (2016) proved Guerra's bound at every N; Lelarge and Miolane (2019) proved the formula for every prior with a finite second moment; Barbier and Macris (2019) introduced the adaptive interpolation, which Barbier et al. (2019) carried, in continuous form, to generalized linear models (Chapter 7); and Barbier (2021) proved the concentration of the overlap matrix in optimal Bayesian inference, and Barbier and Panchenko (2022) that of all multioverlaps. The interpolation goes back to Guerra and Toninelli (2002) and Guerra (2003); below the threshold the spiked and null laws are contiguous, so no test separates them with vanishing error (El Alaoui et al. 2020; Perry et al. 2018).

The proof of Theorem 4.2 assembles standard pieces, following Barbier and Macris (2019) and Barbier et al. (2019); the gauge argument for the variance, the constants of Proposition 4.9 and the finite-N form of Proposition 4.7 are the book's own. The uniqueness of the maximizer is due to Deshpande et al. (2016); the representation of the binary mmse as a mixture of exponentials in Lemma 4.10, and the proof of uniqueness through it, are the book's own: the representation is elementary, but we did not find it in the literature.

TRANSLATING THE LITERATURE. Lelarge and Miolane (2019) observe $Y_{ij} = \sqrt{\lambda/n}\, X_i X_j + Z_{ij}$, $i < j$: their λ is our λ^2, their n, X and $x \cdot X$ are our N, x^* and R_{1*}, their $F(\lambda, q)$ is our $\phi_{\mathrm{RS}}(\sqrt{\lambda}, q)$, and their free energy $F_n = \frac{1}{n}\mathbb{E}\log Z_n$ is our free entropy up to $O(1/N)$ from the diagonal. Barbier and Macris (2019) write $\Delta = 1/\lambda^2$ for the noise variance, $f_n = -\frac{1}{n}\mathbb{E}\ln \mathcal{Z}$, and $f_{\mathrm{RS}}(m; \Delta) = -\phi_{\mathrm{RS}}(\lambda, m)$, so their minimum over m is our maximum over q. Deshpande et al. (2016) write λ for our λ^2, keep $\gamma = \lambda^2 q$, and write the limit

of $\frac{1}{N}I$ as $\Psi(\gamma_*, \lambda) = \frac{\lambda}{4} + \frac{\gamma_*^2}{4\lambda} - \frac{\gamma_*}{2} + I(\gamma_*)$ with $I(\gamma) = \frac{\gamma}{2} - \psi(\gamma)$. Lesieur et al. (2017) have $\Delta/\rho^2 = 1/\lambda^2$, with $\rho = \mathbb{E}X^2$; Montanari and Venkataramanan (2021) use our λ. Physicists' $P(q)$ and $P(m)$ are the laws of R_{12} and R_{1*} under $\mathbb{E}\langle\cdot\rangle$.

References

Aizenman, M., R. Sims, and S. L. Starr (2003). "Extended variational principle for the Sherrington–Kirkpatrick spin-glass model". In: *Physical Review B* 68.21, p. 214403. arXiv: cond-mat/0306386.

Barbier, J. (2021). "Overlap matrix concentration in optimal Bayesian inference". In: *Information and Inference: A Journal of the IMA* 10.2, pp. 597–623. arXiv: 1904.02808.

Barbier, J., M. Dia, N. Macris, F. Krzakala, T. Lesieur, and L. Zdeborová (2016). "Mutual information for symmetric rank-one matrix estimation: A proof of the replica formula". In: *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pp. 424–432. arXiv: 1606.04142.

Barbier, J., F. Krzakala, N. Macris, L. Miolane, and L. Zdeborová (2019). "Optimal errors and phase transitions in high-dimensional generalized linear models". In: *Proceedings of the National Academy of Sciences* 116.12, pp. 5451–5460. arXiv: 1708.03395.

Barbier, J. and N. Macris (2019). "The adaptive interpolation method: a simple scheme to prove replica formulas in Bayesian inference". In: *Probability Theory and Related Fields* 174.3-4, pp. 1133–1185. arXiv: 1705.02780.

Barbier, J. and D. Panchenko (2022). "Strong Replica Symmetry in High-Dimensional Optimal Bayesian Inference". In: *Communications in Mathematical Physics* 393.3, pp. 1199–1239. arXiv: 2005.03115.

Celentano, M., Z. Fan, and S. Mei (2023). "Local convexity of the TAP free energy and AMP convergence for $\mathbb{Z}_2$-synchronization". In: *The Annals of Statistics* 51.2, pp. 519–546. arXiv: 2106.11428.

Chen, W.-K. (2014). "On the mixed even-spin Sherrington–Kirkpatrick model with ferromagnetic interaction". In: *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 50.1, pp. 63–83. arXiv: 1105.2604.

Deshpande, Y., E. Abbe, and A. Montanari (2016). "Asymptotic mutual information for the balanced binary stochastic block model". In: *Information and Inference: A Journal of the IMA*. Published online 2016-12-26; in print as volume 6, issue 2 (2017), pages 125–170. arXiv: 1507.08685.

El Alaoui, A., F. Krzakala, and M. I. Jordan (2020). "Fundamental limits of detection in the spiked Wigner model". In: *The Annals of Statistics* 48.2, pp. 863–885. arXiv: 1806.09588.

Guerra, F. (2003). "Broken Replica Symmetry Bounds in the Mean Field Spin Glass Model". In: *Communications in Mathematical Physics* 233.1, pp. 1–12.

Guerra, F. and F. L. Toninelli (2002). "The Thermodynamic Limit in Mean Field Spin Glass Models". In: *Communications in Mathematical Physics* 230.1, pp. 71–79.

Iba, Y. (1999). "The Nishimori line and Bayesian statistics". In: *Journal of Physics A: Mathematical and General* 32.21, pp. 3875–3888.

Korada, S. B. and N. Macris (2009). "Exact Solution of the Gauge Symmetric p-Spin Glass Model on a Complete Graph". In: *Journal of Statistical Physics* 136.2, pp. 205–230.

Krzakala, F., J. Xu, and L. Zdeborová (2016). "Mutual Information in Rank-One Matrix Estimation". In: *2016 IEEE Information Theory Workshop (ITW)*. IEEE, pp. 71–75. arXiv: 1603.08447.

Lelarge, M. and L. Miolane (2019). "Fundamental limits of symmetric low-rank matrix estimation". In: *Probability Theory and Related Fields* 173.3-4, pp. 859–929.

Lesieur, T., F. Krzakala, and L. Zdeborová (2017). "Constrained low-rank matrix estimation: phase transitions, approximate message passing and applications". In: *Journal of Statistical Mechanics: Theory and Experiment* 2017.7, p. 073403. arXiv: 1701.00858.

Montanari, A. and R. Venkataramanan (2021). "Estimation of low-rank matrices via approximate message passing". In: *The Annals of Statistics* 49.1, pp. 321–345. arXiv: 1711.01682.

Nishimori, H. (1981). "Internal Energy, Specific Heat and Correlation Function of the Bond-Random Ising Model". In: *Progress of Theoretical Physics* 66.4, pp. 1169–1181.

Nishimori, H. (2001). *Statistical Physics of Spin Glasses and Information Processing: An Introduction*. Vol. 111. International Series of Monographs on Physics. Oxford: Oxford University Press.

Nishimori, H. and D. Sherrington (2001). "Absence of replica symmetry breaking in a region of the phase diagram of the Ising spin glass". In: *Disordered and Complex Systems*. Vol. 553. AIP Conference Proceedings. AIP, pp. 67–72. arXiv: cond-mat/0008139.

Perry, A., A. S. Wein, A. S. Bandeira, and A. Moitra (2018). "Optimality and sub-optimality of PCA I: Spiked random matrix models". In: *The Annals of Statistics* 46.5, pp. 2416–2451. arXiv: 1609.05573.

Sherrington, D. and S. Kirkpatrick (1975). "Solvable Model of a Spin-Glass". In: *Physical Review Letters* 35.26, pp. 1792–1796.

Toninelli, F. L. (2002). "About the Almeida–Thouless transition line in the Sherrington–Kirkpatrick mean-field spin glass model". In: *Europhysics Letters* 60.5, pp. 764–767. arXiv: cond-mat/0207296.

Zdeborová, L. and F. Krzakala (2016). "Statistical physics of inference: thresholds and algorithms". In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

5 Spiked Matrices

PCA, the BBP transition and the information threshold

THE DICTIONARY —— zero-temperature replicas ↔ resolvents

IN THIS CHAPTER. Plant a vector x of strength λ in a symmetric Gaussian matrix, $A = \frac{\lambda}{N} xx^\top + W$. Principal component analysis sees it exactly when $\lambda > 1$: the top eigenvalue leaves the semicircle at $\lambda + 1/\lambda$, and the top eigenvector has squared overlap $1 - 1/\lambda^2$ with x. Physicists read both numbers off the ground state of a spherical spin glass, by replicas at zero temperature; we prove them with the resolvent and the secular equation of a rank-one perturbation, and the eigenvector limit comes out as the derivative of the eigenvalue limit, as an order parameter is the derivative of a free energy. The Bayes estimator beats PCA for every non-Gaussian prior, and for sparse priors its overlap can jump: the transition is first order below the density $\rho_{\text{tri}} = 0.0964$, while the information threshold falls below the spectral one only below $\rho_* = 0.0856$, two densities that the value $\rho \approx 0.09$ in the literature does not separate. Simulations show the transition, its finite-size smearing, and a window in which message passing stays far below the Bayes overlap; whether an efficient algorithm can do better there is open.

5.1 The question: when does the top eigenvector see the spike?

Principal component analysis replaces a data matrix by its top eigenvector. The simplest model in which to ask when this works is the *spiked Wigner model*: an unknown $x \in \mathbb{R}^N$ observed through the symmetric matrix

$$A = \frac{\lambda}{N} xx^\top + W, \qquad W \sim \mathrm{GOE}(N), \tag{5.1}$$

where W has independent entries $W_{ij} \sim \mathcal{N}(0, 1/N)$ for $i < j$ and $W_{ii} \sim \mathcal{N}(0, 2/N)$, and the entries of x are independent draws from a prior P_0 with $\mathbb{E}X^2 = 1$, independent of W. The noise alone has the semicircle law on $[-2, 2]$ as its limiting spectral distribution (Theorem A.15); the signal adds a rank-one matrix with eigenvalue $\lambda\|x\|^2/N \approx \lambda$.[1] We write $\mu_1(M) \geq \mu_2(M) \geq \cdots$ for the eigenvalues of a symmetric matrix M.

Two priors run through the chapter. The Rademacher prior, $x_i = \pm 1$ with probability $\frac{1}{2}$ each, is $\mathbb{Z}_2$ *synchronization*: recover N signs from noisy measurements of all their products $x_i x_j$.[2] The sparse Rademacher prior,

[1] Other books scale the GOE so that the semicircle lives on $[-\sqrt{2}, \sqrt{2}]$ or on $[-2\sqrt{N}, 2\sqrt{N}]$; with this book's scaling the edge is 2 and the BBP threshold is $\lambda = 1$.

[2] The name refers to the group $\mathbb{Z}_2 = \{\pm 1\}$: the task is to recover group elements from noisy measurements of their ratios $x_i x_j^{-1} = x_i x_j$.



$x_i = 0$ with probability $1-\rho$ and $x_i = \pm\rho^{-1/2}$ with probability $\rho/2$ each, is the simplest model of *sparse PCA*.

PCA estimates the direction of x by a unit top eigenvector v_1 of A, and its quality is the squared overlap $\langle v_1, x\rangle^2/\|x\|^2$: 1 when $v_1 = \pm x/\|x\|$, of order $1/N$ for a random direction. The Bayes estimator uses the whole posterior. Chapter 4 describes it by the *overlap* $q_*(\lambda)$, which a probabilist would call the limit of the normalized inner product of two independent samples from the posterior: the matrix minimum mean-square error tends to $1 - q_*(\lambda)^2$, and, for priors with bounded support and all but countably many λ, q_* is the largest squared cosine with x that an estimator can achieve (Montanari and Venkataramanan 2021, Proposition 2.2). *Weak recovery* means $q_* > 0$.

THE PHYSICAL PREDICTION · spiked Wigner matrices. Let $N \to \infty$ in (5.1).

(i) The top eigenvalue tends to $\lambda + 1/\lambda$ if $\lambda > 1$ and to the edge 2 if $\lambda \le 1$; the squared overlap of the top eigenvector with $x/\|x\|$ tends to $1 - 1/\lambda^2$ if $\lambda > 1$ and to 0 if $\lambda \le 1$.

(ii) The Bayes overlap $q_*(\lambda)$ maximizes the potential $\phi_{\mathrm{RS}}(\lambda, q)$ of (5.3). For the Rademacher prior it is continuous at $\lambda = 1$ and exceeds $1 - 1/\lambda^2$ above it. For the sparse Rademacher prior it jumps at an information-theoretic threshold $\lambda_{\mathrm{IT}}(\rho)$ when $\rho < \rho_{\mathrm{tri}} \approx 0.0964$, and $\lambda_{\mathrm{IT}} < 1$ when $\rho < \rho_* \approx 0.0856$.

(iii) Message passing reaches q_* except in a hard window $(\lambda_{\mathrm{IT}}, \lambda_{\mathrm{alg}})$, below $\lambda = 1$ or just above it, which no polynomial-time algorithm can enter.

Part (i) comes from the ground state of a spin glass, parts (ii) and (iii) from the replica-symmetric free entropy of the posterior (Lesieur et al. 2017). How do physicists derive them, and what is proved? Part (i) completely, here; part (ii) as a theorem about a variational formula, evaluated numerically; part (iii) for message passing only.

5.2 The physics: the top eigenvalue as a ground state

5.2.1 The spherical model

For a symmetric matrix A let $H(\sigma) = \frac{1}{2}\langle \sigma, A\sigma\rangle$ on the sphere $S_N = \{\sigma \in \mathbb{R}^N : \|\sigma\|^2 = N\}$. Its maximum is $\frac{N}{2}\mu_1(A)$, at $\sigma = \pm\sqrt{N}\, v_1$, so the top eigenvalue is twice the *ground-state energy* per spin of the *spherical model* with Gibbs weight $e^{\beta H}$. A physicist computes a ground state as the zero-temperature limit of a free entropy: with $\mathrm{d}\sigma$ the uniform probability measure on S_N,

$$Z_N(\beta) = \int_{S_N} e^{\beta H(\sigma)}\, \mathrm{d}\sigma, \qquad \phi(\beta) = \lim_{N\to\infty} \frac{1}{N}\mathbb{E}\log Z_N(\beta),$$

and $\lim_{\beta\to\infty} \phi(\beta)/\beta$ should be the limit of $\mu_1(A)/2$. At fixed N the limit $\beta \to \infty$ gives $\mu_1(A)/2$ exactly, by Laplace's method; taking $N \to \infty$ first is the first uncontrolled step. With $m(\sigma) = \langle\sigma\rangle x/N$ and $\|x\|^2 = N$ (the

Rademacher prior), the Hamiltonian is $\frac{1}{2}\langle\sigma, W\sigma\rangle + \frac{N\lambda}{2}m(\sigma)^2$: the spherical Sherrington–Kirkpatrick model with a ferromagnetic coupling along x, a Curie–Weiss term after the gauge change $\sigma_i \mapsto x_i\sigma_i$. For $\lambda = 0$ the model was solved in 1976 by Kosterlitz, Thouless and Jones, who used the known spectrum of a large random matrix and found the results "identical to those obtained by the $n \to 0$ trick" (Kosterlitz et al. 1976): an early instance of the translation this chapter carries out.[3]

[3] The spherical constraint can also be imposed with a Lagrange multiplier z, which makes the integral Gaussian. The saddle-point equation, $\frac{1}{N}\operatorname{tr}(z - \beta A)^{-1} = 1$, contains the physicist's Green's function $(z - \beta A)^{-1}$, which is $-R_{\beta A}(z)$ in the notation of Definition A.14 (Exercise 5.8).

5.2.2 Zero-temperature replicas

THE GROUND STATE BY REPLICAS. *Replicate and average.* For an integer n, the moment $\mathbb{E}Z_N^n$ is an integral over replicas $\sigma^1, \dots, \sigma^n$ in S_N. For the GOE, $\mathbb{E}[\langle\sigma, W\sigma\rangle\langle\tau, W\tau\rangle] = \frac{2}{N}\langle\sigma\rangle\tau^2$ (only products of the same entry survive), so $\sum_a \langle\sigma^a, W\sigma^a\rangle$ is centered Gaussian with variance $2N\sum_{a,b} R_{ab}^2$, where $R_{ab} = \langle\sigma^a, \sigma^b\rangle/N$. With $m_a = \langle\sigma^a\rangle x/N$, averaging over W at fixed x gives

$$\mathbb{E}Z_N^n = \int \exp\Big(N\Big[\frac{\beta^2}{4}\sum_{a,b=1}^{n} R_{ab}^2 + \frac{\beta\lambda}{2}\sum_{a=1}^{n} m_a^2\Big]\Big)\prod_{a=1}^{n} \mathrm{d}\sigma^a.$$

Order parameters. Configurations with prescribed overlaps $Q = (R_{ab})$ and $m = (m_a)$ have mass $e^{N\mathcal{S}(Q,m)+o(N)}$, with $\mathcal{S}(Q, m) = \frac{1}{2}\log\det(Q - mm^\top)$, half the log-determinant of the Gram matrix of $x/\sqrt{N}, \sigma^1/\sqrt{N}, \dots, \sigma^n/\sqrt{N}$. Laplace's method gives $\frac{1}{N}\log\mathbb{E}Z_N^n \to \sup_{Q,m}\big[\frac{\beta^2}{4}\sum_{a,b} Q_{ab}^2 + \frac{\beta\lambda}{2}\sum_a m_a^2 + \mathcal{S}(Q, m)\big]$.

Replica symmetry and $n \to 0$. Take $Q_{ab} = q$ for $a \neq b$ and $m_a = m$. Then $Q - mm^\top = (1-q)I + (q-m^2)J$, with J the matrix of ones, has the eigenvalue $1 - q$ with multiplicity $n - 1$ and $1 - q + n(q - m^2)$ once. Dividing the exponent by n and letting $n \to 0$ gives

$$\phi_{\mathrm{sph}}(q, m) = \frac{\beta^2}{4}(1 - q^2) + \frac{\beta\lambda}{2}m^2 + \frac{1}{2}\log(1 - q) + \frac{q - m^2}{2(1 - q)}. \tag{5.2}$$

Saddle points. The equations $\partial_m\phi_{\mathrm{sph}} = \partial_q\phi_{\mathrm{sph}} = 0$ read $m(\beta\lambda - \frac{1}{1-q}) = 0$ and $\beta^2 q = (q - m^2)/(1 - q)^2$. Their solutions are the paramagnet $q = m = 0$; the spin glass $m = 0$, $q = 1 - 1/\beta$, for $\beta > 1$; and the ferromagnet $1 - q = 1/(\beta\lambda)$, $m^2 = (1 - \frac{1}{\beta\lambda})(1 - \frac{1}{\lambda^2})$, for $\lambda > 1$ and $\beta\lambda > 1$.

Zero temperature. As $\beta \to \infty$, $q \to 1$ with $\chi = \beta(1 - q)$ fixed: $\chi = 1$ for the spin glass and $\chi = 1/\lambda$ for the ferromagnet. Dividing (5.2) by β at the solutions gives the ground-state energies $e_{\mathrm{SG}} = 1$ and $e_{\mathrm{F}} = \frac{1}{2}(\lambda + \frac{1}{\lambda})$, and $m^2 \to 1 - 1/\lambda^2$ for the ferromagnet. For $\lambda > 1$ the ferromagnet has the larger energy, so

$$\mu_1(A) \to 2e_{\mathrm{F}} = \lambda + \frac{1}{\lambda}, \qquad \frac{\langle v_1\rangle x^2}{\|x\|^2} = m^2 \to 1 - \frac{1}{\lambda^2} = 2\frac{\partial e_{\mathrm{F}}}{\partial\lambda};$$

for $\lambda \le 1$ only the spin glass is left, and $\mu_1(A) \to 2$, $m \to 0$. ◇

Five steps are not controlled: the exchange of $N \to \infty$ and $\beta \to \infty$; the continuation of integer moments to $n \to 0$, and its exchange with $N \to \infty$

(Chapter 1); the ansatz; the choice among saddle points, although the "supremum" is a minimum in q when $n < 1$; and the identification of the overlap of the actual ground state with the saddle value of m. The mass formula for (Q, m), by contrast, is a rigorous large-deviation statement at fixed n.[4] The identity $m^2 = 2\,\partial e_{\mathrm{F}}/\partial\lambda$ is the physicist's reading of an order parameter as the derivative of the energy in its conjugate field, λ multiplying $\frac{N}{2}m^2$; it returns in the proof as the normalization of the eigenvector and as convexity. At finite temperature (5.2) gives the phase diagram (Exercise 5.5).

[4] For the sphere and a quadratic Hamiltonian the ansatz costs nothing, as Kosterlitz, Thouless and Jones found at $\lambda = 0$. With Ising spins the same computation describes the Sherrington–Kirkpatrick model of Chapter 3, where it fails at low temperature.

5.2.3 What Bayes adds: the replica-symmetric potential

Chapter 4 derives the replica-symmetric free entropy of the posterior and proves it (Theorem 4.1, Lelarge and Miolane): $\frac{1}{N}I(x;A) \to \frac{\lambda^2}{4} - \sup_{q\ge 0}\phi_{\mathrm{RS}}(\lambda, q)$, with the *replica-symmetric potential*

$$\phi_{\mathrm{RS}}(\lambda, q) = \psi(\lambda^2 q) - \frac{\lambda^2 q^2}{4}, \qquad \psi(\gamma) = \mathbb{E}\log\int e^{\sqrt{\gamma}Zy + \gamma y X - \gamma y^2/2}\,P_0(\mathrm{d}y), \tag{5.3}$$

where $X \sim P_0$ and $Z \sim \mathcal{N}(0,1)$ are independent (ψ is the ψ_{P_0} of Chapter 4); for all but countably many λ the maximizer $q_*(\lambda)$ is unique and the matrix MMSE tends to $1 - q_*(\lambda)^2$. Here ψ is the free entropy of the scalar channel $Y = \sqrt{\gamma}X + Z$, and by the I-MMSE relation (Theorem 1.12 for bounded priors, Guo et al. (2005) in general) its derivative is half the *overlap map*

$$G(\gamma) = 2\psi'(\gamma) = 1 - \mathrm{mmse}(\gamma) = \mathbb{E}\big[X\,\mathbb{E}(X \mid Y)\big]. \tag{5.4}$$

The stationary points of $\phi_{\mathrm{RS}}(\lambda,\cdot)$ solve $q = G(\lambda^2 q)$, the fixed-point equation of state evolution (Theorem 6.4): the iteration $q_{t+1} = G(\lambda^2 q_t)$ is the physicist's picture of an algorithm climbing the potential. A stationary point with $q > 0$ is determined by $\gamma = \lambda^2 q$, through $q = G(\gamma)$ and $\lambda^2 = L(\gamma) := \gamma/G(\gamma)$, so all of them lie on the graph of one function. Since $\partial_q^2\phi_{\mathrm{RS}} = \frac{\lambda^2}{2}(\lambda^2 G'(\lambda^2 q) - 1)$ and $L' = (G - \gamma G')/G^2$, a stationary point is a local maximum of $\phi_{\mathrm{RS}}(\lambda,\cdot)$, and a stable fixed point of the iteration, exactly when L increases at γ. As $\lambda^2 q^2 = \gamma G(\gamma)$ and $\psi(\gamma) = \frac{1}{2}\int_0^\gamma G$, its free entropy relative to $q = 0$ is

$$\phi_{\mathrm{RS}}\big(\lambda, G(\gamma)\big) - \phi_{\mathrm{RS}}(\lambda, 0) = \frac{1}{2}\int_0^\gamma G(t)\,\mathrm{d}t - \frac{\gamma\,G(\gamma)}{4} \qquad \big(\lambda^2 = L(\gamma)\big). \tag{5.5}$$

For the Gaussian prior $G(\gamma) = \gamma/(1+\gamma)$ and $L(\gamma) = 1 + \gamma$, so for $\lambda > 1$ the only stationary point with $q > 0$ is $q = 1 - 1/\lambda^2$, the overlap of PCA. Near $q = 0$ the prior enters to second order only through its third moment.

COMPUTATION 5.1 · THE OVERLAP MAP NEAR $\gamma = 0$

1 *Posterior weights.* Let P_0 have bounded support and moments $m_1 = 0$, $m_2 = 1$, m_3, m_4, and $s = \sqrt{\gamma}$. Given $Y = y$ the posterior has density $e^{syx - s^2x^2/2} = \sum_{k\ge 0}\mathrm{He}_k(y)(sx)^k/k!$ with respect to P_0, up to normalization; here $\mathrm{He}_1 = y$, $\mathrm{He}_2 = y^2 - 1$, $\mathrm{He}_3 = y^3 - 3y$.

2 *The posterior mean.* The numerator is $sy + \frac{s^2 m_3}{2}\mathrm{He}_2(y) + \frac{s^3 m_4}{6}\mathrm{He}_3(y) + O(s^4)$, the denominator $1 + \frac{s^2}{2}\mathrm{He}_2(y) + O(s^3)$, so

$$\mathbb{E}(X \mid Y = y) = sy + \frac{s^2 m_3}{2}\mathrm{He}_2(y) + s^3\Big(\frac{m_4}{6}\mathrm{He}_3(y) - \frac{y^3 - y}{2}\Big) + O(s^4).$$

3 *Average over the data.* With $Y = sX + Z$: $\mathbb{E}[X \cdot sY] = s^2$, $\mathbb{E}[X\,\mathrm{He}_2(Y)] = s^2 m_3$, and $\mathbb{E}[Xh(Y)] = s\,\mathbb{E}h'(Z) + O(s^2)$, which is 0 for $h = \mathrm{He}_3$ and s for $h = (y^3 - y)/2$. As G is a function of s^2 (replace (s, Z) by $(-s, -Z)$),

$$G(\gamma) = \gamma - \Big(1 - \frac{m_3^2}{2}\Big)\gamma^2 + O(\gamma^3).$$

4 *The bifurcation at $\lambda = 1$.* Hence $L(\gamma) = 1 + (1 - \frac{m_3^2}{2})\gamma + O(\gamma^2)$. If $m_3^2 < 2$, the stationary points born at $q = 0$ when $\lambda = 1$ exist for $\lambda > 1$: the branch starts continuously. If $m_3^2 > 2$, they exist for $\lambda < 1$: the transition is first order and $\lambda_{\mathrm{IT}} < 1$. This is the criterion $\langle x_0^3\rangle^2 > 2\langle x_0^2\rangle^3$ of Lesieur et al. (2017). Symmetric priors have $m_3 = 0$, so for them a first-order transition can only come from another branch.

5 *The next order.* For symmetric priors $G(\gamma) = \gamma - \gamma^2 + c_3\gamma^3 + O(\gamma^4)$ with $c_3 = \frac{m_4^2}{6} - m_4 + \frac{5}{2}$ (Exercise 5.7), which is $(1 - 6\rho + 15\rho^2)/(6\rho^2)$ for the sparse Rademacher prior. Truncated, $L(\gamma) \approx 1 + \gamma - (c_3 - 1)\gamma^2$ peaks at $1 + \frac{1}{4(c_3-1)}$, so the branch born at $\lambda = 1$ turns back at $\lambda_{\mathrm{alg}} \approx 1 + \frac{3\rho^2}{4(1-3\rho)^2}$: 1.00260 at $\rho = 0.05$ and 1.000340 at $\rho = 0.02$, against 1.002700 and 1.000341 (Table 5.1).

5.2.4 Sparse priors: three branches and a hard phase

For the sparse Rademacher prior with small ρ the curve L rises from $L(0+) = 1$ to a local maximum λ_{alg}^2, falls to a local minimum λ_{dyn}^2, and rises again, so a level λ^2 meets it up to three times (Figure 5.3). The low branch, born at $q = 0$ when $\lambda = 1$, ends at the *spinodal* λ_{alg}, where it merges with the unstable middle branch; the high branch exists above λ_{dyn}. The Bayes overlap lies on the branch of larger free entropy (5.5), the competitor being $q = 0$ when $\lambda \le 1$, and jumps where the two are equal, at the *information-theoretic threshold* λ_{IT} (*Maxwell's construction*).[5] This is the phase diagram of Lesieur et al. (2017) (Figure 5.1).

The curve stops being monotone below the tricritical density $\rho_{\mathrm{tri}} = 0.0964$ of Lesieur, Krzakala and Zdeborová (we recompute 0.096416), where the three thresholds merge at $\lambda = 1.0200$. The information threshold crosses the spectral one at the smaller density $\rho_* = 0.085593$ (λ_{dyn} at 0.086914): for $\rho_* < \rho < \rho_{\mathrm{tri}}$ the overlap becomes positive continuously at $\lambda = 1$ and jumps above it, while for $\rho < \rho_*$ it jumps from 0 below $\lambda = 1$. The value $\rho \approx 0.09$ that Krzakala et al. (2016) found numerically for the separation

[5] The name comes from Maxwell's equal-area rule for the liquid–gas transition of van der Waals. The analogy is close: the low and high branches are the gas and the liquid, λ_{alg} and λ_{dyn} the spinodals, λ_{IT} the coexistence point.

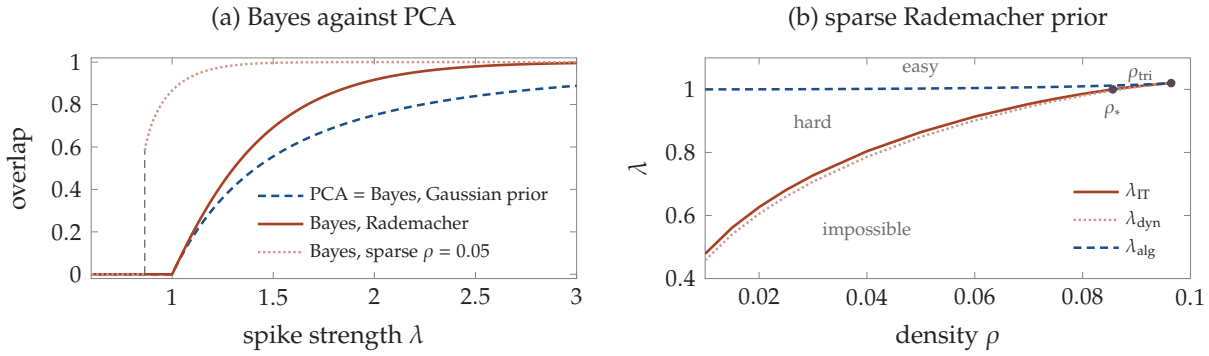


Figure 5.1. (a) The squared cosine with x achieved by PCA, $(1 - 1/\lambda^2)_+$ (Theorem 5.1), which is also the Bayes overlap for the Gaussian prior (Proposition 5.6), and the Bayes overlap $q_*(\lambda)$ for the Rademacher prior and for the sparse Rademacher prior with $\rho = 0.05$, which jumps from 0 to 0.583 at $\lambda_{\mathrm{IT}} = 0.8645$ (gray). (b) Sparse Rademacher prior: weak recovery is impossible below λ_{IT}, message passing is suboptimal between λ_{IT} and λ_{alg} and optimal above; λ_{alg} exceeds the spectral threshold $\lambda = 1$ by at most 0.02, and the dots mark ρ_* and ρ_{tri}. All curves evaluate the potential (5.3), a theorem (Theorem 4.1). Script: `figures/scripts/05-spiked-phase.py`.

of the information threshold from their "algorithmic" one, which is the spectral threshold $\lambda = 1$ (not our λ_{alg}), and that Perry et al. (2018) quote as the critical density for recovery below the spectral threshold, is ρ_*; the first-order transition begins at ρ_{tri}.

Between λ_{IT} and λ_{alg} physicists see a *hard phase*: the posterior sits on the high branch, but message passing started without information climbs ϕ_{RS} from $q = 0$ and stays on the low branch, or at $q = 0$ when $\lambda < 1$. Lesieur et al. (2017) conjecture that no polynomial-time algorithm beats message passing there, and remark that a proof in full generality would imply P ≠ NP. For $\rho < \rho_*$ the window contains $(\lambda_{\mathrm{IT}}, 1]$, where the top eigenvector sees nothing and the posterior sees the spike.

5.3 The stone

PHYSICS · *the zero-temperature replica computation*

1 · The quantity

The top eigenvalue is twice a ground-state energy, the limit of $2\phi_N(\beta)/\beta$ as $N \to \infty$ and then $\beta \to \infty$, in this order and without control.

PROOF · *the resolvent and a rank-one perturbation*

A maximum, $\mu_1(A) = \max_{\|v\|=1} \langle v, Av \rangle$ (Courant–Fischer): no temperature, no limit in β.

2 · Average over the noise

Replicate and average over W; the Gaussian average gives $e^{N\beta^2 \sum_{a,b} R_{ab}^2/4}$, a function of the overlaps alone.	No average: condition on W and use two spectral facts, the semicircle law and $\|W\| \to 2$ (Theorem A.15); every statement holds almost surely.

3 · Order parameters

The overlaps m and q; in the ferromagnetic phase the saddle point gives the susceptibility $\chi = \beta(1-q) \to 1/\lambda$.	The secular equation, the vanishing of the Sherman–Morrison denominator: $z > \mu_1(W)$ is an eigenvalue of $W + \theta uu^\top$ iff $1 + \theta\langle u, R_W(z)u\rangle = 0$ (Lemma 5.3). One unknown, the outlier, replaces q and m, and $\chi = -s(\lambda + 1/\lambda)$.

4 · Self-averaging

The average over W makes all directions alike; the saddle point is the typical value.	Isotropy: for u independent of W, $\langle u, R_W(z)u\rangle \to s(z)$, by orthogonal invariance and a fourth moment (Lemma 5.4). No replica counterpart: the average did this work.

5 · Which saddle

Keep the solution of largest energy: the ferromagnet when $\lambda > 1$, with $e = \frac{1}{2}(\lambda + 1/\lambda)$.	Does $s(z) = -1/\theta$ have a root above the edge? As s increases from $s(2) = -1$, iff $\theta > 1$, and the root is $\theta + 1/\theta$ (Lemma 5.2).

6 · The order parameter

$m^2 = 2\,\partial e/\partial\lambda$: the overlap is the derivative of the energy in its conjugate field.	Normalization: by (A.7), $\langle u, v_1\rangle^2 = 1/(\theta^2\langle u, R_W(\mu_1)^2 u\rangle) \to 1/(\theta^2 s'(\theta + 1/\theta)) = 1 - \theta^{-2}$, the derivative of the eigenvalue limit; for $\theta \le 1$, the supporting lines of the convex function $t \mapsto \mu_1(W + tuu^\top)$ (Lemma 5.5).

7 · Beyond the top eigenvector

The potential $\phi_{\mathrm{RS}}(\lambda, q)$; its S-shaped branches for sparse priors; message passing follows its branch, and nothing efficient does better.	The potential is a theorem (Theorem 4.1), and so is the branch of message passing (Montanari and Venkataramanan 2021); the claim about all efficient algorithms is a conjecture.

The proof follows the derivation closely. The saddle point is the secular equation in another form: the zero-temperature susceptibility, $\chi = 1/\lambda$ for the ferromagnet and 1 for the spin glass, is minus the Stieltjes transform at the top eigenvalue, $-s(\lambda + 1/\lambda)$ or $-s(2)$. The proof replaces the average over W by two spectral facts and adds isotropy, which the replicas never need. The overlap as a derivative of the energy becomes the normalization of the eigenvector, whose limit is the derivative of the outlier (Exercise 5.2),

and, for $\theta \le 1$, the supporting line of a convex function, the device by which Lelarge and Miolane (2019) obtain the MMSE from the mutual information. The routes part at the Bayes estimator, whose potential is proved by interpolation (Chapter 4), not by anything resembling replicas.

5.4 The proof: the resolvent and a rank-one perturbation

Throughout this section $W = W_N \sim \mathrm{GOE}(N)$ and $u = u_N$ is a unit vector in $\mathbb{R}^N$, deterministic or random, independent of W_N. For $\theta \ge 0$ put $A_N(\theta) = W_N + \theta u_N u_N^{\top}$. We use three facts about the GOE: (S1) almost surely, the empirical spectral distribution μ_N of W_N converges weakly to the *semicircle law* $\sigma_{\mathrm{sc}}(\mathrm{d}x) = \frac{1}{2\pi}\sqrt{4-x^2}\,\mathrm{d}x$ on $[-2,2]$; (S2) almost surely, $\mu_1(W_N) \to 2$ and $\|W_N\| \to 2$ (both in Theorem A.15); (S3) the law of W_N, with density proportional to $e^{-N\,\mathrm{tr}(W^2)/4}$, is invariant under $W \mapsto OWO^{\top}$ for every orthogonal O (Definition A.12). We use the resolvent $R_W(z) = (W - z)^{-1}$ and the *Stieltjes transform* of Definition A.14; for the semicircle law and real $z > 2$, (A.8) gives

$$s(z) = \int \frac{\sigma_{\mathrm{sc}}(\mathrm{d}x)}{x - z} = \frac{-z + \sqrt{z^2-4}}{2}, \qquad s'(z) = \int \frac{\sigma_{\mathrm{sc}}(\mathrm{d}x)}{(x-z)^2}. \tag{5.6}$$

Theorem 5.1 (The BBP transition for spiked Wigner). *Fix $\theta \ge 0$; let $\Lambda(\theta) = \theta + 1/\theta$ for $\theta > 1$, $\Lambda(\theta) = 2$ for $\theta \le 1$. Almost surely, as $N \to \infty$:*

(i) $\mu_1(A_N(\theta)) \to \Lambda(\theta)$;

(ii) $\mu_2(A_N(\theta)) \to 2$;

(iii) $\langle u_N, v_N\rangle^2 \to 1 - \theta^{-2}$ *if* $\theta > 1$, *and* $\to 0$ *if* $\theta \le 1$, *for every choice of unit eigenvectors v_N of $A_N(\theta)$ with eigenvalue $\mu_1(A_N(\theta))$.*

The same holds for the spiked Wigner matrix (5.1), with $\theta = \lambda$ and $u_N = x/\|x\|$, when x consists of the first N terms of an i.i.d. sequence with law P_0, $\mathbb{E}X^2 = 1$, independent of $(W_N)_{N\ge1}$.

The *BBP transition* is the change at $\theta = 1$: one eigenvalue leaves the bulk, and its eigenvector acquires a macroscopic overlap with the spike.[6] The proof rests on four lemmas, of which only the third is probabilistic.

[6] The initials are those of Baik, Ben Arous and Péché, who proved the transition for sample covariance matrices (Baik et al. 2005).

Lemma 5.2 (The Stieltjes transform above the edge). *On $(2,\infty)$, s is continuous and strictly increasing, from -1 at 2^+ to 0 at ∞, and $s(z) + 1/s(z) = -z$. For $\theta > 1$, $s(z) = -1/\theta$ has the unique solution $z = \theta + 1/\theta$ in $(2,\infty)$, where $s'(z) = 1/(\theta^2-1)$; for $0 < \theta \le 1$ it has none.*

Proof. Since $s(z) = -2/(z+\sqrt{z^2-4})$, it is continuous and strictly increasing with the stated limits, and $1/s(z) = -(z+\sqrt{z^2-4})/2$, so $s + 1/s = -z$. If $s(z) = -1/\theta$ with $z > 2$, then $z = -s - 1/s = \theta + 1/\theta$, and $s(z) > -1$ forces $\theta > 1$. Conversely, for $\theta > 1$ the point $z = \theta + 1/\theta$ exceeds 2, $\sqrt{z^2-4} = \theta - 1/\theta$, and $s(z) = -1/\theta$. Differentiating $s^2 + zs + 1 = 0$ gives $s' = -s/(2s+z)$, which there is $\frac{1}{\theta}/(\theta - \frac{1}{\theta}) = 1/(\theta^2-1)$. ∎

Lemma 5.3 (The secular equation). *Let W be a real symmetric $N \times N$ matrix, u a unit vector, $\theta > 0$ and $A = W + \theta uu^\top$. For $z > \mu_1(W)$ let $r(z) = \langle u, R_W(z)u\rangle$, with derivative $r'(z) = \langle u, R_W(z)^2 u\rangle$.*

(i) A real $z > \mu_1(W)$ is an eigenvalue of A if and only if $1 + \theta r(z) = 0$, and then every unit eigenvector v for z satisfies $\langle u\rangle v^2 = 1/(\theta^2 r'(z))$.

(ii) On $(\mu_1(W), \infty)$, r is continuous and strictly increasing and r' is decreasing. If $1 + \theta r = 0$ has a root $z_0 > \mu_1(W)$, then $\mu_1(A) = z_0$; otherwise $\mu_1(A) = \mu_1(W)$.

(iii) $\mu_1(W) \le \mu_1(A) \le \mu_1(W) + \theta$ and $\mu_2(W) \le \mu_2(A) \le \mu_1(W)$.

Proof. The plan is to take the eigenvalue equation from the rank-one formula, the rest from the spectral decomposition. Part (i) is Definition A.14 (v), with (A.7). For (ii), write $W = \sum_k \mu_k \phi_k \phi_k^\top$ and $c_k = \langle u, \phi_k\rangle^2$, so that $r(z) = \sum_k c_k/(\mu_k - z)$ and $r'(z) = \sum_k c_k/(\mu_k - z)^2$, with $c_k \ge 0$ and $\sum_k c_k = 1$; on $(\mu_1(W), \infty)$ each term of r increases, strictly if $c_k > 0$, and each term of r' decreases. Hence $1 + \theta r = 0$ has at most one root above $\mu_1(W)$. By (iii), $\mu_1(A) \ge \mu_1(W)$; if $\mu_1(A) > \mu_1(W)$ it is a root by (i), and if a root z_0 exists it is an eigenvalue, so $\mu_1(A) \ge z_0 > \mu_1(W)$ and $\mu_1(A) = z_0$. For (iii), $0 \le \theta uu^\top \le \theta I$, so Courant–Fischer gives $\mu_k(W) \le \mu_k(A) \le \mu_k(W) + \theta$ for every k (Weyl's inequalities), and $\mu_2(A) \le \max_{v \perp u, \|v\|=1} \langle v, Av\rangle = \max_{v \perp u, \|v\|=1} \langle v, Wv\rangle \le \mu_1(W)$. ∎

The *secular equation* $1 + \theta r(z) = 0$ reduces the outlier to one scalar equation, but r depends on all of W. Along a direction independent of W, the resolvent sees only the spectrum.[7]

Lemma 5.4 (The resolvent along an independent direction). *Let $W_N \sim \mathrm{GOE}(N)$ and let the unit vectors u_N be independent of W_N. For each $z > 2$, almost surely $\langle u_N, R_{W_N}(z)u_N\rangle \to s(z)$ and $\langle u_N, R_{W_N}(z)^2 u_N\rangle \to s'(z)$; the left sides are defined once $\|W_N\| < z$, true for all large N by (S2).*

Proof. The plan is to compare each quadratic form with a normalized trace, which (S1) controls, after replacing u_N by a uniformly random direction. Fix $z > 2$ and $p \in \{1, 2\}$, let $\delta = (z-2)/2$, $K = \delta^{-p}$ and $E_N = \{\|W_N\| \le 2 + \delta\}$. On E_N the matrix $M = R_{W_N}(z)^p$ exists and $\|M\| \le K$.

The trace. On E_N, $\frac{1}{N}\operatorname{tr} M = \int f \,\mathrm{d}\mu_N$ with $f(x) = (\min\{x, 2+\delta\} - z)^{-p}$, which is bounded and continuous and equals $(x - z)^{-p}$ on $[-2, 2]$. By (S1) and (5.6), $\int f \,\mathrm{d}\mu_N$ tends almost surely to $s(z)$ if $p = 1$ and to $s'(z)$ if $p = 2$; and by (S2) E_N holds for all large N.

A uniform direction. Let $\xi \sim \mathcal{N}(0, I_N)$ be independent of (W_N, u_N), $w = \xi/\|\xi\|$, and O the Householder reflection exchanging w and u_N ($O = I - 2(w-u_N)(w-u_N)^\top/\|w-u_N\|^2$, or I if $w = u_N$), an orthogonal symmetric function of (u_N, w). Given (u_N, w), the matrix $W' = OW_N O$ is GOE by (S3), so W' is GOE and independent of (u_N, w), and (W', u_N) has the law of (W_N, u_N). As $\langle u_N, R_{W'}(z)^p u_N\rangle = \langle w, Mw\rangle$, $\|W'\| = \|W_N\|$ and $\operatorname{tr} R_{W'}(z)^p = \operatorname{tr} M$, for every $\varepsilon > 0$

$$\mathbb{P}\big(E_N,\ |\langle u_N, Mu_N\rangle - \tfrac{1}{N}\operatorname{tr} M| > \varepsilon\big) = \mathbb{P}\big(E_N,\ |\langle w, Mw\rangle - \tfrac{1}{N}\operatorname{tr} M| > \varepsilon\big).$$

[7] The proof rotates u_N into a uniformly random direction. A Haar-distributed rotation would serve as well; a reflection that exchanges two vectors avoids the Haar measure on the orthogonal group.

A fourth moment. Condition on $W_N \in E_N$, so that M is a fixed symmetric matrix with $\|M\| \le K$, independent of ξ. In an orthonormal eigenbasis of M, with eigenvalues d_i, the coordinates η_i of ξ are i.i.d. standard normal, and $\langle \xi, M\xi\rangle - \operatorname{tr} M = \sum_i d_i(\eta_i^2 - 1)$ is a sum of independent centered terms. With $\mathbb{E}(\eta^2-1)^2 = 2$ and $\mathbb{E}(\eta^2-1)^4 = 60$,

$$\mathbb{E}\big[(\langle \xi, M\xi\rangle - \operatorname{tr} M)^4 \,\big|\, W_N\big] = 60\sum_i d_i^4 + 12\sum_{i\neq j} d_i^2 d_j^2 \le 72N^2K^4.$$

By Markov's inequality, $a = \frac{1}{N}(\langle \xi, M\xi\rangle - \operatorname{tr} M)$ satisfies $\mathbb{P}(|a| > \eta \mid W_N) \le 72K^4/(\eta^4N^2)$, and $b = \|\xi\|^2/N - 1$ (the case $M = I$) satisfies $\mathbb{P}(|b| > \eta) \le 72/(\eta^4N^2)$. Since $\langle w, Mw\rangle - \frac{1}{N}\operatorname{tr} M = (a - b\cdot\frac{1}{N}\operatorname{tr} M)/(1+b)$ and $|\frac{1}{N}\operatorname{tr} M| \le K$, on $\{|a| \le \eta, |b| \le \eta\}$ with $\eta \le \frac{1}{2}$ this difference is at most $2\eta(1+K)$. With $\eta = \min\{\frac{1}{2}, \varepsilon/(2+2K)\}$,

$$\mathbb{P}\big(E_N,\ |\langle w, Mw\rangle - \tfrac{1}{N}\operatorname{tr} M| > \varepsilon\big) \le \frac{72(K^4+1)}{\eta^4N^2},$$

which is summable in N. By the Borel–Cantelli lemma and the equality above, almost surely, for all large N, either E_N fails or $|\langle u_N, Mu_N\rangle - \frac{1}{N}\operatorname{tr} M| \le \varepsilon$; and E_N holds for all large N. Taking $\varepsilon = 1/k$ for all $k \ge 1$ and adding the trace part proves the lemma. ∎

Lemma 5.5 (Supporting lines). *Let W be real symmetric, u a unit vector, and $F(t) = \mu_1(W + tuu^\top)$ for $t \in \mathbb{R}$. Then F is 1-Lipschitz, and for every t, every unit eigenvector v of $W + tuu^\top$ with eigenvalue $F(t)$, and every $h > 0$,*

$$\frac{F(t) - F(t-h)}{h} \le \langle u\rangle v^2 \le \frac{F(t+h) - F(t)}{h}.$$

Proof. For every $\tau \in \mathbb{R}$, Courant–Fischer gives $F(\tau) \ge \langle v, (W + \tau uu^\top)v\rangle = F(t) + (\tau - t)\langle u\rangle v^2$; take $\tau = t \pm h$. The Lipschitz bound is Weyl's inequality, as $\|uu^\top\| = 1$. ∎

So F is convex and $\langle u\rangle v^2$ is a subgradient: the Hellmann–Feynman formula $\mathrm{d}\mu_1/\,\mathrm{d}t = \langle u, v_1\rangle^2$ without differentiability.[8]

[8] In quantum mechanics the derivative of a simple eigenvalue of $H(t)$ is $\langle \psi, H'(t)\psi\rangle$ for the normalized eigenvector ψ; here $H'(t) = uu^\top$.

Proof of Theorem 5.1. The plan is to locate the outlier by the secular equation at finitely many points, bound the second eigenvalue by interlacing, read the overlap above the threshold off (A.7), and bound it for $\theta \le 1$ by convexity. Write $r_N(z) = \langle u_N, R_{W_N}(z)u_N\rangle$, with derivative r_N'. For $\theta = 0$, part (i) is (S2), and parts (ii) and (iii) are covered by Steps 3 and 5.

Step 1: the outlier, $\theta > 1$. Let $z^* = \theta + 1/\theta$ and fix $\varepsilon \in (0, z^* - 2)$. By Lemma 5.2, $s(z^* - \varepsilon) < -1/\theta < s(z^* + \varepsilon)$. By (S2) and Lemma 5.4 at $z^* \pm \varepsilon$, almost surely, for all large N, $\|W_N\| < z^* - \varepsilon$ and $1 + \theta r_N(z^* - \varepsilon) < 0 < 1 + \theta r_N(z^* + \varepsilon)$. As r_N is continuous and increasing on $(\mu_1(W_N), \infty)$, $1 + \theta r_N = 0$ has a root in $(z^* - \varepsilon, z^* + \varepsilon)$, which is $\mu_1(A_N(\theta))$ by Lemma 5.3(ii).

Step 2: no outlier, $0 < \theta \le 1$. By Lemma 5.3(iii) and (S2), $\liminf \mu_1(A_N(\theta)) \ge 2$. Fix $\varepsilon > 0$. As $s(2+\varepsilon) > -1 \ge -1/\theta$, $1 + \theta s(2+\varepsilon) > 0$, so by (S2) and Lemma 5.4, almost surely, for all large N, $\mu_1(W_N) < 2 + \varepsilon$ and

$1 + \theta r_N(2+\varepsilon) > 0$. As r_N increases, every root of $1 + \theta r_N = 0$ lies below $2+\varepsilon$, and by Lemma 5.3(ii), $\mu_1(A_N(\theta)) < 2 + \varepsilon$.

Step 3: the second eigenvalue. By Lemma 5.3(iii), $\mu_2(W_N) \le \mu_2(A_N(\theta)) \le \mu_1(W_N) \to 2$. For $\varepsilon > 0$, by (S1) and the portmanteau theorem the fraction of eigenvalues of W_N above $2-\varepsilon$ has lim inf at least $\sigma_{\mathrm{sc}}((2-\varepsilon, 2]) > 0$, so for all large N at least two of them exceed $2-\varepsilon$.

Step 4: the overlap, $\theta > 1$. By Step 1, almost surely, for all large N, $\mu_1(W_N) < z^* - \varepsilon < \mu_1(A_N(\theta)) < z^* + \varepsilon$, so Lemma 5.3(i) gives $\langle u_N, v_N\rangle^2 = 1/(\theta^2 r_N'(\mu_1(A_N(\theta))))$. As r_N' decreases, this lies between $1/(\theta^2 r_N'(z^* - \varepsilon))$ and $1/(\theta^2 r_N'(z^* + \varepsilon))$, which tend to $1/(\theta^2 s'(z^* \mp \varepsilon))$ by Lemma 5.4. Letting $\varepsilon \to 0$ along a sequence, $\langle u_N, v_N\rangle^2 \to 1/(\theta^2 s'(z^*)) = 1 - \theta^{-2}$ by Lemma 5.2.

Step 5: the overlap, $\theta \le 1$. Part (i), proved in Steps 1–2 and by (S2), holds almost surely at once for the countably many $t \in \{\theta\} \cup \{\theta + 1/k : k \ge 1\}$. With $F_N(t) = \mu_1(W_N + t u_N u_N^{\top})$ and $h = 1/k$, Lemma 5.5 gives $\limsup_N \langle u_N, v_N\rangle^2 \le (\Lambda(\theta+h) - \Lambda(\theta))/h$. As $\Lambda(\theta+h) - \Lambda(\theta) = (\theta + h - 1)_+^2/(\theta + h)$, which is at most h^2 since $\theta \le 1$, the overlap tends to 0.

The spiked Wigner matrix. Here $A = W_N + \theta_N u_N u_N^{\top}$ with $u_N = x/\|x\|$ (a fixed unit vector if $x = 0$) independent of W_N, and $\theta_N = \lambda\|x\|^2/N \to \lambda$ almost surely by the strong law ($x \neq 0$ for all large N, as $\mathbb{P}(X = 0) < 1$). Weyl's inequality, $|\mu_k(W_N + \theta_N u_N u_N^{\top}) - \mu_k(W_N + \lambda u_N u_N^{\top})| \le |\theta_N - \lambda|$, transfers (i) and (ii). For (iii) with $\lambda > 1$, Step 4 applies with θ_N in the overlap formula and μ_1 located by (i); for $\lambda \le 1$, the bound of Lemma 5.5 at $t = \theta_N$ differs from that at λ by at most $2|\theta_N - \lambda|/h$, as F_N is 1-Lipschitz. ∎

WHAT THE GENERAL THEOREMS ADD. The proof used the GOE three times, the last to make u_N look uniformly random. For Wigner matrices whose entries have a symmetric law satisfying a Poincaré inequality, Capitaine et al. (2009) prove the same eigenvalue limits and show that the fluctuations of the outlier are not universal for a localized spike; for a delocalized spike they are Gaussian of order $N^{-1/2}$ above the threshold and Tracy–Widom of order $N^{-2/3}$ below (Féral and Péché 2007; Péché 2006). Benaych-Georges and Nadakuditi (2011) treat a general spectral law μ with edge b, one summand being orthogonally invariant: the outlier ζ solves $s_\mu(\zeta) = -1/\theta$ when $\theta > -1/s_\mu(b^+)$, with squared overlap $1/(\theta^2 s_\mu'(\zeta))$ (their Theorems 2.1–2.3; Exercise 5.2). For sample covariance matrices the threshold is $1 + \sqrt{N/M}$, for M samples in dimension N (Baik et al. 2005, Theorem 1.1).

5.5 The proof: what Bayes adds, and what message passing reaches

Proposition 5.6 (Bayes against PCA). *Let P_0 satisfy $\mathbb{E}X^2 = 1$. For every $\lambda > 1$, every maximizer q_* of $\phi_{\mathrm{RS}}(\lambda, \cdot)$ on $[0,\infty)$ satisfies $q_* \ge 1 - \lambda^{-2}$, with equality only if $P_0 = \mathcal{N}(0,1)$. For $P_0 = \mathcal{N}(0,1)$ the maximizer is unique and equals $(1-\lambda^{-2})_+$ for every $\lambda > 0$.*

PROOF. The plan is to compare the overlap map with that of the best linear estimator. *Step 1.* The scalar channel has mutual information

$I(\gamma) = \frac{\gamma}{2} - \psi(\gamma) \in [0, \frac{\gamma}{2}]$, so ψ is continuous on $[0,\infty)$ with $\psi(0) = 0$, and $\psi' = G/2$ on $(0,\infty)$ by the I-MMSE relation (Theorem 1.12 for bounded support, Guo et al. (2005) for finite variance). So $\phi_{\mathrm{RS}}(\lambda,\cdot)$ is continuous, $\phi_{\mathrm{RS}}(\lambda,q) \le \lambda^2(q/2 - q^2/4) \to -\infty$, maximizers exist, and $\partial_q \phi_{\mathrm{RS}}(\lambda,q) = \frac{\lambda^2}{2}(G(\lambda^2 q) - q)$ for $q > 0$.

Step 2. Whatever the prior, the estimator cY with $c = \sqrt{\gamma}/(1+\gamma)$ has mean-square error $1 - 2c\sqrt{\gamma} + c^2(1+\gamma) = 1/(1+\gamma)$, so $\mathrm{mmse}(\gamma) \le 1/(1+\gamma)$ and $G(\gamma) \ge \gamma/(1+\gamma)$. With $q_G = 1 - \lambda^{-2}$, for $0 < q < q_G$,

$$\partial_q \phi_{\mathrm{RS}}(\lambda,q) \ge \frac{\lambda^2}{2}\Big(\frac{\lambda^2 q}{1+\lambda^2 q} - q\Big) = \frac{\lambda^2 q\,(\lambda^2 - 1 - \lambda^2 q)}{2(1+\lambda^2 q)} > 0,$$

so $\phi_{\mathrm{RS}}(\lambda,\cdot)$ is strictly increasing on $[0,q_G]$ and no maximizer lies in $[0,q_G)$.

Step 3. At q_G, where $\lambda^2 q_G = \gamma_G := \lambda^2 - 1 > 0$, $\partial_q \phi_{\mathrm{RS}}(\lambda,q_G) = \frac{\lambda^2}{2}(\frac{1}{1+\gamma_G} - \mathrm{mmse}(\gamma_G)) \ge 0$. If q_G is a maximizer this vanishes, so cY is the conditional mean $\mathbb{E}(X \mid Y)$ at $\gamma = \gamma_G$, and $\mathbb{E}[(X - cY)e^{itY}] = 0$ for all real t. With ω the characteristic function of X, differentiable as $\mathbb{E}|X| < \infty$, we have $\mathbb{E}[Xe^{itY}] = -i\omega'(\sqrt{\gamma}t)e^{-t^2/2}$ and $\mathbb{E}[Ye^{itY}] = -i\frac{\mathrm{d}}{\mathrm{d}t}(\omega(\sqrt{\gamma}t)e^{-t^2/2})$; substituting $s = \sqrt{\gamma}t$ gives $(1 - c\sqrt{\gamma})\omega'(s) = -\frac{c}{\sqrt{\gamma}} s\omega(s)$. As $1 - c\sqrt{\gamma} = c/\sqrt{\gamma} = 1/(1+\gamma)$, $\omega' = -s\omega$ with $\omega(0) = 1$, so $\omega(s) = e^{-s^2/2}$ and $P_0 = \mathcal{N}(0,1)$.

Step 4. For $P_0 = \mathcal{N}(0,1)$, $G(\gamma) = \gamma/(1+\gamma)$, so Step 2 holds with equality and $\partial_q \phi_{\mathrm{RS}}$ has the sign of $\lambda^2 - 1 - \lambda^2 q$ on $(0,\infty)$. ∎

With Theorem 4.1 this is a statement about estimators.[9] Where q_* is unique, the matrix MMSE $1 - q_*^2$ is at most $1 - (1-\lambda^{-2})^2$, the limit of the error of the best multiple of $v_1 v_1^\top$, since $\|xx^\top/N - cv_1 v_1^\top\|_F^2 \to 1 - 2c(1-\lambda^{-2}) + c^2$ by Theorem 5.1. For the Rademacher prior the gain is large: at $\lambda = 1.5$, $q_* = 0.6923$ against 0.5556, and the matrix MMSE is 0.5207 against 0.6914 (Figure 5.1). Yet $\lambda = 1$ stays the threshold: for the Rademacher prior the MMSE tends to 1 exactly when $\lambda \le 1$ (Theorem 4.2; Deshpande et al. (2016, Theorem 1.4 of arXiv v1) prove it for the two-group block model), no test detects the spike below $\lambda = 1$ (Perry et al. 2018), and for i.i.d. priors detection and estimation share their threshold (El Alaoui et al. 2020).

[9] For a symmetric prior the posterior is invariant under $x \mapsto -x$, so the posterior mean of x is 0; the natural targets are $xx^\top$ and the direction of x up to sign.

Whether an algorithm reaches q_* is the question of Chapter 6. For spiked matrices the answer is the following theorem, in this chapter's notation.

Theorem 5.7 (Montanari and Venkataramanan 2021, Theorem 2 and Corollary 2.3). *Let $x = x(N)$ satisfy $\|x\|^2/N \to 1$, with empirical distributions converging weakly to P_0, $\mathbb{E}X^2 = 1$. Let $\lambda > 1$, and assume that $y \mapsto \mathbb{E}(X \mid \sqrt{\gamma}X + Z = y)$ is Lipschitz for every $\gamma \in (0,\lambda^2]$, as when P_0 has bounded support. Let γ_{ALG} be the smallest positive solution of $\gamma = \lambda^2 G(\gamma)$, and $q_{\mathrm{ALG}} = \gamma_{\mathrm{ALG}}/\lambda^2$. Then Bayes-optimal approximate message passing started from the top eigenvector of A produces estimates $\hat{x}^t$ with*

$$\lim_{t\to\infty}\lim_{N\to\infty} \frac{\langle \hat{x}^t, x\rangle^2}{\|\hat{x}^t\|^2\|x\|^2} = q_{\mathrm{ALG}}(\lambda) \qquad \textit{almost surely.}$$

Moreover, $q_{\mathrm{ALG}}(\lambda) = q_*(\lambda)$ if and only if the global maximizer of $\phi_{\mathrm{RS}}(\lambda,\cdot)$ on $(0,\infty)$ is its smallest positive stationary point.

ρ	λ_{IT}	λ_{dyn}	λ_{alg}	jump of q_*	Δ/ρ^2 at λ_{IT}
0.0909	1.011 198	1.009 590	1.015 268	$0.032 \to 0.290$	0.97798
0.0863	1.001 634	0.998 398	1.012 570	$0.003 \to 0.344$	0.99674
0.0800	0.985 607	0.980 175	1.009 744	$0 \to 0.400$	1.02942
0.0600	0.913 717	0.901 649	1.004 285	$0 \to 0.530$	1.19778
0.0500	0.864 471	0.849 363	1.002 700	$0 \to 0.583$	1.33813
0.0300	0.727 309	0.707 276	1.000 824	$0 \to 0.682$	1.89044
0.0200	0.626 622	0.605 267	1.000 341	$0 \to 0.734$	2.54677

Table 5.1. Thresholds of spiked Wigner with the sparse Rademacher prior: the information threshold, the appearance of the high branch, the end of the low branch, the jump of the Bayes overlap at λ_{IT}, and the noise level $\Delta/\rho^2 = 1/\lambda_{\mathrm{IT}}^2$ of Lesieur et al. (2017), whose densities 0.0909, 0.0863 and 0.08 show the three orderings of λ_{dyn}, λ_{IT} and 1. Critical densities: $\rho_{\mathrm{tri}} = 0.096416$ (where $\lambda_{\mathrm{IT}} = \lambda_{\mathrm{dyn}} = \lambda_{\mathrm{alg}} = 1.019951$), $\rho_* = 0.085593$ ($\lambda_{\mathrm{IT}} = 1$), 0.086914 ($\lambda_{\mathrm{dyn}} = 1$). Script: `figures/scripts/05-spiked-phase.py`.

The proof extends state evolution (Theorem 6.4) to a start that depends on W. Their Lemma B.3 describes the law of A given its top eigenvector and eigenvalue, approximately, as the known outlier term plus the spike and an independent GOE matrix, both projected away from that eigenvector: the spectral analogue of the exact conditioning on WM in Lemma A.13 (i). With Theorem 4.1 the theorem settles where message passing is optimal; and since every positive stationary point satisfies $q = G(\lambda^2 q) \geq \lambda^2 q/(1+\lambda^2 q)$, that is $q \geq 1 - \lambda^{-2}$, message passing never does worse than PCA. For the sparse Rademacher prior, q_{ALG} lies on the low branch and q_* on the high one for $\lambda \in (\max\{1, \lambda_{\mathrm{IT}}\}, \lambda_{\mathrm{alg}})$, so message passing is strictly suboptimal there; and for $\lambda \in (\lambda_{\mathrm{IT}}, 1]$, not empty when $\rho < \rho_*$, weak recovery is possible while the top eigenvector sees nothing. The thresholds of Table 5.1 are uncertified floating-point evaluations of L and (5.5): these consequences are theorems about the variational formula, with numerical inputs not verified by interval arithmetic.

IS THE WINDOW HARD? As of 2026 the evidence covers restricted classes. Below $\lambda = 1$ the low-degree likelihood ratio stays bounded for degrees $o(N/\log N)$, for priors that include the sparse Rademacher prior (Kunisky et al. 2022, Theorem 3.9 of arXiv v1). First-order methods with Lipschitz updates and a fixed number of iterations cannot beat Bayes-optimal message passing (Celentano et al. 2020), but a spectral start needs about $\log N$ iterations; and that constant-degree polynomials cannot beat message passing is proved (Montanari and Wein 2025) only for priors with nonzero mean.

5.6 Measuring it

The script `05-spiked-bbp.py` computes the top eigenpair of A by Lanczos iteration[10] at eleven values of λ. It first checks Lemma 5.3 on one matrix: at $\mu = \mu_1(A)$ the secular equation holds to 10^{-15}, and the overlap formula (A.7)

[10] The routine `scipy.sparse.linalg.eigsh`: a fraction of a second at $N = 2000$ on the machine used, against tens of seconds for a dense eigendecomposition.

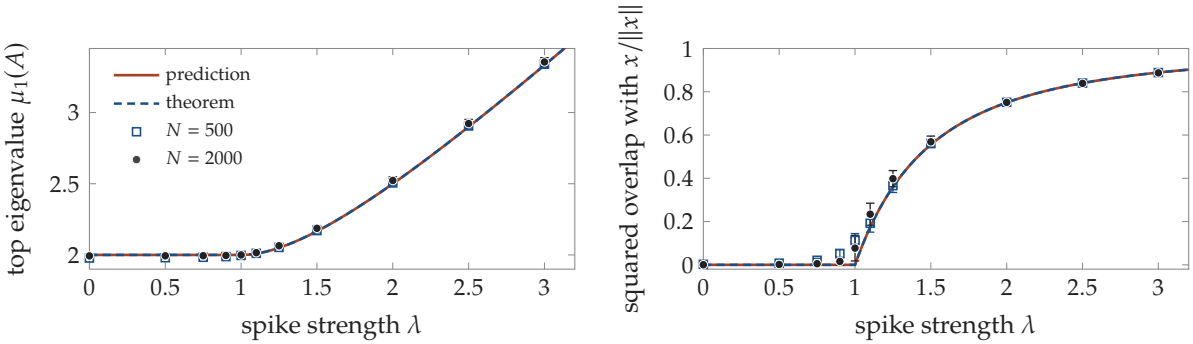


Figure 5.2. The BBP transition at finite size, Rademacher spike. Points: mean top eigenvalue (left) and mean squared overlap of the top eigenvector with x (right) over 40 matrices at $N = 500$ (squares) and 10 at $N = 2000$ (dots), with Student-t 95% intervals; the same matrices serve all eleven values of λ, so errors along a curve are correlated (seeds $5000 + 100N + r$). Carnelian: the replica prediction; lapis dashed: the limits proved in Theorem 5.1, which coincide with it. Script: `figures/scripts/05-spiked-bbp.py`.

matches the eigenvector to ten digits. Away from $\lambda = 1$ the measurements agree with the limits (Figure 5.2): at $N = 2000$ and $\lambda = 2$ the top eigenvalue is 2.523 ± 0.026 (limit 2.5) and the squared overlap 0.751 ± 0.012 (0.75). Near $\lambda = 1$ the transition is smeared: the outlier $\lambda + 1/\lambda - 2 \approx (\lambda - 1)^2$ shows only when it exceeds the edge fluctuations, of order $N^{-2/3}$ (Féral and Péché 2007), that is when $|\lambda - 1| \gtrsim N^{-1/3}$ (0.13 at $N = 500$, 0.08 at $N = 2000$). So the squared overlap at $\lambda = 1$ is 0.113 ± 0.032 at $N = 500$ and 0.077 ± 0.059 at $N = 2000$, and at $\lambda = 1.1$ it is 0.19 ± 0.04 and 0.23 ± 0.05 (limit 0.174).

The script `05-spiked-phase.py` evaluates G by the trapezoid rule (agreeing with the kit's quadrature to 10^{-8}), finds the extrema of L from $G'(\gamma) = \mathbb{E}[\mathrm{Var}(X \mid Y)^2]$ (Gaussian integration by parts; checked against finite differences), and solves Maxwell's construction with (5.5). It reproduces the tricritical point $(\Delta, \Delta/\rho^2, \rho) = (0.008935, 0.9612, 0.09641)$ of Lesieur et al. (2017) as $(0.008936, 0.9612, 0.096416)$. Figure 5.1(b) shows how thin the hard window above $\lambda = 1$ is, and Computation 5.1 why: its width vanishes like $\frac{3}{4}\rho^2$.

The script `05-spiked-amp.py` runs Bayes-optimal message passing in the normalization of Montanari and Venkataramanan (2021), with the signal-to-noise ratio estimated from the data and a start independent of W, so that the state evolution of Bayati and Montanari (2011) applies. The spike has exactly $\rho N = 150$ nonzero coordinates: with i.i.d. coordinates $\|x\|^2/N$, and with it the effective λ, would fluctuate by about $(\rho N)^{-1/2} \approx 8\%$. Figure 5.3 shows the hysteresis of the physical picture. From the strong start the runs follow the high branch: the mean squared cosine is 0.770 ± 0.062 at $\lambda = 0.92$ (state evolution after 50 steps: 0.755) and 0.931 ± 0.010 at $\lambda = 1.1$ (0.934), with 95% intervals over the six matrices. From the weak start they stay near 0 (mean at most 0.04) up to $\lambda = 0.94$, where the Bayes overlap exceeds 0.58; closer to 1, where $q = 0$ is only marginally stable, a growing fraction escape within 50 iterations (1, 2, 4, 4, 5 runs of six at $\lambda = 0.96, 0.98, 1, 1.04, 1.1$).

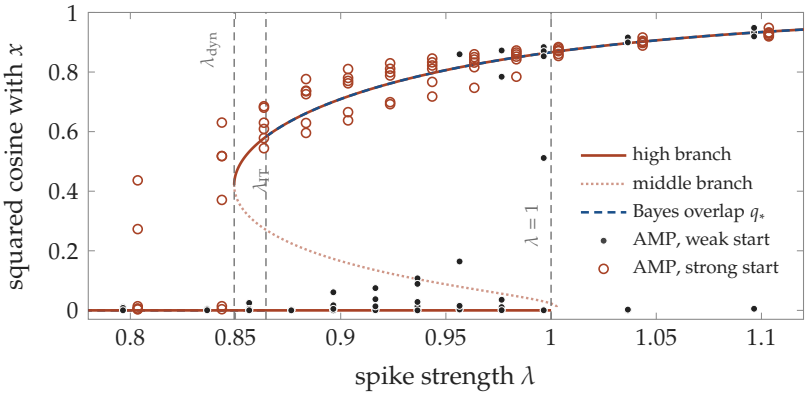


Figure 5.3. The hard phase of sparse PCA at $\rho = 0.05$, seen by Bayes-optimal message passing. Each point is one run at $N = 3000$ with exactly 150 nonzero coordinates, after 50 iterations from side information $g_0 x + \sqrt{g_0}\, G$ independent of W, with $g_0 = 0.01$ (dots, shifted left by 0.0035) or $g_0 = 3$ (circles, shifted right); six matrices per λ (seeds 700–705). Carnelian: the stable branches of the potential ($q = 0$ for $\lambda < 1$, and the high branch) and the unstable middle branch (dotted); lapis dashed: the Bayes overlap, with its jump at $\lambda_{\rm IT}$. Script: `figures/scripts/05-spiked-amp.py`.

Two finite-size effects are visible: near $\lambda_{\rm dyn} = 0.849$ the strong start beats state evolution (0.621 ± 0.059 against 0.557 at $\lambda = 0.86$, and four runs of six still high at $\lambda = 0.84$, where state evolution has collapsed); and the hard window above $\lambda = 1$, of width 0.0027, lies far inside the finite-size window $N^{-1/3} \approx 0.07$.

5.7 The status board

The limits for the top eigenvector are proved, for Wigner matrices well beyond the GOE, and so are the errors of the Bayes estimator and of message passing; the thresholds of sparse priors are theorems about a variational formula evaluated in floating point. The claims about semidefinite relaxations and about all efficient algorithms remain conjectures.

STATUS BOARD · zero-temperature replicas ↔ resolvents

PREDICTION	STATUS	PROOF OR EVIDENCE
Top eigenvalue $\lambda + 1/\lambda$ and squared overlap $1 - 1/\lambda^2$ above $\lambda = 1$; edge 2 and overlap 0 below	proved	Féral and Péché (2007) and Péché (2006); Capitaine et al. (2009); Benaych-Georges and Nadakuditi (2011), Thms 2.1–2.3; Theorem 5.1
Replica-symmetric formula for the mutual information and the MMSE	proved	Lelarge and Miolane (2019), Thm 1; Theorem 4.1

PCA is Bayes-optimal for the Gaussian prior only	proved	Proposition 5.6, from the RS formula
Weak recovery and detection start at $\lambda = 1$ for the Rademacher prior	proved	Theorem 4.2; Deshpande et al. (2016) (block model); Perry et al. (2018); El Alaoui et al. (2020)
Sparse Rademacher: first order for $\rho < \rho_{\text{tri}} = 0.0964$; $\lambda_{\text{IT}} < 1$ for $\rho < \rho_* = 0.0856$	proved under	RS formula proved (Theorem 4.1); read off it by uncertified floating-point evaluation on a grid of densities
Bayes-AMP with spectral start reaches the smallest positive stationary point; optimal iff it is the global maximizer	proved under	Montanari and Venkataramanan (2021), Thm 2, Cor 2.3: Lipschitz posterior mean, $\lambda > 1$, $t \to \infty$ after $N \to \infty$; bounded support for optimality
No polynomial-time algorithm beats message passing in the hard window	conjectured	Lesieur et al. (2017); evidence for restricted classes: low degree, first-order methods
Error curve of the semidefinite relaxation of $\mathbb{Z}_2$ synchronization above $\lambda = 1$	conjectured	Javanmard et al. (2016), cavity method and numerics

5.8 Where the translation fails

WHAT WAS ASSUMED. Theorem 5.1 uses orthogonal invariance to make an independent direction look uniformly random. The physical derivation needs only the second moments of the noise and gives the same answer for every Wigner matrix, which proofs reach only with heavier tools: Capitaine et al. (2009) for the eigenvalue, Benaych-Georges and Nadakuditi (2011) for the eigenvector when the direction of the spike is uniformly random. The spike must be independent of the noise, of finite rank, and of strength of order one. The Bayes statements assume a known prior, i.i.d. coordinates (relaxed in Theorem 5.7), Gaussian noise of known variance, and, for message passing, a Lipschitz posterior mean and $N \to \infty$ before $t \to \infty$. Our message-passing runs fix the number of nonzero coordinates, which Theorem 5.7 allows and Theorem 4.1 does not cover.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. For this model the physical predictions about the top eigenvector, the Bayes estimator and message passing have all been proved, and none is known to fail. The derivation is right for a reason specific to the sphere, where the quadratic model is replica symmetric; on the hypercube $\{\pm 1\}^N$ the same zero-temperature computation describes the Sherrington–Kirkpatrick ground state, where replica symmetry is broken (Chapter 3). The physics says nothing exact at simulated sizes: the transition is smeared over $|\lambda - 1| \lesssim N^{-1/3}$, and the hard window above $\lambda = 1$ at $\rho = 0.05$ stays invisible unless N is far beyond 10^7.

OPEN PROBLEMS. The conjecture of Lesieur et al. (2017) that no polynomial-time algorithm beats message passing in the hard window is open; a proof for all algorithms would need average-case lower bounds beyond current complexity theory. Even the low-degree version for symmetric priors is open: the theorem of Montanari and Wein (2025) needs a nonzero mean, and their Remark 1.2 extends to symmetric priors, with a sign-invariant loss, only the formulas for the AMP and Bayes errors. Whether first-order methods with a spectral start or a growing number of iterations beat Bayes-optimal message passing is open. The error curve of the semidefinite relaxation of $\mathbb{Z}_2$ synchronization above $\lambda = 1$, predicted by Javanmard et al. (2016) with the cavity method, is, to our knowledge, not proved.

Exercises

5.1 CONSOLIDATION. (The Stieltjes transform from moments.) The semicircle law has moments $\int x^{2k}\sigma_{\mathrm{sc}}(\mathrm{d}x) = C_k$, the Catalan numbers, and vanishing odd moments. (a) Show that $s(z) = -\sum_{k\ge 0} C_k z^{-2k-1}$ for $z > 2$. (b) From $C_{k+1} = \sum_{i=0}^{k} C_i C_{k-i}$ deduce $s^2 + zs + 1 = 0$ and (5.6). (c) Check $s'(z) = -s(z)/\sqrt{z^2-4}$.

5.2 CONSOLIDATION. (One derivative, two routes.) (a) For $\theta > 1$, differentiate $1 + \theta s(\Lambda(\theta)) = 0$ to show $\Lambda'(\theta) = 1/(\theta^2 s'(\Lambda(\theta)))$: the limit in Step 4 of the proof of Theorem 5.1 is the derivative that the convexity route of Exercise 5.9 produces. (b) Replace σ_{sc} by a compactly supported law μ with upper edge b and $s_\mu(b^+) > -\infty$. Show that $1 + \theta s_\mu(z) = 0$ has a root $z(\theta) > b$ if and only if $\theta > -1/s_\mu(b^+)$, that it is unique, and that $z'(\theta) = 1/(\theta^2 s_\mu'(z(\theta)))$; compare with the outlier and the overlap of Benaych-Georges and Nadakuditi (2011).

5.3 CONSOLIDATION. (The other edge.) Show that the lowest eigenvalue of $A_N(\theta)$ tends to -2, and that for $\theta > 1$ that of $W_N - \theta u_N u_N^{\mathsf{T}}$ tends to $-(\theta + 1/\theta)$.

5.4 CONSOLIDATION. (The Gaussian prior.) For $P_0 = \mathcal{N}(0,1)$ show $\psi(\gamma) = \frac{\gamma}{2} - \frac{1}{2}\log(1+\gamma)$ and that $\frac{1}{N} I(x;A)$ tends to $\frac{\lambda^2}{4}$ for $\lambda \le 1$ and to $\frac{1}{4\lambda^2} + \log\lambda$ for $\lambda > 1$, a function continuously differentiable at $\lambda = 1$; check that the matrix MMSE equals the error $1 - (1-\lambda^{-2})^2$ of the best multiple of $v_1 v_1^{\mathsf{T}}$.

5.5 BY HAND. (The spherical model at finite temperature.) From (5.2) compute the free entropies $\frac{\beta^2}{4}$ (paramagnet), $\beta - \frac{1}{2}\log\beta - \frac{3}{4}$ (spin glass) and $\frac{\beta}{2}(\lambda + \frac{1}{\lambda}) - \frac{1}{2} - \frac{1}{2}\log(\beta\lambda) - \frac{1}{4\lambda^2}$ (ferromagnet). (a) Show that they agree on the lines $\beta = 1$ ($\lambda \le 1$), $\beta\lambda = 1$ ($\lambda \ge 1$) and $\lambda = 1$ ($\beta \ge 1$), which bound the three phases. (b) Show $\partial_\lambda \phi_{\mathrm{F}} = \beta m^2/2$, and deduce that the ferromagnet has the larger free entropy wherever it and the spin glass both exist.

5.6 BY HAND. (A two-point prior.) Let $X = \sqrt{(1-p)/p}$ with probability p and $X = -\sqrt{p/(1-p)}$ otherwise. (a) Check $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$,

$m_3 = (1-2p)/\sqrt{p(1-p)}$. (b) Show that $m_3^2 > 2$ (Computation 5.1) exactly when $p < \frac{1}{2} - \frac{1}{2\sqrt{3}} \approx 0.211$ or $p > \frac{1}{2} + \frac{1}{2\sqrt{3}}$, the first-order threshold that Lesieur et al. (2017) find for two unequal communities.

5.7 BY HAND. ★ (The next order.) Carry Computation 5.1 one order further for a symmetric prior: $G(\gamma) = \gamma - \gamma^2 + (\frac{m_4^2}{6} - m_4 + \frac{5}{2})\gamma^3 + O(\gamma^4)$, which is $\gamma - \gamma^2 + \frac{5}{3}\gamma^3$ for the Rademacher prior. Deduce $\lambda_{\mathrm{alg}} \approx 1 + \frac{3\rho^2}{4(1-3\rho)^2}$ for the sparse Rademacher prior and compare with Table 5.1. *(Hint: the order s^6 needs* He_4 *and* He_5*; a computer algebra system is welcome.)*

5.8 TRANSLATION. (The Lagrange multiplier is the resolvent.) Impose $\|\sigma\|^2 = N$ with a multiplier z: then $\frac{1}{N}\log Z_N(\beta) \approx \frac{z-1}{2} - \frac{1}{2N}\sum_i \log(z - \beta\mu_i(A))$ at the saddle point $\frac{1}{N}\sum_i (z-\beta\mu_i(A))^{-1} = 1$. (a) For $A = W$ show that the saddle point is $z = 1 + \beta^2$ when $\beta \le 1$, and recover $\phi(\beta) = \beta^2/4$ with $\int \log(t-x)\sigma_{\mathrm{sc}}(\mathrm{d}x) = -\frac{ts(t)}{2} - \log(-s(t)) - \frac{1}{2}$ for $t \ge 2$. (b) Show that the saddle-point equation becomes $s(z/\beta) = -\beta$ and has no solution above the edge for $\beta > 1$: the measure condenses on the top eigenvector. Which step of the proof of Theorem 5.1 plays this role?

5.9 TRANSLATION. (Griffiths' lemma.) Let f_N be convex on an open interval, converging pointwise to f, differentiable at t. Show that every choice of subgradients $d_N \in \partial f_N(t)$ converges to $f'(t)$. Deduce (iii) of Theorem 5.1 from (i), and state the physical identity this lemma replaces.

5.10 MEASUREMENT. (The window.) Run the script `05-spiked-bbp.py` at $N = 250, 500, 1000$ and 2000 for $\lambda = 1$ and 1.1, with 40 matrices each. Check that the squared overlap at $\lambda = 1$ decreases roughly like $N^{-1/3}$, describe how the one at $\lambda = 1.1$ approaches 0.174, and explain both by the scale $|\lambda - 1| \approx N^{-1/3}$.

5.11 MEASUREMENT. (The basin of the high branch.) At $\rho = 0.05$ and $\lambda = 0.92$, state evolution from γ_0 reaches the high branch exactly when $\gamma_0 > \lambda^2 q_{\mathrm{mid}} \approx 0.109$, the unstable fixed point ($q_{\mathrm{mid}} = 0.1293$). Vary the strength g_0 of the start in the script `05-spiked-amp.py`, at $N = 1500$ and 3000: how sharp is this threshold at finite N?

5.12 RESEARCH. (Low degree and symmetric priors.) Formulate a sign-invariant version, for the sparse Rademacher prior, of the equivalence of Montanari and Wein (2025) between low-degree polynomial estimators and message passing, and test numerically whether a low-degree polynomial in the entries of A can beat q_{ALG} for $\lambda \in (1, \lambda_{\mathrm{alg}})$.

Notes and References

The spike entered random-matrix physics through spin glasses. Kosterlitz et al. (1976) solved the spherical spin glass with the semicircle law and checked it against replicas, and Jones et al. (1978) re-derived, against a 1977 claim to the contrary, the prediction of two earlier calculations that a single eigenvalue can split off from the semicircle when the entries of a random symmetric matrix have a nonzero mean; Füredi and Komlós (1981) proved it for a fixed mean $\mu > 0$: the top eigenvalue is close to $n\mu + \sigma^2/\mu$, our $\lambda + 1/\lambda$ far above the threshold. In the statistical mechanics of learning the PCA transition appeared as a critical number of examples needed for the search of the direction of largest variance (Biehl and Mietzner 1994). Hoyle and Rattray (2004) derived by replicas the spectrum of sample covariance matrices with symmetry-breaking directions, one eigenvalue leaving the bulk at each transition; their paper appeared on 27 February 2004, and Baik et al. (2005) posted their proof for complex Gaussian data on 1 March 2004.

The proof of Theorem 5.1 follows the resolvent route of Benaych-Georges and Nadakuditi (2011) for the GOE, the overlap above the threshold included; proving isotropy by a reflection and a fourth moment, and treating the overlap at and below the threshold by convexity, are our choices among standard devices. Proposition 5.6 and its proof are ours; we have not found the statement published, but both ingredients are classical: $\mathrm{mmse}(\gamma) \le 1/(1+\gamma)$, with equality only for a Gaussian input. On the Bayes side the physics came first: Lesieur et al. (2017) derived the potential, the typology of transitions and the hard phase. The formula was proved for $\mathbb{Z}_2$ synchronization by Deshpande et al. (2016), bracketed by Krzakala et al. (2016), and proved for general priors by Lelarge and Miolane (2019) (Chapter 4 gives the history).

TRANSLATING THE LITERATURE. Lelarge and Miolane (2019) and Deshpande et al. (2016) write $Y = \sqrt{\lambda/n}\, XX^\top + Z$: their λ is our λ^2, their overlap $1 - 1/\lambda$, their $F(\lambda, q)$ our $\phi_{\mathrm{RS}}(\sqrt{\lambda}, q)$. Montanari and Venkataramanan (2021) and Perry et al. (2018) use our normalization, the first with γ_t for our $\lambda^2 q_t$; the functional of their Proposition 2.2, as printed in arXiv v4, should read $\gamma^2/(4\lambda^2)$ for $\gamma^2/(4\lambda)$ and is minimized, being a mutual information (so read "minimum" in their Corollary 2.3; `05-spiked-phase.py` checks that only the corrected form is stationary at the fixed point). Lesieur et al. (2017) take $x_i \in \{0, \pm 1\}$, $\mathbb{E}x_i^2 = \rho$, and noise variance Δ, so $\Delta/\rho^2 = 1/\lambda^2$; their Δ_{Alg}, Δ_{Dyn}, Δ_{IT}, Δ_c are our λ_{alg}, λ_{dyn}, λ_{IT}, $\lambda = 1$. Benaych-Georges and Nadakuditi (2011) write θ for our λ and G_μ for $-s_\mu$, with s_μ as in Definition A.14.

References

Baik, J., G. Ben Arous, and S. Péché (2005). "Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices". In: *The Annals of Probability* 33.5, pp. 1643–1697. arXiv: `math/0403022`.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: `1001.3448`.

Benaych-Georges, F. and R. R. Nadakuditi (2011). "The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices". In: *Advances in Mathematics* 227.1, pp. 494–521. arXiv: 0910.2120.

Biehl, M. and A. Mietzner (1994). "Statistical mechanics of unsupervised structure recognition". In: *Journal of Physics A: Mathematical and General* 27.6, pp. 1885–1897.

Capitaine, M., C. Donati-Martin, and D. Féral (2009). "The largest eigenvalues of finite rank deformation of large Wigner matrices: convergence and nonuniversality of the fluctuations". In: *The Annals of Probability* 37.1, pp. 1–47. arXiv: 0706.0136.

Celentano, M., A. Montanari, and Y. Wu (2020). "The estimation error of general first order methods". In: *Proceedings of the 33rd Conference on Learning Theory (COLT)*. Vol. 125. Proceedings of Machine Learning Research, pp. 1078–1141. arXiv: 2002.12903.

Deshpande, Y., E. Abbe, and A. Montanari (2016). "Asymptotic mutual information for the balanced binary stochastic block model". In: *Information and Inference: A Journal of the IMA*. Published online 2016-12-26; in print as volume 6, issue 2 (2017), pages 125–170. arXiv: 1507.08685.

El Alaoui, A., F. Krzakala, and M. I. Jordan (2020). "Fundamental limits of detection in the spiked Wigner model". In: *The Annals of Statistics* 48.2, pp. 863–885. arXiv: 1806.09588.

Féral, D. and S. Péché (2007). "The Largest Eigenvalue of Rank One Deformation of Large Wigner Matrices". In: *Communications in Mathematical Physics* 272.1, pp. 185–228. arXiv: math/0605624.

Füredi, Z. and J. Komlós (1981). "The eigenvalues of random symmetric matrices". In: *Combinatorica* 1.3, pp. 233–241.

Guo, D., S. Shamai, and S. Verdú (2005). "Mutual Information and Minimum Mean-Square Error in Gaussian Channels". In: *IEEE Transactions on Information Theory* 51.4, pp. 1261–1282.

Hoyle, D. C. and M. Rattray (2004). "Principal-component-analysis eigenvalue spectra from data with symmetry-breaking structure". In: *Physical Review E* 69.2, p. 026124.

Javanmard, A., A. Montanari, and F. Ricci-Tersenghi (2016). "Phase transitions in semidefinite relaxations". In: *Proceedings of the National Academy of Sciences* 113.16, E2218–E2223. arXiv: 1511.08769.

Jones, R. C., J. M. Kosterlitz, and D. J. Thouless (1978). "The eigenvalue spectrum of a large symmetric random matrix with a finite mean". In: *Journal of Physics A: Mathematical and General* 11.3, pp. L45–L48.

Kosterlitz, J. M., D. J. Thouless, and R. C. Jones (1976). "Spherical Model of a Spin-Glass". In: *Physical Review Letters* 36.20, pp. 1217–1220.

Krzakala, F., J. Xu, and L. Zdeborová (2016). "Mutual Information in Rank-One Matrix Estimation". In: *2016 IEEE Information Theory Workshop (ITW)*. IEEE, pp. 71–75. arXiv: 1603.08447.

Kunisky, D., A. S. Wein, and A. S. Bandeira (2022). "Notes on Computational Hardness of Hypothesis Testing: Predictions Using the Low-Degree Likelihood Ratio". In: *Mathematical Analysis, its Applications and Computation*. Springer Proceedings in Mathematics & Statistics. Springer, pp. 1–50. arXiv: 1907.11636.

Lelarge, M. and L. Miolane (2019). "Fundamental limits of symmetric low-rank matrix estimation". In: *Probability Theory and Related Fields* 173.3-4, pp. 859–929.

Lesieur, T., F. Krzakala, and L. Zdeborová (2017). "Constrained low-rank matrix estimation: phase transitions, approximate message passing and applications".

In: *Journal of Statistical Mechanics: Theory and Experiment* 2017.7, p. 073403. arXiv: 1701.00858.

Montanari, A. and R. Venkataramanan (2021). "Estimation of low-rank matrices via approximate message passing". In: *The Annals of Statistics* 49.1, pp. 321–345. arXiv: 1711.01682.

Montanari, A. and A. S. Wein (2025). "Equivalence of approximate message passing and low-degree polynomials in rank-one matrix estimation". In: *Probability Theory and Related Fields* 191.1-2, pp. 181–233. arXiv: 2212.06996.

Péché, S. (2006). "The largest eigenvalue of small rank perturbations of Hermitian random matrices". In: *Probability Theory and Related Fields* 134.1, pp. 127–173. arXiv: math/0411487.

Perry, A., A. S. Wein, A. S. Bandeira, and A. Moitra (2018). "Optimality and sub-optimality of PCA I: Spiked random matrix models". In: *The Annals of Statistics* 46.5, pp. 2416–2451. arXiv: 1609.05573.

6 Message Passing

The cavity, TAP and AMP

THE DICTIONARY — the cavity ↔ Gaussian conditioning

IN THIS CHAPTER. Bayes-optimal inference in $\mathbb{Z}_2$ synchronization asks for the magnetizations of a planted Sherrington–Kirkpatrick model. The cavity method derives the TAP equations they satisfy, with their Onsager reaction term, turns them into approximate message passing (AMP), and predicts its iterates through a scalar recursion, state evolution. We derive all three as a physicist does, then prove state evolution for every fixed number of iterations by Bolthausen's conditioning, which removes the past iterates instead of a spin and makes the Gaussian cavity field and the reaction term exact. Simulations at $N = 3000$ confirm the predicted overlaps at every step, show what goes wrong without the Onsager term and with a wrongly scaled spectral start, and reach the Bayes-optimal overlap. Open: state evolution over many more than $\log N/\log\log N$ iterations for general denoisers, and the global landscape of the TAP free energy.

6.1 The question: a posterior mean in polynomial time

In $\mathbb{Z}_2$ *synchronization* a uniform vector of signs $x \in \{-1,+1\}^N$ is observed through the spiked matrix of Chapter 5,

$$A = \frac{\lambda}{N}\, xx^\top + W, \qquad W \sim \mathrm{GOE}(N),$$

with independent entries $W_{ij} \sim \mathcal{N}(0, 1/N)$ above the diagonal and $W_{ii} \sim \mathcal{N}(0, 2/N)$ on it: each entry is a noisy report on whether x_i and x_j agree.[1] Since $\sigma_i^2 = 1$, the likelihood $\exp(-\frac{N}{2}\sum_{i<j}(A_{ij} - \lambda\sigma_i\sigma_j/N)^2)$ depends on σ only through $\lambda\sum_{i<j} A_{ij}\sigma_i\sigma_j$ (the diagonal carries no information), and the posterior is the Gibbs measure

$$\mathbb{P}(x = \sigma \mid A) = \frac{1}{Z_N(A)} \exp\Big(\lambda \sum_{i<j} A_{ij}\sigma_i\sigma_j\Big), \qquad \sigma \in \{-1,+1\}^N. \tag{6.1}$$

With $g_{ij} = \sqrt{N}\, W_{ij}$ its exponent is the Sherrington–Kirkpatrick Hamiltonian of Chapter 3 at $\beta = \lambda$, without field, plus the planted term $\frac{\lambda^2}{N}\sum_{i<j} x_i x_j \sigma_i\sigma_j$; the temperature is tied to the signal, as on the Nishimori line of Chapter 4. The posterior is invariant under $\sigma \mapsto -\sigma$, so its mean is zero. The targets are

[1] The name comes from synchronizing clocks, phases or rotations from noisy measurements of their pairwise offsets; $\mathbb{Z}_2 = \{\pm 1\}$ is the simplest group.

the posterior mean of $xx^\top$ and, in the physicist's language, the *magnetizations* $m_i = \langle \sigma_i \rangle$ inside one of the two symmetric halves of the posterior, the half that an algorithm selects by its start. Here $m \in \mathbb{R}^N$ is a vector of magnetizations or of their estimates, as in the TAP literature, and its overlap with the truth is the number $\frac{1}{N}\langle m, x \rangle$.

How well can they be estimated? For the scalar channel $Y = \sqrt{\gamma} X + Z$ with X uniform on $\{\pm 1\}$, let $q(\gamma) = 1 - \mathrm{mmse}(\gamma) = \mathbb{E} \tanh(\gamma + \sqrt{\gamma}\, Z)$. By the theorem of Lelarge and Miolane (Chapter 4), the Bayes-optimal mean-square error for $xx^\top$ is $1 - q_*^2$, with $q_* = \gamma_*/\lambda^2$ and γ_* the largest solution of

$$\gamma = \lambda^2 q(\gamma) \tag{6.2}$$

(Theorem 4.1), while the top eigenvector of A reaches only the squared cosine $1 - 1/\lambda^2$ with x (Theorem 5.1). At $\lambda = 1.5$ these numbers are 0.692 and 0.556. But the Bayes estimator averages over 2^N configurations. Is there a cheap algorithm that computes the magnetizations, and can its accuracy be predicted at every step?

THE PHYSICAL PREDICTION · Thouless, Anderson and Palmer 1977; Donoho, Maleki and Montanari 2009. Inside one half of the posterior the magnetizations solve the TAP equations

$$m_i = \tanh\Big(\lambda \sum_j A_{ij} m_j - \lambda^2 (1 - q)\, m_i\Big), \qquad q = \frac{1}{N} \sum_j m_j^2,$$

whose last term is the Onsager reaction term. Iterated with the correct time indices they become approximate message passing,

$$y^{t+1} = A m^t - b_t m^{t-1}, \qquad m^t = \tanh(\lambda y^t), \qquad b_t = \lambda\Big(1 - \frac{1}{N} \sum_i (m_i^t)^2\Big),$$

and at every iteration the field λy_i^t is distributed over the coordinates as $\gamma_t x_i + \sqrt{\gamma_t}\, Z_i$, with Z_i standard Gaussian and $\gamma_{t+1} = \lambda^2 q(\gamma_t)$. This state evolution converges to γ_*, so the iteration reaches the Bayes-optimal overlap q_*.

Thouless et al. (1977) proposed the equations for the Sherrington–Kirkpatrick model. The iteration, *approximate message passing*, and its state evolution are the form that Donoho et al. (2009) gave to message passing in compressed sensing,[2] and for $\mathbb{Z}_2$ synchronization the iteration appears in Deshpande et al. (2016) and Montanari and Venkataramanan (2021). How do physicists derive the box, and what is proved?

6.2 The physics: removing a spin

The *cavity method* of Mézard et al. (1986) computes a property of N spins from the response of the system to one added spin.[3] We apply it to (6.1), with couplings $J_{ij} = \lambda A_{ij}$.

[2] They stated state evolution as an empirical "Finding 2": in their simulations it predicted the mean-square error, the false alarms, the missed detections and the convergence of the algorithm.

[3] The title of the cavity paper, *SK model: the replica solution without replicas*, states its program: to recover the replica results by an argument about one added spin.

THE CAVITY DERIVATION OF THE TAP EQUATIONS. Write $\langle\cdot\rangle^{(i)}$ for the Gibbs measure without spin i and $m_j^{(i)}$ for its magnetizations. Spin i sees the others only through its *cavity field* $h_i = \sum_{j\neq i} J_{ij}\sigma_j$, so summing over $\sigma_i = \pm 1$ gives the exact identity $m_i = \langle \sinh h_i\rangle^{(i)} / \langle \cosh h_i\rangle^{(i)}$. *First assumption: the cavity field is Gaussian.* The couplings J_{ij} do not enter $\langle\cdot\rangle^{(i)}$, so h_i is a sum of many terms with independent random coefficients; inside a single pure state, where correlations decay, take it Gaussian with mean $\sum_j J_{ij} m_j^{(i)}$ and some variance v. Since $\mathbb{E}\sinh(\mu + \sqrt{v}Z) = e^{v/2}\sinh\mu$ and $\mathbb{E}\cosh(\mu+\sqrt{v}Z) = e^{v/2}\cosh\mu$, the variance cancels and $m_i = \tanh(\sum_j J_{ij} m_j^{(i)})$. *Second assumption: linear response.* Putting spin i back adds the field $J_{ki}\sigma_i$ to every spin k. Keep only the response of m_j to its own added field, through the susceptibility $1 - m_j^2$, and drop the off-diagonal responses, small inside a pure state: $m_j \approx m_j^{(i)} + (1-m_j^2)J_{ji}m_i$. Hence

$$\sum_j J_{ij} m_j^{(i)} \approx \sum_j J_{ij} m_j - m_i \sum_j J_{ij}^2 (1 - m_j^2).$$

Third assumption: the reaction self-averages. Each $J_{ij}^2 = \lambda^2 A_{ij}^2$ has mean close to λ^2/N; replacing the sum by its average, $\sum_j J_{ij}^2(1-m_j^2) \approx \lambda^2(1-q)$, gives the *TAP equations*

$$m_i = \tanh\big(\lambda(Am)_i - \lambda^2(1-q)\,m_i\big), \qquad q = \frac{1}{N}\sum_j m_j^2. \quad \diamond \tag{6.3}$$

The last term is the *Onsager reaction term*, the part of the field on spin i that spin i has itself induced in the others.[4] It is a sum of N terms of order $1/N$ and survives because $\sum_j A_{ij}^2 \approx 1$, which is why Bolthausen (2014) calls it an Itô-type correction. Without it one gets the *naive mean-field* equations $m = \tanh(\lambda A m)$, which are wrong (Sections 6.6 and 6.8).

[4] The name recalls the reaction field of Onsager (1936)'s theory of polar liquids, the field that a molecule induces in its surroundings and that acts back on it. Donoho et al. (2009) kept the physicists' name for the corresponding term of their algorithm.

FROM BELIEF PROPAGATION TO APPROXIMATE MESSAGE PASSING. On a tree, *belief propagation* computes the marginals exactly by passing, for each ordered pair (i,j), the magnetization $m_{i\to j}$ of spin i without the edge ij: $m_{i\to j}^{t+1} = \tanh\big(\sum_{k\neq j} \operatorname{artanh}(\tanh(J_{ik}) m_{k\to i}^t)\big)$. On the complete graph $J_{ik} = O(N^{-1/2})$, so $\operatorname{artanh}(\tanh(J)m) = Jm + O(J^3)$, and the cubic terms add up to $O(N^{-1/2})$. With $h_i^{t+1} = \sum_k J_{ik} m_{k\to i}^t$ and $m_i^{t+1} = \tanh(h_i^{t+1})$, an expansion to first order in J_{ij} gives $m_{i\to j}^{t+1} \approx m_i^{t+1} - (1-(m_i^{t+1})^2) J_{ij} m_j^t$. Inserting this expansion of $m_{k\to i}^t$ into h_i^{t+1} and replacing J_{ik}^2 by λ^2/N,

$$h_i^{t+1} \approx \sum_k J_{ik} m_k^t - m_i^{t-1} \sum_k J_{ik}^2 \big(1 - (m_k^t)^2\big) \approx \lambda (A m^t)_i - \lambda^2 (1 - q_t)\, m_i^{t-1}.$$

The reaction carries the index $t-1$ because the message from k to i at time t was computed from the message from i to k at time $t-1$. With $y^t = h^t/\lambda$ this is the AMP of the box: N numbers per iteration instead of N^2 messages. $\diamond$

STATE EVOLUTION. Assume that each message $m^t_{k\to i}$ is independent of A_{ik}, as on a tree. Then $h_i^{t+1} = \lambda \sum_k A_{ik} m^t_{k\to i}$ is the signal part $\lambda^2 \mu_t x_i$, with $\mu_t = \frac{1}{N}\langle x, m^t\rangle$, plus $\lambda \sum_k W_{ik} m^t_{k\to i}$, a sum of N independent terms given the messages, hence Gaussian with variance $\lambda^2 q_t$, $q_t = \frac{1}{N}|m^t|^2$. If m^t is a Bayes posterior mean, the Nishimori identity (Lemma 1.10) gives $\mu_t = q_t$, so $h^{t+1} \approx \gamma_{t+1} x + \sqrt{\gamma_{t+1}} Z$ with $\gamma_{t+1} = \lambda^2 q_t$. In this scalar channel the posterior mean of x_i is $\tanh(h_i^{t+1}) = m_i^{t+1}$, so m^{t+1} is again a posterior mean, with overlap $q(\gamma_{t+1})$. This closes the recursion $\gamma_{t+1} = \lambda^2 q(\gamma_t)$, called *state evolution*. ◇

The fixed points of state evolution solve (6.2), the stationary points of the replica-symmetric free entropy of Chapter 4; that the algorithm stops at the Bayes-optimal one is the physicist's reason to trust it. The uncontrolled steps were Gaussianity inside one pure state, linear response, self-averaging of the reaction, independence of messages and couplings at all times, and the order of limits, $N \to \infty$ at fixed t, then $t \to \infty$.

COMPUTATION 6.1 · STATE EVOLUTION OF BAYES-AMP AT $\lambda = 1.5$

1 *The scalar channel.* Let X be uniform on $\{\pm 1\}$ and $Y = \gamma X + \sqrt{\gamma} Z$. The posterior of X is proportional to e^{XY}, so $\mathbb{E}[X \mid Y] = \tanh Y$. Given $X = x$, $xY = \gamma + \sqrt{\gamma}\, xZ$ has the law of $\gamma + \sqrt{\gamma} Z$, and $x \tanh Y = \tanh(xY)$, so $\mathbb{E}[X \tanh Y] = q(\gamma)$; by the tower property $\mathbb{E}\tanh^2 Y = \mathbb{E}[X\,\mathbb{E}(X \mid Y)] = q(\gamma)$ too. Overlap and squared norm agree: the Nishimori identity in its simplest form. Since $\frac{\mathrm{d}}{\mathrm{d}y}\tanh(\lambda y) = \lambda(1 - \tanh^2(\lambda y))$, when λy^t has the law of $\gamma_t X + \sqrt{\gamma_t} Z$ the average derivative is $\lambda(1 - q(\gamma_t))$, the Onsager coefficient b_t of the box.

2 *The numbers.* Gauss–Hermite quadrature (Section A.8) gives, from the value $\gamma_0 = \lambda^2 - 1 = 1.25$ of a spectral start (Computation 6.2), $\gamma_1 = 1.3991$, $\gamma_2 = 1.4807$, $\gamma_3 = 1.5215$, $\gamma_4 = 1.5409$, converging to $\gamma_* = 1.5577$, so that $q_* = 0.6923$ and the cosine with x is $\sqrt{q_*} = 0.8320$, against $\sqrt{1 - 1/\lambda^2} = 0.7454$ for the top eigenvector. The distances $\gamma_* - \gamma_t$ (0.308, 0.159, 0.077, 0.036, 0.017) roughly halve at each step: the slope of $\gamma \mapsto \lambda^2 q(\gamma)$ at γ_* is 0.457. At the fixed point $b = \lambda(1 - q_*) = 0.462$.

THE TAP FREE ENERGY. The TAP equations are the stationarity conditions of the *TAP free entropy* on $(-1, 1)^N$,

$$\Phi_{\mathrm{TAP}}(m) = \frac{\lambda}{2N}\langle m, Am\rangle + \frac{1}{N}\sum_{i=1}^{N} \mathsf{h}(m_i) + \frac{\lambda^2}{4}\Big(1 - \frac{|m|^2}{N}\Big)^2,$$

with $\mathsf{h}(m) = -\frac{1+m}{2}\log\frac{1+m}{2} - \frac{1-m}{2}\log\frac{1-m}{2}$: energy and entropy of a product measure with means m, plus the Onsager correction. Since $\mathsf{h}' = -\operatorname{artanh}$, its critical points solve (6.3). Plefka (1982) obtained it by expanding the free entropy at fixed magnetizations to second order in the couplings, and gave a condition for the expansion to converge; for SK at inverse temperature β,

Plefka's condition reads $\beta^2 \frac{1}{N} \sum_i (1 - m_i^2)^2 < 1$, and at the replica-symmetric solution it becomes the strict de Almeida–Thouless condition. Physically, the posterior is a mixture of pure states centered at local maximizers of Φ_{TAP}, the informative one being the posterior mean of its half. Mathematicians minimize $F_{\mathrm{TAP}} = -\Phi_{\mathrm{TAP}}$.

6.3 The stone

PHYSICS · *the cavity method and state evolution*	PROOF · *Bolthausen's conditioning*
1 · Remove a spin, or remove the past	
Remove spin i: the couplings A_{ij} do not enter $\langle\cdot\rangle^{(i)}$, so they are independent of the $m_j^{(i)}$.	Condition on $\mathcal{F}_t = \sigma(u, s^1, \dots, s^t)$: the iterates reveal only $W\Pi_t$, and $\Pi_t^\perp W \Pi_t^\perp$ is a fresh GOE block (Lemma 6.7).
2 · The cavity field is Gaussian	
A central limit theorem under $\langle\cdot\rangle^{(i)}$, assuming one pure state.	Exact at every N: given $\mathcal{F}_t$, the new part $\Pi_t^\perp W \Pi_t^\perp m^t$ is Gaussian (Lemma 6.7).
3 · The Onsager term and its time index	
Linear response gives $-\lambda^2(1-q)m_i$; a BP message excludes its recipient, so the correction multiplies m_i^{t-1}.	Gaussian integration by parts gives $\frac{1}{N}\langle s^q, m^t\rangle \to \mathrm{Cov}(Z_q, Z_t)\, b_t$ (6.6); as $\mathbb{E} Z_q Z_t = \mathbb{E} M_{q-1} M_{t-1}$, only m^{t-1} survives.
4 · State evolution	
Messages independent of couplings at all times, as on a tree.	Induction on t, at fixed t as $N \to \infty$, plus non-degeneracy $\rho_t > 0$ (Theorem 6.4).
5 · One pure state, and $t = \infty$	
A fixed point inside a pure state; choice of state and convergence assumed.	No counterpart at fixed t; the posterior needs Lelarge–Miolane, Montanari–Venkataramanan and Celentano–Fan–Mei.
6 · The TAP free energy, and universality	
Plefka's expansion; a pure state's mean maximizes Φ_{TAP}. Only a central limit theorem was used.	Chen–Panchenko's representation; a local maximizer at the posterior mean for every $\lambda > 1$ (Theorem 6.11). Non-Gaussian entries need moments or interpolation.

Both routes isolate a part of the system, show that the field on the rest is Gaussian once the correlated part is subtracted, and identify that part as a self-reaction proportional to an average susceptibility. Removing the

past instead of a site makes three assumptions exact: Gaussianity holds at every N, the reaction comes from Gaussian integration by parts, and nothing is assumed independent, because projections carry the dependence on the past. The price is a fixed number of iterations and a Gaussian matrix; the steps without counterpart concern the Gibbs measure. The idea is Bolthausen's: he presented it at a EURANDOM workshop in September 2009, as Bayati and Montanari (2011) record,[5] and proved the case of the SK iteration (Bolthausen 2014).[6]

[5] The workshop was titled *Order, disorder and double disorder*.

[6] Bolthausen stresses that his paper proves nothing about the SK Gibbs measure itself: its subject is the iteration.

6.4 The proof: conditioning on the past

We prove state evolution for an AMP driven by a GOE matrix, with Lipschitz denoisers that may use side information, for every fixed number of iterations; Section 6.5 deduces $\mathbb{Z}_2$ synchronization.

NOTATION. For $a \in \mathbb{R}^N$, $\|a\|_N = (|a|^2/N)^{1/2}$, and for an array of rows $a_1, \dots, a_N \in \mathbb{R}^d$, $\|a\|_N^2 = \frac{1}{N}\sum_i |a_i|^2$. Functions of scalars act coordinate by coordinate, functions of several vectors row by row. A function $\psi\colon \mathbb{R}^d \to \mathbb{R}$ is *pseudo-Lipschitz of order 2*, $\psi \in \mathrm{PL}(2)$, if for some L and all a, b

$$|\psi(a) - \psi(b)| \le L\big(1 + |a| + |b|\big)|a - b|. \tag{6.4}$$

Then $|\psi(a)| \le |\psi(0)| + L(|a| + |a|^2)$. Lipschitz functions, products of two Lipschitz functions, and compositions of functions in PL(2) with linear maps are in PL(2). We write $\xrightarrow{\mathbb{P}}$ for convergence in probability as $N \to \infty$.

Assumption 6.1 (Symmetric AMP with side information).

(i) $W \sim \mathrm{GOE}(N)$.

(ii) The side information is an array $u = (u_1, \dots, u_N)$ of rows $u_i \in \mathbb{R}^k$, independent of W, and there is a random vector $U \in \mathbb{R}^k$ with $\mathbb{E}|U|^2 < \infty$ such that $\frac{1}{N}\sum_i \psi(u_i) \xrightarrow{\mathbb{P}} \mathbb{E}\psi(U)$ for every $\psi \in \mathrm{PL}(2)$ on $\mathbb{R}^k$.

(iii) The denoisers $\eta_0\colon \mathbb{R}^k \to \mathbb{R}$ and $\eta_t\colon \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$, $t \ge 1$, are Lipschitz.

(iv) $\mathbb{E}\eta_0(U)^2 > 0$, and for every $t \ge 1$ the function $s \mapsto \eta_t(s, U)$ is not constant with positive probability.

Part 2 holds, by the weak law of large numbers, when the u_i are independent copies of U. The iteration is

$$m^0 = \eta_0(u), \quad s^1 = Wm^0; \qquad m^t = \eta_t(s^t, u), \quad s^{t+1} = Wm^t - b_t m^{t-1}, \tag{6.5}$$

for $t \ge 1$, with deterministic Onsager coefficients b_t, defined with the limit:

Definition 6.2 (State evolution). Let $M_0 = \eta_0(U)$ and $\Gamma_{00} = \mathbb{E}M_0^2$. For $t \ge 1$, given $Z_1, \dots, Z_{t-1}$ and Γ_{rs} for $r, s \le t-1$, let Z_t be such that $(Z_1, \dots, Z_t)$ is a centered Gaussian vector, independent of U, with $\mathbb{E}Z_r Z_s = \Gamma_{r-1,s-1}$; set $M_t = \eta_t(Z_t, U)$, $\Gamma_{rt} = \Gamma_{tr} = \mathbb{E}M_r M_t$ for $r \le t$, and

$b_t = \mathbb{E}\,\partial_s \eta_t(Z_t, U)$. Let $\Gamma_t = (\Gamma_{rs})_{0\le r,s\le t-1}$.

The prescribed covariance is a Gram matrix in L^2, so Z_t exists, as a linear combination of $Z_1, \dots, Z_{t-1}$ plus an independent Gaussian. The derivative $\partial_s \eta_t(s,u)$ exists for almost every s and is bounded, because $\eta_t(\cdot, u)$ is Lipschitz, and Z_t has positive variance (Lemma 6.3), so b_t is well defined.[7]

[7] A measurable version is $\limsup_n n(\eta_t(s+1/n,u) - \eta_t(s,u))$. Gaussian integration by parts gives the derivative-free form $b_t = \mathbb{E}[Z_t M_t]/\Gamma_{t-1,t-1}$.

Lemma 6.3 (Non-degeneracy). *Under parts 2–4 of Assumption 6.1, Γ_t is positive definite for every $t \ge 1$.*

Proof. The plan is an induction on t: a linear relation among $M_0, \dots, M_t$ would make $\eta_t(Z_t, U)$ a function of the past, which the fresh Gaussian part of Z_t forbids. For $t = 1$, $\Gamma_1 = \mathbb{E}M_0^2 > 0$. Suppose Γ_t positive definite and $\mathbb{E}(\sum_{r\le t} a_r M_r)^2 = 0$. If $a_t = 0$, then $a = 0$. Otherwise $M_t = F(Z_1, \dots, Z_{t-1}, U)$ almost surely for a measurable F. As Γ_t is the covariance of $(Z_1, \dots, Z_t)$, $Z_t = \sum_{r<t} c_r Z_r + \zeta$ with $\zeta \sim \mathcal{N}(0, v)$, $v > 0$, independent of $(Z_1, \dots, Z_{t-1}, U)$. By Fubini's theorem, for almost every value (z, u) of $(Z_1, \dots, Z_{t-1}, U)$, $\eta_t(\sum_r c_r z_r + \zeta, u) = F(z, u)$ for almost every ζ, so the continuous function $\eta_t(\cdot, u)$ is constant. Since U is independent of $(Z_1, \dots, Z_{t-1})$, this holds for almost every value u of U, contradicting part 4. ∎

Theorem 6.4 (State evolution). *Assume Assumption 6.1, and let $s^1, s^2, \dots$ be the iterates (6.5). For every fixed $t \ge 1$ and every $\psi \in \mathrm{PL}(2)$ on $\mathbb{R}^t \times \mathbb{R}^k$,*

$$\frac{1}{N}\sum_{i=1}^{N} \psi\big(s_i^1, \dots, s_i^t, u_i\big) \xrightarrow{\mathbb{P}} \mathbb{E}\,\psi(Z_1, \dots, Z_t, U) \qquad (N \to \infty).$$

The randomness is that of W and u, the limit is deterministic, and t is fixed as $N \to \infty$: the rows $(s_i^1, \dots, s_i^t)$ look like independent draws of the Gaussian vector of state evolution, independent of the side information. With $\psi(s,u) = s_q\,\eta_r(s_r, u)$ and Gaussian integration by parts given U (Theorem A.1, which extends to the Lipschitz $\eta_r(\cdot, u)$ by mollification, Z_r having a density), the theorem gives

$$\frac{1}{N}\langle s^q, m^r\rangle \xrightarrow{\mathbb{P}} \mathbb{E}Z_q \eta_r(Z_r, U) = \Gamma_{q-1,r-1}\, b_r \qquad (1 \le q, r \le t), \tag{6.6}$$

and $\frac{1}{N}\langle s^q, m^0\rangle \xrightarrow{\mathbb{P}} \mathbb{E}Z_q M_0 = 0$. This identity is where the Onsager coefficient comes from.

Lemma 6.5 (Two estimates). *Let $\psi \in \mathrm{PL}(2)$ on $\mathbb{R}^d$ with constant L.*

(i) For arrays a, a' of rows in $\mathbb{R}^d$, $\left|\frac{1}{N}\sum_i \psi(a_i) - \frac{1}{N}\sum_i \psi(a_i')\right| \le \sqrt{3}\,L\,\big(1 + \|a\|_N^2 + \|a'\|_N^2\big)^{1/2}\|a - a'\|_N$.

(ii) If $d = d' + 1$ and $\zeta \sim \mathcal{N}(0,1)$, then $\bar\psi(a) = \mathbb{E}\psi(a, \zeta)$ is in $\mathrm{PL}(2)$ on $\mathbb{R}^{d'}$, and $\mathbb{E}(\psi(a,\zeta) - \psi(a,0))^2 \le 12L^2(1 + |a|^2)$.

PROOF. Part 1: apply (6.4) to each pair, then Cauchy–Schwarz and $(1+x+y)^2 \le 3(1+x^2+y^2)$. Part 2: $|\bar\psi(a)-\bar\psi(b)| \le L\,\mathbb{E}(1+|a|+|b|+2|\zeta|)|a-b| \le L(1+2\mathbb{E}|\zeta|)(1+|a|+|b|)|a-b|$, and $|\psi(a,\zeta)-\psi(a,0)| \le L(1+2|a|+|\zeta|)|\zeta|$, whose square has mean at most $3L^2\mathbb{E}[(1+4|a|^2+\zeta^2)\zeta^2] = 12L^2(1+|a|^2)$. ∎

The next lemma combines parts (i) and (iii) of the conditioning lemma (Lemma A.13) in the form the proof uses; its proof is a covariance computation.

Lemma 6.6 (A fresh Gaussian block). *Let Π be an orthogonal projection of $\mathbb{R}^N$, $v \neq 0$ in its range, $\hat v = v/|v|$, $\Pi' = \Pi - \hat v\hat v^\top$, and $\tilde W \sim \mathrm{GOE}(N)$. Then $\Pi\tilde W v$ and $\Pi'\tilde W\Pi'$ are independent, and $\Pi\tilde W v$ is centered Gaussian with covariance $\frac{|v|^2}{N}(\Pi + \hat v\hat v^\top)$.*

PROOF. Both are linear in the Gaussian entries of $\tilde W$, hence jointly Gaussian, and it suffices to compute covariances. For the GOE, $\mathbb{E}\tilde W_{ab}\tilde W_{cd} = \frac{1}{N}(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$. Since $\Pi' v = 0$ and $\Pi v = v$,

$$\mathbb{E}(\Pi\tilde W v)_i(\Pi'\tilde W\Pi')_{kl} = \tfrac{1}{N}\big((\Pi\Pi')_{ik}(\Pi' v)_l + (\Pi\Pi')_{il}(\Pi' v)_k\big) = 0,$$
$$\mathbb{E}(\Pi\tilde W v)_i(\Pi\tilde W v)_k = \tfrac{1}{N}\big(\Pi_{ik}|v|^2 + v_i v_k\big).$$

Uncorrelated jointly Gaussian vectors are independent. ∎

Part 2 of the next lemma is part (ii) of Lemma A.13, with side information and without its rank condition.

Lemma 6.7 (Conditioning on the past). *Let $\mathcal{F}_t = \sigma(u, s^1, \dots, s^t)$, let Π_t be the orthogonal projection onto the span of $m^0, \dots, m^{t-1}$ ($\Pi_0 = 0$), and $\Pi_t^\perp = I - \Pi_t$. For every $t \ge 0$:*

(i) $W\Pi_t$ and $s^{t+1} - \Pi_t^\perp W\Pi_t^\perp m^t$ are $\mathcal{F}_t$-measurable;

(ii) conditionally on $\mathcal{F}_t$, $\Pi_t^\perp W\Pi_t^\perp$ has the law of $\Pi_t^\perp\tilde W\Pi_t^\perp$, with $\tilde W \sim \mathrm{GOE}(N)$ independent of $\mathcal{F}_t$;

(iii) conditionally on $\mathcal{F}_t$, $\Pi_t^\perp W\Pi_t^\perp m^t$ is centered Gaussian with covariance $\frac{1}{N}|m_\perp^t|^2(\Pi_t^\perp + \hat m\hat m^\top)$, where $m_\perp^t = \Pi_t^\perp m^t$ and $\hat m = m_\perp^t/|m_\perp^t|$ ($\hat m = 0$ if $m_\perp^t = 0$).

PROOF. The plan is to show that the iterates reveal W only through $W\Pi_t$, and to peel off one direction per iteration with Lemma 6.6. *Part 1.* For $r < t$ the vectors m^r and $Wm^r = s^{r+1} + b_r m^{r-1}$ ($m^{-1} = 0$) are $\mathcal{F}_t$-measurable. Gram–Schmidt gives an orthonormal basis $e_1, \dots, e_k$ of $\mathrm{span}(m^0, \dots, m^{t-1})$ whose vectors are combinations of the m^r with $\mathcal{F}_t$-measurable coefficients, so $W\Pi_t = \sum_j (We_j)e_j^\top$ and $\Pi_t W = (W\Pi_t)^\top$ are $\mathcal{F}_t$-measurable. Splitting $m^t = \Pi_t m^t + m_\perp^t$ and $Wm_\perp^t = \Pi_t Wm_\perp^t + \Pi_t^\perp Wm_\perp^t$,

$$s^{t+1} = W\Pi_t m^t + \Pi_t Wm_\perp^t - b_t m^{t-1} + \Pi_t^\perp W\Pi_t^\perp m^t, \tag{6.7}$$

and the first three terms are $\mathcal{F}_t$-measurable. *Part 2.* For $t = 0$ this is the independence of W and u. Suppose it holds at t; write $B = \Pi_t^\perp W\Pi_t^\perp$ and

$v = m_\perp^t$, both $\Pi_t^\perp$ and v being $\mathcal{F}_t$-measurable. By (6.7), $\mathcal{F}_{t+1} = \mathcal{F}_t \vee \sigma(Bv)$. If $v = 0$, then $\mathcal{F}_{t+1} = \mathcal{F}_t$ and $\Pi_{t+1} = \Pi_t$. Otherwise $\Pi_{t+1}^\perp = \Pi_t^\perp - \hat{v}\hat{v}^\top$ and $\Pi_{t+1}^\perp W \Pi_{t+1}^\perp = \Pi_{t+1}^\perp B \Pi_{t+1}^\perp$. Given $\mathcal{F}_t$, the pair $(Bv, \Pi_{t+1}^\perp B \Pi_{t+1}^\perp)$ has the law of $(\Pi_t^\perp \tilde{W} v, \Pi_{t+1}^\perp \tilde{W} \Pi_{t+1}^\perp)$, whose components are independent by Lemma 6.6. So they are conditionally independent given $\mathcal{F}_t$, and conditioning further on Bv does not change the conditional law of $\Pi_{t+1}^\perp B \Pi_{t+1}^\perp$, by the characterization of conditional independence; that law is the law of $\Pi_{t+1}^\perp \tilde{W} \Pi_{t+1}^\perp$. *Part 3.* By part 2, given $\mathcal{F}_t$ the vector Bv has the law of $\Pi_t^\perp \tilde{W} v$, which Lemma 6.6 identifies. ∎

Proof of Theorem 6.4. The plan is an induction on t. Let H_t be the statement of the theorem at t, and H_0 part 2 of Assumption 6.1. Assuming H_t, $t \geq 0$, we write s^{t+1} as a fixed combination of past iterates plus an independent Gaussian vector plus a remainder that vanishes in $\|\cdot\|_N$, and read off H_{t+1}. For $t = 0$ the sums over r below are empty, $\Pi_0 = 0$ and $R = 0$.

Step 1: inner products. Let $Q_t = [m^0 | \cdots | m^{t-1}]$ and $S_t = [s^1 | \cdots | s^t]$, and $\hat{\Gamma}_t = Q_t^\top Q_t / N$. By H_t, $\hat{\Gamma}_t \xrightarrow{\mathbb{P}} \Gamma_t$, $\frac{1}{N} Q_t^\top m^t \xrightarrow{\mathbb{P}} \Gamma_{\cdot t} = (\Gamma_{rt})_{r<t}$, $\|m^t\|_N^2 \xrightarrow{\mathbb{P}} \Gamma_{tt}$, (6.6) holds, and the norms $\|s^q\|_N$ are bounded in probability. By Lemma 6.3, $E_N = \{\lambda_{\min}(\hat{\Gamma}_t) \geq \frac{1}{2}\lambda_{\min}(\Gamma_t)\}$ has probability tending to one. On E_N, $a = \hat{\Gamma}_t^{-1} Q_t^\top m^t / N$ satisfies $\Pi_t m^t = Q_t a$, $a \xrightarrow{\mathbb{P}} a^\infty = \Gamma_t^{-1} \Gamma_{\cdot t}$, and $\|m_\perp^t\|_N^2 = \|m^t\|_N^2 - a^\top \hat{\Gamma}_t a \xrightarrow{\mathbb{P}} \Gamma_{tt} - \Gamma_{\cdot t}^\top \Gamma_t^{-1} \Gamma_{\cdot t} = \rho_t^2$, a Schur complement of Γ_{t+1}, positive by Lemma 6.3.

Step 2: the old part, and the Onsager cancellation. On E_N, in (6.7), $W \Pi_t m^t = \sum_{r<t} a_r (s^{r+1} + b_r m^{r-1})$, and $\Pi_t W m_\perp^t = Q_t \hat{\Gamma}_t^{-1} (W Q_t)^\top m_\perp^t / N = Q_t \hat{\Gamma}_t^{-1} c$ with $c = S_t^\top m_\perp^t / N$, because the columns of $W Q_t$ are $s^{r+1} + b_r m^{r-1}$ and $m^{r-1} \perp m_\perp^t$. Index c like the columns of Q_t, $c_j = \frac{1}{N}\langle s^{j+1}, m_\perp^t \rangle$ for $0 \leq j < t$, and define $d \in \mathbb{R}^t$ by $Q_t d = b_t m^{t-1} - \sum_{r=1}^{t-1} a_r b_r m^{r-1}$, that is, $d_j = b_t \mathbf{1}\{j = t-1\} - a_{j+1} b_{j+1} \mathbf{1}\{j \leq t-2\}$. Then

$$s^{t+1} = \sum_{r<t} a_r s^{r+1} + \Pi_t^\perp W \Pi_t^\perp m^t + R, \qquad R = Q_t \big(\hat{\Gamma}_t^{-1} c - d\big).$$

Since $\|Q_t w\|_N^2 = w^\top \hat{\Gamma}_t w$, we get $\|R\|_N^2 = (c - \hat{\Gamma}_t d)^\top \hat{\Gamma}_t^{-1} (c - \hat{\Gamma}_t d) \leq 2|c - \hat{\Gamma}_t d|^2 / \lambda_{\min}(\Gamma_t)$. By Step 1 and (6.6), $c_j = \frac{1}{N}\langle s^{j+1}, m^t \rangle - \sum_r a_r \frac{1}{N}\langle s^{j+1}, m^r \rangle$ converges to $\Gamma_{j,t-1} b_t - \sum_{r=1}^{t-1} \Gamma_{j,r-1} b_r a_r^\infty$, which is also the limit of $(\hat{\Gamma}_t d)_j$. So $\|R\|_N \xrightarrow{\mathbb{P}} 0$: in the limit, the Onsager term $-b_t m^{t-1}$ cancels the part of $\Pi_t W m_\perp^t$ that is not a combination of past iterates.

Step 3: the new part. By part 3 of Lemma 6.7, given $\mathcal{F}_t$ the vector $\Pi_t^\perp W \Pi_t^\perp m^t$ has the conditional law of $\|m_\perp^t\|_N \Pi_t^\perp z + N^{-1/2} m_\perp^t \xi$, where $z \sim \mathcal{N}(0, I_N)$ and $\xi \sim \mathcal{N}(0,1)$ are independent of each other and of $\mathcal{F}_t$ (on an enlarged space): both have covariance $\|m_\perp^t\|_N^2 \Pi_t^\perp + m_\perp^t m_\perp^{t\top} / N$. As the rest of s^{t+1} is $\mathcal{F}_t$-measurable, this replacement leaves the law of $(u, s^1, \ldots, s^{t+1})$, which is all that H_{t+1} concerns, unchanged; we make it. With $\hat{s} = \sum_{r<t} a_r^\infty s^{r+1} + \rho_t z$, on E_N

$$s^{t+1} - \hat{s} = \sum_{r<t} (a_r - a_r^\infty) s^{r+1} + R + \big(\|m_\perp^t\|_N - \rho_t\big) z - \|m_\perp^t\|_N \Pi_t z + N^{-1/2} m_\perp^t \xi.$$

Each term tends to zero in $\|\cdot\|_N$: by Steps 1 and 2, because $\|z\|_N \xrightarrow{\mathbb{P}} 1$, because $\mathbb{E}[|\Pi_t z|^2 \mid \mathcal{F}_t] = \operatorname{rank}\Pi_t \le t$, and because the last norm is $\|m_\perp^t\|_N|\xi|/\sqrt{N}$. So $\|s^{t+1} - \hat{s}\|_N \xrightarrow{\mathbb{P}} 0$.

Step 4: H_{t+1}. Let $\psi \in \mathrm{PL}(2)$ on $\mathbb{R}^{t+1} \times \mathbb{R}^k$, $y_i = (s_i^1, \dots, s_i^t, u_i)$, and $\varphi(y, \zeta) = \psi(s_1, \dots, s_t, \sum_{r<t} a_r^\infty s_{r+1} + \rho_t \zeta, u)$, which is in PL(2), with constant L say. By part 1 of Lemma 6.5 and Step 3, whose norms are bounded in probability by H_t and part 2 of Assumption 6.1, $\frac{1}{N}\sum_i \psi(s_i^1, \dots, s_i^{t+1}, u_i) - \frac{1}{N}\sum_i \varphi(y_i, z_i) \xrightarrow{\mathbb{P}} 0$. With $\bar\varphi(y) = \mathbb{E}\varphi(y, \zeta)$, in PL(2) by part 2, H_t gives

$$\frac{1}{N}\sum_i \bar\varphi(y_i) \xrightarrow{\mathbb{P}} \mathbb{E}\,\bar\varphi(Z_1, \dots, Z_t, U) = \mathbb{E}\,\psi\Big(Z_1, \dots, Z_t, \sum_{r<t} a_r^\infty Z_{r+1} + \rho_t\zeta, U\Big),$$

which equals $\mathbb{E}\psi(Z_1, \dots, Z_{t+1}, U)$: the vector $(Z_1, \dots, Z_t, \sum_r a_r^\infty Z_{r+1} + \rho_t\zeta)$, with ζ independent, is centered Gaussian, independent of U, with $\mathbb{E}Z_q \sum_r a_r^\infty Z_{r+1} = (\Gamma_t a^\infty)_{q-1} = \Gamma_{q-1,t}$ and $a^{\infty\top}\Gamma_t a^\infty + \rho_t^2 = \Gamma_{tt}$, the covariance of $(Z_1, \dots, Z_{t+1})$. Finally $D_N = \frac{1}{N}\sum_i(\varphi(y_i, z_i) - \bar\varphi(y_i))$ has conditional mean zero given $\mathcal{F}_t$ and, the z_i being independent, conditional variance

$$\frac{1}{N^2}\sum_i \operatorname{Var}\varphi(y_i, \zeta) \le \frac{1}{N^2}\sum_i \mathbb{E}\big(\varphi(y_i, \zeta) - \varphi(y_i, 0)\big)^2 \le \frac{12L^2}{N}\big(1 + \|y\|_N^2\big)$$

by part 2 of Lemma 6.5. The conditional Chebyshev inequality gives $\mathbb{P}(|D_N| > \epsilon \mid \mathcal{F}_t) \le \min\{1, 12L^2(1 + \|y\|_N^2)/(N\epsilon^2)\} \xrightarrow{\mathbb{P}} 0$, and taking expectations $D_N \xrightarrow{\mathbb{P}} 0$. This is H_{t+1}. ∎

Gaussianity of W entered only through Lemma 6.6, the Lipschitz property through the estimates of Lemma 6.5, non-degeneracy through E_N and $\rho_t > 0$, and the Onsager coefficient through (6.6). Denoisers with memory need Onsager terms $\sum_{r\le t} b_{t,r} m^{r-1}$ (Exercise 6.9).

6.5 The proof: from the theorem to $\mathbb{Z}_2$ synchronization

Corollary 6.8 (State evolution with a spike). *Let $A = \frac{\lambda}{N}xx^\top + W$ with $W \sim \mathrm{GOE}(N)$, and let the side information $u_i = (x_i, \tilde u_i) \in \mathbb{R}^k$, whose first coordinate is the signal, satisfy part 2 of Assumption 6.1 with limit $U = (X, \tilde U)$. Let $f_0 \colon \mathbb{R}^k \to \mathbb{R}$ be Lipschitz with $\mathbb{E}f_0(U)^2 > 0$ and $f_t \colon \mathbb{R} \to \mathbb{R}$, $t \ge 1$, Lipschitz and not constant. Let $Z \sim \mathcal{N}(0,1)$ be independent of U, $\mu_1 = \lambda\mathbb{E}[X f_0(U)]$, $\sigma_1^2 = \mathbb{E}f_0(U)^2$, and for $t \ge 1$*

$$\mu_{t+1} = \lambda\mathbb{E}\big[X f_t(\mu_t X + \sigma_t Z)\big], \qquad \sigma_{t+1}^2 = \mathbb{E}\big[f_t(\mu_t X + \sigma_t Z)^2\big], \tag{6.8}$$

and $b_t = \mathbb{E}f_t'(\mu_t X + \sigma_t Z)$. Let $m^0 = f_0(u)$, $y^1 = Am^0$, and for $t \ge 1$

$$m^t = f_t(y^t), \qquad y^{t+1} = Am^t - b_t m^{t-1}. \tag{6.9}$$

Then for every $t \ge 1$ and $\psi \in \mathrm{PL}(2)$ on $\mathbb{R} \times \mathbb{R}^k$, $\frac{1}{N}\sum_i \psi(y_i^t, u_i) \xrightarrow{\mathbb{P}} \mathbb{E}\psi(\mu_t X + \sigma_t Z, U)$. If moreover each f_t, $t \ge 1$, has a bounded Lipschitz

derivative, the same holds for the iterates $\hat{y}^t$ of (6.9) with b_t replaced by the empirical $\hat{b}_t = \frac{1}{N}\sum_i f_t'(\hat{y}_i^t)$.

PROOF. The plan is to compare with an AMP driven by W alone whose denoisers absorb the spike. Let $\eta_0 = f_0$ and $\eta_t(s,u) = f_t(s+\mu_t x)$, with x the first coordinate of u. They satisfy parts 3 and 4 of Assumption 6.1; by induction, Definition 6.2 gives $\Gamma_{t-1,t-1} = \sigma_t^2$ and $\mathbb{E}\partial_s\eta_t(Z_t,U) = b_t$. By Theorem 6.4, $\frac{1}{N}\sum_i \psi(s_i^t + \mu_t x_i, u_i) \xrightarrow{\mathbb{P}} \mathbb{E}\psi(\mu_t X + \sigma_t Z, U)$, so by part 1 of Lemma 6.5 it suffices that $e^t = y^t - s^t - \mu_t x$ satisfies $\|e^t\|_N \xrightarrow{\mathbb{P}} 0$. Set $e^0 = 0$. First, $e^1 = (\frac{\lambda}{N}\langle x, m^0\rangle - \mu_1)x \to 0$, by part 2 of Assumption 6.1. With $\tilde{m}^r = \eta_r(s^r, u)$ ($\tilde{m}^0 = m^0$) and L a Lipschitz constant of $f_1,\dots,f_t$,

$$e^{t+1} = \Big(\frac{\lambda}{N}\langle x, m^t\rangle - \mu_{t+1}\Big)x + W(m^t - \tilde{m}^t) - b_t(m^{t-1} - \tilde{m}^{t-1}),$$

with $\|m^r - \tilde{m}^r\|_N \le L\|e^r\|_N$. By the theorem $\frac{1}{N}\langle x, \tilde{m}^t\rangle \xrightarrow{\mathbb{P}} \mathbb{E}X f_t(Z_t + \mu_t X) = \mu_{t+1}/\lambda$, and $\frac{1}{N}|\langle x, m^t - \tilde{m}^t\rangle| \le L\|x\|_N\|e^t\|_N$. Since $\|W\| \to 2$ (Theorem A.15), induction gives

$$\|e^{t+1}\|_N \le \lambda\|x\|_N\Big|\frac{1}{N}\langle x, m^t\rangle - \frac{\mu_{t+1}}{\lambda}\Big| + L\|W\|\|e^t\|_N + L|b_t|\|e^{t-1}\|_N \xrightarrow{\mathbb{P}} 0.$$

For the empirical coefficients we show $\|\hat{y}^t - y^t\|_N \xrightarrow{\mathbb{P}} 0$ by the same kind of induction, from $\hat{y}^1 = y^1$. With $\hat{m}^r = f_r(\hat{y}^r)$, $\hat{y}^{t+1} - y^{t+1} = A(\hat{m}^t - m^t) - (\hat{b}_t - b_t)\hat{m}^{t-1} - b_t(\hat{m}^{t-1} - m^{t-1})$, where $\|A\| \le \lambda\|x\|_N^2 + \|W\|$ and $\|\hat{m}^{t-1}\|_N$ are bounded in probability, $\|\hat{m}^r - m^r\|_N \le L\|\hat{y}^r - y^r\|_N$, and $|\hat{b}_t - b_t| \le \mathrm{Lip}(f_t')\|\hat{y}^t - y^t\|_N + |\frac{1}{N}\sum_i f_t'(y_i^t) - b_t| \xrightarrow{\mathbb{P}} 0$, the last term by the first part applied to $f_t' \in \mathrm{PL}(2)$. ∎

Corollary 6.9 (Bayes-AMP for $\mathbb{Z}_2$ synchronization). *Let $x_1,\dots,x_N$ be independent and uniform on $\{\pm 1\}$, let $\gamma_0 > 0$, and let the side observation $\tilde{y} = \sqrt{\gamma_0}\,x + g$, $g \sim \mathcal{N}(0, I_N)$, be independent of W. Run (6.9) with $m^0 = \tanh(\sqrt{\gamma_0}\,\tilde{y})$, $f_t(y) = \tanh(\lambda y)$, and Onsager coefficients b_t or $\lambda(1-\|m^t\|_N^2)$, and let $\gamma_{t+1} = \lambda^2 q(\gamma_t)$. Then for every $t \ge 0$, $\frac{1}{N}\langle x, m^t\rangle \xrightarrow{\mathbb{P}} q(\gamma_t)$ and $\|m^t\|_N^2 \xrightarrow{\mathbb{P}} q(\gamma_t)$, and for $t \ge 1$ the empirical law of $(\lambda y_i^t, x_i)$ converges, in the sense of Corollary 6.8, to the law of $(\gamma_t X + \sqrt{\gamma_t}Z, X)$.*

PROOF. Apply Corollary 6.8 with $u_i = (x_i, g_i)$, independent copies of (X, G), and $f_0(u) = \tanh(\gamma_0 x + \sqrt{\gamma_0}\,g)$; note that $\frac{1}{N}\sum_i f_t'(\hat{y}_i^t) = \lambda(1 - \|\hat{m}^t\|_N^2)$. By step 1 of Computation 6.1, $\mu_1 = \lambda q(\gamma_0)$ and $\sigma_1^2 = q(\gamma_0)$; and if $\mu_t = \lambda q(\gamma_{t-1})$, $\sigma_t^2 = q(\gamma_{t-1})$, then $\lambda(\mu_t X + \sigma_t Z)$ has the law of $\gamma_t X + \sqrt{\gamma_t}Z$, so $\mu_{t+1} = \lambda q(\gamma_t)$ and $\sigma_{t+1}^2 = q(\gamma_t)$. The limits follow from the functions $x\tanh(\lambda y)$ and $\tanh^2(\lambda y)$, in PL(2), for $t \ge 1$, and from part 2 of Assumption 6.1 for $t = 0$. ∎

The scalar recursion does the rest. Since q is increasing, (γ_t) is monotone and bounded by λ^2, hence converges to a fixed point. The function q is

strictly concave on $[0,\infty)$, with $q(\gamma) = \gamma - \gamma^2 + O(\gamma^3)$: Chapter 4 proves it by writing $1-q$ as a mixture of decaying exponentials, and Deshpande et al. (2016, App. B.2) from $q'(\gamma) = \mathbb{E}\,\mathrm{sech}^4(\gamma + \sqrt{\gamma}Z)$ (Exercise 6.4). So $q(\gamma)/\gamma$ decreases strictly from 1 to 0: for $\lambda > 1$ the only positive fixed point is γ_*, and $\gamma_t \to \gamma_*$ from every $\gamma_0 > 0$; for $\lambda \le 1$, $\gamma_t \to 0$. Hence $\lim_{t\to\infty}\lim_{N\to\infty} \frac{1}{N}\langle x, m^t\rangle = q_*$ in probability, the Bayes-optimal overlap, with the limits in this order. The warm start uses side information; the spectral start removes it.

COMPUTATION 6.2 · PCA AS MESSAGE PASSING, AND THE SPECTRAL START

1 *Linear AMP.* With $f_t(y) = y/c$ in (6.9), $b_t = 1/c$ and $y^{t+1} = Ay^t/c - y^{t-1}/c^2$, whose fixed points satisfy $Ay = (c + c^{-1})y$; for $c = \lambda$ the eigenvalue is the BBP value $\lambda + 1/\lambda$ (Theorem 5.1). By (6.8), $\mu_{t+1} = \lambda\mu_t/c$ and $\sigma^2_{t+1} = (\mu_t^2 + \sigma_t^2)/c^2$, so $r_t = \sigma_t^2/\mu_t^2$ satisfies $r_{t+1} = (1+r_t)/\lambda^2$ and tends to $1/(\lambda^2-1)$ at the rate λ^{-2}: normalized, $\mu^2 = 1 - \lambda^{-2}$, the squared cosine of the top eigenvector; the scale is stationary exactly when $c = \lambda$.

2 *The spectral start.* Montanari and Venkataramanan (2021) start from $y^0 = \sqrt{N}\varphi_1$, φ_1 the top eigenvector of A, read as the fixed point of step 1 with $c = \lambda$: the previous estimate is $f_{-1}(y^{-1}) = y^0/\lambda$, the convention $\mathbb{E}\{Lf(\tilde{X}_{-1}; -1)\} = 1/\lambda$ of their proof. As y^0 behaves as $\mu_0 X + \sigma_0 Z$ with $\mu_0 = \sqrt{1-\lambda^{-2}}$, $\sigma_0 = \lambda^{-1}$, the Bayes estimate is $m^0 = \tanh(\lambda\sqrt{\lambda^2-1}\, y^0)$, at the signal-to-noise ratio $\gamma_0 = \mu_0^2/\sigma_0^2 = \lambda^2 - 1$, and $y^1 = Am^0 - b_0 y^0/\lambda$ with $b_0 = \lambda\sqrt{\lambda^2-1}\,(1 - \|m^0\|_N^2)$.

3 *In the variables of Celentano, Fan and Mei.* With $h^0 = \lambda\sqrt{\lambda^2-1}\, y^0 = \sqrt{\lambda^2(\lambda^2-1)N}\,\varphi_1$ and $h^t = \lambda y^t$ for $t \ge 1$, so that $m^t = \tanh(h^t)$, the first step reads $h^1 = \lambda Am^0 - \lambda^2(1 - \|m^0\|_N^2)\, m^{-1}$ with $m^{-1} = h^0/\lambda^2$. Version 3 of their arXiv paper prints $m^{-1} = \lambda h^0$, in its definition (SI) and in the proof of its Lemma A.7, which derives it from the same fixed-point reading of linear AMP; this multiplies the first Onsager correction by λ^3, and Section 6.6 shows the effect.

The spectral vector depends on all of W, so Theorem 6.4 does not apply; Montanari and Venkataramanan show that given the top eigenpair, A is close in total variation to the eigenpair part plus the spiked matrix with a fresh GOE, projected on the orthogonal complement of φ_1, and argue as we did on the modified matrix:

Theorem 6.10 (Montanari and Venkataramanan 2021, Theorems 1 and 2). *Let $\lambda > 1$, $\|x\|_N \to 1$, and let the empirical law of the coordinates of x converge weakly to a law ν with unit second moment. Run $y^{t+1} = Af_t(y^t) - b_t f_{t-1}(y^{t-1})$, $b_t = \frac{1}{N}\sum_i f_t'(y_i^t)$, with Lipschitz f_t, from $y^0 = \sqrt{N}\varphi_1$, where φ_1 is a top eigenvector of A with $\langle\varphi_1, x\rangle \ge 0$ and the first Onsager term is read as in step 2 of Computation 6.2. Then for $t \ge 0$ and $\psi \in \mathrm{PL}(2)$, almost*

surely $\frac{1}{N}\sum_i \psi(x_i, y_i^t) \to \mathbb{E}\psi(X, \mu_t X + \sigma_t Z)$, *with* $X \sim \nu$, $\mu_0 = \sqrt{1-\lambda^{-2}}$, $\sigma_0 = \lambda^{-1}$ *and* (6.8). *For the Rademacher prior and the Bayes denoisers,* λy^t, $t \geq 1$, *behaves as* $\gamma_t X + \sqrt{\gamma_t} Z$ *with* $\gamma_0 = \lambda^2 - 1$.

6.5.1 What the general theory adds

OTHER MATRICES, DENOISERS AND TIMES. Bayati and Montanari (2011) proved state evolution for compressed sensing and for a general rectangular AMP; their symmetric Theorem 4, our setting with one denoiser for all t, no side information and empirical Onsager coefficients, is stated with its proof omitted as similar. Javanmard and Montanari (2013) treat vector iterates and block-structured matrices with Lipschitz nonlinearities (spatial coupling; they state that generalized AMP is covered, without the reduction), Berthier et al. (2020) non-separable denoisers, and Fan (2022) rotationally invariant matrices, whose Onsager terms involve all past estimates through free cumulants, as Opper et al. (2016) had derived. Universality is proved by moments for polynomial denoisers (Bayati et al. 2015, Thms. 3–4) and by interpolation for sub-Gaussian entries and Lipschitz denoisers, with Lipschitz derivatives once Onsager terms are present, including a spike and the spectral start (Chen and Lam 2021, Thms. 2.1–2.3). For compressed sensing, Rush and Venkataramanan (2018, Thm. 1) keep state evolution for $t = o(\log N / \log\log N)$ iterations; for $\mathbb{Z}_2$ synchronization the preprint of Li and Wei (2022, v2, Thm. 4) tracks a normalized AMP from the spectral start for $o(N(\lambda-1)^{10}/\log^7 N)$ iterations when $1 + \log N / N^{1/16} < \lambda \leq 1.2$.

THE TAP FREE ENERGY. For Ising mixed p-spin models the free energy is the limit of the maximum of the TAP functional over magnetizations whose self-overlap lies to the right of the support of the Parisi measure (Chen and Panchenko 2018, Thm. 1), and, with a generalized Onsager correction, at every point of that support (Chen et al. 2023). For $\mathbb{Z}_2$ synchronization, Fan et al. (2021) bound the number of critical points of F_{TAP} by a Kac–Rice computation (Chapter 13) and show, for λ above an absolute constant λ_0, that its low critical points are near the posterior mean while those of naive mean field are not (their Theorems 1.2, 1.3). Locally, at every $\lambda > 1$:

Theorem 6.11 (Celentano et al. 2023, Theorem 2.1 and Corollary 2.2). *Fix* $\lambda > 1$. *There are* $\varepsilon, \tau > 0$ *and* $r < 1$ *such that for every* $\iota > 0$, *with probability tending to one,* F_{TAP} *has a local minimizer* $m_\star$ *with* $N^{-2}\|m_\star m_\star^\top - \mathbb{E}[xx^\top \mid A]\|_F^2 < \iota$, *unique up to sign among such critical points for small* ι; $\lambda_{\min}(N\nabla^2 F_{\mathrm{TAP}}(m)) > \tau$ *when* $|m - m_\star| \leq \sqrt{\varepsilon N}$; *and the Jacobian of the AMP map at* $(m_\star, m_\star)$ *has spectral radius below* r. *If* $\lambda > \lambda_0$, $\pm m_\star$ *are the global minimizers.*

AMP from the spectral start followed by natural gradient descent converges to $\pm m_\star$ for every $\lambda > 1$, and either alone does when $\lambda > \lambda_0$ (their Theorems 2.3 and 2.4).

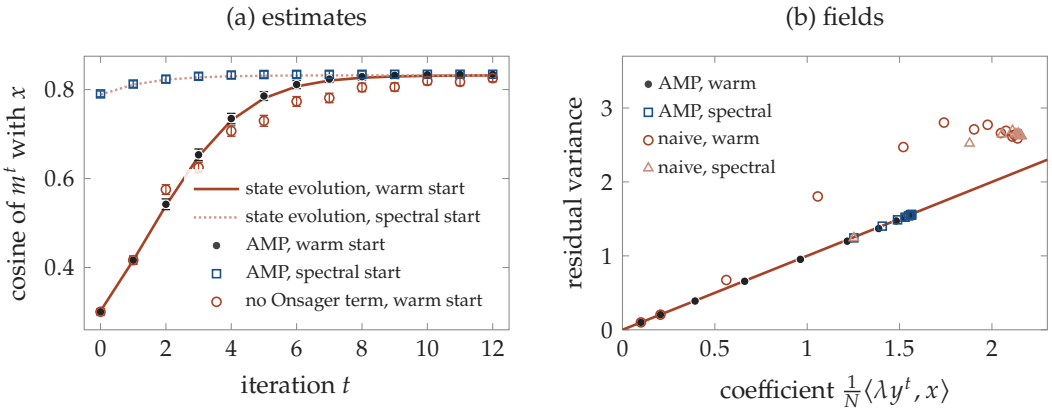


Figure 6.1. Bayes-AMP against state evolution, $N = 3000$, $\lambda = 1.5$, 24 matrices (seeds 6000+r). (a) Cosine of m^t with x, with Student-t 95% intervals, against $\sqrt{q(\gamma_t)}$ (carnelian: the cavity prediction, a theorem by Corollary 6.9 and Theorem 6.10); warm start $\gamma_0 = 0.1$, spectral start $m^{-1} = h^0/\lambda^2$; hollow circles: no Onsager term. (b) Coefficient $\hat{\mu}_t = \frac{1}{N}\langle \lambda y^t, x\rangle$ of the field against its residual variance $\|\lambda y^t - \hat{\mu}_t x\|_N^2$, $t \le 12$, means over the matrices, for AMP and for the naive iteration; state evolution puts every iterate on the diagonal, at (γ_t, γ_t). Script: `figures/scripts/06-amp-iterations.py`.

6.6 Measuring it

`06-amp-iterations.py` runs Bayes-AMP at $N = 3000$, $\lambda = 1.5$, on 24 matrices (seeds $6000 + r$), five ways per matrix: a warm start with $\gamma_0 = 0.1$ (Corollary 6.9), the same without Onsager term, and the spectral start with $m^{-1} = h^0/\lambda^2$ (Theorem 6.10), with $m^{-1} = \lambda h^0$, and without Onsager term. Its iteration is the kit's `amp_z2` (the script checks this); it runs in 28 s. `06-amp-lambda.py` compares AMP with the top eigenvector at $N = 2000$ (43 s), and `06-amp-finite.py` measures the approach to the limit and a non-Gaussian matrix (49 s). Intervals are Student-t 95% intervals over matrices.

Iteration by iteration. In Figure 6.1(a) the state-evolution value lies inside the interval at all 13 iterations, from both starts: from the warm start the cosine climbs from 0.30 to 0.83 (0.653 ± 0.013 against 0.650 at $t = 3$), and from the spectral start it goes from 0.790 ± 0.006 (predicted 0.789) to 0.834 ± 0.007 (predicted 0.832). Figure 6.1(b) shows what state evolution asserts about the field: coefficient and variance both equal γ_t (0.662 and 0.656 against $\gamma_3 = 0.654$ from the warm start).

Without the Onsager term. The naive iteration $y^{t+1} = Am^t$ agrees with AMP for one step and leaves state evolution at the second: from the warm start its cosine at $t = 2$ is 0.575 ± 0.011 against 0.539, and its field has coefficient 0.562 and variance 0.673, off the diagonal, where Corollary 6.8 itself predicts 0.559 and 0.677 (Exercise 6.8). Its cosine then climbs in alternating long and short steps; from the spectral start it settles at 0.824 with coefficient 2.16 and variance 2.62. On the same matrices AMP beats it at $t = 12$ by 0.010 ± 0.003 (spectral start) and 0.009 ± 0.004 (warm start),

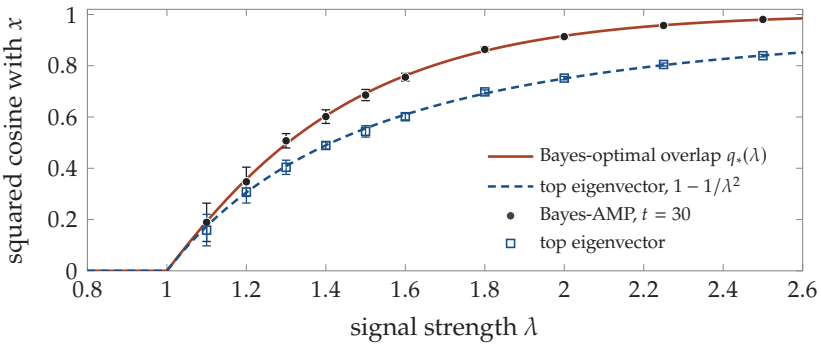


Figure 6.2. Squared cosine with x of Bayes-AMP after 30 iterations from the spectral start, and of the top eigenvector, $N = 2000$, ten matrices per λ (seeds $7000 + 100j + r$), Student-t 95% intervals. Carnelian: the fixed point $q_*(\lambda)$ of (6.2), Bayes-optimal by Theorem 4.1. Lapis dashed: $1 - 1/\lambda^2$ (Theorem 5.1). Script: `figures/scripts/06-amp-lambda.py`.

paired: the finite-λ face of the inconsistency of naive mean field (Fan et al. 2021).

THE SPECTRAL CONVENTION. With $m^{-1} = h^0/\lambda^2$ the iterates follow state evolution from the first step (Figure 6.1(a): 0.812 ± 0.007 against 0.811 at $t = 1$). With $m^{-1} = \lambda h^0$, as printed in version 3 of the arXiv paper of Celentano, Fan and Mei, the first step is wrong: the field at $t = 1$ has coefficient 0.284 and variance 0.163, against $\gamma_1 = 1.399$ for both, and the cosine falls from 0.790 to 0.580 ± 0.028 at $t = 1$ and to -0.472 ± 0.040 at $t = 2$. The iteration still settles, near $-m_\star$ (cosine -0.834 ± 0.007 at $t = 12$), so the wrong scaling is easy to miss; their proofs take state evolution from the theorem of Montanari and Venkataramanan, whose proof uses $m^{-1} = h^0/\lambda^2$.

WHAT AMP REACHES. In Figure 6.2 AMP after 30 iterations agrees with $q_*(\lambda)$, and the top eigenvector with $1 - 1/\lambda^2$, within the intervals at all ten values of λ (at $\lambda = 2$: 0.913 ± 0.008 against 0.917, and 0.752 ± 0.007 against 0.750). The intervals widen near $\lambda = 1$, where $\lambda - 1$ approaches the width $N^{-1/3}$ of the window of Chapter 5 over which the transition is smeared, and AMP inherits the fluctuations of its start. The gain of AMP is largest near $\lambda = 1.9$ (0.17 in squared cosine); near $\lambda = 1$ the curves agree to first order (Exercise 6.10).

THE APPROACH TO THE LIMIT, AND UNIVERSALITY. Theorem 6.4 has no rate. At the fixed iteration $t = 3$ from the warm start, the mean overlap lies within its interval of $q(\gamma_3) = 0.4222$ for $N = 500, 1000, 2000, 4000$ (200, 100, 60, 30 matrices), and $\sqrt{N}$ times its root-mean-square deviation is 1.57, 1.53, 1.37, 1.92: fluctuations of order $N^{-1/2}$. With independent entries $\pm 1/\sqrt{N}$ in place of W ($N = 3000$, 20 matrices) the predictions lie inside all 13 intervals: our proof does not cover that matrix, but the theorem of Chen and Lam (2021) does.

6.7 The status board

State evolution at fixed t is a theorem for Gaussian matrices; universality holds under extra hypotheses; AMP reaches the Bayes-optimal overlap in $\mathbb{Z}_2$ synchronization, $t \to \infty$ after $N \to \infty$. Long times and the global landscape are open.

STATUS BOARD · the cavity ↔ Gaussian conditioning

PREDICTION	STATUS	PROOF OR EVIDENCE
State evolution at fixed t: GOE, Lipschitz denoisers	proved	Bolthausen 2014; Bayati–Montanari 2011; Javanmard–Montanari 2013; Theorem 6.4
The same with a spectral start, non-separable denoisers, or rotationally invariant noise	proved	Montanari–Venkataramanan 2021; Berthier et al. 2020; Fan 2022
Universality: non-Gaussian independent entries	proved under	polynomial denoisers (Bayati et al. 2015); sub-Gaussian entries (Chen–Lam 2021)
State evolution for t growing with N	proved under	$t = o(\log N / \log\log N)$, compressed sensing (Rush–Venkataramanan 2018)
$\mathbb{Z}_2$ synchronization up to $N/\mathrm{polylog}\, N$ iterations	preprint	Li–Wei, arXiv:2208.03313v2, Thm. 4: a normalized AMP, $1 + \log N / N^{1/16} < \lambda \le 1.2$
TAP equations at high temperature; Bolthausen's iteration below the AT line	proved under	Talagrand 2011, §1.7; Bolthausen 2014, Thm. 2.1 ($h > 0$); Chen–Tang 2021
TAP free energy = free energy (Ising mixed p-spin)	proved	Chen–Panchenko 2018; Chen–Panchenko–Subag 2023 (generalized correction)
Bayes-AMP reaches the Bayes risk in $\mathbb{Z}_2$ synchronization ($t \to \infty$ after $N \to \infty$), at a local TAP minimizer	proved	Deshpande et al. 2016; Montanari–Venkataramanan 2021 with Lelarge–Miolane 2019; Celentano et al. 2023
That minimizer is global for every $\lambda > 1$	conjectured	numerics (Celentano et al. 2023, Rem. 4.5); proved for $\lambda > \lambda_0$
Naive mean field finds the posterior mean	false	Fan et al. 2021, Thm. 1.3 ($\lambda > \lambda_0$); observed at $\lambda = 1.5$

6.8 Where the translation fails

What was assumed. The theorem needs a Gaussian matrix, where the physics needs only independent entries of the right variance; a fixed number of iterations, where the physics speaks of fixed points; Lipschitz denoisers, an initialization independent of W, and a non-degeneracy condition that the physics never meets. Bayes-AMP needs the prior and λ: with a misspecified prior state evolution still holds, but the Nishimori relations and the one-dimensional recursion are lost. The model is idealized: a rank-one signal, Gaussian noise, $N \to \infty$.

Where the physics is wrong, or not yet right. The naive mean-field equations are wrong: their critical points stay away from the posterior mean when $\lambda > \lambda_0$ (Fan et al. 2021, Thm. 1.3), and at $\lambda = 1.5$ the measured naive iteration settles below AMP. At low temperature the derivation fails as stated, because the Gibbs measure is not one pure state: the TAP equations have many solutions (Bray and Moore 1980), and Bolthausen's iteration, whose overlap recursion loses stability exactly at the de Almeida–Thouless line (Exercise 6.6), does not converge beyond it. For the pieces of an approximately ultrametric decomposition of the Gibbs measure, the TAP equations hold at every site for generic models whose Parisi measure has a jump at the top of its support (Auffinger and Jagannath 2019), and the generalized Onsager correction reduces to the classical one only where Plefka's condition holds (Chen et al. 2023). The physicist's $t = \infty$ took Celentano et al. (2023) a separate argument; and the wrongly scaled spectral start of Section 6.6 shows how delicate the bookkeeping of the reaction term is.

Open problems. State evolution for many more than $\log N / \log\log N$ iterations, for general denoisers and an initialization independent of W. The conjectures of Celentano et al. (2023, Remarks 2.5 and 4.5) that AMP alone converges to $m_\star$, and that $m_\star$ minimizes F_{TAP} globally, for every $\lambda > 1$; for a normalized AMP and $\lambda \le 1.2$, the preprint of Li and Wei (2022) shows that no refinement stage is needed. A proof of the TAP representation of the free energy that avoids the Parisi formula, which Chen and Panchenko (2018) ask for, and of the sufficiency of Plefka's condition.

Exercises

6.1 consolidation. (Blocks of a GOE matrix.) Check that $\mathbb{E} W_{ab} W_{cd}$ equals $\frac{1}{N}(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$, and show that for a fixed projection Π of rank k the block $\Pi^\perp W \Pi^\perp$, in an orthonormal basis of the range of $\Pi^\perp$, is $\sqrt{(N-k)/N}$ times a GOE$(N-k)$ matrix.

6.2 consolidation. (The Bayes denoiser.) For $Y = \mu X + \sigma Z$ with X uniform on $\{\pm 1\}$, show $\mathbb{E}[X \mid Y] = \tanh(\mu Y/\sigma^2)$; deduce that $\tanh(\lambda y)$ is the posterior mean in Corollary 6.9 because $\mu_t = \lambda \sigma_t^2$, and evaluate the limit of $\lambda(1 - \|m^t\|_N^2)$ at the fixed point for $\lambda = 1.5$.

6.3 CONSOLIDATION. (Linear denoisers.) Verify step 1 of Computation 6.2. Then show that for a Gaussian input, $X \sim \mathcal{N}(0,1)$, the error is $\mathrm{mmse}(\gamma) = 1/(1+\gamma)$, so that $\gamma_{t+1} = \lambda^2\gamma_t/(1+\gamma_t)$; find the fixed point and the slope there, and conclude that Bayes-AMP then does no better than the top eigenvector.

6.4 CONSOLIDATION. (The concavity of q.) For $T = \tanh(\gamma + \sqrt{\gamma}Z)$, show that $\mathbb{E}T^{2k-1} = \mathbb{E}T^{2k}$ for $k \geq 1$, and, by Gaussian integration by parts, that $q'(\gamma) = \mathbb{E}(1-T^2)^2 = \mathbb{E}\,\mathrm{sech}^4(\gamma + \sqrt{\gamma}Z)$. Deduce that q is strictly concave on $[0,\infty)$: $|\gamma + \sqrt{\gamma}Z|$ is stochastically increasing in γ, and sech^4 is even and decreasing on $[0,\infty)$.

6.5 BY HAND. (TAP for SK.) For the Hamiltonian of Chapter 3 with field h, derive $m_i = \tanh(h + \frac{\beta}{\sqrt{N}}\sum_j g_{ij}m_j - \beta^2(1-q)m_i)$ by the cavity method, and check that Bolthausen's iteration $m^{(k+1)} = \tanh(h + \beta g m^{(k)} - \beta^2(1-q)m^{(k-1)})$, with couplings $g_{ij}/\sqrt{N}$, is an AMP.

6.6 BY HAND. ⋆ (Bolthausen's iteration and the AT line.) For $h > 0$ let $\mathrm{Th}(x) = \tanh(h+\beta x)$, let q solve $q = \mathbb{E}\,\mathrm{Th}(\sqrt{q}Z)^2$, and let

$$\psi(t) = \mathbb{E}\,\mathrm{Th}\big(\sqrt{t}Z + \sqrt{q-t}Z'\big)\,\mathrm{Th}\big(\sqrt{t}Z + \sqrt{q-t}Z''\big), \qquad t \in [0,q].$$

(a) Show $\psi(q) = q$ and, by Gaussian integration by parts, $\psi'(q) = \beta^2\mathbb{E}\cosh^{-4}(h + \beta\sqrt{q}Z)$. (b) Bolthausen shows that the overlaps of his iterates follow $\rho_{k+1} = \psi(\rho_k)$ from a point of $(0,q)$, and that ψ is increasing and convex with $\psi(0) > 0$. Show that $\rho_k \to q$ under the AT condition $\psi'(q) \leq 1$, and that otherwise ρ_k tends to a fixed point below q. (c) Check: at $\beta = 0.8$, $h = 0.3$, $q = 0.1404$, $\psi'(q) = 0.486$; at $\beta = 1.5$, $h = 0.3$, $q = 0.3880$, $\psi'(q) = 1.059$, $\rho_k \to 0.2932$. *(Hint: compare ψ with its tangent at q.)*

6.7 TRANSLATION. (The second iteration by hand.) Take $t = 1$ in the proof of Theorem 6.4: show that $s^2 = a_0 s^1 + \Pi_1^{\perp} W \Pi_1^{\perp} m^1 + R$ with $a_0 = \langle m^0, m^1\rangle/|m^0|^2$, $R = (c_0/\|m^0\|_N^2 - b_1)m^0$ and $c_0 = \frac{1}{N}\langle s^1, m_\perp^1\rangle \to \Gamma_{00}b_1$, and name the line of the cavity derivation that each step replaces.

6.8 TRANSLATION. (The naive iteration after one step.) Show that without Onsager term the second iterate is $y^2 + b_1 m^0$, with empirical law tending to that of $\mu_2 X + \sigma_2 Z + b_1 f_0(U)$, not of the form $\mu X + \sigma Z$. For the warm start of Corollary 6.9, show that its field has coefficient $\gamma_2 + \lambda^2(1-q_1)q_0$ and residual variance $\gamma_2 + \lambda^4(1-q_1)^2 q_0(1-q_0)$, $q_r = q(\gamma_r)$, and check 0.559 and 0.677 at $\lambda = 1.5$, $\gamma_0 = 0.1$.

6.9 TRANSLATION. ⋆ (Denoisers with memory.) For $m^t = \eta_t(s^1,\dots,s^t,u)$, show that the proof of Theorem 6.4 goes through for $s^{t+1} = Wm^t - \sum_{r=1}^t b_{t,r}m^{r-1}$, $b_{t,r} = \mathbb{E}\partial_r\eta_t(Z_1,\dots,Z_t,U)$. What replaces (6.6)?

6.10 BY HAND. (Near the threshold.) From $q(\gamma) = \gamma - \gamma^2 + \frac{5}{3}\gamma^3 + O(\gamma^4)$, show $q_* = \delta + \frac{2}{3}\delta^2 + O(\delta^3)$ with $\delta = 1 - 1/\lambda^2$; at $\lambda = 1.05$, $\delta + \frac{2}{3}\delta^2 = 0.09873$ and $q_* = 0.09842$.

6.11 MEASUREMENT. (Universality and its limits.) In `06-amp-finite.py`, say which step of the proof fails for entries $\pm 1/\sqrt{N}$, then try heavy-tailed entries (Student t with three degrees of freedom, scaled to variance $1/N$) at $N = 500$, 1000, 2000 and 3000, and fit the mean deviation at $t = 3$ to $CN^{-\theta}$.

6.12 RESEARCH. (Long times.) The preprint of Li and Wei (2022) tracks a normalized AMP from the spectral start for $N/\operatorname{polylog} N$ iterations when $\lambda \le 1.2$. Decide whether the Bayes-AMP of Corollary 6.9, with its deterministic Onsager coefficients and its warm start, follows state evolution for $t \le N^c$ iterations at every $\lambda > 1$, by adapting their decomposition or by finding where it fails (Celentano et al. 2023).

Notes and References

Thouless et al. (1977) proposed the TAP equations, Plefka (1982) derived them and the TAP functional from an expansion in the couplings, and Mézard et al. (1986) gave the cavity derivation. Kabashima (2003) reduced dense belief propagation to N numbers for CDMA detection; Donoho et al. (2009) derived approximate message passing for compressed sensing, named its correction after the Onsager reaction term, and introduced state evolution as an empirical finding of extensive simulations. Mézard and Montanari (2009) is a textbook account of belief propagation and the cavity method; Zdeborová and Krzakala (2016) review their use in inference.

Bolthausen's conditioning (Bolthausen 2014) became a general proof in Bayati and Montanari (2011); the tutorial of Feng et al. (2022) unifies its extensions, and Montanari and Sen (2024) and the 2024 lecture notes of Krzakala and Zdeborová (2024) set the cavity derivation next to Bolthausen's argument, which Montanari and Sen sketch. The TAP equations are theorems at high temperature (Talagrand 2011), the TAP free energy a variational formula for the free energy (Chen and Panchenko 2018), and its landscape in $\mathbb{Z}_2$ synchronization is partly mapped (Celentano et al. 2023; Fan et al. 2021); Chen and Tang (2021) showed that Bolthausen's iteration converges to the magnetizations when the overlap concentrates. Conditioning along an iteration is also a rigorous route to dynamical mean-field theory (Chapter 14); AMP for the LASSO and for generalized linear models appears in Chapters 7 and 10.

The organization of the proof of Theorem 6.4 around one induction hypothesis, with side information and the variance bound of Step 4, is this book's arrangement of the arguments of Bolthausen and of Bayati and Montanari. Celentano, Fan and Mei read the spectral start as a fixed point of linear AMP in the proof of their Lemma A.7; Computation 6.2 carries out that reading with the scaling of Montanari and Venkataramanan's proof and obtains $m^{-1} = h^0/\lambda^2$, where version 3 of their arXiv paper prints λh^0. We checked this against their text and by simulation, not against the published version.

TRANSLATING THE LITERATURE. Bolthausen (2014) writes the SK Hamiltonian with a minus sign and couplings $g_{ij} \sim \mathcal{N}(0, 1/N)$, our $g_{ij}/\sqrt{N}$, with

zero diagonal; his $\langle x, y\rangle$ is our $\frac{1}{N}\langle x, y\rangle$. In the symmetric case of Bayati and Montanari (2011), $G = A + A^*$ with $A_{ij} \sim \mathcal{N}(0, 1/(2N))$ is our W, and, as their iteration starts from $m^0 = 0$, their h^{t+1}, m^t, λ_t, τ_t^2 are our s^t, m^{t-1}, empirical b_{t-1}, $\Gamma_{t-1,t-1}$. Montanari and Venkataramanan (2021) write x^t for our y^t, but their Bayes-AMP builds λ into $f_t = \lambda F(\cdot; \gamma_t)$, so its iterate is our λy^t, the h^t of Celentano et al. (2023), who write Y for A, $Q(m)$ for $\|m\|_N^2$, and minimize $F_{\mathrm{TAP}} = -\Phi_{\mathrm{TAP}}$. Lelarge and Miolane (2019) use the signal-to-noise ratio λ^2.

References

Auffinger, A. and A. Jagannath (2019). "Thouless–Anderson–Palmer equations for generic p-spin glasses". In: *The Annals of Probability* 47.4, pp. 2230–2256.

Bayati, M., M. Lelarge, and A. Montanari (2015). "Universality in polytope phase transitions and message passing algorithms". In: *The Annals of Applied Probability* 25.2, pp. 753–822. arXiv: 1207.7321.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: 1001.3448.

Berthier, R., A. Montanari, and P.-M. Nguyen (2020). "State evolution for approximate message passing with non-separable functions". In: *Information and Inference: A Journal of the IMA* 9.1, pp. 33–79. arXiv: 1708.03950.

Bolthausen, E. (2014). "An Iterative Construction of Solutions of the TAP Equations for the Sherrington–Kirkpatrick Model". In: *Communications in Mathematical Physics* 325.1, pp. 333–366. arXiv: 1201.2891.

Bray, A. J. and M. A. Moore (1980). "Metastable states in spin glasses". In: *Journal of Physics C: Solid State Physics* 13.19, pp. L469–L476.

Celentano, M., Z. Fan, and S. Mei (2023). "Local convexity of the TAP free energy and AMP convergence for $\mathbb{Z}_2$-synchronization". In: *The Annals of Statistics* 51.2, pp. 519–546. arXiv: 2106.11428.

Chen, W.-K. and W.-K. Lam (2021). "Universality of approximate message passing algorithms". In: *Electronic Journal of Probability* 26, pp. 1–44. arXiv: 2003.10431.

Chen, W.-K. and D. Panchenko (2018). "On the TAP free energy in the mixed p-spin models". In: *Communications in Mathematical Physics* 362.1, pp. 219–252.

Chen, W.-K., D. Panchenko, and E. Subag (2023). "Generalized TAP Free Energy". In: *Communications on Pure and Applied Mathematics* 76.7, pp. 1329–1415.

Chen, W.-K. and S. Tang (2021). "On Convergence of the Cavity and Bolthausen's TAP Iterations to the Local Magnetization". In: *Communications in Mathematical Physics* 386.2, pp. 1209–1242. arXiv: 2011.00495.

Deshpande, Y., E. Abbe, and A. Montanari (2016). "Asymptotic mutual information for the balanced binary stochastic block model". In: *Information and Inference: A Journal of the IMA*. Published online 2016-12-26; in print as volume 6, issue 2 (2017), pages 125–170. arXiv: 1507.08685.

Donoho, D. L., A. Maleki, and A. Montanari (2009). "Message-passing algorithms for compressed sensing". In: *Proceedings of the National Academy of Sciences* 106.45, pp. 18914–18919. arXiv: 0907.3574.

Fan, Z. (2022). "Approximate Message Passing algorithms for rotationally invariant matrices". In: *The Annals of Statistics* 50.1, pp. 197–224. arXiv: 2008.11892.

Fan, Z., S. Mei, and A. Montanari (2021). "TAP free energy, spin glasses and variational inference". In: *The Annals of Probability* 49.1, pp. 1–45.

Feng, O. Y., R. Venkataramanan, C. Rush, and R. J. Samworth (2022). "A Unifying Tutorial on Approximate Message Passing". In: *Foundations and Trends in Machine Learning* 15.4, pp. 335–536. arXiv: 2105.02180.

Javanmard, A. and A. Montanari (2013). "State evolution for general approximate message passing algorithms, with applications to spatial coupling". In: *Information and Inference* 2.2, pp. 115–144. arXiv: 1211.5164.

Kabashima, Y. (2003). "A CDMA multiuser detection algorithm on the basis of belief propagation". In: *Journal of Physics A: Mathematical and General* 36.43, pp. 11111–11121.

Krzakala, F. and L. Zdeborová (2024). *Statistical Physics Methods in Optimization and Machine Learning: An Introduction to Replica, Cavity & Message-Passing techniques*. Lecture notes, https://sphinxteam.github.io/EPFLDoctoralLecture2021/Notes.pdf. Unpublished draft, version of 9 June 2024.

Lelarge, M. and L. Miolane (2019). "Fundamental limits of symmetric low-rank matrix estimation". In: *Probability Theory and Related Fields* 173.3-4, pp. 859–929.

Li, G. and Y. Wei (2022). *A Non-Asymptotic Framework for Approximate Message Passing in Spiked Models*. Preprint, version 2 (2023). arXiv: 2208.03313.

Mézard, M., G. Parisi, and M. A. Virasoro (1986). "SK Model: The Replica Solution without Replicas". In: *Europhysics Letters* 1.2, pp. 77–82.

Mézard, M. and A. Montanari (2009). *Information, Physics, and Computation*. Oxford Graduate Texts. Oxford: Oxford University Press.

Montanari, A. and S. Sen (2024). "A Friendly Tutorial on Mean-Field Spin Glass Techniques for Non-Physicists". In: *Foundations and Trends in Machine Learning* 17.1, pp. 1–173. arXiv: 2204.02909.

Montanari, A. and R. Venkataramanan (2021). "Estimation of low-rank matrices via approximate message passing". In: *The Annals of Statistics* 49.1, pp. 321–345. arXiv: 1711.01682.

Onsager, L. (1936). "Electric Moments of Molecules in Liquids". In: *Journal of the American Chemical Society* 58.8, pp. 1486–1493.

Opper, M., B. Çakmak, and O. Winther (2016). "A theory of solving TAP equations for Ising models with general invariant random matrices". In: *Journal of Physics A: Mathematical and Theoretical* 49.11, p. 114002.

Plefka, T. (1982). "Convergence condition of the TAP equation for the infinite-ranged Ising spin glass model". In: *Journal of Physics A: Mathematical and General* 15.6, pp. 1971–1978.

Rush, C. and R. Venkataramanan (2018). "Finite Sample Analysis of Approximate Message Passing Algorithms". In: *IEEE Transactions on Information Theory* 64.11, pp. 7264–7286. arXiv: 1606.01800.

Talagrand, M. (2011). *Mean Field Models for Spin Glasses: Volume I: Basic Examples*. Vol. 54. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Thouless, D. J., P. W. Anderson, and R. G. Palmer (1977). "Solution of 'Solvable model of a spin glass'". In: *Philosophical Magazine* 35.3, pp. 593–601.

Zdeborová, L. and F. Krzakala (2016). "Statistical physics of inference: thresholds and algorithms". In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

7 Generalized Linear Models

The teacher–student perceptron and the hard phase

THE DICTIONARY —— the replica formula ↔ adaptive interpolation

IN THIS CHAPTER. A teacher perceptron with weights ±1 labels $n = \alpha d$ Gaussian patterns, and a Bayesian student tries to recover the weights. Replicas predict that the error falls continuously until $\alpha_{\mathrm{IT}} = 1.2447$ and then jumps to zero, while message passing stays on a metastable branch until $\alpha_{\mathrm{AMP}} = 1.4926$. Barbier et al. (2019) proved the formula behind the first prediction by adaptive interpolation, and Celentano et al. (2020) the second for first-order methods. We prove the identity on which their proof rests (free entropy equals replica potential plus a remainder that vanishes when the overlap concentrates), the bounds $1 \le \alpha_{\mathrm{IT}} \le 1.4480$ by counting the version space, and that overlap concentration yields the physicists' Gibbs and Bayes errors and exact recovery above α_{IT}. We recompute α_{IT} in high precision (the paper prints 1.249, which its formula does not give) and measure GAMP against state evolution at $d = 2000$. Whether some efficient algorithm learns between α_{IT} and α_{AMP} is open.

7.1 The question: how many examples does a student need?

A perceptron with weights $w \in \mathbb{R}^d$ labels a pattern $x \in \mathbb{R}^d$ by the sign of $\langle x \rangle w$. In the *teacher–student scenario*, which Barbier et al. (2019) credit to Gardner and Derrida (1989) for the perceptron, a teacher w^* labels n random patterns, and a student who sees patterns and labels tries to recover w^*. How many examples does the student need, and how well does it then label a new pattern? Statisticians call the model a *generalized linear model*: the labels depend on w^* only through the projections $\langle x_\mu, w^* \rangle$. We take

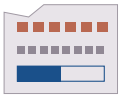
$$w^* \sim P_0^{\otimes d}, \qquad x_\mu \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_\mu \sim P_{\mathrm{out}}\Big(\cdot \,\Big|\, \frac{\langle x_\mu, w^* \rangle}{\sqrt{d}}\Big), \tag{7.1}$$

for $\mu \le n$, with $n/d \to \alpha$. The kernel P_{out} is the *output channel*; equivalently $y_\mu = \varphi_{\mathrm{out}}(\langle x_\mu, w^* \rangle / \sqrt{d}, A_\mu)$ with independent noise A_μ. The running example is the noiseless perceptron, $y_\mu = \operatorname{sgn}\langle x_\mu, w^* \rangle$, with two teachers: the *Ising teacher*, w^* uniform on $\{-1, 1\}^d$, and the *spherical teacher*, $w^* \sim \mathcal{N}(0, I_d)$.[1]

The student is Bayes-optimal: it knows P_0 and P_{out} and computes the posterior $P(w \mid X, y) \propto P_0^{\otimes d}(w) \prod_\mu P_{\mathrm{out}}(y_\mu \mid \langle x_\mu \rangle w / \sqrt{d})$, whose normalizer $\mathcal{Z}(X, y)$, the evidence, is the partition function. For the Ising teacher and

[1] Physicists put the spherical teacher on the sphere of radius $\sqrt{d}$; the Gaussian prior, which Barbier et al. (2019) use, gives the same limits because $\|w^*\|^2/d \to 1$.

PHYSICS
NUMERICS
PROVED IN PART



the noiseless channel the posterior is uniform on the *version space* $\mathcal{V}$, the set of $w \in \{-1,1\}^d$ with $\operatorname{sgn}\langle x_\mu\rangle w = y_\mu$ for every $\mu \le n$, and $\mathcal{Z} = |\mathcal{V}|/2^d$. Physicists call sampling from it *Gibbs learning* at zero temperature (Seung et al. 1992); with the teacher's prior, it is exact Bayesian inference.

The *free entropy* $f_d = \frac{1}{d}\mathbb{E}\log\mathcal{Z}(X,y)$, the expectation over teacher, patterns and labels, is minus the mutual information per weight between teacher and data when the channel is noiseless;[2] for the Ising teacher $f_d + \log 2 = \frac{1}{d}\mathbb{E}\log|\mathcal{V}|$ is the entropy of the version space. The *overlap* $Q = \frac{1}{d}\langle w, w^*\rangle$ of a posterior sample w with the teacher is the book's m. For a fresh pattern, the *Gibbs error* ε_{G} is the probability that a posterior sample labels it differently from the teacher, and the *Bayes error* ε_{B} the smallest probability of error of a prediction from the data. Learning is *perfect* when the overlap tends to one.

[2] We write f_d, as Barbier et al. (2019) do, rather than the book's generic ϕ, because ϕ and Φ denote the standard normal density and distribution function in this chapter.

THE PHYSICAL PREDICTION · Gardner and Derrida 1989; Györgyi 1990; Sompolinsky, Tishby and Seung 1990; Opper and Haussler 1991. The overlap concentrates on a number $q_*(\alpha)$, the solution of the variational problem (7.2), and $\varepsilon_{\mathrm{G}} = \frac{1}{\pi}\arccos q_*$, $\varepsilon_{\mathrm{B}} = \frac{1}{\pi}\arccos\sqrt{q_*}$. For the Ising teacher the overlap rises continuously along one branch until $\alpha_{\mathrm{GD}} = 1.245$, where the student freezes into the teacher and the error drops from a positive value to zero: a first-order transition to perfect generalization. The branch survives as a metastable state up to a spinodal near $\alpha = 1.49$. For the spherical teacher the errors decay like $1/\alpha$, and the Gibbs error exceeds the Bayes error by a factor that tends to $\sqrt{2}$.

Györgyi's abstract attributes 1.245 to Gardner and Derrida (1989); the spinodal 1.493 is Györgyi's, as Barbier et al. (2019) report (of these early papers we have read only the abstracts). How do physicists derive these statements, and what is proved? And between the transition and the spinodal, where the data determine the teacher but the algorithms of physics do not find it, is the obstacle in those algorithms or in every efficient one?

7.2 The physics: replicas with a teacher

Györgyi (1990) and Sompolinsky et al. (1990) computed the entropy of the version space and the errors with replicas, and Opper and Haussler (1991) the error of the Bayes-optimal classifier. We follow the modern form for a general channel (Barbier et al. 2019; Zdeborová and Krzakala 2016), in which the teacher is one more replica (Chapter 4).

THE REPLICA COMPUTATION WITH A TEACHER. *Replicate.* For discrete outputs the probability of the labels is the evidence itself, $P(y \mid X) = \mathcal{Z}(X,y)$,

so for an integer $k \geq 1$

$$\mathbb{E}\mathcal{Z}^k = \mathbb{E}_X \sum_y \mathcal{Z}(X,y)^{k+1}$$
$$= \mathbb{E}_{w^0,\dots,w^k} \prod_{\mu=1}^{n} \mathbb{E}_{x_\mu} \sum_y \prod_{a=0}^{k} P_{\text{out}}\Big(y \;\Big|\; \frac{\langle x_\mu, w^a\rangle}{\sqrt{d}}\Big),$$

with $w^0, \dots, w^k$ independent draws from the prior: the teacher has become replica 0.

Order parameters. Given the replicas, the fields $\langle x_\mu, w^a\rangle/\sqrt{d}$ are Gaussian with covariance $Q_{ab} = \langle w^a, w^b\rangle/d$, with $Q_{aa} = 1$ for the Ising prior, so each pattern contributes the factor $\mathcal{I}_k(Q) = \sum_y \mathbb{E}_{s\sim\mathcal{N}(0,Q)} \prod_a P_{\text{out}}(y \mid s^a)$. Fixing Q by delta functions with conjugate parameters $\hat{Q}_{ab}$, and writing $\mathbb{E}$ for the average over $k+1$ independent signs, Laplace's method in d gives $\frac{1}{d}\log \mathbb{E}\mathcal{Z}^k \approx \operatorname{extr} \Lambda_k(Q, \hat{Q})$ with

$$\Lambda_k = -\sum_{a<b} \hat{Q}_{ab} Q_{ab} + \log \mathbb{E} e^{\sum_{a<b} \hat{Q}_{ab} w^a w^b} + \alpha \log \mathcal{I}_k(Q).$$

Replica symmetry with a teacher. Teacher and replicas are exchangeable, $k+1$ samples from the posterior, so take $Q_{ab} = q$ and $\hat{Q}_{ab} = r$ for all $a < b$. Then $s^a = \sqrt{q}\, V + \sqrt{1-q}\, u^a$ with independent standard normals, and linearizing $\sum_{a<b} w^a w^b = \frac{1}{2}(\sum_a w^a)^2 - \frac{k+1}{2}$ with a Gaussian Z gives $\mathbb{E} e^{r\sum_{a<b} w^a w^b} = \mathbb{E}\mathcal{Z}_0(Z)^{k+1}$ and $\mathcal{I}_k = \mathbb{E}_V \sum_y \mathcal{Z}_{\text{out}}(y,V)^{k+1}$, with $\mathcal{Z}_0(Z) = e^{-r/2}\cosh(\sqrt{r}Z)$ and $\mathcal{Z}_{\text{out}}(y,V) = \mathbb{E}_u P_{\text{out}}(y \mid \sqrt{q}V + \sqrt{1-q}\,u)$.

Continue to $k \to 0$. Both are normalizations ($\mathbb{E}\mathcal{Z}_0 = 1$, $\mathbb{E}_V \sum_y \mathcal{Z}_{\text{out}} = 1$), so to first order they are $1 + k\mathbb{E}[\mathcal{Z}_0 \log \mathcal{Z}_0]$ and $1 + k\mathbb{E}_V \sum_y \mathcal{Z}_{\text{out}} \log \mathcal{Z}_{\text{out}}$. With $\frac{k(k+1)}{2}$ pairs and $f_d \approx \frac{1}{kd}\log \mathbb{E}\mathcal{Z}^k$, the free entropy is the value at a stationary point of

$$f_{\text{RS}}(q,r) = \psi(r) + \alpha\Psi(q) - \frac{rq}{2}, \tag{7.2}$$

where, by the shift $\mathbb{E}[\mathcal{Z}_0(Z) g(Z)] = \mathbb{E} g(Z + \sqrt{r}\, w^*)$ for a uniform sign w^*,

$$\psi(r) = \mathbb{E}\log\cosh(r + \sqrt{r}Z) - \frac{r}{2},$$
$$\Psi(q) = \mathbb{E}_V \sum_y \mathcal{Z}_{\text{out}}(y,V) \log \mathcal{Z}_{\text{out}}(y,V). \quad \diamond \tag{7.3}$$

Both functions are free entropies of scalar problems: ψ that of a sign w^* seen through the Gaussian channel $\sqrt{r}\, w^* + Z$ of Chapter 1, and Ψ that of the output channel when a part $\sqrt{q}\, V$ of the field is known. The stationarity conditions are

$$q = 2\psi'(r) = \mathbb{E}\tanh\big(r + \sqrt{r}Z\big), \qquad r = 2\alpha\Psi'(q), \tag{7.4}$$

the first by the I-MMSE relation and the Nishimori identity (Theorem 1.12 and Lemma 1.10). Five steps are not justified: the continuation from

integers to $k \to 0$, the exchange of $k \to 0$ with $d \to \infty$, Laplace's method with a non-integer number of replicas, the replica-symmetric ansatz, and the choice among stationary points, where physicists take the largest free entropy; the rigorous form of this rule is a supremum over q of an infimum over r (Theorem 7.5).

COMPUTATION 7.1 · THE SIGN CHANNEL

1 For $y = \operatorname{sgn} s$ and $a = \sqrt{q/(1-q)}$, $\mathcal{Z}_{\text{out}}(y, V) = \mathbb{P}_u(y(\sqrt{q}V + \sqrt{1-q}\,u) > 0) = \Phi(yaV)$. Summing over y and using $V \mapsto -V$, $\Psi(q) = 2\mathbb{E}_V[\Phi(aV)\log\Phi(aV)]$, with $\Psi(0) = -\log 2$ and $\Psi(1) = 0$.

2 Let $G(x) = \Phi(x)\log\Phi(x)$. As $\frac{\mathrm{d}}{\mathrm{d}q}a^2 = (1-q)^{-2}$, Stein's identity $\mathbb{E}[Vh(V)] = \mathbb{E}h'(V)$ gives $\Psi'(q) = (1-q)^{-2}\mathbb{E}_V[G''(aV)]$, and $G''(x) = \phi(x)^2/\Phi(x) - x\phi(x)(1 + \log\Phi(x))$.

3 By symmetry $\mathbb{E}_V[aV\phi(aV)] = 0$, and Stein's identity once more gives $X = \mathbb{E}_V[aV\phi(aV)\log\Phi(aV)] = -a^2X + a^2\mathbb{E}_V[\phi(aV)^2/\Phi(aV)]$, so $X = q\mathbb{E}_V[\phi(aV)^2/\Phi(aV)]$, as $a^2/(1+a^2) = q$.

4 Hence $\mathbb{E}_V[G''(aV)] = (1-q)\mathbb{E}_V[\phi(aV)^2/\Phi(aV)]$ and, with $\phi(x)^2 = e^{-x^2}/(2\pi)$,

$$2\Psi'(q) = \frac{1}{\pi(1-q)}\,\mathbb{E}_V\Big[\frac{e^{-a^2V^2}}{\Phi(aV)}\Big], \qquad 2\Psi'(0) = \frac{2}{\pi}.$$

5 The value at $q = 0$ is Hebb's rule: knowing nothing, the student still gains the signal-to-noise ratio $2\alpha/\pi$, because the average of $y_\mu x_\mu$ points to the teacher. So $q = 0$ is never a fixed point for the sign channel, unlike for channels that are even in s.

THE ISING TEACHER. As ψ is convex, $\inf_r[\psi(r) - rq/2]$ is attained where $\mathbb{E}\tanh(r + \sqrt{r}Z) = q$, and $s(q;\alpha) = \log 2 + \inf_r f_{\text{RS}}(q, r)$ is, in the replica picture, the entropy of the part of the version space at overlap q; its stationary points are the fixed points of (7.4) (Exercise 7.2), and $s(1;\alpha) = 0$, the teacher alone. In Figure 7.1(a), for $\alpha < 1.2447$ an interior maximum lies above 0 ($q_* = 0.5607$, $\varepsilon_{\text{B}} = 0.2306$ at $\alpha = 1$). At $\alpha_{\text{IT}} = 1.2447$ it reaches 0, at $q = 0.6938$; beyond, it is negative, the global maximum moves to $q = 1$, and the Bayes error jumps from 0.1867 to 0. A set of sign vectors cannot have negative entropy, so past this point the version space is the teacher and a set too small to carry entropy.[3] The interior maximum survives, with negative entropy, until the spinodal $\alpha_{\text{AMP}} = 1.4926$, at $q = 0.8937$.[4]

MESSAGE PASSING. Iterating (7.4) from $q^0 = 0$,

$$q^{t+1} = \mathbb{E}\tanh\big(r^t + \sqrt{r^t}Z\big), \qquad r^t = 2\alpha\Psi'(q^t), \tag{7.5}$$

is the cavity method run as an algorithm: the *state evolution* of the generalized approximate message passing algorithm (GAMP) of Rangan (2011), the AMP of Chapter 6 for generalized linear models.[5] With $F = X/\sqrt{d}$, $\hat{w}^0 = 0$

[3] Krauth and Mézard (1989) used the same criterion, a replica-symmetric entropy that vanishes, for the capacity of the Ising perceptron without a teacher (Chapter 8).

[4] According to Barbier et al. (2019), Sompolinsky, Tishby and Seung argued that metastable states exist at least up to $\alpha \approx 1.628$, beyond the spinodal of the replica-symmetric branch.

[5] Rangan stated it without a complete proof. Javanmard and Montanari (2013) need Lipschitz nonlinearities, and sign labels are discontinuous; Barbier et al. (2019) keep it as their Claim 1

and $g^{-1} = 0$, Bayes-optimal GAMP alternates $\omega^t = F\hat{w}^t - V^t g^{t-1}$, with the Onsager term $-V^t g^{t-1}$, the output step $g^t = \partial_\omega \log \Phi(y\,\omega^t/\sqrt{V^t})$, and the input step $\hat{w}^{t+1} = \tanh(F^\top g^t + A^t \hat{w}^t)$, componentwise, where V^t is the mean of $1 - (\hat{w}^t)^2$ and $A^t = -\frac{1}{d}\sum_\mu \partial_\omega g^t_\mu$; state evolution predicts $\frac{1}{d}\langle \hat{w}^t, w^*\rangle \to q^t$. The map (7.5) is increasing, as Ψ is convex (Barbier et al. 2019), so from $q^0 = 0$ it climbs to the smallest fixed point: the branch of Figure 7.1(a) for $\alpha < \alpha_{\text{AMP}}$, and $q = 1$ above. Between α_{IT} and α_{AMP} the data determine the teacher, but GAMP stops on the metastable branch, at an error falling from 0.187 to 0.106. This is the *hard phase*, and the physicists' conjecture (Barbier et al. 2019) is that no polynomial-time algorithm does better there.

THE ERRORS. In the replica-symmetric state the fields of a fresh pattern are $\sqrt{q}\,V + \sqrt{1-q}\,U$ for the teacher and $\sqrt{q}\,V + \sqrt{1-q}\,u$ for a posterior sample. Two Gaussian fields with correlation ρ have different signs with probability $\frac{1}{\pi}\arccos\rho$, so a posterior sample errs with probability $\frac{1}{\pi}\arccos q$, and the best prediction, $\operatorname{sgn} V$ (the sign of the posterior mean field, the "center of mass"), with probability $\frac{1}{\pi}\arccos\sqrt{q}$.

COMPUTATION 7.2 · THE SPHERICAL TEACHER

1. For the Gaussian prior $\psi(r) = \frac{r}{2} - \frac{1}{2}\log(1+r)$, so $q = r/(1+r)$ in (7.4). The fixed point is unique (numerically, for $\frac{1}{4} \le \alpha \le 20$), and the errors decrease smoothly, $\varepsilon_{\text{B}} = 0.2544$ and $\varepsilon_{\text{G}} = 0.3384$ at $\alpha = 1$ (Figure 7.1(b)).
2. As $q \to 1$, $a \to \infty$; substituting $x = aV$ in Computation 7.1, $\mathbb{E}_V[e^{-a^2V^2}/\Phi(aV)] \simeq I/a$ with $I = \int_{\mathbb{R}} \frac{e^{-x^2}}{\Phi(x)}\frac{\mathrm{d}x}{\sqrt{2\pi}} = 2.263980$.
3. So $r \simeq \alpha I/(\pi\sqrt{1-q})$, and $1 - q = 1/(1+r) \simeq 1/r$ gives $\sqrt{1-q} \simeq \pi/(\alpha I)$.
4. Since $\arccos\sqrt{q} \simeq \sqrt{1-q}$ and $\arccos q \simeq \sqrt{2(1-q)}$, $\alpha\varepsilon_{\text{B}} \to 1/I = 0.44170$ and $\alpha\varepsilon_{\text{G}} \to \sqrt{2}/I = 0.62466$: the ratio $\sqrt{2}$ of Opper and Haussler (1991), and the constant 0.4417 that Aubin et al. (2020) print. At $\alpha = 100$ the products are 0.44164 and 0.62456.

7.3 The stone

PHYSICS · *the replica formula*

1 · Replicate

$\mathbb{E}\mathcal{Z}^k = \mathbb{E}\sum_y \mathcal{Z}^{k+1}$: the teacher is replica 0; continue $k \to 0$.

PROOF · *adaptive interpolation*

A path from the GLM ($t = 0$) to two scalar channels ($t = 1$) along which the student's measure stays a posterior (Definition 7.2). Nothing is continued in k.

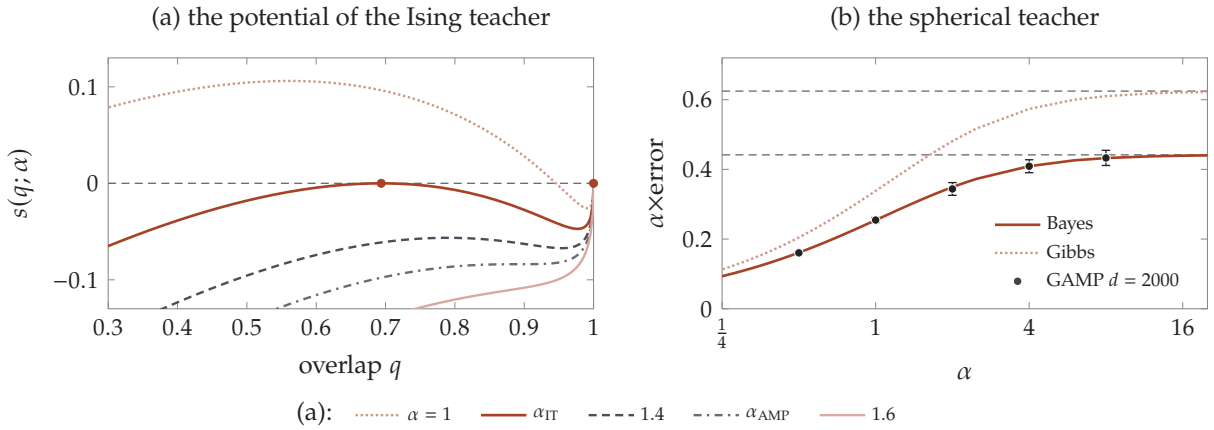


Figure 7.1. (a) The potential $s(q;\alpha) = \log 2 + \inf_r f_{\mathrm{RS}}(q,r)$ of the Ising teacher. At α_{IT} the interior maximum (dot at $q = 0.694$) is level with the value 0 of the teacher alone ($q = 1$); above, the maximum that message passing reaches has negative entropy; at α_{AMP} it disappears. (b) The spherical teacher: α times the Bayes and Gibbs errors of the formula, their limits $1/I$ and $\sqrt{2}/I$ (dashed), and α times the error of Bayes-optimal GAMP at $d = 2000$ (3 instances per point, 95% Student-t intervals, seeds $9000 + 100k + s$). Scripts: `07-glm-potential.py`, `07-glm-gamp.py`.

2 · Gaussian fields

The fields $\langle x, w^a\rangle/\sqrt{d}$ are Gaussian with covariance Q_{ab}; replica symmetry makes them $\sqrt{q}\,V + \sqrt{1-q}\,u^a$.

Gaussian integration by parts in x_μ, V_μ, U_μ. At $t = 1$ the field is $\sqrt{R_2}\,V + \sqrt{1-R_2}\,U$: the symmetric decomposition is the end of the path.

3 · The saddle point

Stationarity of $f_{\mathrm{RS}}(q,r)$ in the order parameter and its conjugate.

The sum rule (Theorem 7.3): for all path functions $q(t)$, $r(t)$, the free entropy is the potential plus a remainder, exactly at every d.

4 · Replica symmetry

$Q_{ab} = q$ for all pairs, teacher included.

The remainder, a covariance, vanishes if the overlap concentrates and $q(t) = \mathbb{E}\langle Q\rangle_t$ (Corollary 7.4); concentration is proved under small perturbations.

5 · The extremum

Among stationary points, the largest free entropy.

Two adaptive paths give matching bounds; convex duality turns $\sup_r \inf_q$ into $\sup_q \inf_r$.

6 · Annealed approximation

Replace $\mathbb{E}\log\mathcal{Z}$ by $\log\mathbb{E}\mathcal{Z}$.	Jensen's inequality and a first moment: $\alpha_{\mathrm{IT}} \le 1.4480$; counting gives $\alpha_{\mathrm{IT}} \ge 1$ (Theorem 7.7).

7 · The spinodal

The metastable branch ends at α_{AMP}; the iterated cavity equations are trapped on it until then.	State evolution for a fixed number of iterations: no first-order method beats it, and one comes within every ϵ of it (Celentano et al. 2020); so such methods learn perfectly if and only if $\alpha > \alpha_{\mathrm{AMP}}$. For GAMP itself it is a claim.

8 · The hard phase

Between α_{IT} and α_{AMP} no efficient algorithm learns.	No counterpart: a conjecture, proved for first-order methods with boundedly many iterations (Celentano et al. 2020).

The rigorous route keeps the two scalar channels and the replica-symmetric decomposition of the fields as the end of the interpolation, and replaces the replicas by a path along which the student's measure stays a true posterior, so that the Nishimori identity, the physical reason for replica symmetry, holds at every step. What the proof adds is adaptivity: in the spiked models of Chapters 4 and 5 the remainder is a square and a straight path gives a bound, while here it is a covariance between the overlap and an output statistic, and the path must be chosen, by a differential equation, so that it vanishes. The routes part at the extremum, which the proof derives from convexity, and at the hard phase, where the proof reaches only restricted classes of algorithms.

7.4 The proof: the interpolation sum rule

We prove the identity behind adaptive interpolation for the Ising teacher and a smooth binary channel, and deduce one direction of the replica formula when the overlap concentrates.

Assumption 7.1 (A smooth binary channel). The labels are ± 1 and $P_{\mathrm{out}}(y \mid s) = \sigma(ys)$ for a twice continuously differentiable $\sigma\colon \mathbb{R} \to (0,1)$ with $\sigma(s) + \sigma(-s) = 1$ and $|(\log\sigma)'(s)| + |(\log\sigma)''(s)| \le C(1+|s|)$.

The probit channel $\sigma(s) = \Phi(s/\sqrt{\Delta})$, which adds Gaussian noise of variance $\Delta > 0$ to the field before taking the sign, and the logistic channel satisfy the assumption; the noiseless channel is their limit. For such a channel Ψ is the negative entropy of a two-point law, so $-\log 2 \le \Psi \le 0$.

Definition 7.2 (The interpolating model). Fix $d, n \geq 1$ and continuous $q\colon [0,1] \to [0,1]$, $r\colon [0,1] \to [0,\infty)$; let $R_1(t) = \int_0^t r$ and $R_2(t) = \int_0^t q$, so $0 \leq R_2(t) \leq t$. Let w^* be uniform on $\{-1,1\}^d$, and $x_\mu \sim \mathcal{N}(0, I_d)$, $V_\mu, U_\mu, Z_i \sim \mathcal{N}(0,1)$ independent ($\mu \leq n$, $i \leq d$). At time t the data are the patterns, the V_μ, $Y_{t,\mu} \sim P_{\text{out}}(\cdot \mid S_{t,\mu})$ independently given the rest, and $Y'_{t,i} = \sqrt{R_1(t)}\, w^*_i + Z_i$, where

$$S_{t,\mu} = \sqrt{1-t}\,\frac{\langle x_\mu, w^*\rangle}{\sqrt{d}} + \sqrt{R_2(t)}\,V_\mu + \sqrt{t - R_2(t)}\,U_\mu.$$

The posterior of (w^*, U) is the Gibbs measure $\langle\cdot\rangle_t$ on $(w,u) \in \{-1,1\}^d \times \mathbb{R}^n$ with weight

$$2^{-d}\prod_\mu P_{\text{out}}(Y_{t,\mu} \mid s_{t,\mu})\phi(u_\mu)\prod_i e^{-\frac{1}{2}(Y'_{t,i} - \sqrt{R_1(t)}\,w_i)^2},$$

$s_{t,\mu}$ being $S_{t,\mu}$ with (w, u_μ) in place of (w^*, U_μ), and normalizer $\mathcal{Z}_t$; let $f_d(t) = \frac{1}{d}\mathbb{E}\log\mathcal{Z}_t$. Write $Q = \frac{1}{d}\langle w, w^*\rangle$, $u_y(s) = \log P_{\text{out}}(y \mid s)$, and

$$A_t = \frac{1}{d}\sum_{\mu=1}^{n} u'_{Y_{t,\mu}}(S_{t,\mu})\,u'_{Y_{t,\mu}}(s_{t,\mu}).$$

Theorem 7.3 (The sum rule). *Assume Assumption 7.1 and the Rademacher prior. For every $d, n \geq 1$ and all path functions q, r as in Definition 7.2,*

$$\begin{aligned} f_d = \psi\big(R_1(1)\big) + \frac{n}{d}\,\Psi\big(R_2(1)\big) - \frac{1}{2}\int_0^1 r(t)q(t)\,\mathrm{d}t \\ + \frac{1}{2}\int_0^1 \mathbb{E}\left\langle \big(A_t - r(t)\big)\big(Q - q(t)\big)\right\rangle_t \,\mathrm{d}t, \end{aligned} \tag{7.6}$$

with f_d the free entropy of (7.1) and ψ, Ψ as in (7.3). For constant paths the first three terms are $f_{\text{RS}}(q,r)$ with n/d in place of α.

Proof. The plan is to compute the free entropy at the two ends of the path, to differentiate along it, and to remove the teacher from the derivative with the Nishimori identity and Gaussian integration by parts. Assume first that q takes values in $[\delta, 1-\delta]$ and r in $[\delta,\infty)$ for some $\delta > 0$, so that R_1, R_2 and $t - R_2$ are positive on $(0,1]$ and the square roots below are differentiable there. The growth condition gives $|u_y(s)| \leq C'(1+s^2)$ and $|u'_y(s)| \leq C(1+|s|)$, so every function below is bounded by a polynomial in the Gaussian variables; this justifies differentiation under the expectation and Gaussian integration by parts (Theorem A.1).

The endpoints. At $t = 0$, $R_1 = R_2 = 0$, the u_μ drop out, and the side channel contributes $\prod_i e^{-Z_i^2/2}$, which does not depend on w; so $f_d(0) = f_d - \frac{1}{2}$. At $t = 1$ the patterns drop out and the measure factorizes. The output factors give $\frac{n}{d}\Psi(R_2(1))$. With $R = R_1(1)$, a side-channel factor

gives $\mathbb{E}\log\sum_{w=\pm1}\frac{1}{2}e^{-\frac{1}{2}(Y'-\sqrt{R}w)^2} = -\frac{1+R}{2} - \frac{R}{2} + \mathbb{E}\log\cosh(R+\sqrt{R}Z)$, because $\mathbb{E}Y'^2 = 1+R$, $\sqrt{R}Y' = Rw^* + \sqrt{R}Z$ and $\log\cosh$ is even. Hence $f_d(1) = \psi(R_1(1)) - \frac{1}{2}(1+R_1(1)) + \frac{n}{d}\Psi(R_2(1))$.

Differentiation. Given the patterns and V, the data (Y_t, Y_t') have density $(2\pi)^{-d/2}\mathcal{Z}_t(Y,Y')$ with respect to counting measure times Lebesgue measure, as integrating the joint law of (w^*, U, Y_t, Y_t') over (w^*, U) shows. So $f_d(t) = \frac{1}{d}\mathbb{E}_{X,V}\sum_Y\int(2\pi)^{-d/2}\mathcal{Z}_t\log\mathcal{Z}_t\,\mathrm{d}Y'$, and as $\sum_Y\int\partial_t\mathcal{Z}_t\,\mathrm{d}Y' = 0$, with ∂_t at fixed data, $f_d'(t) = \frac{1}{d}\mathbb{E}[\mathcal{Z}_t^{-1}\partial_t\mathcal{Z}_t\log\mathcal{Z}_t] = \frac{1}{d}\mathbb{E}[\langle L_t\rangle_t\log\mathcal{Z}_t]$. Here L_t is the derivative at fixed data of the logarithm of the weight of (w,u), with $\dot{s} = \partial_t s$:

$$L_t = \sum_\mu \dot{s}_{t,\mu}u'_{Y_{t,\mu}}(s_{t,\mu}) + \frac{r(t)}{2\sqrt{R_1(t)}}\sum_i w_i\big(Y'_{t,i} - \sqrt{R_1(t)}w_i\big).$$

Given the data, (w^*, U) is one more sample from $\langle\cdot\rangle_t$, so the Nishimori identity (Lemma 1.10) replaces (w,u) by (w^*,U), for which $Y'_{t,i} - \sqrt{R_1}w_i^* = Z_i$. Thus $f_d'(t) = T_1 + T_2$ with

$$T_1 = \frac{1}{d}\sum_\mu \mathbb{E}\big[\dot{S}_{t,\mu}u'_{Y_{t,\mu}}(S_{t,\mu})\log\mathcal{Z}_t\big], \qquad T_2 = \frac{r(t)}{2d\sqrt{R_1(t)}}\sum_i \mathbb{E}\big[w_i^*Z_i\log\mathcal{Z}_t\big].$$

The side channel. Z_i enters $\log\mathcal{Z}_t$ only through $Y'_{t,i}$, and $\partial_{Y_i'}\log\mathcal{Z}_t = -\left\langle Y'_{t,i} - \sqrt{R_1}w_i\right\rangle_t$. Integration by parts in Z_i, with $\mathbb{E}[w_i^*Z_i] = 0$ and $(w_i^*)^2 = 1$, gives $\mathbb{E}[w_i^*Z_i\log\mathcal{Z}_t] = -\sqrt{R_1}(1 - \mathbb{E}[w_i^*\langle w_i\rangle_t])$, so $T_2 = -\frac{r(t)}{2}(1 - \mathbb{E}\langle Q\rangle_t)$.

The output channel. Let L_y be $\log\mathcal{Z}_t$ with $Y_{t,\mu}$ set to y. Averaging over $Y_{t,\mu}$ given the rest turns the μ-th term of T_1 into $\frac{1}{d}\mathbb{E}[\dot{S}_{t,\mu}G_\mu]$, with

$$G_\mu = \sum_{y=\pm1}\partial_s P_{\mathrm{out}}(y \mid S_{t,\mu})\,L_y.$$

The variables $x_{\mu1},\dots,x_{\mu d}, V_\mu, U_\mu$ are independent of the rest; they enter G_μ through $S_{t,\mu}$, and all but U_μ also through $s_{t,\mu}$ in L_y. Integrate by parts in each, using

$$\dot{S}_{t,\mu} = -\frac{\langle x_\mu, w^*\rangle}{2\sqrt{(1-t)d}} + \frac{q(t)V_\mu}{2\sqrt{R_2(t)}} + \frac{(1-q(t))U_\mu}{2\sqrt{t-R_2(t)}}.$$

The terms with $\partial_s^2 P_{\mathrm{out}}(y \mid S_{t,\mu})L_y$ have total coefficient $-\frac{\|w^*\|^2}{2d} + \frac{q(t)}{2} + \frac{1-q(t)}{2} = 0$, since $\|w^*\|^2 = d$. The terms where the derivative falls on L_y give $\partial_s P_{\mathrm{out}}(y \mid S_{t,\mu})$ times

$$-\frac{1}{2d}\sum_i w_i^*\left\langle w_i u'_y(s_{t,\mu})\right\rangle_{t,y} + \frac{q(t)}{2}\left\langle u'_y(s_{t,\mu})\right\rangle_{t,y} = -\frac{1}{2}\left\langle (Q-q(t))u'_y(s_{t,\mu})\right\rangle_{t,y},$$

$\langle\cdot\rangle_{t,y}$ being the Gibbs measure at $Y_{t,\mu} = y$. Undoing the average over $Y_{t,\mu}$ and summing over μ, $T_1 = -\frac{1}{2}\mathbb{E}\left\langle A_t(Q-q(t))\right\rangle_t$.

Assembling. Writing $1 - \mathbb{E}\langle Q\rangle_t = (1-q(t)) - \mathbb{E}\left\langle Q - q(t)\right\rangle_t$,

$$f_d'(t) = -\frac{r(t)(1-q(t))}{2} - \frac{1}{2}\mathbb{E}\left\langle (A_t - r(t))(Q-q(t))\right\rangle_t, \qquad 0 < t < 1.$$

This is bounded on $(0,1)$ and f_d is continuous on $[0,1]$, so $f_d = f_d(0) + \frac{1}{2} = f_d(1) - \int_0^1 f_d' + \frac{1}{2}$; the endpoints and $\frac{1}{2}\int_0^1 r(1-q) = \frac{1}{2}R_1(1) - \frac{1}{2}\int_0^1 rq$ give (7.6). The left side does not depend on the path and the right side is continuous in (q,r) for uniform convergence, by dominated convergence, so approximation removes the restriction to $[\delta, 1-\delta]\times[\delta,\infty)$. ∎

The identity holds at every d, without limits or concentration (Section 7.6 checks it by Monte Carlo): the replica potential is exact up to a remainder that measures how far the overlap is from concentrating. In the spiked Wigner model the remainder is a positive multiple of $\mathbb{E}\left\langle (Q-q)^2\right\rangle$ (Chapter 4); here it has no sign, being nonnegative at $q = r = 0$ (Exercise 7.4) and negative along other paths. A bound needs a path along which it vanishes.

Corollary 7.4 (One bound from overlap concentration). *Assume Assumption 7.1, the Rademacher prior, $n/d \to \alpha \in (0,\infty)$, and*

$$\delta_d = \sup \int_0^1 \mathbb{E}\left\langle \left(Q - \mathbb{E}\langle Q\rangle_t\right)^2\right\rangle_t \,\mathrm{d}t \to 0, \tag{7.7}$$

the supremum over the paths of Definition 7.2 with constant r. Then $\liminf_d f_d \geq \sup_{r\geq 0}\inf_{q\in[0,1]} f_{\mathrm{RS}}(q,r)$.

Proof. The plan is to choose the path so that $q(t) = \mathbb{E}\langle Q\rangle_t$ and to bound the remainder by the Cauchy–Schwarz inequality. The field $S_{t,\mu}$ is standard normal, its variance being $(1-t)\|w^*\|^2/d + R_2 + t - R_2 = 1$, so $\mathbb{E}u'_{Y_{t,\mu}}(S_{t,\mu})^4 \leq C^4\mathbb{E}(1+|Z|)^4 =: C_4$, and by the Nishimori identity so is $\mathbb{E}\left\langle u'_{Y_{t,\mu}}(s_{t,\mu})^4\right\rangle_t$. With $(\frac{1}{d}\sum_\mu a_\mu)^2 \leq \frac{n}{d^2}\sum_\mu a_\mu^2$ and Cauchy–Schwarz, $\mathbb{E}\left\langle A_t^2\right\rangle_t \leq (n/d)^2 C_4$ for every t and path.

The model at time t depends on the path only through $(R_1(t), R_2(t))$; let $F_d(t,R_1,R_2) = \mathbb{E}\langle Q\rangle_t$ for $0\leq t\leq 1$, $R_1\geq 0$, $0\leq R_2\leq t$. It is continuous by dominated convergence, with values in $[0,1]$ since $\mathbb{E}\langle Q\rangle_t = \frac{1}{d}\mathbb{E}\|\langle w\rangle_t\|^2$ by the Nishimori identity. Fix $r\geq 0$. By Peano's theorem, $R_2'(t) = F_d(t, rt, \min(\max(R_2(t),0),t))$, $R_2(0) = 0$, has a solution on $[0,1]$; its right side lies in $[0,1]$, so $0\leq R_2(t)\leq t$ and the clamp is inactive. The path $r(t) = r$, $q(t) = R_2'(t)$ then satisfies $q(t) = \mathbb{E}\langle Q\rangle_t$.

Along it the remainder in (7.6) is $\frac{1}{2}\int_0^1 \mathbb{E}\langle (A_t - r)(Q - \mathbb{E}\langle Q\rangle_t)\rangle_t \,\mathrm{d}t$, at most $\frac{1}{2}(\sqrt{2}\frac{n}{d}\sqrt{C_4} + \sqrt{2}r)\delta_d^{1/2}$ in absolute value by Cauchy–Schwarz. Hence

$$f_d \geq \psi(r) + \frac{n}{d}\Psi\big(R_2(1)\big) - \frac{r}{2}R_2(1) - o(1) \geq \inf_{q\in[0,1]}\left[\psi(r) + \frac{n}{d}\Psi(q) - \frac{rq}{2}\right] - o(1).$$

As $-\log 2 \leq \Psi \leq 0$, replacing n/d by α moves the infimum by at most $|n/d - \alpha|\log 2$. Let $d\to\infty$ and take the supremum over r. ∎

The corollary is replica symmetry turned into a hypothesis: if the overlap does not fluctuate, one half of the replica formula follows from an identity. The hypothesis (7.7) is not known as stated. What is proved is the following.

Theorem 7.5 (Barbier et al. 2019, Theorems 1–3 and Corollary 3). *Consider (7.1) with $n/d \to \alpha$ under hypotheses (h1)–(h5): the prior has a finite third moment and at least two points in its support; the outputs have a bounded moment of order $2+\gamma$; the pattern entries are independent, centered, of variance 1 and bounded third moment; φ_{out} is continuous almost everywhere; and either Gaussian noise $\Delta > 0$ is added to the outputs or the outputs take values in $\mathbb{N}$. Let $\rho = \mathbb{E}_{P_0} X^2$ and $f_{\text{RS}}(q, r) = \psi_{P_0}(r) + \alpha\Psi(q; \rho) - \frac{rq}{2}$, with the scalar free entropies of $\sqrt{r}X_0 + Z$ and of $P_{\text{out}}(\cdot \mid \sqrt{q}V + \sqrt{\rho - q}\,W)$.*

(i) $f_d \to \sup_{q\in[0,\rho]} \inf_{r\geq 0} f_{\text{RS}}(q, r)$, which equals the supremum of f_{RS} over the fixed points of $q = 2\psi'_{P_0}(r)$, $r = 2\alpha\Psi'(q)$ ($r = \infty$ allowed at $q = \rho$); for an informative channel both are attained at the same points.

(ii) If the channel is informative and P_0 has all moments, then for α in a set D^* with countable complement the maximizer $q_*(\alpha)$ is unique, $|Q| \to q_*$ in probability under $\mathbb{E}\langle\cdot\rangle$, and the Bayes mean squared error of a new label tends to that of the scalar channel at q_*.

(iii) For the noiseless perceptron with the Rademacher prior, $\frac{1}{d}\mathbb{E}\log|\mathcal{V}| \to \log 2 + \sup_{q<1} \inf_r f_{\text{RS}}(q, r)$.

Both teachers satisfy the hypotheses, with $\Delta = 0$ after relabeling the outputs as $\{0, 1\}$. Their proof adds four things to Corollary 7.4. Concentration in averaged form: the initial values are perturbed to $R_i(0) = \epsilon_i \in [s_d, 2s_d]$, $s_d = \frac{1}{2}d^{-1/16}$, and the average over ϵ and t of the overlap variance is at most $Cd^{-1/8}$ (their Proposition 4), controlling thermal fluctuations by the curvature of the free entropy in a side-channel signal-to-noise ratio and the Nishimori identity, and fluctuations of the data by concentration of the free entropy; the average survives adaptive paths because the flow $\epsilon \mapsto (R_1(t), R_2(t))$ has Jacobian at least 1, by Liouville's formula (their Propositions 6–7). The upper bound, along $r(t) = 2\alpha\Psi'(q(t))$, by Jensen's inequality for the convex ψ and Ψ (Proposition 8). Convex duality, $\sup_r \inf_q = \sup_q \inf_r$ (Appendix D). And approximation from bounded priors and noisy smooth channels to (h1)–(h5) (Appendix C).

7.5 The proof: counting the version space

The second proof is elementary and specific to the Ising teacher. A sign vector w at Hamming distance j from w^* makes the angle $\theta_j = \arccos(1 - 2j/d)$ with it, and a Gaussian pattern separates the two with probability θ_j/π: the projection of the pattern on their plane has a uniform direction, and the signs differ when the orthogonal line separates them. Let

$$g_\alpha(m) = h\Big(\frac{1-m}{2}\Big) + \alpha \log\Big(1 - \frac{\arccos m}{\pi}\Big), \qquad m \in (-1, 1],$$

with $h(p) = -p\log p - (1-p)\log(1-p)$, so $g_\alpha(1) = 0$ and $g_\alpha \to -\infty$ at -1; let α_{ann} be the infimum of the α with $g_\alpha < 0$ on $(-1, 1)$. As $g_\alpha(m)$ decreases strictly in α for $m < 1$, $g_\alpha < 0$ on $(-1, 1)$ for every $\alpha > \alpha_{\text{ann}}$; numerically $\alpha_{\text{ann}} = 1.4479664$.

Lemma 7.6 (No near neighbors). *For every $\alpha > 0$ there is $\epsilon \in (0, e^{-1}]$ such that if $n \geq \alpha d$, then the expected number of $w \in \mathcal{V} \setminus \{w^*\}$ within Hamming distance ϵd of w^* is at most $d\, e^{-\alpha\sqrt{d}/\pi}$.*

PROOF. The plan is a first moment at each distance j. A fixed w at distance j lies in $\mathcal{V}$ when no pattern separates it from w^*, with probability p_j^n, $p_j = 1 - \theta_j/\pi$. As $\cos\theta \geq 1 - \theta^2/2$, $\cos(2\sqrt{j/d}) \geq \cos\theta_j$ with both angles in $[0, \pi]$, so $\theta_j \geq 2\sqrt{j/d}$ and $p_j^n \leq e^{-\frac{2n}{\pi}\sqrt{j/d}}$. With $\binom{d}{j} \leq (ed/j)^j$, the expected number at distance j is at most $\exp(\sqrt{jd}\,[\sqrt{j/d}\log(ed/j) - \frac{2n}{\pi d}])$. The function $x \mapsto \sqrt{x}\log(e/x)$ increases on $(0, e^{-1}]$ and tends to 0 at 0; choose $\epsilon \leq e^{-1}$ with $\sqrt{\epsilon}\log(e/\epsilon) \leq \alpha/\pi$. For $1 \leq j \leq \epsilon d$ the bracket is at most $-\alpha/\pi$, so each of the at most d terms is at most $e^{-\alpha\sqrt{d}/\pi}$. ∎

Theorem 7.7 (Counting the version space). *For the Ising teacher and the noiseless perceptron, for every d and n,*

$$\Big(1 - \frac{n}{d}\Big)_+ \log 2 \leq \frac{1}{d}\mathbb{E}\log|\mathcal{V}| \leq \frac{1}{d}\log\mathbb{E}|\mathcal{V}| \leq \frac{\log(d+1)}{d} + \sup_m g_{\frac{n}{d}}(m). \tag{7.8}$$

If $\liminf n/d > \alpha_{\text{ann}}$, then $\mathbb{P}(\mathcal{V} = \{w^*\}) \to 1$. So the entropy of the version space stays above $(1-\alpha)\log 2$ when $n/d \to \alpha < 1$, and it tends to 0, with a posterior that is the point mass at the teacher with probability tending to one, when $n/d \to \alpha > \alpha_{\text{ann}}$.

PROOF. The plan is an information count below, and Jensen's inequality with a first moment above. Given the data the posterior is uniform on $\mathcal{V}$, so $\mathbb{E}\log|\mathcal{V}|$ is the conditional entropy $H(w^* \mid X, y) = H(w^*) - I(w^*; y \mid X) \geq d\log 2 - H(y \mid X) \geq (d-n)\log 2$; and $|\mathcal{V}| \geq 1$ since $w^* \in \mathcal{V}$. The middle inequality is Jensen's. Next, $\mathbb{E}|\mathcal{V}| = \sum_{j=0}^{d}\binom{d}{j}p_j^n$ as in Lemma 7.6, and $\binom{d}{j} \leq e^{dh(j/d)}$; with $m = 1 - 2j/d$, each of the $d+1$ terms is at most $e^{dg_{n/d}(m)}$.

Let $\liminf n/d > \alpha_{\text{ann}}$ and fix $\alpha' \in (\alpha_{\text{ann}}, \liminf n/d)$, so that $n \geq \alpha' d$ for large d, and let ϵ be given by Lemma 7.6 for α'. Vectors within distance ϵd contribute at most $d\, e^{-\alpha'\sqrt{d}/\pi}$ to $\mathbb{E}|\mathcal{V} \setminus \{w^*\}|$. For $j > \epsilon d$, $m \in [-1, 1-2\epsilon)$, and as $\log p_j \leq 0$ the term is at most $e^{dg_{\alpha'}(m)}$. The function $g_{\alpha'}$ is continuous on $(-1, 1]$, tends to $-\infty$ at -1 and is negative on $(-1, 1)$, so $c = -\sup_{[-1,1-2\epsilon]} g_{\alpha'} > 0$ and these terms contribute at most $d\, e^{-cd}$. Hence $\mathbb{P}(\mathcal{V} \neq \{w^*\}) \leq \mathbb{E}|\mathcal{V} \setminus \{w^*\}| \to 0$. The last sentence follows, as $\frac{1}{d}\mathbb{E}\log|\mathcal{V}| \leq \mathbb{P}(\mathcal{V} \neq \{w^*\})\log 2$. ∎

Let α_{IT} be the infimum of the α such that $\frac{1}{d}\mathbb{E}\log|\mathcal{V}| \to 0$ whenever $n/d \to \alpha$; as patterns only shrink $\mathcal{V}$, the entropy tends to 0 for every $\alpha > \alpha_{\text{IT}}$. By Theorem 7.5(iii), α_{IT} is the smallest α with $\log 2 + \sup_{q<1}\inf_r f_{\text{RS}}(q, r) = 0$, numerically 1.2447. So $1 \leq \alpha_{\text{IT}} \leq \alpha_{\text{ann}} < 1.4480$ (Section 7.6). The upper bound in (7.8) is the zero-temperature annealed approximation of Seung et al. (1992), and it is not exact: 0.1569 at $\alpha = 1$, against 0.1061 from Theorem 7.5. The counting bound is the one that Barbier et al. (2019) mention for exact

recovery with discrete outputs. The physicists' errors need one more step, from the overlap to predictions; write $\bar{R}(u,v) = \langle u\rangle v/(\|u\|\|v\|)$.

Proposition 7.8 (Errors from the overlap). *Consider the noiseless perceptron with the Rademacher or the standard Gaussian prior, a fresh pattern $x \sim \mathcal{N}(0, I_d)$, and $p = \langle \mathbf{1}\{\langle x\rangle w > 0\}\rangle$, the posterior probability of the label $+1$.*

(i) For every d, $\varepsilon_{\mathrm{G}} = \frac{1}{\pi}\mathbb{E}\left\langle \arccos \bar{R}(w,w^*)\right\rangle = \mathbb{E}[2p(1-p)]$ and $\varepsilon_{\mathrm{B}} = \mathbb{E}\min(p,1-p)$, so $\varepsilon_{\mathrm{B}} \le \varepsilon_{\mathrm{G}} \le 2\varepsilon_{\mathrm{B}}$.

(ii) If $Q \to q \in [0,1]$ in probability under $\mathbb{E}\langle\cdot\rangle$, then $\varepsilon_{\mathrm{G}} \to \frac{1}{\pi}\arccos q$ and $\varepsilon_{\mathrm{B}} \to \frac{1}{\pi}\arccos\sqrt{q}$.

(iii) For $\alpha \in D^*$, part 2 applies with $q = q_*(\alpha)$ of Theorem 7.5.

Proof. The plan is to write both errors through overlaps of replicas, which the Nishimori identity ties to Q, and to identify the limit law of p by its moments. Part 1: given the data and a posterior sample, x is independent of both, and the angle argument gives the first formula. Given the data and x, the teacher is one more posterior sample (Lemma 1.10), so the label and the prediction of an independent sample are independent signs, each $+1$ with probability p; they differ with probability $2p(1-p)$, the best prediction errs with probability $\min(p,1-p)$, and $\min(p,1-p) \le 2p(1-p) \le 2\min(p,1-p)$.

Part 2. For the Gaussian prior $\|w^*\|^2/d \to 1$ in probability, and so does $\|w\|^2/d$ under $\mathbb{E}\langle\cdot\rangle$, by the Nishimori identity; hence $\bar{R}(w,w^*) \to q$, and by the same identity $\bar{R}(w^a,w^b) \to q$ for independent posterior samples (for the Rademacher prior $\bar{R} = Q$). Bounded convergence gives the limit of ε_{G}, and if $q = 1$, then $\varepsilon_{\mathrm{B}} \le \varepsilon_{\mathrm{G}} \to 0$. Let $q < 1$. Given the data and k samples, $(\langle x, w^a\rangle/\|w^a\|)_{a\le k}$ is Gaussian with correlation matrix $\bar{R}^{(k)}$, so $\mathbb{E}p^k = \mathbb{E}\left\langle G_k(\bar{R}^{(k)})\right\rangle$, where $G_k(\Sigma)$ is the probability that a $\mathcal{N}(0,\Sigma)$ vector has positive entries. In probability $\bar{R}^{(k)} \to \Sigma_q = (1-q)I + q\mathbf{1}\mathbf{1}^\top$, positive definite, at which G_k is continuous; so $\mathbb{E}p^k \to G_k(\Sigma_q) = \mathbb{E}\Phi(bV)^k$, $b = \sqrt{q/(1-q)}$, writing $g_a = \sqrt{q}V + \sqrt{1-q}u_a$. Moments determine laws on $[0,1]$, so $p \to \Phi(bV)$ in law and $\varepsilon_{\mathrm{B}} \to \mathbb{E}\Phi(-b|V|)$: the probability that a standard Gaussian vector of the plane falls in a cone of opening angle $\pi - 2\arctan b$, that is, $\frac{1}{2} - \frac{1}{\pi}\arctan b = \frac{1}{\pi}\arccos\sqrt{q}$.

Part 3. By Theorem 7.5 and the Nishimori identity, $|\bar{R}(w^1,w^2)| \to q_*$. The Bayes predictor of a new label is $2p-1$, and as $\mathbb{E}[y(2p-1)] = \mathbb{E}(2p-1)^2$, its mean squared error is $1 - \mathbb{E}(2p-1)^2 = 1 - \frac{2}{\pi}\mathbb{E}\left\langle \arcsin \bar{R}(w^1,w^2)\right\rangle$, by the angle formula for two replicas; the same computation in the scalar channel gives $1 - \frac{2}{\pi}\arcsin q_*$. So $\mathbb{E}\left\langle \arcsin \bar{R}(w^1,w^2)\right\rangle \to \arcsin q_*$. Every fixed point of (7.4) has $r = 2\alpha\Psi'(q) > 0$, by the closed form of Computation 7.1, hence $q_* > 0$, and the probability that $\bar{R}(w^1,w^2) < 0$ tends to 0. So $\bar{R}(w^1,w^2) \to q_*$, and by the Nishimori identity and the norm argument of part 2, $Q \to q_*$. ∎

For the Ising teacher and $\alpha > \alpha_{\mathrm{IT}}$, Theorem 7.5(iii) makes $q = 1$, where $\inf_r f_{\mathrm{RS}}(q,r) = -\log 2$, a maximizer, so $q_*(\alpha) = 1$ when $\alpha \in D^*$, and with Lemma 7.6 the proposition sharpens Theorem 7.7.

Corollary 7.9 (Exact recovery above the transition). *For the Ising teacher, if $n/d \to \alpha > \alpha_{\mathrm{IT}}$ with $\alpha \in D^*$, then $\mathbb{P}(\mathcal{V} = \{w^*\}) \to 1$.*

PROOF. Take ϵ from Lemma 7.6 for $\alpha/2$, and let K_{near} and K_{far} count the vectors of $\mathcal{V} \setminus \{w^*\}$ within distance ϵd of w^* and beyond it, where $Q < 1 - 2\epsilon$. By the lemma $\mathbb{P}(K_{\mathrm{near}} \geq 1) \to 0$. If $K_{\mathrm{near}} = 0$, the posterior mass of $\{Q < 1 - 2\epsilon\}$ is $K_{\mathrm{far}}/(1 + K_{\mathrm{far}}) \geq \frac{1}{2}\mathbf{1}\{K_{\mathrm{far}} \geq 1\}$, and $\mathbb{E}\,\langle \mathbf{1}\{Q < 1 - 2\epsilon\}\rangle \to 0$ by Proposition 7.8 with $q_* = 1$. Hence $\mathbb{P}(K_{\mathrm{far}} \geq 1) \to 0$. ∎

So exhaustive search recovers the teacher exactly from α_{IT} on, while first-order methods with boundedly many iterations stop at an error of order 0.1 up to α_{AMP}; for $\alpha \in (1.4480, 1.4926)$ both sides of this gap need only Theorem 7.7 and Celentano et al. (2020), with a floating-point evaluation of (7.5).

7.6 Measuring it

THE TRANSITION, RECOMPUTED. The script `07-glm-potential.py` evaluates the potential of the Ising teacher by quadrature, with the closed forms of Computation 7.1 (checked against finite differences to 10^{-9}), and solves the stationarity conditions together with the vanishing of the branch entropy, in double precision and in 20-digit arithmetic (residuals below 10^{-21}).[6] Both give $\alpha_{\mathrm{IT}} = 1.2446840$; the branch entropy is $+3.31 \times 10^{-5}$ at $\alpha = 1.2446$ and -6.31×10^{-6} at 1.2447. In the same way $\alpha_{\mathrm{AMP}} = 1.4926318$ and $\alpha_{\mathrm{ann}} = 1.4479664$, and a grid bound certifies $g_{1.4480} < 0$ on $(-1, 1)$, so $\alpha_{\mathrm{ann}} < 1.4480$.[7] Barbier et al. (2019) print $\alpha_{\mathrm{IT}} \approx 1.249$ in the main text and 1.249(1) in the supplement. At $\alpha = 1.249$ their formula, that of Theorem 7.5, gives the branch the entropy -1.698×10^{-3}, so 1.249 lies inside the phase of perfect learning that the formula defines; Györgyi's 1.245 agrees with the zero, and their spinodal 1.493(1) with ours. The theorem concerns the formula, not its evaluation, and is unaffected.

THE VERSION SPACE. In Figure 7.2(a) every enumerated entropy lies between the finite-d bounds (7.8), as it must. Below $\alpha = 1$ the enumeration is close to the formula at $d = 18$ (0.368 ± 0.006 against 0.372 at $\alpha = 0.5$); near and above the transition it is not (0.127 ± 0.009 against 0.106 at $\alpha = 1$, 0.017 ± 0.004 against 0 at $\alpha = 1.5$). One source: the d neighbors at Hamming distance 1 survive on average $d(1 - \theta_1/\pi)^n$ times, 0.21 at $d = 18$, $n = 27$, a correction that decays only like $e^{-2\alpha\sqrt{d}/\pi}$. At $\alpha = 1.5$ the probability that the version space is the teacher alone grows from 0.58 at $d = 10$ to 0.69 at $d = 18$, as Corollary 7.9 predicts.

GAMP AGAINST STATE EVOLUTION. The script `07-glm-gamp.py` runs Bayes-optimal GAMP at $d = 2000$ from the prior mean, 8 instances at each of 11 values of α, stopping when the iterates stop moving, when GAMP's variance estimate falls below 10^{-8} (the estimate has frozen to a sign vector), or after 150 iterations.[8] Up to $\alpha = 1.47$ it stops on the branch, and the mean error of $\mathrm{sgn}\langle x, \hat{w}\rangle$ agrees with state evolution within the intervals at every point but $\alpha = 0.8$ (0.2605 ± 0.0035 against 0.2644): 0.2295 ± 0.0052 against

[6] Near $q = 1$ the scale $a = \sqrt{q/(1-q)}$ is large, and quadrature in V fails; the substitution $x = aV$ keeps the integrands of Computation 7.1 on the unit scale.

[7] On each of 1.8×10^7 intervals of $[-0.9, 0.9]$, g_α is at most its entropy term at the point nearest 0 plus its second term at the right end, as the first increases up to $m = 0$ and then decreases and the second increases; this gives $g_{1.4480} \leq -9.7 \times 10^{-6}$. Near $m = 1$ the estimates of Lemma 7.6, and near $m = -1$ the bound $h \leq \log 2$, do the rest.

[8] The second rule matters. Once V^t is tiny, no pattern lies within $\sqrt{V^t}$ of the decision boundary, the empirical A^t collapses, and the iteration falls back to the prior mean; we saw this in a test run at $\alpha = 1.47$.

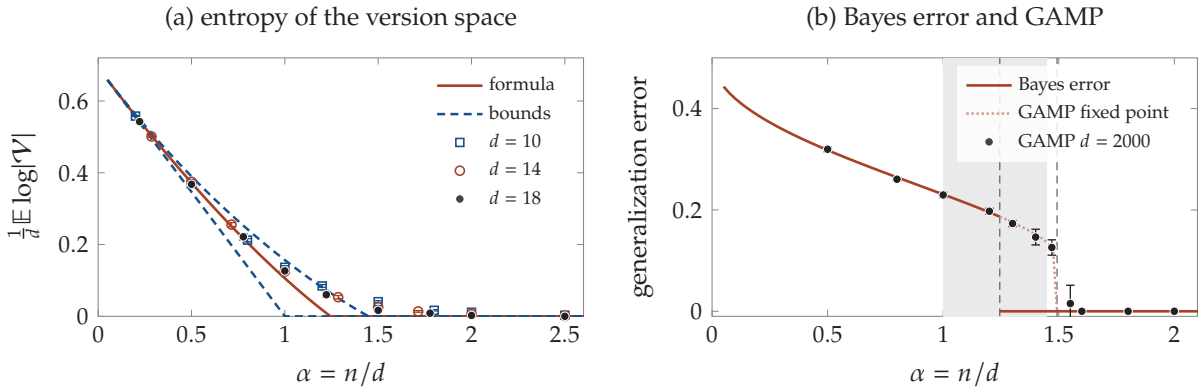


Figure 7.2. The Ising teacher. (a) Entropy of the version space: the formula of Theorem 7.5 (solid), the limits as $d \to \infty$ of the bounds of Theorem 7.7 (dashed: annealed upper bound and counting lower bound; the enumerations obey the finite-d bounds), and exact enumeration at $d = 10, 14, 18$ with 1500, 600, 250 instances per point, at $\alpha = n/d$, with Student-t 95% intervals (seeds $50000 + 1000j + 10k$). (b) The Bayes error of the formula (solid, jumping at α_{IT}), the error at GAMP's fixed point from an uninformative start where it differs (dotted, up to α_{AMP}), and Bayes-optimal GAMP at $d = 2000$: mean of 8 instances, 95% intervals, at most 150 iterations (seeds $7000 + 100k + s$). Shaded: the proved bracket $[1, \alpha_{\mathrm{ann}}]$ for α_{IT}; dashed lines at α_{IT} and α_{AMP}. Scripts: `07-glm-versionspace.py`, `07-glm-gamp.py`.

0.2306 at $\alpha = 1$, and 0.1259 ± 0.0151 against 0.1272 at $\alpha = 1.47$, in the hard phase, where no instance found the teacher. At $\alpha = 1.55$ seven of eight instances recovered w^* exactly, from 1.6 on all. Through the iterations the overlap follows state evolution: 0.708 after 5 iterations at $\alpha = 1.3$ (prediction 0.708), 0.925 after 10 at $\alpha = 1.6$ (0.926). Near the spinodal the fixed point is soft, single instances spread by about ± 0.05, and in exploratory runs with other seeds an occasional instance escaped to the teacher at $\alpha = 1.47$: finite-size effects that state evolution, a statement for a fixed number of iterations as $d \to \infty$, does not describe. With the Gaussian prior GAMP reaches the Bayes error (Figure 7.1(b)): 0.2546 ± 0.0073 against 0.2544 at $\alpha = 1$.

THE SUM RULE. The script `07-glm-sumrule.py` checks Theorem 7.3 by Monte Carlo with exact enumeration at $d = 6$, $n = 7$, for the probit channel with $\Delta = \frac{1}{2}$ (1.4×10^5 samples for f_d, 1.8×10^5 per path, six-point quadrature in t, seed 2026): $f_d = -0.7552 \pm 0.0007$. Along $q = r = 0$ the potential part is -0.8087 and the remainder $+0.0532$; along $q = 0.7$, $r = 0.3$, -0.6876 and -0.0681; along $q = 0.2 + 0.5t$, $r = 1 - 0.6t$, -0.7414 and -0.0143. The totals, -0.7555, -0.7556, -0.7557, agree: the remainder takes both signs and absorbs the choice of path.

7.7 The status board

The predictions of the physics box are theorems for Bayes-optimal learning from Gaussian patterns and all but countably many α, except that 1.2447 and 1.4926 are floating-point evaluations of proved formulas, and that the spinodal's meaning is proved for first-order methods with a fixed number of iterations, GAMP's own state evolution being a claim. That no efficient algorithm beats GAMP in the hard phase is a conjecture.

STATUS BOARD · the replica formula ↔ adaptive interpolation

PREDICTION	STATUS	PROOF OR EVIDENCE
Replica formula for the free entropy and mutual information of a GLM	proved under	(h1)–(h5), Bayes-optimality; Barbier et al. (2019), Thm 1
Overlap concentration (replica symmetry); errors of the scalar channel	proved under	$\alpha \in D^*$; averaged over a small perturbation; Barbier et al. (2019), Prop. 4, Thms 2–3
Ising teacher: first-order transition to perfect learning at $\alpha_{\mathrm{IT}} = 1.2447$	proved under	a floating-point evaluation, not certified, of Barbier et al. (2019), Cor. 3, Thm 2, which cover the noiseless sign channel (printed 1.249); Corollary 7.9
$1 \le \alpha_{\mathrm{IT}} \le 1.4480$, by counting and a first moment	proved	Theorem 7.7, elementary; grid certificate of $\alpha_{\mathrm{ann}} < 1.4480$
The annealed approximation of the learning curve is exact	false	an upper bound strictly above the formula: 0.157 against 0.106 at $\alpha = 1$
Gibbs error $\frac{1}{\pi}\arccos q_*$, Bayes error $\frac{1}{\pi}\arccos\sqrt{q_*}$; spherical teacher $0.442/\alpha$, $0.625/\alpha$	proved under	$\alpha \in D^*$; Proposition 7.8 with Barbier et al. (2019), Thms 2–3; constants: Computation 7.2
Message passing from an uninformative start learns perfectly if and only if $\alpha > \alpha_{\mathrm{AMP}} = 1.4926$	proved under	the best first-order method with t iterations as $d \to \infty$, then $t \to \infty$ (Celentano et al. 2020, Thm. 1 of arXiv v2, for every measurable channel of linear growth); a floating-point evaluation of (7.5); for GAMP itself, Barbier et al.'s Claim 1
No polynomial-time algorithm learns in the hard phase $(\alpha_{\mathrm{IT}}, \alpha_{\mathrm{AMP}})$	conjectured	physics (Barbier et al. 2019); Bayes-AMP optimal among first-order methods with boundedly many iterations (Celentano et al. 2020)

7.8 Where the translation fails

WHAT WAS ASSUMED. Independent patterns with independent entries: (h3) allows non-Gaussian entries, not the correlations of real data (Chapter 12). A Bayes-optimal student who knows prior and channel exactly: every step, from the derivative of the free entropy to the error formulas, uses the Nishimori identity, and Barbier et al. stress that their formula fails outside this setting. A student that minimizes a convex loss is the subject of Chapter 11 (Aubin et al. 2020); a Bayesian student with the wrong prior is a spin glass, and nothing here applies to it. A proportional limit and a product prior; our sum rule also needs the Rademacher prior, whose norm is exact, and a smooth channel. And the algorithmic statements concern first-order methods with a fixed number of iterations as $d \to \infty$ (for GAMP itself, a claim).

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. Outside the Bayes-optimal setting replica symmetry fails in neighboring models: for the Ising perceptron without a teacher the replica-symmetric capacity $4/\pi$, which Gardner and Derrida (1988) met (Chapter 8), exceeds the first-moment bound 1, and the correct value, Krauth and Mézard's zero-entropy point, is proved under numerical conditions (Ding and Sun 2025; Huang 2024) and, without them, in a computer-assisted preprint whose author states that it has not been independently reproduced or peer reviewed (Shmalo 2026). The rigorous paper prints 1.249, a value its formula does not give, where the physics literature had 1.245. And near the transition the limits are approached slowly (Section 7.6).

OPEN PROBLEMS. The one that Barbier et al. (2019) leave open: does some polynomial-time algorithm recover the Ising teacher for $\alpha_{\mathrm{IT}} < \alpha < \alpha_{\mathrm{AMP}}$? The evidence for hardness specific to this model is the physics of the metastable branch and the optimality of Bayes-AMP among first-order methods with boundedly many iterations (Celentano et al. 2020; Montanari and Wu 2024); low-degree bounds (Kunisky et al. 2022) and overlap-gap arguments (Gamarnik et al. 2022) exist for spiked matrices and the symmetric binary perceptron, and we did not find a result of either kind for the planted Ising perceptron. Second, the rate of convergence of $\frac{1}{d}\mathbb{E}\log|\mathcal{V}|$, slow near α_{IT} in our measurements. Third, the free entropy of a mismatched Bayesian student, where replica symmetry breaking is expected.

Exercises

7.1 CONSOLIDATION. (The sign channel.) Carry out Computation 7.1, check $2\Psi'$ against a finite difference of Ψ at $q = 0.2, 0.8, 0.95$, and interpret $\Psi(0) = -\log 2$ and $\Psi(1) = 0$.

7.2 CONSOLIDATION. (Stationary points.) For the Rademacher prior show that $\psi'(r) = \frac{1}{2}\mathbb{E}\tanh(r + \sqrt{r}Z)$ by Stein's and the Nishimori identities, that ψ is convex, and that $\frac{\mathrm{d}}{\mathrm{d}q}s(q;\alpha) = \alpha\Psi'(q) - \frac{1}{2}r(q)$, with $r(q)$ the inverse of $r \mapsto \mathbb{E}\tanh(r + \sqrt{r}Z)$; conclude that the stationary points of

s are the fixed points of (7.4). For the Gaussian prior, eliminate r and find the potential $\frac{1}{2}\log(1-q)+\alpha\Psi(q)+\frac{q}{2}$ of Barbier et al.'s Corollary 3.

7.3 CONSOLIDATION. (Gibbs and Bayes.) Show that $\arccos q/\arccos\sqrt{q}$ increases from 1 at $q=0$ to $\sqrt{2}$ as $q\to 1$; compare with part 1 of Proposition 7.8.

7.4 CONSOLIDATION. (The remainder at the origin.) Show that at $q=r=0$ the remainder in (7.6) is $\frac{1}{2d^2}\sum_\mu\int_0^1 \mathbb{E}\|\left\langle u'_{Y_{t,\mu}}(s_{t,\mu})\,w\right\rangle_t\|^2\,\mathrm{d}t\ge 0$. *(Hint: use the Nishimori identity with two replicas.)* Which bound on f_d follows?

7.5 BY HAND. (Linear regression.) For $y=s+\sqrt{\Delta}z$, show that $\Psi(q)=-\frac{1}{2}\log(2\pi e(\Delta+1-q))$. With the Gaussian prior, show that the error $m=1-q$ solves $m^2+(\Delta+\alpha-1)m-\Delta=0$, and that $m\to\max(0,1-\alpha)$ as $\Delta\to 0$. Compare with the bound $\alpha_{\mathrm{IT}}\ge 1$.

7.6 BY HAND. (The first moment.) Derive $g_\alpha(m)$, show that $g_\alpha(0)=(1-\alpha)\log 2$ and that $g_\alpha<0$ near $m=1$ for every $\alpha>0$, and compute $\alpha_{\mathrm{ann}}=1.4480$, with the maximizer at $m=0.6976$. Why does the annealed bound miss the transition at 1.2447?

7.7 BY HAND. ⋆ (A symmetric channel.) For $y=\mathrm{sgn}(|s|-\kappa)$ with $\kappa=0.6745$, so that both labels are equally likely, show that $q=0$ is a fixed point of (7.4) for every α, stable when $\alpha\cdot 2\Psi''(0)\cdot 2\psi''(0)<1$, that $2\psi''(0)=1$ and $2\Psi''(0)=16\kappa^2\phi(\kappa)^2$, and so that state evolution started from a small overlap returns to 0 for $\alpha<1.3605$. *(Hint: expand both scalar channels to second order, and use* $\int_{-\kappa}^{\kappa}(z^2-1)\phi(z)\,\mathrm{d}z=-2\kappa\phi(\kappa)$*.)*

7.8 TRANSLATION. (Replica symmetry is concentration.) Show with the Nishimori identity that the overlap of two posterior samples has the law of Q, and deduce that if $Q\to q$ in probability, then the overlap matrix of k samples and the teacher tends to the replica-symmetric matrix. Which step of the replica computation does this replace?

7.9 TRANSLATION. (The conjugate parameter.) Show, by the Nishimori identity, that at $t=1$ the mean of A_1 is $\frac{n}{d}$ times $2\Psi'(R_2(1))$. Conclude that the physicist's conjugate parameter $\hat{q}=r$ is the mean squared derivative of the output log-likelihood.

7.10 MEASUREMENT. (The spinodal at finite size.) Run `07-glm-gamp.py` at $d=500, 1000, 4000$ for $\alpha=1.47$ and 1.55, and record the fraction of instances that recover w^*. Why is GAMP's transition smeared around α_{AMP}, and why should the smearing shrink with d?

7.11 RESEARCH. (Noise.) For the probit channel with $\Delta>0$, show that $q_*<1$ for every α, and locate the critical Δ_c below which $q_*(\alpha)$ still jumps. Is there a hard phase for $\Delta<\Delta_c$?

7.12 RESEARCH. (The hard phase.) Find a polynomial-time algorithm that recovers the Ising teacher for some $\alpha<\alpha_{\mathrm{AMP}}$, or prove a low-degree lower bound for the planted Ising perceptron in $(\alpha_{\mathrm{IT}},\alpha_{\mathrm{AMP}})$.

Notes and References

The teacher–student perceptron entered physics with the "three unfinished works" of Gardner and Derrida (1989), as Barbier et al. (2019) recount. Györgyi (1990) found the first-order transition of the Ising teacher; Sompolinsky et al. (1990) and Seung et al. (1992) developed Gibbs learning, with its annealed and high-temperature approximations; Opper and Haussler (1991) computed the Bayes-optimal error. Engel and Van den Broeck (2001) review the physics, and Zdeborová and Krzakala (2016) place the model among inference problems with algorithmic thresholds.

Korada and Macris (2009) used interpolation on the Nishimori line for a gauge-symmetric spin glass; the replica formula for random linear estimation was proved by Reeves and Pfister (2016) and Barbier et al. (2020); Barbier and Macris (2019) introduced adaptive interpolation, and Barbier et al. (2019) used it for generalized linear models, inspired, they note, by Talagrand's interpolation paths for the perceptron without a teacher; Barbier (2021) developed the concentration of the overlap. GAMP is due to Rangan (2011), who stated its state evolution; Javanmard and Montanari (2013) proved state evolution for a general class of AMP algorithms with Lipschitz nonlinearities, building on Bayati and Montanari (2011); Celentano et al. (2020) and Montanari and Wu (2024) proved that Bayes-AMP is optimal among first-order methods with boundedly many iterations.

The sum rule (Theorem 7.3) is Proposition 3 of Barbier et al. (2019) made exact for the Rademacher prior and a smooth binary channel; the proof through the density of the data and Corollary 7.4 are the book's presentation. Lemma 7.6, Theorem 7.7, Proposition 7.8 and Corollary 7.9 are the book's own: the counting and first-moment arguments are classical in spirit, but we did not find them stated for this model, and Barbier et al. state their generalization result for the mean squared error of the label, not for classification errors.

TRANSLATING THE LITERATURE. Barbier et al. (2019) write n for our d, m for our n, Φ for the data matrix, X^* for w^*, ρ for $\mathbb{E}w_i^{*2}$ (here 1), $N(x)$ for our $\Phi(x)$, and ψ_{P_0}, $\Psi_{P_{\text{out}}}$, f_{RS}, f_n as we do; their $\alpha = m/n$ is ours. The physics of learning writes N for the number of weights and $P = \alpha N$ for the number of examples (Seung et al. 1992), and, in the review of Engel and Van den Broeck (2001), J and T for student and teacher, R for the teacher–student overlap (our Q) and q for the overlap of two students (our R_{12}); its generalization error $\frac{1}{\pi}\arccos R$ of a single student is our Gibbs error. In Rangan's GAMP the pairs $(\hat{p}, \tau^p)$, $(\hat{s}, \tau^s)$, $(\hat{r}, \tau^r)$, $(\hat{x}, \tau^x)$ are our (ω, V), $(g, -\partial_\omega g)$, $(\hat{w} + F^\top g/A, 1/A)$ and $(\hat{w}, v)$.

References

Aubin, B., F. Krzakala, Y. M. Lu, and L. Zdeborová (2020). "Generalization error in high-dimensional perceptrons: Approaching Bayes error with convex optimization". In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pp. 12199–12210. arXiv: 2006.06560.

Barbier, J. (2021). "Overlap matrix concentration in optimal Bayesian inference". In: *Information and Inference: A Journal of the IMA* 10.2, pp. 597–623. arXiv: 1904.02808.

Barbier, J., F. Krzakala, N. Macris, L. Miolane, and L. Zdeborová (2019). "Optimal errors and phase transitions in high-dimensional generalized linear models". In: *Proceedings of the National Academy of Sciences* 116.12, pp. 5451–5460. arXiv: 1708.03395.

Barbier, J. and N. Macris (2019). "The adaptive interpolation method: a simple scheme to prove replica formulas in Bayesian inference". In: *Probability Theory and Related Fields* 174.3-4, pp. 1133–1185. arXiv: 1705.02780.

Barbier, J., N. Macris, M. Dia, and F. Krzakala (2020). "Mutual Information and Optimality of Approximate Message-Passing in Random Linear Estimation". In: *IEEE Transactions on Information Theory* 66.7, pp. 4270–4303. arXiv: 1701.05823.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: 1001.3448.

Celentano, M., A. Montanari, and Y. Wu (2020). "The estimation error of general first order methods". In: *Proceedings of the 33rd Conference on Learning Theory (COLT)*. Vol. 125. Proceedings of Machine Learning Research, pp. 1078–1141. arXiv: 2002.12903.

Ding, J. and N. Sun (2025). "Capacity lower bound for the Ising perceptron". In: *Probability Theory and Related Fields* 193.3-4, pp. 627–715.

Engel, A. and C. Van den Broeck (2001). *Statistical Mechanics of Learning*. Cambridge University Press.

Gamarnik, D., E. C. Kizildag, W. Perkins, and C. Xu (2022). "Algorithms and Barriers in the Symmetric Binary Perceptron Model". In: *2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, pp. 576–587. arXiv: 2203.15667.

Gardner, E. and B. Derrida (1988). "Optimal storage properties of neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 271–284.

Gardner, E. and B. Derrida (1989). "Three unfinished works on the optimal storage capacity of networks". In: *Journal of Physics A: Mathematical and General* 22.12, pp. 1983–1994.

Györgyi, G. (1990). "First-order transition to perfect generalization in a neural network with binary synapses". In: *Physical Review A* 41.12, pp. 7097–7100.

Huang, B. (2024). "Capacity Threshold for the Ising Perceptron". In: *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, pp. 1126–1136. arXiv: 2404.18902v2 [math.PR].

Javanmard, A. and A. Montanari (2013). "State evolution for general approximate message passing algorithms, with applications to spatial coupling". In: *Information and Inference* 2.2, pp. 115–144. arXiv: 1211.5164.

Korada, S. B. and N. Macris (2009). "Exact Solution of the Gauge Symmetric p-Spin Glass Model on a Complete Graph". In: *Journal of Statistical Physics* 136.2, pp. 205–230.

Krauth, W. and M. Mézard (1989). "Storage capacity of memory networks with binary couplings". In: *Journal de Physique* 50.20, pp. 3057–3066.

Kunisky, D., A. S. Wein, and A. S. Bandeira (2022). "Notes on Computational Hardness of Hypothesis Testing: Predictions Using the Low-Degree Likelihood Ratio". In: *Mathematical Analysis, its Applications and Computation*. Springer Proceedings in Mathematics & Statistics. Springer, pp. 1–50. arXiv: 1907.11636.

Montanari, A. and Y. Wu (2024). "Statistically optimal first-order algorithms: a proof via orthogonalization". In: *Information and Inference: A Journal of the IMA* 13.4, iaae027. arXiv: 2201.05101.

Opper, M. and D. Haussler (1991). "Generalization performance of Bayes optimal classification algorithm for learning a perceptron". In: *Physical Review Letters* 66.20, pp. 2677–2680.

Rangan, S. (2011). "Generalized approximate message passing for estimation with random linear mixing". In: *2011 IEEE International Symposium on Information Theory Proceedings*, pp. 2168–2172. arXiv: 1010.5141.

Reeves, G. and H. D. Pfister (2016). "The replica-symmetric prediction for compressed sensing with Gaussian matrices is exact". In: *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, pp. 665–669. arXiv: 1607.02524.

Seung, H. S., H. Sompolinsky, and N. Tishby (1992). "Statistical mechanics of learning from examples". In: *Physical Review A* 45.8, pp. 6056–6091.

Shmalo, Y. (2026). *The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions*. arXiv preprint, v1 of 13 July 2026; unrefereed. arXiv: 2607.11010v1 [math.PR].

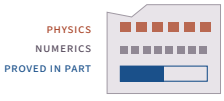
Sompolinsky, H., N. Tishby, and H. S. Seung (1990). "Learning from examples in large neural networks". In: *Physical Review Letters* 65.13, pp. 1683–1686.

Zdeborová, L. and F. Krzakala (2016). "Statistical physics of inference: thresholds and algorithms". In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

Part III

Comparison and Convexity

Many learning problems are convex, and for convex problems with Gaussian data one comparison inequality, Gordon's, replaces the whole replica calculation. The replica-symmetric saddle point and Gordon's auxiliary problem turn out to be the same equations, which is why the physicists were right; where the problem stops being convex, Gordon gives only a bound, and the replicas break.



8 Gaussian Comparison

Gordon's theorem and the capacity of the perceptron

THE DICTIONARY — Gardner's volume ↔ Gordon's min-max

IN THIS CHAPTER. How many random patterns can a perceptron store with margin κ? Gardner (1988) computed the answer from the replica-symmetric volume of solutions: $\alpha_c(\kappa) = 1/\mathbb{E}(Z+\kappa)_+^2$ patterns per dimension, which is 2 at zero margin. We derive her formula and test the stability of her saddle point, then prove Gordon's comparison inequality and the convex Gaussian min–max theorem completely, by Gaussian interpolation; with them the capacity for $\kappa \geq 0$ reduces to a comparison of two Gaussian norms, $\|(g+\kappa\mathbf{1})_+\|$ against $\|h\|$, with explicit bounds at every size. Exact maximal margins at $d = 100$ to 400 agree with Gardner's curve to about 0.01. Below zero margin the problem is not convex and the comparison gives only an upper bound; physics predicts that the true capacity needs full replica-symmetry breaking, and it is open.

8.1 The question: how many patterns fit on a sphere?

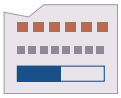
A perceptron with weights $w \in \mathbb{R}^d$ gives the input x the label $\operatorname{sgn}\langle w\rangle x$. Show it n inputs $x_\mu \sim \mathcal{N}(0, I_d)$ with random labels $y_\mu = \pm 1$: can one weight vector reproduce all the labels, and with how much room to spare? The labels can be absorbed, because the *patterns* $g_\mu = y_\mu x_\mu$ are again independent $\mathcal{N}(0, I_d)$ vectors. In the book's normalization the weights lie on $\sqrt{d}\, S^{d-1}$, and every pattern must clear the *margin* $\kappa \in \mathbb{R}$, which physicists call the stability:

$$\frac{\langle g_\mu, w\rangle}{\sqrt{d}} \geq \kappa \qquad (\mu = 1, \dots, n), \qquad w \in \sqrt{d}\, S^{d-1}. \tag{8.1}$$

In the proofs we write $\theta = w/\sqrt{d} \in S^{d-1}$, so that the constraints read $\langle g_\mu\rangle\theta \geq \kappa$, and stack the patterns as the rows of $G \in \mathbb{R}^{n\times d}$, which has i.i.d. $\mathcal{N}(0,1)$ entries. Let $\mathcal{F}_\kappa$ be the event that (8.1) has a solution. The *capacity* $\alpha_c(\kappa)$ is the critical load: as $n, d \to \infty$ with $n/d \to \alpha$, $\mathbb{P}(\mathcal{F}_\kappa) \to 1$ for $\alpha < \alpha_c(\kappa)$ and $\mathbb{P}(\mathcal{F}_\kappa) \to 0$ for $\alpha > \alpha_c(\kappa)$. It is the memorization threshold of a linear classifier, and the largest margin at a given load is that of the hard-margin support vector machine through the origin.

For $\kappa > 0$ each constraint keeps a cap of the sphere, and the solution set is the trace of a convex polyhedron that avoids the origin. For $\kappa < 0$ each constraint removes a cap, and the solution set, a sphere with n holes, is not convex. This difference will decide everything. At $\kappa = 0$ the answer is exact at every size: some unit vector has $\langle g_\mu\rangle\theta > 0$ for all μ with probability

PHYSICS
NUMERICS
PROVED IN PART



$2^{1-n}\sum_{k<d}\binom{n-1}{k}$ (Cover 1965; Wendel 1962), which is $\frac{1}{2}$ at $n = 2d$ (Figure P.1). For general margins the formula comes from physics.

THE PHYSICAL PREDICTION · Gardner 1988. For $n = \alpha d$ Gaussian patterns, the typical volume of the set of weights that satisfy (8.1) is $e^{d\min_q \phi_{\mathrm{RS}}(q)+o(d)}$, with ϕ_{RS} the replica-symmetric free entropy (8.3); it shrinks to a point as α increases to the capacity

$$\alpha_c(\kappa) = \Big(\int_{-\kappa}^{\infty}(z+\kappa)^2\,\mathrm{D}z\Big)^{-1} = \frac{1}{(1+\kappa^2)\Phi(\kappa)+\kappa\phi(\kappa)}, \qquad \alpha_c(0) = 2.$$

We write $\varrho(\kappa) = \mathbb{E}(Z+\kappa)_+^2$, so that $\alpha_c = 1/\varrho$; for instance $\alpha_c(1) = 0.5196$ and $\alpha_c(-1) = 13.273$. How do physicists derive the formula, and for which margins is it a theorem? Section 8.2 follows Gardner's computation and its stability test; Sections 8.4 and 8.5 prove Gordon's inequality and the convex Gaussian min–max theorem, which the rest of Part III uses, and deduce the capacity for $\kappa \geq 0$ and an upper bound for $\kappa < 0$ (Theorem 8.8).

8.2 The physics: Gardner's volume of solutions

Gardner asked not what a learning rule does but what the space of all good weight vectors looks like (Gardner 1988). Its volume is a partition function at zero temperature, in which every solution has weight one and every other weight vector weight zero. With σ_d the uniform probability measure on $\sqrt{d}\,S^{d-1}$ and Θ the indicator of $[0,\infty)$, the *free entropy of the solution set* is

$$\frac{1}{d}\mathbb{E}\log V, \qquad V(G) = \int\prod_{\mu=1}^{n}\Theta\Big(\frac{\langle g_\mu, w\rangle}{\sqrt{d}} - \kappa\Big)\sigma_d(\mathrm{d}w), \tag{8.2}$$

the logarithm per dimension of the fraction of the sphere that satisfies the constraints; the capacity is where it tends to $-\infty$.

REPLICAS FOR THE VOLUME. *The replica trick.* Write $\mathbb{E}\log V$ as the limit of $\frac{1}{r}\log\mathbb{E}V^r$ as $r \to 0$, and compute $\mathbb{E}V^r$ for integer r as an integral over r independent copies $w^1,\dots,w^r$ of the weights, the *replicas*. The patterns are independent copies of $g \sim \mathcal{N}(0, I_d)$, so

$$\mathbb{E}V^r = \int\prod_{a=1}^{r}\sigma_d(\mathrm{d}w^a)\;\mathbb{P}_g\big(\langle g, w^a\rangle \geq \kappa\sqrt{d},\ a \leq r\big)^n.$$

The overlaps. For fixed replicas the fields $\lambda_a = \langle g, w^a\rangle/\sqrt{d}$ are centered Gaussian with covariance the *overlap matrix* $Q_{ab} = \langle w^a, w^b\rangle/d$, $Q_{aa} = 1$; so the probability is $P_\kappa(Q) = \mathbb{P}(\lambda \geq \kappa\mathbf{1})$ for $\lambda \sim \mathcal{N}(0, Q)$. The Gram matrix of r independent uniform points of the sphere has a density proportional to $\det(Q)^{(d-r-1)/2}$, and Laplace's method gives

$$\frac{1}{d}\log\mathbb{E}V^r \longrightarrow \sup_{Q}\Big\{\tfrac{1}{2}\log\det Q + \alpha\log P_\kappa(Q)\Big\},$$

the entropy of r directions with prescribed overlaps plus the energy of the constraints.

Replica symmetry. Assume that the supremum is attained at $Q_{ab} = q$ for all $a \neq b$. Then $\det Q = (1-q)^{r-1}(1-q+rq)$, and the fields can be written $\lambda_a = \sqrt{q}\, z + \sqrt{1-q}\, \xi_a$ with $z, \xi_1, \dots, \xi_r$ independent standard normal, so that $P_\kappa(Q) = \mathbb{E}_z\big[H\big(\frac{\kappa-\sqrt{q}z}{\sqrt{1-q}}\big)^r\big]$, with $H(x) = \mathbb{P}(Z > x)$.

The limit $r \to 0$. To first order in r, $\frac{1}{2}\log\det Q = \frac{r}{2}\big(\log(1-q) + \frac{q}{1-q}\big)$ and $\log P_\kappa(Q) = r\,\mathbb{E}\log H\big(\frac{\kappa-\sqrt{q}Z}{\sqrt{1-q}}\big)$. Divide by r. For $r < 1$ a matrix has a negative number $r(r-1)/2$ of off-diagonal entries, and, by the usual inversion of the replica method, the supremum over q becomes an infimum:

$$\begin{aligned} \frac{1}{d}\mathbb{E}\log V &\longrightarrow \min_{0\le q<1} \phi_{\mathrm{RS}}(q), \\ \phi_{\mathrm{RS}}(q) &= \frac{1}{2}\log(1-q) + \frac{q}{2(1-q)} + \alpha\,\mathbb{E}\log H\Big(\frac{\kappa-\sqrt{q}\,Z}{\sqrt{1-q}}\Big). \end{aligned} \tag{8.3}$$

This is the replica-symmetric free entropy of the solution set. ◇

Four steps are uncontrolled: the continuation to $r \to 0$, which integer moments do not determine (Chapter 1); the exchange of $r \to 0$ with $d \to \infty$; the ansatz; and the choice of a minimum. At the minimizer, q is the typical overlap of two solutions, which a probabilist would call the limit in probability of $\langle \theta^1, \theta^2 \rangle$ for two independent uniform points of the random set $\{\theta \in S^{d-1} : G\theta \ge \kappa\mathbf{1}\}$. Differentiating (8.3) with the heat-equation identity of Exercise 8.7 gives Gardner's equation

$$\frac{q}{1-q} = \alpha\,\mathbb{E}\Big[R\Big(\frac{\kappa-\sqrt{q}\,Z}{\sqrt{1-q}}\Big)^2\Big], \qquad R(y) = \frac{\phi(y)}{H(y)}, \tag{8.4}$$

with R the inverse Mills ratio. As α grows the solutions become fewer and more alike, and q increases.

COMPUTATION 8.1 · THE CAPACITY FROM $q \to 1$

1 *The tail function.* As $y \to \infty$, $\log H(y) = -\frac{y^2}{2} - \log(y\sqrt{2\pi}) + o(1)$, and $\log H(y) \to 0$ as $y \to -\infty$; hence $\varepsilon \log H(x/\sqrt{\varepsilon}) \to -\frac{1}{2}x_+^2$ as $\varepsilon \downarrow 0$, for every $x \neq 0$.

2 *The divergence of ϕ_{RS}.* Put $\varepsilon = 1-q$. The entropic terms of (8.3) satisfy $\varepsilon(\frac{1}{2}\log\varepsilon + \frac{1-\varepsilon}{2\varepsilon}) \to \frac{1}{2}$, and step 1 with dominated convergence ($0 \le -\log H(y) \le 2 + y_+^2$) gives $\varepsilon\,\mathbb{E}\log H(\frac{\kappa-\sqrt{q}Z}{\sqrt{\varepsilon}}) \to -\frac{1}{2}\mathbb{E}(\kappa - Z)_+^2 = -\frac{1}{2}\varrho(\kappa)$. So, as $q \to 1$,

$$\phi_{\mathrm{RS}}(q) = \frac{1-\alpha\varrho(\kappa)}{2(1-q)} + o\Big(\frac{1}{1-q}\Big).$$

3 *The capacity.* If $\alpha\varrho(\kappa) < 1$, then $\phi_{\mathrm{RS}} \to +\infty$ as $q \to 1$, the minimum is attained at some $q^* < 1$, and the volume has a finite exponent.

If $\alpha\varrho(\kappa) > 1$, then $\phi_{\mathrm{RS}} \to -\infty$ and no volume is left. So $\alpha_c = 1/\varrho$, and $q^* \to 1$ as $\alpha \uparrow \alpha_c$.

4 *The closed form.* Integration by parts gives $\int_{-\kappa}^{\infty} z^2\phi = \Phi(\kappa) - \kappa\phi(\kappa)$ and $\int_{-\kappa}^{\infty} z\phi = \phi(\kappa)$, so

$$\varrho(\kappa) = \int_{-\kappa}^{\infty} (z+\kappa)^2 \phi(z)\,\mathrm{d}z = (1+\kappa^2)\Phi(\kappa) + \kappa\phi(\kappa), \tag{8.5}$$

which is $\frac{1}{2}$ at $\kappa = 0$.

Keeping the logarithmic terms of step 1 gives $1-q^* \simeq (1-\alpha/\alpha_c)/(1+\alpha\Phi(\kappa))$ near capacity (Exercise 8.4); the script `08-gordon-numbers.py` solves (8.4) and finds $1 - q^* = 5.10 \times 10^{-4}$ at $\kappa = 0$, $\alpha = 0.999\,\alpha_c$, against 5.00×10^{-4}. At a fixed load the formula gives Gardner's *maximal stability*, the largest margin with a positive volume: the $\kappa^*(\alpha)$ that solves $\alpha_c(\kappa^*) = \alpha$.

8.2.1 Above capacity: the replica-symmetric ground state

Above capacity the physicist asks for the weights that violate the constraints least, with the energy $E(w) = \frac{1}{2}\sum_\mu (\kappa - \lambda_\mu)_+^2$, $\lambda_\mu = \langle g_\mu \rangle w/\sqrt{d}$, the harmonic cost that Franz et al. (2017) use for jamming (Gardner and Derrida (1988) count the violated constraints instead).[1]

[1] The standard zero-temperature computation, written out for this book; Chapter 9 does the same for least squares.

THE GROUND STATE AT ZERO TEMPERATURE. Replace $\Theta(\lambda - \kappa)$ by $e^{-\beta V(\lambda)}$, $V(\lambda) = \frac{1}{2}(\kappa - \lambda)_+^2$. As $\beta \to \infty$ the replicas collapse onto the minimizer with the *susceptibility* $\chi = \beta(1-q)$ fixed, and Laplace's method in the individual noise gives a Moreau envelope,

$$\frac{1}{\beta}\log \mathbb{E}_\xi e^{-\beta V(h+\sqrt{\chi/\beta}\,\xi)} \longrightarrow -\min_y \Big\{ V(y) + \frac{(y-h)^2}{2\chi} \Big\} = -\frac{(\kappa-h)_+^2}{2(1+\chi)},$$

and the entropic term $\frac{q}{2(1-q)}$, divided by β, tends to $\frac{1}{2\chi}$. The ground-state energy is minus the minimum over χ:

$$\frac{1}{d}\min_w E(w) \longrightarrow -\min_{\chi>0}\Big\{ \frac{1}{2\chi} - \frac{\alpha\varrho(\kappa)}{2(1+\chi)} \Big\} = \frac{1}{2}\big(\sqrt{\alpha\varrho(\kappa)} - 1\big)_+^2, \tag{8.6}$$

attained at $\chi^* = 1/(\sqrt{\alpha\varrho} - 1)$ when $\alpha\varrho > 1$; for $\alpha\varrho \le 1$ the infimum 0 is approached as $\chi \to \infty$. ◇

Section 8.5 meets the same number from the other side: Gordon's auxiliary problem has the value $(\|(g + \kappa\mathbf{1})_+\| - \|h\|)_+$, whose square over $2d$ tends to (8.6).

8.2.2 Is the saddle point stable?

Replica symmetry is an ansatz, and its first test is local: does the saddle point survive perturbations δQ in the *replicon* directions, symmetric with zero diagonal and zero row sums, which break the symmetry between replicas? For the Sherrington–Kirkpatrick model this test gives the de

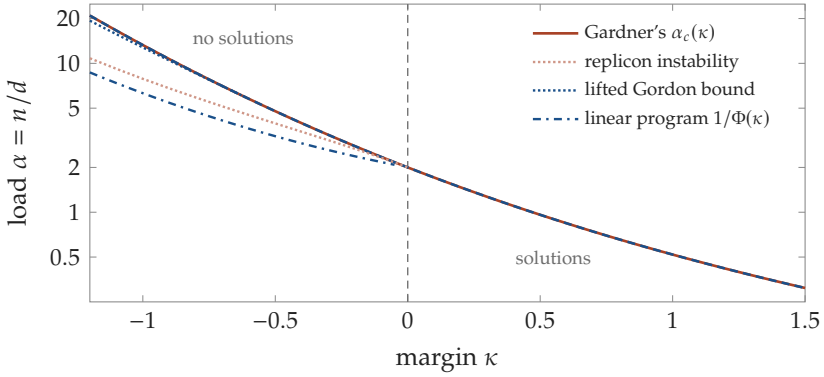


Figure 8.1. The phase diagram of the spherical perceptron. Carnelian: Gardner's $\alpha_c(\kappa)$, and the replicon instability $\lambda_R = 1$ of (8.7), computed on the replica-symmetric branch at $\kappa = -1.2, -1.1, \dots, -0.1, -0.05$. Lapis, what is proved: the capacity is $\alpha_c(\kappa)$ for $\kappa \geq 0$ and at most $\alpha_c(\kappa)$ for $\kappa < 0$ (dashed, Theorem 8.8); the lifted bound is smaller below $\kappa = -0.622$ (dotted, barely visible: 12.784 against 13.273 at $\kappa = -1$); a linear program finds solutions below $1/\Phi(\kappa)$ (dash-dotted, Montanari et al. (2024)). Script: `08-gordon-numbers.py`.

Almeida–Thouless line (Chapter 3). Since $Q^{-1}\delta Q = \delta Q/(1-q)$ on these directions, the entropic term has second variation $-\frac{1}{4(1-q)^2}\sum_{a\neq b}\delta Q_{ab}^2$; the energetic term contributes $\frac{\alpha}{4}\mathbb{E}[(\partial_h^2 \log \mathcal{Z}_q(h))^2]\sum_{a\neq b}\delta Q_{ab}^2$ at $h = \sqrt{q}Z$, as for the Sherrington–Kirkpatrick model with $\mathcal{Z}_q(h) = H((\kappa - h)/\sqrt{1-q})$ in place of cosh. The saddle point is stable when the entropic term wins, that is, with $y = (\kappa - \sqrt{q}Z)/\sqrt{1-q}$, when

$$\lambda_R = \alpha(1-q)^2\,\mathbb{E}\left[(\partial_h^2 \log \mathcal{Z}_q(h))^2\right]_{h=\sqrt{q}Z} = \alpha\,\mathbb{E}\left[R'(y)^2\right] \leq 1, \tag{8.7}$$

the condition written by Franz et al. (2017, eq. (52)); here $R' = R(R-y) \in (0,1)$. As $q \to 1$, $R'(y) \to 1$ where $Z < \kappa$ and $R'(y) \to 0$ where $Z > \kappa$, so at capacity $\lambda_R \to \alpha_c(\kappa)\Phi(\kappa)$, and by (8.5)

$$\alpha_c(\kappa)\Phi(\kappa) \leq 1 \iff \kappa^2\Phi(\kappa) + \kappa\phi(\kappa) \geq 0 \iff \kappa\,\mathbb{E}(Z+\kappa)_+ \geq 0 \iff \kappa \geq 0.$$

The value, 0.437 at $\kappa = 1$, 1 at $\kappa = 0$ and 2.106 at $\kappa = -1$, counts the patterns that touch the margin per dimension, the isostaticity index of Franz et al. (2017). For $\kappa < 0$ the instability sets in before capacity, on the line $\alpha_{\mathrm{dAT}}(\kappa)$ of Figure 8.1 (3.951 at $\kappa = -0.5$, against $\alpha_c = 4.770$). Gardner and Derrida (1988, Figure 1) already found the replica-symmetric saddle point stable along the critical line for $\kappa > 0$ and unstable for $\kappa < 0$; Franz et al. (2017) confirmed that replica symmetry is stable for every $\kappa > 0$, and found that full replica-symmetry breaking covers the whole jamming line for $\kappa < 0$. Physics predicts its own failure: $\alpha_c(\kappa)$ is the capacity for $\kappa \geq 0$ and not for $\kappa < 0$.

8.2.3 Binary weights: when $q \to 1$ is the wrong criterion

For the *Ising perceptron*, $w \in \{\pm 1\}^d$, only the entropic term changes, and the replica-symmetric value of $\frac{1}{d}$ log of the number of solutions is (Aubin et al. 2019, eq. (2))

$$s_{\mathrm{RS}} = \operatorname*{extr}_{q,\hat{q}} \Big\{ -\frac{\hat{q}(1-q)}{2} + \mathbb{E}\log\big(2\cosh(\sqrt{\hat{q}}\,Z)\big) + \alpha\,\mathbb{E}\log H\Big(\frac{\kappa - \sqrt{q}\,Z}{\sqrt{1-q}}\Big)\Big\}, \quad (8.8)$$

with stationarity equations $q = \mathbb{E}\tanh^2(\sqrt{\hat{q}}Z)$ and $\hat{q} = \frac{\alpha}{1-q}\mathbb{E}R(y)^2$. Gardner's criterion $q \to 1$ gives $1 - q \simeq \sqrt{2/(\pi\hat{q})}$ and $\hat{q}(1-q)^2 \to \alpha\varrho$, hence $\alpha = 2/(\pi\varrho(\kappa))$, which is $4/\pi = 1.273$ at $\kappa = 0$ (Exercise 8.5). But the expected number of solutions at zero margin is 2^{d-n}, so the capacity is at most 1; Gardner and Derrida (1988), who derived $4/\pi$, pointed out the inconsistency themselves. Krauth and Mézard (1989) argued that the solutions are isolated points, so that q never approaches 1 continuously (their one-step replica-symmetry-breaking solution has $q_1 = 1$), and that the capacity is where s_{RS} vanishes: the script finds $s_{\mathrm{RS}} = 0.314$, 0.035 and -0.019 at $\alpha = 0.5$, 0.8 and 0.85, and the zero at $\alpha_\star = 0.8330786$.

8.3 The stone

PHYSICS · *Gardner's replica computation*

1 · What is computed

The typical log-volume of the solution set, a free entropy at zero temperature.

2 · Average over the patterns

Replicate and average: the patterns decouple, and the replicas interact only through the overlaps Q_{ab}.

3 · Replica symmetry

Ansatz $Q_{ab} = q$: each pattern sees a field $\sqrt{q}\,z$ common to the replicas plus independent noise.

4 · The saddle point

$\phi_{\mathrm{RS}} \simeq \frac{1-\alpha\varrho}{2(1-q)}$ as $q \to 1$; above capacity the ground state is $\frac{1}{2}(\sqrt{\alpha\varrho}-1)_+^2$, at $\chi^* = 1/(\sqrt{\alpha\varrho}-1)$.

PROOF · *Gordon's comparison and the CGMT*

The sign of the min–max ξ_S of (8.12), zero exactly when a solution exists. No volume is computed.

Interpolation compares $\lambda^\top G\theta$ with $\|\theta\|g^\top\lambda + \|\lambda\|h^\top\theta$; the covariances differ by a product of two Cauchy–Schwarz gaps (Lemma 8.2).

One Gaussian vector for the patterns, one for the weights, nothing to break; optimizing over directions leaves two norms (Lemma 8.6).

The value $(\|(g+\kappa\mathbf{1})_+\| - \|h\|)_+$: its sign is that of $\alpha\varrho - 1$, and its square over $2d$ tends to $\frac{1}{2}(\sqrt{\alpha\varrho}-1)_+^2$.

5 · Self-averaging	
Assumed: $\log V$ concentrates.	Proved for two 1-Lipschitz norms, with bounds at every size (Lemma 8.7).
6 · Below capacity	
$q^* < 1$: the volume is positive, and nothing more is needed.	No counterpart in physics: the reverse inequality. For $\kappa > 0$, relax to the ball, exchange min and max, apply Gordon to the dual (Theorem 8.3(ii)).
7 · Negative margins	
The replicon exceeds 1 at capacity: replica symmetry breaks.	The ball contains $\theta = 0$, the set is not convex, only Gordon's direction survives: an upper bound, improved for $\kappa < -0.622$.

WHAT THE TRANSLATION SHOWS. Two counterparts are identities: the coefficient of the divergence of ϕ_{RS} is the limit of $(\|h\|^2 - \|(g + \kappa\mathbf{1})_+\|^2)/(2d)$, and the replica-symmetric ground state is the squared auxiliary value over $2d$, with $\chi^* = \|h\|/(\|(g + \kappa\mathbf{1})_+\| - \|h\|)$. One overlap and two norms carry the same information because both computations decouple the patterns from the weights, by averaging or by comparison. The proof never continues in r and never computes a volume, and it needs one idea that physics lacks, the reverse inequality, which convexity provides. Where convexity fails the stability test fails too: the two languages agree about where each can be trusted.

8.4 The proof: Gordon's inequality and the convex Gaussian min–max theorem

This section proves the comparison theorem behind Part III: Gordon's inequality, its min–max form, and the convex Gaussian min–max theorem as Thrampoulidis et al. (2018) state it. The only tool is the interpolation lemma of the appendix.

Theorem 8.1 (Gordon's comparison inequality). *Let (X_{ij}) and (Y_{ij}), $1 \le i \le I$, $1 \le j \le J$, be centered Gaussian vectors such that*

(i) $\mathbb{E}X_{ij}^2 = \mathbb{E}Y_{ij}^2$ *for all i, j;*

(ii) $\mathbb{E}X_{ij}X_{ik} \ge \mathbb{E}Y_{ij}Y_{ik}$ *for all i and all j, k;*

(iii) $\mathbb{E}X_{ij}X_{lk} \le \mathbb{E}Y_{ij}Y_{lk}$ *for all $i \ne l$ and all j, k.*

Then, for all real numbers λ_{ij},

$$\mathbb{P}\Big(\bigcap_{i=1}^{I}\bigcup_{j=1}^{J}\{Y_{ij} \ge \lambda_{ij}\}\Big) \ge \mathbb{P}\Big(\bigcap_{i=1}^{I}\bigcup_{j=1}^{J}\{X_{ij} \ge \lambda_{ij}\}\Big). \tag{8.9}$$

In particular $\mathbb{P}(\min_i \max_j Y_{ij} \ge c) \ge \mathbb{P}(\min_i \max_j X_{ij} \ge c)$ *for every* c, *and* $\mathbb{E}\min_i \max_j Y_{ij} \ge \mathbb{E}\min_i \max_j X_{ij}$.

Proof. The plan is to write the event as a smooth product of the coordinates, to check that every mixed second derivative has the sign that makes the interpolation of Lemma A.2 monotone, and to let the smooth functions tend to indicators. Only the laws of X and Y enter, so we take them independent.

A smooth version of the event. For $\eta \in (0,1]$ let $f_\eta(s) = \Phi(s/\eta + \eta^{-1/2})$: it is infinitely differentiable and nondecreasing, with values in $(0,1)$ and bounded derivatives of all orders, and $f_\eta(s) \to \mathbf{1}\{s \ge 0\}$ for every s as $\eta \downarrow 0$. For $x = (x_{ij})$ put $u_{ij} = f_\eta(x_{ij} - \lambda_{ij})$, $A_i = 1 - \prod_{p=1}^{J}(1 - u_{ip})$ and $F(x) = \prod_{i=1}^{I} A_i$. As a polynomial in the u_{ij}, F is infinitely differentiable with bounded derivatives, and $0 \le F \le 1$.

The signs of the second derivatives. Write $f'_{ij} = f'_\eta(x_{ij} - \lambda_{ij}) \ge 0$. Then $\partial_{ij}F = f'_{ij}\prod_{p\neq j}(1-u_{ip})\prod_{l\neq i}A_l$. Differentiating once more, for $k \neq j$ in the same row and for $l \neq i$ in another row,

$$\partial_{ik}\partial_{ij}F = -f'_{ij}f'_{ik}\prod_{p\neq j,k}(1-u_{ip})\prod_{l\neq i}A_l \le 0,$$

$$\partial_{lk}\partial_{ij}F = f'_{ij}f'_{lk}\prod_{p\neq j}(1-u_{ip})\prod_{p\neq k}(1-u_{lp})\prod_{m\neq i,l}A_m \ge 0,$$

because all the factors u, $1-u$ and A lie in $[0,1]$.

Interpolation. By Lemma A.2, with $Z_t = \sqrt{t}\,X + \sqrt{1-t}\,Y$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}F(Z_t) = \frac{1}{2}\sum_{(i,j),(l,k)}\big(\mathbb{E}X_{ij}X_{lk} - \mathbb{E}Y_{ij}Y_{lk}\big)\,\mathbb{E}\big[\partial_{ij}\partial_{lk}F(Z_t)\big].$$

We check the sign term by term. The diagonal terms vanish by hypothesis (i). For $l = i$ and $k \neq j$ the covariance difference is nonnegative by (ii) and the derivative nonpositive; for $l \neq i$ the difference is nonpositive by (iii) and the derivative nonnegative. So $\mathbb{E}F(X) = \mathbb{E}F(Z_1) \le \mathbb{E}F(Z_0) = \mathbb{E}F(Y)$.

The limit. As $\eta \downarrow 0$, $u_{ij} \to \mathbf{1}\{x_{ij} \ge \lambda_{ij}\}$, so $F(x)$ tends to $\prod_i(1 - \prod_j \mathbf{1}\{x_{ij} < \lambda_{ij}\})$, the indicator of the event in (8.9), at every x; bounded convergence gives (8.9). With $\lambda_{ij} = c$ the events are $\{\min_i \max_j Y_{ij} \ge c\}$ and $\{\min_i \max_j X_{ij} \ge c\}$, and the expectations compare because $\mathbb{E}W = \int_0^\infty \mathbb{P}(W \ge c)\,\mathrm{d}c - \int_{-\infty}^0 \mathbb{P}(W < c)\,\mathrm{d}c$ for an integrable W. ∎

With one row ($I = 1$), hypothesis (iii) is empty and the theorem is Slepian's lemma, Theorem A.4(i); Gordon's inequality lets the two levels compare in opposite directions. Gordon (1985) proved it as a generalization of the inequalities of Slepian and Fernique; the interpolation proof is standard. For learning, the bilinear form matters most.

Lemma 8.2 (Gordon's min–max inequality). *Let* $G \in \mathbb{R}^{n\times d}$, $g \in \mathbb{R}^n$, $h \in \mathbb{R}^d$ *and* $z \in \mathbb{R}$ *be independent, with i.i.d.* $\mathcal{N}(0,1)$ *entries. Let* $S_w \subset \mathbb{R}^d$ *and* $S_u \subset \mathbb{R}^n$ *be nonempty compact sets and* ψ *a continuous function on*

$S_w \times S_u$. *Define*

$$\mathcal{P}(G,z) = \min_{w\in S_w}\max_{u\in S_u}\big(u^\top G w + z\|w\|\|u\| + \psi(w,u)\big),$$
$$\mathcal{A}(g,h) = \min_{w\in S_w}\max_{u\in S_u}\big(\|w\|\, g^\top u + \|u\|\, h^\top w + \psi(w,u)\big).$$

Then $\mathbb{P}(\mathcal{P}(G,z) < c) \le \mathbb{P}(\mathcal{A}(g,h) \le c)$ *for every* $c \in \mathbb{R}$.

Proof. The plan is to apply Theorem 8.1 on finite subsets, with rows indexed by w and columns by u, and to pass to the compact sets by approximation.

Finite sets. For finite $W \subset S_w$, $U \subset S_u$ and $(w,u) \in W \times U$, let $X_{wu} = \|w\|\, g^\top u + \|u\|\, h^\top w$ and $Y_{wu} = u^\top G w + z\|w\|\|u\|$. Since $\mathbb{E}(u^\top G w)(u'^\top G w') = \langle u,u'\rangle\langle w,w'\rangle$, $\mathbb{E}(g^\top u)(g^\top u') = \langle u,u'\rangle$ and $\mathbb{E}(h^\top w)(h^\top w') = \langle w,w'\rangle$, expanding the products gives

$$\mathbb{E}X_{wu}X_{w'u'} - \mathbb{E}Y_{wu}Y_{w'u'} = -\big(\|w\|\|w'\| - \langle w,w'\rangle\big)\big(\|u\|\|u'\| - \langle u,u'\rangle\big).$$

Both factors are nonnegative by the Cauchy–Schwarz inequality, and the first vanishes when $w = w'$; so hypotheses (i) and (ii) of Theorem 8.1 hold with equality, and (iii) holds. With $\lambda_{wu} = c - \psi(w,u)$ the theorem gives $\mathbb{P}(\mathcal{P}_{W,U} \ge c) \ge \mathbb{P}(\mathcal{A}_{W,U} \ge c)$ for the values of the problems restricted to $W \times U$, that is, $\mathbb{P}(\mathcal{P}_{W,U} < c) \le \mathbb{P}(\mathcal{A}_{W,U} < c)$.

Compact sets. For $k \ge 1$ let $W_k \subset S_w$ and $U_k \subset S_u$ be finite sets within distance $1/k$ of every point of S_w and of S_u. For a fixed realization the objective $F(w,u) = u^\top G w + z\|w\|\|u\| + \psi(w,u)$ is uniformly continuous on the compact $S_w \times S_u$, with a modulus of continuity ω. Then $M(w) = \max_{S_u} F(w,\cdot)$ and $M_k(w) = \max_{U_k} F(w,\cdot)$ satisfy $|M_k - M| \le \omega(1/k)$, and M has modulus ω, so $|\min_{W_k} M_k - \min_{S_w} M| \le 2\omega(1/k)$. Hence $\mathcal{P}_k = \mathcal{P}_{W_k,U_k} \to \mathcal{P}(G,z)$ for every realization, and likewise $\mathcal{A}_k \to \mathcal{A}(g,h)$. If $\mathcal{P}(G,z) < c$, then $\mathcal{P}_k < c$ for all large k; if $\mathcal{A}_k < c$ for infinitely many k, then $\mathcal{A}(g,h) \le c$. Fatou's lemma, applied to indicators, gives

$$\mathbb{P}(\mathcal{P}(G,z) < c) \le \liminf_k \mathbb{P}(\mathcal{P}_k < c) \le \limsup_k \mathbb{P}(\mathcal{A}_k < c) \le \mathbb{P}(\mathcal{A}(g,h) \le c).$$

∎

The term $z\|w\|\|u\|$ only equalizes the variances. It can be removed at the price of a factor 2, and convexity then buys the reverse inequality.

Theorem 8.3 (convex Gaussian min–max theorem; Thrampoulidis et al. 2018, Theorem 6.1). *Let* $G \in \mathbb{R}^{n\times d}$, $g \in \mathbb{R}^n$ *and* $h \in \mathbb{R}^d$ *have i.i.d.* $\mathcal{N}(0,1)$ *entries, let* $S_w \subset \mathbb{R}^d$ *and* $S_u \subset \mathbb{R}^n$ *be compact sets, and let* ψ *be continuous on* $S_w \times S_u$. *Define the primary and the auxiliary optimization problems*

$$\mathcal{P}(G) = \min_{w\in S_w}\max_{u\in S_u}\big(u^\top G w + \psi(w,u)\big), \tag{8.10}$$
$$\mathcal{A}(g,h) = \min_{w\in S_w}\max_{u\in S_u}\big(\|w\|\, g^\top u + \|u\|\, h^\top w + \psi(w,u)\big), \tag{8.11}$$

and let $w_{\mathcal{P}}(G)$ *denote a minimizer in* (8.10). *Then:*

(i) for every $c \in \mathbb{R}$, $\mathbb{P}(\mathcal{P}(G) < c) \le 2\mathbb{P}(\mathcal{A}(g,h) \le c)$;

(ii) if, in addition, S_w and S_u are convex and ψ is convex–concave on $S_w \times S_u$, then for every $c \in \mathbb{R}$, $\mathbb{P}(\mathcal{P}(G) > c) \le 2\mathbb{P}(\mathcal{A}(g,h) \ge c)$; in particular, for all $\mu \in \mathbb{R}$ and $t > 0$, $\mathbb{P}(|\mathcal{P}(G) - \mu| > t) \le 2\mathbb{P}(|\mathcal{A}(g,h) - \mu| \ge t)$;

(iii) under the assumptions of (ii), let S be an open subset of S_w and $S^c = S_w \setminus S$, and let $\mathcal{P}_{S^c}(G)$ and $\mathcal{A}_{S^c}(g,h)$ be the optimal values of (8.10) and (8.11) with the minimization restricted to $w \in S^c$. If there are constants $\bar{a}$, $\bar{a}_{S^c}$ and $\eta > 0$ such that (a) $\bar{a}_{S^c} \ge \bar{a} + 3\eta$, (b) $\mathcal{A}(g,h) < \bar{a} + \eta$ with probability at least $1 - p$, and (c) $\mathcal{A}_{S^c}(g,h) > \bar{a}_{S^c} - \eta$ with probability at least $1 - p$, then $\mathbb{P}(w_{\mathcal{P}}(G) \in S) \ge 1 - 4p$.

Corollary 8.4 (asymptotic CGMT; Thrampoulidis et al. 2018, Corollary 6.1). *Consider a sequence of problems as in Theorem 8.3(iii), indexed by d, with $n = n(d)$. If there are constants $\bar{a} < \bar{a}_{S^c}$ such that $\mathcal{A}(g,h) \to \bar{a}$ and $\mathcal{A}_{S^c}(g,h) \to \bar{a}_{S^c}$ in probability, then $\mathbb{P}(w_{\mathcal{P}}(G) \in S) \to 1$.*

Proof of Theorem 8.3 and Corollary 8.4. The plan is to remove the term $z\|w\|\|u\|$ of Lemma 8.2 by conditioning on its sign, to obtain (ii) by applying (i) to the dual problem, and (iii) by comparing two optimal values.

(i) Let $z \sim \mathcal{N}(0,1)$ be independent of G. On $\{z \le 0\}$ the objective of $\mathcal{P}(G,z)$ is pointwise at most that of $\mathcal{P}(G)$, so $\mathcal{P}(G,z) \le \mathcal{P}(G)$; by independence and Lemma 8.2,

$$\tfrac{1}{2}\mathbb{P}(\mathcal{P}(G) < c) = \mathbb{P}(\mathcal{P}(G) < c,\ z \le 0) \le \mathbb{P}(\mathcal{P}(G,z) < c) \le \mathbb{P}(\mathcal{A}(g,h) \le c).$$

(ii) The function $(w,u) \mapsto u^\top G w + \psi(w,u)$ is continuous on the compact convex set $S_w \times S_u$, convex in w and concave in u, so Sion's minimax theorem (Sion 1958) gives $\mathcal{P}(G) = \max_u \min_w (u^\top G w + \psi(w,u))$, that is,

$$-\mathcal{P}(G) = \min_{u \in S_u} \max_{w \in S_w} \big(w^\top (-G^\top) u - \psi(w,u) \big),$$

a primary problem with the roles of w and u exchanged, the matrix $-G^\top$ with i.i.d. standard normal entries, and the continuous function $-\psi$. Its auxiliary problem, with standard Gaussian vectors $\tilde{g} \in \mathbb{R}^d$ and $\tilde{h} \in \mathbb{R}^n$, is $\min_u \max_w (\|u\|\, \tilde{g}^\top w + \|w\|\, \tilde{h}^\top u - \psi(w,u))$. Since $(\tilde{g}, \tilde{h})$ has the law of $(-h, -g)$, this has the law of

$$-\max_{u \in S_u} \min_{w \in S_w} \big(\|w\|\, g^\top u + \|u\|\, h^\top w + \psi(w,u) \big).$$

Part (i) at the level $-c$ gives $\mathbb{P}(\mathcal{P}(G) > c) \le 2\mathbb{P}(\max_u \min_w (\cdots) \ge c)$, and the weak minimax inequality $\max_u \min_w \le \min_w \max_u$, which needs no convexity, bounds the right side by $2\mathbb{P}(\mathcal{A}(g,h) \ge c)$. For the last claim, add (i) at $c = \mu - t$ and (ii) at $c = \mu + t$.

(iii) If S^c is empty there is nothing to prove. Otherwise S^c is compact, so (i) applies to the restricted problems and, with (c), gives $\mathbb{P}(\mathcal{P}_{S^c}(G) < \bar{a}_{S^c} - \eta) \le 2p$; by (ii) and (b), $\mathbb{P}(\mathcal{P}(G) > \bar{a} + \eta) \le 2p$. Outside these two events, so with probability at least $1 - 4p$, assumption (a) gives

$$\mathcal{P}_{S^c}(G) \ge \bar{a}_{S^c} - \eta \ge \bar{a} + 2\eta > \bar{a} + \eta \ge \mathcal{P}(G).$$

A minimizer $w_{\mathcal{P}}$ in S^c would give $\mathcal{P}(G) = \max_u(u^\top G w_{\mathcal{P}} + \psi(w_{\mathcal{P}}, u)) \geq \mathcal{P}_{S^c}(G)$, which this chain excludes; so every minimizer lies in S. For the corollary put $\eta = (\bar{a}_{S^c} - \bar{a})/3$: for every $p > 0$, (b) and (c) hold with probability at least $1 - p$ for all large d, and (iii) applies. ∎

Remark 8.5 (how the theorem is used). Part (i) needs no convexity: the auxiliary value is a high-probability lower bound for the primary one. Part (ii) adds the upper bound, so that if $\mathcal{A}(g, h)$ concentrates, so does $\mathcal{P}(G)$. Part (iii) localizes minimizers, typically in a shell $\{|\|w\| - a_*| < \varepsilon\}$; it is proved from (ii), so the convexity hypotheses are part of it, as in Thrampoulidis et al. (2015, Theorem 3); Thrampoulidis et al. (2018) do not repeat them in the statement of (iii). In applications one first confines the minimizer to a large ball (Chapter 9). The idea of combining Gordon's inequality with convex duality is Stojnic's (Stojnic 2013a); Thrampoulidis et al. (2015) first stated parts (i)–(ii), and Thrampoulidis et al. (2018) gave (iii) in this generality and credit Stojnic explicitly.

8.5 The proof: the capacity for nonnegative margins

We prove Gardner's formula for $\kappa \geq 0$ and the upper bound for every κ, following Stojnic (2013b) with the CGMT in place of his direct use of Gordon's inequality, and with bounds at every size. Write $\Lambda = \{\lambda \in \mathbb{R}^n : \lambda \geq 0,\ \|\lambda\| \leq 1\}$, B^d for the closed unit ball, and, for independent $g \sim \mathcal{N}(0, I_n)$ and $h \sim \mathcal{N}(0, I_d)$, $D_\kappa = \|(g + \kappa\mathbf{1})_+\| - \|h\|$. For $v \in \mathbb{R}^n$ and $\lambda \in \Lambda$, $\lambda^\top v \leq \lambda^\top v_+ \leq \|\lambda\|\|v_+\|$, with equality at $\lambda = v_+/\|v_+\|$ when $v_+ \neq 0$; so $\max_{\lambda\in\Lambda} \lambda^\top v = \|v_+\|$, and $\mathcal{F}_\kappa$ is the event that

$$\xi_S = \min_{\theta\in S^{d-1}} \max_{\lambda\in\Lambda} \lambda^\top(\kappa\mathbf{1} - G\theta) = \min_{\theta\in S^{d-1}} \|(\kappa\mathbf{1} - G\theta)_+\| \tag{8.12}$$

vanishes: a primary problem with $w = \theta$, $u = \lambda$, the matrix $-G$ and $\psi(\theta, \lambda) = \kappa\mathbf{1}^\top\lambda$.

Lemma 8.6 (the auxiliary problems of the perceptron). *Let $\kappa \in \mathbb{R}$, $g \in \mathbb{R}^n$, $h \in \mathbb{R}^d$, $v = g + \kappa\mathbf{1}$ and $D = \|v_+\| - \|h\|$. Let $\mathcal{A}_S$ and $\mathcal{A}_B$ be the values of*

$$\min_\theta \max_{\lambda\in\Lambda} \big(\|\theta\|\, g^\top\lambda + \|\lambda\|\, h^\top\theta + \kappa\mathbf{1}^\top\lambda\big)$$

with θ ranging over S^{d-1} and over B^d. If $v_+ \neq 0$, then $\mathcal{A}_S \geq D_+$; in all cases $\{\mathcal{A}_S \leq 0\} \subseteq \{D \leq 0\}$ and $\mathcal{A}_B \leq D_+$.

Proof. For $\theta \in S^{d-1}$ the objective is $\lambda^\top v + \|\lambda\|\, h^\top\theta$. If $v_+ \neq 0$, the choice $\lambda = v_+/\|v_+\|$ gives $\|v_+\| + h^\top\theta \geq D$ and $\lambda = 0$ gives 0, so the inner maximum is at least D_+ for every θ. Then $\mathcal{A}_S \leq 0$ forces $D \leq 0$; and if $v_+ = 0$, then $D = -\|h\| \leq 0$ anyway. For $\mathcal{A}_B$ take $\theta = -h/\|h\|$ (an arbitrary unit vector if $h = 0$); for every $\lambda \in \Lambda$ the objective is $\lambda^\top v - \|\lambda\|\|h\| \leq \|\lambda\|(\|v_+\| - \|h\|) \leq D_+$. ∎

Lemma 8.7 (two Gaussian norms). *Let $\varrho = \varrho(\kappa)$. Then* $\operatorname{Var} D_\kappa \leq 2$. *If $n\varrho > d + 1$, then $\mathbb{P}(D_\kappa \leq 0) \leq 2/(\sqrt{n\varrho - 1} - \sqrt{d})^2$; if $n\varrho < d - 1$, then*

$$\mathbb{P}(D_\kappa > 0) \le 2/(\sqrt{d-1} - \sqrt{n\varrho})^2.$$

PROOF. The maps $g \mapsto \|(g+\kappa\mathbf{1})_+\|$ and $h \mapsto \|h\|$ are 1-Lipschitz, so each has variance at most 1 by Theorem A.6(ii); they are independent, so $\operatorname{Var} D_\kappa \le 2$. Since $\mathbb{E}\|(g+\kappa\mathbf{1})_+\|^2 = n\varrho$, $(\mathbb{E}\|(g+\kappa\mathbf{1})_+\|)^2 = n\varrho - \operatorname{Var}\|(g+\kappa\mathbf{1})_+\| \in [n\varrho-1, n\varrho]$, and likewise $\mathbb{E}\|h\| \in [\sqrt{d-1}, \sqrt{d}]$. If $n\varrho > d+1$, then $\mathbb{E}D_\kappa \ge \sqrt{n\varrho-1} - \sqrt{d} > 0$, and Chebyshev's inequality bounds $\mathbb{P}(D_\kappa \le 0)$ by $\operatorname{Var} D_\kappa/(\mathbb{E}D_\kappa)^2$. If $n\varrho < d-1$, then $-\mathbb{E}D_\kappa \ge \sqrt{d-1} - \sqrt{n\varrho} > 0$, and the same inequality bounds $\mathbb{P}(D_\kappa > 0)$. ∎

Theorem 8.8 (the capacity of the spherical perceptron). *Let $G \in \mathbb{R}^{n\times d}$ have i.i.d. $\mathcal{N}(0,1)$ rows $g_1, \dots, g_n$; for $\kappa \in \mathbb{R}$ let $\mathcal{F}_\kappa$ be the event that some $\theta \in S^{d-1}$ satisfies $\langle g_\mu\rangle\theta \ge \kappa$ for all $\mu \le n$; and let $\varrho(\kappa) = \mathbb{E}(Z+\kappa)_+^2$ and $\alpha_c = 1/\varrho$.*

(i) (Every margin.) For every $\kappa \in \mathbb{R}$, $\mathbb{P}(\mathcal{F}_\kappa) \le 2\mathbb{P}(D_\kappa \le 0)$; and if $n\varrho(\kappa) > d+1$, then $\mathbb{P}(\mathcal{F}_\kappa) \le 4/(\sqrt{n\varrho(\kappa)-1} - \sqrt{d})^2$.

(ii) (Nonnegative margins.) For every $\kappa \ge 0$, $\mathbb{P}(\mathcal{F}_\kappa^c) \le 2\mathbb{P}(D_\kappa > 0)$; and if $n\varrho(\kappa) < d-1$, then $\mathbb{P}(\mathcal{F}_\kappa^c) \le 4/(\sqrt{d-1} - \sqrt{n\varrho(\kappa)})^2$.

Consequently, if $n, d \to \infty$ with $n/d \to \alpha \in (0,\infty)$, then $\mathbb{P}(\mathcal{F}_\kappa) \to 0$ for every κ with $\alpha > \alpha_c(\kappa)$, and $\mathbb{P}(\mathcal{F}_\kappa) \to 1$ for every $\kappa \ge 0$ with $\alpha < \alpha_c(\kappa)$. In particular the capacity at zero margin is 2.

PROOF. The plan is to read $\mathcal{F}_\kappa$ as the vanishing of a primary problem, to bound its probability from above with part (i) of the CGMT on the sphere and from below with part (ii) on the ball, and to evaluate both auxiliary problems with Lemmas 8.6 and 8.7.

(i) The value ξ_S of (8.12) is the primary value $\mathcal{P}(-G)$ of Theorem 8.3 with the compact sets $S_w = S^{d-1}$ and $S_u = \Lambda$ and the continuous function $\psi(\theta,\lambda) = \kappa\mathbf{1}^\top\lambda$; the matrix $-G$ has i.i.d. standard normal entries, and on the sphere $\|\theta\| = 1$, so the auxiliary problem is $\mathcal{A}_S$. Since $\xi_S \ge 0$, part (i) gives, for every $c > 0$,

$$\mathbb{P}(\mathcal{F}_\kappa) = \mathbb{P}(\xi_S \le 0) \le \mathbb{P}(\xi_S < c) \le 2\mathbb{P}(\mathcal{A}_S \le c).$$

Letting $c \downarrow 0$ and using Lemma 8.6 gives $\mathbb{P}(\mathcal{F}_\kappa) \le 2\mathbb{P}(\mathcal{A}_S \le 0) \le 2\mathbb{P}(D_\kappa \le 0)$, and Lemma 8.7 gives the explicit bound.

(ii) Let first $\kappa > 0$. Then $\mathcal{F}_\kappa$ is the event that some $\theta \in B^d$ has $G\theta \ge \kappa\mathbf{1}$: such a θ is nonzero, and $\theta/\|\theta\|$ satisfies $G\theta/\|\theta\| \ge \kappa\mathbf{1}/\|\theta\| \ge \kappa\mathbf{1}$. So $\mathcal{F}_\kappa = \{\xi_B = 0\}$ with $\xi_B = \min_{\theta\in B^d}\max_{\lambda\in\Lambda}\lambda^\top(\kappa\mathbf{1} - G\theta)$, a primary problem with compact convex sets and a bilinear-plus-linear, hence convex–concave, objective. Part (ii) of the CGMT and Lemma 8.6 give, for every $c > 0$,

$$\mathbb{P}(\xi_B > c) \le 2\mathbb{P}(\mathcal{A}_B \ge c) \le 2\mathbb{P}(D_\kappa \ge c).$$

As $c \downarrow 0$ the left side increases to $\mathbb{P}(\mathcal{F}_\kappa^c)$ and the right side decreases to $2\mathbb{P}(D_\kappa > 0)$. For $\kappa = 0$, note that $\mathcal{F}_0 \supseteq \mathcal{F}_{\kappa'}$ for every $\kappa' > 0$, so $\mathbb{P}(\mathcal{F}_0^c) \le 2\mathbb{P}(D_{\kappa'} > 0)$. As $\kappa' \downarrow 0$, $D_{\kappa'}$ decreases to D_0, so the events $\{D_{\kappa'} > 0\}$

decrease to an event contained in $\{D_0 \geq 0\}$; and $\mathbb{P}(D_0 = 0) = 0$, because $\|h\|$ has a density and is independent of g. Hence $\mathbb{P}(\mathcal{F}_0^c) \leq 2\mathbb{P}(D_0 > 0)$, and Lemma 8.7 gives the explicit bound.

For the limits: if $\alpha > \alpha_c(\kappa)$ then $(\sqrt{n\varrho - 1} - \sqrt{d})^2/d \to (\sqrt{\alpha\varrho} - 1)^2 > 0$, and if $\kappa \geq 0$ and $\alpha < \alpha_c(\kappa)$ then $(\sqrt{d-1} - \sqrt{n\varrho})^2/d \to (1 - \sqrt{\alpha\varrho})^2 > 0$; in both cases the bound is $O(1/d)$. ∎

The bounds locate the transition within $|n\varrho(\kappa) - d| = O(\sqrt{d})$, the width of Cover's window at $\kappa = 0$; with the sharp concentration inequality, Theorem A.7(ii), applied to the $\sqrt{2}$-Lipschitz function D_κ, they become exponential, for instance $\mathbb{P}(\mathcal{F}_\kappa) \leq 2\exp(-(\sqrt{n\varrho - 1} - \sqrt{d})^2/4)$.

Corollary 8.9 (the largest margin). *Let $\kappa_{\max}(G) = \max_{\theta \in S^{d-1}} \min_\mu \langle g_\mu \rangle \theta$, and for $\alpha > 0$ let $\kappa^*(\alpha)$ solve $\alpha_c(\kappa^*) = \alpha$. If $n, d \to \infty$ with $n/d \to \alpha$, then $\mathbb{P}(\kappa_{\max} \geq \kappa^*(\alpha) + \varepsilon) \to 0$ for every $\varepsilon > 0$; and if $\alpha < 2$, then $\kappa_{\max} \to \kappa^*(\alpha) > 0$ in probability.*

Proof. The function ϱ is continuous and strictly increasing, with $\varrho'(\kappa) = 2\mathbb{E}(Z + \kappa)_+ > 0$ and limits 0 and ∞, so κ^* is well defined and $\alpha_c(\kappa^* + \varepsilon) < \alpha < \alpha_c(\kappa^* - \varepsilon)$. The maximum defining $\kappa_{\max}$ is attained, so $\{\kappa_{\max} \geq \kappa\} = \mathcal{F}_\kappa$. Theorem 8.8(i) at $\kappa^* + \varepsilon$ gives the first claim. If $\alpha < 2$, then $\kappa^* > 0$, and for $\varepsilon < \kappa^*$ part (ii) at $\kappa^* - \varepsilon$ gives $\mathbb{P}(\kappa_{\max} < \kappa^* - \varepsilon) \to 0$. ∎

Gardner's maximal stability is therefore the largest margin of random labels at every load below 2; and Gardner's $q \to 1$ is the auxiliary norm reaching the boundary of the ball (Exercise 8.8).

Remark 8.10 (what the comparison does not see). Shcherbina and Tirozzi (2003, Theorem 3) proved, for ± 1 patterns, that for $\kappa \geq 0$ and $\alpha \leq \alpha_c(\kappa)$ the truncated log-volume is self-averaging and converges to $\min_q \phi_{\mathrm{RS}}(q)$, Gardner's whole free entropy; Talagrand (2011, Chapter 8) also derives the Gardner formula for the sphere. Both work with the volume, a partition function at positive temperature, which the min–max argument never meets. Part (i) does turn the ground state (8.6) into a lower bound, $\xi_S/\sqrt{d} \geq (\sqrt{\alpha\varrho} - 1)_+ - \varepsilon$ with probability tending to one (Exercise 8.9).

8.5.1 Beyond convexity: negative margins and binary weights

For $\kappa < 0$ the ball contains the feasible point $\theta = 0$, and $-1/\kappa_{\max}$ is the largest norm in the polytope $\{G\theta \geq -\mathbf{1}\}$ (Montanari et al. 2024). Part (i) of Theorem 8.8 still gives Gardner's value as an upper bound, which, as Stojnic (2013b, Section 2.3) reports, answers a conjecture of Talagrand (his Conjecture 8.4.4, which we have not seen). Stojnic (2013c) improved it by comparing an exponential of the cost, and Montanari et al. (2024, Definition 2, Theorem 4.2) proved the resulting *lifted bound*: with $\ell_\kappa(u) = -\log \mathbb{E} e^{-u(\kappa - Z)_+^2}$, a load α exceeds the capacity if some $s > 0$ satisfies

$$\frac{1}{\sqrt{s^2+4}+s} + \frac{1}{s}\log\frac{\sqrt{s^2+4}+s}{2} < \inf_{t>0}\Big\{\frac{1}{4t} + \frac{\alpha}{s}\,\ell_\kappa(st)\Big\}. \tag{8.13}$$

As $s \to 0$ this becomes Gardner's condition $\alpha\varrho(\kappa) > 1$. Evaluated numerically, (8.13) reproduces the tables of Stojnic (2013c) (12.784 against 13.273 at $\kappa = -1$), which show no improvement at $\kappa = -0.5$ and -0.6 and one of 1.3×10^{-4} at -0.63; Stojnic presents them as an illustration and states no threshold. The bound improves on Gardner's value exactly for $\kappa < \kappa_\times = -0.6219$. A second-order expansion in s (Exercise 8.6) shows that at $\alpha = \alpha_c(\kappa)$ small s succeed exactly when $\mathrm{Var}[(Z+\kappa)_+^2] < 2\,\mathbb{E}[(Z+\kappa)_+^2]$, which holds exactly for $\kappa < \kappa_\times$ (the difference changes sign once on $(-\infty, 0)$); for $\kappa_\times < \kappa < 0$ we found no s that succeeds below α_c (margins every 0.02, values of s from 10^{-4} to 30). Montanari et al. (2024) also proved that the capacity is $(1 + o(1))\log|\kappa|/\Phi(\kappa)$ as $\kappa \to -\infty$, while Gardner's value is $(1 + o(1))\kappa^2/(2\Phi(\kappa))$, and that their linear program, which maximizes $\langle v\rangle\theta$ over $\{\theta : G\theta \geq \kappa\mathbf{1},\ \|\theta\| \leq 1\}$ for a random direction v, returns a unit vector with high probability exactly when $\alpha < 1/\Phi(\kappa)$ (their Theorem 4.3).

Binary weights. For $w \in \{\pm 1\}^d$ a conditional first moment, outlined by Krauth and Mézard (1989, Section 5) and completed by Altschuler and Tikhomirov (2024), beats the first-moment bound 1. A binary solution is a point of the spherical solution set at $\kappa = 0$, and a random rotation makes it uniform; so, when the spherical log-volume is near its typical value $d\,\mathrm{GD}(\alpha)$, with $\mathrm{GD}(\alpha) = \min_q \phi_{\mathrm{RS}}(q)$ at $\kappa = 0$, there are at most $2^d e^{d(\mathrm{GD}(\alpha)+\varepsilon)}$ binary solutions on average, and the capacity is at most the root of $\mathrm{GD}(\alpha) = -\log 2$, 0.846557. At $q = \frac{1}{2}$, where $\mathbb{E}\log H(U) = -1$ because $H(U)$ is uniform, $\phi_{\mathrm{RS}}(\frac{1}{2}) = -\alpha + \frac{1}{2}(1 - \log 2)$, so $\alpha_c \leq (1+\log 2)/2 = 0.846574$, a bound that needs no numerical evaluation. Krauth and Mézard read $\alpha \approx 0.85$ and $q \approx 0.5$ off Gardner's results, and Altschuler and Tikhomirov evaluate $\phi_{\mathrm{RS}}(\frac{1}{2})$ in this way at $\alpha = 0.847$; neither writes the closed form, which we record as the book's observation. Their Proposition 1 states $\mathrm{GD}(0.847) < -\log 2 - 0.002$, but the value at $q = \frac{1}{2}$ is $-\log 2 - 0.00043$ and the minimum over q is $-\log 2 - 0.00045$; the margin should read 0.0004, which leaves their bound 0.847 intact, since their argument needs only a positive margin.

The value $\alpha_\star = 0.8330786$ of Krauth and Mézard is proved under numerical conditions: Ding and Sun (2019) proved the lower bound with positive probability under a one-variable condition, sharp-threshold results (Nakajima and Sun 2023; Xu 2021) upgrade it to high probability, and Huang (2024) proved the upper bound under a two-variable condition. Shmalo (2026) verified both conditions by a computer-assisted proof (arXiv version 1, July 2026); his paper states that the proof has not yet been independently reproduced or peer reviewed, and its declarations say: "Most of the work reported here was done by two AI systems" (Fable 5 of Anthropic and Codex of OpenAI). For the symmetric binary perceptron, with constraints $|\langle g_\mu\rangle w|/\sqrt{d} \leq \kappa$, the capacity is the first-moment value $-\log 2/\log \mathbb{P}(|Z| \leq \kappa)$, derived by Aubin et al. (2019) and proved by Abbe et al. (2022).

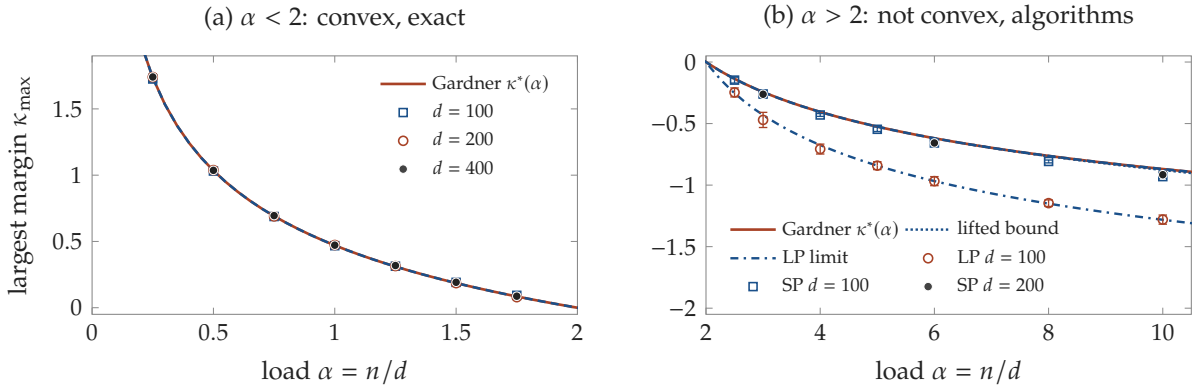


Figure 8.2. The largest margin of $n = \alpha d$ Gaussian patterns. Carnelian with lapis dashes: Gardner's $\kappa^*(\alpha)$, the limit for $\alpha < 2$ and an upper bound for $\alpha > 2$ (Corollary 8.9). (a) Exact margins at $d = 100, 200, 400$ (200, 100, 40 instances; seeds $80900 + d$). (b) Lower bounds from the linear program of Montanari et al. (2024) (LP, $d = 100$, 16 instances), with its proved limit $\Phi^{-1}(1/\alpha)$ (dash-dotted), and from a soft-minimum ascent followed by one linear program (SP, $d = 100$ with 16 instances, $d = 200$ with 6); dotted, the lifted upper bound. Seeds $81000 + 100\alpha + d$. Student-t 95% intervals. Scripts: `08-gordon-margin.py`, `08-gordon-negative.py`.

8.6 Measuring it

Three scripts test the theorem and the physics at finite size. For $\alpha < 2$, `08-gordon-margin.py` computes the largest margin exactly as $1/\|\theta^*\|$, for the minimum-norm solution θ^* of $G\theta \geq \mathbf{1}$, a least-distance problem that Lawson and Hanson's reduction turns into nonnegative least squares.[2] At all seven loads and three sizes of Figure 8.2(a) the mean margin differs from $\kappa^*(\alpha)$ by at most 0.0101, and 20 of the 21 intervals contain it; the exception, at $d = 100$ and $\alpha = 1.75$, exceeds it by 0.0085 with half-width 0.0072, and is gone at $d = 200$ and 400. There 5 of the 200 instances have no positive margin and are counted as 0; their margins lie above -0.04 (by the soft-minimum algorithm below), so this raises the mean by less than 0.0005.

For $\alpha > 2$, `08-gordon-negative.py` runs two polynomial-time algorithms, which give lower bounds. The linear program agrees with its proved limit $\Phi^{-1}(1/\alpha)$ at all seven loads (-0.968 ± 0.037 against -0.967 at $\alpha = 6$). The soft-minimum ascent comes much closer to Gardner's curve: its intervals contain $\kappa^*(\alpha)$ at $\alpha = 2.5$ and 3, and from $\alpha = 4$ on it stays below by more than its interval, by 0.025 at $\alpha = 4$ and 0.062 at $\alpha = 10$; at $d = 200$, $\alpha = 10$ it reaches -0.916 ± 0.024, below both Gardner's -0.869 and the lifted bound -0.878. No measurement contradicts a proved bound, and none locates the capacity: the gap may come from the algorithm, from replica-symmetry breaking or from the finite size.

At zero margin Cover's formula is exact, and `08-gordon-cover.py` uses it to judge the simulation and the comparison. For each instance it finds by bisection the largest n for which the first n patterns are separable (the least-distance program agrees with a linear-programming test on 40 of

[2] Minimize $\|Eu - f\|$ over $u \geq 0$, where E is $G^\top$ with a row of ones appended and f is the last unit vector; if the residual $r = Eu - f$ is nonzero, then $\theta^* = -(r_1, \ldots, r_d)/r_{d+1}$, and if it is zero the constraints are infeasible.

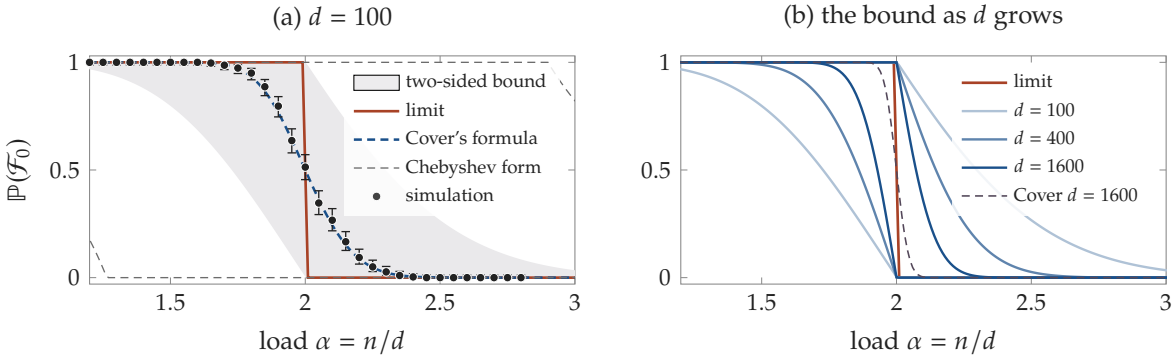


Figure 8.3. Zero margin. (a) At $d = 100$: the probability that $n = \alpha d$ Gaussian patterns admit a unit vector with $G\theta \geq 0$, by Cover's formula (lapis dashed) and in 300 simulated instances (points, 95% Clopper–Pearson intervals, seed 80900), with the two-sided bound $1 - 2\mathbb{P}(D_0 > 0) \leq \mathbb{P}(\mathcal{F}_0) \leq 2\mathbb{P}(D_0 \leq 0)$ of Theorem 8.8, computed exactly (shaded), its Chebyshev form (gray dashed), and the limit (carnelian). (b) The two-sided bound at $d = 100$, 400, 1600 (lapis, darker for larger d) closes in on the step and on Cover's curve (granite). Script: `08-gordon-cover.py`.

40 instances). Cover's formula makes the mean of this instance capacity exactly $2d$ (Exercise 8.3); the measured means are 2.018 ± 0.013, 1.999 ± 0.014, 2.006 ± 0.016 and 2.015 ± 0.018 times d at $d = 25$, 50, 100, 200 (95% half-widths). The first is 2.8 standard errors high; six further runs at $d = 25$ with other seeds, 6000 instances in all, give 2.001 ± 0.007, so we read it as a fluctuation. At $d = 100$ the simulated probability is 0.950 [0.919, 0.972], 0.513 [0.455, 0.571] and 0.093 [0.063, 0.132] at $\alpha = 1.8$, 2, 2.2, against Cover's 0.9326, 0.5, 0.0882.

The two-sided bound of the proof is valid but loose at this size: it guarantees $\mathbb{P}(\mathcal{F}_0) \geq 0.685$ at $\alpha = 1.6$, where the truth is 0.9993, and $\mathbb{P}(\mathcal{F}_0) \leq 0.370$ at $\alpha = 2.4$, where it is 0.0048. Two things are lost, the factor 2 of the CGMT and the shape of the transition: $\mathbb{P}(D_0 \leq 0)$ crosses $\frac{1}{2}$ at $\alpha = 2$, as Cover's curve does, but its transition is three times wider, since $\operatorname{Var} D_0 \to \frac{9}{8}$ (Exercise 8.10). The comparison controls where the transition is, not its shape. As d grows the bound closes in on the step: at $d = 1600$ it gives $\mathbb{P}(\mathcal{F}_0) \geq 0.947$ at $\alpha = 1.8$ and ≤ 0.066 at $\alpha = 2.2$.

8.7 The status board

For nonnegative margins the capacity is settled, with three proofs. For negative margins Gardner's formula is a proved upper bound, known to fail for sufficiently negative margins and, according to physics, for every negative margin; the capacity is open. For binary weights the replica-symmetric criterion is wrong, and the Krauth–Mézard value is proved under numerical conditions that a 2026 preprint verifies by computer.

STATUS BOARD · Gardner's volume ↔ Gordon's min-max

PREDICTION	STATUS	PROOF OR EVIDENCE
Capacity $1/\mathbb{E}(Z+\kappa)_+^2$ for $\kappa \geq 0$; 2 at $\kappa = 0$	proved	Shcherbina and Tirozzi (2003); Talagrand (2011); Theorem 8.8
Gardner's free entropy (8.3), $\kappa \geq 0$	proved	Shcherbina and Tirozzi (2003) (± 1 patterns)
Upper bound $\alpha_c(\kappa)$ for $\kappa < 0$	proved	Stojnic (2013b) (arXiv); Theorem 8.8(i)
$\alpha_c(\kappa)$ is the capacity for $\kappa < 0$	false	lifted bound (8.13) (Montanari et al. 2024; Stojnic 2013c): for $\kappa < -0.6219$ (Exercise 8.6, with a numerical root), and as $\kappa \to -\infty$
It fails for every $\kappa < 0$ (full RSB)	conjectured	Franz et al. (2017)
The capacity for $\kappa < 0$	open	leading order as $\kappa \to -\infty$ (Montanari et al. 2024)
RS stable at capacity iff $\kappa \geq 0$	heuristic	replicon (8.7); Gardner and Derrida (1988)
Ising perceptron: $4/\pi$ from $q \to 1$	false	first-moment bound 1
Ising perceptron: $\alpha_\star = 0.8330786$	proved under	numerical conditions (Ding and Sun 2019; Huang 2024), verified by computer in Shmalo (2026) (arXiv, unrefereed, AI-assisted)
Symmetric binary perceptron	proved	Abbe et al. (2022)

8.8 Where the translation fails

WHAT WAS ASSUMED. Gaussian patterns with random labels: Shcherbina and Tirozzi (2003) treat ± 1 patterns, but the CGMT needs a Gaussian matrix, and a teacher changes the question (Chapter 11). The limit is proportional, and the bounds of Theorem 8.8 locate the transition to $O(\sqrt{d})$ patterns without describing its shape. The theorem concerns feasibility, not the volume, and its convex half rests on relaxing the sphere to the ball, which is exact only because $\kappa > 0$ lets a solution be rescaled onto the sphere.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. The replica-symmetric capacity is wrong for $\kappa < 0$, as the physics itself says: the replicon exceeds 1 at capacity, and Franz et al. (2017) place the jamming line in a phase with full replica-symmetry breaking. The rigorous confirmation is partial: the lifted bound proves the failure for $\kappa < -0.6219$, by an expansion and the numerical root of an explicit function, and Montanari et al. (2024) prove that Gardner's value is too large by a factor of order $\kappa^2/\log|\kappa|$ as $\kappa \to -\infty$; between -0.622 and 0 nobody has proved it wrong. The caption of their

Figure 1 calls their upper bound strictly better than Gardner's for all negative κ; our evaluation of their Definition 2 finds no improvement for $\kappa > -0.622$, in agreement with Stojnic's tables. For binary weights the criterion $q \to 1$ gives the impossible $4/\pi$.

OPEN PROBLEMS. Prove that the replica-symmetric value fails for every $\kappa < 0$, in particular for $-0.622 < \kappa < 0$, and identify the capacity that full replica-symmetry breaking predicts (Franz et al. 2017); a comparison inequality that is tight beyond convexity would do both. Montanari et al. (2024) raise the question of how algorithms other than their linear program behave; Figure 8.2(b) suggests that simple heuristics come within a few hundredths of Gardner's value at moderate loads. For binary weights, an independent check or a conceptual proof of the conditions behind $\alpha_\star$ remains to be given.

Exercises

8.1 CONSOLIDATION. (The function ϱ.) Show that $\varrho'(\kappa) = 2(\kappa\Phi(\kappa) + \phi(\kappa))$ and $\varrho''(\kappa) = 2\Phi(\kappa)$; that $\alpha_c(\kappa)(1+\kappa^2) \to 1$ as $\kappa \to \infty$ and $2\Phi(\kappa)\alpha_c(\kappa)/\kappa^2 \to 1$ as $\kappa \to -\infty$; and check $\alpha_c(1) = 0.5196$ and $\alpha_c(2) = 0.2002$.

8.2 CONSOLIDATION. (The smallest singular value.) For $n \geq d$, write $\sigma_{\min}(G) = \min_{\theta \in S^{d-1}} \max_{\|u\| \leq 1} u^\top G\theta$, show that its auxiliary value is $(\|g\| - \|h\|)_+$, and deduce $\mathbb{P}(\sigma_{\min}(G) < \sqrt{n-1} - \sqrt{d} - t) \leq 4/t^2$ for $0 < t < \sqrt{n-1} - \sqrt{d}$.

8.3 CONSOLIDATION. (Cover's formula.) (a) Show that almost surely some unit vector has $G\theta \geq 0$ if and only if some has $G\theta > 0$. *(Hint: almost surely every d rows of G are linearly independent; perturb a weak solution along a vector with inner product 1 with the rows where it is tight.)* (b) Let n^* be the largest n for which the first n patterns of an infinite sequence are separable. Show that $\mathbb{E}n^* = 2d$. *(Hint: $\mathbb{P}(n^* \geq n) = \mathbb{P}(\mathrm{Bin}(n-1, \frac{1}{2}) \leq d-1)$; count fair coin flips until the dth head.)*

8.4 BY HAND. (Near capacity.) Keep the logarithmic term of $\log H$ in Computation 8.1 to show that $\phi_{\mathrm{RS}} = \frac{1-\alpha\varrho}{2\varepsilon} + \frac{1+\alpha\Phi(\kappa)}{2}\log\varepsilon + O(1)$ with $\varepsilon = 1-q$, hence $1 - q^* \simeq (1-\alpha/\alpha_c)/(1+\alpha\Phi(\kappa))$.

8.5 BY HAND. (Binary weights.) Show that $q \to 1$ in the stationarity equations of (8.8) forces $\alpha = 2/(\pi\varrho(\kappa))$. The first moment bounds the capacity by $\log 2/(-\log H(\kappa))$; show that the replica-symmetric value exceeds it at $\kappa = 0.5$ (0.612 against 0.589) but not at $\kappa = 1$ (0.331 against 0.377).

8.6 BY HAND. ★ (Where the lifted bound starts.) Let $X = (Z+\kappa)_+^2$ have mean ϱ and variance v. Show that the left side of (8.13) is $1 - \frac{s}{4} + O(s^2)$ and the right side $\sqrt{\alpha\varrho} - \frac{sv}{8\varrho} + O(s^2)$, so that at $\alpha = \alpha_c(\kappa)$ small

s succeed when $v < 2\varrho$; locate $\kappa_\times = -0.6219$ numerically. *(Hint: $\ell_\kappa(u) = u\varrho - \frac{u^2}{2}v + O(u^3)$; evaluate the correction at $t_0 = 1/(2\sqrt{\alpha\varrho})$.)*

8.7 TRANSLATION. (The heat-equation step.) For $0 < q < 1$ let $\mathcal{Z}_q(h) = H((\kappa - h)/\sqrt{1-q})$. Prove with Theorem A.1 that

$$\frac{\mathrm{d}}{\mathrm{d}q}\mathbb{E}\log\mathcal{Z}_q(\sqrt{q}Z) = -\frac{1}{2}\mathbb{E}\big[(\partial_h \log\mathcal{Z}_q)(\sqrt{q}Z)^2\big],$$

and derive (8.4). Which step of the replica computation does this justify, and which not?

8.8 TRANSLATION. (The auxiliary norm.) For $\kappa > 0$, show that the minimum-norm solution of $G\theta \geq \kappa\mathbf{1}$ has norm $\kappa/\kappa_{\max}$, that $\mathcal{A}_B = \min_{0\leq t\leq 1}(\|(tg + \kappa\mathbf{1})_+\| - t\|h\|)_+$, and that the smallest t at which this vanishes tends to $\kappa/\kappa^*(\alpha)$. Translate Gardner's $q \to 1$.

8.9 TRANSLATION. ★ (The ground state as a theorem.) Show that $\xi_S/\sqrt{d} \geq (\sqrt{\alpha\varrho(\kappa)} - 1)_+ - \varepsilon$ with probability tending to one, for every κ and $\varepsilon > 0$. Why would a matching upper bound need part (ii) of the CGMT, and why does the ball relaxation fail for it when $\alpha > \varrho(\kappa)/\Phi(\kappa)^2$, even for $\kappa > 0$? *(Hint: differentiate $t(\|(g + \kappa\mathbf{1}/t)_+\| - \|h\|)$ at $t = 1$; $\mathbb{E}[Z(Z+\kappa)_+] = \Phi(\kappa)$.)*

8.10 MEASUREMENT. (The auxiliary window.) Show that $\mathrm{Var}\|g_+\| \to \frac{5}{8}$ and $\mathrm{Var}\|h\| \to \frac{1}{2}$ at $n = 2d$, so that $\mathbb{P}(D_0 \leq 0) \approx \Phi(-(\alpha-2)\sqrt{d}/4.24)$, while Cover's probability is about $\Phi(-(\alpha-2)\sqrt{d}/\sqrt{2})$; check both against `08-gordon-cover.py` at $d = 100$. Then edit the sizes in `08-gordon-negative.py` to run it at $d = 50, 100, 200$: does the gap between the soft-minimum algorithm and Gardner's curve shrink?

8.11 RESEARCH. (Between -0.622 and 0.) Prove that the capacity is strictly below $\alpha_c(\kappa)$ for some $\kappa \in (-0.622, 0)$. What would a Gordon-type inequality with a hierarchical comparison process, the analogue of replica-symmetry breaking, have to look like?

Notes and References

Gardner (1988) introduced the space of interactions and computed its volume by replicas; Gardner and Derrida (1988) extended the method to the minimal fraction of errors above capacity, tested the stability of the saddle point, and treated binary weights, and Krauth and Mézard (1989) found the right criterion for the latter. Franz and Parisi (2016) and Franz et al. (2017) connected the negative perceptron with jamming. The zero-temperature computation of Section 8.2.1 and the replicon at capacity are written out for this book.

Gordon (1985) extended Slepian's inequality to the two-level comparison of Theorem 8.1, and Gordon (1988) proved the bilinear form and escape through a mesh. Stojnic's random duality theory (Stojnic 2013a) added matching upper bounds by convex duality and treated the spherical

perceptron (Stojnic 2013b,c). The convex Gaussian min–max theorem is due to Thrampoulidis et al. (2015) and, in the form of Theorem 8.3, to Thrampoulidis et al. (2018); our proofs follow theirs in substance. The presentation of Theorem 8.8 through the CGMT with bounds at every size is the book's, and so are the closed form $(1+\log 2)/2$, the corrected margin in Proposition 1 of Altschuler and Tikhomirov (2024), the crossover $\kappa_\times = -0.6219$, and our reading of Figure 1 of Montanari et al. (2024), all computed by `08-gordon-numbers.py`.

TRANSLATING THE LITERATURE. Gardner (1988) writes N for our d, $p = \alpha N$ for n and J for w. Stojnic (2013b,c) write $H \in \mathbb{R}^{m\times n}$ for our G, with $m = \alpha n$, x for θ, ξ_n for our ξ_S (with $\|\lambda\| = 1$ in place of $\|\lambda\| \le 1$), and $f_{\mathrm{gar}}(\kappa)$ for $\varrho(\kappa)$. Thrampoulidis et al. (2015) and Thrampoulidis et al. (2018) write $G \in \mathbb{R}^{m\times n}$, $g \in \mathbb{R}^m$, $h \in \mathbb{R}^n$ for our $G \in \mathbb{R}^{n\times d}$, $g \in \mathbb{R}^n$, $h \in \mathbb{R}^d$, and $\Phi(G)$, $\phi(g,h)$, $\bar\phi$ for our $\mathcal{P}(G)$, $\mathcal{A}(g,h)$, $\bar a$; we avoid their letters because Φ and ϕ are the Gaussian distribution function and density here. Theorem, corollary and section numbers of Thrampoulidis et al. (2018) are those of its arXiv version 1, which we read; we did not check them against the journal version. Montanari et al. (2024) write δ for α; Franz et al. (2017) write σ for κ, and their $\Theta(x/\sqrt{2})$ is our $\Phi(x)$; Altschuler and Tikhomirov (2024) write $\mathrm{GD}(\alpha)$ for $\min_q \phi_{\mathrm{RS}}(q)$ at $\kappa = 0$.

References

Abbe, E., S. Li, and A. Sly (2022). "Proof of the Contiguity Conjecture and Lognormal Limit for the Symmetric Perceptron". In: *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, pp. 327–338. arXiv: 2102.13069.

Altschuler, D. J. and K. Tikhomirov (2024). *A note on the capacity of the binary perceptron*. arXiv: 2401.15092.

Aubin, B., W. Perkins, and L. Zdeborová (2019). "Storage capacity in symmetric binary perceptrons". In: *Journal of Physics A: Mathematical and Theoretical* 52.29, p. 294003. arXiv: 1901.00314.

Cover, T. M. (1965). "Geometrical and Statistical Properties of Systems of Linear Inequalities with Applications in Pattern Recognition". In: *IEEE Transactions on Electronic Computers* EC-14.3, pp. 326–334.

Ding, J. and N. Sun (2019). "Capacity lower bound for the Ising perceptron". In: *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*. ACM, pp. 816–827. arXiv: 1809.07742v1 [math.PR].

Franz, S. and G. Parisi (2016). "The simplest model of jamming". In: *Journal of Physics A: Mathematical and Theoretical* 49.14, p. 145001. arXiv: 1501.03397.

Franz, S., G. Parisi, M. Sevelev, P. Urbani, and F. Zamponi (2017). "Universality of the SAT-UNSAT (jamming) threshold in non-convex continuous constraint satisfaction problems". In: *SciPost Physics* 2.3, p. 019. arXiv: 1702.06919v2 [cond-mat.dis-nn].

Gardner, E. (1988). "The space of interactions in neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 257–270.

Gardner, E. and B. Derrida (1988). "Optimal storage properties of neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 271–284.

Gordon, Y. (1988). "On Milman's inequality and random subspaces which escape through a mesh in $\mathbb{R}^n$". In: *Geometric Aspects of Functional Analysis (1986/87)*. Vol. 1317. Lecture Notes in Mathematics. Springer, pp. 84–106.

Gordon, Y. (1985). "Some inequalities for Gaussian processes and applications". In: *Israel Journal of Mathematics* 50.4, pp. 265–289.

Huang, B. (2024). "Capacity Threshold for the Ising Perceptron". In: *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, pp. 1126–1136. arXiv: 2404.18902v2 [math.PR].

Krauth, W. and M. Mézard (1989). "Storage capacity of memory networks with binary couplings". In: *Journal de Physique* 50.20, pp. 3057–3066.

Montanari, A., Y. Zhong, and K. Zhou (2024). "Tractability from overparametrization: the example of the negative perceptron". In: *Probability Theory and Related Fields* 188.3-4, pp. 805–910. arXiv: 2110.15824v3 [cs.LG].

Nakajima, S. and N. Sun (2023). "Sharp threshold sequence and universality for Ising perceptron models". In: *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. Society for Industrial and Applied Mathematics, pp. 638–674. arXiv: 2204.03469.

Shcherbina, M. and B. Tirozzi (2003). "Rigorous Solution of the Gardner Problem". In: *Communications in Mathematical Physics* 234.3, pp. 383–422. arXiv: math-ph/0112003v1 [math-ph].

Shmalo, Y. (2026). *The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions*. arXiv preprint, v1 of 13 July 2026; unrefereed. arXiv: 2607.11010v1 [math.PR].

Sion, M. (1958). "On general minimax theorems". In: *Pacific Journal of Mathematics* 8.1, pp. 171–176.

Stojnic, M. (2013a). *A framework to characterize performance of LASSO algorithms*. arXiv: 1303.7291.

Stojnic, M. (2013b). *Another look at the Gardner problem*. arXiv preprint. arXiv: 1306.3979v1 [math.OC].

Stojnic, M. (2013c). *Negative spherical perceptron*. arXiv: 1306.3980.

Talagrand, M. (2011). *Mean Field Models for Spin Glasses: Volume II: Advanced Replica-Symmetry and Low Temperature*. Vol. 55. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). "Precise Error Analysis of Regularized M-Estimators in High Dimensions". In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Thrampoulidis, C., S. Oymak, and B. Hassibi (2015). "Regularized Linear Regression: A Precise Analysis of the Estimation Error". In: *Proceedings of The 28th Conference on Learning Theory*. Vol. 40. Proceedings of Machine Learning Research, pp. 1683–1709.

Wendel, J. G. (1962). "A Problem in Geometric Probability". In: *Mathematica Scandinavica* 11, pp. 109–112.

Xu, C. (2021). "Sharp threshold for the Ising perceptron model". In: *The Annals of Probability* 49.5, pp. 2399–2415. arXiv: 1905.05978.

9 Ridge Regression

Three derivations of double descent

THE DICTIONARY —— replicas ↔ random matrices and the CGMT

IN THIS CHAPTER. Fit n noisy linear observations in dimension d by ridge regression and let $n, d \to \infty$ with $n/d \to \alpha$. The excess risk tends to an explicit function of α and the ridge λ; without ridge it diverges at the interpolation threshold $\alpha = 1$, and with the optimal ridge it decreases in α. Physicists obtain this learning curve from a replica-symmetric saddle point at zero temperature. We prove it completely by random matrices, where the replicas' susceptibility is the trace of the resolvent and the cavity field is the leave-one-out prediction, and in outline by Gordon's comparison, whose auxiliary problem has the same stationarity equations as the replica saddle point. A simulation at $n = 1000$ agrees with the curves (the three training-error intervals that miss them do so by fluctuation, not bias); everything about the isotropic Gaussian case is proved, and what remains open concerns real data and nearby models.

9.1 The question: the test error of ridge regression when $n \approx d$

Least squares with an ℓ_2 penalty is one of the oldest regularized estimators in statistics.[1] With d fixed and n growing, the excess risk of least squares is about $\sigma^2 d/n$. With n and d of the same order it is not monotone in n/d: it blows up at $n = d$, where the data are first fitted exactly, and comes down on the other side. Belkin et al. (2019) named the shape *double descent*; for the linear model, physicists had drawn it nearly thirty years earlier (Krogh and Hertz 1991; Loog et al. 2020).

[1] The name is from Hoerl and Kennard (1970). In scikit-learn's `Ridge` the parameter `alpha` is $n\lambda$ in our normalization, and the default intercept must be switched off to match our model.

THE MODEL. We use the conventions of Part III. The inputs $x_1, \dots, x_n \in \mathbb{R}^d$, the rows of the design $X \in \mathbb{R}^{n \times d}$, are independent $\mathcal{N}(0, I_d)$ vectors, and $y_i = \langle x_i, \beta^* \rangle + \sigma \varepsilon_i$ with a teacher of norm $\|\beta^*\| = r$ and noise of mean 0 and variance 1. The *ridge estimator* with penalty $\lambda > 0$ is

$$\hat{\beta}_\lambda = \arg\min_{\beta \in \mathbb{R}^d} \left\{ \frac{1}{2n} \|y - X\beta\|^2 + \frac{\lambda}{2} \|\beta\|^2 \right\} = (X^\top X + n\lambda I)^{-1} X^\top y, \tag{9.1}$$

and its limit as $\lambda \downarrow 0$ is the *minimum-norm interpolator* $\hat{\beta}_0 = X^+ y$, which fits the data exactly when $d \geq n$. For a fresh input $x \sim \mathcal{N}(0, I_d)$ the excess risk is $R(\hat{\beta}) = \mathbb{E}_x(\langle x, \hat{\beta} \rangle - \langle x, \beta^* \rangle)^2 = \|\hat{\beta} - \beta^*\|^2$, the test error is $R + \sigma^2$, and the

training error is $T(\hat\beta) = \frac{1}{n}\|y - X\hat\beta\|^2$. The proportional limit takes $n, d \to \infty$ with $n/d \to \alpha$, the load,[2] and the *interpolation threshold* is $\alpha = 1$.

[2] This is Gardner's α and Donoho's δ; Hastie et al. (2022) and Dobriban and Wager (2018) write $\gamma = d/n = 1/\alpha$.

THE PHYSICAL PREDICTION · replica symmetry at zero temperature. Let $\chi > 0$ solve $\lambda\chi + \dfrac{\chi}{1+\chi} = \dfrac{1}{\alpha}$ and put $v = \dfrac{\alpha\chi^2}{(1+\chi)^2}$. In the proportional limit the excess risk and the training error of ridge regression are

$$R = \frac{r^2(\alpha\lambda\chi)^2 + \sigma^2 v}{1 - v}, \qquad T = \frac{R + \sigma^2}{(1+\chi)^2}.$$

As $\lambda \to 0$, $R \to \sigma^2/(\alpha - 1)$ for $\alpha > 1$ and $R \to r^2(1-\alpha) + \sigma^2\alpha/(1-\alpha)$ for $\alpha < 1$: the risk of the minimum-norm interpolator diverges at $\alpha = 1$. The ridge $\lambda^* = \sigma^2/(\alpha r^2)$ is optimal, its risk is $\sigma^2\chi$, and its curve has no peak.

Figure 9.1 draws the prediction. Below $\alpha = 1$ the interpolator fits the noise exactly, and its risk rises to the peak as $\alpha \to 1$, so that there more data hurt (Nakkiran 2019); a small ridge caps the peak, and the optimal ridge removes it. How do physicists derive these curves, and what is proved? The second answer is short: all of it, in the isotropic Gaussian case, by two rigorous routes whose steps can be set against the physicist's one by one.

9.2 The physics: replicas at zero temperature

A physicist reads an estimator defined by minimization as the ground state of a Gibbs measure, and computes its typical properties from the free entropy.[3]

[3] Since β denotes the regression coefficients, we write b for the inverse temperature.

REPLICAS AT ZERO TEMPERATURE. *Set up.* In physics units the weights are $w = \sqrt{d}\,\beta$ and the teacher is $w^* = \sqrt{d}\,\beta^*$, so that $\|w^*\|^2/d = r^2$. The local fields of sample μ are $\zeta_\mu = \langle x_\mu, w\rangle/\sqrt{d}$ and $\zeta^*_\mu = \langle x_\mu, w^*\rangle/\sqrt{d}$, and $y_\mu = \zeta^*_\mu + \sigma\varepsilon_\mu$. Multiplying the ridge objective (9.1) by n gives the energy

$$H(w) = \frac{1}{2}\sum_{\mu=1}^{n}(y_\mu - \zeta_\mu)^2 + \frac{\alpha\lambda}{2}\|w\|^2, \tag{9.2}$$

whose Gibbs measure $e^{-bH(w)}\,\mathrm{d}w/Z$ concentrates on $\sqrt{d}\,\hat\beta_\lambda$ as $b \to \infty$. For a sample w put $Q = \|w\|^2/d$ and $m = \langle w, w^*\rangle/d$, and for two independent samples, two *replicas*, $q = \langle w^1, w^2\rangle/d$; at $b = \infty$ the replicas coincide, and $R = q - 2m + r^2$.

Replicate. The free entropy $\frac{1}{d}\mathbb{E}\log Z$ is computed from $\mathbb{E}Z^k$ at integer k through $\mathbb{E}\log Z = \lim_{k\to 0}\frac{1}{k}\log\mathbb{E}Z^k$. Given k replicas, the fields $(\zeta^*, \zeta^1, \dots, \zeta^k)$ of one sample are centered Gaussian with covariances $Q_{ac} = \langle w^a, w^c\rangle/d$, $m_a = \langle w^a, w^*\rangle/d$ and r^2, and the samples are independent. With $e^{dS(Q,m)}$ the volume of the replica configurations with prescribed overlaps and $I(Q,m) = \mathbb{E}\exp\{-\frac{b}{2}\sum_a(\zeta^* + \sigma\varepsilon - \zeta^a)^2\}$,

$$\mathbb{E}Z^k = \int \exp\Big\{d\Big[S(Q,m) - \frac{b\alpha\lambda}{2}\sum_a Q_{aa} + \alpha\log I(Q,m)\Big]\Big\}\,\mathrm{d}Q\,\mathrm{d}m,$$

which for large d is dominated by its saddle point. The components of the replicas orthogonal to w^* have Gram matrix $d(Q - mm^\top/r^2)$, and the volume of k vectors in $\mathbb{R}^{d-1}$ with Gram matrix C is proportional to $\det(C)^{(d-k-2)/2}$ (the Wishart density), so $S = \frac{1}{2}\log\det(Q - mm^\top/r^2)$ up to constants and terms of order $1/d$.

Replica symmetry. Assume that the saddle point does not distinguish the replicas: $Q_{aa} = Q$, $Q_{ac} = q$ for $a \neq c$, $m_a = m$. Then $Q - mm^\top/r^2$ has the eigenvalue $Q-q$ with multiplicity $k-1$ and the eigenvalue $Q-q+k(q-m^2/r^2)$, so that $\frac{1}{k}S = \frac{1}{2}\log(Q-q) + \frac{q-m^2/r^2}{2(Q-q)} + O(k)$. For the energy write $\zeta^a = \omega + \sqrt{Q-q}\, z_a$, with (ω, ζ^*) centered Gaussian, $\mathbb{E}\omega^2 = q$, $\mathbb{E}\omega\zeta^* = m$, and $z_1, \dots, z_k$ independent standard Gaussians; this reproduces the covariances, and $\frac{1}{k}\log I = \mathbb{E}_{\omega,y}\log \mathbb{E}_z e^{-\frac{b}{2}(y-\omega-\sqrt{Q-q}\,z)^2} + O(k)$. The shared part ω of the fields is the *cavity field*.

Zero temperature. As $b \to \infty$ the replicas collapse onto the ground state, with $Q - q = \chi/b$ for a finite *susceptibility* χ, which a probabilist would read as b times the variance per coordinate of the Gibbs measure. The z-average is dominated by its saddle point (for the square loss it is a Gaussian integral, and the formula is exact up to a factor $(1+\chi)^{-1/2}$):

$$\begin{aligned} \mathbb{E}_z e^{-\frac{b}{2}(y-\omega-\sqrt{\chi/b}\,z)^2} &\asymp \exp\Big\{-b \min_\zeta \Big[\frac{(\zeta-\omega)^2}{2\chi} + \frac{(y-\zeta)^2}{2}\Big]\Big\} \\ &= \exp\Big\{-\frac{b\,(y-\omega)^2}{2(1+\chi)}\Big\}, \end{aligned} \tag{9.3}$$

the minimum being attained at $\hat\zeta = (\omega + \chi y)/(1+\chi)$. Keeping the terms of order b and using $\mathbb{E}(y-\omega)^2 = r^2 + \sigma^2 - 2m + q$, we get $\frac{1}{dk}\log \mathbb{E}Z^k \approx b\,\Psi(m, q, \chi)$ with the replica-symmetric potential

$$\Psi(m, q, \chi) = \frac{q - m^2/r^2}{2\chi} - \frac{\alpha\lambda q}{2} - \frac{\alpha(r^2 + \sigma^2 - 2m + q)}{2(1+\chi)}.$$

The ground-state energy per coordinate, $\min H/d$, is $-\Psi$ at the stationary point, where $\partial_q\Psi = \partial_m\Psi = \partial_\chi\Psi = 0$:

$$\frac{1}{\chi} = \alpha\lambda + \frac{\alpha}{1+\chi}, \qquad m = \frac{\alpha r^2 \chi}{1+\chi}, \qquad \frac{q - m^2/r^2}{\chi^2} = \frac{\alpha(r^2+\sigma^2-2m+q)}{(1+\chi)^2}. \tag{9.4}$$

The training residuals are $y - \hat\zeta = (y-\omega)/(1+\chi)$, so the training error is $(R+\sigma^2)/(1+\chi)^2$. ◇

COMPUTATION 9.1 · SOLVING THE REPLICA-SYMMETRIC EQUATIONS

1 *The susceptibility.* Multiplying the first equation of (9.4) by χ/α gives $\lambda\chi + \chi/(1+\chi) = 1/\alpha$, that is, $\alpha\lambda\chi^2 + (\alpha\lambda + \alpha - 1)\chi - 1 = 0$, with one positive root

$$\chi = \frac{1 - \alpha - \alpha\lambda + \sqrt{(1-\alpha-\alpha\lambda)^2 + 4\alpha\lambda}}{2\alpha\lambda}.$$

2 *The overlap with the teacher.* The same equation reads $1 - \alpha\lambda\chi =$

$\alpha\chi/(1+\chi)$, so the second equation gives $m = r^2(1-\alpha\lambda\chi)$.

3 *The risk.* Since $q = R + 2m - r^2$, the third equation reads $R - (r^2 - m)^2/r^2 = \nu(R+\sigma^2)$ with $\nu = \alpha\chi^2/(1+\chi)^2$, and $(r^2-m)^2/r^2 = r^2(\alpha\lambda\chi)^2$; hence $R(1-\nu) = r^2(\alpha\lambda\chi)^2 + \sigma^2\nu$, the formula of the physics box.

4 *No ridge.* Let $\lambda \to 0$. If $\alpha > 1$, step 1 gives $\chi/(1+\chi) \le 1/\alpha$, so $\chi \le 1/(\alpha-1)$, $\lambda\chi \to 0$, $\chi \to 1/(\alpha-1)$, $\nu \to 1/\alpha$, $R \to \sigma^2/(\alpha-1)$ and $T \to \sigma^2(1-1/\alpha)$. If $\alpha < 1$, it gives $\lambda\chi > 1/\alpha - 1 > 0$, so $\chi \to \infty$, $\alpha\lambda\chi \to 1-\alpha$, $\nu \to \alpha$, $R \to r^2(1-\alpha) + \sigma^2\alpha/(1-\alpha)$ and $T \to 0$.

5 *The optimal ridge.* At $\alpha\lambda = \sigma^2/r^2$ try $q = m$. Then $R = r^2 - m = \sigma^2\chi$, and the third equation becomes $\sigma^2\chi(1-\sigma^2\chi/r^2) = \nu\sigma^2(1+\chi)$, that is, $1 - \alpha\lambda\chi = \alpha\chi/(1+\chi)$, which is step 2. So at $\lambda^* = \sigma^2/(\alpha r^2)$ the risk is $\sigma^2\chi$, and the overlap of two replicas equals their overlap with the teacher.

WHAT WAS NOT CONTROLLED. The continuation of $\mathbb{E}Z^k$ from integer k to $k \to 0$, which the integer moments do not determine (Chapter 1); the exchange of $d \to \infty$ with $k \to 0$ and $b \to \infty$; Laplace's method as $k \to 0$, where the number $k(k-1)/2$ of off-diagonal overlaps becomes negative and the saddle point need not be a maximum; the replica-symmetric ansatz; and the concentration of the overlaps. Here the energy is a strictly convex quadratic, so the Gibbs measure is a Gaussian centered at $\sqrt{d}\,\hat\beta_\lambda$, nothing can break the symmetry between two draws from it, and a physicist would call replica symmetry exact. The rest of the chapter shows that the answer is right, although no proof continues anything in k.

THE OPTIMAL RIDGE AS A BAYES ESTIMATOR. For the prior $\beta^* \sim \mathcal{N}(0, \frac{r^2}{d} I_d)$ and noise $\mathcal{N}(0,\sigma^2)$, the posterior mean of β^* is $(X^\top X + \frac{\sigma^2 d}{r^2} I)^{-1} X^\top y$, the ridge estimator at λ^*. So step 5 is the Nishimori line of Chapter 4: the Nishimori identity (Lemma 1.10) gives $m = q$, and a posterior mean minimizes the mean squared error. In physics units the optimal weight decay is $\alpha\lambda^* = \sigma^2/r^2$, whatever α.

HOW THE CURVE WAS FIRST FOUND. For regression, not with replicas. Hertz et al. (1989) treated the learning dynamics of the one-layer perceptron with random patterns, and Krogh and Hertz (1992) solved the gradient dynamics of the linear perceptron trained by a noisy teacher, finding that a weight decay of the size of the noise variance improves generalization. Krogh and Hertz (1991) wrote these dynamics in the eigenbasis of the input correlation matrix, found the optimal decay σ^2/σ_u^2 (our $\alpha\lambda^*$), noted that it does not depend on the spectrum of that matrix, and plotted the learning curve for random patterns against α: without decay it has the peak at $\alpha = 1$, with the optimal decay none.[4] These are averages over the spectrum of a random matrix. Dunmur and Wallace (1993) then computed the linear perceptron with weight decay and noisy data by the Gardner (replica) method at finite temperature.

[4] Their Figure 1 has $r^2 = 1$ and $\sigma^2 = 0.2$: the noiseless curve falls from 1 to 0 at $\alpha = 1$, and the decay 0.2 of the full line is optimal. Read off the figure, the optimally decayed curve passes through 0.36 at $\alpha = 1$ and about 0.15 at $\alpha = 2$, where our formula gives 0.358 and 0.148; the curve without decay follows part (ii) of Theorem 9.2 below.

9.3 The stone

The proof column holds both rigorous routes, random matrices (Section 9.4) and comparison (Section 9.5); in both, the susceptibility is a concrete random quantity and the replica equations reappear unchanged.

PHYSICS · *replicas at zero temperature*	PROOF · *random matrices; Gordon's comparison*
1 · The object	
A Gibbs measure $e^{-bH(w)}$ at $b \to \infty$, and the overlaps Q, q, m of its samples.	The estimator $\hat\beta_\lambda = (X^\top X + n\lambda I)^{-1} X^\top y$, linear in y; no temperature, no free entropy.
2 · Average over the data	
$\mathbb{E}Z^k$ at integer k: the fields of each sample are Gaussian with covariance (Q, m).	Average the noise and the teacher's direction exactly (Lemmas 9.5 and 9.6); traces of $G = (X^\top X/n + \lambda I)^{-1}$ remain. Comparison: $v^\top X w \to \|w\| g^\top v + \|v\| h^\top w$ (Theorem 8.3).
3 · Replica symmetry, $b \to \infty$	
$Q_{ac} = q + \delta_{ac}\chi/b$: the measure shrinks onto the ground state, with width $\chi = b(Q - q)$.	Strict convexity: one minimizer. The Gibbs measure is Gaussian, and $b(Q-q) = \operatorname{tr}(X^\top X + n\lambda I)^{-1} =: \chi_n$ exactly (Exercise 9.7).
4 · The cavity field	
$\zeta^a = \omega + \sqrt{\chi/b}\, z_a$, and (9.3) gives $\hat\zeta = (\omega + \chi y)/(1+\chi)$.	Leave one out (Lemma 9.4): $x_i^\top \hat\beta_\lambda = (x_i^\top \hat\beta_{(i)} + \tau_i y_i)/(1+\tau_i)$, $\tau_i \to \chi$; the cavity field is the leave-one-out prediction.
5 · $\partial_q \Psi = 0$: the susceptibility	
$\lambda\chi + \dfrac{\chi}{1+\chi} = \dfrac{1}{\alpha}.$	Trace of $\lambda G + G X^\top X/n = I$: $\lambda\chi_n + \frac{1}{n}\sum_i \ell_i = \frac{d}{n}$, with leverages $\ell_i \to \chi/(1+\chi)$ (Proposition 9.8). Comparison: $\partial_\theta \mathcal{G} = \partial_a \mathcal{G} = 0$ with $1 + \chi = \sqrt{a^2+\sigma^2}/\theta$ (Proposition 9.9).
6 · $\partial_m \Psi = \partial_\chi \Psi = 0$: the risk	
$m = \alpha r^2 \chi/(1+\chi)$, $q - \frac{m^2}{r^2} = \frac{\alpha\chi^2(R+\sigma^2)}{(1+\chi)^2}$; so $R(1-v) = r^2(\alpha\lambda\chi)^2 + \sigma^2 v$.	$R \to -\alpha r^2 \lambda^2 \chi' + \sigma^2(\chi + \lambda\chi')$ with $\chi' = -\alpha\chi^2/(1-v)$: the same formula. Comparison: the direction of w gives m, its component along h gives $q - m^2/r^2$.

7 · The limits

$k \to 0$ after $d \to \infty$, then $b \to \infty$; then $\lambda \to 0$ in the formula.

No counterpart for $k \to 0$. The exchange of $\lambda \to 0$ and $n \to \infty$ is Theorem 9.2(ii): monotone comparison and an inverse Wishart mean.

WHAT THE TRANSLATION SHOWS. Every step except the continuation in k has a rigorous counterpart, and three are identities rather than analogies: the susceptibility is the trace of the *resolvent*, in the limit $1/\alpha$ times the Marchenko–Pastur Stieltjes transform at $-\lambda$; the cavity field is the leave-one-out prediction; and the replica equations are the stationarity equations of Gordon's scalar problem, after the change of variables $1 + \chi = \sqrt{a^2 + \sigma^2}/\theta$. The routes then part: the random-matrix proof uses the closed form of the estimator but extends to non-Gaussian designs, while the comparison proof needs Gaussian designs but works for every convex loss and penalty (Chapters 10 and 11). And the physicist's ridgeless curve sets $\lambda = 0$ after $d \to \infty$, an exchange of limits that the proof must justify and that at $\alpha = 1$ matters at every finite size (Section 9.8).

9.4 The proof: random matrices

Assumption 9.1 (Isotropic Gaussian ridge regression). For each n let $d = d_n$ with $n/d_n \to \alpha \in (0, \infty)$. The design $X \in \mathbb{R}^{n \times d}$ has independent $\mathcal{N}(0,1)$ entries; the noise $\varepsilon \in \mathbb{R}^n$ is independent of X and has independent entries with mean 0, variance 1 and $\mathbb{E}\varepsilon_i^4 \le c_4$ for a constant c_4; the teacher $\beta^* \in \mathbb{R}^d$ is deterministic with $\|\beta^*\| = r$ for a fixed $r \ge 0$; $\sigma \ge 0$ is fixed; and $y = X\beta^* + \sigma\varepsilon$.

Theorem 9.2 (The proportional-limit ridge risk). *Assume Assumption 9.1. For $\lambda > 0$ let $\chi = \chi_\alpha(\lambda)$ be the unique positive solution of*

$$\lambda\chi + \frac{\chi}{1+\chi} = \frac{1}{\alpha}, \tag{9.5}$$

let $\nu = \alpha\chi^2/(1+\chi)^2$, and define

$$\mathcal{R}(\alpha,\lambda) = \frac{r^2(\alpha\lambda\chi)^2 + \sigma^2\nu}{1-\nu}, \qquad \mathcal{T}(\alpha,\lambda) = \frac{\mathcal{R}(\alpha,\lambda) + \sigma^2}{(1+\chi)^2}. \tag{9.6}$$

Then $0 < \nu < 1$, and as $n \to \infty$ the following hold in probability.

(i) For every $\lambda > 0$, $R(\hat\beta_\lambda) \to \mathcal{R}(\alpha,\lambda)$ and $T(\hat\beta_\lambda) \to \mathcal{T}(\alpha,\lambda)$.

(ii) If $\alpha > 1$, then $R(\hat\beta_0) \to \sigma^2/(\alpha-1)$; if $\alpha < 1$, then $R(\hat\beta_0) \to r^2(1-\alpha) + \sigma^2\alpha/(1-\alpha)$; if $\alpha = 1$, then $R(\hat\beta_0) \to \infty$ when $\sigma > 0$ and $R(\hat\beta_0) \to 0$ when $\sigma = 0$. In each case the limit equals $\lim_{\lambda\downarrow 0}\mathcal{R}(\alpha,\lambda)$.

(iii) If $r > 0$ and $\sigma > 0$, then $\lambda \mapsto \mathcal{R}(\alpha,\lambda)$ is strictly decreasing on $(0,\lambda^*]$ and strictly increasing on $[\lambda^*,\infty)$, where $\lambda^* = \sigma^2/(\alpha r^2)$; moreover

> *$\mathcal{R}(\alpha,\lambda^*) = \sigma^2\chi_\alpha(\lambda^*)$, and this optimally tuned risk is a strictly decreasing function of α. If $\sigma = 0 < r$, $\lambda \mapsto \mathcal{R}(\alpha,\lambda)$ is strictly increasing, and if $r = 0 < \sigma$ it is strictly decreasing.*

The theorem is the physics box. In other normalizations, and for the risk averaged over the noise, part (i) is due to Dicker (2016) at λ^*, to Dobriban and Wager (2018) for a random teacher, and to Hastie et al. (2022, Corollary 6); part (ii) for $\alpha \neq 1$ is Proposition 2 ($\alpha > 1$) and Theorem 1 ($\alpha < 1$) of Hastie et al. (2022); the optimal ridge of part (iii) and its risk are in all three. The proof below is assembled from standard tools so that each step answers a step of the replica computation.

REMARK 9.3 (The Marchenko–Pastur form). By Proposition 9.8 below, $\chi_\alpha(\lambda) = \lim \frac{1}{n}\operatorname{tr}(X^\top X/n + \lambda I)^{-1}$; since the spectral distribution of $X^\top X/n$ converges to the *Marchenko–Pastur law* MP with ratio $1/\alpha$ (Theorem A.16), its Stieltjes transform m (Definition A.14) satisfies $m(-\lambda) = \alpha\chi_\alpha(\lambda)$, and (9.6) reads

$$\mathcal{R}(\alpha,\lambda) = \int \frac{r^2\lambda^2 + \sigma^2 s/\alpha}{(s+\lambda)^2}\,\mathrm{MP}(\mathrm{d}s) = r^2\lambda^2 m'(-\lambda) + \frac{\sigma^2}{\alpha}\big(m(-\lambda) - \lambda m'(-\lambda)\big),$$

Corollary 6 of Hastie et al. (2022) with their $\gamma = 1/\alpha$. With the *effective ridge* $\lambda_{\mathrm{eff}} = \lambda(1+\chi)$ of Chapter 12, $\mathcal{R} = (r^2\lambda_{\mathrm{eff}}^2 + \sigma^2/\alpha)/((1+\lambda_{\mathrm{eff}})^2 - 1/\alpha)$, the flat-spectrum kernel learning curve.

9.4.1 Four lemmas

The first lemma is the rigorous cavity: what removing one sample does to the estimator.[5]

[5] It rests on the Sherman–Morrison formula $(B + xx^\top)^{-1} = B^{-1} - B^{-1}xx^\top B^{-1}/(1 + x^\top B^{-1}x)$, valid for invertible B with $1 + x^\top B^{-1}x \neq 0$.

Lemma 9.4 (Leave one out). *Let $\lambda > 0$, $A = X^\top X + n\lambda I$, and for a sample i let $A_{(i)} = A - x_i x_i^\top$, $\tau_i = x_i^\top A_{(i)}^{-1} x_i$, and $\hat\beta_{(i)} = A_{(i)}^{-1}(X^\top y - y_i x_i)$, the ridge estimator computed without sample i and with the same penalty $n\lambda$. Then the* leverage *of sample i and its fitted value are*

$$\ell_i = x_i^\top A^{-1} x_i = \frac{\tau_i}{1+\tau_i}, \qquad x_i^\top \hat\beta_\lambda = \frac{x_i^\top \hat\beta_{(i)} + \tau_i y_i}{1+\tau_i},$$

and $0 \le \operatorname{tr} A_{(i)}^{-1} - \operatorname{tr} A^{-1} \le 1/(n\lambda)$.

PROOF. The Sherman–Morrison formula gives $A^{-1} = A_{(i)}^{-1} - A_{(i)}^{-1}x_i x_i^\top A_{(i)}^{-1}/(1+\tau_i)$. Applied to x_i it gives $A^{-1}x_i = A_{(i)}^{-1}x_i/(1+\tau_i)$, hence $\ell_i = \tau_i/(1+\tau_i)$. With $b_i = X^\top y - y_i x_i$ we have $\hat\beta_\lambda = A^{-1}(b_i + y_i x_i)$, so $x_i^\top\hat\beta_\lambda = (A^{-1}x_i)^\top b_i + \ell_i y_i = (x_i^\top\hat\beta_{(i)} + \tau_i y_i)/(1+\tau_i)$. Taking traces, $\operatorname{tr} A_{(i)}^{-1} - \operatorname{tr} A^{-1} = x_i^\top A_{(i)}^{-2} x_i/(1+\tau_i)$, which lies between 0 and $\|A_{(i)}^{-1}\|\tau_i/(1+\tau_i) \le 1/(n\lambda)$, because $A_{(i)} \ge n\lambda I$. ∎

Lemma 9.5 (Quadratic forms). *Let M be a fixed symmetric $k \times k$ matrix.*

(i) If u is uniform on the unit sphere of $\mathbb{R}^k$, then $\mathbb{E}\,u^\top M u = \operatorname{tr} M/k$ and $\operatorname{Var}(u^\top M u) \le 2\|M\|^2/(k+2)$.

(ii) If $\xi \in \mathbb{R}^k$ has independent entries with mean 0, variance 1 and $\mathbb{E}\xi_j^4 \le c_4$, then $\mathbb{E}\xi^\top M\xi = \operatorname{tr} M$ and $\operatorname{Var}(\xi^\top M\xi) \le (1+c_4)\operatorname{tr} M^2$; if ξ is standard Gaussian, $\operatorname{Var}(\xi^\top M\xi) = 2\operatorname{tr} M^2$.

PROOF. The plan is to expand the second moment for part 2 and to reduce part 1 to the Gaussian case. In $\mathbb{E}(\xi^\top M\xi)^2 = \sum M_{jl}M_{st}\mathbb{E}\xi_j\xi_l\xi_s\xi_t$ only the terms whose indices match in pairs survive, so $\operatorname{Var}(\xi^\top M\xi) = 2\sum_{j\neq l} M_{jl}^2 + \sum_j M_{jj}^2(\mathbb{E}\xi_j^4 - 1)$, which is at most $\max(2, c_4-1)\operatorname{tr} M^2 \le (1+c_4)\operatorname{tr} M^2$, because $c_4 \ge (\mathbb{E}\xi_j^2)^2 = 1$, and equals $2\operatorname{tr} M^2$ when $\mathbb{E}\xi_j^4 = 3$. For part 1, write $u = z/\|z\|$ with $z \sim \mathcal{N}(0, I_k)$, so that u is independent of $\|z\|$ and $z^\top Mz = \|z\|^2 u^\top Mu$. Taking means with $\mathbb{E}\|z\|^2 = k$ gives $\mathbb{E}u^\top Mu = \operatorname{tr} M/k$, and taking second moments with $\mathbb{E}\|z\|^4 = k(k+2)$ and $\mathbb{E}(z^\top Mz)^2 = (\operatorname{tr} M)^2 + 2\operatorname{tr} M^2$ gives

$$\begin{aligned}\operatorname{Var}(u^\top Mu) &= \frac{(\operatorname{tr} M)^2 + 2\operatorname{tr} M^2}{k(k+2)} - \frac{(\operatorname{tr} M)^2}{k^2} \\ &= \frac{2}{k+2}\left(\frac{\operatorname{tr} M^2}{k} - \left(\frac{\operatorname{tr} M}{k}\right)^2\right) \le \frac{2\|M\|^2}{k+2}.\end{aligned}$$

■

Lemma 9.6 (Rotation invariance). *For $\lambda \ge 0$ write $R_\lambda(X, \varepsilon; \beta^*)$ and $T_\lambda(X, \varepsilon; \beta^*)$ for the excess risk and the training error of $\hat\beta_\lambda$ computed from $y = X\beta^* + \sigma\varepsilon$. For every orthogonal matrix O, $R_\lambda(XO, \varepsilon; \beta^*) = R_\lambda(X, \varepsilon; O\beta^*)$, and the same holds for T_λ. Consequently, under Assumption 9.1, the law of $R_\lambda(X, \varepsilon; \beta^*)$ is that of $R_\lambda(X, \varepsilon; ru)$ with u uniform on the unit sphere and independent of (X, ε), and the same holds for T_λ.*

PROOF. The plan is to check the identities for a fixed rotation and then to average over a random one. The response $y = XO\beta^* + \sigma\varepsilon$ is that of the design X and the teacher $O\beta^*$, and the estimator from (XO, y) is $O^\top\tilde\beta$, with $\tilde\beta$ the estimator from (X, y), because $((XO)^\top XO + n\lambda I)^{-1}(XO)^\top = O^\top(X^\top X + n\lambda I)^{-1}X^\top$ and $(XO)^+ = O^\top X^+$. Hence $\|O^\top\tilde\beta - \beta^*\| = \|\tilde\beta - O\beta^*\|$ and $\|y - XOO^\top\tilde\beta\| = \|y - X\tilde\beta\|$. Since XO has the law of X and is independent of ε, the law of $R_\lambda(X, \varepsilon; O\beta^*)$ is that of $R_\lambda(X, \varepsilon; \beta^*)$ for every fixed O; taking O uniform on the orthogonal group and independent of (X, ε) gives the last claim. ■

Lemma 9.7 (An inverse Wishart mean). *Let $Z \in \mathbb{R}^{N\times k}$ have independent $\mathcal{N}(0,1)$ entries, with $N \ge k+2$. Then $Z^\top Z$ is invertible almost surely and $\mathbb{E}\operatorname{tr}(Z^\top Z)^{-1} = k/(N-k-1)$.*

PROOF. The columns of Z are almost surely linearly independent, since a Gaussian vector falls in a fixed proper subspace with probability 0. Let z_1 be the first column and P the orthogonal projection onto the span of the others. The Schur complement formula gives $((Z^\top Z)^{-1})_{11} = 1/(z_1^\top(I-P)z_1)$, the inverse squared distance from z_1 to the other columns: the random-matrix cavity. Given the other columns, z_1 is standard Gaussian and $I - P$ is a

projection of rank $N-k+1$, so $z_1^\top(I-P)z_1$ has the chi-square law with $m = N-k+1 \geq 3$ degrees of freedom, and

$$\mathbb{E}\frac{1}{\chi^2_m} = \int_0^\infty \frac{1}{x}\,\frac{x^{m/2-1}e^{-x/2}}{2^{m/2}\Gamma(m/2)}\,\mathrm{d}x = \frac{2^{m/2-1}\Gamma(m/2-1)}{2^{m/2}\Gamma(m/2)} = \frac{1}{m-2}.$$

So $\mathbb{E}((Z^\top Z)^{-1})_{11} = 1/(N-k-1)$, and the columns are exchangeable. ∎

9.4.2 The susceptibility

The next proposition is the rigorous form of the first replica equation, and its proof is the leave-one-out version of the cavity argument.[6]

[6] The sum of the leverages, $\operatorname{tr} X(X^\top X + n\lambda I)^{-1}X^\top$, is the effective number of parameters of ridge regression; (9.8) below says that it equals $d - n\lambda\chi_n$.

Proposition 9.8 (The susceptibility is the trace of the resolvent). *Let X be as in Assumption 9.1, $S = X^\top X/n$, and for $\lambda > 0$ let $G = (S+\lambda I)^{-1}$,*

$$\chi_n(\lambda) = \frac{1}{n}\operatorname{tr} G = \operatorname{tr}(X^\top X + n\lambda I)^{-1}, \qquad \psi_n(\lambda) = \frac{1}{n}\operatorname{tr} G^2 = -\chi_n'(\lambda).$$

Then $\mathbb{E}(\chi_n(\lambda) - \chi_\alpha(\lambda))^2 \to 0$ and $\psi_n(\lambda) \to -\chi_\alpha'(\lambda)$ in probability, and

$$\chi_\alpha'(\lambda) = -\frac{\alpha\chi^2}{1-\nu}, \qquad 1-\alpha\lambda\chi = \frac{\alpha\chi}{1+\chi}, \qquad 0<\nu<1. \tag{9.7}$$

Proof. The plan is to show that χ_n concentrates, that its mean nearly solves the self-consistent equation (9.5) because of the leave-one-out identity, and that the equation has one positive solution; the derivative then follows by convexity. Recall $\|G\| \leq 1/\lambda$.

Concentration. The function $f(X) = \operatorname{tr} G = n\chi_n$ is smooth, and $\mathrm{d}G = -G(\mathrm{d}S)G$ with $\mathrm{d}S = (\mathrm{d}X^\top X + X^\top \mathrm{d}X)/n$ gives $\partial f/\partial X_{ik} = -\frac{2}{n}(XG^2)_{ik}$. Hence $\|\nabla f\|^2 = \frac{4}{n}\operatorname{tr}(G^2SG^2) \leq \frac{4d}{n\lambda^3}$, because each eigenvalue $s/(s+\lambda)^4$ of G^2SG^2 is at most λ^{-3}. The Gaussian Poincaré inequality (Theorem A.6) gives $\operatorname{Var} f \leq \mathbb{E}\|\nabla f\|^2$, so $\operatorname{Var}\chi_n \leq 4d/(n^3\lambda^3)$.

Leave one out. Taking the trace of $\lambda G + GS = I$, with $GS = \frac{1}{n}\sum_i Gx_ix_i^\top$ and $G = nA^{-1}$, Lemma 9.4 gives

$$\lambda\chi_n + \frac{1}{n}\sum_{i=1}^n \ell_i = \frac{d}{n}, \qquad \ell_i = \frac{1}{n}x_i^\top G x_i = \frac{\tau_i}{1+\tau_i}, \tag{9.8}$$

where $\tau_i = \frac{1}{n}x_i^\top G_{(i)}x_i$ with $G_{(i)} = nA_{(i)}^{-1}$ independent of x_i. Let $c_n = \mathbb{E}\chi_n$. Given $G_{(i)}$, Lemma 9.5 (Gaussian case) gives $\mathbb{E}(x_i^\top G_{(i)}x_i - \operatorname{tr} G_{(i)})^2 = 2\mathbb{E}\operatorname{tr} G_{(i)}^2 \leq 2d/\lambda^2$; Lemma 9.4 gives $0 \leq \operatorname{tr} G_{(i)} - \operatorname{tr} G \leq 1/\lambda$; and $\mathbb{E}|\chi_n - c_n| \leq (\operatorname{Var}\chi_n)^{1/2}$. Together,

$$\mathbb{E}|\tau_i - c_n| \leq \frac{\sqrt{2d}}{n\lambda} + \frac{1}{n\lambda} + \frac{2\sqrt{d}}{n^{3/2}\lambda^{3/2}} =: \epsilon_n \to 0.$$

The equation. The map $t \mapsto t/(1+t)$ is 1-Lipschitz on $[0,\infty)$, so taking means in (9.8) gives $|\varphi(c_n) - d/n| \leq \epsilon_n$ for $\varphi(c) = \lambda c + c/(1+c)$. The function φ is continuous and strictly increasing from $\varphi(0) = 0$ to ∞, so it

has a continuous inverse on $[0,\infty)$, the equation $\varphi(c) = 1/\alpha$ has exactly one solution $\chi_\alpha(\lambda) > 0$, and $c_n \to \chi_\alpha(\lambda)$ because $d/n \to 1/\alpha$. With the concentration bound, $\mathbb{E}(\chi_n - \chi_\alpha)^2 = \operatorname{Var}\chi_n + (c_n - \chi_\alpha)^2 \to 0$.

The derivative. The second identity in (9.7) is (9.5) multiplied by α; it gives $\alpha\chi/(1+\chi) < 1$, hence $\nu = \frac{\alpha\chi}{1+\chi}\cdot\frac{\chi}{1+\chi} < 1$. Differentiating (9.5) (the implicit function theorem applies, since $\varphi' > 0$) gives $\chi'(\lambda + (1+\chi)^{-2}) = -\chi$, and by (9.5), $\lambda + (1+\chi)^{-2} = \frac{1}{\alpha\chi} - \frac{\chi}{(1+\chi)^2} = \frac{1-\nu}{\alpha\chi}$; this is the first identity. Finally, $\chi_n(\lambda) = \frac{1}{n}\sum_j (s_j+\lambda)^{-1}$, with s_j the eigenvalues of S, is convex in λ, so for $0 < h < \lambda$

$$\frac{\chi_n(\lambda) - \chi_n(\lambda - h)}{h} \le \chi_n'(\lambda) \le \frac{\chi_n(\lambda + h) - \chi_n(\lambda)}{h}.$$

Both bounds converge in probability to the difference quotients of χ_α, which tend to $\chi_\alpha'(\lambda)$ as $h \downarrow 0$; hence $\chi_n'(\lambda) \to \chi_\alpha'(\lambda)$ in probability. ∎

9.4.3 Proof of the theorem

Proof of Theorem 9.2, part (i). The plan is to write the risk and the training error as quadratic forms in the noise and in the teacher, average them with Lemma 9.5, and recognize the traces that remain. By Lemma 9.6 we may assume that $\beta^* = ru$ with u uniform on the unit sphere and independent of (X,ε). With S, G, χ_n and ψ_n as in Proposition 9.8, $\frac{1}{n}\operatorname{tr}(SG^2) = \frac{1}{n}\operatorname{tr}(G - \lambda G^2) = \chi_n - \lambda\psi_n$.

The risk. Since $\hat\beta_\lambda = GX^\top y/n$ and $GS = I - \lambda G$, $\hat\beta_\lambda - \beta^* = -\lambda G\beta^* + \frac{\sigma}{n}GX^\top\varepsilon$, so $R(\hat\beta_\lambda) = B + C + V$ with $B = \lambda^2\beta^{*\top}G^2\beta^*$, $C = -\frac{2\lambda\sigma}{n}\beta^{*\top}G^2X^\top\varepsilon$ and $V = \frac{\sigma^2}{n^2}\varepsilon^\top XG^2X^\top\varepsilon$. Given X: part 1 of Lemma 9.5 with $M = G^2$ gives $\mathbb{E}[B \mid X] = \lambda^2 r^2\frac{n}{d}\psi_n$ and $\operatorname{Var}(B \mid X) \le 2r^4/(d+2)$; the cross term has $\mathbb{E}[C \mid X,u] = 0$ and $\mathbb{E}[C^2 \mid X,u] = \frac{4\lambda^2\sigma^2}{n}\beta^{*\top}G^2SG^2\beta^* \le \frac{4\sigma^2r^2}{n\lambda}$; and part 2 with $M = XG^2X^\top/n^2$ gives $\mathbb{E}[V \mid X] = \sigma^2(\chi_n - \lambda\psi_n)$ and $\operatorname{Var}(V \mid X) \le (1 + c_4)\sigma^4\frac{1}{n^2}\operatorname{tr}(S^2G^4) \le (1+c_4)\frac{\sigma^4 d}{n^2\lambda^2}$, because $s^2/(s+\lambda)^4 \le \lambda^{-2}$. By Chebyshev's inequality given X and by Proposition 9.8,

$$R(\hat\beta_\lambda) \to -\alpha r^2\lambda^2\chi' + \sigma^2(\chi + \lambda\chi') \tag{9.9}$$

in probability. By (9.7), $-\alpha r^2\lambda^2\chi' = r^2(\alpha\lambda\chi)^2/(1-\nu)$ and $\chi + \lambda\chi' = \chi(1 - \nu - \alpha\lambda\chi)/(1-\nu) = \chi(\frac{\alpha\chi}{1+\chi} - \nu)/(1-\nu) = \nu/(1-\nu)$, so the limit is $\mathcal{R}(\alpha,\lambda)$.

The training error. Let $K = XX^\top/n$. The push-through identity $X(X^\top X + n\lambda I)^{-1} = (XX^\top + n\lambda I)^{-1}X$ gives $y - X\hat\beta_\lambda = \lambda(K+\lambda I)^{-1}y$, so that $T(\hat\beta_\lambda) = \frac{\lambda^2}{n}\|(K+\lambda)^{-1}(X\beta^* + \sigma\varepsilon)\|^2$. By the singular value decomposition $X^\top(K+\lambda)^{-2}X = nSG^2$, and $\frac{1}{n}\operatorname{tr}(K+\lambda)^{-2} = \psi_n + (1 - \frac{d}{n})\lambda^{-2}$ because K and S have the same nonzero eigenvalues. The same averaging as for the risk (bias mean $\lambda^2 r^2\frac{n}{d}(\chi_n - \lambda\psi_n)$, cross term of mean 0, conditional variances $O(1/n)$) gives

$$T(\hat\beta_\lambda) \to \alpha r^2\lambda^2(\chi + \lambda\chi') + \sigma^2\Big(1 - \frac{1}{\alpha} - \lambda^2\chi'\Big).$$

Put $s = 1/(1+\chi)$; by (9.7), $\alpha\lambda\chi = 1 - \alpha(1-s)$ and $\nu = \alpha(1-s)^2$. The bias coefficient is $\alpha\lambda^2\nu/(1-\nu) = (\alpha\lambda\chi)^2 s^2/(1-\nu)$, and expanding the squares,

the noise coefficient is

$$1-\frac{1}{\alpha}+\frac{\alpha\lambda^2\chi^2}{1-\nu}=\frac{(1-\frac{1}{\alpha})(1-\alpha(1-s)^2)+\frac{1}{\alpha}(1-\alpha(1-s))^2}{1-\nu}=\frac{s^2}{1-\nu}.$$

So $T(\hat\beta_\lambda)\to s^2(r^2(\alpha\lambda\chi)^2+\sigma^2)/(1-\nu)=s^2(\mathcal{R}+\sigma^2)=\mathcal{T}(\alpha,\lambda)$. ∎

Proof of Theorem 9.2, part (ii). The plan is to split the risk of $\hat\beta_0$ into a projection of β^* and a noise term that dominates the noise term of ridge, and to match the noise term's mean with Lemma 9.7. Let $P=X^+X$, the orthogonal projection onto the row space of X, of rank $\min(n,d)$ almost surely. Since $\hat\beta_0=P\beta^*+\sigma X^+\varepsilon$ and $X^+\varepsilon$ lies in the row space, $R(\hat\beta_0)=\|(I-P)\beta^*\|^2+\sigma^2\|X^+\varepsilon\|^2$.

The bias. If $n\ge d$ then $P=I$. If $d>n$, Lemma 9.6 with $\sigma=0$ shows that the bias has the law of $r^2\|(I-P)u\|^2$ with u uniform and independent of X, and given X, Lemma 9.5 with $M=I-P$, of trace $d-n$, gives mean $r^2(1-n/d)$ and variance at most $2r^4/(d+2)$. So the bias tends to $r^2(1-\alpha)_+$ in probability.

The noise. If $n\ge d+2$, then $\mathbb{E}[\|X^+\varepsilon\|^2\mid X]=\operatorname{tr}(X^\top X)^{-1}$, of mean $d/(n-d-1)$ by Lemma 9.7; if $d\ge n+2$, then $X^+=X^\top(XX^\top)^{-1}$ and the mean is $\mathbb{E}\operatorname{tr}(XX^\top)^{-1}=n/(d-n-1)$. These tend to $N_0=1/(\alpha-1)$ if $\alpha>1$ and $N_0=\alpha/(1-\alpha)$ if $\alpha<1$. For a lower bound, write $X=\sum_j s_ju_jv_j^\top$ over its positive singular values; then $X^+\varepsilon=\sum_j s_j^{-1}(u_j^\top\varepsilon)v_j$, while $(X^\top X+n\lambda I)^{-1}X^\top\varepsilon=\sum_j s_j(s_j^2+n\lambda)^{-1}(u_j^\top\varepsilon)v_j$ with $s_j/(s_j^2+n\lambda)<1/s_j$, so for every $\lambda>0$

$$\|X^+\varepsilon\|^2\ge Q_\lambda:=\|(X^\top X+n\lambda I)^{-1}X^\top\varepsilon\|^2.$$

The variance term of part (i), with $\sigma=1$, shows $Q_\lambda\to N(\lambda):=\chi+\lambda\chi'=\nu/(1-\nu)$ in probability. As $\lambda\downarrow 0$, step 4 of Computation 9.1, which uses only (9.5), gives $\nu\to 1/\alpha$ if $\alpha>1$ and $\nu\to\alpha$ if $\alpha<1$, so $N(\lambda)\to N_0$; if $\alpha=1$, (9.5) reads $\lambda\chi(1+\chi)=1$, so $\chi\to\infty$, $\nu\to 1$ and $N(\lambda)\to\infty$.

Conclusion. Let $\alpha\ne 1$, $Y_n=\|X^+\varepsilon\|^2$ and $\delta>0$, and choose λ with $N(\lambda)>N_0-\delta/2$. Then $\mathbb{P}(Y_n<N_0-\delta)\le\mathbb{P}(Q_\lambda<N(\lambda)-\delta/2)\to 0$. Since $|Y_n-N_0|=(Y_n-N_0)+2(N_0-Y_n)_+$, $\mathbb{E}Y_n\to N_0$ and $\mathbb{E}(N_0-Y_n)_+\le\delta+N_0\mathbb{P}(Y_n<N_0-\delta)$, this lower bound and the matching mean give $\mathbb{E}|Y_n-N_0|\to 0$; with the bias, this is the claim for $\alpha\ne 1$. If $\alpha=1$ and $\sigma>0$, then $R(\hat\beta_0)\ge\sigma^2Q_\lambda$ for every λ and $N(\lambda)\to\infty$, so $R(\hat\beta_0)\to\infty$; if $\alpha=1$ and $\sigma=0$, $R(\hat\beta_0)$ is the bias, of mean $r^2(1-n/d)_+\to 0$. Finally, $\mathcal{R}(\alpha,\lambda)=r^2(\alpha\lambda\chi)^2/(1-\nu)+\sigma^2N(\lambda)$, and the same limits show that its first term tends to 0 if $\alpha>1$ and to $r^2(1-\alpha)$ if $\alpha<1$; if $\alpha=1$ it equals $r^2/(1+2\chi)\to 0$, because there $\lambda\chi=1/(1+\chi)$ and $1-\nu=(1+2\chi)/(1+\chi)^2$. ∎

Proof of Theorem 9.2, part (iii). The plan is to differentiate (9.9) and to show that one factor of the derivative has a fixed sign. Differentiating,

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\lambda}=-\alpha r^2\lambda(2\chi'+\lambda\chi'')+\sigma^2(2\chi'+\lambda\chi'')=(\sigma^2-\alpha r^2\lambda)(2\chi'+\lambda\chi'').$$

To find the sign of $2\chi'+\lambda\chi''$, invert (9.5): $\lambda=u-v$ with $u=1/(\alpha\chi)$ and $v=1/(1+\chi)$, a smooth function of χ with

$$\lambda_\chi=-\frac{u}{\chi}+v^2=-\frac{1-\nu}{\alpha\chi^2}<0,\qquad \lambda_{\chi\chi}=\frac{2u}{\chi^2}-2v^3.$$

Then $\chi' = 1/\lambda_\chi$, $\chi'' = -\lambda_{\chi\chi}/\lambda_\chi^3$, and $2\chi' + \lambda\chi'' = (2\lambda_\chi^2 - \lambda\lambda_{\chi\chi})/\lambda_\chi^3$. Expanding,

$$2\lambda_\chi^2 - \lambda\lambda_{\chi\chi} = 2uv\Big(\frac{1}{\chi} - v\Big)^2 = \frac{2}{\alpha\chi^3(1+\chi)^3} > 0,$$

so $2\chi' + \lambda\chi'' < 0$ for every $\lambda > 0$, and the derivative of $\mathcal{R}$ has the sign of $\alpha r^2\lambda - \sigma^2$, which proves the monotonicity in λ in each case. At λ^*, $\alpha r^2\lambda^* = \sigma^2$, and (9.9) gives $\mathcal{R}(\alpha, \lambda^*) = \sigma^2\chi + \lambda^*\chi'(\sigma^2 - \alpha r^2\lambda^*) = \sigma^2\chi$. For the monotonicity in α, put $c = \sigma^2/r^2$: at $\lambda = c/\alpha$, (9.5) multiplied by α reads $c\chi + \alpha\chi/(1+\chi) = 1$, so $c\chi < 1$ and $\alpha = (1 - c\chi)(1 + 1/\chi)$, a product of two positive, strictly decreasing functions of $\chi \in (0, 1/c)$ that maps $(0, 1/c)$ onto $(0, \infty)$. Its inverse $\alpha \mapsto \chi_\alpha(\lambda^*)$ is therefore strictly decreasing, and so is $\sigma^2\chi_\alpha(\lambda^*)$. ∎

The *optimal ridge* is optimal at every finite size. For Gaussian designs and the risk averaged over the design and the noise, this is Proposition 1 of Dicker (2016); in fact, with $\beta^* = ru$ and s_j the eigenvalues of $X^\top X/n$, $\mathbb{E}[R(\hat\beta_\lambda) \mid X] = \frac{1}{d}\sum_j (r^2\lambda^2 + \sigma^2\frac{d}{n}s_j)/(s_j + \lambda)^2$ is minimized at $\lambda = \sigma^2 d/(nr^2)$ for every design, with minimum $\sigma^2\chi_n$ (Exercise 9.3).

9.5 The proof: Gordon's auxiliary problem

The second rigorous route writes the ridge problem as a Gaussian min–max and replaces the design by two Gaussian vectors, with the *convex Gaussian min–max theorem* (CGMT) of Chapter 8.[7] We compute the auxiliary problem as a physicist would, prove that its stationarity equations are the replica equations, and then say what the CGMT adds.

[7] The idea of combining Gordon's comparison with convex duality to obtain exact asymptotics is Stojnic's (Stojnic 2013 for the LASSO); Thrampoulidis et al. (2015) and Thrampoulidis et al. (2018) turned it into the CGMT.

COMPUTATION 9.2 · GORDON'S AUXILIARY PROBLEM FOR RIDGE

1. *The primary problem.* With $w = \beta - \beta^*$ the residual is $\sigma\varepsilon - Xw$, and $\frac{1}{2n}\|Xw - \sigma\varepsilon\|^2 = \max_{v\in\mathbb{R}^n}\{v^\top(Xw - \sigma\varepsilon) - \frac{n}{2}\|v\|^2\}$, attained at $v = (Xw - \sigma\varepsilon)/n$. So the minimum of the ridge objective is

$$\min_{w\in\mathbb{R}^d}\max_{v\in\mathbb{R}^n}\Big\{v^\top Xw - \sigma v^\top\varepsilon - \frac{n}{2}\|v\|^2 + \frac{\lambda}{2}\|\beta^* + w\|^2\Big\},$$

   a bilinear form in the Gaussian matrix X plus a function convex in w and concave in v.

2. *The auxiliary problem.* The CGMT compares it with the problem in which $v^\top Xw$ is replaced by $\|w\|g^\top v + \|v\|h^\top w$, with $g \sim \mathcal{N}(0, I_n)$ and $h \sim \mathcal{N}(0, I_d)$ independent of everything else.

3. *The directions.* For $\|v\| = \theta/\sqrt{n}$, the terms linear in v are largest for v parallel to $\|w\|g - \sigma\varepsilon$, with value $\frac{\theta}{\sqrt{n}}\|\|w\|g - \sigma\varepsilon\| \approx \theta\sqrt{\|w\|^2 + \sigma^2}$. For $\|w\| = a$, the terms linear in w, $w^\top(\theta h/\sqrt{n} + \lambda\beta^*)$, are smallest for w antiparallel to $\theta h/\sqrt{n} + \lambda\beta^*$, with value $\approx -a\sqrt{\theta^2/\alpha + \lambda^2 r^2}$, since $\|h\|^2/n \approx 1/\alpha$ and $\langle h, \beta^*\rangle/\sqrt{n} \approx 0$.

4. *The scalar problem.* Exchanging the minimum over the direction of w with the maximum over θ, which the convexity of the limit

allows, and adding $-\theta^2/2$ and $\frac{\lambda}{2}(r^2+a^2)$, the auxiliary problem reduces to $\min_{a\ge 0}\max_{\theta\ge 0}\mathcal{G}(a,\theta)$ with

$$\mathcal{G}(a,\theta)=\theta\sqrt{a^2+\sigma^2}-a\sqrt{\theta^2/\alpha+\lambda^2 r^2}-\frac{\theta^2}{2}+\frac{\lambda}{2}(r^2+a^2). \quad (9.10)$$

5 *Reading the solution.* At the saddle point (a_*,θ_*): $a_*=\|\hat\beta_\lambda-\beta^*\|$, so $R=a_*^2$; $\theta_*^2=n\|v_*\|^2=T$; the optimal direction gives $\langle\hat\beta_\lambda,\beta^*\rangle=r^2-a_*\lambda r^2/\sqrt{\theta_*^2/\alpha+\lambda^2r^2}$; and the component of $\hat\beta_\lambda-\beta^*$ along h has squared length $a_*^2(\theta_*^2/\alpha)/(\theta_*^2/\alpha+\lambda^2r^2)$.

The vector h plays the part of the Gaussian fluctuation of the replicas: the estimator is a multiple of β^* plus an independent Gaussian vector, as the overlaps m and $q-m^2/r^2$ describe. The correspondence is exact.

Proposition 9.9 (Three systems, one solution). *Let $\alpha,\lambda,r,\sigma>0$, let $\chi=\chi_\alpha(\lambda)$ and v be as in Theorem 9.2, and let $\mathcal{G}$ be the function (9.10) on $[0,\infty)^2$.*

(i) The function $\mathcal{G}$ is strictly convex in a and strictly concave in θ, and it has exactly one stationary point (a_*,θ_*) in $(0,\infty)^2$; it is a saddle point, and

$$1+\chi=\frac{\sqrt{a_*^2+\sigma^2}}{\theta_*},\qquad a_*^2=\mathcal{R}(\alpha,\lambda),\qquad \theta_*^2=\mathcal{T}(\alpha,\lambda).$$

(ii) The replica-symmetric equations (9.4) have exactly one solution (m,q,χ) with $\chi>0$. Its χ is $\chi_\alpha(\lambda)$, $q-2m+r^2=\mathcal{R}(\alpha,\lambda)$, and m and $q-m^2/r^2$ are the quantities read off the auxiliary problem in step 5 of Computation 9.2.

In particular the replica prediction, the auxiliary problem and Theorem 9.2 give the same susceptibility, risk and training error.

Proof. The plan is to solve the stationarity equations of $\mathcal{G}$ by introducing χ through the first formula of part 1. Convexity holds term by term: $\theta\sqrt{a^2+\sigma^2}$ is convex in a and linear in θ, $-a\sqrt{\theta^2/\alpha+\lambda^2r^2}$ is linear in a and concave in θ, and $\frac{\lambda}{2}a^2$ and $-\frac{\theta^2}{2}$ are strictly convex and strictly concave. The stationarity equations are

$$\partial_\theta\mathcal{G}=\sqrt{a^2+\sigma^2}-\frac{a\theta/\alpha}{\sqrt{\theta^2/\alpha+\lambda^2r^2}}-\theta=0,$$
$$\partial_a\mathcal{G}=\frac{\theta a}{\sqrt{a^2+\sigma^2}}-\sqrt{\theta^2/\alpha+\lambda^2r^2}+\lambda a=0.$$

Let $(a,\theta)\in(0,\infty)^2$ solve them and define χ by $1+\chi=\sqrt{a^2+\sigma^2}/\theta$. The first equation says $\theta\chi=(a\theta/\alpha)/\sqrt{\theta^2/\alpha+\lambda^2r^2}$, so $\chi>0$ and $\sqrt{\theta^2/\alpha+\lambda^2r^2}=a/(\alpha\chi)$. Dividing the second equation by a and inserting this gives $\frac{1}{1+\chi}-\frac{1}{\alpha\chi}+\lambda=0$, which is (9.5), so $\chi=\chi_\alpha(\lambda)$. Squaring $\sqrt{\theta^2/\alpha+\lambda^2r^2}=$

$a/(\alpha\chi)$, inserting $\theta^2 = (a^2+\sigma^2)/(1+\chi)^2$ and multiplying by $\alpha^2\chi^2$ gives $\nu(a^2+\sigma^2) + r^2(\alpha\lambda\chi)^2 = a^2$, that is, $a^2 = \mathcal{R}(\alpha,\lambda)$, and then $\theta^2 = \mathcal{T}(\alpha,\lambda)$. So there is at most one stationary point. Conversely, with $\chi = \chi_\alpha(\lambda)$, $a_* = \mathcal{R}^{1/2} > 0$ and $\theta_* = \sqrt{a_*^2+\sigma^2}/(1+\chi)$, the computation runs backward and both equations hold; a stationary point of a function convex in a and concave in θ is a saddle point. This proves part 1.

For part 2, Computation 9.1 shows that (9.4) has exactly one solution with $\chi > 0$ (the first equation determines χ, the second $m = r^2(1-\alpha\lambda\chi)$, and the third is linear in q with coefficient $1-\nu \neq 0$) and that $q - 2m + r^2 = \mathcal{R}$. In the auxiliary problem, $\sqrt{\theta_*^2/\alpha + \lambda^2 r^2} = a_*/(\alpha\chi)$ turns the overlap of step 5 into $r^2 - r^2\alpha\lambda\chi = m$, and the squared component along h into $\frac{\theta_*^2}{\alpha}\alpha^2\chi^2 = \nu(a_*^2+\sigma^2) = \mathcal{R} - r^2(\alpha\lambda\chi)^2 = q - m^2/r^2$. ∎

Thrampoulidis et al. (2018, Remark 5.2.2) solve the same two equations for ridge regression with a Gaussian design, with their κ for our χ.[8] By Exercise 9.4, $-\Psi$ at the replica saddle point is also $\alpha\,\mathcal{G}(a_*,\theta_*)$.

[8] Their Remark 5.2.3 notes that the optimally tuned ridge attains the minimum mean squared error when the signal is Gaussian: the Bayes reading of λ^* in Section 9.2.

PROOF SKETCH OF THEOREM 9.2(I) BY THE CGMT. The plan is to localize the minimizer of the auxiliary problem and transfer the localization. Since $\|(X^\top X + n\lambda I)^{-1}X^\top\| \le (4n\lambda)^{-1/2}$ and $\|y\|^2/n \to r^2+\sigma^2$, the minimizer and the maximizing v lie in fixed balls with probability tending to one, so the CGMT applies on compact convex sets. Conditionally on ε, the auxiliary objective converges to $\mathcal{G}$ uniformly on compacts; since $\mathcal{G}$ is strictly convex in a, restricting it to $|\|w\| - a_*| \ge \epsilon$ raises its value by a deterministic gap, and the localization part of the CGMT (Thrampoulidis et al. 2018, Theorem 6.1(iii)) transfers the gap, so that $\|\hat\beta_\lambda - \beta^*\| \to a_*$. Omitted: the uniform convergence and the verification of the localization hypotheses, which Thrampoulidis et al. (2018) carry out for separable convex losses and regularizers (their Theorem 4.1, proved in their Section 6 and Appendix A). ∎

WHAT THE GENERAL THEOREMS ADD. Hastie et al. (2022) prove part (ii) for $\alpha \neq 1$, for the risk averaged over the noise, for designs with i.i.d. entries: with a finite fourth moment for $\alpha > 1$ (their Proposition 2, a result they quote from the random-matrix literature) and a finite moment of order $4+\eta$ for $\alpha < 1$ (their Theorem 1); they also give non-asymptotic deterministic equivalents for general covariance (Theorems 2–3 and 5–6); Cheng and Montanari (2024) give dimension-free ones, and the formula is the kernel learning curve that Canatar et al. (2021) derived by replicas (Chapter 12). Dobriban and Wager (2018, Theorem 2.1) prove that for a random isotropic teacher the optimal ridge is $\sigma^2/(\alpha r^2)$ for every covariance, the rigorous form of Krogh and Hertz's remark. Bartlett et al. (2020, Theorems 1 and 2) characterize, up to constants, when the interpolator's risk tends to zero; the isotropic proportional regime fails their conditions, as part (ii) requires. And Hastie et al. (2022, Theorem 7) prove, for a random isotropic teacher and i.i.d. entries with a finite moment of order $4+\eta$, that leave-one-out cross-validation and *generalized cross-validation* (Golub et al. 1979) estimate the test error uniformly on compact sets of $\lambda > 0$ and so select the optimal

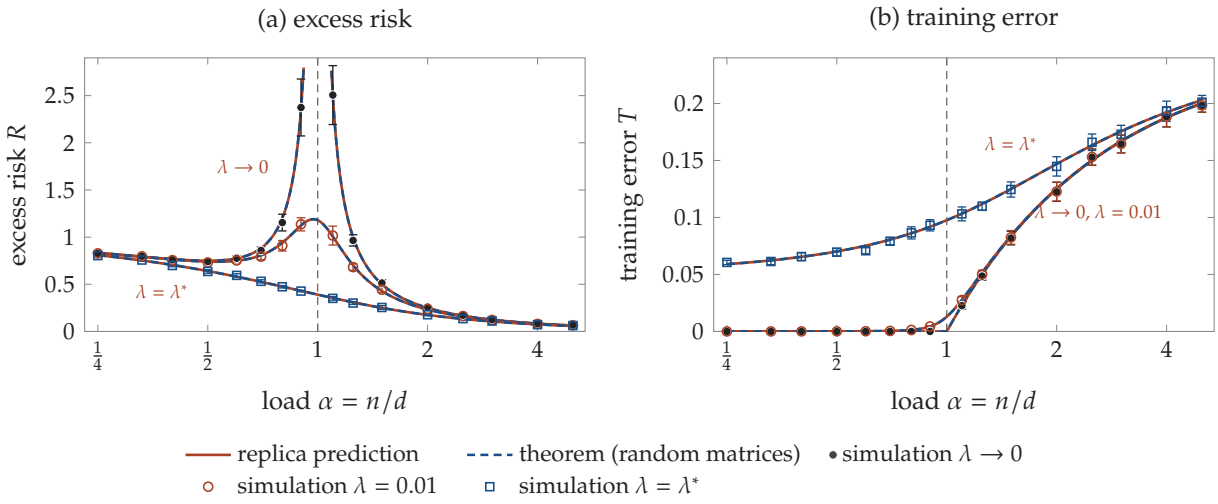


Figure 9.1. Double descent and its cure, with $r^2 = 1$ and $\sigma^2 = 0.25$. Curves: the replica prediction (carnelian) and Theorem 9.2 (lapis dashed), which coincide, for the minimum-norm interpolator ($\lambda \to 0$, divergent at $\alpha = 1$), a small ridge $\lambda = 0.01$ (the middle curve in (a)) and the optimal ridge $\lambda^* = \sigma^2/(\alpha r^2)$. Points: means over 8 independent data sets with $n = 1000$ and $d = \text{round}(n/\alpha)$ at 16 loads, with Student-t 95% intervals (seeds $90000 + 100k + j$ for load k and data set j). All 48 intervals in (a) contain the curves; in (b), 37 of the 40 intervals of positive width do (below $\alpha = 1$ the interpolator's training error is exactly 0), and the three misses are explained in the text. Scripts: `figures/scripts/09-ridge-routes.py` and `09-ridge-sim.py`.

ridge; for GCV this is, in this chapter's language, the replica formula for the training error (Exercise 9.8).

9.6 Measuring it

Three scripts measure the chapter's claims. `09-ridge-routes.py` computes $\mathcal{R}$ at 21 pairs (α, λ) from the replica equations (9.4) by fixed-point iteration, from (9.6), from the Stieltjes form of Remark 9.3, and from the auxiliary problem (9.10) by nested optimization. The first three agree to 6×10^{-14}, the last to 3×10^{-8}, the accuracy of the nested one-dimensional optimizer; by Proposition 9.9 this checks the programs, not the physics.

`09-ridge-sim.py` draws data sets with $n = 1000$ at 16 loads, with the teacher uniform on the unit sphere, noise variance 0.25 and 8 data sets per load, in under two minutes.[9] All 48 risk intervals in Figure 9.1 contain the curves, and 29 of the 32 training-error intervals at $\lambda = 0.01$ and λ^* contain $\mathcal{T}$. The three misses, at $\alpha = 0.6$ and 0.7, are 6–8% low (1.0 to 1.7 half-widths), and they come from the teacher and the noise, not from a bias at finite n: for the same data sets the script also computes $\mathbb{E}[T \mid X]$, in which the direction of the teacher and the noise are averaged exactly, and its mean lies within 0.2% of $\mathcal{T}$ there (within 0.9% at every load). Given X, the training error is a quadratic form in the teacher and the noise whose weights are largest at the smallest eigenvalues of $XX^\top$, so below $\alpha = 1$ it fluctuates by

[9] The estimators are computed by Cholesky solves of the smaller Gram matrix; for $d > n$ the push-through identity gives $\hat{\beta}_\lambda = X^\top(XX^\top + n\lambda I)^{-1}y$.

several percent from one data set to the next; with 32 intervals at 95%, one or two misses are expected, and two of the three share their data sets. The susceptibility, measured as the resolvent trace $\chi_n = \operatorname{tr}(X^\top X + n\lambda^* I)^{-1}$ of Proposition 9.8, is the best-measured quantity: over the 16 loads its relative half-widths are 4×10^{-5} to 4×10^{-3}, and its mean differs from χ by at most 0.3%, mostly upward, a bias of order $1/n$ that the narrow intervals detect at 4 loads. The peak is the hardest place to measure: at $\alpha = 0.9$ and 1.1 the interpolator's risk is 2.37 ± 0.30 and 2.51 ± 0.31, against limits 2.35 and 2.50 and exact finite-n means (Lemma 9.7) 2.37 and 2.53, relative half-widths of 13% and 12% against 2% and 6% for the optimal ridge, because $\operatorname{tr}(X^\top X)^{-1}$ has a heavy tail near $n = d$ (Exercise 9.6).

`09-ridge-universal.py` repeats the simulation at $n = 500$ with independent ± 1 entries and with Gaussian ones (12 data sets, $\alpha \in \{0.5, 2\}$, $\lambda \in \{0, \lambda^*\}$). Seven of the eight risk intervals contain the limits (the miss is Gaussian, 1.4 half-widths high), and the conditional means $\mathbb{E}[R \mid X]$, computable for every design because the teacher is uniform, agree with the limits within 0.7% for both designs (0.6406 ± 0.0001 Gaussian and 0.6403 ± 0.0001 for ± 1 entries at $\alpha = 0.5$ and λ^*, limit 0.6404), although every Gaussian step of our proof fails for ± 1 entries (Exercise 9.9).

9.7 The status board

As of October 2026 every prediction in the physics box is a theorem for isotropic Gaussian designs, and its risk formulas, averaged over the noise, also for designs with i.i.d. entries (Hastie et al. 2022); the replica computation itself remains a heuristic.

STATUS BOARD · replicas ↔ random matrices and the CGMT

PREDICTION	STATUS	PROOF OR EVIDENCE
Replica-symmetric equations and formulas for R, T	heuristic	Section 9.2; finite temperature: Dunmur and Wallace (1993)
Ridge risk and training error for $\lambda > 0$, isotropic Gaussian design	proved	Theorem 9.2(i); risk: Hastie et al. (2022, Cor. 6); random teacher: Dobriban and Wager (2018, Thm. 2.1)
Ridgeless limits, divergence at $\alpha = 1$, exchange of $\lambda \to 0$ and $n \to \infty$	proved	Theorem 9.2(ii); i.i.d. entries, $\alpha \neq 1$: Hastie et al. (2022, Prop. 2, Thm. 1)
Optimal ridge $\lambda^* = \sigma^2/(\alpha r^2)$, risk $\sigma^2\chi$, no peak	proved	Theorem 9.2(iii); every Σ, random teacher: Dobriban and Wager (2018, Thm. 2.1); finite n: Dicker (2016, Prop. 1)
Replica equations = auxiliary-problem equations = resolvent equation	proved	Propositions 9.8 and 9.9; same system: Thrampoulidis et al. (2018, Rem. 5.2.2)

Cross-validation and GCV select the optimal ridge	proved	isotropic design and random isotropic teacher, Hastie et al. (2022, Thm. 7)
The learning curve for general covariance (replicas: Canatar et al. (2021))	proved under	$x = \Sigma^{1/2}z$, z with independent entries, bounded spectrum (Hastie et al. 2022, Thms. 2–3, 5–6); or z with convex concentration (Cheng and Montanari 2024)
Conditions for benign overfitting of the interpolator	proved	up to constants, Bartlett et al. (2020, Thms. 1, 2)
The deep-network peak is a jamming transition	conjectured	numerics, Spigler et al. (2019); Geiger et al. (2019)

9.8 Where the translation fails

WHAT WAS ASSUMED. Both the replica computation and the theorem take isotropic Gaussian inputs, a well-specified linear model and noise independent of the inputs, which for the proof has a finite fourth moment, used only for the variance term. The proof uses Gaussianity in rotation invariance, the Poincaré inequality and the inverse Wishart mean; the comparison route needs it throughout. The exchange of limits is not assumed: part (ii) proves it.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. In the proportional limit nothing in the physics box is wrong; what fails is its reading at finite size and for a single teacher. Self-averaging fails near the threshold: for Gaussian designs and $\sigma > 0$ the interpolator's risk has infinite mean when $|n - d| \le 1$ and infinite variance when $|n - d| \le 3$ (Exercise 9.6). Krogh and Hertz's spectrum-independent optimal decay holds on average over an isotropic teacher, as their derivation and Dobriban and Wager (2018) assume; for a fixed teacher and an anisotropic covariance the optimal ridge depends on how the teacher aligns with the covariance, and it can be zero or negative (Kobak et al. 2020). And the proposal that the interpolation peak of deep networks is a jamming transition (Geiger et al. 2019; Spigler et al. 2019) is a conjecture supported by numerics, which the linear model, whose peak comes from small eigenvalues of $X^\top X/n$, neither confirms nor refutes.

OPEN PROBLEMS. The isotropic model is closed; its neighbors are not. First, real data: Loureiro et al. (2022) conjecture that the errors of estimators trained on generic feature maps of realistic data are those of a Gaussian model with the same covariances (their Conjecture 1); for ridge regression they find it accurate on real data, while for the classification error on real images it fails. Cheng and Montanari (2024) prove the Gaussian-design ridge formula when the whitened features have independent coordinates or satisfy a convex concentration property, hypotheses that real data need not satisfy (Chapter 12). Second, for random-features ridge regression the

exchange of limits proved here in part (ii) was left as a conjecture by Mei and Montanari (2022, Remark 1). Third, a question of our own: here the integer moments $\mathbb{E}Z^k$ are explicit random-matrix integrals, so this is a natural model on which to test whether their continuation to $k \to 0$ reproduces the free entropy (Exercise 9.10); we know of no such computation.

Exercises

9.1 CONSOLIDATION. (The optimal ridge on the Nishimori line.) (a) Prove the Bayes reading of $\hat{\beta}_{\lambda^*}$ in Section 9.2, and deduce $m = q$ from the Nishimori identity. (b) Give a second proof that $\alpha \mapsto \mathcal{R}(\alpha, \lambda^*(\alpha))$ is non-increasing, without computing it. *(Hint: for fixed d the Bayes risk does not increase with n.)*

9.2 CONSOLIDATION. (Facts in the load.) Let $\mathrm{SNR} = r^2/\sigma^2$. Show that least squares ($\alpha > 1$) beats the null estimator $\hat{\beta} = 0$, of risk r^2, exactly when $\alpha > 1 + 1/\mathrm{SNR}$, and the interpolator ($\alpha < 1$) exactly when $\alpha < 1 - 1/\mathrm{SNR}$; and that if $\sigma < r$ the interpolator's risk on $(0, 1)$ is smallest at $\alpha = 1 - \sigma/r$, where it equals $\sigma(2r - \sigma)$.

9.3 CONSOLIDATION. (Finite-size optimality.) For $\beta^* = ru$ with u uniform and independent of X, prove the formula for $\mathbb{E}[R(\hat{\beta}_\lambda) \mid X]$ stated after the proof of Theorem 9.2 and its minimization at $\lambda = \sigma^2 d/(nr^2)$. Why does only the second moment of the prior matter?

9.4 BY HAND. (Energy and auxiliary value.) Show that at the replica saddle point $-\Psi = \frac{\alpha\lambda q}{2} + \frac{a}{2}\mathcal{T}$, the minimum of H/d, and that this equals $\alpha\,\mathcal{G}(a_*, \theta_*)$. *(Hint: use the third equation of (9.4) to eliminate $(q - m^2/r^2)/\chi$.)*

9.5 BY HAND. (Gordon for least squares.) For $\lambda = 0$ and $\alpha > 1$ solve $\min_a \max_\theta \{\theta\sqrt{a^2 + \sigma^2} - a\theta/\sqrt{\alpha} - \theta^2/2\}$ by hand and recover $R = \sigma^2/(\alpha - 1)$ and $T = \sigma^2(1 - 1/\alpha)$. What goes wrong for $\alpha < 1$?

9.6 BY HAND. ★ (Heavy tails at the threshold.) Let $\sigma > 0$ and the noise be Gaussian. Show that $\mathbb{E}R(\hat{\beta}_0) < \infty$ exactly when $|n - d| \geq 2$, and $\mathrm{Var}\, R(\hat{\beta}_0) < \infty$ exactly when $|n - d| \geq 4$. *(Hint: let W be the smaller of $X^\top X$ and $XX^\top$, of size $k = \min(n, d)$; $\mathbb{E}[(\chi^2_m)^{-2}] = 1/((m-2)(m-4))$ for $m > 4$ and is infinite for $m \leq 4$; use $(\mathrm{tr}\, W^{-1})^2 \geq (W^{-1})^2_{11}$ and, for the converse, $\mathrm{tr}\, W^{-2} \leq (\mathrm{tr}\, W^{-1})^2 \leq k \sum_j (W^{-1})^2_{jj}$.)*

9.7 TRANSLATION. (The susceptibility of a Gaussian Gibbs measure.) Show that the Gibbs measure of (9.2) at inverse temperature b is $\mathcal{N}(\sqrt{d}\,\hat{\beta}_\lambda, (bH'')^{-1})$ with $H'' = X^\top X/d + \alpha\lambda I$, so that $b(\langle Q \rangle - \langle q \rangle) = \mathrm{tr}(X^\top X + n\lambda I)^{-1}$ exactly. Which replica steps does this make unnecessary?

9.8 TRANSLATION. (The training error is generalized cross-validation.) Let $H_\lambda = X(X^\top X + n\lambda I)^{-1}X^\top$. Show that $\mathrm{tr}\, H_\lambda/n = \frac{d}{n} - \lambda\chi_n \to \chi/(1 + \chi)$,

and deduce that $\mathrm{GCV}(\lambda) = T(\hat{\beta}_\lambda)/(1-\mathrm{tr}\, H_\lambda/n)^2 \to \mathcal{R}(\alpha,\lambda)+\sigma^2$. Which replica formula is this?

9.9 MEASUREMENT. (Universality.) Run `09-ridge-universal.py` and compare both designs with the limits. Why does the conditional mean of Exercise 9.3, valid for every design, fluctuate far less than the risk? Which steps of the proof of Theorem 9.2 fail for ±1 entries?

9.10 RESEARCH. (The replica trick on a solvable model.) For the energy (9.2), $Z = e^{-b\min H}(2\pi/b)^{d/2}\det(H'')^{-1/2}$. Compute $\lim_d \frac{1}{dk}\log \mathbb{E}Z^k$ for integer k, a large-deviation problem for $(\min H/d, \frac{1}{d}\log\det H'')$, and decide whether its continuation to $k \to 0$ gives $\lim_d \frac{1}{d}\mathbb{E}\log Z$.

Notes and References

Ridge regression is due to Hoerl and Kennard (1970) and generalized cross-validation to Golub et al. (1979). The prehistory of double descent, from the pseudo-inverse rule of 1989–1990, is told by Loog et al. (2020), its name is from Belkin et al. (2019), and Mei and Montanari (2022) proved it for random-features regression (Chapter 12).

The rigorous theory of the isotropic case was built in steps: Marčenko and Pastur (1967) for the spectrum, Dicker (2016) for the optimally tuned risk, Dobriban and Wager (2018) for general covariance with a random teacher, and Hastie et al. (2022) for the ridgeless limits, general covariance and cross-validation. Gordon's comparison (Gordon 1988) gives the second proof through the CGMT. The leave-one-out method, the rigorous form of the cavity method, was used by El Karoui (2013) for ridge-regularized robust regression.

The replica derivation of Section 9.2 is the standard zero-temperature computation, and the proof of Theorem 9.2 is assembled from standard tools: the leave-one-out proof of Proposition 9.8 is a version of the usual Schur-complement route to the Marchenko–Pastur law, and the treatment of the ridgeless limit by a monotone comparison and the exact inverse Wishart mean is, as far as we know, the book's own. Proposition 9.9 and the dictionary of the stone are the book's formulation; for the Gaussian covariate model, which contains ridge regression, Loureiro et al. (2022, Appendix C) showed that the replica and Gordon equations coincide.

TRANSLATING THE LITERATURE. Result numbers follow the journals for Hastie et al. (2022) (whose arXiv v5 numbers the isotropic ridge corollary 5) and Bartlett et al. (2020) (whose arXiv v3 numbers Theorems 1 and 2 as 4 and 6), and the arXiv versions for the papers of Dobriban and Wager (v2), Thrampoulidis et al. (v1), Loureiro et al. (v3) and Mei and Montanari (v5), whose journal numbering we could not check. Hastie et al. (2022) write p for our d and $\gamma = p/n = 1/\alpha$; their objective $\frac{1}{n}\|y - Xb\|^2 + \lambda\|b\|^2$ has our minimizer, their $m(z)$ is the Marchenko–Pastur Stieltjes transform with $m(-\lambda) = \alpha\chi$, and in their Theorem 7 the letter α means $r^2/(\sigma^2\gamma)$. Dobriban and Wager (2018) use $\gamma = p/n$, our λ, noise variance 1, signal strength α^2 for r^2/σ^2, a risk that includes the noise, and a companion transform

v with $\lambda v(-\lambda) = 1/(1+\chi)$. Dicker (2016) writes $(X^\top X + \frac{d}{t^2} I)^{-1} X^\top y$, so that $\lambda = d/(nt^2)$. Thrampoulidis et al. (2018) have m measurements of n unknowns with $A_{ij} \sim \mathcal{N}(0, 1/n)$, so their δ is our α and their x our $w = \sqrt{d}\,\beta$; their ridge parameter is $\alpha\lambda$, their κ our χ, their α our $\sqrt{R}$, and their σ_x^2, σ_z^2 our r^2, σ^2. Krogh and Hertz (1991) write N inputs, $p = \alpha N$ patterns, the decay $\alpha\lambda$, σ_u^2 (their u^2) for r^2, F for R, and A for $X^\top X/d$. Canatar et al. (2021) write κ for $n\lambda_{\text{eff}} = n\lambda(1+\chi)$.

References

Bartlett, P. L., P. M. Long, G. Lugosi, and A. Tsigler (2020). "Benign overfitting in linear regression". In: *Proceedings of the National Academy of Sciences* 117.48, pp. 30063–30070. arXiv: 1906.11300.

Belkin, M., D. Hsu, S. Ma, and S. Mandal (2019). "Reconciling modern machine-learning practice and the classical bias–variance trade-off". In: *Proceedings of the National Academy of Sciences* 116.32, pp. 15849–15854. arXiv: 1812.11118.

Canatar, A., B. Bordelon, and C. Pehlevan (2021). "Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks". In: *Nature Communications* 12, p. 2914. arXiv: 2006.13198.

Cheng, C. and A. Montanari (2024). "Dimension free ridge regression". In: *The Annals of Statistics* 52.6, pp. 2879–2912. arXiv: 2210.08571.

Dicker, L. H. (2016). "Ridge regression and asymptotic minimax estimation over spheres of growing dimension". In: *Bernoulli* 22.1, pp. 1–37. arXiv: 1601.03900.

Dobriban, E. and S. Wager (2018). "High-dimensional asymptotics of prediction: Ridge regression and classification". In: *The Annals of Statistics* 46.1, pp. 247–279. arXiv: 1507.03003.

Dunmur, A. P. and D. J. Wallace (1993). "Learning and generalization in a linear perceptron stochastically trained with noisy data". In: *Journal of Physics A: Mathematical and General* 26.21, pp. 5767–5779.

El Karoui, N. (2013). *Asymptotic behavior of unregularized and ridge-regularized high-dimensional robust regression estimators: rigorous results*. arXiv: 1311.2445.

Geiger, M., S. Spigler, S. d'Ascoli, L. Sagun, M. Baity-Jesi, G. Biroli, and M. Wyart (2019). "Jamming transition as a paradigm to understand the loss landscape of deep neural networks". In: *Physical Review E* 100.1, p. 012115.

Golub, G. H., M. Heath, and G. Wahba (1979). "Generalized Cross-Validation as a Method for Choosing a Good Ridge Parameter". In: *Technometrics* 21.2, pp. 215–223.

Gordon, Y. (1988). "On Milman's inequality and random subspaces which escape through a mesh in $\mathbb{R}^n$". In: *Geometric Aspects of Functional Analysis (1986/87)*. Vol. 1317. Lecture Notes in Mathematics. Springer, pp. 84–106.

Hastie, T., A. Montanari, S. Rosset, and R. J. Tibshirani (2022). "Surprises in high-dimensional ridgeless least squares interpolation". In: *The Annals of Statistics* 50.2, pp. 949–986. arXiv: 1903.08560.

Hertz, J. A., A. Krogh, and G. I. Thorbergsson (1989). "Phase transitions in simple learning". In: *Journal of Physics A: Mathematical and General* 22.12, pp. 2133–2150.

Hoerl, A. E. and R. W. Kennard (1970). "Ridge Regression: Biased Estimation for Nonorthogonal Problems". In: *Technometrics* 12.1, pp. 55–67.

Kobak, D., J. Lomond, and B. Sanchez (2020). "Optimal ridge penalty for real-world high-dimensional data can be zero or negative due to the implicit ridge

regularization". In: *Journal of Machine Learning Research* 21.169, pp. 1–16. arXiv: 1805.10939.

Krogh, A. and J. A. Hertz (1991). "A Simple Weight Decay Can Improve Generalization". In: *Advances in Neural Information Processing Systems*. Vol. 4, pp. 950–957.

Krogh, A. and J. A. Hertz (1992). "Generalization in a linear perceptron in the presence of noise". In: *Journal of Physics A: Mathematical and General* 25.5, pp. 1135–1147.

Loog, M., T. Viering, A. Mey, J. H. Krijthe, and D. M. J. Tax (2020). "A brief prehistory of double descent". In: *Proceedings of the National Academy of Sciences* 117.20, pp. 10625–10626. arXiv: 2004.04328.

Loureiro, B., C. Gerbelot, H. Cui, S. Goldt, F. Krzakala, M. Mézard, and L. Zdeborová (2022). "Learning curves of generic features maps for realistic datasets with a teacher-student model". In: *Journal of Statistical Mechanics: Theory and Experiment* 2022.11, p. 114001. arXiv: 2102.08127.

Marčenko, V. A. and L. A. Pastur (1967). "Distribution of eigenvalues for some sets of random matrices". In: *Mathematics of the USSR-Sbornik* 1.4, pp. 457–483.

Mei, S. and A. Montanari (2022). "The Generalization Error of Random Features Regression: Precise Asymptotics and the Double Descent Curve". In: *Communications on Pure and Applied Mathematics* 75.4, pp. 667–766. arXiv: 1908.05355.

Nakkiran, P. (2019). *More Data Can Hurt for Linear Regression: Sample-wise Double Descent*. arXiv: 1912.07242.

Spigler, S., M. Geiger, S. d'Ascoli, L. Sagun, G. Biroli, and M. Wyart (2019). "A jamming transition from under- to over-parametrization affects generalization in deep learning". In: *Journal of Physics A: Mathematical and Theoretical* 52.47, p. 474001.

Stojnic, M. (2013). *A framework to characterize performance of LASSO algorithms*. arXiv: 1303.7291.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). "Precise Error Analysis of Regularized M-Estimators in High Dimensions". In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Thrampoulidis, C., S. Oymak, and B. Hassibi (2015). "Regularized Linear Regression: A Precise Analysis of the Estimation Error". In: *Proceedings of The 28th Conference on Learning Theory*. Vol. 40. Proceedings of Machine Learning Research, pp. 1683–1709.

The LASSO and Robust Regression

10

Four routes to one fixed point

THE DICTIONARY —— the replica fixed point ↔ AMP and the CGMT

IN THIS CHAPTER. When a sparse vector is estimated from Gaussian linear measurements whose number is proportional to the dimension, the LASSO behaves, coordinate by coordinate, like soft thresholding of the truth seen in Gaussian noise, with a noise level and a threshold fixed by two scalar equations. Physicists derived this fixed point with zero-temperature replicas; Bayati and Montanari proved it with approximate message passing, and Gordon's comparison proves it again, through an auxiliary problem whose scalar form is the replica-symmetric potential after a change of variables. We prove the limit of the LASSO's error by the convex Gaussian min-max theorem and the noiseless Donoho–Tanner threshold by computing a statistical dimension, and we show that the replica, state-evolution and geometric formulas for that threshold are one function. Simulations of the LASSO, of AMP, of ℓ_1 minimization and of Huber regression agree with every prediction; the open questions concern non-Gaussian designs, non-convex penalties and finite-size corrections.

10.1 The question: what does sparsity buy?

A statistician with n observations and $d > n$ candidate predictors cannot fit least squares: the minimum-norm interpolant of Chapter 9 keeps only the fraction n/d of the signal. If few coefficients are nonzero, the *LASSO* of Tibshirani (1996) adds an ℓ_1 penalty and sets many coordinates exactly to zero,[1] and its noiseless limit, ℓ_1 minimization or basis pursuit (Chen et al. 1998), can recover a sparse vector from fewer measurements than unknowns. How many measurements are needed, and how large is the error when they are noisy, if n and d are large and comparable?

We use the scaling of Bayati and Montanari (2012), in which the unknown has entries of order one. The design $A \in \mathbb{R}^{n\times d}$ has independent $\mathcal{N}(0, 1/n)$ entries, so $A = X/\sqrt{n}$ for the design X of Chapter 9; the unknown is $\theta_0 = \sqrt{n}\,\beta^*$; the noise w has independent $\mathcal{N}(0, \sigma^2)$ entries (other chapters write it $\sigma\varepsilon$; here ε is a sparsity); $y = A\theta_0 + w$ and $\alpha = n/d$. The LASSO is

$$\hat\theta \in \arg\min_{\theta\in\mathbb{R}^d} F(\theta), \qquad F(\theta) = \tfrac{1}{2}\|y - A\theta\|^2 + \lambda\|\theta\|_1, \quad \lambda > 0, \tag{10.1}$$

and its error is $\mathrm{MSE} = \|\hat\theta - \theta_0\|^2/d$. Since $\hat\theta = \sqrt{n}\,\hat\beta$, the excess risk of the other regression chapters is $R = \|\hat\beta - \beta^*\|^2 = \mathrm{MSE}/\alpha$; this is the only

[1] Tibshirani wrote the penalty as a constraint, $\|\beta\|_1 \le t$; by convex duality the constrained and penalized forms trace the same path of solutions.

translation the chapter needs.[2] The entries of θ_0 are independent with law P_{Θ_0}, and $\varepsilon = \mathbb{P}(\Theta_0 \neq 0)$. The running example is the Bernoulli–Gaussian signal $(1-\varepsilon)\delta_0 + \varepsilon\mathcal{N}(0,1)$ with $\varepsilon = 0.1$, at $\alpha = 0.5$ and $\sigma^2 = 0.05$.

Soft thresholding, $\eta(u;\zeta) = \operatorname{sgn}(u)(|u|-\zeta)_+$, minimizes $\frac{1}{2}(x-u)^2 + \zeta|x|$: for $x \neq 0$ the condition $x - u + \zeta \operatorname{sgn} x = 0$ has a solution exactly when $|u| > \zeta$. If the columns of A were orthonormal, which needs $n \geq d$, the LASSO would be $\eta(A^\top y;\lambda)$ coordinatewise, and $A^\top y = \theta_0 + A^\top w$ is the truth plus Gaussian noise of variance σ^2. The physical prediction is that this picture survives at $n < d$, once the noise and the threshold are renormalized.

[2] In the per-sample normalization $\frac{1}{2n}\|y - X\beta\|^2 + \lambda'\|\beta\|_1$ of Chapter 9 the same estimator has $\lambda' = \lambda/\sqrt{n}$, which tends to zero; hence the scaling used here.

THE PHYSICAL PREDICTION · replicas and cavities, 2009–2016. Let $n, d \to \infty$ with $n/d \to \alpha$. (i) The empirical distribution of the pairs $(\hat\theta_j, \theta_{0j})$ tends to the law of $(\eta(\Theta_0 + \tau Z;\zeta), \Theta_0)$, with $Z \sim \mathcal{N}(0,1)$ independent of Θ_0, where $\tau > 0$ and $\zeta > 0$ solve

$$\tau^2 = \sigma^2 + \frac{1}{\alpha}\mathbb{E}\big[(\eta(\Theta_0+\tau Z;\zeta)-\Theta_0)^2\big], \qquad \lambda = \zeta\Big(1 - \frac{\mathbb{P}(|\Theta_0 + \tau Z| > \zeta)}{\alpha}\Big). \tag{10.2}$$

So MSE $\to \alpha(\tau^2 - \sigma^2)$, and the fraction of nonzero coordinates of $\hat\theta$ tends to $\mathbb{P}(|\Theta_0 + \tau Z| > \zeta)$. (ii) Without noise, ℓ_1 minimization recovers θ_0 when $\alpha > \psi(\varepsilon)$ and fails when $\alpha < \psi(\varepsilon)$, where $D(t) = \mathbb{E}(|Z| - t)_+^2$ and

$$\psi(\varepsilon) = \min_{t \geq 0} J_\varepsilon(t), \qquad J_\varepsilon(t) = \varepsilon(1+t^2) + (1-\varepsilon)D(t). \tag{10.3}$$

(iii) The M-estimator $\arg\min_\theta \sum_i \rho(y_i - \langle a_i\rangle\theta)$, with a convex loss ρ, the rows a_i of A, noise $w_i \overset{\text{iid}}{\sim} P_W$ and $\alpha > 1$, has $R \to r^2$, where, with $\hat z = W + rZ$, $W \sim P_W$ independent of Z, and $\operatorname{prox}_c(x) = \arg\min_u\{c\rho(u) + \frac{1}{2}(u-x)^2\}$,

$$\mathbb{E}\big[\operatorname{prox}_c'(\hat z)\big] = 1 - \frac{1}{\alpha}, \qquad \frac{r^2}{\alpha} = \mathbb{E}\big[(\hat z - \operatorname{prox}_c(\hat z))^2\big]. \tag{10.4}$$

Prediction (i), the *decoupling principle*, is the replica-symmetric result of Rangan et al. (2012); (10.2) is the fixed point of Bayati and Montanari (2012). Prediction (ii) was derived by replicas (Kabashima et al. 2009) and by state evolution (Donoho et al. 2009), and earlier, as a bound, from the faces of random polytopes (Donoho and Tanner 2009; Donoho 2006); the curve is the *Donoho–Tanner phase transition*, plotted in the coordinates $(\delta, \rho) = (\alpha, \varepsilon/\alpha)$. Prediction (iii) concerns the *M-estimators* of robust regression (Huber 1964); Huber (1973) already let the number of parameters grow with n, and the system was derived by a probabilistic heuristic (El Karoui et al. 2013) and by replicas (Advani and Ganguli 2016). It says that the estimator sees the noise inflated by an independent Gaussian of variance r^2; for least squares it gives $R = \sigma^2/(\alpha - 1)$, which a robust loss can beat under heavy tails.

How do physicists derive the fixed point, and what is proved? Every item of the box is a theorem for Gaussian designs; we prove the error of the LASSO and the noiseless threshold in full. Four routes reach the same

fixed point: replicas, the state evolution of approximate message passing, Gordon's comparison and, without noise, the geometry of convex cones.

10.2 The physics: replicas at zero temperature

The derivation is the standard one, in this book's normalization: Kabashima et al. (2009) gave it for noiseless ℓ_p reconstruction (their equation (7) is the noiseless form of (10.5) below), Rangan et al. (2012) with noise, Advani and Ganguli (2016) for general convex losses and penalties. The replica trick is introduced in Chapter 1, and its zero-temperature version for ridge in Chapter 9.

THE REPLICATED LASSO. *Step 1: the design.* Let $Z_\beta = \int e^{-\beta F(\theta)}\,\mathrm{d}\theta$; as $\beta\to\infty$ the Gibbs measure $e^{-\beta F}/Z_\beta$ concentrates on the minimizers, and $e_0 = \lim_d \frac{1}{d}\min F$ is the limit of $-\frac{1}{\beta d}\mathbb{E}\log Z_\beta$, computed from the moments $\mathbb{E}Z_\beta^k$ continued to $k\to 0$. With replicas $\theta^1,\dots,\theta^k$ and $v^a = \theta^a - \theta_0$, the identity $y - A\theta^a = w - Av^a$ shows that the design enters only through the fields $u_i^a = \langle a_i, v^a\rangle$, independent over i and centered Gaussian with covariance $\langle v^a, v^b\rangle/n = (q_{ab} - m_a - m_b + \mathbb{E}\Theta_0^2)/\alpha$, where $q_{ab} = \langle\theta^a,\theta^b\rangle/d$ and $m_a = \langle\theta^a,\theta_0\rangle/d$ are the overlaps.

Step 2: replica symmetry. Assume $q_{aa} = Q$, $q_{ab} = q$ $(a\neq b)$ and $m_a = m$. Then $u_i^a = \sqrt{\omega}\,\xi_0 + \sqrt{(Q-q)/\alpha}\,\xi_a$ with independent standard Gaussians ξ_a and $\omega = (q - 2m + \mathbb{E}\Theta_0^2)/\alpha$, and with $s = w_i - \sqrt{\omega}\,\xi_0 \sim \mathcal{N}(0,\sigma^2+\omega)$ and $\Delta_\beta = \beta(Q-q)/\alpha$ a Gaussian integral gives the rows' contribution, per coordinate,

$$\alpha\log\mathbb{E}_s\Big[\big(\mathbb{E}_\xi e^{-\beta(s-\sqrt{(Q-q)/\alpha}\,\xi)^2/2}\big)^k\Big] = -\frac{\alpha k}{2}\log(1+\Delta_\beta) - \frac{\alpha}{2}\log\Big(1+\frac{k\beta(\sigma^2+\omega)}{1+\Delta_\beta}\Big).$$

Step 3: the entropic term. The volume of replica configurations with these overlaps, weighted by $e^{-\beta\lambda\sum_a\|\theta^a\|_1}$, is written with conjugate variables scaled to survive $\beta\to\infty$: $\beta\hat{Q}/2$ for the norms, $\beta^2\hat{\chi}/2$ for the off-diagonal overlaps, $\beta\hat{m}$ for m. The identity $e^{\beta^2\hat{\chi}(\sum_a x_a)^2/2} = \mathbb{E}_z e^{\beta\sqrt{\hat{\chi}}z\sum_a x_a}$ decouples replicas and coordinates, and each coordinate sees the single-site weight $e^{-\beta[\hat{Q}x^2/2 - x(\hat{m}\Theta_0 + \sqrt{\hat{\chi}}Z) + \lambda|x|]}$.

Step 4: zero temperature. Let $\beta\to\infty$ with $\chi = \beta(Q-q)$ fixed and $k\to 0$, keep the terms of order βk, and use Laplace's method (Theorem A.11). With $\mathcal{E} = Q - 2m + \mathbb{E}\Theta_0^2$, the mean squared error,

$$e_0 = \operatorname{extr}\Big\{ -\frac{\hat{Q}Q}{2} + \frac{\hat{\chi}\chi}{2} + \hat{m}m + \frac{\alpha\sigma^2+\mathcal{E}}{2(1+\chi/\alpha)} + \mathbb{E}\min_{x\in\mathbb{R}}\Big[\frac{\hat{Q}}{2}x^2 - x\big(\hat{m}\Theta_0 + \sqrt{\hat{\chi}}Z\big) + \lambda|x|\Big]\Big\}, \tag{10.5}$$

the extremum being over $Q,\chi,m,\hat{Q},\hat{\chi},\hat{m}$: the fraction is the residual energy, the minimum the entropic term of one coordinate. ◇

COMPUTATION 10.1 · THE SADDLE POINT OF THE REPLICA POTENTIAL

1 *Fields.* The potential is affine in Q and m; stationarity gives $\hat{Q} = \hat{m} = \alpha/(\alpha+\chi)$. Stationarity in χ gives $\hat{\chi} = \alpha(\alpha\sigma^2+\mathcal{E})/(\alpha+\chi)^2$.

2 *Single site.* As $\hat{m} = \hat{Q}$ and $\eta(cu; c\zeta) = c\,\eta(u;\zeta)$ for $c > 0$, the minimizer is $\hat{x} = \eta(\Theta_0 + \tau Z; \zeta)$ with $\tau = \sqrt{\hat{\chi}}/\hat{Q}$ and $\zeta = \lambda/\hat{Q}$.

3 *Order parameters.* The derivative of a minimum is that at the minimizer, so stationarity in $\hat{Q}$, $\hat{m}$ gives $Q = \mathbb{E}\hat{x}^2$, $m = \mathbb{E}\hat{x}\Theta_0$, hence $\mathcal{E} = \mathbb{E}(\hat{x} - \Theta_0)^2$, and step 1 gives $\tau^2 = \hat{\chi}/\hat{Q}^2 = \sigma^2 + \mathcal{E}/\alpha$: the first equation of (10.2).

4 *Susceptibility.* Stationarity in $\hat{\chi}$ gives $\chi = \mathbb{E}[\hat{x}Z]/\sqrt{\hat{\chi}} = s/\hat{Q}$, with $s = \mathbb{P}(|\Theta_0 + \tau Z| > \zeta)$, by Gaussian integration by parts (Theorem A.1); with step 1, $\hat{Q} = 1 - s/\alpha$.

5 *Calibration.* So $\lambda = \zeta\hat{Q} = \zeta(1 - s/\alpha)$, the second equation, and MSE $= \mathcal{E} = \alpha(\tau^2 - \sigma^2)$. The same computation gives the training error $(\sigma^2 + \mathcal{E}/\alpha)/(1 + \chi/\alpha)^2 = (\tau\hat{Q})^2$.

This is prediction (i). Steps 1 and 4 of the computation also give $\chi = s/\hat{Q} = \alpha s/(\alpha - s)$: the susceptibility diverges as the fraction s of selected coordinates approaches α, that is, as the number of nonzero coordinates of $\hat{\theta}$ approaches the number of measurements. The prediction is reached through five uncontrolled steps: the continuation to $k \to 0$ (Chapter 1); the order of the limits $d \to \infty$, $k \to 0$, $\beta \to \infty$; the *replica-symmetric ansatz*, supported by the convexity of F and checked by the local stability of the solution, the *de Almeida–Thouless condition*, which Kabashima et al. (2009) verified for noiseless ℓ_1 recovery; the choice of extremum hidden in "extr", since the conjugate integrals run along imaginary axes; and the concentration of the overlaps.

THE NOISELESS LIMIT. Take $\sigma = 0$, $\lambda \to 0$ and $\zeta = t\tau$. Exact recovery needs the map $\tau^2 \mapsto \alpha^{-1}\mathbb{E}(\eta(\Theta_0 + \tau Z; t\tau) - \Theta_0)^2$ to contract near 0. As $\tau \to 0$, $(\eta(\theta + \tau Z; t\tau) - \theta)/\tau \to Z - t\,\operatorname{sgn}\theta$ for $\theta \neq 0$, and $\eta(\tau Z; t\tau) = \tau\eta(Z; t)$; so the slope of the map at 0 is $J_\varepsilon(t)/\alpha$ for every law with $\mathbb{P}(\Theta_0 \neq 0) = \varepsilon$, and zero error is stable for some t exactly when $\min_t J_\varepsilon < \alpha$: prediction (ii), as Donoho et al. (2009) derived it. For very sparse signals $\psi(\varepsilon) \sim 2\varepsilon\log(1/\varepsilon)$ (Exercise 10.5): about $2k\log(d/k)$ measurements recover k nonzero entries. Kabashima et al. (2009) found the curve as the stability boundary of their "success solution".[3]

ROBUST REGRESSION. For $F(\theta) = \sum_i \rho(y_i - \langle a_i\rangle\theta)$, step 2 yields the Moreau envelope $\min_u\{(u-s)^2/(2c) + \rho(u)\}$, $c = \chi/\alpha$,[4] and the saddle point is (10.4) with $r^2 = \mathcal{E}/\alpha$ (Exercise 10.4): the susceptibility is the proximal step size. For $\rho(u) = u^2/2$ it gives the ridgeless risk $\sigma^2/(\alpha - 1)$ of Theorem 9.2. For Huber's loss, $\frac{1}{2}u^2$ for $|u| \le k$ and $k|u| - \frac{1}{2}k^2$ otherwise, $\operatorname{prox}_c(x)$ is $x/(1+c)$ for $|x| \le k(1+c)$ and $x - ck\operatorname{sgn}x$ otherwise.

[3] After submission they noticed that it coincides with the weak threshold of Donoho and Tanner's polytope computations, and added a note.

[4] By Laplace's method, $-\beta^{-1}$ log of the convolution of $e^{-\beta\rho}$ with a Gaussian of variance c/β tends to the Moreau envelope as $\beta \to \infty$.

10.3 The stone

The rigorous route closest to the replica computation is Gordon's comparison in its convex form (Chapter 8); the proof of Theorem 10.5 follows the rows.

PHYSICS · *the replica computation*	PROOF · *Gordon's comparison (the CGMT)*
1 · Replicate, or dualize	
Write $\min F$ as the ground state of $e^{-\beta F}$; compute $\mathbb{E}Z_\beta^k$ for k replicas; $\beta \to \infty$ at the end.	Write $\frac{1}{2}\|r\|^2 = \max_u\{u^\top r - \frac{1}{2}\|u\|^2\}$: the LASSO is a min–max of a bilinear form in a Gaussian matrix. No replicas, no temperature: nothing corresponds to $k \to 0$.
2 · Average over the design	
The fields $\langle a_i, \theta^a - \theta_0\rangle$ are Gaussian with covariance given by the overlaps.	Gordon's comparison replaces $u^\top G\tilde{v}$ by $\|\tilde{v}\|g^\top u + \|u\|h^\top\tilde{v}$ with independent Gaussian vectors g, h (Theorem 8.3).
3 · The energetic term	
A Gaussian integral over the fields gives the residual energy $(\alpha\sigma^2 + \mathcal{E})/(2(1+\chi/\alpha))$.	Maximizing over the direction of u leaves $b = \|u\|$, the residual's norm over $\sqrt{n}$, and the term $b\sqrt{\|v\|^2/n + \sigma^2}$.
4 · The entropic term	
A Gaussian field $\sqrt{\hat{\chi}}z$ decouples the coordinates; each solves $\min_x \frac{\hat{Q}}{2}x^2 - x(\hat{m}\theta_{0j} + \sqrt{\hat{\chi}}z) + \lambda\|x\|$.	The identity $\sqrt{x} = \min_{\tau>0}\{\frac{x}{2\tau} + \frac{\tau}{2}\}$ separates the coordinates; each solves $\min_x \frac{b}{2\tau}(x-\theta_{0j})^2 - bh_j(x-\theta_{0j}) + \lambda\|x\|$: the same, with $\hat{Q} = \hat{m} = b/\tau$, $\sqrt{\hat{\chi}} = b$, $z = h_j$.
5 · The saddle point	
Extremize over six order parameters (Computation 10.1); the method does not say which extremum.	Eliminating Q and m turns the potential into $\alpha\Psi(b,\tau)$ (Proposition 10.6); the auxiliary problem tends to $\max_b \min_\tau \Psi$, attained at (b_*, τ_*) only (Lemma 10.3).
6 · Replica symmetry	
Assume one dominant overlap, supported by convexity and checked by stability.	No ansatz. Convexity gives the matching upper bound (part (ii) of the CGMT) and, as strong convexity, a gap at every v of the wrong norm: "the overlap concentrates".

7 · The decoupling principle

The cavity field of coordinate j is θ_{0j} plus Gaussian noise of variance τ^2, for every statistic.

AMP produces such fields with an exact state evolution (Theorem 6.4), and its iterates converge to the LASSO (Theorem 10.1).

8 · The noiseless limit

The success solution $\tau = 0$ is stable exactly when $\alpha > \psi(\varepsilon)$.

The ℓ_1 descent cone has statistical dimension $d\psi(s/d)$ up to $2\sqrt{d/s}$ (Theorem 10.7); the kinematic formula gives the threshold (Corollary 10.8).

The proof does not merely confirm the replica answer; it reproduces the replica potential, and gives its conjugate fields meanings: $\sqrt{\hat{\chi}}$ is the residual's norm over $\sqrt{n}$, $1/\hat{Q}$ a proximal step size, the Hubbard–Stratonovich field Gordon's vector h. The continuation in k and the limit $\beta \to \infty$ disappear because the comparison is made at zero temperature, and convexity decides the extremum. The routes part where convexity or Gaussianity fails: for ℓ_0 the replica computation gives an unstable answer and Gordon's comparison one inequality, and for non-Gaussian designs the comparison has no counterpart.

10.4 The proof: the LASSO risk by Gordon's comparison

The first proof analyzed an algorithm. Approximate message passing iterates $x^{t+1} = \eta(x^t + A^\top z^t; \zeta_t)$, $z^t = y - Ax^t + \alpha^{-1} z^{t-1} \langle \eta' \rangle$ (Chapter 6); by state evolution (Theorem 6.4; Bayati and Montanari (2011)) the field $x^t + A^\top z^t$ behaves like $\theta_0 + \tau_t Z$, with τ_{t+1}^2 the right side of the first equation of (10.2) at τ_t, and Bayati and Montanari proved that the iterates converge to the LASSO.

Theorem 10.1 (The LASSO risk; Bayati and Montanari 2012, Theorem 1.5 and Corollary 1.6). *Let $n/d \to \alpha \in (0, \infty)$, let A have independent $\mathcal{N}(0, 1/n)$ entries, and let $\theta_0 \in \mathbb{R}^d$ and $w \in \mathbb{R}^n$ be deterministic, with empirical distributions converging weakly, with their second moments, to laws P_{Θ_0} and P_W such that $\mathbb{P}(\Theta_0 \neq 0) > 0$ and $\sigma^2 = \mathbb{E}W^2 > 0$. Let $\lambda > 0$. Then (10.2) has a unique solution (τ_*, ζ_*) with $\tau_*, \zeta_* > 0$, and for every $\varphi : \mathbb{R}^2 \to \mathbb{R}$ with $|\varphi(a) - \varphi(b)| \le L(1 + \|a\| + \|b\|)\|a - b\|$, almost surely,*

$$\lim_{d\to\infty} \frac{1}{d} \sum_{j=1}^{d} \varphi(\hat{\theta}_j, \theta_{0j}) = \mathbb{E}\, \varphi\big(\eta(\Theta_0 + \tau_* Z; \zeta_*), \Theta_0\big);$$

in particular $\|\hat{\theta} - \theta_0\|^2/d \to \alpha(\tau_*^2 - \sigma^2)$ almost surely.

The noise need not be Gaussian; Bayati and Montanari remark that the

Gaussian design is necessary for their technique. We prove the statement about the error by the second route, under two simplifications that keep the idea.[5]

[5] For designs with a continuous distribution the LASSO minimizer is unique almost surely (Tibshirani 2013); the proof below does not need this, because it controls all minimizers at once.

Assumption 10.2 (Gaussian design and noise, independent signal). As $d \to \infty$, $n = n(d)$ with $n/d \to \alpha \in (0, \infty)$; A has independent $\mathcal{N}(0, 1/n)$ entries; w has independent $\mathcal{N}(0, \sigma^2)$ entries, $\sigma > 0$; θ_0 has independent entries with a law P_{Θ_0}, $\mathbb{E}\Theta_0^2 < \infty$; A, w, θ_0 are independent; $\lambda > 0$ is fixed.

10.4.1 The scalar problem

The limit of Gordon's auxiliary problem is a convex problem over random variables, whose minimizer is the decoupled estimate and whose first-order conditions are (10.2).

Lemma 10.3 (The scalar problem). *Let $\sigma, \lambda, \alpha > 0$, $\mathbb{E}\Theta_0^2 < \infty$, $Z \sim \mathcal{N}(0,1)$ independent of Θ_0, and let $\mathcal{H}$ be the Hilbert space of square-integrable functions of (Θ_0, Z), with $\|V\|^2 = \mathbb{E}V^2$. For $V \in \mathcal{H}$ and $b, \tau > 0$ put*

$$M(V) = \sqrt{\sigma^2 + \|V\|^2/\alpha} - \mathbb{E}[VZ]/\alpha,$$

$$\mathcal{F}(V) = \tfrac{1}{2}M(V)_+^2 + \tfrac{\lambda}{\alpha}\mathbb{E}|\Theta_0 + V|, \tag{10.6}$$

$$\Psi(b, \tau) = \frac{b\sigma^2}{2\tau} + \frac{b\tau}{2} - \frac{b^2}{2} + \frac{1}{\alpha}\mathbb{E}\,q_b(\tau; \Theta_0, Z), \tag{10.7}$$

$$q_b(\tau; \theta, z) = \min_{x \in \mathbb{R}} \left\{ \frac{b}{2\tau}(x - \theta)^2 - bz(x - \theta) + \lambda|x| \right\}.$$

(i) $\mathcal{F}$ has a unique minimizer V_*, and $b_* := M(V_*) > 0$.

(ii) With $\tau_* = \sqrt{\sigma^2 + \|V_*\|^2/\alpha}$ and $\zeta_* = \lambda\tau_*/b_*$: $\Theta_0 + V_* = \eta(\Theta_0 + \tau_* Z; \zeta_*)$ almost surely, (τ_*, ζ_*) is the only solution of (10.2) with $\tau, \zeta > 0$, and $b_* = \tau_*(1 - \mathbb{P}(|\Theta_0 + \tau_* Z| > \zeta_*)/\alpha)$.

(iii) $\Psi(b_*, \cdot)$ is strictly convex on $(0, \infty)$, with unique minimizer τ_* and minimum $J_* := \mathcal{F}(V_*) = \frac{1}{2}b_*^2 + \frac{\lambda}{\alpha}\mathbb{E}|\eta(\Theta_0 + \tau_* Z; \zeta_*)|$, and b_* is the only maximizer of $b \mapsto \inf_{\tau > 0} \Psi(b, \tau)$ on $(0, \infty)$.

Proof. The plan is to obtain V_* from convexity and coercivity, to read the fixed point off its first-order conditions, and to recover Ψ by undoing $\sqrt{x} = \min_\tau\{\frac{x}{2\tau} + \frac{\tau}{2}\}$.

Part 1. The map $V \mapsto \sqrt{\sigma^2 + \|V\|^2/\alpha}$ is the norm of the affine map $V \mapsto (V/\sqrt{\alpha}, \sigma)$, so M, and with it $\mathcal{F}$, is convex and continuous. Choose $b_0 > 0$ with $D(\lambda/b_0) \le \alpha/4$. The bounds $\frac{1}{2}m_+^2 \ge b_0 m - \frac{1}{2}b_0^2$ and $VZ - \frac{\lambda}{b_0}|V| \le |V|(|Z| - \frac{\lambda}{b_0})_+$, and the Cauchy–Schwarz inequality, give $\mathcal{F}(V) \ge \frac{b_0}{2\sqrt{\alpha}}\|V\| - \frac{1}{2}b_0^2 - \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0|$. A convex, continuous, coercive function on a Hilbert space attains its minimum, since its sublevel sets are weakly compact and it is weakly lower semicontinuous. If a minimizer V had $M(V) \le 0$, then $V \ne -\Theta_0$, because $M(-\Theta_0) = \sqrt{\sigma^2 + \mathbb{E}\Theta_0^2/\alpha} > 0$, so $\mathcal{F}(V) = \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0 + V| > 0$; the points $V_s = (1 - s)V - s\Theta_0$ have $\mathbb{E}|\Theta_0 + V_s| = (1 - s)\mathbb{E}|\Theta_0 + V|$ and

$M(V_s) \le sM(-\Theta_0)$, so $\mathcal{F}(V_s) \le (1-s)\mathcal{F}(V) + \frac{1}{2}s^2 M(-\Theta_0)^2 < \mathcal{F}(V)$ for small $s > 0$, a contradiction. Two distinct minimizers would give, at their midpoint, a value of M below the average of theirs, because equality in the triangle inequality forces positively proportional vectors and $(V/\sqrt{\alpha}, \sigma)$ has the fixed coordinate $\sigma > 0$; as $m \mapsto \frac{1}{2}m_+^2$ is convex and increasing on $[0,\infty)$, $\mathcal{F}$ would be smaller there than its minimum.

Part 2. Let $X_* = \Theta_0 + V_*$ and $T = b_*(V_*/\tau_* - Z)$. Near V_* the function $\frac{1}{2}M_+^2 = \frac{1}{2}M^2$ has gradient T/α, and the one-sided derivative of $s \mapsto \mathbb{E}|X_* + s\Delta|$ at 0 is $\mathbb{E}[\operatorname{sgn}(X_*)\Delta; X_* \neq 0] + \mathbb{E}[|\Delta|; X_* = 0]$. Minimality makes the one-sided derivative of $\mathcal{F}$ nonnegative in every direction Δ; the directions $-(T + \lambda \operatorname{sgn} X_*)\mathbf{1}\{X_* \neq 0\}$ and $\pm\mathbf{1}_B$ with $B \subset \{X_* = 0\}$ give $T = -\lambda \operatorname{sgn} X_*$ on $\{X_* \neq 0\}$ and $|T| \le \lambda$ on $\{X_* = 0\}$. Since $T = (b_*/\tau_*)(X_* - \Theta_0 - \tau_* Z)$, these are the conditions for $X_* = \eta(\Theta_0 + \tau_* Z; \zeta_*)$. The definition of τ_* is the first equation of (10.2). Conditionally on Θ_0, $z \mapsto \eta(\Theta_0 + \tau_* z; \zeta_*)$ is Lipschitz with derivative $\tau_* \mathbf{1}\{|\Theta_0 + \tau_* z| > \zeta_*\}$ almost everywhere, so Gaussian integration by parts (Theorem A.1) gives $\mathbb{E}[V_* Z] = \tau_* \mathbb{P}(|\Theta_0 + \tau_* Z| > \zeta_*)$; this is the formula for $b_* = M(V_*)$, and $\lambda = \zeta_* b_*/\tau_*$ is the second equation. Conversely, if (τ, ζ) with $\tau, \zeta > 0$ solves (10.2), then $V = \eta(\Theta_0 + \tau Z; \zeta) - \Theta_0$ has $M(V) = b := \lambda\tau/\zeta$ by the same integration by parts, and $T = (b/\tau)(V - \tau Z) = -\lambda S$ for a pointwise subgradient S of $|\cdot|$ at $\Theta_0 + V$. The gradient inequality for $\frac{1}{2}M_+^2$ and the subgradient inequality for $|\cdot|$ give $\mathcal{F}(V') \ge \mathcal{F}(V) + \frac{1}{\alpha}\mathbb{E}[(T + \lambda S)(V' - V)] = \mathcal{F}(V)$ for all V', so $V = V_*$ and $(\tau, \zeta) = (\tau_*, \zeta_*)$.

Part 3. For fixed (θ, z) the function $(x, \tau) \mapsto \frac{b}{2\tau}(x - \theta)^2$ is jointly convex (a perspective), so $q_b(\cdot; \theta, z)$ and its expectation are convex in τ, and $b\sigma^2/(2\tau)$ is strictly convex: $\Psi(b_*, \cdot)$ is strictly convex. The minimizer in q_b is $\eta(\theta + \tau z; \lambda\tau/b)$, square integrable at (Θ_0, Z), so with $V = x - \Theta_0$

$$\Psi(b_*, \tau) = \min_{V \in \mathcal{H}} \left\{ b_* \left(\frac{\sigma^2 + \|V\|^2/\alpha}{2\tau} + \frac{\tau}{2} \right) - \frac{b_*}{\alpha}\mathbb{E}[VZ] + \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0 + V| \right\} - \frac{b_*^2}{2}.$$

The arithmetic–geometric mean inequality bounds the inner parenthesis below by $\sqrt{\sigma^2 + \|V\|^2/\alpha}$, so $\Psi(b_*, \tau) \ge \min_V G(V)$ with $G(V) = b_* M(V) - \frac{1}{2}b_*^2 + \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0 + V|$. The convex function $b_* M$ has gradient T/α at V_*, as $\frac{1}{2}M^2$ has, so the subgradient argument of part 2, with $S = -T/\lambda$, shows that V_* minimizes G; and $G(V_*) = \frac{1}{2}b_*^2 + \frac{\lambda}{\alpha}\mathbb{E}|X_*| = \mathcal{F}(V_*)$. At $\tau = \tau_*$ the inequality is an equality at $V = V_*$, which is the pointwise minimizer there because $\lambda\tau_*/b_* = \zeta_*$; so $\Psi(b_*, \tau_*) = \mathcal{F}(V_*)$, and τ_* is the unique minimizer. For $b > 0$ the display holds with b for b_*, and minimizing over τ first gives $\inf_\tau \Psi(b, \tau) = \inf_V G_b(V) \le G_b(V_*) = bb_* - \frac{1}{2}b^2 + \frac{\lambda}{\alpha}\mathbb{E}|X_*|$, where G_b is G with b for b_*; this is below J_* unless $b = b_*$. ∎

10.4.2 Gordon's comparison for the LASSO

The Gaussian noise can be absorbed into the matrix.[6] Let $\tilde{G} = [\sqrt{n}A,\ w/\sigma] \in \mathbb{R}^{n \times (d+1)}$, which has independent $\mathcal{N}(0,1)$ entries, independent of θ_0, and $\tilde{v} = (-v, \sigma\sqrt{n})$ for $v \in \mathbb{R}^d$. Then $\tilde{G}\tilde{v} = \sqrt{n}(w - Av)$, and

$$\max_{u \in \mathbb{R}^n} \left\{ u^\top \tilde{G}\tilde{v} - \frac{n}{2}\|u\|^2 \right\} = \frac{1}{2}\|w - Av\|^2 = F(\theta_0 + v) - \lambda\|\theta_0 + v\|_1, \qquad (10.8)$$

[6] The device is used by Thrampoulidis et al. (2015) and Miolane and Montanari (2021). Without it the auxiliary problem contains $\|\|v\|g/\sqrt{n} - w\|$, which is not convex in v.

attained at $u = (w - Av)/\sqrt{n}$. For compact $S \subset \mathbb{R}^d$, $C > 0$ and $\Upsilon(v,u) = -\frac{n}{2}\|u\|^2 + \lambda\|\theta_0 + v\|_1$ define the primary problem and the *auxiliary optimization*

$$\mathcal{P}(S) = \min_{v\in S}\max_{\|u\|\le C}\{u^\top \tilde{G}\tilde{v} + \Upsilon(v,u)\},$$
$$\mathcal{A}(S) = \min_{v\in S}\max_{\|u\|\le C}\{\|\tilde{v}\| g^\top u + \|u\|\,\tilde{h}^\top \tilde{v} + \Upsilon(v,u)\},$$

with $g \in \mathbb{R}^n$ and $\tilde{h} = (h, h_0) \in \mathbb{R}^d \times \mathbb{R}$ independent standard Gaussian vectors, independent of θ_0. The convex Gaussian min-max theorem (Theorem 8.3), in the form of Theorem 6.1 of Thrampoulidis et al. (2018), whose parts (i) and (ii) are those of Theorem 3 of Thrampoulidis et al. (2015), applies conditionally on θ_0 to the compact set $\{\tilde{v} : v \in S\}$, and gives, for every $c \in \mathbb{R}$: (i) $\mathbb{P}(\mathcal{P}(S) < c) \le 2\mathbb{P}(\mathcal{A}(S) \le c)$, which is Gordon's inequality (Gordon 1988; Gordon 1985) and needs no convexity; and (ii) if S is convex, $\mathbb{P}(\mathcal{P}(S) > c) \le 2\mathbb{P}(\mathcal{A}(S) \ge c)$, because Υ is convex in v and concave in u. In $\mathcal{A}(S)$ the maximum over the direction of u at fixed $\|u\| = b$ puts u along g, and $\|\tilde{v}\| = \sqrt{n}\sqrt{\|v\|^2/n + \sigma^2}$, so

$$\frac{\mathcal{A}(S)}{n} = \min_{v\in S}\max_{0\le b\le C} \ell_b(v), \qquad \ell_b(v) = bX(v) - \frac{b^2}{2} + \frac{\lambda}{n}\|\theta_0 + v\|_1, \tag{10.9}$$

with $X(v) = \sqrt{\|v\|^2/n + \sigma^2}\,\|g\|/\sqrt{n} - \langle h\rangle v/n + \sigma h_0/\sqrt{n}$. As every claim below concerns laws, we take the entries of θ_0, g, h from fixed infinite independent sequences, so that the strong law of large numbers applies.

Lemma 10.4 (A priori bounds). *Assume Assumption 10.2. There is K_0, depending on $\alpha, \sigma, \lambda, \mathbb{E}|\Theta_0|$, such that for $K \ge K_0$ and $C \ge 2\sigma + (3 + \alpha^{-1/2})K$, with probability tending to one, every minimizer of F lies in $\{\theta : \|\theta - \theta_0\| < K\sqrt{n}\}$, and $\|w - Av\| \le C\sqrt{n}$ whenever $\|v\| \le K\sqrt{n}$.*

Proof. The plan is to bound A and w, and to show by Gordon's inequality that F is large on the sphere of radius $K\sqrt{n}$ around θ_0; convexity then traps the minimizers inside. By the Sudakov–Fernique inequality (Theorem A.4; Vershynin (2018, §7.3)) $\mathbb{E}\|\sqrt{n}A\|_{\mathrm{op}} \le \sqrt{n} + \sqrt{d}$, and Gaussian concentration (Theorem A.7) gives $\|A\|_{\mathrm{op}} \le 2 + \sqrt{d/n} \le 3 + \alpha^{-1/2}$ with probability tending to one; with $\|w\| \le 2\sigma\sqrt{n}$ this gives the second claim.

Let $F_0 = \frac{1}{2}\sigma^2 + \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0|$, the limit of $F(\theta_0)/n$; take $b_0 \in (0,1]$ with $D(\lambda/b_0) \le \alpha/4$ and $K_0 = \frac{2}{b_0}(F_0 + 3 + \frac{1}{2}b_0^2 + \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0|)$, so that $b_0 \le 1 \le C$. On $S_K = \{v : \|v\| = K\sqrt{n}\}$, the bound $\lambda|v_j| - b_0 h_j v_j \ge -b_0(|h_j| - \lambda/b_0)_+|v_j|$ and the Cauchy–Schwarz inequality give

$$\ell_{b_0}(v) \ge b_0 K\frac{\|g\|}{\sqrt{n}} - \frac{b_0\sigma|h_0|}{\sqrt{n}} - \frac{b_0^2}{2} - \frac{\lambda}{n}\|\theta_0\|_1 - b_0 K\frac{\|(|h| - \lambda/b_0)_+\|}{\sqrt{n}}.$$

By the law of large numbers the right side tends to a limit at least $\frac{1}{2}b_0 K - \frac{1}{2}b_0^2 - \frac{\lambda}{\alpha}\mathbb{E}|\Theta_0| \ge F_0 + 3$, since $\|(|h| - \lambda/b_0)_+\|^2/n \to D(\lambda/b_0)/\alpha \le \frac{1}{4}$. So $\mathcal{A}(S_K)/n \ge F_0 + 2$ with probability tending to one, and part (i) of the CGMT, which needs only compactness, gives $\mathbb{P}(\mathcal{P}(S_K)/n < F_0 + 2) \to 0$. Restricting

u lowers the maximum in (10.8), so $F(\theta_0+v)\ge\mathcal{P}(S_K)$ on S_K, and with probability tending to one $\min_{S_K}F(\theta_0+\cdot)>F(\theta_0)$. Then no minimizer $\hat\theta$ has $\|\hat\theta-\theta_0\|\ge K\sqrt{n}$: the segment from θ_0 to $\hat\theta$ would meet the sphere at a point where, by convexity, $F\le\max\{F(\theta_0),F(\hat\theta)\}=F(\theta_0)$. ∎

Theorem 10.5 (The LASSO risk by Gordon's comparison). *Assume Assumption 10.2, and let τ_*,ζ_*,b_*,J_* be as in Lemma 10.3. For every $\gamma>0$ the probability that some minimizer $\hat\theta$ of F has $|\|\hat\theta-\theta_0\|^2/d-\alpha(\tau_*^2-\sigma^2)|\ge\gamma$ tends to zero, and $\min F/n\to J_*=\frac12 b_*^2+\frac{\lambda}{\alpha}\mathbb{E}|\eta(\Theta_0+\tau_*Z;\zeta_*)|$ in probability.*

Proof. The plan is to show that the auxiliary problem tends to J_*, with an upper bound from a candidate built on the fixed point and a lower bound from fixing $b=b_*$, and that it exceeds J_* by a fixed amount where the norm of v is wrong; the CGMT transfers both facts.

Step 1: reduction. Let $a_*=\sqrt{\tau_*^2-\sigma^2}$; as $n/d\to\alpha$, it suffices to control $\|\hat\theta-\theta_0\|/\sqrt{n}-a_*$. Fix $\gamma>0$, $K\ge\max\{K_0,a_*+1\}$, $C\ge\max\{b_*+1,2\sigma+(3+\alpha^{-1/2})K\}$, the ball $B=\{v:\|v\|\le K\sqrt{n}\}$ and the compact set $S^c=\{v\in B:|\|v\|/\sqrt{n}-a_*|\ge\gamma\}$. On the event Ω_1 of Lemma 10.4 the constraint $\|u\|\le C$ does not bind in (10.8) on B, so $\mathcal{P}(B)=\min F$, $\mathcal{P}(S^c)=\min_{S^c}F(\theta_0+\cdot)$, and a minimizer with $\hat\theta-\theta_0\in S^c$ forces $\mathcal{P}(S^c)\le\mathcal{P}(B)$. It suffices to show $\mathbb{P}(\mathcal{P}(S^c)\le\mathcal{P}(B))\to0$.

Step 2: upper bound. Let $x_j^*=\eta(\theta_{0j}+\tau_*h_j;\zeta_*)$, $v^*=x^*-\theta_0$ and $X_*=\Theta_0+V_*$. By the strong law of large numbers and integration by parts, almost surely, $\|v^*\|^2/n\to a_*^2<K^2$, $\langle h,v^*\rangle/n\to\tau_*\mathbb{P}(|\Theta_0+\tau_*Z|>\zeta_*)/\alpha$ and $\frac{\lambda}{n}\|x^*\|_1\to\frac{\lambda}{\alpha}\mathbb{E}|X_*|$; so $X(v^*)\to b_*$ by Lemma 10.3, and

$$\limsup_d \mathcal{A}(B)/n\le\limsup_d\max_{b\ge0}\ell_b(v^*)=\tfrac12 b_*^2+\tfrac{\lambda}{\alpha}\mathbb{E}|X_*|=J_*\qquad\text{almost surely.}$$

Step 3: lower bound. As $b_*\le C$, $\mathcal{A}(S)/n\ge\min_S\ell_{b_*}$ for $S\subset B$. With $\omega_d=\|g\|/\sqrt{n}$, the identity $\omega_d\sqrt{x}=\min_{\tau>0}\{\frac{x}{2\tau}+\frac{\omega_d^2\tau}{2}\}$ and completing squares coordinatewise give $\min_{\mathbb{R}^d}\ell_{b_*}=\min_{\tau>0}\Gamma_d(\tau)$, with

$$\Gamma_d(\tau)=\frac{b_*\sigma^2}{2\tau}+\frac{b_*\omega_d^2\tau}{2}+\frac{b_*\sigma h_0}{\sqrt{n}}-\frac{b_*^2}{2}+\frac1n\sum_{j=1}^d q_{b_*}(\tau;\theta_{0j},h_j),$$

with q_b as in Lemma 10.3; at fixed τ the minimum over v is attained at $v_{\tau,j}=\eta(\theta_{0j}+\tau h_j;\lambda\tau/b_*)-\theta_{0j}$. Each $q_{b_*}(\cdot;\theta,z)$ is convex (part 3 of the proof of Lemma 10.3), and $|q_{b_*}|\le\lambda|\theta|+\frac12 b_*\tau z^2$, so Γ_d is convex and $\Gamma_d(\tau)\to\Psi(b_*,\tau)$ almost surely, simultaneously for all rational $\tau>0$; convex functions converging on a dense set converge uniformly on compacts (Rockafellar 1970, Theorem 10.8). As $\Psi(b_*,\tau_*\pm\delta)>J_*$ for $\delta\in(0,\tau_*)$, for large d $\Gamma_d(\tau_*\pm\delta)>\Gamma_d(\tau_*)$, so by convexity Γ_d attains its minimum, at points $\hat\tau_d\in(\tau_*-\delta,\tau_*+\delta)$. Hence, almost surely, $\hat\tau_d\to\tau_*$ and $\min\Gamma_d\to J_*$. The strictly convex ℓ_{b_*} has the unique minimizer $\hat v=v_{\hat\tau_d}$, and since η is 1-Lipschitz in both arguments, $\|v_\tau-v_{\tau'}\|\le|\tau-\tau'|(\|h\|+\lambda\sqrt{d}/b_*)$; with $v_{\tau_*}=v^*$ this gives $\|\hat v\|/\sqrt{n}\to a_*$ almost surely.

Step 4: the gap. On B, $f(v) = \sqrt{\|v\|^2/n + \sigma^2}$ has Hessian $\frac{1}{nf}(I - \frac{vv^\top}{nf^2}) \geq \frac{\sigma^2}{nf^3} I \geq \frac{\sigma^2}{n(K^2+\sigma^2)^{3/2}} I$; with $\mu = b_*\sigma^2/(2(K^2+\sigma^2)^{3/2})$ and $\omega_d \geq \frac{1}{2}$, $\ell_{b_*} - \frac{\mu}{2n}\|\cdot\|^2$ is convex on B. For $\hat{v}, v \in B$ and $s \in (0,1]$, convexity along the segment gives $\ell_{b_*}(\hat{v}) \leq \ell_{b_*}(\hat{v}+s(v-\hat{v})) \leq (1-s)\ell_{b_*}(\hat{v}) + s\ell_{b_*}(v) - \frac{\mu}{2n}s(1-s)\|v-\hat{v}\|^2$, and $s \to 0$ gives $\ell_{b_*}(v) \geq \ell_{b_*}(\hat{v}) + \frac{\mu}{2n}\|v-\hat{v}\|^2$. For large d, $\hat{v} \in B$ and $|\|\hat{v}\|/\sqrt{n} - a_*| < \gamma/2$, so each $v \in S^c$ has $\|v - \hat{v}\| \geq \frac{\gamma}{2}\sqrt{n}$, and with $\Delta = \mu\gamma^2/8$

$$\mathcal{A}(S^c)/n \geq \min_{S^c} \ell_{b_*} \geq \min \Gamma_d + \Delta \to J_* + \Delta.$$

Step 5: transfer. Part (i) of the CGMT and step 4 give $\mathbb{P}(\mathcal{P}(S^c)/n < J_* + \frac{\Delta}{2}) \leq 2\mathbb{P}(\mathcal{A}(S^c)/n \leq J_* + \frac{\Delta}{2}) \to 0$; part (ii), on the convex set B, and step 2 give $\mathbb{P}(\mathcal{P}(B)/n > J_* + \frac{\Delta}{4}) \leq 2\mathbb{P}(\mathcal{A}(B)/n \geq J_* + \frac{\Delta}{4}) \to 0$. So $\mathbb{P}(\mathcal{P}(S^c) \leq \mathcal{P}(B)) \to 0$, and step 1 proves the first claim. For the second, steps 2 and 3 give $\mathcal{A}(B)/n \to J_*$ in probability, parts (i) and (ii) give $\mathbb{P}(|\mathcal{P}(B)/n - J_*| > t) \leq 2\mathbb{P}(|\mathcal{A}(B)/n - J_*| \geq t) \to 0$, and $\mathcal{P}(B) = \min F$ on Ω_1. ∎

The same gap controls every function of v that is Lipschitz for $\|\cdot\|/\sqrt{n}$ and converges at $\hat{v}$, such as the overlap m (Exercise 10.7); the empirical distribution of the $(\hat{\theta}_j, \theta_{0j})$ needs a stability argument (Miolane and Montanari 2021).

10.4.3 The replica potential is the auxiliary problem

Proposition 10.6 (The replica potential and Ψ). *Let Φ_{RS} be the function in braces in (10.5). Stationarity in Q and m holds exactly when $\hat{Q} = \hat{m} = \alpha/(\alpha+\chi)$, and under these relations the change of variables $\hat{Q} = b/\tau$, $\hat{\chi} = b^2$, $\chi = \alpha(\tau/b - 1)$, with $b, \tau > 0$, gives $\Phi_{\mathrm{RS}} = \alpha\Psi(b,\tau)$.*

Proof. The plan is to substitute. The terms containing Q or m are $-\frac{1}{2}\hat{Q}Q + \hat{m}m + (Q-2m)/(2(1+\chi/\alpha))$; their derivatives vanish exactly when $\hat{Q} = \hat{m} = 1/(1+\chi/\alpha)$, and then they cancel, leaving $\frac{1}{2}\hat{\chi}\chi + \frac{1}{2}\hat{Q}(\alpha\sigma^2 + \mathbb{E}\Theta_0^2)$ plus the expected minimum. With $\hat{Q} = b/\tau$ and $\sqrt{\hat{\chi}} = b$ the bracket is $\frac{b}{2\tau}(x-\Theta_0)^2 - bZ(x-\Theta_0) + \lambda|x| - \frac{b}{2\tau}\Theta_0^2 - bZ\Theta_0$, with $\mathbb{E}[Z\Theta_0] = 0$, and $\frac{1}{2}\hat{\chi}\chi = \frac{1}{2}\alpha b(\tau - b)$. The terms in $\mathbb{E}\Theta_0^2$ cancel, and what remains is $\frac{1}{2}\alpha b\tau - \frac{1}{2}\alpha b^2 + \frac{\alpha b\sigma^2}{2\tau} + \mathbb{E}\min_x[\cdots] = \alpha\Psi(b,\tau)$. ∎

With the computation's $\hat{Q} = 1 - s/\alpha$, the relation $\hat{Q} = b/\tau$ is the formula for b_*, and the replica training error $(\tau\hat{Q})^2$ is b_*^2, the squared norm of the dual variable, the residual over $\sqrt{n}$. The proof uses no property of the penalty, so the identification holds for every separable penalty, the ridge penalty of Chapter 9 included.

10.4.4 What the general proofs add

Theorem 10.1 gives every pseudo-Lipschitz statistic, for deterministic signals and general noise. Thrampoulidis et al. (2018) treat general convex losses and separable penalties by our route (their Theorem 4.1 and §5.4). Miolane and Montanari (2021) prove Wasserstein convergence of the empirical

distribution uniformly over $\lambda \in [\lambda_{\min}, \lambda_{\max}]$ and over ℓ_p balls; their max–min (8) is our Ψ up to a constant, and their Proposition 3.1 our Lemma 10.3; general Gaussian designs are treated by Celentano et al. (2023).

For robust regression with design $x_i \overset{\text{iid}}{\sim} \mathcal{N}(0, I_p)$ and $n/p \to \alpha > 1$, the system (10.4) gives the limit of $\|\hat{\beta} - \beta^*\|^2$ by AMP for strongly convex smooth losses (Donoho and Montanari 2016, Theorem 4.1), by the CGMT for convex losses under conditions that Huber's loss meets even with Cauchy noise (Thrampoulidis et al. 2018, Theorem 4.1, §5.1), and, for designs with independent non-Gaussian entries and strongly convex losses, by leave-one-out arguments in the unrefereed preprint of El Karoui (2013), who notes that such ideas "seem to be known in Physics under the name 'cavity method' "; its refereed revision (El Karoui 2018) keeps only ridge-regularized estimators, with elliptical-like designs, as does the universality theorem of Han and Shen (2023). Donoho and Montanari also show that the system forces $R \geq 1/((\alpha-1)I(P_W))$, I the Fisher information of the noise: a *variance inflation* that no loss avoids (Exercise 10.3).

10.5 The proof: the Donoho–Tanner threshold as a statistical dimension

Without noise, the LASSO tends as $\lambda \to 0$ to ℓ_1 minimization, $\min\|\theta\|_1$ subject to $A\theta = y$, which succeeds if θ_0 is its unique solution. The *descent cone* of a convex f at x is $\mathcal{D}(f, x) = \{v : f(x + sv) \leq f(x) \text{ for some } s > 0\}$, and ℓ_1 minimization succeeds exactly when $\ker A \cap \mathcal{D}(\|\cdot\|_1, \theta_0) = \{0\}$: a nonzero v in the intersection makes $\theta_0 + sv$ a competitor, and a competitor θ puts $\theta - \theta_0$ in the intersection. The *statistical dimension* of a closed convex cone $C \subset \mathbb{R}^d$ is $\delta(C) = \mathbb{E}\|\Pi_C(g)\|^2$, with $g \sim \mathcal{N}(0, I_d)$ and Π_C the projection, and that of a convex cone is that of its closure; a k-dimensional subspace has $\delta = k$, and the orthant $\delta = d/2$, which gives the prologue's number two.[7]

[7] Up to one, $\delta(C)$ is the squared Gaussian width of $C \cap S^{d-1}$ (Amelunxen et al. 2014, Proposition 10.2), the quantity of Gordon's escape-through-a-mesh argument.

Theorem (Amelunxen et al. 2014, Theorem II). *Fix $p \in (0,1)$, $a_p = \sqrt{8\log(4/p)}$, a fixed $x_0 \in \mathbb{R}^d$, a proper convex f, and $A \in \mathbb{R}^{n\times d}$ with independent standard normal entries. If $n \leq \delta(\mathcal{D}(f, x_0)) - a_p\sqrt{d}$, then* $\min f(x)$ *subject to* $Ax = Ax_0$ *succeeds with probability at most p; if $n \geq \delta(\mathcal{D}(f, x_0)) + a_p\sqrt{d}$, with probability at least $1 - p$.*

Its proof combines the optimality condition with an approximate *kinematic formula* for a random subspace meeting a cone; what is particular to ℓ_1 is the statistical dimension.

Theorem 10.7 (The ℓ_1 descent cone; Amelunxen et al. 2014, Proposition 4.5). *If $x \in \mathbb{R}^d$ has s nonzero entries, $1 \leq s \leq d-1$, and $\mathcal{D} = \mathcal{D}(\|\cdot\|_1, x)$, then $d\,\psi(s/d) - 2\sqrt{d/s} \leq \delta(\mathcal{D}) \leq d\,\psi(s/d)$.*

Proof. The plan, which is the argument of Amelunxen et al. (2014, Proposition 4.1 and Appendix C.2) specialized to ℓ_1, is to write $\delta(\mathcal{D})$ as an expected squared distance to the cone generated by the subdifferential,

to exchange expectation and minimum for the upper bound, and to bound the cost of the exchange by two variances. The subdifferential $S = \{u : u_i = \operatorname{sgn} x_i \text{ if } x_i \neq 0,\ |u_i| \le 1 \text{ otherwise}\}$, and with it $\mathcal{D}$, depends only on the signs of x, and signed permutations preserve the law of g; so we take $x = (1, \dots, 1, 0, \dots, 0)$ with s ones, for which $\langle u \rangle x = s$ on S.

By Moreau's decomposition $\|\Pi_C(g)\| = \operatorname{dist}(g, C^\circ)$, and the polar of the descent cone at a point where the subdifferential is compact and does not contain 0 is the cone it generates (Rockafellar 1970, Corollary 23.7.1), here the closed cone $\mathcal{K} = \bigcup_{t \ge 0} tS$. Computing the nearest point of tS coordinatewise,

$$\delta(\mathcal{D}) = \mathbb{E} \min_{t \ge 0} F_g(t), \qquad F_g(t) = \operatorname{dist}^2(g, tS) = \sum_{i \le s} (g_i - t)^2 + \sum_{i > s} (|g_i| - t)_+^2.$$

Since $\mathbb{E} F_g(t) = s(1 + t^2) + (d - s)D(t) = dJ_{s/d}(t)$, exchanging expectation and minimum gives $\delta(\mathcal{D}) \le d\psi(s/d)$.

For the lower bound write $J = J_{s/d}$. From $D(t) = 2[(1 + t^2)\Phi(-t) - t\phi(t)]$, $D'(t) = -4[\phi(t) - t\Phi(-t)]$ and $D''(t) = 4\Phi(-t) > 0$; so J is strictly convex with $J'(0) < 0$ and $J'(\infty) = \infty$, and has a unique minimizer $t_\star > 0$. Each F_g is convex and differentiable, $F_g'(t) = 2st - 2\sum_{i \le s} g_i - 2\sum_{i > s}(|g_i| - t)_+$, and $\mathbb{E} F_g'(t) = dJ'(t)$. With t_g the minimizer of F_g on $[0, \infty)$, the tangent at $t_\star$ gives $F_g(t_g) \ge F_g(t_\star) + (t_g - t_\star)F_g'(t_\star)$, and since $\mathbb{E} F_g'(t_\star) = 0$,

$$\delta(\mathcal{D}) \ge d\psi(s/d) + \operatorname{Cov}\big(t_g, F_g'(t_\star)\big) \ge d\psi(s/d) - \sqrt{\operatorname{Var}(t_g)\operatorname{Var}(F_g'(t_\star))}.$$

The Gaussian Poincaré inequality (Theorem A.6) gives $\operatorname{Var} f(g) \le L^2$ for L-Lipschitz f. The gradient of $g \mapsto F_g'(t_\star)$ has entries -2 and $-2\operatorname{sgn}(g_i)\mathbf{1}\{|g_i| > t_\star\}$, so $\operatorname{Var} F_g'(t_\star) \le 4d$. The nearest point of $\mathcal{K}$ to g is that of $t_g S$, by uniqueness of projections, so $t_g = \langle \Pi_{\mathcal{K}}(g) \rangle x / s$; as $\Pi_{\mathcal{K}}$ is 1-Lipschitz, $|t_g - t_{g'}| \le \|x\| \|g - g'\| / s = \|g - g'\| / \sqrt{s}$ and $\operatorname{Var} t_g \le 1/s$. Hence $\delta(\mathcal{D}) \ge d\psi(s/d) - 2\sqrt{d/s}$. ∎

For a fixed signal this gives the *weak threshold* of Donoho and Tanner, as opposed to the strong threshold, at which all k-sparse vectors are recovered at once.

Corollary 10.8 (The weak threshold). *Let $s/d \to \varepsilon \in (0, 1)$ and $n/d \to \alpha \in (0, 1)$, let θ_0 have s nonzero entries, and let A be Gaussian. If $\alpha > \psi(\varepsilon)$, ℓ_1 minimization succeeds with probability tending to one; if $\alpha < \psi(\varepsilon)$, it fails with probability tending to one. The function ψ is continuous and increasing from $\psi(0) = 0$ to $\psi(1) = 1$, so the threshold is $\varepsilon_c(\alpha) = \psi^{-1}(\alpha)$.*

Proof. The plan is to insert Theorem 10.7 into Theorem II. For each t, $J_\varepsilon(t)$ is affine in ε with slope $1 + t^2 - D(t)$, positive for $t > 0$; the minimizing t is positive for $\varepsilon < 1$, because $J_\varepsilon'(0) < 0$. So ψ is concave and strictly increasing, with $\varepsilon \le \psi(\varepsilon) \le J_\varepsilon(t)$ for every t, whence $\psi(0) = \inf_t D(t) = 0$, $\psi(1) = 1$ and continuity at the endpoints. Hence $\delta(\mathcal{D}) = d\psi(\varepsilon) + o(d)$, while $a_p\sqrt{d} = o(d)$: for every fixed p the condition of Theorem II holds for large d, in the direction of success if $\alpha > \psi(\varepsilon)$ and of failure if $\alpha < \psi(\varepsilon)$. ∎

Proposition 10.9 (Three formulas, one curve). *For $\alpha \in (0,1)$ the following coincide:* (a) $\psi^{-1}(\alpha)$; (b) $\alpha\rho_{\mathrm{SE}}(\alpha)$, *with the state-evolution formula of Donoho et al. (2009, eq. (5), signed case)*

$$\rho_{\mathrm{SE}}(\alpha) = \max_{z\ge 0} \frac{1-(2/\alpha)[(1+z^2)\Phi(-z)-z\phi(z)]}{1+z^2-2[(1+z^2)\Phi(-z)-z\phi(z)]};$$

(c) *the replica threshold of Kabashima et al. (2009, eqs. (11)–(12)): the ε for which some $t>0$ has $\alpha = J_\varepsilon(t)$ and $\alpha = 2(1-\varepsilon)\Phi(-t)+\varepsilon$, with $t=\hat\chi^{-1/2}$ in their notation.*

Proof. The plan is to write each formula through J_ε. In (b) the bracket is $D(z)/2$. Since ψ is continuous and increasing and its minimum is attained, $\psi^{-1}(\alpha)$ is the supremum of the ε with $J_\varepsilon(t)\le\alpha$ for some t, and for $t>0$ that condition reads $\varepsilon \le (\alpha - D(t))/(1+t^2-D(t))$, a number below one; exchanging the suprema gives (b). In (c), multiplying equation (11) of Kabashima et al. by $t^2 = 1/\hat\chi$ turns it into $\alpha = J_\varepsilon(t)$, and $J_\varepsilon'(t) = 2\varepsilon t - 4(1-\varepsilon)[\phi(t) - t\Phi(-t)]$ gives the identity

$$J_\varepsilon(t) - \big[2(1-\varepsilon)\Phi(-t)+\varepsilon\big] = \tfrac{t}{2}J_\varepsilon'(t);$$

so for $t>0$ the two equations say $J_\varepsilon'(t)=0$ and $\alpha = J_\varepsilon(t)$, that is, $\alpha=\psi(\varepsilon)$. ∎

The formulas appeared separately: Donoho et al. (2009) reported the agreement of state evolution with the polytope curves as a numerical finding; Amelunxen et al. note that their upper bound is equivalent to Donoho and Tanner's asymptotic bound and the lower bound is new (their Remark 4.6); Bayati et al. (2015) proved the transition asymptotically by AMP, for independent sub-Gaussian entries.

10.6 Measuring it

The LASSO. `10-lasso-risk.py`[8] solves (10.2) by bisection in τ^2 at each multiplier $t=\zeta/\tau$ above $t_{\min}=0.405234$, the root of $D(t)=\alpha$ (Bayati and Montanari 2012, Proposition 1.3), and independently computes $\max_b \min_\tau \Psi$; the two agree to eight digits ($\tau_*=0.31103735$, $b_*=0.17155008$). The optimal $\lambda_*=0.23378$ gives $\mathrm{MSE}_*=0.023372$ ($R_*=0.046744$) and keeps 22.4% of the coordinates, twice the 10% that are nonzero. In Figure 10.1(a,b) all ten means lie within their intervals of the prediction, and at λ_* so do the means for $d=250$ to 4000 (80 to 5 data sets, 0.0229 to 0.0239, against 0.02337). In panel (c) AMP matches state evolution to 10^{-4} after ten iterations; its iterates end within 10^{-7} (in $\|\cdot\|^2/d$) of the LASSO at the data-dependent penalty $\zeta_t(1-\langle\eta'\rangle/\alpha)$ of their fixed point, at most 5.6% from λ_*, and AMP and the LASSO at λ_* give 0.02334 ± 0.00157 and 0.02331 ± 0.00157.

The noiseless threshold. `10-lasso-transition.py` evaluates the three formulas of Proposition 10.9 at 104 values of α; they agree to 1.4×10^{-14}. At $\alpha=0.1,\,0.25,\,0.5,\,0.75,\,0.9$ the threshold is $\rho_c=0.18942937$, 0.26738355, 0.38568967, 0.53371168, 0.67816938, and $\psi(0.5)=0.83129991$, which Kabashima et al. (2009) print as 0.83129. A Monte Carlo estimate of

[8] Each of the three scripts runs in about a minute and prints every number quoted here. The LASSO is computed by coordinate descent in scikit-learn, whose objective $\frac{1}{2n}\|y-A\theta\|^2 + a\|\theta\|_1$ has $a=\lambda/n$ in our scaling; tolerance 10^{-10}, warm starts along the grid of λ.

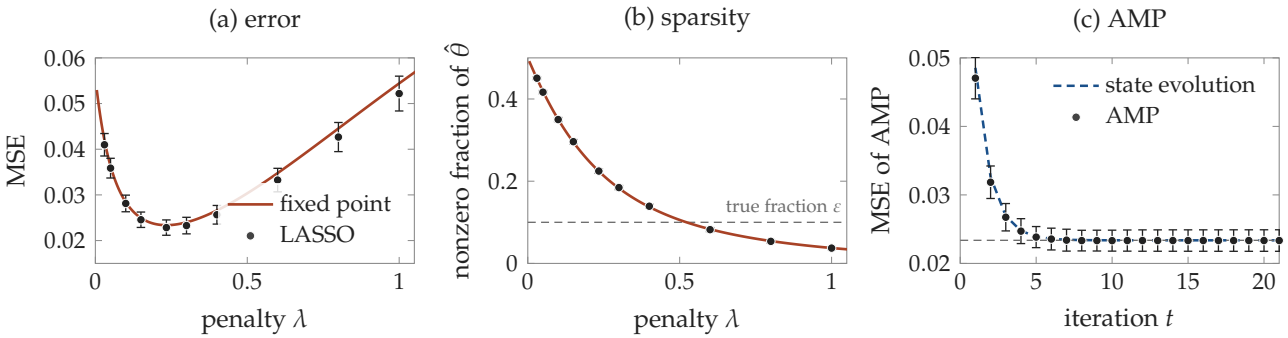


Figure 10.1. The LASSO for the Bernoulli–Gaussian signal, $\varepsilon = 0.1$, $\alpha = 0.5$, $\sigma^2 = 0.05$. (a), (b) Error and fraction of nonzero coordinates against the penalty: carnelian, the fixed point (10.2), which Theorems 10.1 and 10.5 prove; dots, the LASSO by coordinate descent at $d = 1000$, $n = 500$, mean of 20 data sets (seeds 1000–1019, the same for every λ) with Student-t 95% intervals. (c) AMP with thresholds $t_*\tau_t$ at the optimal $\lambda_* = 0.23378$, $d = 4000$, mean of 5 data sets (seeds 3000–3004) with 95% intervals, against state evolution (lapis dashed); gray, the fixed point 0.023372. Script: `figures/scripts/10-lasso-risk.py`.

$\delta(\mathcal{D})$ at $d = 100$ (20000 draws) lies between the bounds of Theorem 10.7, 0.5 to 0.7 below $d\psi(s/d)$ for $s = 5$ to 50 (at $s = 20$, 50.53 ± 0.16 against 51.11), where the bound allows 2.8 to 8.9. In Figure 10.2 the transition sharpens with d, and the 50% points lie on or slightly above the curve, by at most 0.03 in ρ (two half-widths, at $\alpha = 0.85$); with $n = 15$ to 85 measurements this is the expected direction, since $\delta(\mathcal{D})$ is below $d\psi(s/d)$.

ROBUST REGRESSION. `10-lasso-robust.py` solves (10.4) for Huber's loss (Section 10.2); expectations over t_3 noise use a 400-point Gauss–Legendre rule after $w = \sqrt{3}\tan u$. Under Gaussian noise R decreases in k to the least-squares value 3/7, which is also the information bound; the customary $k = 1.345$ costs 3.5% ($R = 0.44370$). Under t_3 noise the best threshold is $k = 1.138$, with $R = 0.77818$, 39% below least squares (9/7) and 21% above the bound 0.64286. The simulations agree with the system within their intervals at all twelve values of k and for least squares. No measurement contradicts a theorem of the chapter.

10.7 The status board

For Gaussian designs every prediction of the physics box is a theorem, most with two proofs or more; replica symmetry is right because the problems are convex, and the physics rejects its own symmetric solution only for non-convex ℓ_0 reconstruction.

STATUS BOARD · the replica fixed point ↔ AMP and the CGMT

PREDICTION	STATUS	PROOF OR EVIDENCE

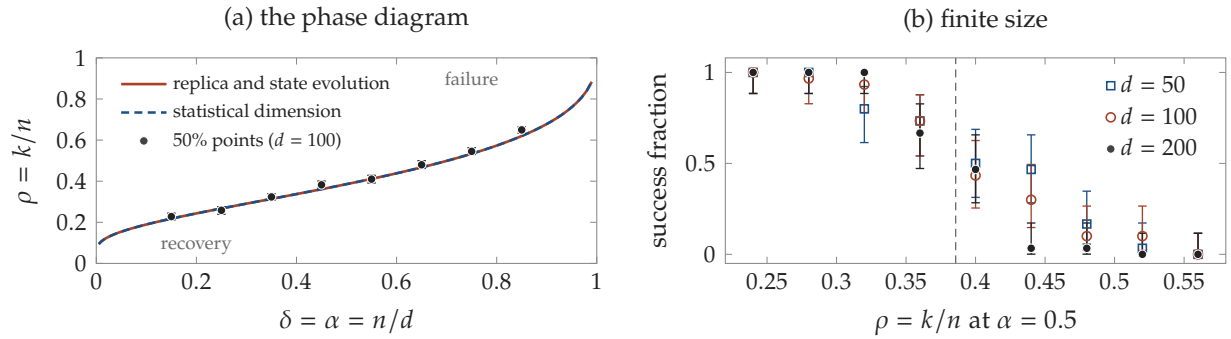


Figure 10.2. The Donoho–Tanner transition. (a) The threshold $\rho = \psi^{-1}(\alpha)/\alpha$: the replica and state-evolution formulas (carnelian) and the statistical-dimension curve of Corollary 10.8 (lapis dashed) are one function (Proposition 10.9). Dots: 50% points of the success probability of ℓ_1 minimization at $d = 100$, from logistic fits to 7 values of ρ with 24 problems each (seeds $20000 + 100j + i$), with 95% parametric-bootstrap intervals (400 resamples). (b) Success fraction at $\alpha = 0.5$, 30 problems per point (seeds $10000 + 1000d/50 + i$), Clopper–Pearson 95% intervals; vertical line, $\rho_c(0.5) = 0.38568967$. Nonzero entries are standard normal; success means $\|\hat{\theta} - \theta_0\| \le 10^{-4}\|\theta_0\|$ for the linear program's solution. Script: `figures/scripts/10-lasso-transition.py`.

LASSO risk and law of $(\hat{\theta}_j, \theta_{0j})$: the fixed point (10.2)	proved	Bayati–Montanari 2012, Thm 1.5; Thrampoulidis et al. 2018, §5.4; Miolane–Montanari 2021; Theorem 10.5
Replica-symmetric derivations (LASSO, ℓ_1, M-estimation)	heuristic	Kabashima et al. 2009; Rangan et al. 2012; Advani–Ganguli 2016; they agree with the theorems (Propositions 10.6 and 10.9)
ℓ_1 weak threshold $\alpha = \psi(\varepsilon)$ (Donoho–Tanner)	proved	Amelunxen et al. 2014, Thm II, Prop. 4.5; Bayati et al. 2015
State evolution, replica and statistical dimension give one curve	proved	Proposition 10.9; numerical in Donoho et al. 2009
Replica-symmetric description of ℓ_0 reconstruction	false	success solution AT-unstable for every α (Kabashima et al. 2009, §4.1); $\alpha = \varepsilon$ is the counting bound
LASSO formulas, independent non-Gaussian entries	proved under	Han–Shen 2023 (sub-Gaussian; large λ if $\alpha < 1$); ℓ_1: Bayati et al. 2015
Gaussian formulas for general isotropic designs	false	Han–Shen 2023: least squares, rows $U_i Z_i$; ridge, LASSO in simulations

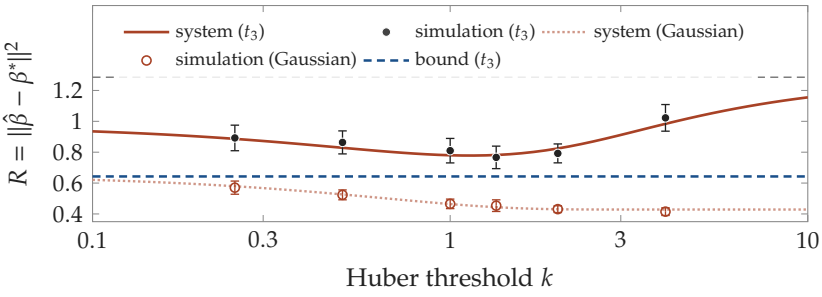


Figure 10.3. Huber regression, $p/n = 0.3$, noise t_3 (variance 3) or standard Gaussian. Curves: the system (10.4). Points: $n = 1000$, $p = 300$, mean of 10 data sets per point (seeds $4000 + 1000i_{\text{noise}} + 100i_k + r$), Student-$t$ 95% intervals. Lapis dashed: the information bound $1/((\alpha - 1)I) = 0.643$ for t_3; for Gaussian noise it equals the least-squares risk 3/7. Gray: least squares under t_3, 9/7. Script: `figures/scripts/10-lasso-robust.py`.

M-estimation system (10.4), Gaussian design, $\alpha > 1$	proved	Donoho–Montanari 2016, Thm 4.1 (smooth, strongly convex ρ); Thrampoulidis et al. 2018, Thm 4.1 (convex ρ)
The same, independent non-Gaussian entries	preprint	El Karoui, arXiv:1311.2445v1, Thms 1.1, 6.1 (strongly convex ρ); refereed only with a ridge penalty (El Karoui 2018; Han–Shen 2023)
Variance inflation $R \geq 1/((\alpha - 1)I(P_W))$	proved	Donoho–Montanari 2016, Cor. 4.3; replicas: Advani–Ganguli 2016

10.8 Where the translation fails

WHAT WAS ASSUMED. The derivation and every proof here assume independent Gaussian entries: the replica computation when it treats the fields as Gaussian, the CGMT for Gordon's comparison, AMP for its conditioning lemma. Kabashima et al. (2009) argue that their result holds for independent entries of fixed variance; this is a theorem for the ℓ_1 transition (Bayati et al. 2015) and, for sub-Gaussian entries and (if $\alpha < 1$) a large enough λ, for the LASSO (Han and Shen 2023). Both sides fix $\lambda, \sigma > 0$ and a converging empirical distribution of the signal, so the statements are pointwise in the signal; uniformity over ℓ_p balls holds, and over ℓ_0 balls fails for the empirical law (Miolane and Montanari 2021, Remark 2). Our theorem adds Gaussian noise and an independent signal and controls only the norm of the error.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. For ℓ_0 reconstruction the replica-symmetric success solution is unstable for every α, because hard thresholding is discontinuous (Kabashima et al. 2009, §4.1); its threshold $\alpha = \varepsilon$ is the counting bound, which exhaustive search attains (Exercise 10.2), but the symmetric description is wrong, and for non-convex penalties

Gordon's comparison gives one inequality only. The fields need not be Gaussian: for isotropic rows $U_i Z_i$, U_i a scalar that is rarely large, Han and Shen (2023) prove that the least-squares risk can exceed its Gaussian value by an arbitrary factor, and see the same for ridge and the LASSO in simulations. The thresholds concern a fixed signal; recovering all k-sparse vectors at once, the strong threshold, needs more measurements, and a typical-case computation does not address it. Finally, a computation made at $d = \infty$ is silent about the finite-size shifts of Figure 10.2.

OPEN PROBLEMS. First, the error of the statistical-dimension recipe, which Amelunxen et al. (2014) leave open (their Remark 4.4); for ℓ_1 our Monte Carlo suggests that $\delta(\mathcal{D}) - d\psi(s/d)$ stays bounded (Exercise 10.10). Second, which designs share the Gaussian LASSO risk: independent entries suffice under conditions, general isotropic rows do not, and Han and Shen (2023) leave open whether i.i.d. rows whose sample covariance obeys the Marchenko–Pastur law suffice. Third, the typical performance of non-convex penalties (ℓ_0, ℓ_p with $p < 1$) with noise: the symmetry-breaking computation that Kabashima et al. (2009) announced, and a rigorous counterpart. Fourth, a refereed proof of the unregularized robust-regression system for non-Gaussian designs; the refereed results need a ridge penalty.

Exercises

10.1 CONSOLIDATION. (Ridge as a special case.) Replace $\lambda\|\theta\|_1$ by $\frac{\lambda}{2}\|\theta\|^2$. (a) Show that the single site of Computation 10.1 becomes $s(\Theta_0 + \tau Z)$ with $s = 1/(1 + \lambda/\hat{Q})$ and $\hat{Q} = 1 - s/\alpha$. (b) Show that λ is then exactly the per-sample ridge penalty of Chapter 9, and that with $\alpha = 2$, $\mathbb{E}\Theta_0^2 = 2$ (so $r^2 = 1$), $\sigma^2 = 0.25$, $\lambda = 0.1$ the fixed point and Theorem 9.2 both give $R = 0.17678$.

10.2 CONSOLIDATION. (The counting bound.) Let $\theta_0 \in \mathbb{R}^d$ be fixed with k nonzero entries and $A \in \mathbb{R}^{n\times d}$ Gaussian with $n \geq k + 1$. Show that, almost surely, θ_0 is the only solution of $A\theta = A\theta_0$ with at most k nonzero entries. *(Hint: for $T \neq \operatorname{supp}\theta_0$ with $|T| \leq k$, the part of $A\theta_0$ carried by columns outside T is independent of the columns in T.)*

10.3 CONSOLIDATION. ⋆ (Variance inflation.) (a) From (10.4) with $\rho(u) = u^2/2$ derive $R = \sigma^2/(\alpha - 1)$. (b) With $\psi_c(z) = z - \operatorname{prox}_c(z)$ and f_r the density of $\hat{z}$, show $\mathbb{E}\psi_c'(\hat{z}) = 1/\alpha$ and $\mathbb{E}\psi_c(\hat{z})^2 = r^2/\alpha$, and deduce $r^2 \geq 1/(\alpha I(f_r))$. (c) With Stam's inequality $1/I(f_r) \geq 1/I(P_W) + r^2$ (Stam 1959) conclude $R \geq 1/((\alpha - 1)I(P_W))$; evaluate it for t_3 noise, of Fisher information $\frac{2}{3}$. *(Hint: $\mathbb{E}\psi_c'(\hat{z}) = -\mathbb{E}[\psi_c(\hat{z})(\log f_r)'(\hat{z})]$, and the Cauchy–Schwarz inequality.)*

10.4 BY HAND. ⋆ (Replicas for robust regression.) Carry out the computation of Section 10.2 for $F(\theta) = \sum_i \rho(y_i - \langle a_i\rangle\theta)$: the energetic term becomes $\alpha\mathbb{E}M_{\chi/\alpha}(W + \sqrt{\mathcal{E}/\alpha}\,Z)$, the entropic term is that of a Gaussian channel, and the saddle point is (10.4) with $r^2 = \mathcal{E}/\alpha$, $c = \chi/\alpha$. *(Hint: $\partial_s M_c(s) = (s - \operatorname{prox}_c(s))/c$, $\partial_c M_c(s) = -(s - \operatorname{prox}_c(s))^2/(2c^2)$, and $\hat{Q} = \hat{m} = 1/(\alpha c)$ at the saddle.)*

10.5 BY HAND. (Very sparse signals.) Show that $\psi(\varepsilon) = 2\varepsilon\log(1/\varepsilon)(1+o(1))$ as $\varepsilon \to 0$, so that $\rho_c(\delta) \sim 1/(2\log(1/\delta))$ as $\delta \to 0$, as Donoho et al. (2009) report. The ratio $\psi(\varepsilon)/(2\varepsilon\log(1/\varepsilon))$ is only 0.66 at $\varepsilon = 10^{-2}$ and 0.79 at 10^{-8}; explain. *(Hint: $D(t) \approx 4\phi(t)/t^3$, and the minimizer has $t^2 = 2\log(1/\varepsilon) - O(\log\log(1/\varepsilon))$.)*

10.6 BY HAND. (Nonnegative signals.) For $\|\cdot\|_1$ restricted to the nonnegative orthant and $x \geq 0$ with s positive entries, show $\delta(\mathcal{D}) \leq d\psi_+(s/d)$, $\psi_+(\varepsilon) = \min_t\{\varepsilon(1+t^2) + (1-\varepsilon)\mathbb{E}(Z-t)_+^2\}$, and check numerically that $\psi_+^{-1}(\alpha)/\alpha$ is the formula of Proposition 10.9 with 2 replaced by 1: 0.240976, 0.558228, 0.902657 at $\alpha = 0.1, 0.5, 0.9$.

10.7 TRANSLATION. (The overlap concentrates.) Let $|f(v) - f(v')| \leq L\|v - v'\|/\sqrt{n}$ and $f(\hat{v}) \to \bar{f}$ in probability for the minimizer $\hat{v}$ of ℓ_{b_*}. Show by steps 4 and 5 of the proof of Theorem 10.5 that $f(\hat{\theta} - \theta_0) \to \bar{f}$, and deduce that the replica overlaps $m = \langle\hat{\theta}, \theta_0\rangle/d$ and $Q = \|\hat{\theta}\|^2/d$ tend to $\mathbb{E}[X_*\Theta_0]$ and $\mathbb{E}X_*^2$.

10.8 TRANSLATION. (The cavity field of AMP.) Show that a fixed point (x, z) of AMP with threshold ζ satisfies the optimality conditions of the LASSO with $\lambda = \zeta(1 - \langle\eta'(x + A^\top z; \zeta)\rangle/\alpha)$, and identify the cavity field with $x + A^\top z$ and $1 - \langle\eta'\rangle/\alpha$ with $\hat{Q}$.

10.9 MEASUREMENT. (Noise sensitivity.) Modify `10-lasso-risk.py` to minimize the predicted error over λ at $\alpha = 0.5$ for $\sigma^2 = 0.05, \ldots, 0.00005$. For $\varepsilon = 0.1 < \varepsilon_c(0.5)$, MSE_*/σ^2 tends to about 0.96; for $\varepsilon = 0.25$ the error tends to about 0.022. Explain; compare Donoho et al. (2011).

10.10 RESEARCH. (The constant in the statistical dimension.) Estimate $\delta(\mathcal{D}) - d\psi(s/d)$ by Monte Carlo for $s/d = 0.1$ and $d = 50$ to 800. Does it converge, and to what? Amelunxen et al. (2014, Remark 4.4) pose the general question of an optimal error estimate.

Notes and References

The proportional theory of ℓ_1 recovery began with polytope geometry: Donoho (2006) related exact recovery to the neighborliness of random projections of the cross-polytope, and Donoho and Tanner (2009) proved exact thresholds when the projection lowers the dimension radically. Stojnic (2009) also obtained the threshold, in an arXiv paper, as a sufficient condition for recovery, from Gordon's escape through a mesh (Gordon 1988) and linear-programming duality; his later combination of Gordon's comparison with convex duality (Chapter 8) is the idea that Thrampoulidis, Oymak and Hassibi turned into the CGMT (arXiv 2014; COLT 2015) (Thrampoulidis et al. 2014, 2015).

The proof of Theorem 10.5 assembles, in the book's own form, ingredients of Thrampoulidis et al. (2015), Thrampoulidis et al. (2018) and Miolane and Montanari (2021): the absorption of the noise, the function Ψ, the strong-convexity gap. The functional $\mathcal{F}$ of Lemma 10.3 and the change of variables of Proposition 10.6 are the book's own; Proposition 10.9 collects

elementary identities that we do not claim as new; the proof of Theorem 10.7 is that of Amelunxen, Lotz, McCoy and Tropp.

TRANSLATING THE LITERATURE. Bayati and Montanari write N for our d, δ for α, x_0 for θ_0, α for our multiplier t, and θ for our threshold ζ. Donoho, Maleki and Montanari use the same δ and write $\rho = k/n$, our ε/α. Kabashima, Wadayama and Tanaka write P, N for our n, d, ρ for ε, and F, with entries of variance $1/N$, for A; their conjugate fields are ours, and their $\phi_p(h; \hat{Q})$ is our single site with $\lambda = 1$. Amelunxen et al. write m for our n, s for k, τ for t. Miolane and Montanari write β for our b and $s_{\max}(\delta)$ for $\varepsilon_c(\alpha)$. El Karoui writes $\kappa = p/n$, r_ρ, c_ρ, and Donoho and Montanari $\delta = n/p$, τ, b, for our $1/\alpha$, r, c. Theorem and section numbers of Thrampoulidis et al. (2018) are those of its arXiv version 1.

References

Advani, M. and S. Ganguli (2016). "Statistical Mechanics of Optimal Convex Inference in High Dimensions". In: *Physical Review X* 6.3, p. 031034. arXiv: 1601.04650.

Amelunxen, D., M. Lotz, M. B. McCoy, and J. A. Tropp (2014). "Living on the edge: phase transitions in convex programs with random data". In: *Information and Inference: A Journal of the IMA* 3.3, pp. 224–294. arXiv: 1303.6672.

Bayati, M., M. Lelarge, and A. Montanari (2015). "Universality in polytope phase transitions and message passing algorithms". In: *The Annals of Applied Probability* 25.2, pp. 753–822. arXiv: 1207.7321.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: 1001.3448.

Bayati, M. and A. Montanari (2012). "The LASSO Risk for Gaussian Matrices". In: *IEEE Transactions on Information Theory* 58.4, pp. 1997–2017. arXiv: 1008.2581.

Celentano, M., A. Montanari, and Y. Wei (2023). "The Lasso with General Gaussian Designs with Applications to Hypothesis Testing". In: *The Annals of Statistics* 51.5, pp. 2194–2220.

Chen, S. S., D. L. Donoho, and M. A. Saunders (1998). "Atomic Decomposition by Basis Pursuit". In: *SIAM Journal on Scientific Computing* 20.1, pp. 33–61.

Donoho, D. and A. Montanari (2016). "High Dimensional Robust M-Estimation: Asymptotic Variance via Approximate Message Passing". In: *Probability Theory and Related Fields* 166.3-4, pp. 935–969. arXiv: 1310.7320.

Donoho, D. and J. Tanner (2009). "Counting Faces of Randomly Projected Polytopes when the Projection Radically Lowers Dimension". In: *Journal of the American Mathematical Society* 22.1, pp. 1–53.

Donoho, D. L. (2006). "High-Dimensional Centrally Symmetric Polytopes with Neighborliness Proportional to Dimension". In: *Discrete & Computational Geometry* 35.4, pp. 617–652.

Donoho, D. L., A. Maleki, and A. Montanari (2009). "Message-passing algorithms for compressed sensing". In: *Proceedings of the National Academy of Sciences* 106.45, pp. 18914–18919. arXiv: 0907.3574.

Donoho, D. L., A. Maleki, and A. Montanari (2011). "The Noise-Sensitivity Phase Transition in Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.10, pp. 6920–6941.

El Karoui, N. (2013). *Asymptotic behavior of unregularized and ridge-regularized high-dimensional robust regression estimators: rigorous results*. arXiv: 1311.2445.

El Karoui, N. (2018). "On the Impact of Predictor Geometry on the Performance on High-Dimensional Ridge-Regularized Generalized Robust Regression Estimators". In: *Probability Theory and Related Fields* 170.1-2, pp. 95–175.

El Karoui, N., D. Bean, P. J. Bickel, C. Lim, and B. Yu (2013). "On Robust Regression with High-Dimensional Predictors". In: *Proceedings of the National Academy of Sciences* 110.36, pp. 14557–14562.

Gordon, Y. (1988). "On Milman's inequality and random subspaces which escape through a mesh in $\mathbb{R}^n$". In: *Geometric Aspects of Functional Analysis (1986/87)*. Vol. 1317. Lecture Notes in Mathematics. Springer, pp. 84–106.

Gordon, Y. (1985). "Some inequalities for Gaussian processes and applications". In: *Israel Journal of Mathematics* 50.4, pp. 265–289.

Han, Q. and Y. Shen (2023). "Universality of Regularized Regression Estimators in High Dimensions". In: *The Annals of Statistics* 51.4, pp. 1799–1823. arXiv: 2206.07936.

Huber, P. J. (1964). "Robust Estimation of a Location Parameter". In: *The Annals of Mathematical Statistics* 35.1, pp. 73–101.

Huber, P. J. (1973). "Robust Regression: Asymptotics, Conjectures and Monte Carlo". In: *The Annals of Statistics* 1.5, pp. 799–821.

Kabashima, Y., T. Wadayama, and T. Tanaka (2009). "A Typical Reconstruction Limit for Compressed Sensing Based on L_p-Norm Minimization". In: *Journal of Statistical Mechanics: Theory and Experiment* 2009.09, p. L09003. arXiv: 0907.0914.

Miolane, L. and A. Montanari (2021). "The distribution of the Lasso: Uniform control over sparse balls and adaptive parameter tuning". In: *The Annals of Statistics* 49.4, pp. 2313–2335. arXiv: 1811.01212.

Rangan, S., A. K. Fletcher, and V. K. Goyal (2012). "Asymptotic Analysis of MAP Estimation via the Replica Method and Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 58.3, pp. 1902–1923. arXiv: 0906.3234.

Rockafellar, R. T. (1970). *Convex Analysis*. Princeton: Princeton University Press.

Stam, A. J. (1959). "Some Inequalities Satisfied by the Quantities of Information of Fisher and Shannon". In: *Information and Control* 2.2, pp. 101–112.

Stojnic, M. (2009). *Various thresholds for ℓ_1-optimization in compressed sensing*. arXiv: 0907.3666.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). "Precise Error Analysis of Regularized M-Estimators in High Dimensions". In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Thrampoulidis, C., S. Oymak, and B. Hassibi (2014). *The Gaussian min-max theorem in the presence of convexity*. arXiv: 1408.4837.

Thrampoulidis, C., S. Oymak, and B. Hassibi (2015). "Regularized Linear Regression: A Precise Analysis of the Estimation Error". In: *Proceedings of The 28th Conference on Learning Theory*. Vol. 40. Proceedings of Machine Learning Research, pp. 1683–1709.

Tibshirani, R. (1996). "Regression Shrinkage and Selection via the Lasso". In: *Journal of the Royal Statistical Society Series B* 58.1, pp. 267–288.

Tibshirani, R. J. (2013). "The Lasso Problem and Uniqueness". In: *Electronic Journal of Statistics* 7, pp. 1456–1490.

Vershynin, R. (2018). *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press.

Logistic Regression and Max-Margin Classifiers

11

The capacity problem with a teacher

THE DICTIONARY — Gardner with a teacher ↔ the CGMT

IN THIS CHAPTER. Logistic regression with n examples in d dimensions has no maximum-likelihood estimate when the data are linearly separable. For Gaussian inputs labeled by a teacher of signal strength r this happens exactly when $n/d < \alpha_{\mathrm{MLE}}(r) = 1/\min_t \mathbb{E}(Z - tV)_+^2$, which is Cover's 2 when the labels are random. Physics derives this curve, the largest margin of separable data and the error of the max-margin classifier from Gardner's volume of solutions, sliced by the overlap with the teacher; we prove all three for isotropic Gaussian inputs with the one-sided Gordon inequality, applied to the primal problem and to its dual. Beyond the threshold the estimate exists but is inflated, by factors that Sur and Candès derived and proved; we know of no earlier replica computation of them. Linear programs, Newton's method and support vector machines agree with the predictions; open are non-Gaussian inputs and the negative margins of non-separable data.

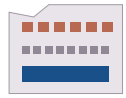
11.1 The question: when do labeled data stop being separable?

Logistic regression is the first model a statistician fits to binary labels. Given examples (x_i, y_i) with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, the maximum-likelihood estimate (MLE) minimizes

$$L_n(b) = \sum_{i=1}^{n} \ell\big(y_i \langle x_i, b\rangle\big), \qquad \ell(t) = \log\big(1 + e^{-t}\big).$$

When n is much larger than d, the MLE is consistent and asymptotically normal, and the likelihood-ratio statistic of a null coefficient is asymptotically χ_1^2. When d is a fixed fraction of n all three statements fail. The MLE may not exist: if some θ has $y_i\langle x_i, \theta\rangle > 0$ for all i, then $L_n(s\theta) \to 0 = \inf L_n$ as $s \to \infty$.[1] When it exists it overestimates the coefficients, and the χ_1^2 calibration makes p-values too small. For labels independent of the inputs, Cover (1965) computed the probability of separability through the origin exactly, $2^{1-n}\sum_{k<d}\binom{n-1}{k}$, which tends to 1 or to 0 as n/d tends to $\alpha < 2$ or to $\alpha > 2$ (the Prologue). Labels from a teacher are easier to separate, and the threshold moves: this is Gardner's capacity problem with a teacher.

[1] Albert and Anderson (1984) characterized existence: the logistic MLE exists if and only if the data are separated neither completely nor quasi-completely (with weak inequalities). For Gaussian inputs the boundary case has probability zero, because it makes some $k \le d$ of the vectors $y_i x_i$ linearly dependent.



Definition 11.1 (The teacher model). The dimension d tends to infinity, and $n = n_d$ with $n/d \to \alpha \in (0, \infty)$. The inputs $x_1, \dots, x_n$ are independent $\mathcal{N}(0, I_d)$ vectors. A fixed vector $\beta^* = \beta_d^* \in \mathbb{R}^d$ of norm $r \geq 0$, the teacher, and a measurable link $f : \mathbb{R} \to [0, 1]$ generate labels $y_i \in \{-1, +1\}$ that are conditionally independent given the inputs, with $\mathbb{P}(y_i = 1 \mid x_i) = f(\langle x_i, \beta^* \rangle)$.

Three links recur: the logistic link $f(t) = 1/(1 + e^{-t})$, for which logistic regression is well specified; the noiseless link $f(t) = \mathbf{1}\{t > 0\}$, the teacher perceptron of physics; and $r = 0$, random labels. For separability and the MLE, a covariance Σ reduces to the identity by $x \mapsto \Sigma^{-1/2} x$, with $r = \langle \beta^*, \Sigma \beta^* \rangle^{1/2}$; the maximal margin does not, since the map changes lengths. Let $U \sim \mathcal{N}(0, 1)$ stand for the teacher's field $\langle x, \beta^* \rangle / r$ of a fresh input, Y for a label with $\mathbb{P}(Y = 1 \mid U) = f(rU)$, and $Z \sim \mathcal{N}(0, 1)$ for a variable independent of both, and let $V = YU$ be the *signed teacher field*, positive when the teacher agrees with the label.

The data are *strictly separable* if some θ has $y_i \langle x_i, \theta \rangle > 0$ for all i. Their *maximal margin*, Gardner's stability,

$$\kappa_n = \max_{\|\theta\| = 1} \min_{i \leq n} y_i \langle x_i, \theta \rangle,$$

is positive exactly for strictly separable data, and its maximizer $\hat{\theta}_n$ is the *max-margin classifier*, the perceptron of optimal stability and the hard-margin support vector machine through the origin, to which gradient descent on the logistic loss converges in direction on separable data (Soudry et al. 2018). With $e^* = \beta^* / r$, the test error of a unit vector θ on an independent fresh pair (x, y) is

$$\mathrm{Err}(\theta) = \mathbb{P}\big(y \neq \operatorname{sgn}\langle x, \theta \rangle\big) = \mathcal{E}\big(\langle \theta, e^* \rangle\big), \qquad \mathcal{E}(c) = \mathbb{P}\big(cV + \sqrt{1 - c^2}\, Z < 0\big),$$

which is $\arccos(c)/\pi$ for the noiseless teacher (Theorem 11.9).

THE PHYSICAL PREDICTION · Gardner 1988, with a teacher. For Gaussian inputs labeled by a teacher of signal strength r, with $n = \alpha d$ examples:

(i) the data are separable when $\alpha < \alpha_{\mathrm{MLE}}(r)$ and not when $\alpha > \alpha_{\mathrm{MLE}}(r)$, where

$$\alpha_{\mathrm{MLE}}(r) = \frac{1}{h_{\mathrm{MLE}}(r)}, \qquad h_{\mathrm{MLE}}(r) = \inf_{t \in \mathbb{R}} \mathbb{E}\big[(Z - tV)_+^2\big]; \quad (11.1)$$

so the logistic MLE exists exactly when $\alpha > \alpha_{\mathrm{MLE}}(r)$, and $\alpha_{\mathrm{MLE}}(0) = 2$;

(ii) when $\alpha < \alpha_{\mathrm{MLE}}(r)$, the maximal margin tends to the largest κ with

$$\max_{|c| \leq 1} \big[(1 - c^2) - \alpha\, \mathbb{E}\big(\kappa - cV - \sqrt{1 - c^2}\, Z\big)_+^2\big] \geq 0, \quad (11.2)$$

the max-margin classifier has the maximizing overlap c_* with the teacher, and its error is $\mathcal{E}(c_*)$.

For the logistic link the threshold is 2.2782, 2.8991, 3.0714 and 5.4039 at $r = 1, 2, \sqrt{5}$ and 5, and $\alpha_{\mathrm{MLE}}(r)/r \to 0.985$ as $r \to \infty$ (Exercise 11.5); the probit link $f = \Phi$, steeper at the origin, gives 2.6953 at $r = 1$ and 4.5174 at $r = \sqrt{5}$ (`figures/scripts/11-classification-threshold.py`). The order of discovery is not the box's. Its second statement is, for the noiseless teacher, the replica computation of Opper et al. (1990), and for random labels Gardner's formula (Gardner 1988). Its first was found without physics: at $\beta^* = 0$ it is Cover's count, and Candès and Sur (2020) derived and proved the curve by conic integral geometry (arXiv, 2018); we know of no physics paper that computed it before them, and the derivation in Section 11.2 is the book's. Beyond the threshold, too, Sur and Candès (2019) derived and proved the bias and variance of the MLE; we know of no replica computation of them before theirs, and Aubin et al. (2020) later wrote the replica fixed point of convex losses with a teacher in a form that contains them.[2] How does a physicist derive these predictions, and how much of the derivation survives as a proof? For isotropic Gaussian inputs, all of it.

[2] Physics came first for four of the chapter's predictions only: Gardner's capacity, the error of the optimal-stability perceptron, the Bayes-optimal perceptron and the classification of clustered patterns (Notes and references). Everything about the logistic MLE was derived and proved directly by statisticians.

11.2 The physics: Gardner's volume, sliced by the teacher

Gardner's question is geometric. With the labeled inputs $z_i = y_i x_i$ and a margin κ, the *version space* $\Omega_\kappa = \{\theta \in S^{d-1} : \langle z_i, \theta\rangle \geq \kappa \text{ for } i \leq n\}$ is the set of unit weight vectors that classify every example with margin at least κ; the maximal margin is the largest κ for which it is not empty. Gardner (1988) computed the typical value of $\frac{1}{d}\log \sigma(\Omega_\kappa)$, with σ the uniform probability on the sphere, and located the capacity where it tends to $-\infty$.[3]

[3] Gardner treated random patterns and labels. The teacher–student version of her method is the subject of the book of Engel and Van den Broeck (2001), who write R for the overlap with the teacher.

SLICES. Let $V_i = y_i\langle x_i, e^*\rangle$, a copy of V, and $\tilde{z}_i = y_i(x_i - \langle x_i, e^*\rangle e^*)$, a standard Gaussian vector in the orthogonal complement of e^*, independent of V_i (Lemma 11.2). A unit vector $\theta = c\,e^* + \sqrt{1-c^2}\,\omega$, with $\omega \perp e^*$ a unit vector, has $\langle z_i, \theta\rangle = cV_i + \sqrt{1-c^2}\,\langle \tilde{z}_i, \omega\rangle$. So the slice of Ω_κ at teacher overlap c is the version space of a spherical perceptron in $d-1$ dimensions with quenched random margins

$$\kappa_i(c) = \frac{\kappa - cV_i}{\sqrt{1-c^2}}, \tag{11.3}$$

of both signs: easy where the teacher agrees strongly with the label, hard where it disagrees. The slice has measure proportional to $(1-c^2)^{(d-3)/2}\,\mathrm{d}c$, so on the exponential scale $\frac{1}{d}\log\sigma(\Omega_\kappa) \approx \max_c[\frac{1}{2}\log(1-c^2) + s(c)]$, with $s(c)$ the typical log-volume per dimension of the slice relative to its own sphere: the largest slice wins.

REPLICAS FOR ONE SLICE. Fix c, let X be the relative volume of the slice, and use $\mathbb{E}\log X = \lim_{m\to 0}(\mathbb{E}X^m - 1)/m$. For integer m, $\mathbb{E}X^m$ is the probability that m independent uniform points ω^a of the slice's sphere all satisfy the constraints; given their overlaps $Q_{ab} = \langle \omega^a, \omega^b\rangle$ the fields $\langle \tilde{z}_i, \omega^a\rangle$ are $\mathcal{N}(0, Q)$, independently over i, so with $\lambda \sim \mathcal{N}(0, Q)$

$$\mathbb{E}X^m = \int \exp\Big\{d\Big[\tfrac{1}{2}\log\det Q + \alpha\log\mathbb{P}\big(\lambda^a \geq \kappa_1(c),\ a \leq m\big)\Big] + o(d)\Big\}\,\mathrm{d}Q,$$

the probability including the average over V_1 and $\frac{1}{2}\log\det Q$ being the entropy of m-tuples with Gram matrix Q. Laplace's method selects a saddle point, and the replica-symmetric ansatz $Q_{ab} = q$ $(a \neq b)$ makes it explicit: the eigenvalues of Q are $1-q$, $m-1$ times, and $1-q+mq$, so $\frac{1}{2}\log\det Q = \frac{m}{2}[\log(1-q)+\frac{q}{1-q}]+O(m^2)$; and with $\lambda^a = \sqrt{q}\,z+\sqrt{1-q}\,\xi_a$ for independent standard normals, the probability is $\mathbb{E}[H((\kappa_1-\sqrt{q}\,z)/\sqrt{1-q})^m] = 1 + m\mathbb{E}\log H(\cdot) + O(m^2)$, with $H(x) = \mathbb{P}(Z > x)$. As $m \to 0$,

$$s(c) = \min_{0\le q<1}\Big[\tfrac{1}{2}\log(1-q) + \frac{q}{2(1-q)} + \alpha\,\mathbb{E}\log H\Big(\frac{\kappa_1(c)-\sqrt{q}\,Z}{\sqrt{1-q}}\Big)\Big], \qquad (11.4)$$

the expectation being over Z and V: Gardner's formula with random margins (Exercise 11.6 writes it with the overlap of two whole weight vectors). The extremum in q is a minimum, as in her computation: the typical log-volume lies below its average over the fields with the margins fixed, which is the value at $q = 0$. ◇

THE CAPACITY OF A SLICE. As the load grows, the typical solutions of a slice approach each other and the minimizing q tends to 1. Since $\log H(x) = -\frac{x^2}{2} + O(\log x)$ as $x \to +\infty$ and $\log H(x) \to 0$ as $x \to -\infty$, the bracket of (11.4) is $[1-\alpha\mathbb{E}(\kappa_1(c)-Z)_+^2+o(1)]/(2(1-q))$ as $q \to 1$. If $\alpha\mathbb{E}(\kappa_1(c)-Z)_+^2 < 1$, the minimum sits at some $q < 1$ and the slice has a finite log-volume; if it is larger than 1, the minimum is $-\infty$ and the slice is empty. By (11.3) the *capacity of the slice* is

$$\alpha_\kappa(c) = \frac{1}{\mathbb{E}(\kappa_1(c)-Z)_+^2} = \frac{1-c^2}{\mathbb{E}\big(\kappa - cV - \sqrt{1-c^2}\,Z\big)_+^2}, \qquad (11.5)$$

and the version space is not empty while $\alpha < \max_c \alpha_\kappa(c)$, which is (11.2). ◇

Three readings give the physicsbox. For random labels, $cV + \sqrt{1-c^2}Z$ is standard normal for every c, the best slice is $c = 0$, and the capacity is Gardner's $1/\mathbb{E}(Z+\kappa)_+^2$ (Chapter 8). For a teacher and $\kappa = 0$, dividing by $1-c^2$ and putting $t = c/\sqrt{1-c^2}$ gives $\alpha_0(c) = 1/\mathbb{E}(-tV-Z)_+^2 = 1/\mathbb{E}(Z-tV)_+^2$, and maximizing over c is minimizing over t: this is (11.1). For $\kappa > 0$, the largest margin the load allows, with its best slice c_*, is the second statement, and for the noiseless teacher the computation of Opper et al. (1990).

A CHECK BY COUNTING. At the threshold the best slice shrinks to one direction, which d active constraints pin down, so about d examples should lie on the margin. Gaussian integration by parts in Z (Theorem A.1) gives $\mathbb{E}[Z(Z-tV)_+] = \mathbb{P}(Z > tV)$, hence

$$\mathbb{E}(Z-tV)_+^2 = \mathbb{P}(Z > tV) - t\,\mathbb{E}\big[V(Z-tV)_+\big],$$

and the last expectation is $-\frac{1}{2}$ times the derivative of $\mathbb{E}(Z-tV)_+^2$ in t, which vanishes at the minimizer t_{MLE}. So $h_{\mathrm{MLE}} = \mathbb{P}(Z > t_{\mathrm{MLE}}V)$: the fraction of examples on the margin is $1/\alpha_{\mathrm{MLE}}$, and their number is d. The threshold is *isostatic*, the boundary case of the perceptron viewed as a model of jamming, which has fewer contacts than dimensions at positive margin and is "isostatic and critical" at negative margin (Franz and Parisi 2016).

ZERO-TEMPERATURE REPLICAS FOR THE MLE. When $\alpha > \alpha_{\mathrm{MLE}}(r)$ the MLE exists, and the Gibbs measure $e^{-\beta L_n(b)}\,\mathrm{d}b$ concentrates on it as $\beta \to \infty$. Its replicas have the order parameters $a = \langle b, e^* \rangle$, $Q_0 = \|b\|^2$ and $q = \langle b^1, b^2 \rangle$; replica symmetry at zero temperature means $q \to Q_0$ with the *susceptibility* $\chi = \beta(Q_0 - q)$ fixed, and $\tau^2 = q - a^2$ tends to the squared norm of the part of b orthogonal to the teacher. The labeled field of a fresh example under replica b^a is $aV + \tau Z + \sqrt{\chi/\beta}\,\xi_a$, so the energy term is a Laplace integral,

$$\alpha\,\mathbb{E}\log\int e^{-\beta\ell(aV+\tau Z+\sqrt{\chi/\beta}\,\xi)}\,\mathrm{D}\xi = -\alpha\beta\,\mathbb{E}\,e_\ell(aV+\tau Z;\chi) + o(\beta),$$

with the Moreau envelope $e_\ell(x;\chi) = \min_v[\ell(v) + (v-x)^2/(2\chi)]$; the entropy term, from the eigenvalues of the Gram matrix of the parts orthogonal to the teacher as before, is $\frac{1}{2}\log\frac{\chi}{\beta} + \frac{\beta\tau^2}{2\chi}$. The minimal loss per example is the stationary value of

$$\Psi(a,\tau,\chi) = \mathbb{E}\,e_\ell(aV+\tau Z;\chi) - \frac{\tau^2}{2\alpha\chi}. \tag{11.6}$$

Let $p = \mathrm{prox}_{\chi\ell}(aV+\tau Z)$ be the minimizer in e_ℓ, the *proximal map*; then $\partial_x e_\ell = \ell'(p)$, $\partial_\chi e_\ell = -\frac{1}{2}\ell'(p)^2$ and $\partial_x p = 1/(1+\chi\ell''(p))$, and with Gaussian integration by parts for the derivative in τ, the stationarity conditions in a, χ and τ are

$$\mathbb{E}\big[V\ell'(p)\big] = 0, \qquad \tau^2 = \alpha\chi^2\,\mathbb{E}\big[\ell'(p)^2\big], \qquad 1 = \alpha\chi\,\mathbb{E}\Big[\frac{\ell''(p)}{1+\chi\ell''(p)}\Big]. \quad \diamond \tag{11.7}$$

This is the system (5)–(7) of Sur and Candès (2019) in the book's units, with their $(\alpha_\star, \sigma_\star, \lambda_\star)$ our $(a/r, \tau\sqrt{\alpha}, \chi)$ (Exercise 11.2). The MLE is $\mu\beta^*$ plus a vector of norm τ orthogonal to β^*, with the *bias factor* $\mu = a/r > 1$: the zero-temperature student is overconfident. The equations read as a cavity computation: the margin of example i under the fit without it is Gaussian, $aV + \tau Z$, its fitted margin is the proximal map of that, and χ measures how far an example pulls the fit toward itself (Exercise 11.8).

WHAT WAS NOT CONTROLLED. The derivations take five steps that a mathematician would not: the continuation of integer moments to $m \to 0$; the exchange of $m \to 0$ with $d \to \infty$; the replica-symmetric ansatz; the choice of the extremum, a minimum in q and a maximum over slices; and the limit $q \to 1$ inside the replica-symmetric formula. The slices add a sixth: their margins take both signs, and for the spherical perceptron with negative margin replica symmetry breaks at capacity (Chapter 8).

11.3 The stone

The proof follows the derivation slice by slice, but it never computes a volume.

PHYSICS · *Gardner's volume with a teacher*	PROOF · *Gordon's inequality, primal and dual*
1 · Slice by the teacher	
Cut the version space by the overlap c with the teacher: each slice is a spherical perceptron with quenched margins $(\kappa - cV_i)/\sqrt{1-c^2}$.	Rotate the teacher to e_1 and condition on V; the rest of the data is Gaussian (Lemma 11.2). In place of a volume, the smallest violation $\|(\kappa\mathbf{1} - Z\theta)_+\|$ at angles in a set T.
2 · Replicate, and let the volume shrink to a point	
$\mathbb{E}X^m$, Gaussian fields with covariance Q_{ab}, Laplace's method, $Q_{ab} = q$ and $m \to 0$ give (11.4); as $q \to 1$ the coefficient of $1/(1-q)$ changes sign at the slice capacity $\alpha_\kappa(c)$.	No replicas, no ansatz, no temperature: Gordon's comparison (Lemma 11.4), the auxiliary problem in closed form (Computation 11.1) and its limit $\Gamma_\kappa(c)$ (Lemma 11.5), positive exactly when $\alpha > \alpha_\kappa(c)$; then the slice is empty.
3 · Pick the best slice	
The version space is not empty while $\alpha < \max_c \alpha_\kappa(c)$; the best slice gives κ_* and c_*.	Gordon's inequality is one-sided; duality gives the other side, Gordon's inequality for the distance from the origin to the convex hull of the z_i at the best slice (Lemmas 11.3 and 11.6): the scalar problem of Montanari, Ruan, Sohn and Yan.
4 · Trust replica symmetry	
Assumed for every slice, although the margins take both signs.	Not needed: the primal bound holds for every slice; the best slice is unique by strict convexity (Lemma 11.7).
5 · Beyond the threshold: the cavity field	
Zero temperature, $\chi = \beta(Q_0 - q)$ fixed; the fitted margin is $\mathrm{prox}_{\chi\ell}$ of the Gaussian cavity margin $aV + \tau Z$; stationarity of Ψ.	Sur and Candès: message passing with fixed point (11.7), and leave-one-out (Theorem 11.10); or the CGMT, whose auxiliary problem reduces to Ψ (Salehi et al. 2019).

WHAT THE TRANSLATION SHOWS. Both sides compute the same numbers: the slice capacity $\alpha_\kappa(c)$ is where $\Gamma_\kappa(c)$, the limit of Gordon's auxiliary problem, changes sign. The proof justifies the replica-symmetric capacity of every slice as an upper bound, with no stability analysis, and replaces the choice of the largest slice by duality, the one place where convexity enters. The routes part beyond the threshold, where the maximal margin is negative and the problem is not convex: Gordon's inequality gives only a bound there, and replica symmetry is expected to fail (Section 11.8).

11.4 The proof: Gordon's inequality, primal and dual

This section proves the physicsbox for the teacher model: the threshold (Theorem 11.8), and the maximal margin with the error of the max-margin classifier (Theorem 11.9). The tool is the one-sided half of the convex Gaussian min–max theorem (CGMT) of Chapter 8; the arrangement, applying it once to the primal problem and once to its dual, is that of Stojnic (2013a) for random labels, and Montanari et al. (2025) reach the same scalar problem for the max-margin classifier with the two-sided CGMT.

11.4.1 Reductions

Lemma 11.2 (The labeled inputs). *In the teacher model, let $e^* = \beta^*/r$ (a fixed unit vector if $r = 0$) and $z_i = y_i x_i$. Then $z_i = V_i e^* + \tilde{z}_i$, where $V_1, \dots, V_n$ are independent copies of V, the vectors $\tilde{z}_i$ are independent $\mathcal{N}(0, I_d - e^* e^{*\top})$, and the two families are independent. The density of V is $(f(rv) + 1 - f(-rv))\phi(v) \le 2\phi(v)$, and $\mathbb{E}V^2 = 1$.*

Proof. Write $x_i = U_i e^* + P x_i$ with $U_i = \langle x_i, e^* \rangle$ and $P = I_d - e^* e^{*\top}$. The Gaussian variables U_i and $P x_i$ are uncorrelated, hence independent, and y_i depends on x_i only through U_i and independent randomness; so given (U_i, y_i) both $P x_i$ and $\tilde{z}_i = y_i P x_i$ are $\mathcal{N}(0, P)$, and $\tilde{z}_i$ is independent of $V_i = y_i U_i$. The density follows from $\mathbb{P}(V \in B) = \mathbb{E}[\mathbf{1}\{U \in B\} f(rU)] + \mathbb{E}[\mathbf{1}\{-U \in B\}(1 - f(rU))]$, and $V^2 = U^2$. ∎

Lemma 11.3 (Separability, the MLE and the maximal margin). *Let $z_1, \dots, z_n \in \mathbb{R}^d$, $\kappa_n = \max_{\|\theta\|=1} \min_i \langle z_i, \theta \rangle$, $\mathcal{S}_n = \{\lambda \in \mathbb{R}^n : \lambda \ge 0,\ \sum_i \lambda_i = 1\}$, $D_n = \min_{\lambda \in \mathcal{S}_n} \|\sum_i \lambda_i z_i\|$, the distance from the origin to the convex hull of the z_i, and $L_n(b) = \sum_i \ell(\langle z_i, b \rangle)$. Then $(\kappa_n)_+ = D_n$. If $\kappa_n > 0$, then L_n has no minimizer. If $\kappa_n < 0$, then L_n has a minimizer, which is unique if the z_i span $\mathbb{R}^d$.*

Proof. The plan is to compare κ_n with D_n by averaging and by the nearest point of the hull, and to follow L_n along rays. For every unit θ and $\lambda \in \mathcal{S}_n$, $\min_i \langle z_i, \theta \rangle \le \sum_i \lambda_i \langle z_i, \theta \rangle \le \|\sum_i \lambda_i z_i\|$, so $\kappa_n \le D_n$. If $D_n > 0$, let p be the point of the compact convex hull K nearest to the origin. For $k \in K$ and $s \in (0, 1]$, $\|p + s(k - p)\|^2 \ge \|p\|^2$; dividing by s and letting $s \downarrow 0$ gives $\langle p \rangle k \ge \|p\|^2$, so $\theta = p/\|p\|$ has $\langle z_i, \theta \rangle \ge D_n$ for all i, and $\kappa_n \ge D_n$. Hence $(\kappa_n)_+ = D_n$. If $\kappa_n > 0$ with maximizer $\hat{\theta}_n$, then $L_n(s\hat{\theta}_n) \le n\ell(s\kappa_n) \to 0$ as $s \to \infty$, while $L_n > 0$. If $\kappa_n < 0$, every b has $\min_i \langle z_i, b \rangle \le \kappa_n \|b\|$, and since ℓ is decreasing with $\ell(t) \ge -t$, $L_n(b) \ge |\kappa_n| \|b\|$: the continuous function L_n is coercive and has a minimizer. Its Hessian $\sum_i \ell''(\langle z_i, b \rangle) z_i z_i^\top$ is positive definite when the z_i span $\mathbb{R}^d$, because $\ell'' > 0$, and the minimizer is then unique. ∎

When $\kappa_n > 0$ the maximizer $\hat{\theta}_n$ is unique: the normalized average of two maximizers would have a larger margin.

11.4.2 Gordon's inequality and the primal bound

We use the CGMT of Theorem 8.3 only in its one-sided form, which needs no convexity, writing $\mathcal{P}$ and $\mathcal{A}$ for the values that Thrampoulidis et al. (2018) call Φ and ϕ, letters we keep for the Gaussian distribution function and density.

> **Theorem** (The one-sided CGMT). *Let $G \in \mathbb{R}^{m\times p}$, $g \in \mathbb{R}^m$ and $h \in \mathbb{R}^p$ have independent standard normal entries, let $S_w \subset \mathbb{R}^p$ and $S_u \subset \mathbb{R}^m$ be non-empty compact sets, and let ψ be continuous on $S_w \times S_u$. Let*
>
> $$\mathcal{P}(G) = \min_{w\in S_w}\max_{u\in S_u}\big[u^\top G w + \psi(w,u)\big],$$
> $$\mathcal{A}(g,h) = \min_{w\in S_w}\max_{u\in S_u}\big[\|w\|\, g^\top u + \|u\|\, h^\top w + \psi(w,u)\big].$$
>
> *Then $\mathbb{P}(\mathcal{P}(G) < s) \le 2\mathbb{P}(\mathcal{A}(g,h) \le s)$ for every $s \in \mathbb{R}$.*

This is Gordon's comparison inequality (Gordon 1985) in the form of Thrampoulidis et al. (2018, Theorem 6.1(i)). We apply it conditionally on $V = (V_1, \dots, V_n)$, which is independent of the Gaussian parts, with a function ψ that depends on V.

Let Z be the $n \times d$ matrix with rows $z_i^\top$ and $\mathbf{1} \in \mathbb{R}^n$ the vector of ones. For $\kappa \in \mathbb{R}$ and a closed set $T \subset [0,\pi]$ of angles, the smallest violation of the margin κ by a unit vector at an angle in T from the teacher, and its auxiliary counterpart, are

$$\Delta_T(\kappa) = \min\Big\{\|(\kappa\mathbf{1} - Z\theta)_+\| :\ \|\theta\| = 1,\ \arccos\langle\theta, e^*\rangle \in T\Big\},$$
$$\delta_T(\kappa) = \min_{t\in T}\Big(\|(\kappa\mathbf{1} + \sin t\, g - \cos t\, V)_+\| - \sin t\, \|h\|\Big)_+,$$

with $g \sim \mathcal{N}(0, I_n)$ and $h \sim \mathcal{N}(0, I_{d-1})$ independent of each other and of V. The violation $\Delta_T(\kappa)$ vanishes exactly when some unit vector at an angle in T has margin at least κ.

> **Lemma 11.4** (The primal bound). *In the teacher model, $\mathbb{P}(\Delta_T(\kappa) < s) \le 4\mathbb{P}(\delta_T(\kappa) \le s)$ for all $\kappa, s \in \mathbb{R}$ and every closed set $T \subset [0,\pi]$.*

Proof. The plan is to write each half of $\Delta_T(\kappa)$ as a Gaussian min–max. Rotate so that $e^* = e_1$; then $Z = [V \mid \tilde{Z}]$ with $\tilde{Z} \in \mathbb{R}^{n\times(d-1)}$ standard Gaussian and independent of V (Lemma 11.2). Split T into $T_+ = T \cap [0, \frac{\pi}{2}]$ and $T_- = T \cap [\frac{\pi}{2}, \pi]$, so that $\Delta_T = \min(\Delta_{T_+}, \Delta_{T_-})$, a minimum over the empty set being $+\infty$. A unit vector at an angle in T_+ is $\theta = (\sqrt{1-\|w\|^2}, w)$ with w in the compact set $W_+ = \{w : \|w\| \in \sin T_+\}$, as sin is one-to-one on $[0, \frac{\pi}{2}]$. With $\mathcal{U} = \{u \in \mathbb{R}^n : u \ge 0,\ \|u\| \le 1\}$ and $\max_{u\in\mathcal{U}} u^\top a = \|a_+\|$ (step 2 of Computation 11.1),

$$\|(\kappa\mathbf{1} - Z\theta)_+\| = \max_{u\in\mathcal{U}}\Big[u^\top(-\tilde{Z})w + \kappa\, u^\top\mathbf{1} - \sqrt{1-\|w\|^2}\, u^\top V\Big].$$

So $\Delta_{T_+}(\kappa) = \mathcal{P}(-\tilde{Z})$ with $S_w = W_+$, $S_u = \mathcal{U}$ and a continuous ψ, and $\mathbb{P}(\Delta_{T_+} < s) \le 2\mathbb{P}(\mathcal{A} \le s)$, where $\mathcal{A}$ is the minimum over T_+ in $\delta_T(\kappa)$ by

Computation 11.1. The half over T_- is the same with $\theta_1 = -\sqrt{1-\|w\|^2}$. Both auxiliary halves are at least δ_T, and adding the bounds for the two halves gives the factor 4. ∎

COMPUTATION 11.1 · THE AUXILIARY PROBLEM OF THE MARGIN CONSTRAINTS

1 The auxiliary problem of the half over T_+ is $\min_{w\in W_+}\max_{u\in\mathcal{U}}[\|w\|\, g^\top u + \|u\|\, h^\top w + \kappa\, u^\top \mathbf{1} - \sqrt{1-\|w\|^2}\, u^\top V]$. Fix w with $\|w\| = \sin t$, $t \in T_+$, and put $a = \kappa\mathbf{1} + \sin t\, g - \cos t\, V$ and $b = h^\top w$; the objective is $u^\top a + \|u\|\, b$.

2 Write $u = \rho\bar{u}$ with $\rho \in [0,1]$ and $\bar{u} \geq 0$ a unit vector. If a has a positive entry, $\bar{u}^\top a \leq \bar{u}^\top a_+ \leq \|a_+\|$, with equality at $\bar{u} = a_+/\|a_+\|$; otherwise $\max_{\bar{u}} \bar{u}^\top a = \max_i a_i \leq 0$. Calling this maximum $a^\sharp$, the maximum over ρ is $(a^\sharp + b)_+$.

3 Over the sphere $\|w\| = \sin t$, b takes every value in $[-\sin t\|h\|, \sin t\|h\|]$ and $(a^\sharp + b)_+$ is nondecreasing in b, so the minimum is $(a^\sharp - \sin t\|h\|)_+ = (\|a_+\| - \sin t\|h\|)_+$, in both cases of step 2.

4 Minimizing over $t \in T_+$ gives the half of $\delta_T(\kappa)$ over T_+; the half over T_- is the same with $\cos t \leq 0$.

11.4.3 The limit, and the dual bound

The limit of the auxiliary problem is the replica criterion for an empty slice. For $\kappa \in \mathbb{R}$ and $c \in [-1,1]$ let

$$\Gamma_\kappa(c) = \Big(\mathbb{E}\big(\kappa - cV - \sqrt{1-c^2}\, Z\big)_+^2\Big)^{1/2} - \sqrt{\frac{1-c^2}{\alpha}}; \tag{11.8}$$

by (11.5), $\Gamma_\kappa(c) > 0$ exactly when $\alpha > \alpha_\kappa(c)$.

Lemma 11.5 (The auxiliary problem concentrates). *For every $\kappa \in \mathbb{R}$, the supremum over $t \in [0,\pi]$ of $\big|n^{-1/2}\big(\|(\kappa\mathbf{1} + \sin t\, g - \cos t\, V)_+\| - \sin t\, \|h\|\big) - \Gamma_\kappa(\cos t)\big|$ tends to 0 in probability. Consequently $n^{-1/2}\delta_T(\kappa) \to \min_{t\in T}\Gamma_\kappa(\cos t)_+$ in probability, for every closed set $T \subset [0,\pi]$.*

PROOF. The plan is the law of large numbers at each angle and a Lipschitz bound for uniformity. Call the normalized difference $A_n(t)$. For fixed t, $\frac{1}{n}\|(\kappa\mathbf{1} + \sin t\, g - \cos t\, V)_+\|^2$ is an average of n independent copies of $(\kappa + \sin t\, Z - \cos t\, V)_+^2$, so by the law of large numbers it tends to $\mathbb{E}(\kappa - \cos t\, V - \sin t\, Z)_+^2$ (replace Z by $-Z$), and $\|h\|^2/n \to 1/\alpha$; so $A_n(t) \to \Gamma_\kappa(\cos t)$. Since $|\|x_+\| - \|y_+\|| \leq \|x - y\|$, A_n is Lipschitz with constant $L_n = (\|g\| + \|V\| + \|h\|)/\sqrt{n} \to 2 + \alpha^{-1/2}$, and by Minkowski's inequality $t \mapsto \Gamma_\kappa(\cos t)$ is $(2+\alpha^{-1/2})$-Lipschitz. On a grid of mesh ε, with probability tending to one $L_n \leq 3 + \alpha^{-1/2}$ and $|A_n - \Gamma_\kappa(\cos\cdot)| \leq \varepsilon$ at the grid points, and then everywhere $|A_n - \Gamma_\kappa(\cos\cdot)| \leq (6 + 2\alpha^{-1/2})\varepsilon$. The minimum over T and the positive part are 1-Lipschitz in the supremum norm. ∎

The primal bound can show that no unit vector has margin κ, never that one has. For that we use $(\kappa_n)_+ = D_n$: a lower bound on the minimum D_n is again a statement about a lower tail.

Lemma 11.6 (The dual bound). *In the teacher model, fix $c \in (-1,1)$, let $R = \sqrt{1-c^2}$, let $g \sim \mathcal{N}(0, I_{d-1})$ and $h \sim \mathcal{N}(0, I_n)$ be independent of each other and of V, and put*

$$\varkappa_n(c) = \max\Big\{\kappa \in \mathbb{R} : \ \|(\kappa \mathbf{1} - Rh - cV)_+\| \le R\|g\|\Big\}.$$

Then $\mathbb{P}(D_n < s) \le 2\mathbb{P}(\varkappa_n(c) \le s)$ for every $s \in \mathbb{R}$, and $\varkappa_n(c) \to \varkappa(c)$ in probability, where $\varkappa(c)$ is the unique κ with $\Gamma_\kappa(c) = 0$.

Proof. The plan is to restrict the inner maximum in D_n to the slice at c and to bound the auxiliary problem below. Rotate so that $e^* = e_1$. Since $\|Z^\top \lambda\| = \max_{\|\theta\|\le 1} \lambda^\top Z\theta$ and every $\theta = (c, w)$ with $\|w\| \le R$ has norm at most one, $D_n \ge \min_{\lambda\in\mathcal{S}_n} \max_{\|w\|\le R}[w^\top \tilde{Z}^\top \lambda + c\,\lambda^\top V]$, which is $\mathcal{P}(\tilde{Z}^\top)$ with $S_w = \mathcal{S}_n$, $S_u = \{w : \|w\| \le R\}$ and $\psi(\lambda, w) = c\lambda^\top V$. The maximum over w in its auxiliary problem is attained along g at norm R or at $w = 0$, so

$$\mathcal{A} = \min_{\lambda\in\mathcal{S}_n}\Big[R\big(\|g\|\|\lambda\| + h^\top\lambda\big)_+ + c\lambda^\top V\Big].$$

Let $\kappa = \varkappa_n(c)$ and $\lambda \in \mathcal{S}_n$. Since $\sum_i \lambda_i = 1$, $\lambda \ge 0$ and $x \le x_+$, Cauchy–Schwarz gives

$$\begin{aligned} R\|g\|\|\lambda\| + \lambda^\top(Rh + cV) &= \kappa + R\|g\|\|\lambda\| - \lambda^\top(\kappa\mathbf{1} - Rh - cV) \\ &\ge \kappa + \|\lambda\|\big(R\|g\| - \|(\kappa\mathbf{1} - Rh - cV)_+\|\big) \ge \kappa, \end{aligned}$$

so $\mathcal{A} \ge \varkappa_n(c)$, and the one-sided CGMT gives the inequality. For the limit, $k(\kappa) = \mathbb{E}(\kappa - cV - RZ)_+^2$ is continuous and strictly increasing, since $\mathbb{P}(\kappa - cV - RZ > 0) > 0$ for every κ, with limits 0 and ∞, so the root $\varkappa(c)$ of $k(\kappa) = R^2/\alpha$ is unique. For $\kappa' < \varkappa(c) < \kappa''$, the law of large numbers gives $\frac{1}{n}\|(\kappa'\mathbf{1} - Rh - cV)_+\|^2 \to k(\kappa') < R^2/\alpha$, the same for κ'' with the limit $k(\kappa'') > R^2/\alpha$, and $R^2\|g\|^2/n \to R^2/\alpha$; the norm being nondecreasing in κ, $\kappa' \le \varkappa_n(c) < \kappa''$ with probability tending to one. ∎

11.4.4 The scalar problem and the theorems

Lemma 11.7 (The scalar problem). *In the teacher model, let $M(\kappa) = \min_{|c|\le 1} \Gamma_\kappa(c)$ and $F_\kappa(t) = \mathbb{E}\big(\kappa\sqrt{1+t^2} - tV - Z\big)_+^2$.*

(i) For $|c| < 1$ and $t = c/\sqrt{1-c^2}$, $\Gamma_\kappa(c) = \sqrt{1-c^2}\,\big(F_\kappa(t)^{1/2} - \alpha^{-1/2}\big)$; and $F_0(t) = \mathbb{E}(Z - tV)_+^2$.

(ii) For $\kappa \ge 0$, F_κ is strictly convex; for $\kappa > 0$, $F_\kappa(t) \to \infty$ as $|t| \to \infty$.

(iii) If $\alpha > \alpha_{\mathrm{MLE}}(r)$, then $M(0) > 0$.

(iv) If $\alpha < \alpha_{\mathrm{MLE}}(r)$, there is a unique $\kappa_* > 0$ with $M(\kappa_*) = 0$; $M < 0$ on $[0, \kappa_*)$ and $M > 0$ on (κ_*, ∞). The function Γ_{κ_*} is nonnegative

and vanishes at exactly one point $c_* = t_*/\sqrt{1+t_*^2}$, where t_* is the minimizer of F_{κ_*}; and $F_{\kappa_*}(t_*) = 1/\alpha$.

Proof. The plan is to pass from the overlap c to the slope t, where convexity is visible. For (i), $(\kappa - cV - \sqrt{1-c^2}Z)/\sqrt{1-c^2} = \kappa\sqrt{1+t^2} - tV - Z$; square the positive parts and take expectations, and for F_0 replace Z by $-Z$.

For (ii), $A(t) = \kappa\sqrt{1+t^2} - tV - Z$ is convex in t when $\kappa \geq 0$, and $x \mapsto x_+^2$ is convex and nondecreasing, so for $t_0 \neq t_1$ and $t_s = (1-s)t_0 + st_1$ with $s \in (0,1)$,

$$A(t_s)_+^2 \leq \big((1-s)A(t_0) + sA(t_1)\big)_+^2 \leq (1-s)A(t_0)_+^2 + sA(t_1)_+^2.$$

The second inequality is strict when $A(t_0) \neq A(t_1)$ and $\max(A(t_0), A(t_1)) > 0$, by the strict convexity of x^2 if both are positive and because $s^2 < s$ and $(1-s)^2 < 1-s$ if one is not; this event has positive probability, because $A(t_0) - A(t_1)$ is a nonconstant affine function of V, which has a density, and Z is unbounded below given V. If $\kappa > 0$ and $t \to +\infty$, then $A(t) \geq t(\kappa - V) - Z \to \infty$ on the event $\{V < \kappa/2\}$, of positive probability since $|V| = |U|$, and Fatou's lemma gives $F_\kappa(t) \to \infty$; $t \to -\infty$ is the same with $\{V > -\kappa/2\}$.

For (iii), if $\mathbb{P}(V < 0) = 0$, then $V > 0$ almost surely and $\mathbb{E}(Z - tV)_+^2 \to 0$ as $t \to \infty$, so $\alpha_{\mathrm{MLE}} = \infty$; likewise if $\mathbb{P}(V > 0) = 0$. So $\alpha > \alpha_{\mathrm{MLE}}$ forces V to take both signs; then $\Gamma_0(\pm 1) = (\mathbb{E}(\mp V)_+^2)^{1/2} > 0$, and for $|c| < 1$, by (i), $\Gamma_0(c)$ has the sign of $F_0(t) - 1/\alpha \geq h_{\mathrm{MLE}} - 1/\alpha > 0$. As Γ_0 is continuous on $[-1,1]$, $M(0) > 0$.

For (iv), by Minkowski's inequality $\Gamma_\kappa(c)$ is 1-Lipschitz and nondecreasing in κ, so M is continuous and nondecreasing. For $\kappa > 0$, $\mathbb{P}(\kappa - cV - \sqrt{1-c^2}Z > 0) > 0$ for every c (for $c = \pm 1$ because $\mathbb{P}(|V| < \kappa) = \mathbb{P}(|U| < \kappa) > 0$), so $\Gamma_\kappa(c)$ is strictly increasing in $\kappa \geq 0$; with c' a minimizer at $\kappa' > \kappa \geq 0$, $M(\kappa') = \Gamma_{\kappa'}(c') > \Gamma_\kappa(c') \geq M(\kappa)$. Some t has $F_0(t) < 1/\alpha$, so $M(0) < 0$ by (i); and $M(\kappa) \geq (\mathbb{E}(\kappa - |V| - |Z|)_+^2)^{1/2} - \alpha^{-1/2} \to \infty$. So M has exactly one zero κ_*, with the stated signs. Then $\Gamma_{\kappa_*} \geq 0$ with $\Gamma_{\kappa_*}(\pm 1) > \Gamma_0(\pm 1) \geq 0$, so by (i) its zeros are the points where $F_{\kappa_*} = 1/\alpha$, while $F_{\kappa_*} \geq 1/\alpha$: they are the minimizers of F_{κ_*}, which exist by (ii) and are unique by strict convexity. ∎

Theorem 11.8 (The separability threshold). *In the teacher model, let $\alpha_{\mathrm{MLE}}(r) = 1/h_{\mathrm{MLE}}(r)$ with $h_{\mathrm{MLE}}(r) = \inf_{t\in\mathbb{R}} \mathbb{E}(Z - tV)_+^2$.*

(i) If $\alpha < \alpha_{\mathrm{MLE}}(r)$, then with probability tending to one the data are strictly separable and the logistic MLE does not exist.

(ii) If $\alpha > \alpha_{\mathrm{MLE}}(r)$, then with probability tending to one every unit vector misclassifies some example, $\kappa_n < 0$, and the logistic MLE exists and is unique.

(iii) $\alpha_{\mathrm{MLE}}(r) \geq 2$, with equality if $r = 0$; and $\alpha_{\mathrm{MLE}}(r) > 2$ if $\mathbb{E}V \neq 0$, which holds for every increasing nonconstant link when $r > 0$.

Proof. The plan is to turn the sign of the scalar problem into probabilities with Lemmas 11.4 and 11.6. For (i), some t has $\mathbb{E}(Z - tV)_+^2 < 1/\alpha$; with

$c = t/\sqrt{1+t^2}$, Lemma 11.7(i) gives $\Gamma_0(c) < 0$, so the zero $\varkappa(c)$ of the increasing function $\kappa \mapsto \Gamma_\kappa(c)$ is positive, and Lemma 11.6 gives $\mathbb{P}(D_n < \varkappa(c)/2) \le 2\mathbb{P}(\varkappa_n(c) \le \varkappa(c)/2) \to 0$. On the complement $\kappa_n = D_n > 0$ (Lemma 11.3): the data are strictly separable and the MLE does not exist.

For (ii), $\eta = M(0) > 0$ by Lemma 11.7(iii), so by Lemmas 11.4 and 11.5 with $T = [0,\pi]$, $\mathbb{P}(\Delta_{[0,\pi]}(0) < \eta\sqrt{n}/2) \le 4\mathbb{P}(n^{-1/2}\delta_{[0,\pi]}(0) \le \eta/2) \to 0$. On the complement every unit θ has $\|(-Z\theta)_+\| > 0$, that is, some $\langle z_i, \theta\rangle < 0$; so the maximum κ_n of the continuous function $\theta \mapsto \min_i\langle z_i,\theta\rangle$ over the sphere is negative, and the MLE exists (Lemma 11.3). Since $\alpha > \alpha_{\mathrm{MLE}} \ge 2$ by (iii), eventually $n \ge d$, and the z_i span $\mathbb{R}^d$ almost surely, which gives uniqueness.

For (iii), $h_{\mathrm{MLE}} \le \mathbb{E}Z_+^2 = \frac{1}{2}$. If $r = 0$, V is standard normal and $\mathbb{E}(Z - tV)_+^2 = \frac{1}{2}(1+t^2)$ is smallest at $t = 0$. If $\mathbb{E}V \ne 0$, the derivative $-2\mathbb{E}[VZ_+] = -2\mathbb{E}V\,\mathbb{E}Z_+$ of this convex function at $t = 0$ is not zero, so its minimum is below $\frac{1}{2}$; and $\mathbb{E}V = 2\,\mathrm{Cov}(U, f(rU)) > 0$ for an increasing nonconstant link and $r > 0$. ∎

This is Theorems 1 and 2 of Candès and Sur (2020) for the logistic link, who allow every nonsingular covariance and an intercept, and bound the transition window by $O(n^{-1/2})$, by the approximate kinematic formula of Amelunxen et al. (2014), which they describe as a modern form of Gordon's escape through a mesh. For continuous links under which V is unbounded in both directions it is Theorem 3(a) of Montanari et al. (2025) (their Assumption 3), and for the logistic model with selected features and for Gaussian mixtures it is Proposition 3.1 of Deng et al. (2022).[4]

[4] We cite theorem and assumption numbers of Montanari et al. (2025) from their arXiv version 3 (2023), which we read; we could not check the journal's numbering.

Theorem 11.9 (The max-margin classifier). *In the teacher model with $\alpha < \alpha_{\mathrm{MLE}}(r)$, let κ_* and c_* be as in Lemma 11.7(iv). Then, in probability,*

$$\kappa_n \to \kappa_*, \qquad \langle \hat\theta_n, e^*\rangle \to c_*, \qquad \mathrm{Err}(\hat\theta_n) \to \mathcal{E}(c_*),$$

with $\hat\theta_n$ defined on the event $\{\kappa_n > 0\}$, whose probability tends to one. For random labels $c_* = 0$ and $\mathbb{E}(Z+\kappa_*)_+^2 = 1/\alpha$; for the noiseless teacher $\alpha_{\mathrm{MLE}} = \infty$ and $\mathcal{E}(c_*) = \arccos(c_*)/\pi$.

Proof. The plan is to squeeze κ_n between the dual bound at the best slice and the primal bound over all slices, and to apply the primal bound once more to the slices far from c_*. Fix $\varepsilon \in (0, \kappa_*)$. Since $\varkappa(c_*) = \kappa_*$, Lemma 11.6 gives $\kappa_n = D_n \ge \kappa_* - \varepsilon$ with probability tending to one. Since $M(\kappa_* + \varepsilon) > 0$, Lemmas 11.4 and 11.5 with $T = [0,\pi]$ give $\Delta_{[0,\pi]}(\kappa_* + \varepsilon) > 0$, so that no unit vector has margin $\kappa_* + \varepsilon$, with probability tending to one.

For the direction, let $C = \{c \in [-1,1] : |c - c_*| \ge \varepsilon\}$, assumed non-empty, and $T = \{t \in [0,\pi] : \cos t \in C\}$. The continuous function Γ_{κ_*} has a positive minimum η on the compact set C; with $\delta = \min(\eta/2, \varepsilon)$ and the Lipschitz dependence on κ, $\Gamma_{\kappa_*-\delta} \ge \eta/2$ on C. So with probability tending to one $\Delta_T(\kappa_* - \delta) > 0$ and $\kappa_n > \kappa_* - \delta$, and then $\hat\theta_n$, whose margin exceeds $\kappa_* - \delta$, is not at an angle in T: $|\langle\hat\theta_n, e^*\rangle - c_*| < \varepsilon$.

For the error, Lemma 11.2 applied to the fresh pair shows that $y\langle x\rangle\theta$ has the law of $cV + \sqrt{1-c^2}Z$ with $c = \langle\theta, e^*\rangle$, and $\mathbb{P}(\langle x\rangle\theta = 0) = 0$, so $\mathrm{Err}(\theta) = \mathcal{E}(c)$; the function $\mathcal{E}$ is continuous on $[-1,1]$, at ±1 because V has no

atom at 0, and the continuous mapping theorem gives the limit. For random labels, $F_\kappa(t) = (1+t^2)\mathbb{E}(\kappa - Z)_+^2$ is smallest at $t = 0$. For the noiseless teacher $V = |U| > 0$, so $\alpha_{\mathrm{MLE}} = \infty$, and $\mathcal{E}(c) = 2\mathbb{P}(U > 0,\ cU + \sqrt{1-c^2}Z < 0)$ is twice the Gaussian measure of a wedge of angle $\arccos c$. ■

For random labels this is Theorem 8.8 read at fixed load; for the noiseless teacher it confirms Opper et al. (1990). Montanari et al. (2025) prove the three limits for general covariances under an assumption (their Assumption 3) that excludes the noiseless link, and Aubin et al. (2020) prove, for the noiseless teacher, the learning curves of ridge-penalized hinge and logistic losses, which tend to the max-margin curve as the penalty tends to 0; the exchange of limits is not stated as a theorem.[5]

[5] We have not found the noiseless case stated as a theorem in the sources we read. The proof is the book's arrangement of the arguments of Stojnic (2013a) and Montanari et al. (2025); in the isotropic case the strict convexity of F_κ gives the uniqueness of the best slice, which Montanari et al. (2025) prove for general covariances by a convex problem in infinite dimension.

11.5 The proof, continued: the MLE beyond the threshold

Beyond the threshold the MLE exists, and Sur and Candès proved the prediction (11.7) for it. We state their theorem in the book's units.

Theorem 11.10 (Sur and Candès 2019, in book units). *In the teacher model with the logistic link, let $\alpha > \alpha_{\mathrm{MLE}}(r)$, and let the empirical distribution of the coordinates of $\sqrt{n}\,\beta^*$ converge to the law of a variable B, with second moments. Sur and Candès state that (11.7) then has a unique solution (a, τ, χ); let $\mu = a/r$.*

(i) Almost surely, for every pseudo-Lipschitz function ψ of order two, $\frac{1}{d}\sum_{j=1}^{d} \psi\big(\sqrt{n}(\hat\beta_j - \mu\beta_j^*), \sqrt{n}\,\beta_j^*\big) \to \mathbb{E}\,\psi(\tau\sqrt{\alpha}\,Z, B)$, with Z independent of B; in particular $\|\hat\beta - \mu\beta^*\| \to \tau$.

(ii) If $\beta_j^* = 0$, then $\sqrt{d}\,\hat\beta_j \to \mathcal{N}(0, \tau^2)$ in law, and the likelihood-ratio statistic $2\Lambda_j$ of the hypothesis $\beta_j^* = 0$ converges in law to $(\tau^2/\chi)\,\chi_1^2$.

Proof sketch. These are Theorems 2–4 of Sur and Candès (2019), translated by $\kappa_{\mathrm{SC}} = 1/\alpha$, $\gamma = r$, $\beta_{\mathrm{SC}} = \sqrt{n}\,\beta$ and $(\alpha_\star, \sigma_\star, \lambda_\star) = (\mu, \tau\sqrt{\alpha}, \chi)$. Sur and Candès define a generalized approximate message-passing iteration whose state evolution, a theorem for a fixed number of iterations (Javanmard and Montanari 2013) (Theorem 6.4 is the symmetric case), has the fixed point (11.7); a leave-one-out argument after Sur et al. (2019), with their bound on the norm of the MLE in place of strong convexity, shows that the iterates converge to the MLE. The details are in the article's supplementary appendix. Salehi et al. (2019) reach the same equations by the CGMT, as the zero-penalty case of ridge-regularized logistic regression, omitting the reduction to compact sets and assuming a unique fixed point. ■

The factor τ^2/χ exceeds one and diverges at the threshold, so the classical test rejects true nulls too often; at $d/n = 0.2$, $r^2 = 5$ it is 1.4867. Zhao et al. (2022) allow an arbitrary covariance. With a perceptron teacher, Aubin et al. (2020) proved the learning curves of ridge-penalized convex losses by Gordon's inequality, with optimally regularized logistic and hinge losses

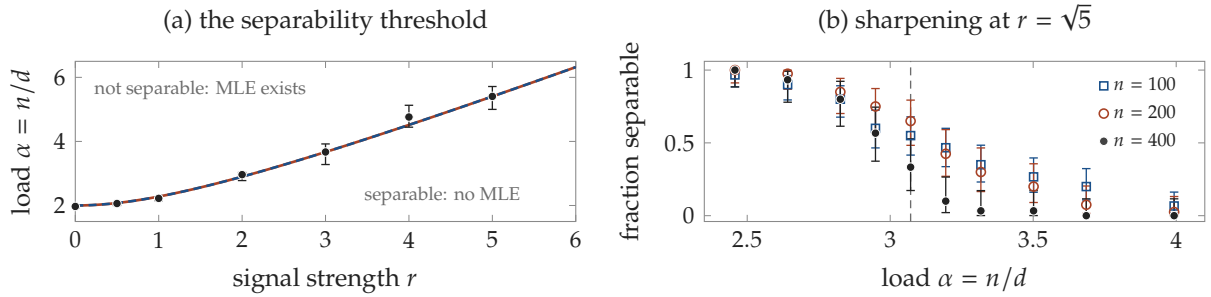


Figure 11.1. The separability threshold of logistic data. (a) The curve $\alpha_{\mathrm{MLE}}(r)$: the replica prediction (carnelian) and Theorem 11.8 (lapis dashed), which coincide. Points: medians, over 40 data sets with $n = 200$ examples and 200 features (the teacher first), of the critical load n/d_c; distribution-free 95% intervals from order statistics; seed 11001. (b) The fraction of 60, 40 and 30 data sets at $n = 100, 200, 400$ that are separable at load α, for $r = \sqrt{5}$, with Clopper–Pearson 95% intervals; seed 11002; dashed: $\alpha_{\mathrm{MLE}}(\sqrt{5}) = 3.0714$. Scripts: `figures/scripts/11-classification-threshold.py` and `-sharpness.py`.

close to the Bayes-optimal error of Chapter 7 (Theorem 7.5); Mignacco et al. (2020) treated Gaussian mixtures, and Deng et al. (2022) double descent.

11.6 Measuring it

The threshold script computes $\alpha_{\mathrm{MLE}}(r)$ by quadrature over the density of V and decides separability by linear programming: the data restricted to the first d features are strictly separable if and only if some b has $y_i\langle x_i, b\rangle \geq 1$ for all i. Separability can only appear as features are added, so each data set has a critical dimension d_c, the smallest number of leading features for which it is separable, found by bisection. In Figure 11.1(a) six of the seven intervals contain the prediction, for instance 3.670 in $[3.279, 3.922]$ against 3.669 at $r = 3$; the seventh, $[2.151, 2.273]$ at $r = 1$, ends at 200/88, the attainable value just below 2.278; at $r = 0$, where Cover's formula gives $\mathbb{P}(d_c \leq n/2) = \frac{1}{2}$, we measure 1.970, the midpoint of 200/101 and 200/102. In panel (b) the standard deviation of the critical load falls from 0.72 to 0.37 and 0.21 as n doubles, and the medians, 3.125, 3.125 and 2.974, have intervals containing 3.0714; at $n = 400$ the fraction separable at the threshold is 0.33, a shift inside the window of width $O(n^{-1/2})$ (Candès and Sur 2020) about which the theorem says nothing.

The MLE script solves (11.7) by Gauss quadrature and reproduces the values that Sur and Candès (2019) print: $(\alpha_\star, \sigma_\star, \lambda_\star) = (1.1678, 3.3466, 0.9605)$ at $d/n = 0.1$, and $(\alpha_\star, \sigma_\star) = (1.4994, 4.7436)$ at $d/n = 0.2$ against their $(1.499, 4.744)$. In Figure 11.2 nine of the ten intervals contain the prediction, for instance $\hat{\mu} = 1.506 \pm 0.054$ against 1.499 at $d/n = 0.2$; the tenth, 1.339 ± 0.037 against 1.299 at $d/n = 0.15$, misses by 0.003, and the estimates of μ sit slightly above the limit, as in the simulations of Sur and Candès. The classical values fail already at $d/n = 0.05$, where the noise norm is 0.651 ± 0.024, against 0.661 predicted and 0.595 classically. For 400 null

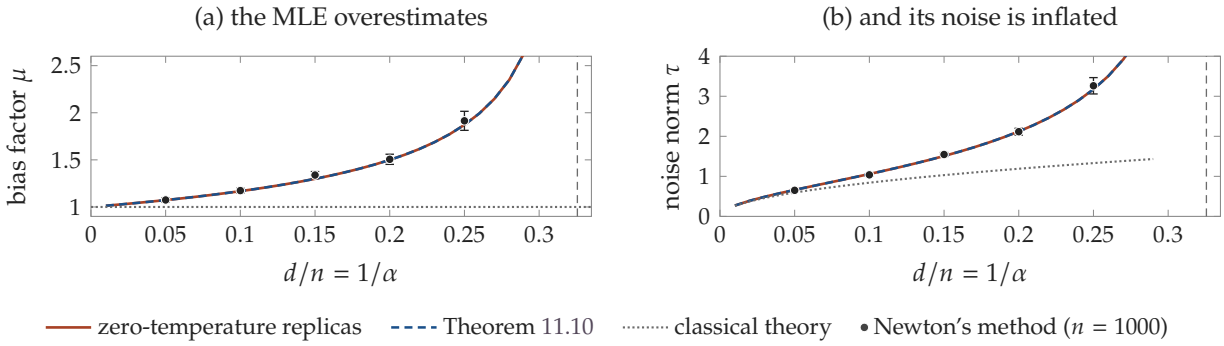


Figure 11.2. Bias and noise of the logistic MLE for the logistic teacher with $r = \sqrt{5}$. Curves: $\mu = a/r$ and τ from (11.7) (carnelian; lapis dashed for Theorem 11.10), up to $d/n = 0.29$; both diverge at $h_{\mathrm{MLE}}(\sqrt{5}) = 0.3256$ (dashed). Dotted: the classical $\mu = 1$ and $\tau^2 = (d/n)/\mathbb{E}f'(rU)$, $\mathbb{E}f'(rU) = 0.1412$. Points: $\langle\hat{\beta}, \beta^*\rangle/r^2$ and the norm of the part of $\hat{\beta}$ orthogonal to β^*, for the MLE (Newton's method), means over 40 data sets with $n = 1000$, Student-t 95% intervals; seed 11003. Script: `figures/scripts/11-classification-mle.py`.

coordinates at $d/n = 0.2$, ten per data set, the mean of $2\Lambda_j$ is 1.36 ± 0.20 against $\tau^2/\chi = 1.487$, and at level 5% the χ^2_1 test rejects 42 (10.5%, interval $[7.7\%, 13.9\%]$), the rescaled test 22 (5.5%, $[3.5\%, 8.2\%]$).

For the noiseless teacher all ten intervals in Figure 11.3 contain the limits; at $\alpha = 1$ the margin is 0.768 ± 0.030 against $\kappa_* = 0.7865$, and the error 0.265 ± 0.008 against $\mathcal{E}(c_*) = 0.2657$. At large load $\alpha\mathcal{E}(c_*) \to 0.5005$ (its value at $\alpha = 200$; the limit is 0.50049), against 0.4417 for the Bayes-optimal classifier, the constants that Aubin et al. (2020) print as 0.500 (0.5005 in their supplement) and 0.4417. For the logistic teacher nine of the ten intervals contain the limits; near the threshold, at $\alpha = 2.5$, the error is 0.3245 ± 0.0087 against 0.3341. The predicted error decreases to $0.3256 = h_{\mathrm{MLE}}(\sqrt{5})$ at the threshold (Exercise 11.4), far above the Bayes error 0.2054: the classifier fits the label noise.

11.7 The status board

For convex estimators on Gaussian data everything in this chapter is proved; universality beyond Gaussian inputs is not, and replica symmetry fails on the non-convex side, for errors beyond capacity and for negative margins.

STATUS BOARD · Gardner with a teacher ↔ the CGMT

PREDICTION	STATUS	PROOF OR EVIDENCE
Random labels are separable if and only if $\alpha < 2$	proved	Cover (1965), at every n
Separability (MLE existence) threshold $\alpha_{\mathrm{MLE}}(r)$, Gaussian inputs	proved	Candès and Sur (2020, Thms. 1–2); Montanari et al. (2025, Thm. 3(a)); Theorem 11.8

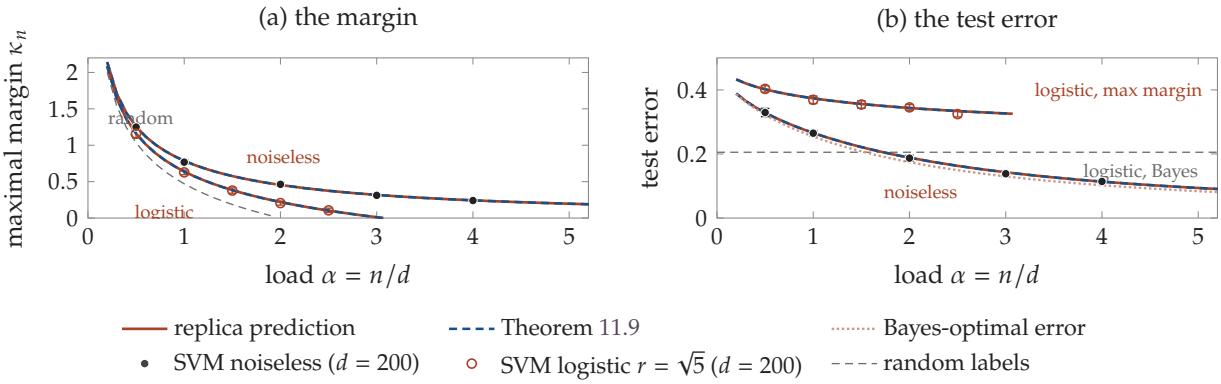


Figure 11.3. The max-margin classifier with a teacher. Curves: the limits of Theorem 11.9, which are the replica predictions (carnelian, lapis dashed on top), for the noiseless teacher and for the logistic teacher with $r = \sqrt{5}$, which end at $\alpha_{\text{MLE}}(\sqrt{5}) = 3.0714$; in (a) random labels (dashed gray, Gardner's formula); in (b) the Bayes-optimal error of the noiseless teacher (Barbier et al. 2019; Opper and Haussler 1991) (dotted) and the error $\mathbb{E} f(-r|U|) = 0.2054$ of the logistic teacher itself (dashed gray). Points: hard-margin support vector machines through the origin, solved in the dual, at $d = 200$ and $n = \alpha d$; means over 10 data sets with Student-t 95% intervals; seed 11004; errors computed exactly from the cosine with the teacher. Script: `figures/scripts/11-classification-maxmargin.py`.

The threshold from Gardner's volume sliced by the teacher	heuristic	Section 11.2 (the book's)
The same threshold for non-Gaussian inputs, $r > 0$	conjectured	Candès and Sur (2020, Sec. 5)
MLE bias, noise, null law, rescaled likelihood-ratio test, Gaussian inputs	proved	Sur and Candès (2019, Thms. 2–4); Zhao et al. (2022)
Ridge- and ℓ_1-regularized logistic regression	proved under	Salehi et al. (2019, Thms. 1–2): unique fixed point assumed; the reduction to compact sets omitted
Maximal margin and max-margin error, noisy labels	proved	Montanari et al. (2025, Thm. 3), their Assumptions 1–3; isotropic inputs, every link: Theorem 11.9
Optimal-stability perceptron, noiseless teacher: margin and error ($\approx 0.5005/\alpha$ at large α)	proved	Theorem 11.9, the book's proof; prediction Opper et al. (1990)
Convex ERM: perceptron teacher, Gaussian mixtures, double descent; Bayes error $\approx 0.4417/\alpha$	proved	Aubin et al. (2020, Thm. 2.2) ($\lambda > 0$); Mignacco et al. (2020, Thms. 1–2); Deng et al. (2022); Barbier et al. (2019)

Replica symmetry for the fewest errors beyond capacity; the replica-symmetric margin of non-separable data	false	Gardner and Derrida (1988): above a line; $r = 0$: for $\kappa < -0.622$ by the lifted bound (Montanari et al. 2024; Stojnic 2013b) (Chapter 8)

11.8 Where the translation fails

WHAT WAS ASSUMED. The theorems assume Gaussian inputs, used through rotation invariance and Gordon's inequality; the physics needs only Gaussian fields $\langle x_i, \theta \rangle$, which the central limit theorem provides for many designs. Both assume the proportional limit, a fixed signal strength and classifiers through the origin, and both describe exact minimizers, not algorithms: gradient descent reaches the max-margin direction only as its iterations tend to infinity (Soudry et al. 2018). The theory of the MLE assumes the logistic model; the threshold holds for every link.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. Replica symmetry fails on the non-convex side. Gardner and Derrida (1988) found the replica-symmetric solution for the fewest training errors beyond capacity unstable above a line in the plane of margin and load. Beyond the separability threshold the maximal margin is negative and the problem is not convex: the primal bound still gives $\kappa_n \le \bar{\kappa}(\alpha) + o(1)$, the replica-symmetric value (Exercise 11.11), but for random labels that value is too large once it is below -0.622. There the lifted Gordon bound of Stojnic (2013b), stated and proved again as Theorem 4.2 of Montanari et al. (2024), is smaller than Gardner's capacity: exactly for $\kappa < -0.6219$, by the expansion of Chapter 8. Stojnic states no threshold; his tables, which he presents as an illustration, show no gain at -0.5 and -0.6 and a gain of 1.3×10^{-4} at -0.63. Physics predicts full replica-symmetry breaking along the whole jamming line at negative margin (Franz et al. 2017); with a teacher we know of no analysis.[6]

OPEN PROBLEMS. First, universality: prove Theorem 11.8 for $r > 0$ and non-Gaussian inputs, as Candès and Sur (2020, Sec. 5) conjecture, and the theory of the MLE for non-Gaussian designs, which Sur and Candès (2019) call important to establish rigorously. Second, find the limit of the negative maximal margin for $\alpha > \alpha_{\mathrm{MLE}}(r)$; at $r = 0$ this is the capacity of the negative perceptron (Chapter 8). Third, close the two gaps in Salehi et al. (2019).

[6] Two sources need care. Proposition 3.3 of Deng et al. (2022), on the support vector machine, assumes in its arXiv version 2 a ratio p/n below their separability threshold $\kappa_\star$, where it must be above it, since the machine needs separable data; we did not see the journal version. Salehi et al. (2019) state that, for simplicity, they ignore the compactness hypothesis of the CGMT.

Exercises

11.1 CONSOLIDATION. (Cover from the threshold.) Show that $h_{\mathrm{MLE}}(0) = \frac{1}{2}$, that Cover's probability $2^{1-n} \sum_{k<d} \binom{n-1}{k}$ tends to 1 or 0 as $n/d \to \alpha < 2$ or $\alpha > 2$, and that at $r = 0$ the critical dimension of Section 11.6 has $\mathbb{P}(d_c \le n/2) = \frac{1}{2}$ for even n.

11.2 CONSOLIDATION. (Sur and Candès in book units.) Show that (11.7) is the system (5)–(7) of Sur and Candès (2019), with $\{0, 1\}$ labels, $\rho(t) = \log(1 + e^t)$, $Q_1 = \gamma U$ and $Q_2 = -\alpha_\star \gamma U + \sqrt{\kappa_{\mathrm{SC}}}\, \sigma_\star Z$, under

$(\alpha_\star, \sigma_\star, \lambda_\star) = (a/r, \tau\sqrt{\alpha}, \chi)$. *(Hint:* $\mathbb{E}\,g(V) = 2\mathbb{E}[f(rU)g(U)]$ *for a symmetric link;* $\operatorname{prox}_{\chi\ell}(x) = -\operatorname{prox}_{\chi\rho}(-x)$.*)*

11.3 CONSOLIDATION. (The classical limit.) Show that as $\alpha \to \infty$ the solution of (11.7) has $a \to r$ and $\alpha\chi$, $\alpha\tau^2 \to 1/\mathbb{E}\,f'(rU)$, the classical inverse Fisher information of the part of $\hat{\beta}$ orthogonal to the teacher. Check $\kappa_* = 0.7865$, $c_* = 0.6714$ and $\mathcal{E}(c_*) = 0.2657$ for the noiseless teacher at $\alpha = 1$.

11.4 BY HAND. (Counting at the threshold.) Prove $h_{\mathrm{MLE}} = \mathbb{P}(Z > t_{\mathrm{MLE}}V)$, check it at $r = \sqrt{5}$ ($t_{\mathrm{MLE}} = 0.8708$), and show that $\mathcal{E}(c_{\mathrm{MLE}}) = h_{\mathrm{MLE}}$ for $c_{\mathrm{MLE}} = t_{\mathrm{MLE}}/(1+t_{\mathrm{MLE}}^2)^{1/2}$: at the threshold, the error of the max-margin direction is the fraction of examples on the margin.

11.5 BY HAND. ⋆ (Large signal.) For the logistic link, let m_ℓ be the minimum over $s > 0$ of $\int_{\mathbb{R}} \mathbb{E}(Z - sw)_+^2\, f(w)\,\mathrm{d}w$. Show that $\alpha_{\mathrm{MLE}}(r)/r$ tends to $\sqrt{\pi/2}/m_\ell = 0.985$, with $m_\ell = 1.2724$ attained at $s = 0.3786$. *(Hint: put* $t = sr$ *and* $v = w/r$ *in* $\int \mathbb{E}(Z - tv)_+^2\, 2f(rv)\phi(v)\,\mathrm{d}v$.*)*

11.6 BY HAND. (The whole volume.) With $Q = c^2 + (1 - c^2)q$, show that $\frac{1}{2}\log(1 - c^2)$ plus the bracket of (11.4) is $\frac{1}{2}\log(1 - Q) + \frac{Q-c^2}{2(1-Q)} + \alpha\mathbb{E}\log H\big((\kappa - cV - \sqrt{Q - c^2}\,Z)/\sqrt{1 - Q}\big)$, and derive it from the Gram matrix of m students and the teacher.

11.7 TRANSLATION. (Which half of a slice is a theorem.) Show that Lemmas 11.4 and 11.5 with $T = \{\arccos c\}$ prove the slice at overlap c empty when $\alpha > \alpha_\kappa(c)$ and say nothing when $\alpha < \alpha_\kappa(c)$; compare with the negative perceptron of Chapter 8.

11.8 TRANSLATION. ⋆ (The cavity field is a leave-one-out margin.) Let $\hat{\beta}^{(-i)}$ be the MLE without example i and $H_{(-i)}$ the Hessian of its loss there. Expanding the score equation around $\hat{\beta}^{(-i)}$, show that $\langle z_i, \hat{\beta}\rangle \approx \operatorname{prox}_{\chi_i\ell}(\langle z_i, \hat{\beta}^{(-i)}\rangle)$ with $\chi_i = z_i^\top H_{(-i)}^{-1} z_i$, and identify the limit of χ_i with χ of (11.7).

11.9 MEASUREMENT. (Universality.) Modify the sharpness script to draw, at each load, a fresh data set with d features uniform on $[-\sqrt{3}, \sqrt{3}]$ and $\beta^* = r(1, \dots, 1)/\sqrt{d}$, and to decide its separability by one linear program, at $r = 0$ and $r = 2$. (Why would the critical-dimension trick of the scripts be wrong here?) Why does Cover's count make $r = 0$ a theorem, while $r = 2$ is a conjecture?

11.10 MEASUREMENT. (The likelihood-ratio test.) With the MLE script, compute the rejection rates of both tests at $d/n = 0.1$ ($\tau^2/\chi = 1.1660$) and $d/n = 0.25$ (1.8396). Why does the classical test fail more as d/n grows?

11.11 RESEARCH. (Negative margins with a teacher.) For $\alpha > \alpha_{\mathrm{MLE}}(r)$, show from Lemmas 11.4 and 11.5 that κ_n is asymptotically at most the root

$\bar\kappa(\alpha)$ of M, and compute $\bar\kappa(4) = -0.1234$ for $r = \sqrt{5}$. Is the bound tight? The exact limit is open.

Notes and References

The statistical mechanics of learning from examples began with Gardner's space of interactions (Gardner 1988; Gardner and Derrida 1988), and Engel and Van den Broeck (2001) collect its computations. Of the predictions in this chapter, physics came first for four only: Gardner's capacity; the error of the optimal-stability perceptron, computed by replicas by Opper et al. (1990); that of the Bayes-optimal perceptron, by Opper and Haussler (1991), proved by Barbier et al. (2019); and the classification of patterns divided into classes (Del Giudice et al. 1989), a model related to the Gaussian mixtures of Mignacco et al. (2020). The logistic MLE was derived and proved directly by statisticians. In the proportional regime its theory begins, as far as we know, with Sur et al. (2019), for the global null; Candès and Sur (2020) proved the threshold curve with the kinematic formula of Amelunxen et al. (2014), and Sur and Candès (2019) the bias, the variance inflation and the rescaled test, by approximate message passing and the leave-one-out analysis of robust regression (Chapter 10), not by a replica computation.

Stojnic combined Gordon's inequality (Gordon 1985) with duality for the perceptron (Stojnic 2013a,b), an idea that became the CGMT (Chapter 8). The book's own, as far as we know, are the slicing of Gardner's volume for the separability threshold, the counting identity $h_{\mathrm{MLE}} = \mathbb{P}(Z > t_{\mathrm{MLE}} V)$ with its consequence for the error at the threshold, and the arrangement of the proofs of Theorems 11.8 and 11.9 around the one-sided CGMT, which covers the noiseless teacher.

Translating the literature. Candès and Sur (2020) write $\kappa = p/n = 1/\alpha$, γ_0 (or γ) for r, and $V = YX$. Sur and Candès (2019) use $\{0,1\}$ labels and inputs $X_i \sim \mathcal{N}(0, I_p/n)$, so their β is $\sqrt{n}$ times ours, and their $(\alpha_\star, \sigma_\star, \lambda_\star)$ are our $(\mu, \tau\sqrt{\alpha}, \chi)$. Montanari et al. (2025) write $\psi = 1/\alpha$, $\rho = r$, $T = YG$ for our V, and $F_\kappa(c_1, c_2) = (\mathbb{E}(\kappa - c_1 YG - c_2 Z)_+^2)^{1/2}$, so that $\Gamma_\kappa(c) = F_\kappa(c, \sqrt{1-c^2}) - \sqrt{(1-c^2)/\alpha}$. Deng et al. (2022) write $\kappa = p/n$ and (μ, α, λ) for our (a, τ, χ). Aubin et al. (2020) use $w = \sqrt{d}\,\beta$, $\rho_{w^*} = r^2$, unnormalized sums of losses, and $\eta = m^2/(\rho_{w^*} q)$, our squared cosine. The theorem numbers we give for Thrampoulidis et al. (2018) are those of its arXiv version 1.

References

Albert, A. and J. A. Anderson (1984). "On the existence of maximum likelihood estimates in logistic regression models". In: *Biometrika* 71.1, pp. 1–10.

Amelunxen, D., M. Lotz, M. B. McCoy, and J. A. Tropp (2014). "Living on the edge: phase transitions in convex programs with random data". In: *Information and Inference: A Journal of the IMA* 3.3, pp. 224–294. arXiv: 1303.6672.

Aubin, B., F. Krzakala, Y. M. Lu, and L. Zdeborová (2020). "Generalization error in high-dimensional perceptrons: Approaching Bayes error with convex optimization". In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pp. 12199–12210. arXiv: 2006.06560.

Barbier, J., F. Krzakala, N. Macris, L. Miolane, and L. Zdeborová (2019). "Optimal errors and phase transitions in high-dimensional generalized linear models". In: *Proceedings of the National Academy of Sciences* 116.12, pp. 5451–5460. arXiv: 1708.03395.

Candès, E. J. and P. Sur (2020). "The phase transition for the existence of the maximum likelihood estimate in high-dimensional logistic regression". In: *The Annals of Statistics* 48.1, pp. 27–42. arXiv: 1804.09753.

Cover, T. M. (1965). "Geometrical and Statistical Properties of Systems of Linear Inequalities with Applications in Pattern Recognition". In: *IEEE Transactions on Electronic Computers* EC-14.3, pp. 326–334.

Del Giudice, P., S. Franz, and M. A. Virasoro (1989). "Perceptron beyond the limit of capacity". In: *Journal de Physique* 50.2, pp. 121–134.

Deng, Z., A. Kammoun, and C. Thrampoulidis (2022). "A model of double descent for high-dimensional binary linear classification". In: *Information and Inference: A Journal of the IMA* 11.2, pp. 435–495. arXiv: 1911.05822.

Engel, A. and C. Van den Broeck (2001). *Statistical Mechanics of Learning*. Cambridge University Press.

Franz, S. and G. Parisi (2016). "The simplest model of jamming". In: *Journal of Physics A: Mathematical and Theoretical* 49.14, p. 145001. arXiv: 1501.03397.

Franz, S., G. Parisi, M. Sevelev, P. Urbani, and F. Zamponi (2017). "Universality of the SAT-UNSAT (jamming) threshold in non-convex continuous constraint satisfaction problems". In: *SciPost Physics* 2.3, p. 019. arXiv: 1702.06919v2 [cond-mat.dis-nn].

Gardner, E. (1988). "The space of interactions in neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 257–270.

Gardner, E. and B. Derrida (1988). "Optimal storage properties of neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 271–284.

Gordon, Y. (1985). "Some inequalities for Gaussian processes and applications". In: *Israel Journal of Mathematics* 50.4, pp. 265–289.

Javanmard, A. and A. Montanari (2013). "State evolution for general approximate message passing algorithms, with applications to spatial coupling". In: *Information and Inference* 2.2, pp. 115–144. arXiv: 1211.5164.

Mignacco, F., F. Krzakala, Y. M. Lu, P. Urbani, and L. Zdeborová (2020). "The Role of Regularization in Classification of High-dimensional Noisy Gaussian Mixture". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research, pp. 6874–6883. arXiv: 2002.11544.

Montanari, A., F. Ruan, Y. Sohn, and J. Yan (2025). "The generalization error of max-margin linear classifiers: Benign overfitting and high dimensional asymptotics in the overparametrized regime". In: *The Annals of Statistics* 53.2, pp. 822–853. arXiv: 1911.01544.

Montanari, A., Y. Zhong, and K. Zhou (2024). "Tractability from overparametrization: the example of the negative perceptron". In: *Probability Theory and Related Fields* 188.3-4, pp. 805–910. arXiv: 2110.15824v3 [cs.LG].

Opper, M., W. Kinzel, J. Kleinz, and R. Nehl (1990). "On the ability of the optimal perceptron to generalise". In: *Journal of Physics A: Mathematical and General* 23.11, pp. L581–L586.

Opper, M. and D. Haussler (1991). "Generalization performance of Bayes optimal classification algorithm for learning a perceptron". In: *Physical Review Letters* 66.20, pp. 2677–2680.

Salehi, F., E. Abbasi, and B. Hassibi (2019). "The Impact of Regularization on High-dimensional Logistic Regression". In: *Advances in Neural Information Processing Systems*. Vol. 32. arXiv: 1906.03761.

Soudry, D., E. Hoffer, M. S. Nacson, S. Gunasekar, and N. Srebro (2018). "The Implicit Bias of Gradient Descent on Separable Data". In: *Journal of Machine Learning Research* 19.70, pp. 1–57. arXiv: 1710.10345.

Stojnic, M. (2013a). *Another look at the Gardner problem*. arXiv preprint. arXiv: 1306.3979v1 [math.OC].

Stojnic, M. (2013b). *Negative spherical perceptron*. arXiv: 1306.3980.

Sur, P. and E. J. Candès (2019). "A modern maximum-likelihood theory for high-dimensional logistic regression". In: *Proceedings of the National Academy of Sciences* 116.29, pp. 14516–14525. arXiv: 1803.06964.

Sur, P., Y. Chen, and E. J. Candès (2019). "The likelihood ratio test in high-dimensional logistic regression is asymptotically a rescaled Chi-square". In: *Probability Theory and Related Fields* 175.1-2, pp. 487–558. arXiv: 1706.01191.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). "Precise Error Analysis of Regularized M-Estimators in High Dimensions". In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Zhao, Q., P. Sur, and E. J. Candès (2022). "The asymptotic distribution of the MLE in high-dimensional logistic models: Arbitrary covariance". In: *Bernoulli* 28.3, pp. 1835–1861. arXiv: 2001.09351.

12 Kernels and Random Features

Learning curves and Gaussian equivalence

THE DICTIONARY — the Gaussian approximation ↔ universality

IN THIS CHAPTER. Physicists predict the learning curve of kernel ridge regression from the spectrum of the kernel alone: a replica computation, in which the features are replaced by Gaussian vectors with the same covariance, gives an effective ridge λ_{eff} fixed by one scalar equation and a closed formula for the risk. We prove the formula for Gaussian features with finitely many distinct eigenvalues by a leave-one-out argument that turns the physicists' cavity step into a theorem, and state what is proved for real kernels and for random features. We also prove that the formula is wrong at zero ridge in fixed dimension: on the circle it predicts the noise term σ^2, and the truth is $2\sigma^2/3$. Simulations confirm the formula for Gaussian features, for a kernel on the sphere at $n \asymp d$ and $n \asymp d^2$, and for random features, and show where the linear Gaussian equivalent stops working. Open: a criterion that says, for a given kernel and data distribution, when the Gaussian approximation is exact.

12.1 The question: how fast does a kernel learn?

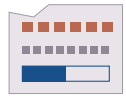
A kernel method fits a linear function of fixed nonlinear features of the input. Take a positive semidefinite kernel K on an input space $\mathcal{X}$, with reproducing kernel Hilbert space $\mathcal{H}$ (RKHS), and n samples: independent inputs x_i of law P_x and labels $y_i = f_*(x_i) + \sigma\varepsilon_i$, the ε_i independent of the inputs, with mean 0 and variance 1. Write $\mathbf{K} = (K(x_i, x_j))_{i,j\le n}$ and $k(x) = (K(x, x_i))_{i\le n}$. *Kernel ridge regression* with the per-sample normalized ridge $\lambda > 0$ of this book is

$$\hat{f}_\lambda = \arg\min_{f\in\mathcal{H}} \frac{1}{2n}\sum_{i=1}^{n}\big(y_i - f(x_i)\big)^2 + \frac{\lambda}{2}\|f\|_{\mathcal{H}}^2, \qquad \hat{f}_\lambda(x) = k(x)^\top(\mathbf{K} + n\lambda I)^{-1}y. \tag{12.1}$$

At $\lambda = 0$ we mean the limit $\lambda \downarrow 0$, the *minimum-norm interpolant*. The quantity of this chapter is the excess risk $R = \|\hat{f}_\lambda - f_*\|^2_{L^2(P_x)}$ (the test error is $R + \sigma^2$); its graph against n is the *learning curve*.

The kernel acts through its *Mercer decomposition*: the integral operator of K on $L^2(P_x)$ has eigenvalues $\eta_k \ge 0$ and orthonormal eigenfunctions e_k, and $K(x, x') = \sum_k \eta_k e_k(x)e_k(x')$.[1] Expand the target as $f_* = \sum_k v_k e_k$, so that v_k^2 is its power in mode k. In the coordinates $u(x) = (\sqrt{\eta_k}\, e_k(x))_k$, a function $f = \langle w, u\rangle$ has $\|f\|_{\mathcal{H}} = \|w\|$ and $\|f\|^2_{L^2} = \langle w, \Lambda w\rangle$ with $\Lambda = \operatorname{diag}(\eta_k)$: kernel

[1] Mercer's theorem gives this expansion, with uniform convergence, for a continuous kernel on a compact space; the examples of this chapter are of this kind.



ridge regression is ridge regression (Chapter 9) on features of covariance Λ, with target weights $w_{*,k} = v_k/\sqrt{\eta_k}$ where $\eta_k > 0$. Two kernels recur. On the circle, $K(\theta,\theta') = 1 + \sum_{k\ge1} 2k^{-2}\cos k(\theta-\theta')$ has $\eta_0 = 1$ and, for $k \ge 1$, $\eta_k = k^{-2}$ twice (on $\cos k\theta$ and $\sin k\theta$). For data uniform on the sphere $S^{d-1}(\sqrt{d})$, an inner-product kernel $h(\langle x, x'\rangle/d)$ has the spherical harmonics as eigenfunctions, and the harmonics of degree ℓ share one eigenvalue, with multiplicity $B(d,\ell) \approx d^\ell/\ell!$.

Random features approximate a kernel by a finite random sum (Rahimi and Recht 2007),[2] $f(x;a) = \sum_{j\le N} a_j \varsigma(\langle\theta_j\rangle x/\sqrt{d})$, with frozen θ_j independent and uniform on $S^{d-1}(\sqrt{d})$ and an activation ς; ridge regression fits a by minimizing $\frac{1}{2n}\|y - Za\|^2 + \frac{\lambda}{2}\|a\|^2$, $Z_{ij} = \varsigma(\langle\theta_j, x_i\rangle/\sqrt{d})$, on data uniform on $S^{d-1}(\sqrt{d})$, with $\alpha = n/d$ and $\psi = N/d$. With $G \sim \mathcal{N}(0,1)$, the activation enters through $\mu_0 = \mathbb{E}\varsigma(G)$, $\mu_1 = \mathbb{E}[G\varsigma(G)]$, $\mu_*^2 = \mathbb{E}\varsigma(G)^2 - \mu_0^2 - \mu_1^2$ and $\zeta = \mu_1/\mu_*$; for the ReLU, $\mu_0 = 1/\sqrt{2\pi}$, $\mu_1 = \frac12$, $\mu_*^2 = \frac14 - \frac{1}{2\pi}$ and $\zeta = 1.6589$. Finally, write $\mathrm{df}_1(t) = \sum_k \eta_k/(\eta_k + t)$ and $\mathrm{df}_2(t) = \sum_k \eta_k^2/(\eta_k + t)^2$ for the two degrees of freedom of the kernel at ridge t.

[2] Random features were introduced to make kernel methods cheap: N features replace the $n \times n$ kernel matrix.

THE PHYSICAL PREDICTION · Sollich 1998; Bordelon, Canatar and Pehlevan 2020; Canatar, Bordelon and Pehlevan 2021. Let the *effective ridge* $\lambda_{\mathrm{eff}} > 0$ solve

$$\lambda = \lambda_{\mathrm{eff}}\Big(1 - \frac{\mathrm{df}_1(\lambda_{\mathrm{eff}})}{n}\Big). \tag{12.2}$$

Then, for large n, the excess risk of kernel ridge regression is

$$R = \frac{\sum_k \lambda_{\mathrm{eff}}^2 v_k^2/(\eta_k + \lambda_{\mathrm{eff}})^2 + \sigma^2\,\mathrm{df}_2(\lambda_{\mathrm{eff}})/n}{1 - \mathrm{df}_2(\lambda_{\mathrm{eff}})/n}. \tag{12.3}$$

For random features, the risk is that of ridge regression on the Gaussian covariates $u = \mu_0\mathbf{1} + \mu_1\Theta x/\sqrt{d} + \mu_* w$, with $w \sim \mathcal{N}(0, I_N)$ independent of x (Gerace et al. 2020; Goldt et al. 2020).

The modes with $\eta_k \ll \lambda_{\mathrm{eff}}$, which the data cannot resolve, act as extra ridge, so $\lambda_{\mathrm{eff}} > \lambda$. The formula contains the phenomenology of kernel learning: modes are learned in decreasing order of η_k, the factor $1/(1-\mathrm{df}_2/n)$ produces interpolation peaks, and on the sphere the risk drops in stages at $n \asymp d^\ell$. The last sentence of the box is the *Gaussian equivalence* principle:[3] nonlinear features may be replaced by Gaussian vectors with the same first two moments. How do physicists derive (12.3), and when is it a theorem? The answer has a twist: the formula is exactly the risk of Gaussian features; for real kernels it is right in high dimension, close in fixed dimension with a large ridge, and wrong there without one.

[3] Goldt et al. (2020) named it the Gaussian Equivalence Property; Xiao et al. (2023) describe the physics derivations of kernel curves as resting on a "Gaussian equivalence conjecture".

12.2 The physics: replicas over Gaussian features

Canatar et al. (2021) derived (12.3) by replicas, after Bordelon et al. (2020) had obtained it from an equation of Sollich (1998). The key step is, in their words, "a Gaussian approximation to the average over the dataset in the

feature space". We redo the computation in the book's units.

REPLICA COMPUTATION OF THE LEARNING CURVE. *Gibbs measure.* In Mercer coordinates, n times the objective (12.1) is $L(w) = \frac{1}{2}\sum_i (y_i - \langle w, u(x_i)\rangle)^2 + \frac{n\lambda}{2}\|w\|^2$. The Gibbs weight $e^{-\beta L(w)}$ concentrates at the estimator as $\beta \to \infty$. Replica a has error $\Delta^a = w^a - w_*$ and excess risk $\langle \Delta^a, \Lambda\Delta^a\rangle$.

Replicas. The samples are independent, so for integer r, with $h^a = \langle \Delta^a, u(x)\rangle$,

$$\mathbb{E}Z^r = \int \prod_{a\le r} \mathrm{d}w^a \, e^{-\frac{\beta n\lambda}{2}\sum_a \|w^a\|^2} \left(\mathbb{E}_{x,\varepsilon} e^{-\frac{\beta}{2}\sum_a (\sigma\varepsilon - h^a)^2}\right)^n .$$

The Gaussian approximation. Treat $(h^a)_{a\le r}$ as a centered Gaussian vector with second moments $Q_{ab} = \mathbb{E}h^a h^b = \langle \Delta^a, \Lambda\Delta^b\rangle$, and ε as standard Gaussian. This is exact when $u(x)$ is Gaussian; for a kernel it is the approximation. With J the all-ones matrix, the average becomes $\det(I_r + \beta(Q + \sigma^2 J))^{-1/2}$, and the data enter only through Q.

Order parameters. Impose $Q_{ab} = \langle \Delta^a, \Lambda\Delta^b\rangle$ with Lagrange multipliers $\hat{Q}_{ab}$. The exponent acquires $-\frac{1}{2}\sum_{ab}\hat{Q}_{ab}Q_{ab}$, and the integral over w factorizes over the modes k, mode k carrying $\exp(\frac{\eta_k}{2}\sum_{ab}\hat{Q}_{ab}\Delta_k^a\Delta_k^b)$.

Replica symmetry at zero temperature. The problem is convex; take $Q_{ab} = q + \frac{\chi}{\beta}\delta_{ab}$ and $\hat{Q}_{ab} = n(-\beta\hat{m}\,\delta_{ab} + \beta^2\hat{s})$. As $r \to 0$ the determinant contributes $-\frac{rn}{2}[\log(1+\chi) + \beta(q+\sigma^2)/(1+\chi)]$ and the multipliers $-\frac{r\beta n}{2}(\hat{s}\chi - \hat{m}q) + O(rn)$. In mode k a Hubbard–Stratonovich variable $z \sim \mathcal{N}(0,1)$ decouples the replicas, and as $\beta \to \infty$ (Laplace) each sees the scalar problem

$$\mathcal{E}_k(z) = \min_w \left\{ \frac{\lambda}{2}w^2 + \frac{\eta_k\hat{m}}{2}(w - w_{*,k})^2 - \sqrt{\frac{\eta_k\hat{s}}{n}}\, z\,(w - w_{*,k}) \right\},$$

with $\mathbb{E}_z\mathcal{E}_k = \lambda\hat{m}v_k^2/(2(\lambda + \eta_k\hat{m})) - \eta_k\hat{s}/(2n(\lambda + \eta_k\hat{m}))$. Collecting the terms of order $r\beta n$, $\mathbb{E}Z^r \approx e^{-r\beta n f}$ with

$$f = \frac{q + \sigma^2}{2(1+\chi)} + \frac{\hat{s}\chi - \hat{m}q}{2} + \sum_k \left[\frac{\lambda\hat{m}\, v_k^2}{2(\lambda + \eta_k\hat{m})} - \frac{\eta_k\hat{s}}{2n(\lambda + \eta_k\hat{m})} \right].$$

Saddle point. Setting the four derivatives of f to zero gives $\hat{m} = 1/(1+\chi)$, $\hat{s} = (q+\sigma^2)/(1+\chi)^2$ and

$$\chi = \frac{1}{n}\sum_k \frac{\eta_k}{\lambda + \eta_k\hat{m}}, \qquad q = \sum_k \frac{\lambda^2 v_k^2}{(\lambda + \eta_k\hat{m})^2} + \frac{\hat{s}}{n}\sum_k \frac{\eta_k^2}{(\lambda + \eta_k\hat{m})^2}.$$

The effective ridge. Put $\lambda_{\mathrm{eff}} = \lambda/\hat{m} = \lambda(1+\chi)$, so that $\lambda + \eta_k\hat{m} = \hat{m}(\lambda_{\mathrm{eff}} + \eta_k)$. The equation for χ becomes $1/(1+\chi) = 1 - \mathrm{df}_1(\lambda_{\mathrm{eff}})/n$, which is (12.2); the equation for q becomes $q = \sum_k \lambda_{\mathrm{eff}}^2 v_k^2/(\eta_k + \lambda_{\mathrm{eff}})^2 + (q+\sigma^2)\mathrm{df}_2/n$. Since $q = Q_{aa}$ at $\beta = \infty$ is the risk of the estimator, this is (12.3). ◇

Five steps are uncontrolled: the Gaussian approximation; the continuation from integer r to $r \to 0$ and its exchange with $n \to \infty$ (Chapter 1); the replica-symmetric ansatz, which convexity makes plausible;[4] the choice of the stationary point; and the limit $\beta \to \infty$ after $n \to \infty$. Only the first depends on the kernel; the rest is the computation of Chapter 9 with a general spectrum.

[4] A strictly convex objective has a unique minimizer, so the replicas collapse onto it as $\beta \to \infty$; for the nonconvex problems of Chapter 8, the perceptron at negative margin, the replicas break.

THE CAVITY VERSION. Sollich's route is closer to the proof.[5] In our normalization: adding sample $n+1$ to the resolvent G of the empirical feature covariance by the *Sherman–Morrison formula* changes G by a rank-one term divided by $1+u_{n+1}^\top G u_{n+1}/n$. Sollich replaced the average of this ratio by the ratio of the averages, so that the denominator becomes $1+\operatorname{tr}(\Lambda G)/n$, took a continuum limit in n, and solved the resulting equation by characteristics (Bordelon et al. 2020, §2.2); Canatar et al. (2021) note that their replica formula agrees with his "lower continuous approximation".

[5] Sollich studied Gaussian process regression; its posterior mean is kernel ridge regression with $n\lambda$ equal to the noise variance of the prior model.

RANDOM FEATURES. One more step reduces random features to Gaussian features.

GAUSSIAN EQUIVALENCE OF RANDOM FEATURES. Fix Θ. For a random input, $t_j = \langle \theta_j \rangle x/\sqrt{d}$ is nearly standard Gaussian, and t_j, t_l are nearly jointly Gaussian with correlation $\rho_{jl} = \langle \theta_j, \theta_l \rangle/d = O(d^{-1/2})$ for $j \neq l$. Expand $\varsigma = \sum_k \mu_k \mathrm{He}_k/k!$ with $\mu_k = \mathbb{E}[\varsigma(G)\mathrm{He}_k(G)]$; Mehler's formula gives the *Hermite expansion*

$$\mathbb{E}\big[\varsigma(t_j)\varsigma(t_l)\big] = \sum_{k\geq 0} \frac{\mu_k^2}{k!}\rho_{jl}^k = \mu_0^2 + \mu_1^2 \rho_{jl} + O(\rho_{jl}^2),$$

and at $\rho_{jj} = 1$ it gives $\mu_0^2 + \mu_1^2 + \mu_*^2$, since $\mu_*^2 = \sum_{k\geq 2} \mu_k^2/k!$. So the feature covariance is $\mu_0^2 \mathbf{1}\mathbf{1}^\top + \mu_1^2 \Theta\Theta^\top/d + \mu_*^2 I$ up to off-diagonal entries of order $1/d$, and by Stein's lemma the covariance with a linear target $\langle \beta \rangle x$ is $\mu_1 \Theta\beta/\sqrt{d}$: the moments of the *Gaussian covariate model* $u = \mu_0 \mathbf{1} + \mu_1 \Theta x/\sqrt{d} + \mu_* w$. Assume that at the trained weights the risk sees only these moments; ridge regression on u is then ridge regression on Gaussian features, and the computation above gives the curve of Mei and Montanari (2022). ◇

The uncontrolled step is the last: moments are matched at fixed weights, and the weights that matter are trained on the features. Montanari and Saeed (2022) point out that the naive argument fails as stated, since the projections of the data on the trained weights are not Gaussian; a proof must compare minimum values, not distributions at the minimizer.

12.3 The stone

PHYSICS · *the replica and cavity computations*

1 · The Gaussian approximation

The local fields $\langle \Delta^a, u(x) \rangle$ are treated as jointly Gaussian; the eigenfunctions disappear, and only (η_k) and (v_k^2) remain.

PROOF · *deterministic equivalents and universality*

A hypothesis of the theorem (Assumption 12.1). For real kernels, a universality theorem with its own hypotheses (Misiakiewicz and Saeed 2024; Xiao et al. 2023); false in fixed dimension at $\lambda = 0$ (Proposition 12.9).

2 · Replicas, order parameters, self-averaging

Overlaps $Q_{ab} = q + \chi\delta_{ab}/\beta$, assumed to concentrate; q is the risk, χ a susceptibility.

No replicas: $\hat\beta - \beta_* = -\lambda Q_\lambda \beta_* + \sigma Q_\lambda X^\top \varepsilon/n$ with $Q_\lambda = (\hat\Sigma + \lambda I)^{-1}$, so the risk is a sum of traces, which concentrate by the Poincaré inequality (Lemma 12.3). The susceptibility is $\delta = \mathbb{E}\,\mathrm{tr}(\Sigma Q_\lambda)/n$; in the limit $1 + \delta = \lambda_{\mathrm{eff}}/\lambda = 1 + \chi$.

3 · The cavity: add one sample

Sherman–Morrison, then the average of a ratio replaced by the ratio of averages: $1 + u^\top G u/n$ becomes $1 + \mathrm{tr}(\Lambda G)/n$.

The *leave-one-out* identity $Qx_1 = Q_1 x_1/(1+d_1)$ is exact, and replacing $d_1 = x_1^\top Q_1 x_1/n$ by δ costs $\mathbb{E}(d_1 - \delta)^2 = O(1/n)$, because Q_1 is independent of x_1 (Proposition 12.5).

4 · The saddle-point equation

$\partial_{\hat s} f = 0$ gives (12.2); its solution is taken as the physical one.

The same equation is an approximate fixed point $\delta \approx F_n(\delta)$; uniqueness and the bound $|\delta - \delta_*| \le U^2|\delta - F(\delta)|$ keep the error of order $n^{-1/2}$ (Lemma 12.4).

5 · The risk from the saddle

$\partial_{\hat m} f = 0$ gives $q = B_0 + (q + \sigma^2)\mathrm{df}_2/n$: noise and unlearned target, fed back through the residuals, give $1/(1 - \mathrm{df}_2/n)$.

The risk is a derivative of traces, $(\lambda h_n)'$ and $-\lambda^2 \partial_t\,\mathrm{tr}(BQ_{\lambda,t})$; traces are convex in λ and t, so derivatives converge (Lemma 12.6); $1/(1-D_2) = \mathrm{d}\lambda_{\mathrm{eff}}/\,\mathrm{d}\lambda$.

6 · Gaussian equivalence of random features

Mehler's expansion matches two moments with the Gaussian covariate model.

Ridge: resolvent algebra (Theorem 12.8); kernel matrices: El Karoui's linearization; convex losses: Lindeberg swapping (Hu and Lu 2023); nonconvex losses inside a domain of Gaussianity: Montanari and Saeed (2022). The linear equivalent is false beyond $n \asymp d$ (Ghorbani et al. 2021).

7 · Zero temperature, then zero ridge

$\beta \to \infty$ after $n \to \infty$; the ridgeless curve is the formula at $\lambda \to 0$.

No counterpart in general: the theorems need $\lambda > 0$ or effective-rank conditions; for random features the limit is conjectured; in fixed dimension the formula's noise term is wrong by the factor 3/2.

The proof follows the cavity derivation, not the replica one: both use the Sherman–Morrison formula, and the proof adds an error budget for

Sollich's one approximation, the replacement of a random denominator by its mean, with a stability estimate that carries the error through the fixed-point equation. The replica-symmetric ansatz has no counterpart, because nothing is guessed. Two steps are not translated: Gaussianity, a hypothesis of both routes that only separate universality theorems remove, and the zero-ridge limit, in which the formula can fail. The physics computes the Gaussian model exactly; everything difficult is the passage from the Gaussian model to the real one.

12.4 The proof: the cavity, made rigorous

We prove the formula where the Gaussian approximation is no approximation, for the simplest spectrum that keeps the structure of (12.3): finitely many distinct eigenvalues, each with multiplicity proportional to n.

Assumption 12.1 (Gaussian features with finitely many distinct eigenvalues). Fix $m \geq 1$, distinct $\eta_1, \dots, \eta_m > 0$, $\pi_1, \dots, \pi_m > 0$ and $b_1, \dots, b_m \geq 0$. For each n let $p_1, \dots, p_m$ be positive integers with $p_j/n \to \pi_j$, $p = \sum_j p_j$, $\Sigma = \operatorname{diag}(\eta_1 I_{p_1}, \dots, \eta_m I_{p_m})$, and P_j the orthogonal projection on the jth block of coordinates.

(i) The rows of $X \in \mathbb{R}^{n \times p}$ are $x_i = \Sigma^{1/2} g_i$, with $g_1, \dots, g_n$ independent $\mathcal{N}(0, I_p)$.

(ii) $y_i = \langle x_i, \beta_* \rangle + \sigma \varepsilon_i$, with $\varepsilon_1, \dots, \varepsilon_n$ independent, of mean 0 and variance 1; $\beta_* = \sum_j b_j u_j$ with u_j uniform on the unit sphere of the range of P_j; X, ε and the u_j are independent.

(iii) $\hat{\beta} = (\hat{\Sigma} + \lambda I)^{-1} X^\top y/n$ with $\hat{\Sigma} = X^\top X/n$ and $\lambda > 0$ fixed.

These are Gaussian Mercer features with m distinct eigenvalues, and a target with power $\eta_j b_j^2$ spread isotropically over the jth eigenspace. Write $R = \langle \hat{\beta} - \beta_*, \Sigma(\hat{\beta} - \beta_*) \rangle$, $\bar{R}_n = \mathbb{E}[R \mid X]$ for its average over the noise and the target, and $D_1(t) = \sum_j \pi_j \eta_j/(\eta_j + t)$, $D_2(t) = \sum_j \pi_j \eta_j^2/(\eta_j + t)^2$ for the limits of $\mathrm{df}_1(t)/n$ and $\mathrm{df}_2(t)/n$.

Theorem 12.2 (The learning curve for Gaussian features). *Under Assumption 12.1, the equation $\lambda = t(1 - D_1(t))$ has exactly one solution $t = \lambda_{\mathrm{eff}}$ with $D_1(t) < 1$, at this solution $D_2(\lambda_{\mathrm{eff}}) < 1$, and, in probability as $n \to \infty$,*

$$\bar{R}_n \longrightarrow R_* = \frac{\sum_j \lambda_{\mathrm{eff}}^2 \eta_j b_j^2/(\eta_j + \lambda_{\mathrm{eff}})^2 + \sigma^2 D_2(\lambda_{\mathrm{eff}})}{1 - D_2(\lambda_{\mathrm{eff}})}.$$

This is (12.3) with the target powers summed over each eigenspace; for one block ($m = 1$, $\eta_1 = 1$, $\pi_1 = \gamma = 1/\alpha$, $b_1 = r$) it is the isotropic ridge risk of Theorem 9.2, $R_* = (r^2 \lambda_{\mathrm{eff}}^2 + \sigma^2 \gamma)/((1 + \lambda_{\mathrm{eff}})^2 - \gamma)$ (Exercise 12.1). It is a special case of Hastie et al. (2022, Theorems 5–6 of arXiv version 5) and of Theorem 12.7 below, which allow general covariances and non-Gaussian features; the special case keeps the structure of the proof.

Throughout, $A = \sum_j a_j P_j$ is block-scalar with $a_j \ge a > 0$, $Q = (\hat\Sigma + A)^{-1}$, $c = \sup_n p/n$ and $\bar\eta = \max_j \eta_j$. Since $\hat\Sigma \ge 0$ and $A \succeq aI$, $\|Q\| \le 1/a$.

Lemma 12.3 (Concentration of traces). *For every deterministic symmetric D, $\operatorname{Var}\big(\operatorname{tr}(DQ)/n\big) \le 4c\bar\eta^2\|D\|^2/(a^4n^2)$.*

Proof. The plan is to apply the Gaussian Poincaré inequality (Theorem A.6) to $f = \operatorname{tr}(DQ)/n$ as a function of the standard Gaussian vector $(g_1, \dots, g_n)$; f is smooth because $\hat\Sigma + A \succeq aI$. Moving g_i in a direction h changes $\hat\Sigma$, to first order, by $\Sigma^{1/2}(hg_i^\top + g_i h^\top)\Sigma^{1/2}/n$ and Q by minus Q times this times Q; since D and Q are symmetric, $\nabla_{g_i} f = -2n^{-2}\Sigma^{1/2}QDQx_i$. The matrix $M = (QDQ)\Sigma(QDQ)$ is positive semidefinite with $\|M\| \le \bar\eta\|D\|^2/a^4$, and $\operatorname{tr}(M\hat\Sigma) \le \|M\| \operatorname{tr}\hat\Sigma$, so

$$\sum_i \|\nabla_{g_i} f\|^2 = \frac{4}{n^4}\sum_i x_i^\top M x_i = \frac{4}{n^3}\operatorname{tr}(M\hat\Sigma) \le \frac{4\bar\eta\|D\|^2}{a^4n^3}\operatorname{tr}\hat\Sigma.$$

Since $\mathbb{E}\operatorname{tr}\hat\Sigma = \operatorname{tr}\Sigma \le c\bar\eta n$, the Poincaré inequality gives $\operatorname{Var} f \le \mathbb{E}\|\nabla f\|^2 \le 4c\bar\eta^2\|D\|^2/(a^4n^2)$. ∎

Lemma 12.4 (A stable fixed point). *Let $a_j > 0$, $c_j \ge 0$ with $\sum_j c_j > 0$, and $F(\delta) = \sum_j c_j/(a_j + \eta_j/(1+\delta))$ for $\delta \ge 0$. Then F has exactly one fixed point δ_*, $0 < \delta_* \le \sum_j c_j/a_j$, and for every $\delta \ge 0$, $|\delta - \delta_*| \le (1+\max(\delta,\delta_*))^2|\delta - F(\delta)|$.*

Proof. Put $u = 1 + \delta$ and $\phi(u) = \sum_j c_j/(a_j u + \eta_j)$, strictly decreasing on $[1,\infty)$. Then $F(\delta) = u\phi(u)$, so $\delta - F(\delta) = u\psi(u)$ with $\psi(u) = 1 - 1/u - \phi(u)$, which is strictly increasing, negative at $u = 1$ and tends to 1; so it has exactly one zero $u_* > 1$, and $\delta_* = u_* - 1$ is the only fixed point. Since $u/(a_j u + \eta_j) < 1/a_j$, $\delta_* = F(\delta_*) \le \sum_j c_j/a_j$. Finally $\psi'(v) = v^{-2} - \phi'(v) \ge v^{-2}$, so by the mean value theorem $|\psi(u)| = |\psi(u) - \psi(u_*)| \ge |u - u_*|/\max(u,u_*)^2$, while $|\psi(u)| = |\delta - F(\delta)|/u \le |\delta - F(\delta)|$; together they give the bound. ∎

Proposition 12.5 (Deterministic equivalent of block traces). *Let δ_n be the fixed point of $F_n(\delta) = \sum_j (p_j\eta_j/n)/(a_j + \eta_j/(1+\delta))$ and $\bar\tau_j = (p_j/n)/(a_j + \eta_j/(1+\delta_n))$. There is a constant C, depending only on c, $\bar\eta$ and a, such that $\mathbb{E}(\operatorname{tr}(P_j Q)/n - \bar\tau_j)^2 \le C/n$ for every j and n. If the a_j do not depend on n, then $\operatorname{tr}(P_j Q)/n \to \pi_j/(a_j + \eta_j/(1+\delta_\infty))$ in probability, with δ_∞ the fixed point of $F_\infty(\delta) = \sum_j \pi_j\eta_j/(a_j + \eta_j/(1+\delta))$.*

This is the *deterministic equivalent* of the resolvent on block traces:[6] $\mathbb{E}Q$ is close to $T = (A + \Sigma/(1+\delta))^{-1}$, as if the data covariance acted through its population value, discounted by $1 + \delta$.

[6] The term comes from random-matrix theory: a deterministic matrix whose normalized traces against bounded test matrices agree, to leading order, with those of a random one.

Proof. The plan is the cavity argument with an error budget: an exact leave-one-out identity, the replacement of a random quadratic form by its mean, and the fixed point. Write $Q_i = (\hat\Sigma - x_i x_i^\top/n + A)^{-1}$, independent of x_i; $d_i = x_i^\top Q_i x_i/n \ge 0$; $\delta = \mathbb{E}\operatorname{tr}(\Sigma Q)/n$; and $T = (A + \Sigma/(1+\delta))^{-1}$, block-scalar. Each of Q, Q_i, T has norm at most $1/a$. We use two facts: for a standard

Gaussian $g \in \mathbb{R}^p$ and a deterministic symmetric M, $\mathbb{E} g^\top M g = \operatorname{tr} M$ and $\operatorname{Var} g^\top M g = 2 \operatorname{tr} M^2 \le 2p\|M\|^2$ (diagonalize M); and $\mathbb{E}\|x_1\|^2 = \operatorname{tr}\Sigma \le c\bar\eta n$, $\mathbb{E}\|x_1\|^4 = (\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}\Sigma^2 \le 3c^2\bar\eta^2 n^2$.

Step 1: an exact identity. Multiplying $Q^{-1} = Q_1^{-1} + x_1 x_1^\top/n$ by $Q_1 x_1$ gives the Sherman–Morrison identity $Q x_1 = Q_1 x_1/(1+d_1)$. From $Q(\hat\Sigma + A) = I$ and the exchangeability of the samples, $I - \mathbb{E}[Q]A = \mathbb{E}[Q\hat\Sigma] = \mathbb{E}[Q x_1 x_1^\top] = \mathbb{E}[Q_1 x_1 x_1^\top/(1+d_1)]$. Insert $1/(1+d_1) = 1/(1+\delta) + (\delta - d_1)/((1+\delta)(1+d_1))$, use $\mathbb{E}[Q_1 x_1 x_1^\top] = \mathbb{E}[Q_1]\Sigma$, write $\mathbb{E}Q_1 = \mathbb{E}Q + \mathbb{E}(Q_1 - Q)$, and multiply on the right by T:

$$T - \mathbb{E}Q = \frac{\mathbb{E}[Q_1 - Q]\Sigma T}{1+\delta} + \frac{\mathbb{E}\left[Q_1 x_1 x_1^\top T\,(\delta - d_1)/(1+d_1)\right]}{1+\delta}. \tag{12.4}$$

Step 2: the two error terms. Take $\operatorname{tr}(D\,\cdot)/n$ of (12.4) for a deterministic D. By the resolvent identity $Q_1 - Q = Q x_1 x_1^\top Q_1/n$, the first term is $\mathbb{E}[x_1^\top Q_1 \Sigma T D Q x_1]/(n^2(1+\delta))$, at most $\bar\eta\|D\|\mathbb{E}\|x_1\|^2/(a^3 n^2) \le c\bar\eta^2\|D\|/(a^3 n)$ in absolute value. In the second, $1 + d_1 \ge 1$ and $|x_1^\top T D Q_1 x_1| \le \|D\|\|x_1\|^2/a^2$, so the Cauchy–Schwarz inequality bounds it by $(\sqrt{3}c\bar\eta\|D\|/a^2)(\mathbb{E}(d_1-\delta)^2)^{1/2}$.

Step 3: the quadratic form concentrates. Split $d_1 - \delta$ into $d_1 - \operatorname{tr}(\Sigma Q_1)/n$, $\operatorname{tr}(\Sigma(Q_1 - Q))/n$ and $\operatorname{tr}(\Sigma Q)/n - \delta$. Given Q_1, the first is $(g_1^\top M g_1 - \operatorname{tr} M)/n$ with $M = \Sigma^{1/2} Q_1 \Sigma^{1/2}$, $\|M\| \le \bar\eta/a$, so its second moment is at most $2c\bar\eta^2/(a^2 n)$. The second equals $x_1^\top Q_1 \Sigma Q x_1/n^2$, with second moment at most $3c^2\bar\eta^4/(a^4 n^2)$. The third has second moment at most $4c\bar\eta^4/(a^4 n^2)$ by Lemma 12.3. Hence $\mathbb{E}(d_1 - \delta)^2 \le C_1/n$ with $C_1 = 6c\bar\eta^2/a^2 + 3(3c^2 + 4c)\bar\eta^4/a^4$, and with Step 2, for every deterministic D,

$$\left|\frac{\operatorname{tr}(DT)}{n} - \frac{\mathbb{E}\operatorname{tr}(DQ)}{n}\right| \le \frac{C_2\|D\|}{\sqrt{n}}, \qquad C_2 = \frac{c\bar\eta^2}{a^3} + \frac{\sqrt{3C_1}\,c\bar\eta}{a^2}. \tag{12.5}$$

Step 4: the fixed point. Because T is block-scalar, $\operatorname{tr}(\Sigma T)/n = F_n(\delta)$, and (12.5) with $D = \Sigma$ gives $|\delta - F_n(\delta)| \le \bar\eta C_2/\sqrt{n}$. Both δ and δ_n are at most $c\bar\eta/a$, so Lemma 12.4 gives $|\delta - \delta_n| \le U^2\bar\eta C_2/\sqrt{n}$ with $U = 1 + c\bar\eta/a$.

Step 5: the block traces. With $D = P_j$, (12.5) gives $|(p_j/n)/(a_j + \eta_j/(1+\delta)) - \mathbb{E}\operatorname{tr}(P_j Q)/n| \le C_2/\sqrt{n}$. The map $\delta \mapsto (p_j/n)/(a_j + \eta_j/(1+\delta))$ is $(c\bar\eta/a^2)$-Lipschitz on $[0,\infty)$, so $|\mathbb{E}\operatorname{tr}(P_j Q)/n - \bar\tau_j| \le C_3/\sqrt{n}$ with $C_3 = C_2(1 + c\bar\eta^2 U^2/a^2)$, and Lemma 12.3 with $D = P_j$ adds a variance of at most $4c\bar\eta^2/(a^4 n^2)$.

Step 6: the limit. If the a_j are fixed, $|F_n - F_\infty| \le \sum_j |p_j/n - \pi_j|\eta_j/a_j \to 0$ uniformly, so $|\delta_\infty - F_n(\delta_\infty)| \to 0$, and Lemma 12.4 for F_n gives $\delta_n \to \delta_\infty$. Then $\bar\tau_j$ converges to the stated limit, and Chebyshev's inequality completes the proof. ∎

The risk is a derivative of traces; derivatives of limits need care.

Lemma 12.6 (Derivatives of convex limits). *Let f_n be random differentiable convex functions on an open interval I with $f_n(y) \to f(y)$ in probability for every $y \in I$, where f is deterministic and differentiable at $x \in I$. Then $f_n'(x) \to f'(x)$ in probability.*

PROOF. Fix $\epsilon > 0$ and choose $h > 0$ with $[x-h, x+h] \subset I$ and both quotients $(f(x \pm h) - f(x))/(\pm h)$ within ϵ of $f'(x)$. By convexity, $(f_n(x) - f_n(x-h))/h \le f_n'(x) \le (f_n(x+h) - f_n(x))/h$, and both bounds converge in probability to numbers within ϵ of $f'(x)$; so $\mathbb{P}(|f_n'(x) - f'(x)| > 2\epsilon) \to 0$. ∎

PROOF OF THEOREM 12.2. The plan is to write $\bar{R}_n$ as a bias and a variance, each the derivative of a first-order trace, and to pass to the limit with Proposition 12.5 and Lemma 12.6.

The effective ridge. Let $\varphi(t) = t(1 - D_1(t))$. The factor $1 - D_1$ increases strictly from $1 - \sum_j \pi_j$ to 1; let t_0 be its zero if $\sum_j \pi_j > 1$ and $t_0 = 0$ otherwise. Then $\varphi \le 0$ on $(0, t_0]$, and on (t_0, ∞) it is positive and strictly increasing to ∞, so λ has exactly one preimage λ_{eff} with $D_1(\lambda_{\text{eff}}) < 1$. Moreover $\varphi'(t) = 1 - D_1(t) + t \sum_j \pi_j \eta_j/(\eta_j + t)^2 = 1 - D_2(t)$, which is positive at λ_{eff} because the middle expression is a sum of two positive terms there. For a ridge $\mu > 0$ let $\delta(\mu)$ be the fixed point of F_∞ with all $a_j = \mu$; with $t = \mu(1 + \delta)$, the equation $\delta = (1 + \delta) \sum_j \pi_j \eta_j/(\mu(1 + \delta) + \eta_j) = (1 + \delta) D_1(t)$ reads $\mu = t(1 - D_1(t))$. So $\mu(1 + \delta(\mu)) = \lambda_{\text{eff}}(\mu)$, the effective ridge of the ridge μ.

Bias and variance. Write $Q_\mu = (\hat{\Sigma} + \mu I)^{-1}$. Since $Q_\lambda \hat{\Sigma} = I - \lambda Q_\lambda$, $\hat{\beta} - \beta_* = -\lambda Q_\lambda \beta_* + \sigma Q_\lambda X^\top \varepsilon/n$. Average over ε, which has mean zero and identity covariance, and over β_*, with $\mathbb{E}\beta_* \beta_*^\top = B = \sum_j (b_j^2/p_j) P_j$:

$$\bar{R}_n = \lambda^2 \operatorname{tr}(B Q_\lambda \Sigma Q_\lambda) + \frac{\sigma^2}{n} \operatorname{tr}(\Sigma Q_\lambda \hat{\Sigma} Q_\lambda) =: B_n + V_n.$$

The variance. The function $h_n(\mu) = \operatorname{tr}(\Sigma Q_\mu)/n$ is convex, because $h_n''(\mu) = 2 \operatorname{tr}(\Sigma Q_\mu^3)/n \ge 0$. From $Q_\lambda \hat{\Sigma} Q_\lambda = Q_\lambda - \lambda Q_\lambda^2$, $V_n = \sigma^2(h_n(\lambda) + \lambda h_n'(\lambda))$. By Proposition 12.5 with $A = \mu I$, $h_n(\mu) = \sum_j \eta_j \operatorname{tr}(P_j Q_\mu)/n$ converges in probability to $F_\infty(\delta(\mu)) = \delta(\mu)$. Since $\varphi' = 1 - D_2 > 0$, the implicit function theorem makes $\lambda_{\text{eff}}(\mu)$ differentiable with derivative $1/(1 - D_2)$, so $\mu\delta(\mu) = \lambda_{\text{eff}}(\mu) - \mu$ has derivative $D_2/(1 - D_2)$. By Lemma 12.6, $h_n'(\lambda) \to \delta'(\lambda)$, and $V_n \to \sigma^2 D_2/(1 - D_2)$ at λ_{eff}.

The bias. For $t \in I = (-\lambda/(2\bar{\eta}), \infty)$ let $Q_{\lambda,t} = (\hat{\Sigma} + \lambda I + t\Sigma)^{-1}$, which is Q for $A = \lambda I + t\Sigma$, with $a_j = \lambda + t\eta_j \ge \lambda/2$, and $f_n(t) = \operatorname{tr}(B Q_{\lambda,t}) = \sum_j b_j^2 (n/p_j) \operatorname{tr}(P_j Q_{\lambda,t})/n$. Then $f_n'(t) = -\operatorname{tr}(B Q_{\lambda,t} \Sigma Q_{\lambda,t})$, so $B_n = -\lambda^2 f_n'(0)$, and $f_n''(t) = 2 \operatorname{tr}(B^{1/2}(Q_{\lambda,t}\Sigma) Q_{\lambda,t} (\Sigma Q_{\lambda,t}) B^{1/2}) \ge 0$. By Proposition 12.5, $f_n(t) \to f(t) = \sum_j b_j^2/(\lambda + t\eta_j + \eta_j/(1 + \tilde{\delta}(t)))$ in probability, where $\tilde{\delta}(t)$ is the fixed point of $F_t(\delta) = \sum_j \pi_j \eta_j/(\lambda + t\eta_j + \eta_j/(1 + \delta))$. At $t = 0$, with $\delta = \tilde{\delta}(0) = \delta(\lambda)$, $\partial_\delta F_0 = \sum_j \pi_j \eta_j^2/(\lambda_{\text{eff}} + \eta_j)^2 = D_2 < 1$ and $\partial_t F_t = -(1 + \delta)^2 D_2$, so the implicit function theorem gives $\tilde{\delta}'(0) = -(1 + \delta)^2 D_2/(1 - D_2)$. Differentiating f, with $\lambda + \eta_j/(1 + \delta) = (\lambda_{\text{eff}} + \eta_j)/(1 + \delta)$ and $\lambda(1 + \delta) = \lambda_{\text{eff}}$,

$$-\lambda^2 f'(0) = \sum_j \frac{\lambda_{\text{eff}}^2 \eta_j b_j^2}{(\eta_j + \lambda_{\text{eff}})^2} \Big(1 - \frac{\tilde{\delta}'(0)}{(1 + \delta)^2}\Big) = \frac{1}{1 - D_2} \sum_j \frac{\lambda_{\text{eff}}^2 \eta_j b_j^2}{(\eta_j + \lambda_{\text{eff}})^2}.$$

Lemma 12.6 gives $B_n \to -\lambda^2 f'(0)$, and adding the limits of B_n and V_n proves the theorem. ∎

WHAT THE GENERAL THEOREMS ADD. Deterministic targets, through bilinear forms $v^\top Q v$; non-Gaussian features, through the Hanson–Wright inequality; infinitely many eigenvalues, through effective ranks; multiplicative errors.

Theorem 12.7 (Cheng and Montanari 2024, Theorem 1, in outline). *Let $x = \Sigma^{1/2} z$ with Σ of finite trace and norm 1, and z with independent sub-Gaussian coordinates of mean 0 and variance 1 (or with a convex concentration property); assume $\|\Sigma^{-1/2}\beta_*\| < \infty$ and an effective-rank condition on Σ. Define λ_{eff} by (12.2) with the eigenvalues of Σ, and B_n, V_n as the two terms of (12.3). Under a condition relating n, the effective rank and $\lambda/\lambda_{\text{eff}}$, for every $D > 0$, with probability $1 - O(n^{-D+1})$, the bias and the variance of ridge regression given X are $B_n(1+\epsilon_n)$ and $V_n(1+\epsilon'_n)$, with explicit $\epsilon_n, \epsilon'_n \to 0$.*

For kernels, Misiakiewicz and Saeed (2024) prove the same equivalent, in an unrefereed preprint, under hypercontractivity of the top eigenfunctions, a relaxed Hanson–Wright inequality, and a high-frequency part of the kernel matrix close to a multiple of the identity, which inner-product kernels on the sphere satisfy in high dimension. On the sphere at $n \asymp d^\ell$ the curves are theorems: Xiao et al. (2023, Theorem 2) for the risk averaged over isotropic random labels, and Misiakiewicz (2022), in a preprint, with the effective regularization $(n\lambda + \mu_{>\ell})/\mu_\ell$, in which the mass of the higher harmonics acts as ridge (Exercise 12.6). Between the stages the kernel learns exactly the projection of f_* on the polynomials of degree at most ℓ (Ghorbani et al. 2021, Thm. 4).

For random features, $\lambda_{\text{MM}} = \lambda/\psi$ is the ridge in Mei and Montanari's units.

Theorem 12.8 (Mei and Montanari 2022, Theorems 1, 2 and 7). *Let x_i, θ_j be independent and uniform on $S^{d-1}(\sqrt{d})$, $N/d \to \psi$, $n/d \to \alpha$ in $(0,\infty)$, and ς weakly differentiable with $|\varsigma(u)|, |\varsigma'(u)| \le c_0 e^{c_1|u|}$ and $\mu_0, \mu_1, \mu_* \neq 0$. Let $y_i = f_d(x_i) + \sigma\varepsilon_i$ with $\mathbb{E}\varepsilon^4 < \infty$ and $f_d(x) = \beta_0 + \langle\beta_1\rangle x + f_d^{\text{NL}}(x)$, where $\|\beta_1\|^2 \to F_1^2$ and f_d^{NL} is a centered isotropic Gaussian process orthogonal to the polynomials of degree at most one, of variance tending to F_*^2. For every $\lambda_{\text{MM}} > 0$ the expected excess risk of random-features ridge regression is, up to $o(1)$,*

$$F_1^2\,\mathscr{B}(\zeta,\psi,\alpha,\lambda_{\text{MM}}/\mu_*^2) + (\sigma^2 + F_*^2)\,\mathscr{V}(\zeta,\psi,\alpha,\lambda_{\text{MM}}/\mu_*^2) + F_*^2,$$

with $\zeta = \mu_1/\mu_*$ and $\mathscr{B}$, $\mathscr{V}$ explicit functions of the solution of two algebraic equations (their Definition 1). For linear f_d the risk converges in probability, and the same limit holds for ridge regression on the Gaussian covariates $u = \mu_0\mathbf{1} + \mu_1\Theta x/\sqrt{d} + \mu_* w$ with $x \sim \mathcal{N}(0, I_d)$ (Theorem 7, whose proof is omitted as a repetition of that of Theorem 2).

At $N, n \asymp d$ random features learn only the linear part of f_*. The proof writes the risk as traces of resolvent products, and its replacement of the feature kernel by $\mu_1^2\Theta\Theta^\top/d + \mu_*^2 I$ inside them is the rigorous form of the Hermite heuristic; El Karoui (2010, Theorem 2.1) had proved the matrix version, a kernel matrix $(f(x_i^\top x_j/p))$ with $n \asymp p$ being in operator norm

$f'(0)XX^\top/p$ plus multiples of $\mathbf{1}\mathbf{1}^\top$ and I. Beyond ridge, Hu and Lu (2023) prove Gaussian equivalence for convex losses, odd activations with three bounded derivatives, strongly convex penalties and Gaussian weights, by a Lindeberg replacement of the samples; Montanari and Saeed (2022) remove convexity for minimizers in a "domain of Gaussianity", by interpolating between the features and Gaussian vectors with the same covariance, for centered activations with three bounded derivatives (their Corollary 2); full proofs are in their arXiv version 3 (2026). Neither theorem covers the ReLU.

12.5 The proof: where the formula fails

In fixed dimension at zero ridge neither Gaussian features nor a nearly scalar high-frequency kernel matrix are available, and the formula is wrong. The cleanest case is the circle kernel of Section 12.1, whose RKHS is the Sobolev space H^1.[7]

[7] The periodic linear spline minimizes $\|f'\|$ among the interpolants of the data; the term $c_0(f)^2$ of the RKHS norm adds a small parabolic correction (Proposition 12.9).

COMPUTATION 12.1 · THE FORMULA AT ZERO RIDGE ON THE CIRCLE

1 *The spectrum.* $\eta_0 = 1$ and $\eta_k = k^{-2}$ twice. With $a = t^{-1/2}$, $1/(1+tk^2) = a^2/(a^2+k^2)$, and the classical sums of $(k^2+a^2)^{-1}$ and $(k^2+a^2)^{-2}$ over $k \geq 1$ give

$$\mathrm{df}_1(t) = \frac{1}{1+t} + \pi a \coth \pi a - 1,$$
$$\mathrm{df}_2(t) = \frac{1}{(1+t)^2} + \frac{\pi a \coth \pi a + \pi^2 a^2 \operatorname{csch}^2 \pi a - 2}{2}.$$

2 *The effective ridge.* At $\lambda = 0$, (12.2) reads $\mathrm{df}_1(\lambda_{\mathrm{eff}}) = n$; df_1 decreases from $+\infty$ to 0, so there is one root λ_n, and $\lambda_n \to 0$. As $a \to \infty$, $\coth \pi a - 1$ and $a^2 \operatorname{csch}^2 \pi a$ vanish exponentially, so $\mathrm{df}_1 = \pi a + o(1)$, $\mathrm{df}_2 = \pi a/2 + o(1)$, $\lambda_n = \pi^2 n^{-2}(1 + O(1/n))$ and $\mathrm{df}_2(\lambda_n)/n \to \frac{1}{2}$.

3 *The noise term.* The variance term of (12.3) at $\lambda = 0$ is $\sigma^2(\mathrm{df}_2/n)/(1 - \mathrm{df}_2/n) \to \sigma^2$; it is $0.98657\sigma^2$ at $n = 16$ and $0.99978\sigma^2$ at $n = 64$.

4 *Other spectra.* For $\eta_k = k^{-b}$, $b > 1$, $\mathrm{df}_2/\mathrm{df}_1 \to 1 - 1/b$ and the noise term tends to $\sigma^2(b-1)$ (Exercise 12.4): the *tempered overfitting* constant of Mallinar et al. (2022) (test error $b\sigma^2$).

Proposition 12.9 (The ridgeless circle). *Let P_x be uniform on the circle $[0, 2\pi)$, K the kernel of Computation 12.1, and $y_i = \sigma\varepsilon_i$ (target 0). For $n \geq 2$ distinct points whose largest gap between circularly consecutive points is G, the minimum-norm interpolant $\hat{f}$ satisfies*

$$\left| \mathbb{E}_\varepsilon \|\hat{f}\|_{L^2}^2 - \frac{2\sigma^2}{3} \right| \leq \sigma^2 \Big(\frac{G^3}{12\pi} + \frac{G^5}{128\pi} \Big).$$

If the points are independent and uniform, then $\mathbb{E}_\varepsilon \|\hat{f}\|_{L^2}^2 \to 2\sigma^2/3$ in probability, while the formula (12.3) at $\lambda = 0$ tends to σ^2: the formula overestimates the ridgeless noise term by the factor $3/2$.

PROOF. The plan is to identify the RKHS norm, show that the interpolant is a linear spline plus a small parabolic correction, and compute the noise of the spline exactly. Norms and inner products are in $L^2(\,\mathrm{d}\theta/2\pi)$, and $c_0(f) = \frac{1}{2\pi}\int_0^{2\pi} f$.

The RKHS norm. For $f = c_0 + \sum_{k\ge1}(a_k\sqrt{2}\cos k\theta + b_k\sqrt{2}\sin k\theta)$, $\|f\|_{\mathcal{H}}^2$ is the sum of the squared coefficients divided by the eigenvalues, $c_0^2 + \sum_k k^2(a_k^2 + b_k^2) = c_0(f)^2 + \|f'\|^2$. So $\hat f$ minimizes $J(f) = c_0(f)^2 + \|f'\|^2$ over H^1 subject to $f(\theta_i) = y_i$; J is the square of a Hilbert norm, so the minimizer is unique.

The interpolant. Let s be the periodic piecewise-linear interpolant of the data; on each gap $[\theta_i, \theta_{i+1}]$, of length g_i, let $q(\theta) = (\theta - \theta_i)(\theta_{i+1} - \theta)$; and put $S_3 = \sum_i g_i^3$. Let $h \in H^1$ vanish at the nodes. Then $\langle s', h'\rangle = 0$, because s' is constant on each gap and h' integrates to zero there, and integration by parts on each gap, with $q'' = -2$, gives $c_0(h) = \frac12\langle h', q'\rangle$. Hence $J(s+h) = (c_0(s) + c_0(h))^2 + \|s'\|^2 + \|h'\|^2$, and by the Cauchy–Schwarz inequality $\|h'\|^2 \ge 4c_0(h)^2/\|q'\|^2$, with equality for multiples of q. Since $\|q'\|^2 = S_3/(6\pi)$, it remains to minimize $(c_0(s)+c)^2 + 24\pi c^2/S_3$ over $c = c_0(h)$, and

$$\hat f = s - \tau q, \qquad \tau = \frac{12\pi\, c_0(s)}{S_3 + 24\pi}.$$

The noise of the spline. On the gap from θ_i to θ_{i+1}, $s = (1-u)y_i + u y_{i+1}$ with u uniform on $[0,1]$ when θ is uniform on the gap, and y_i, y_{i+1} are independent because $n \ge 2$. Hence, for every configuration of points,

$$\mathbb{E}_\varepsilon\|s\|^2 = \frac{\sigma^2}{2\pi}\sum_i g_i \int_0^1 \big((1-u)^2 + u^2\big)\,\mathrm{d}u = \frac{2\sigma^2}{3}.$$

The correction. Write $s = \sum_i y_i\phi_i$, with ϕ_i the hat function at θ_i. Then $c_0(s) = \sum_i w_i y_i$ with $w_i = (g_{i-1} + g_i)/(4\pi) \le G/(2\pi)$ and $\sum_i w_i = 1$, and $\tau = 12\pi c_0(s)/(S_3 + 24\pi)$ with $12\pi/(S_3 + 24\pi) \le \frac12$, so that $\|\hat f\|^2 - \|s\|^2 = -2\tau\langle s\rangle q + \tau^2\|q\|^2$. Integrating q against the two linear pieces of ϕ_i gives $\langle\phi_i\rangle q = (g_{i-1}^3 + g_i^3)/(24\pi)$, and the y_i are uncorrelated, so

$$0 \le \mathbb{E}_\varepsilon\big[c_0(s)\langle s\rangle q\big] = \sigma^2\sum_i w_i\langle\phi_i\rangle q = \frac{\sigma^2}{96\pi^2}\sum_i (g_{i-1}+g_i)(g_{i-1}^3 + g_i^3) \le \frac{\sigma^2 G^3}{12\pi},$$

by $g_{i-1}^3 + g_i^3 \le G^2(g_{i-1}+g_i)$, $g_{i-1}+g_i \le 2G$ and $\sum_i(g_{i-1}+g_i) = 4\pi$. With $|q| \le G^2/4$ and $\mathbb{E}_\varepsilon c_0(s)^2 = \sigma^2\sum_i w_i^2 \le \sigma^2 G/(2\pi)$, the two terms of $\mathbb{E}_\varepsilon\|\hat f\|^2 - \mathbb{E}_\varepsilon\|s\|^2$ are at most $\sigma^2G^3/(12\pi)$ and $\sigma^2G^5/(128\pi)$ in absolute value.

Random points. The gap after θ_i exceeds x only if the other $n-1$ points avoid an arc of length x, so $\mathbb{P}(G > x) \le n(1 - x/2\pi)^{n-1}$; for $x = 4\pi\log n/n$ this is at most $n^{-1+2/n} \to 0$. So $G \to 0$ in probability and the bound tends to 0. The statement about the formula is Computation 12.1. ∎

The order G^3 and its constant are sharp: for equally spaced points the deviation is $\sigma^2G^3/(12\pi)$ up to a factor $1 + O(G^2)$ (Exercise 12.7). With a target $f_* \in \mathcal{H}$ nothing changes: the interpolation error of s is at most $2\sqrt{2\pi G}\|f_*'\|$, the correction is $O(G^2)$, and the cross term vanishes in expectation over the noise, so $\mathbb{E}_\varepsilon R \to 2\sigma^2/3$. The formula matches Gaussian features with the

same spectrum (Figure 12.1): the eigenfunctions make the difference. The Gaussian approximation spreads the unresolved modes as independent noise over every coordinate, while on the circle the interpolant is local, a hat function per sample, and its noise is an average over one gap. In the language of Misiakiewicz and Saeed (2024), the high-frequency part of the kernel matrix is far from a multiple of the identity; a ridge that makes up most of the effective ridge, $\lambda/\lambda_{\text{eff}} \to 1$, makes that part negligible against $n\lambda I$ and restores the formula (Section 12.6).[8]

The ingredients of the example are known. Belkin et al. (2018) note that in one dimension the minimum-norm interpolant of the Laplace kernel $e^{-|x-x'|/w}$ tends to linear interpolation as the bandwidth w grows, and their bound $2\sigma^2/(d+2)$ on the noise term of simplicial interpolation is $2\sigma^2/3$ at $d = 1$. Mallinar et al. (2022), who checked $b\sigma^2$ on Gaussian features, call this tempered overfitting, ask that the universality assumption be checked for each kernel, and leave open whether the Laplace kernel on a sphere reaches the formula's noise term or a fixed fraction of it. For a kernel with RKHS H^1 the circle answers this in one dimension: the fraction is $2/3$. We have not found this comparison in print.

[8] That Gaussian-design predictions can fail in fixed dimension was known in other forms: below a crossover ridge the formula gets the exponent of the learning curve of a Laplace kernel wrong (Tomasini et al. 2022); Gaussian equivalence needs a strong ridge (Cheng et al. 2024); and a kernel on the disk has a ridgeless variance that appears to diverge (Barzilai and Shamir 2024).

12.6 Measuring it

Four scripts measure the formula. Each runs in under two minutes on one core; intervals are Student-t 95% intervals over independent repetitions.

THE THEOREM. [9] The script `12-kernels-theorem.py` takes three blocks with $\eta = (1, 0.3, 0.05)$ and $\pi = (0.3, 0.6, 1.2)$, with $b^2 = (0.5, 1, 2)$, $\sigma^2 = 0.25$ and $\lambda = 0.05$, for which $\lambda_{\text{eff}} = 0.236079$, $D_2 = 0.420909$ and $R_* = 0.431273$. It computes $\bar{R}_n$ for each design by the trace formula of the proof; averaged over 60, 40, 20, 10, 5 designs at $n = 50, 100, 200, 400, 800$ ($p = 2.1n$), it is 0.43426 ± 0.00187, 0.43258 ± 0.00128, 0.43153 ± 0.00060, 0.43170 ± 0.00069 and 0.43127 ± 0.00098: the finite-size bias, 0.0030 at $n = 50$ and 0.0013 at $n = 100$, decays about like $1/n$, faster than the $n^{-1/2}$ of Proposition 12.5. The training error, 0.03062 at $n = 800$, matches $(\lambda/\lambda_{\text{eff}})^2(R_* + \sigma^2) = 0.030559$ (Exercise 12.5).

[9] The scripts for the theorem, the circle and the sphere average over the noise analytically, through traces, which removes the largest source of Monte Carlo error.

THE CIRCLE. Figure 12.1(a) is Proposition 12.9 at finite n. The interpolant's noise term, computed from the closed form $s - \tau q$ (equal to the kernel computation to 10^{-11} at $n = 8, 32$), is 0.6058 at $n = 8$, 0.6524 at $n = 16$, and within 10^{-5} of $2/3$ from $n = 256$ on; Gaussian features give 1.0001 ± 0.0032, 1.0010 ± 0.0019 and 1.0025 ± 0.0033 at $n = 128, 256, 512$, on the formula. With a ridge the agreement returns (Figure 12.1(b)): at $n = 128, 512, 1024$ the kernel gives 0.0852 ± 0.0061, 0.0413 ± 0.0032, 0.0323 ± 0.0036 against 0.0856, 0.0397, 0.0322, though 9 to 19% above the formula at $n \le 64$. Sweeping the ridge at $n = 512$, the ratio of the kernel's noise term to the formula's is 0.665 when $\lambda/\lambda_{\text{eff}} = 0.003$, 0.735 at 0.18, 0.925 at 0.55 and 0.991 at 0.82 (Exercise 12.10).

THE SPHERE. Figure 12.2 uses the ReLU arc-cosine kernel $h(t) = (\sqrt{1-t^2} + (\pi - \arccos t)t)/(2\pi)$ on $S^{19}(\sqrt{20})$. Its eigenvalue on the harmonics of degree k is $\xi_k = \mathbb{E}[h(\tau)P_k(\tau)]$ (Funk–Hecke), with P_k the Gegenbauer polynomial

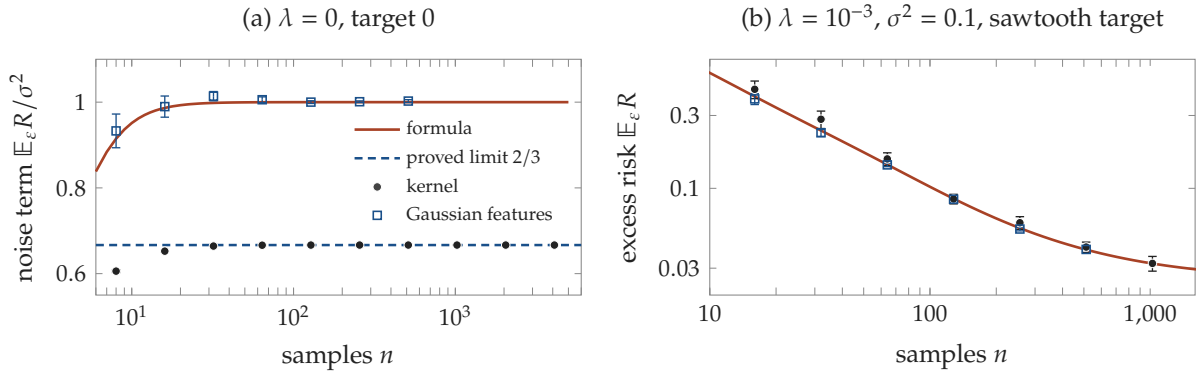


Figure 12.1. Kernel ridge regression on the circle with $\eta_0 = 1$, $\eta_k = k^{-2}$ (dots), Gaussian features with the same spectrum (squares), and the formula (12.3) (solid). (a) Ridgeless, pure noise: the exact noise term of the interpolant, from its closed form, averaged over 400 point sets per n, tends to the proved limit 2/3 (dashed); Gaussian features (80, 24, 12 draws at $n \le 128, 256, 512$; modes up to $k = 4n$ explicit, the rest as a diagonal term) follow the formula, which tends to 1. (b) Ridge $\lambda = 10^{-3}$, $\sigma^2 = 0.1$, target the sawtooth $(\pi - \theta)/\sqrt{2}$, with power k^{-2} on the sine modes: exact $\mathbb{E}_\varepsilon R$ over 100, 40, 12 point sets ($n \le 128, \le 512, 1024$). Seeds 1201, 1202. Script: `figures/scripts/12-kernels-circle.py`.

and τ the inner product of two independent uniform unit vectors; the degree masses $B(d,k)\xi_k$ are 0.163182, 0.25, 0.077336 for $k = 0, 1, 2$, and 0.009482 beyond.[10] The simulated risk is exact in the noise and in the population, through the L^2 Gram kernel $\sum_k \xi_k^2 B(d,k) P_k$ (checked against 10^5 test points). The formula with the full spectrum follows the simulation through both stages, $n \approx d$ and $n \approx B(d,2) = 209$: 0.3551 ± 0.0178 against 0.3413 at $n = 209$, 0.04149 ± 0.00044 against 0.04149 at $n = 800$, 0.02246 ± 0.00005 against 0.02250 at $n = 1600$. The linear Gaussian equivalent, exact at $n \asymp d$ as $d \to \infty$, stalls at the power of the degree-2 target, 0.92 at $n = 1600$: the kernel learns it once n is of order d^2.

[10] The masses of odd degree 3 and higher vanish: the ReLU has no odd Hermite coefficient beyond the first, so h has no odd part beyond $t/4$.

RANDOM FEATURES. Figure 12.3 shows the double descent of random features at $\alpha = 3$: a peak at $N = n$ of height 2.0541 for $\lambda_{\mathrm{MM}} = 10^{-3}$ (simulated 2.0217 ± 0.1290 for random features, 2.0802 ± 0.1195 for Gaussian covariates), removed by the larger ridge. Of the 44 simulated points, 42 contain the formula in their intervals, about what 95% intervals should give, and random features and Gaussian covariates cannot be told apart at this precision. The formula needs $\zeta = \mu_1/\mu_*$, as eq. (9) of Mei and Montanari defines it: with μ_1^2/μ_*^2, as their eqs. (4) and (52) print it, the prediction at $\lambda_{\mathrm{MM}} = 0.1$, $\psi = 2$ is 0.351, against 0.4842 ± 0.0219 (random features), 0.4915 ± 0.0174 (Gaussian covariates) and 0.4912 (formula).

No measurement contradicts a theorem. Two observations go beyond the theorems: at $d = 20$ the formula with the full spectrum is accurate across both learning stages, and on the circle its error is governed by $\lambda/\lambda_{\mathrm{eff}}$.

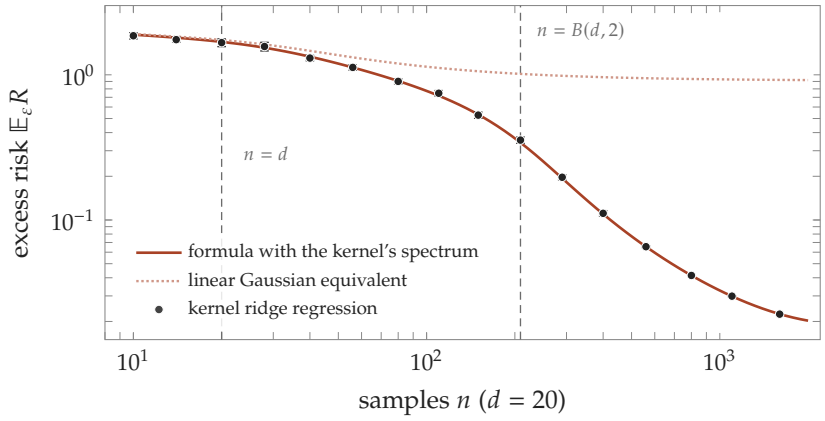


Figure 12.2. Kernel ridge regression on $S^{19}(\sqrt{20})$ with the ReLU arc-cosine kernel, target $x_1 + x_1x_2$ (powers 1 and $d/(d+2)$ in degrees 1 and 2), $\sigma^2 = 0.1$, unnormalized ridge $n\lambda = 10^{-3}$. Dots: exact $\mathbb{E}_\varepsilon R$ over 30, 12, 6 data sets ($n \le 290$, ≤ 800, larger). Solid: the formula with the kernel's spectrum at $d = 20$ (degrees 0 to 16). Dotted: the formula for the linear Gaussian equivalent, in which the mass of degree at least 2 is ridge and the degree-2 target is noise. Seed 1203. Script: `figures/scripts/12-kernels-sphere.py`.

12.7 The status board

The learning-curve formula is a theorem for Gaussian features, for kernels on the sphere at $n \asymp d^\ell$ (with random labels) and for random features with a ridge; in fixed dimension with a ridge it is supported by numerics, and without one it is false. Gaussian equivalence is proved for ridge and for convex losses with smooth odd activations, false for the linear equivalent beyond the proportional regime or after a large step and for the classification error on real data, and conjectured for the ReLU with classification losses.

STATUS BOARD · the Gaussian approximation ↔ universality

PREDICTION	STATUS	PROOF OR EVIDENCE
Learning-curve formula (12.3) for Gaussian features, $\lambda > 0$	proved	Hastie et al. (2022); Cheng and Montanari (2024, Thm. 1); Theorem 12.2
The formula for kernels on the sphere at $n \asymp d^\ell$: learning stages and peaks	proved under	random labels: Xiao et al. (2023, Thm. 2); between stages: Ghorbani et al. (2021, Thm. 4)
The formula for general kernels with concentrated features	preprint	Misiakiewicz and Saeed (2024), arXiv:2403.08938v1, under concentration conditions; real data: numerics

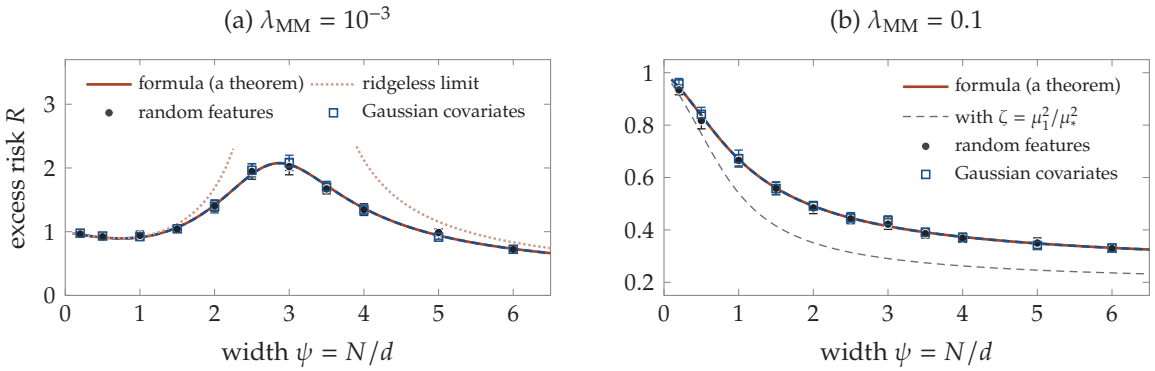


Figure 12.3. Random-features ridge regression with ReLU features, $d = 100$, $n = 300$ ($\alpha = 3$), linear target with $F_1^2 = 1$, noise $\sigma^2 = 0.5$. Solid with dashes over it: Mei and Montanari's formula with $\zeta = \mu_1/\mu_*$, the physical prediction and, for $\lambda_{\mathrm{MM}} > 0$, a theorem (Theorem 12.8). Dots: random features; squares: the Gaussian covariate model; 20 repetitions each. Dotted in (a): the ridgeless formula (their Theorem 3), conjectured to be the risk of the ridgeless estimator. Thin dashed in (b): the formula with $\zeta = \mu_1^2/\mu_*^2$, as eq. (4) of their arXiv version 5 prints it. Seed 1204. Script: `figures/scripts/12-kernels-rf.py`.

The formula at $\lambda = 0$ in fixed dimension	false	circle, $2\sigma^2/3$ against σ^2: Proposition 12.9; the $2\sigma^2/3$ of linear interpolation: Belkin et al. (2018)
Random-features ridge risk (Mei–Montanari), $\lambda > 0$	proved	Mei and Montanari (2022, Thms. 1–2)
Ridgeless random features as the $\lambda \to 0$ limit of the formula	conjectured	numerics; Mei and Montanari (2022, Remark 1)
Random features ≈ Gaussian covariates, ridge	proved	Mei and Montanari (2022, Thm. 7), linear target, proof omitted as a repetition; with a proof, odd smooth activations: Hu and Lu (2023)
Random features ≈ Gaussian covariates, convex losses	proved under	odd smooth activation, strongly convex penalty: Hu and Lu (2023)
The same for ReLU features with logistic or hinge loss	conjectured	numerics: sign features, logistic loss (Gerace et al. 2020); ReLU features, max-margin (Montanari et al. 2025); the universality theorems exclude the ReLU
The Gaussian covariate model for real feature maps on real data	false	classification error, MNIST and CIFAR-10: Loureiro et al. (2022, arXiv v3, Figs. 3, 7); square loss: conjectured

Linear Gaussian equivalence at n or $N \asymp d^\ell$, $\ell \geq 2$, or after one large gradient step	false	Ghorbani et al. (2021) and Mei et al. (2022); Figure 12.2; Ba et al. (2022)

12.8 Where the translation fails

WHAT WAS ASSUMED. The theorem of this chapter assumes Gaussian features, a positive ridge and a target spread isotropically over each eigenspace; the general theorems keep a concentration hypothesis that real feature maps need not satisfy. The kernel theorems assume data uniform on the sphere or the hypercube, and Xiao et al. (2023) average over random labels. Mei and Montanari assume spherical data and features, a linear target plus an isotropic Gaussian process, and $\lambda > 0$; Hu and Lu exclude the ReLU, which is not odd. The physics asserts the formula for every kernel and data law.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. At zero ridge in fixed dimension the Gaussian approximation misjudges the noise term (Proposition 12.9) and can miss its divergence (Cheng et al. 2024). Beyond the proportional regime the linear equivalent declares the nonlinear part of the target unlearnable, while kernels and random features learn degree-ℓ polynomials once n (and N) exceed d^ℓ (Ghorbani et al. 2021; Mei et al. 2022); the refined, degree-ℓ equivalent of the physicists' learning stages is the one that is proved. Even it fails at $n \asymp N \asymp d^2$ for single-index targets and non-quadratic losses, by a preprint of Wen et al. (2025).[11] After one large gradient step the features carry a spike that the plain equivalent misses (Ba et al. 2022). For data from a mixture of Gaussians, one Gaussian with the same covariance can fail when the target is aligned with the means and covariances of the clusters (Pesce et al. 2023). And Loureiro et al. (2022) conjecture that the Gaussian covariate model describes real feature maps on real data: their experiments and their universality theorem for linear teachers support it for the square loss, and their own experiments show that it fails for the classification error on MNIST and CIFAR-10.

[11] Their conditional Gaussian equivalent keeps, along one direction of feature space, the second Hermite polynomial of the target's index, and is Gaussian in the orthogonal complement.

OPEN PROBLEMS. A criterion, for a given kernel and data distribution, for the exactness of the Gaussian approximation; in particular the ridgeless learning curve in fixed dimension, of which the circle gives one constant (Exercise 12.12). The interchange of $\lambda \to 0$ and $d \to \infty$ for random features (Mei and Montanari 2022, Remark 1). Gaussian equivalence for ReLU random features with logistic or hinge loss. And an explanation of the accuracy of the finite-d formula across learning stages (Figure 12.2).

Exercises

12.1 CONSOLIDATION. (The flat spectrum.) Derive the one-block case of Theorem 12.2 and check that it is Theorem 9.2. For $\gamma = \lambda = r^2 = \sigma^2 = 1$, show that λ_{eff} is the golden ratio ϕ and $R_* = 1/\phi = 0.618034$.

12.2 CONSOLIDATION. (Spectral bias.) Show that $\lambda_{\text{eff}} \geq \lambda$, that $\lambda_{\text{eff}} \to \lambda$ as $n \to \infty$ for a fixed summable spectrum, and that the modal error

$\lambda_{\text{eff}}^2/((\eta_k+\lambda_{\text{eff}})^2(1-\mathrm{df}_2/n))$ decreases in η_k.

12.3 CONSOLIDATION. (Hermite data of the ReLU.) Show that $\mu_0=\mu_2=1/\sqrt{2\pi}$, $\mu_1=\frac{1}{2}$, $\mu_*^2=\frac{1}{4}-\frac{1}{2\pi}$ and $\zeta=1.658897$, and that the Taylor coefficients of $h(\rho)=(\sqrt{1-\rho^2}+(\pi-\arccos\rho)\rho)/(2\pi)$ at 0 are $\frac{1}{2\pi}$, $\frac{1}{4}$, $\frac{1}{4\pi}$, which are $\mu_k^2/k!$ for $k=0,1,2$.

12.4 CONSOLIDATION. (Power-law spectra.) For $\eta_k=k^{-b}$, $b>1$, show that $\mathrm{df}_2(t)/\mathrm{df}_1(t)\to 1-1/b$ as $t\to 0$, so that the ridgeless noise term of the formula tends to $\sigma^2(b-1)$. *(Hint: compare sums with integrals;* $\int_0^\infty(1+v^b)^{-2}\,\mathrm{d}v=(1-1/b)\int_0^\infty(1+v^b)^{-1}\,\mathrm{d}v$.*)*

12.5 BY HAND. (The training error.) In the replica computation write $h^a=\omega+\sqrt{\chi/\beta}\,z_a$, with ω Gaussian of variance q, and apply Laplace's method in the z_a: show that the training residual is $(y-\omega)/(1+\chi)$, so that the training error is $(\lambda/\lambda_{\text{eff}})^2(R+\sigma^2)$ (0.03062 against 0.030559 in `12-kernels-theorem.py`). Why is this generalized cross-validation?

12.6 BY HAND. (A block that acts as ridge.) Add to a spectrum p' modes of eigenvalue μ/p' and total target power v^2, and let $p'\to\infty$ at fixed n. Show that in (12.2) the block replaces λ by $\lambda+\mu/n$, and that in (12.3) its target power is unlearned and acts as noise. Relate this to Misiakiewicz's effective regularization $(n\lambda+\mu_{>\ell})/\mu_\ell$.

12.7 BY HAND. (Grid points.) For equally spaced points on the circle, use the closed form of Proposition 12.9 to show that the ridgeless noise term is $\sigma^2(\frac{2}{3}-\frac{2\pi^2}{3n^3}+O(n^{-5}))$; the formula, blind to the points, still predicts $\sigma^2(1+o(1))$.

12.8 TRANSLATION. (Iterating the saddle point.) Show that the map F of Lemma 12.4 is increasing and concave, that $\delta_{k+1}=F(\delta_k)$ from $\delta_0=0$ increases to δ_*, and that for $A=\lambda I$, $F'(\delta_*)=D_2(\lambda_{\text{eff}})<1$: the physicists' iteration converges.

12.9 TRANSLATION. (The susceptibility is observable.) With the hat matrix $S=XQ_\lambda X^\top/n$, use $\operatorname{tr}S=p-\lambda\operatorname{tr}Q_\lambda$ and Proposition 12.5 to prove that $1-\operatorname{tr}S/n\to\lambda/\lambda_{\text{eff}}=1/(1+\chi)$ in probability. Which step of the replica computation does this replace?

12.10 MEASUREMENT. (Turning the ridge on.) At $n=512$ the ratio of the kernel's noise term to the formula's is 0.665, 0.667, 0.735, 0.925, 0.991, 0.999 for $\lambda=10^{-7},\dots,10^{-2}$ (part (c) of `12-kernels-circle.py`). Plot it against $\lambda/\lambda_{\text{eff}}$ for $n=128$, 512, 2048; why is this, not λ, the variable that matters?

12.11 MEASUREMENT. (A misprint, measured.) Set $\zeta=\mu_1^2/\mu_*^2$ in `12-kernels-rf.py` and repeat Figure 12.3. By how many interval widths does each point disagree?

12.12 RESEARCH. ★ (Fixed-dimension learning curves.) Find the ridgeless noise constant of kernel ridge regression with the Laplace kernel on S^d,

$d \geq 2$, at uniform random points, and compare it with the formula's σ^2/d; then find a learning-curve formula exact at zero ridge in fixed dimension. *(Hint: in one dimension the interpolant is local.)*

12.13 RESEARCH. ★★ (Beyond smooth activations.) Prove Gaussian equivalence of the test error of logistic regression on ReLU random features in the proportional regime.

Notes and References

At about the time of Sollich (1998), Dietrich et al. (1999) computed learning curves of support vector machines by replicas, with the Gaussian approximation that Canatar et al. adopt; with Opper and Urbanczik (2001) they found, in Canatar et al.'s summary, limits $n \propto d^\ell$, the learning stages that the rigorous theory later established. Mallinar et al. (2022) used the formula to classify overfitting as benign, tempered or catastrophic.

Random features entered machine learning as a computational device (Rahimi and Recht 2007). Their double descent was computed by Mei and Montanari (2022), building on Hastie et al. (2022), and by Gerace et al. (2020) with replicas and the Gaussian equivalence that Goldt et al. (2020) proposed for the hidden manifold model. Loureiro et al. (2022) solved the Gaussian covariate model for convex losses with the convex Gaussian min-max theorem (Theorem 8.3).

The proof of Theorem 12.2 is the standard leave-one-out derivation of deterministic equivalents, written for this book with explicit constants and with the convexity step of Lemma 12.6. The circle counterexample, Proposition 12.9, was found in the research for this volume and rechecked here; its ingredients are in Belkin et al. (2018) (Section 12.5).

TRANSLATING THE LITERATURE. Canatar et al. (2021) write P for our n, $\lambda_{\mathrm{CBP}} = n\lambda$, $\kappa = n\lambda_{\mathrm{eff}}$, $\gamma = \mathrm{df}_2/n$, $\eta_\rho \bar{w}_\rho^2 = v_k^2$ and $E_g = R$; Bordelon et al. (2020) write p for n and λ_ρ for η_k. Mallinar et al. (2022) write $\delta = n\lambda$, $\kappa = \lambda_{\mathrm{eff}}$ and $E_0 = 1/(1 - \mathrm{df}_2/n)$, and include the noise in the test error. Cheng and Montanari (2024) write $\lambda_\star$ for λ_{eff}; Misiakiewicz (2022) uses the unnormalized ridge $n\lambda$ and $\psi = n/B(d, \ell)$. Mei and Montanari (2022) write σ for our activation ς, τ^2 for our σ^2, $\psi_1 = \psi$, $\psi_2 = \alpha$, and the ridge as $\frac{1}{n}\sum(y - f)^2 + \frac{N\lambda_{\mathrm{MM}}}{d}\|a\|^2$; their arXiv version 5 defines $\zeta = \mu_1^2/\mu_*^2$ in eqs. (4) and (52) but $\zeta = \mu_1/\mu_*$ in eq. (9), and only the latter makes their eq. (15) agree with the equations of their proof (eq. (85)) and with simulation (Figure 12.3).

References

Ba, J., M. A. Erdogdu, T. Suzuki, Z. Wang, D. Wu, and G. Yang (2022). "High-Dimensional Asymptotics of Feature Learning: How One Gradient Step Improves the Representation". In: *Advances in Neural Information Processing Systems 35*, pp. 37932–37946. arXiv: 2205.01445.

Barzilai, D. and O. Shamir (2024). "Generalization in Kernel Regression Under Realistic Assumptions". In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research, pp. 3096–3132. arXiv: 2312.15995.

Belkin, M., D. J. Hsu, and P. P. Mitra (2018). "Overfitting or perfect fitting? Risk bounds for classification and regression rules that interpolate". In: *Advances in Neural Information Processing Systems 31*. arXiv: 1806.05161.

Bordelon, B., A. Canatar, and C. Pehlevan (2020). "Spectrum Dependent Learning Curves in Kernel Regression and Wide Neural Networks". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research, pp. 1024–1034. arXiv: 2002.02561.

Canatar, A., B. Bordelon, and C. Pehlevan (2021). "Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks". In: *Nature Communications* 12, p. 2914. arXiv: 2006.13198.

Cheng, C. and A. Montanari (2024). "Dimension free ridge regression". In: *The Annals of Statistics* 52.6, pp. 2879–2912. arXiv: 2210.08571.

Cheng, T. S., A. Lucchi, A. Kratsios, and D. Belius (2024). "A Comprehensive Analysis on the Learning Curve in Kernel Ridge Regression". In: *Advances in Neural Information Processing Systems 37*, pp. 24659–24723. arXiv: 2410.17796.

Dietrich, R., M. Opper, and H. Sompolinsky (1999). "Statistical Mechanics of Support Vector Networks". In: *Physical Review Letters* 82.14, pp. 2975–2978.

El Karoui, N. (2010). "The spectrum of kernel random matrices". In: *The Annals of Statistics* 38.1, pp. 1–50. arXiv: 1001.0492.

Gerace, F., B. Loureiro, F. Krzakala, M. Mézard, and L. Zdeborová (2020). "Generalisation error in learning with random features and the hidden manifold model". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research, pp. 3452–3462. arXiv: 2002.09339.

Ghorbani, B., S. Mei, T. Misiakiewicz, and A. Montanari (2021). "Linearized two-layers neural networks in high dimension". In: *The Annals of Statistics* 49.2, pp. 1029–1054. arXiv: 1904.12191.

Goldt, S., M. Mézard, F. Krzakala, and L. Zdeborová (2020). "Modeling the Influence of Data Structure on Learning in Neural Networks: The Hidden Manifold Model". In: *Physical Review X* 10.4, p. 041044. arXiv: 1909.11500.

Hastie, T., A. Montanari, S. Rosset, and R. J. Tibshirani (2022). "Surprises in high-dimensional ridgeless least squares interpolation". In: *The Annals of Statistics* 50.2, pp. 949–986. arXiv: 1903.08560.

Hu, H. and Y. M. Lu (2023). "Universality Laws for High-Dimensional Learning With Random Features". In: *IEEE Transactions on Information Theory* 69.3, pp. 1932–1964. arXiv: 2009.07669.

Loureiro, B., C. Gerbelot, H. Cui, S. Goldt, F. Krzakala, M. Mézard, and L. Zdeborová (2022). "Learning curves of generic features maps for realistic datasets with a teacher-student model". In: *Journal of Statistical Mechanics: Theory and Experiment* 2022.11, p. 114001. arXiv: 2102.08127.

Mallinar, N., J. B. Simon, A. Abedsoltan, P. Pandit, M. Belkin, and P. Nakkiran (2022). "Benign, Tempered, or Catastrophic: Toward a Refined Taxonomy of Overfitting". In: *Advances in Neural Information Processing Systems 35*, pp. 1182–1195. arXiv: 2207.06569.

Mei, S., T. Misiakiewicz, and A. Montanari (2022). "Generalization error of random feature and kernel methods: Hypercontractivity and kernel matrix concen-

tration". In: *Applied and Computational Harmonic Analysis* 59, pp. 3–84. arXiv: 2101.10588.

Mei, S. and A. Montanari (2022). "The Generalization Error of Random Features Regression: Precise Asymptotics and the Double Descent Curve". In: *Communications on Pure and Applied Mathematics* 75.4, pp. 667–766. arXiv: 1908.05355.

Misiakiewicz, T. (2022). *Spectrum of inner-product kernel matrices in the polynomial regime and multiple descent phenomenon in kernel ridge regression*. arXiv: 2204.10425.

Misiakiewicz, T. and B. Saeed (2024). *A non-asymptotic theory of Kernel Ridge Regression: deterministic equivalents, test error, and GCV estimator*. arXiv: 2403.08938.

Montanari, A., F. Ruan, Y. Sohn, and J. Yan (2025). "The generalization error of max-margin linear classifiers: Benign overfitting and high dimensional asymptotics in the overparametrized regime". In: *The Annals of Statistics* 53.2, pp. 822–853. arXiv: 1911.01544.

Montanari, A. and B. N. Saeed (2022). "Universality of empirical risk minimization". In: *Proceedings of Thirty Fifth Conference on Learning Theory*. Vol. 178. Proceedings of Machine Learning Research. Extended abstract, pp. 4310–4312. arXiv: 2202.08832.

Opper, M. and R. Urbanczik (2001). "Universal Learning Curves of Support Vector Machines". In: *Physical Review Letters* 86.19, pp. 4410–4413.

Pesce, L., F. Krzakala, B. Loureiro, and L. Stephan (2023). "Are Gaussian Data All You Need? The Extents and Limits of Universality in High-Dimensional Generalized Linear Estimation". In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research, pp. 27680–27708. arXiv: 2302.08923.

Rahimi, A. and B. Recht (2007). "Random Features for Large-Scale Kernel Machines". In: *Advances in Neural Information Processing Systems*. Vol. 20.

Sollich, P. (1998). "Learning Curves for Gaussian Processes". In: *Advances in Neural Information Processing Systems 11*, pp. 344–350.

Tomasini, U. M., A. Sclocchi, and M. Wyart (2022). "Failure and success of the spectral bias prediction for Laplace Kernel Ridge Regression: the case of low-dimensional data". In: *Proceedings of the 39th International Conference on Machine Learning*. Vol. 162. Proceedings of Machine Learning Research, pp. 21548–21583. arXiv: 2202.03348.

Wen, G. G., H. Hu, Y. M. Lu, Z. Fan, and T. Misiakiewicz (2025). *When does Gaussian equivalence fail and how to fix it: Non-universal behavior of random features with quadratic scaling*. arXiv: 2512.03325.

Xiao, L., H. Hu, T. Misiakiewicz, Y. M. Lu, and J. Pennington (2023). "Precise learning curves and higher-order scaling limits for dot-product kernel regression". In: *Journal of Statistical Mechanics: Theory and Experiment* 2023.11, p. 114005. arXiv: 2205.14846.

Part IV

Landscapes and Dynamics

Inference by optimization asks a different question: not what the posterior knows, but where an algorithm goes. Physicists answer by counting critical points and by writing closed equations for the dynamics; mathematicians have proved both, on fixed time horizons and for specific models, and the gap between the energy that dynamics reaches and the energy that exists is one of the cleanest facts the two languages share.



13 Counting Critical Points

Kac–Rice and the spiked tensor model

THE DICTIONARY —— TAP complexity ↔ Kac–Rice and the second moment

IN THIS CHAPTER. Maximum likelihood for tensor PCA minimizes a random polynomial on the sphere whose noise part is the pure spherical p-spin glass. Physicists count its critical points as TAP states, averaging the Kac–Rice integral with a self-averaging Hessian determinant, and find exponentially many local minima between the ground state $-E_0$ and a threshold $-E_\infty$, where minima become marginal and descent is predicted to stop. We prove the mean counts exactly at every size, by a Gaussian regression and an identity that turns the absolute determinant of a GOE matrix into the density of one more eigenvalue, and in the limit by large deviations of the extreme eigenvalue; that the typical count equals the mean below the threshold is Subag's second-moment theorem. Exhaustive enumeration at small N matches the exact formula; gradient descent at $N \le 200$ stops above $-E_\infty$, where the exact formula places almost all minima at these sizes; warm-started descent on the spiked tensor lands on the predicted overlap. As of 2026 the typical landscape near the spike and the endpoint of gradient descent are not proved.

13.1 The question: how rugged is the likelihood of tensor PCA?

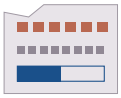
In *tensor PCA* (Montanari and Richard 2014) one observes the p-tensor $Y = \lambda N^{-(p-1)/2}(x^*)^{\otimes p} + G$, where x^* lies on the sphere $S_N = \{\sigma \in \mathbb{R}^N : \|\sigma\|^2 = N\}$, G has independent $\mathcal{N}(0,1)$ entries, and $\lambda \ge 0$ is the signal-to-noise ratio. On S_N the log-likelihood of σ is an increasing affine function of $\langle Y, \sigma^{\otimes p}\rangle$, so the *maximum likelihood estimator* minimizes

$$H_{N,\lambda}(\sigma) = H_N(\sigma) - \lambda N\, m(\sigma)^p, \qquad m(\sigma) = \frac{\langle \sigma, x^*\rangle}{N}, \tag{13.1}$$

where, with $J = -G$, $H_N(\sigma) = N^{-(p-1)/2} \sum_{i_1,\dots,i_p=1}^{N} J_{i_1\cdots i_p}\sigma_{i_1}\cdots\sigma_{i_p}$. The noise part H_N is the *pure spherical p-spin* glass in the normalization of Auffinger et al. (2013) that this book uses: a centered Gaussian process with $\mathbb{E}H_N(\sigma)H_N(\sigma') = N\,R(\sigma,\sigma')^p$, $R(\sigma,\sigma') = \langle\sigma,\sigma'\rangle/N$. The case $\lambda = 0$ is the pure model; energies are per spin.

For $p \ge 3$ the estimator is hard to compute but statistically strong: it correlates with x^* once λ exceeds a constant (Jagannath et al. 2020). Efficient methods need far more: gradient flow and Langevin dynamics from a

PHYSICS
NUMERICS
PROVED IN PART



uniform start with $m > 0$ reach the spike in time of order one when $\lambda = N^{\alpha}$, $\alpha > (p-2)/2$ (Ben Arous et al. 2020, Theorem 2.1), and sum-of-squares relaxations succeed, for odd p, from λ of order $N^{(p-2)/4}$ up to a logarithm (Hopkins et al. 2015). Why does descent fail for a range of λ growing with N, when the minimizer is good? The physicist's answer: the landscape is rugged, with exponentially many local minima, and descent stops in one of them.

A *critical point* of a smooth function on S_N is a zero of its gradient along the sphere; its *index* is the number of negative eigenvalues of its Hessian along the sphere (local minima have index 0). For Borel $B \subset \mathbb{R}$, $\mathrm{Crt}_N(B)$ counts the critical points of H_N with $H_N(\sigma)/N \in B$, and $\mathrm{Crt}_{N,k}(B)$ those of index k. The *annealed complexity* is $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_N(B)$; the *quenched complexity* $\frac{1}{N}\mathbb{E}\log \mathrm{Crt}_N(B)$ describes a typical landscape.[1]

[1] Physicists call the logarithm of the number of states per degree of freedom the complexity, or configurational entropy, of a glass. The word has nothing to do with computational complexity, though this chapter is about a link between the two.

THE PHYSICAL PREDICTION · Crisanti and Sommers 1995; Kurchan and Laloux 1996; Cugliandolo and Kurchan 1993; Ros et al. 2019. Let $N \to \infty$, $E_\infty = 2\sqrt{(p-1)/p}$, and $J(x) = \frac{x}{4}\sqrt{x^2-4} - \log\frac{x+\sqrt{x^2-4}}{2}$ for $x \geq 2$, $J = 0$ on $[0,2]$.

(i) In the pure model the number of critical points with energy near u is, on average and typically, $e^{N\Sigma(u)}$ with

$$\Sigma(u) = \tfrac{1}{2}\log(p-1) - \tfrac{p-2}{4(p-1)}u^2 - J\big(2|u|/E_\infty\big).$$

(ii) Below the *threshold energy* $-E_\infty$ almost all critical points are local minima; above it they are saddles of index proportional to N. Minima fill $[-E_0, -E_\infty]$, where $-E_0$, the zero of Σ, is the ground state; the most numerous sit at $-E_\infty$ and are *marginal*, with Hessian eigenvalues down to 0. Saddles of index k appear only above $-E_k$, where $E_0 > E_1 > \cdots \to E_\infty$. For $p = 3$, $E_0 = 1.65700$ and $E_\infty = 1.63299$.

(iii) Gradient descent from a random start ends at $-E_\infty$, not at the ground state.

(iv) In the spiked model the threshold minima near the equator $m = 0$ survive at every fixed λ. A local minimum correlated with x^* appears at $\lambda_c(p)$ and becomes the global minimum at $\lambda_{\mathrm{ML}}(p) > \lambda_c(p)$; for $p = 3$, $\lambda_c = 2/\sqrt{3}$ and $\lambda_{\mathrm{ML}} \approx 1.21$.

Parts (i) and (ii) are the TAP complexity of Crisanti and Sommers (1995) with the index layers of Kurchan and Laloux (1996), (iii) the dynamical prediction of Cugliandolo and Kurchan (1993), (iv) the replicated Kac–Rice computation of Ros et al. (2019) in our normalization. How do physicists derive them, and what is proved? We prove the mean counts of (i) and (ii) exactly at every N and in the limit; the typical counts are Subag's theorem; (iii) is a conjecture whose Langevin version Chapter 14 discusses; (iv) is proved for mean counts and for the estimator, not for the typical landscape.

13.2 The physics: counting TAP states

13.2.1 States are critical points

At low temperature a physicist describes a spin glass by its *states*, the local minima of the *TAP free energy* of the local magnetizations (Chapter 6). For the spherical p-spin, Kurchan et al. (1993) and Crisanti and Sommers (1995) wrote it for magnetizations $\sqrt{q}\,\sigma$, $\sigma \in S_N$, $0 \le q < 1$; per spin, in our normalization and with the physicists' inverse temperature β (Auffinger et al. 2013, eq. (6.3)),

$$f_{\mathrm{TAP}}(q,\sigma) = \frac{q^{p/2}H_N(\sigma)}{\sqrt{2}\,N} - \frac{\log(1-q)}{2\beta} - \frac{\beta}{4}\big(1+(p-1)q^p - pq^{p-1}\big). \quad (13.2)$$

The direction enters only through $H_N(\sigma)/N$, so at a stationary point σ is a critical point of H_N and q solves a scalar equation in that energy, with exactly two solutions below a temperature-dependent energy (Auffinger et al. 2013, Section 6): counting TAP states is counting critical points, and at zero temperature the states are the local minima of H_N.[2]

[2] In (13.2) the first term is the energy of the magnetization, the second is minus the temperature times the entropy $\frac{1}{2}\log(1-q)$ of the fluctuations around it, and the third is the Onsager reaction term.

13.2.2 The annealed count

THE COUNT BY A DELTA FUNCTION. Write the number of critical points with energy in $[u, u+\mathrm{d}u]$ as $\int_{S_N} \delta(\operatorname{grad} H_N)|\det \operatorname{Hess} H_N|\,\delta(H_N/N-u)\,\mathrm{d}\sigma$ and average over the couplings. The law of H_N is rotation invariant, so the count is the area $|S_N|$ times a one-point quantity. At a point, the covariance $N^{1-p}\langle\sigma,\sigma'\rangle^p$ gives the joint law of the value (variance N), the tangential gradient (independent coordinates of variance p) and the tangential second derivatives G, with $\mathbb{E}G_{ij}G_{kl} = \frac{p(p-1)}{N}(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$; the three are uncorrelated. The sphere adds the Lagrange multiplier pH_N/N on the diagonal, so at energy u the Hessian is

$$\sqrt{p(p-1)}\,\big(A - t(u)I\big), \qquad t(u) = u\sqrt{\tfrac{p}{p-1}} = \frac{2u}{E_\infty},$$

with A a GOE matrix of size $N-1$, semicircle on $[-2,2]$. The gradient has density $(2\pi p)^{-(N-1)/2}$ at 0, the energy $\sqrt{N/(2\pi)}\,e^{-Nu^2/2}$ at u. Now treat the spectrum of A as self-averaging, $\mathbb{E}|\det(A-t)| \approx e^{N\Phi(t)}$ with $\Phi(t) = \int \log|y-t|\,\sigma_{\mathrm{sc}}(\mathrm{d}y)$, and turn the prefactor $|S_N|(2\pi p)^{-(N-1)/2}(p(p-1))^{(N-1)/2}$ into $e^{\frac{N}{2}(1+\log(p-1))+o(N)}$ by Stirling's formula. The complexity at energy u is

$$\Sigma(u) = \tfrac{1}{2}\big(1+\log(p-1)\big) - \tfrac{1}{2}u^2 + \Phi\big(t(u)\big). \quad \diamond \quad (13.3)$$

COMPUTATION 13.1 · THE COMPLEXITY FROM THE SEMICIRCLE

1 *Inside the bulk.* The Stieltjes transform $s(z) = \frac{1}{2}(-z+\sqrt{z^2-4})$ of the semicircle (Theorem A.15) has real part $-t/2$ at $z = t+\mathrm{i}0$, $|t|<2$, so $\Phi'(t) = -\operatorname{Re} s(t+\mathrm{i}0) = t/2$. With $\Phi(0) = \int \log|y|\,\sigma_{\mathrm{sc}}(\mathrm{d}y) = -\frac{1}{2}$ (substitute $y = 2\cos\theta$), $\Phi(t) = \frac{t^2}{4} - \frac{1}{2}$ for $|t| \le 2$.

2 *Outside.* For $t > 2$, $\Phi'(t) = -s(t) = \frac{1}{2}(t-\sqrt{t^2-4})$, so $\Phi(t) =$

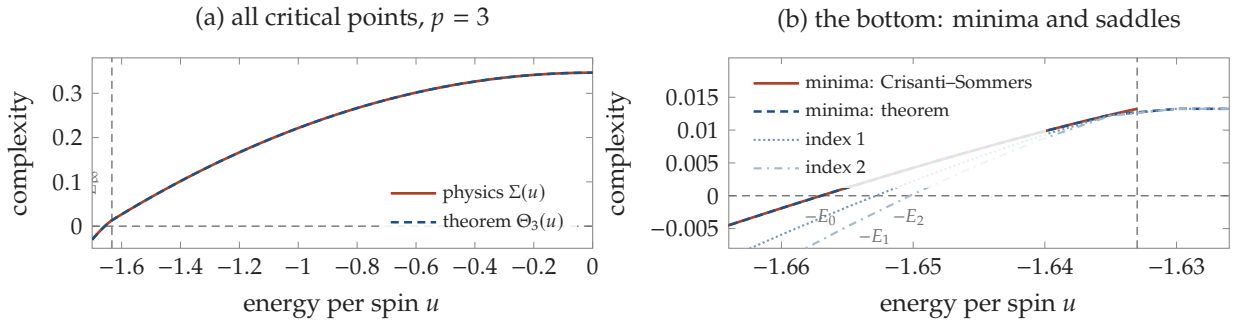


Figure 13.1. The complexity of the pure spherical 3-spin. (a) The physicists' $\Sigma(u)$, (13.3) (carnelian), and the limit $\Theta_3(u)$ of $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_N((-\infty, u))$ of Theorem 13.5 (lapis dashed); they coincide on $u \le 0$. (b) The bottom: the complexity of minima of Crisanti and Sommers (1995), as printed in Auffinger et al. (2013, eq. (2.18)) with $u = \sqrt{2}E$ (carnelian), the theorem for minima (lapis dashed), and saddles of index 1 and 2 (Auffinger et al. 2013, Theorem 2.5). Minima disappear below $-E_0 = -1.65700$, saddles of index k below $-E_k$ ($E_1 = 1.65287$, $E_2 = 1.65025$); above $-E_\infty = -1.63299$ (gray) the fixed-index curves are flat. Script: `13-kacrice-complexity.py`.

$\frac{t^2}{4} - \frac{1}{2} - J(t)$, with $J(t) = \frac{1}{2}\int_2^t \sqrt{y^2-4}\,\mathrm{d}y$ the rate function of the largest GOE eigenvalue (Remark A.17); Φ is even.

3 *The complexity.* Since $t(u)^2 = \frac{p}{p-1}u^2$, (13.3) is the formula of the box. For $p = 3$, $\Sigma(0) = \frac{1}{2}\log 2 = 0.34657$, $\Sigma(-E_\infty) = \frac{1}{2}\log 2 - \frac{1}{3} = 0.013240$, and $\Sigma(u) = 0$ at $u = -1.656998 = -E_0$.

4 *The index.* At energy u the Hessian spectrum fills $\sqrt{p(p-1)}\,[-2 - t, 2 - t]$ with semicircle density; a fraction $\sigma_{\mathrm{sc}}((-2, t))$ of it is negative. For $u < -E_\infty$ the typical critical point is a local minimum; at $u = -E_\infty$ the spectrum touches 0.

Below the threshold the complexity of minima is $\Sigma_0(u) = \Sigma(u)$; above it, minima would need the whole semicircle on the positive side, and physicists conclude that there are none on the exponential scale. A saddle of index k below the threshold needs k eigenvalues of A below t, outside the semicircle, each at the cost $J(|t|)$: its complexity $\Sigma(u) - kJ(|t(u)|)$ vanishes at $-E_k$ (Kurchan and Laloux 1996). Since Σ_0 increases up to $-E_\infty$, the most numerous minima sit at the threshold, where the lowest Hessian eigenvalues touch zero: they are marginal. That is the landscape behind part (iii): descent slows down among marginal minima and stops where they are typical (Figure 13.1).

13.2.3 The typical count and the ground state

The mean count can be dominated by rare landscapes. Physicists compute the typical count with replicas: $\mathbb{E}\log \mathrm{Crt} = \lim_{n\to 0}(\mathbb{E}\mathrm{Crt}^n - 1)/n$, where $\mathbb{E}\mathrm{Crt}^n$ counts n-tuples of critical points interacting through their overlaps q_{ab}. If the dominant tuples are orthogonal ($q_{ab} = 0$ for $a \neq b$), then $\mathbb{E}\mathrm{Crt}^n \approx (\mathbb{E}\mathrm{Crt})^n$ and the quenched complexity equals the annealed one, as physicists

expect for the pure model below the threshold (Crisanti and Sommers 1995); the ground state is then the zero of Σ_0. A second derivation agrees: the zero-temperature limit of the one-step replica-symmetry-breaking solution of Crisanti and Sommers (1992) gives the ground state $-\inf_{c,d>0} \frac{1}{2}[c + pd + \frac{1}{c}\log(1 + \frac{c}{d})]$ (Auffinger et al. 2013, eq. (5.11)); for $p = 3$ the infimum, at $c = 0.62502$, $d = 0.34399$, is 1.6569983635, the zero of Σ_0 to ten digits.[3]

[3] Some computations drop the absolute value of the determinant. Each critical point then counts with the sign $(-1)^{\text{index}}$, and the count over all energies is the Euler characteristic of the sphere (Exercise 13.2); below the threshold, where almost all critical points are minima, the two counts agree on the exponential scale. Cavagna et al. (1998) counted stationary points of every index of the TAP free energy.

13.2.4 The spike

With the spike, the count changes in three ways at a point of overlap m. The energy shifts by $-\lambda m^p$; the gradient shifts by $\lambda p m^{p-1}$ times the tangential part of x^*, which costs $e^{-Np\lambda^2 m^{2p-2}(1-m^2)/2}$ in the density of the gradient at 0; and the Hessian acquires the rank-one term $-\sqrt{p(p-1)}\,\theta(m) ee^\top$ along the tangential direction e of x^*, $\theta(m) = \lambda\sqrt{p(p-1)}\,m^{p-2}(1-m^2)$, which for $\theta > 1$ pulls an eigenvalue of $A - \theta ee^\top$ out of the semicircle to $-(\theta + 1/\theta)$ (the BBP transition of Chapter 5) and can turn a would-be minimum into a saddle. With the entropy $\frac{1}{2}\log(1 - m^2)$ of the points at overlap m, the complexity at overlap m and energy u is

$$\Sigma_\lambda(m,u) = \tfrac{1}{2}\big(1 + \log(p-1)\big) + \tfrac{1}{2}\log(1-m^2) - \tfrac{p}{2}\lambda^2 m^{2p-2}(1-m^2) - \tfrac{1}{2}(u + \lambda m^p)^2 + \Phi\big(t(u)\big). \tag{13.4}$$

For $p \geq 3$ the spike has no gradient and no curvature at the equator, where $\Sigma_\lambda(0,u) = \Sigma(u)$: the threshold minima survive there at every λ. Away from it, $\sup_u \Sigma_\lambda(m,u)$ can touch zero again at an overlap m^* near 1, an isolated minimum correlated with the truth, exactly when $m^{2p-4}(1-m^2) = 1/(p\lambda^2)$ has a root, that is for

$$\lambda \geq \lambda_c(p) = \left(\frac{(p-1)^{p-1}}{p\,(p-2)^{p-2}}\right)^{1/2}, \tag{13.5}$$

$\lambda_c(3) = 2/\sqrt{3} = 1.15470$; m^* is the largest root. Ros et al. (2019) computed the typical complexity by a replica-symmetric replicated Kac–Rice formula. For $p = 3$ a band of minima around the equator grows with λ; at λ_c an isolated minimum detaches from its top, at $\lambda_{\text{ML}} \approx 1.21$ it becomes the global minimum, and the band shrinks but never disappears. The two thresholds are the inverse dynamical and static temperatures of the pure model, $T_d = \sqrt{3}/2$ and $T_s = 0.8288$ in our normalization (Chapter 14).[4]

[4] Ros et al. (2019, eq. (58)) write $r_c = \frac{p}{2}\beta_d$ and $r_{\text{1ST}} = \frac{p}{2}\beta_s$; their spike is rNm^k/k in physicists' energies, so $r = p\lambda/\sqrt{2}$.

The derivations contain five uncontrolled steps: the annealed average, where the typical count is wanted; the replacement of $\mathbb{E}|\det|$ by the exponential of the mean logarithm, which exchanges an expectation with the exponential of an extensive quantity; the continuation $n \to 0$; the zero-overlap ansatz; and the identification of where descent stops with where minima are most numerous.

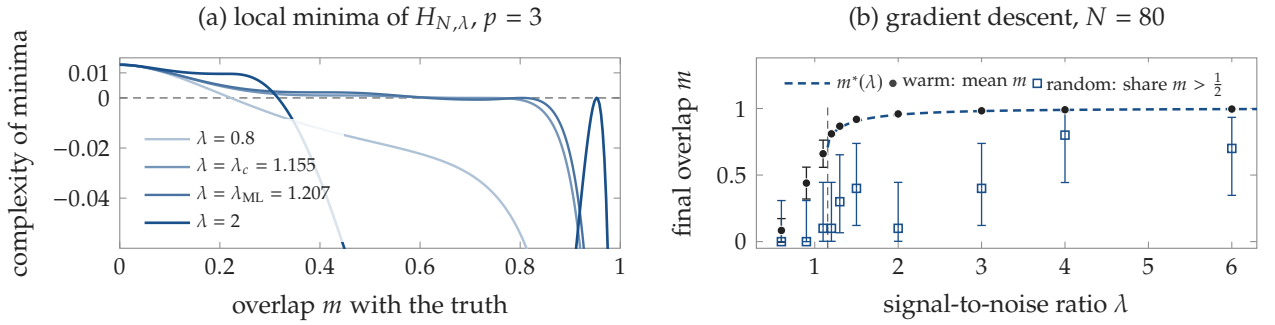


Figure 13.2. The spiked tensor model, $p = 3$. (a) The annealed complexity of local minima at overlap m, maximized over the energy (Theorem 13.10), darker for larger λ. At $m = 0$ every curve equals $\frac{1}{2}\log 2 - \frac{1}{3} = 0.0132$; for $\lambda \geq \lambda_c$ the curve touches zero again at $m^*(\lambda)$: 0.707 at λ_c, 0.803 at λ_{ML}, 0.953 at $\lambda = 2$. (b) Projected gradient descent at $N = 80$ on two noise tensors, five starts of each kind per tensor (seed 1331): from warm starts at $m = 0.5$, the mean final overlap with a 95% Student-t interval over the 10 runs; from uniform random starts, the share of runs ending with $m > \frac{1}{2}$, with 95% Clopper–Pearson intervals. The curve is $m^*(\lambda)$, above λ_{ML} also the overlap of the maximum likelihood estimator (Theorem 13.11); the dashed line marks λ_c. Script: `13-kacrice-spiked.py`.

13.3 The stone

PHYSICS · *the TAP count of Crisanti and Sommers, with replicas*

1 · Count

Count states by $\int \delta(\nabla H)|\det \nabla^2 H|\,\delta(H - Nu)$ over the sphere, averaged over the couplings.

PROOF · *Auffinger, Ben Arous and Černý; Subag; Ben Arous, Mei, Montanari and Nica*

The Kac–Rice formula (Theorem A.18), exact under nondegeneracy and a modulus of continuity.

2 · The Hessian

By isotropy the Hessian is a GOE matrix shifted by p times the energy, independent of the gradient.

Lemma 13.1: value, gradient and second derivatives at a point are independent Gaussians, exactly at every N.

3 · The determinant

Self-averaging: $\mathbb{E}|\det(A - t)| \approx e^{N\Phi(t)}$, Φ the logarithmic potential of the semicircle.

Lemma 13.2: $\mathbb{E}|\det(A - t)|$ is $c_N e^{Nt^2/4}$ times the density of an eigenvalue of a GOE(N) matrix at t, exactly. No physical counterpart; the constant comes from $p = 2$ (Lemma 13.3).

4 · The saddle point

Maximize the exponent over the energy.	The semicircle law inside the bulk, the large deviations of the extreme eigenvalue outside it (Theorem 13.5).

5 · The threshold

Minima need the shifted semicircle on the positive side, $u \le -E_\infty$; at $-E_\infty$ they are marginal.	The rate of the smallest eigenvalue is infinite inside the bulk: minima above $-E_\infty$ are rarer in mean than every exponential (Corollary 13.6).

6 · The typical count

Replicas with zero mutual overlap: $\mathbb{E}\log \mathrm{Crt} = \log \mathbb{E}\mathrm{Crt}$.	Subag's second moment (Theorem 13.8): two replicas suffice, and nothing is continued in n.

7 · The ground state

The zero of the complexity of minima, confirmed by the zero-temperature 1RSB free energy.	The first moment gives $\liminf \mathrm{GS}_N \ge -E_0$ (Corollary 13.6); the matching bound needs the second moment (Subag) or the Parisi formula (ABC, even p).

8 · The spike

Shifted energy and gradient and a rank-one Hessian term whose BBP outlier destabilizes minima; replicated Kac–Rice for the typical landscape.	The same regression (Proposition 13.9) and the outlier's large deviations give the mean landscape (Theorem 13.10), the Parisi formula the estimator (Theorem 13.11); the typical landscape has no counterpart yet.

The translation is closer here than in most chapters: the physicist's integral is the Kac–Rice formula, and the covariance computation behind the shifted GOE is the same on both sides. The proof replaces two steps. Self-averaging becomes an exact identity at every N, stronger than needed; the saddle point over the energy becomes a large-deviation estimate for one eigenvalue, the only place where the proof needs more than the semicircle law, and "no minima above the threshold" becomes the infinite rate of that eigenvalue inside the bulk. The routes part at the typical count: the replicas are continued in their number and assumed orthogonal, the proof computes exactly two, which is all concentration needs; for mixed models the assumption fails (Section 13.8).

13.4 The proof: the mean number of critical points

We prove the mean counts of the pure model exactly at every N (Theorem 13.4; Auffinger et al. (2013), Theorems 2.1–2.2) and their rates for all critical points and for minima (Theorem 13.5; their Theorems 2.8 and 2.5 with $k = 0$), given the Kac–Rice formula, the GOE facts of Chapter A and the joint density of GOE eigenvalues. We follow their proof with two changes of our own: the constant comes from the case $p = 2$, which they use only as a check (their Remark 2.3), instead of Selberg's integral; and we treat only the cases that need the large deviations of the largest eigenvalue alone.

SETTING. Fix integers $p \geq 2$ and $N \geq 2$, and extend H_N to a homogeneous polynomial of degree p on $\mathbb{R}^N$, with covariance $N^{1-p}\langle \sigma, \sigma' \rangle^p$. For smooth F on $\mathbb{R}^N$ and $\sigma \in S_N$, the gradient of F along the sphere is $P_\sigma \nabla F(\sigma)$, $P_\sigma = I - \sigma\sigma^\top/N$, and its Hessian along the sphere is the form on $T_\sigma = \sigma^\perp$

$$\operatorname{Hess} F(\sigma) = P_\sigma \nabla^2 F(\sigma) P_\sigma - \frac{\langle \sigma, \nabla F(\sigma) \rangle}{N} P_\sigma, \tag{13.6}$$

the second derivative of F along the great circle $s \mapsto \sigma \cos(s/\sqrt{N}) + \sqrt{N} v \sin(s/\sqrt{N})$, $v \perp \sigma$ a unit vector. For H_N, Euler's identity gives $\langle \sigma, \nabla H_N(\sigma) \rangle = p H_N(\sigma)$. Let $W \sim \mathrm{GOE}(N)$ (Definition A.12), with ordered eigenvalues $\lambda_{(0)} \leq \cdots \leq \lambda_{(N-1)}$ and empirical spectral measure μ_N, and let ρ_N be the density of $\mathbb{E}\mu_N$. Let A be an $(N-1) \times (N-1)$ symmetric matrix with independent centered Gaussian entries A_{ab}, $a \leq b$, of variances $(1 + \delta_{ab})/N$, that is $A = ((N-1)/N)^{1/2} W'$ with $W' \sim \mathrm{GOE}(N-1)$; its density is proportional to $e^{-N \operatorname{tr}(A^2)/4}$, and we write $A - t$ for $A - tI$. Finally $E_\infty = 2\sqrt{(p-1)/p}$, $t(u) = 2u/E_\infty$, $a = \frac{p-2}{4p}$, and J is the function of the physics box.

THE EIGENVALUE AT THE EDGE. By Remark A.17, $\lambda_{\max}(W)$ satisfies a large deviation principle with speed N and a good rate function that is $+\infty$ on $(-\infty, 2)$ and equal to J on $[2, \infty)$, where J is continuous and strictly increasing with $J(2) = 0$. Applied to closed half-lines and open intervals, the principle gives, for $x \geq 2$, $\delta > 0$ and $y < 2$,

$$\begin{aligned} \limsup_{N\to\infty} \tfrac{1}{N} \log \mathbb{P}\big(\lambda_{\max}(W) \geq x\big) &\leq -J(x), \\ \liminf_{N\to\infty} \tfrac{1}{N} \log \mathbb{P}\big(x < \lambda_{\max}(W) < x + \delta\big) &\geq -J(x), \\ \lim_{N\to\infty} \tfrac{1}{N} \log \mathbb{P}\big(\lambda_{\max}(W) \leq y\big) &= -\infty, \end{aligned} \tag{13.7}$$

and the same for $-\lambda_{(0)}$, because $-W \sim \mathrm{GOE}(N)$.

Lemma 13.1 (Gaussian regression at a point). *Fix $\sigma \in S_N$ and an orthonormal basis $e_1, \dots, e_{N-1}$ of T_σ. The value $H_N(\sigma)$, the gradient $g = (\langle e_i, \nabla H_N(\sigma) \rangle)_{i<N}$ and the matrix $G = (\langle e_i, \nabla^2 H_N(\sigma) e_j \rangle)_{i,j<N}$ are independent, with $H_N(\sigma) \sim \mathcal{N}(0, N)$, $g \sim \mathcal{N}(0, pI_{N-1})$, and G distributed as $\sqrt{p(p-1)}\,A$. In the basis (e_i), $\operatorname{Hess} H_N(\sigma) = \sqrt{p(p-1)}\,(A - t(u)I)$ with $u = H_N(\sigma)/N$ and A independent of (u, g).*

PROOF. The plan is to compute the covariances at one point by differentiating the covariance function. For orthogonal O, $\sigma \mapsto H_N(O\sigma)$ has the covariance, hence the law, of H_N, so we may take $\sigma = \sqrt{N}e_N$ and e_i the standard basis vectors. Since H_N is a polynomial with Gaussian coefficients, its derivatives are jointly Gaussian with $\mathbb{E}[\partial^\alpha H_N(x)\,\partial^\beta H_N(y)] = \partial_x^\alpha \partial_y^\beta C(x,y)$, $C(x,y) = N^{1-p}\langle x\rangle y^p$. At $x = y = \sigma$ we have $\langle\sigma\rangle\sigma = N$ and $\sigma_i = 0$ for $i < N$, so every term with a factor x_i or y_i, $i < N$, vanishes. For $i, j, k, l < N$ this leaves $\mathbb{E}H_N(\sigma)^2 = N$, $\mathbb{E}[\partial_i H_N\,\partial_j H_N] = N^{1-p}pN^{p-1}\delta_{ij} = p\delta_{ij}$ and

$$\mathbb{E}[\partial_{ij}H_N\,\partial_{kl}H_N] = \frac{p(p-1)}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}),$$

while every cross covariance keeps a factor σ_i with $i < N$ and vanishes; for instance $\mathbb{E}[H_N\,\partial_i H_N] = pN^{1-p}\langle\sigma\rangle\sigma^{p-1}\sigma_i = 0$. Jointly Gaussian and uncorrelated, the three blocks are independent, and G has the covariance of $\sqrt{p(p-1)}A$. By (13.6) and Euler's identity $\operatorname{Hess} H_N(\sigma) = G - pu\,I$, and $pu = \sqrt{p(p-1)}\,t(u)$. ∎

Lemma 13.2 (one more eigenvalue). *There is a constant $c_N > 0$ such that for every $k \in \{0, \dots, N-1\}$ and every measurable $\varphi \ge 0$,*

$$\int_{\mathbb{R}} \varphi(t)\, e^{-Nt^2/4}\, \mathbb{E}\big[|\det(A-t)|\mathbf{1}\{i(A-t) = k\}\big]\,\mathrm{d}t = \frac{c_N}{N}\,\mathbb{E}\varphi\big(\lambda_{(k)}\big), \quad (13.8)$$

with $i(M)$ the number of negative eigenvalues of M. Summed over k: $\int \varphi(t)e^{-Nt^2/4}\mathbb{E}|\det(A-t)|\,\mathrm{d}t = c_N \int \varphi\rho_N$.

PROOF. The plan is to recognize $|\det(A-t)|$ as the missing factors of a Vandermonde determinant. By Theorem 2.5.2 of Anderson et al. (2010), rescaled, the eigenvalues of W in a uniformly random order have the density $q_N(x) = Z_N^{-1}|\Delta(x)|\prod_{i=1}^N e^{-Nx_i^2/4}$ on $\mathbb{R}^N$, with $\Delta(x) = \prod_{i<j}(x_j - x_i)$, and those of A have the density $Z'^{-1}_{N-1}|\Delta(y)|\prod_i e^{-Ny_i^2/4}$ on $\mathbb{R}^{N-1}$. For eigenvalues $y_1, \dots, y_{N-1}$ of A, $|\det(A-t)| = \prod_i|y_i - t|$, the event $i(A - t) = k$ says that exactly k of them lie below t, and $|\Delta(y)|\prod_i|y_i - t| = |\Delta(y_1, \dots, y_{N-1}, t)|$. By Tonelli's theorem the left side of (13.8) is therefore Z_N/Z'_{N-1} times $\mathbb{E}[\varphi(\lambda'_N)\mathbf{1}\{\lambda'_N = \lambda_{(k)}\}]$, with $(\lambda'_1, \dots, \lambda'_N)$ the eigenvalues of W in a uniformly random order, almost surely distinct. Given the set of eigenvalues, λ'_N is each of them with probability $1/N$, so the expectation is $\mathbb{E}\varphi(\lambda_{(k)})/N$, and $c_N = Z_N/Z'_{N-1}$. For the sum, $\sum_k \mathbb{E}\varphi(\lambda_{(k)}) = \mathbb{E}\sum_i \varphi(\lambda_i) = N\int\varphi\rho_N$. ∎

Lemma 13.3 (two spins). *For $p = 2$, almost surely H_N has exactly $2N$ critical points on S_N, all nondegenerate.*

PROOF. Here $H_N(\sigma) = \langle\sigma, B\sigma\rangle$ with $B = (J + J^\top)/(2\sqrt{N}) = W/\sqrt{2}$ for a GOE matrix W. By (13.6) the critical points are the eigenvectors of B of norm $\sqrt{N}$. The eigenvalues of W are almost surely distinct (Anderson et al. (2010), after Theorem 2.5.2), so the critical points are the $2N$ points $\pm\sqrt{N}v_i$, and at

$\sqrt{N}v_i$ the Hessian $2(B-\mu_i I)$ on $v_i^{\perp}$ has the nonzero eigenvalues $2(\mu_j-\mu_i)$, $j\neq i$. ∎

Theorem 13.4 (the exact mean count). *For all integers $p\geq 2$, $N\geq 2$ and $k\in\{0,\dots,N-1\}$ and every Borel set $B\subset\mathbb{R}$, with $t(B)=\{t(u):u\in B\}$,*

$$\mathbb{E}\mathrm{Crt}_{N,k}(B)=2\sqrt{\tfrac{2}{p}}\,(p-1)^{N/2}\,\mathbb{E}\big[e^{-Na\lambda_{(k)}^2}\mathbf{1}\{\lambda_{(k)}\in t(B)\}\big],\qquad(13.9)$$

$$\mathbb{E}\mathrm{Crt}_N(B)=2N\sqrt{\tfrac{2}{p}}\,(p-1)^{N/2}\int_{t(B)}e^{-Nat^2}\rho_N(t)\,\mathrm{d}t.$$

Proof. The plan is to apply the Kac–Rice formula, read the integrand off Lemma 13.1, convert the determinant with Lemma 13.2, and find the constant from Lemma 13.3.

The formula on S_N. Theorem A.18 applies to $f(y)=H_N(\sqrt{N}y)$ on the unit sphere. In the chart $y\mapsto(y_1,\dots,y_{N-1})$ near e_N, the first and second partial derivatives of f at e_N are the gradient and the Hessian of Lemma 13.1 (the term $\partial_N f\cdot\partial_{ij}(1-|y|^2)^{1/2}$ supplies $-pf\delta_{ij}$), jointly nondegenerate by the lemma, hence near e_N by continuity; finitely many rotated charts cover the sphere. The chart derivatives are polynomials in y and $(1-|y|^2)^{1/2}$ with Gaussian coefficients, so hypothesis (ii) holds with $\beta=1$, as Auffinger et al. (2013) verify. Under $\sigma=\sqrt{N}y$ the factors N^{N-1} (Hessian), $N^{-(N-1)/2}$ (density of the gradient) and $N^{-(N-1)/2}$ (surface measure) cancel: the formula holds on S_N as written.

The integrand. By rotation invariance it is the same at every point. By Lemma 13.1 the gradient is independent of (u,A), so conditioning on $g=0$ leaves their law unchanged, and g has density $(2\pi p)^{-(N-1)/2}$ at 0. With $g_k(t)=\mathbb{E}[|\det(A-t)|\mathbf{1}\{i(A-t)=k\}]$ and $|S_N|=N^{(N-1)/2}2\pi^{N/2}/\Gamma(N/2)$,

$$\mathbb{E}\mathrm{Crt}_{N,k}(B)=\frac{|S_N|(p(p-1))^{\frac{N-1}{2}}}{(2\pi p)^{\frac{N-1}{2}}}\int_B\sqrt{\tfrac{N}{2\pi}}\,e^{-\frac{N}{2}u^2}g_k\big(t(u)\big)\,\mathrm{d}u.$$

The determinant. Substitute $t=t(u)$: $\mathrm{d}u=\sqrt{(p-1)/p}\,\mathrm{d}t$ and $\frac{N}{2}u^2=\frac{N}{4}t^2+Nat^2$. Lemma 13.2 with $\varphi(t)=e^{-Nat^2}\mathbf{1}\{t\in t(B)\}$ turns the integral into $\sqrt{(p-1)/p}\,\frac{c_N}{N}\mathbb{E}\varphi(\lambda_{(k)})$. The factors that depend on p combine to $(p(p-1))^{\frac{N-1}{2}}p^{-\frac{N-1}{2}}\sqrt{(p-1)/p}=(p-1)^{N/2}/\sqrt{p}$, so $\mathbb{E}\mathrm{Crt}_{N,k}(B)=\gamma_N(p-1)^{N/2}p^{-1/2}\mathbb{E}\varphi(\lambda_{(k)})$, with γ_N independent of p, k and B.

The constant. For $p=2$, $B=\mathbb{R}$, summed over k, the right side is $\gamma_N 2^{-1/2}N$ and the left side is $2N$ by Lemma 13.3. Hence $\gamma_N=2\sqrt{2}$, which is (13.9); summing it over k with Lemma 13.2 gives the second formula. ∎

For $p=2$ the formula says $\mathbb{E}\mathrm{Crt}_{N,k}(B)=2\mathbb{P}(\lambda_{(k)}\in\sqrt{2}B)$: the critical values of $\langle\sigma,B\sigma\rangle$ are the eigenvalues of $W/\sqrt{2}$, each twice. Selberg's integral gives the same constant directly (Exercise 13.3).[5]

Theorem 13.5 (the annealed complexity). *Let $p\geq 2$, $\ell_p=\frac{1}{2}\log(p-1)$, $c_p=\frac{p-2}{4(p-1)}$ and $I_1(u)=J(2|u|/E_\infty)$. For every $u\in$*

[5] Fyodorov (2004) reduced the mean count of an isotropic landscape to a mean absolute determinant of a GOE-type matrix and evaluated it at finite N; the extra-eigenvalue identity is Lemma 3.3 of Auffinger et al. (2013).

$\mathbb{R}$, $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_N((-\infty,u)) \to \Theta_p(u)$ *and* $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_{N,0}((-\infty,u)) \to \Theta_{0,p}(u)$, *where*

$$\Theta_p(u) = \begin{cases} \ell_p - c_p u^2 - I_1(u), & u \le -E_\infty, \\ \ell_p - c_p u^2, & -E_\infty \le u \le 0, \\ \ell_p, & u \ge 0, \end{cases}$$

$$\Theta_{0,p}(u) = \begin{cases} \ell_p - c_p u^2 - I_1(u), & u \le -E_\infty, \\ \ell_p - \frac{p-2}{p}, & u \ge -E_\infty. \end{cases}$$

Proof. The plan is to evaluate the exponential rates of the expectations in (13.9) by bounding the weight e^{-Nat^2} on the event that matters, using (13.7). Let $\tau = t(u)$, so that $a\tau^2 = c_p u^2$, $J(-\tau) = I_1(u)$ for $u \le -E_\infty$, and $4a = \frac{p-2}{p}$. Since $\frac{1}{N}\log(2\sqrt{2/p}\,N(p-1)^{N/2}) \to \ell_p$, Theorem 13.4 reduces the claims to the limits of $\frac{1}{N}\log Q_N$ and $\frac{1}{N}\log Q_N^0$, where

$$Q_N = \int_{-\infty}^{\tau} e^{-Nat^2}\rho_N(t)\,\mathrm{d}t = \mathbb{E}\Big[\frac{1}{N}\sum_i e^{-Na\lambda_i^2}\mathbf{1}\{\lambda_i < \tau\}\Big],$$

$$Q_N^0 = \mathbb{E}\big[e^{-Na\lambda_{(0)}^2}\mathbf{1}\{\lambda_{(0)} < \tau\}\big].$$

All critical points, upper bounds. If $\tau \ge 0$, then $Q_N \le 1$. If $\tau < 0$, every term with $\lambda_i < \tau$ has $\lambda_i^2 > \tau^2$, and such a term exists only if $\lambda_{(0)} < \tau$, so $Q_N \le e^{-Na\tau^2}\mathbb{P}(\lambda_{(0)} < \tau)$. For $-2 \le \tau < 0$ we bound the probability by 1; for $\tau < -2$, (13.7) for $-\lambda_{(0)}$ gives $\limsup \frac{1}{N}\log Q_N \le -a\tau^2 - J(-\tau)$.

All critical points, lower bounds. If $-2 < \tau \le 0$ and $0 < \delta < \tau + 2$, the weight is at least $e^{-Na(\tau-\delta)^2}$ on $(\tau-\delta,\tau)$, so $Q_N \ge e^{-Na(\tau-\delta)^2}\mathbb{E}\mu_N((\tau-\delta,\tau))$, and $\mathbb{E}\mu_N((\tau-\delta,\tau)) \to \sigma_{\mathrm{sc}}((\tau-\delta,\tau)) > 0$ by the semicircle law (Theorem A.15) and bounded convergence. If $\tau > 0$, the interval $(-\delta,0)$ serves in the same way. If $\tau \le -2$, then on $\{\tau - \delta < \lambda_{(0)} < \tau\}$ the sum in Q_N is at least $\frac{1}{N}e^{-Na(\tau-\delta)^2}$, so $Q_N \ge \frac{1}{N}e^{-Na(\tau-\delta)^2}\mathbb{P}(-\tau < -\lambda_{(0)} < -\tau + \delta)$, and (13.7) gives $\liminf \frac{1}{N}\log Q_N \ge -a(\tau-\delta)^2 - J(-\tau)$. Letting $\delta \to 0$ matches the upper bounds.

Minima. For $\tau \le -2$ the same two estimates hold for Q_N^0. For $\tau > -2$ and $0 < \delta < \min(2, \tau+2)$, split according to whether $\lambda_{(0)} \le -2 + \delta$, where $\lambda_{(0)}^2 \ge (2-\delta)^2$:

$$Q_N^0 \le e^{-Na(2-\delta)^2} + \mathbb{P}\big(\lambda_{\max}(W) < 2 - \delta\big),$$

and the last term is negligible by (13.7), so $\limsup \frac{1}{N}\log Q_N^0 \le -a(2-\delta)^2$. Conversely, $(-2-\delta,-2) \subset (-\infty,\tau)$ gives $Q_N^0 \ge e^{-Na(2+\delta)^2}\mathbb{P}(2 < -\lambda_{(0)} < 2+\delta)$, and $\liminf \frac{1}{N}\log Q_N^0 \ge -a(2+\delta)^2 - J(2)$, with $J(2) = 0$. Letting $\delta \to 0$ gives $-4a$. ∎

Both functions are continuous and nondecreasing. On $u \le 0$, $\Theta_p(u)$ is also the rate of the number of critical points near u, the physicist's $\Sigma(u)$, and below the threshold $\Theta_{0,p} = \Theta_p$: almost every critical point below $-E_\infty$ is a

minimum. The weight e^{-Nat^2} favors critical values near 0, the eigenvalue's entropy favors the bulk, and the threshold is where the eigenvalue leaves the bulk.

Corollary 13.6 (the threshold and the ground state). *Let $p \geq 3$.*

(i) For every $u > -E_\infty$, $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_{N,0}((u,\infty)) \to -\infty$.

(ii) $\Theta_{0,p}$ is strictly increasing on $(-\infty, -E_\infty]$ and has a unique zero $-E_0$ there, $E_0 > E_\infty$; almost surely $\liminf_N \mathrm{GS}_N \geq -E_0$, where $\mathrm{GS}_N = \frac{1}{N}\min_{S_N} H_N$.

Proof. The plan is to read (i) off the exact formula and to get (ii) from Markov's inequality. (i) By (13.9) with $B = (u,\infty)$, the mean is at most $2\sqrt{2/p}\,(p-1)^{N/2}$ times $\mathbb{P}(\lambda_{(0)} > t(u)) = \mathbb{P}(\lambda_{\max}(W) < -t(u))$, where $-t(u) < 2$, and (13.7) applies. (ii) On $u < -E_\infty$, $-c_p u^2$ is strictly increasing for $p \geq 3$, and $I_1(u) = J(|t(u)|)$ decreases in u because J increases on $[2,\infty)$; so $\Theta_{0,p}$ increases strictly there and tends to $-\infty$ as $u \to -\infty$. At $-E_\infty$ it equals $\ell_p - \frac{p-2}{p}$, which is 0.0132 at $p = 3$ and has the derivative $\frac{(p-2)^2}{2p^2(p-1)} \geq 0$ in p; so the zero $-E_0$ exists, is unique and lies below $-E_\infty$. A global minimizer of H_N is a critical point of index 0, so for $\varepsilon > 0$, by Markov's inequality and Theorem 13.5,

$$\mathbb{P}\big(\mathrm{GS}_N < -E_0 - \varepsilon\big) \leq \mathbb{E}\mathrm{Crt}_{N,0}\big((-\infty, -E_0 - \varepsilon)\big) = e^{N(\Theta_{0,p}(-E_0-\varepsilon)+o(1))}.$$

The exponent is negative, the bounds are summable, and the Borel–Cantelli lemma gives the claim. ∎

13.4.1 What the general theorems add

Theorem 13.7 (fixed index; Auffinger et al. 2013, Theorems 2.5, 2.14 and 2.15). *For $p \geq 2$ and fixed $k \geq 0$, $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_{N,k}((-\infty,u)) \to \Theta_{k,p}(u)$, which is $\Theta_{0,p}$ with $(k+1)I_1$ in place of I_1; with $-E_k$ its zero, $E_0 > E_1 > \cdots \to E_\infty$. The probability of a critical point of index k above $-E_\infty + \varepsilon$ is at most e^{-cN^2}, and that of one of index at least k below $-E_k - \varepsilon$ decays exponentially.*

Proof sketch. Replace $\lambda_{(0)}$ by $\lambda_{(k)}$ in the proof of Theorem 13.5: by Theorem A.1 of Auffinger et al. (2013), the $(k+1)$-th smallest eigenvalue obeys a large deviation principle with rate $(k+1)J(-\cdot)$, because $k+1$ eigenvalues must leave the bulk. The bound e^{-cN^2} uses the large deviation principle of the empirical measure at speed N^2 (Remark A.17) in place of Corollary 13.6(i); the last claim is Markov's inequality. Omitted: the principle for $\lambda_{(k)}$. ∎

Theorem 13.8 (the typical count; Subag 2017, Theorems 1 and 3, Corollary 2). *Let $p \geq 3$. For $u \in (-E_0, -E_\infty)$, $\mathbb{E}[\mathrm{Crt}_N((-\infty,u))^2]/(\mathbb{E}\mathrm{Crt}_N((-\infty,u)))^2 \to 1$, so the count divided by its mean tends to 1 in L^2 and in probability. For $u > -E_0$ the two moments agree on the exponential scale.*

Proof sketch. A two-point Kac–Rice formula writes the second moment as an integral over pairs of points organized by their overlap; given both gradients vanish, the two Hessians are correlated Gaussian matrices whose determinants Subag controls. Below the threshold, pairs with overlap away from 0 contribute exponentially less than $(\mathbb{E}\mathrm{Crt}_N)^2$, and nearly orthogonal pairs match it up to $1 + o(1)$. Omitted: the joint law of the two Hessians and the control of every range of overlaps (Sections 5–8 and Appendix II of Subag (2017)). ∎

With Corollary 13.6, the second moment gives $\mathrm{GS}_N \to -E_0$ almost surely for every $p \geq 3$ (Subag 2017, Appendix IV); Auffinger et al. (2013, Theorem 2.12) had proved it for even $p \geq 4$ through the Parisi formula for spherical models (Talagrand 2006), and formula (1.7) of Jagannath et al. (2020) gives the same $E_0(3)$ to twelve digits. Above the threshold Subag and Zeitouni (2021) prove concentration for $p \geq 32$. For mixed models the mean can mislead: in some, the mean number of minima below an energy grows exponentially while the probability that a critical value lies below it tends to zero (Auffinger and Ben Arous 2013, Corollary 4.1). Kac–Rice also bounds the TAP complexity of $\mathbb{Z}_2$ synchronization (Fan et al. 2021, Theorem 1.1), placing its low-energy critical points near the Bayes posterior for strong signals (Chapter 6). For even $p \geq 4$, an algorithm with Lipschitz dependence on the couplings reaches energy below $-E_\infty - \varepsilon$ only with exponentially small probability (Huang and Sellke 2025, Theorem 5 and Proposition 2.2 of arXiv v2): for such algorithms the threshold is the limit.

13.5 The proof, continued: the spiked tensor

Proposition 13.9 (the spiked count at finite N; Ben Arous et al. 2019, Lemmas 2–3). *Let $p \geq 3$, $\lambda \geq 0$, and let $\mathrm{Crt}_{N,\lambda}(M, B)$ count the critical points of $H_{N,\lambda}$ with $m(\sigma) \in M$ and $H_{N,\lambda}(\sigma)/N \in B$, for Borel sets $M \subset (-1, 1)$, $B \subset \mathbb{R}$. With $\theta(m) = \lambda\sqrt{p(p-1)}\, m^{p-2}(1-m^2)$ and a fixed unit vector $e \in \mathbb{R}^{N-1}$,*

$$\mathbb{E}\mathrm{Crt}_{N,\lambda}(M, B) = K_N \int_M (1-m^2)^{\frac{N-3}{2}} e^{-\frac{N}{2}p\lambda^2 m^{2p-2}(1-m^2)} \times \int_B e^{-\frac{N}{2}(u+\lambda m^p)^2}\, \mathbb{E}\left|\det\left(A - t(u) - \theta(m) e e^\top\right)\right| \mathrm{d}u\, \mathrm{d}m,$$

with $K_N = |S_N| \frac{\Gamma(N/2)}{\sqrt{\pi}\,\Gamma((N-1)/2)} \left(\frac{p-1}{2\pi}\right)^{\frac{N-1}{2}} \sqrt{\frac{N}{2\pi}}$. For local minima, insert the indicator of $A - t(u) - \theta(m) e e^\top > 0$ in the expectation.

Proof. The plan is the regression of Lemma 13.1, shifted by the deterministic spike. We apply the Kac–Rice formula to $H_{N,\lambda}$, a centered Gaussian field plus the smooth function $-\lambda N m^p$, as Ben Arous et al. (2019, Lemmas 1–3) do. At a point with $m(\sigma) = m$, the spike $F_s(\sigma) = -\lambda N^{1-p}\langle \sigma, x^*\rangle^p$ has gradient $-\lambda p m^{p-1} x^*$, Hessian $-\lambda p(p-1) m^{p-2} x^* x^{*\top}/N$, and $\langle \sigma, \nabla F_s\rangle = -\lambda p N m^p$. The tangential part of x^* has norm $\sqrt{N(1-m^2)}$; call its direction e. By

(13.6), along the sphere F_s has gradient $-\lambda p m^{p-1}\sqrt{N(1-m^2)}\,e$ and Hessian $-\lambda p(p-1)m^{p-2}(1-m^2)ee^\top + \lambda p m^p I$. The gradient of $H_{N,\lambda}$ vanishes when $g \sim \mathcal{N}(0, pI_{N-1})$ equals $\lambda p m^{p-1}\sqrt{N(1-m^2)}\,e$, which has density $(2\pi p)^{-(N-1)/2}e^{-Np\lambda^2 m^{2p-2}(1-m^2)/2}$, and conditioning on it leaves the value and the second derivatives of H_N unchanged. The energy $u = H_N/N - \lambda m^p$ has density $\sqrt{N/(2\pi)}\,e^{-N(u+\lambda m^p)^2/2}$. The Hessian is

$$\sqrt{p(p-1)}A - p(u+\lambda m^p) - \lambda p(p-1)m^{p-2}(1-m^2)ee^\top + \lambda p m^p,$$

which is $\sqrt{p(p-1)}\big(A - t(u) - \theta(m)ee^\top\big)$, and by the rotation invariance of A we may fix e. Finally, m pushes the surface measure of S_N forward to $|S_N|\frac{\Gamma(N/2)}{\sqrt{\pi}\Gamma((N-1)/2)}(1-m^2)^{(N-3)/2}\,\mathrm{d}m$, the law of one coordinate of a uniform point of the unit sphere; collect the constants as in Theorem 13.4. ∎

At $m = 0$ and $p \geq 3$ the spike drops out of the integrand: the equator does not see the signal.

Theorem 13.10 (the landscape of the spiked tensor; Ben Arous et al. 2019, Theorems 1–2 and Propositions 1–2, in our normalization). *Let $p \geq 3$ and $\lambda \geq 0$ be fixed. For Borel sets $M \subset [-1,1]$, $B \subset \mathbb{R}$, $\frac{1}{N}\log \mathbb{E}\mathrm{Crt}_{N,\lambda}(M,B)$ has $\liminf$ at least $\sup \Sigma_\lambda$ over $M^\circ \times B^\circ$ and $\limsup$ at most $\sup \Sigma_\lambda$ over $\bar{M}\times\bar{B}$, with Σ_λ of (13.4). The same holds for local minima with $\Sigma_\lambda(m,u) - L(\theta(m), -t(u))$, where $L(\theta,s) = +\infty$ for $s < 2$; $L(\theta,s) = \frac{1}{2}J(s) - \frac{1}{2}J(b) - \frac{\theta}{2}(s-b) + \frac{1}{8}(s^2-b^2)$ for $\theta > 1, 2 \leq s < b = \theta + \frac{1}{\theta}$; and $L = 0$ otherwise. For $0 \leq m < m_c = ((p-2)/(\lambda\sqrt{p(p-1)}))^{1/p}$, $S(m) = \sup_u \Sigma_\lambda(m,u)$ is*

$$S(m) = \ell_p + \tfrac{1}{2}\log(1-m^2) - \tfrac{p}{2}\lambda^2 m^{2p-2}(1-m^2) + \tfrac{p\lambda^2}{2(p-2)}m^{2p};$$

on $[m_c, 1)$, $S \leq 0$, with equality exactly at the largest root m^* of $m^{2p-4}(1-m^2) = 1/(p\lambda^2)$, which exists if and only if $\lambda \geq \lambda_c(p)$.

Proof sketch. By Proposition 13.9 the count is a Laplace integral in (m,u), whose explicit factors give the first four terms of Σ_λ, while $|S_N|$ and the constants give $\frac{1}{2}(1+\log(p-1))$. A rank-one perturbation changes the determinant by the factor $|1 - \theta\langle e, (A-t)^{-1}e\rangle|$, which is not exponential, so the determinant contributes $\Phi(t)$; for minima the outlier must be pushed above t, at the cost L from the large deviations of the largest eigenvalue of a rank-one perturbation of the GOE (Maïda 2007). Omitted: uniformity in (m,u) and control of the determinant near the outlier (Section 4 of Ben Arous et al. (2019)); the formulas for S are calculus (their Appendix B). ∎

The equatorial band is a theorem about means: $S(0) = \ell_p$, and the complexity of minima at $m = 0$ is $\Theta_{0,p}(-E_\infty)$, for every λ. The estimator itself is understood exactly.

Theorem 13.11 (the maximum likelihood estimator; Jagannath et al. 2020, Theorem 1.2). *Let $p \geq 3$, $f_\lambda(q) = \lambda^2 q^p + \log(1-q) + q$ on*

$[0,1)$, $\lambda_{\mathrm{ML}} = \sup\{\lambda : \sup_q f_\lambda(q) \le 0\}$, and $q_*(\lambda) = 0$ for $\lambda < \lambda_{\mathrm{ML}}$, $q_*(\lambda) = \arg\max f_\lambda$ for $\lambda > \lambda_{\mathrm{ML}}$. If $\hat\sigma$ minimizes $H_{N,\lambda}$ on S_N, then $|m(\hat\sigma)| \to \sqrt{q_*(\lambda)}$ in probability for every $\lambda \neq \lambda_{\mathrm{ML}}$.

Stationarity of f_λ reads $q^{p-2}(1-q) = 1/(p\lambda^2)$, so above λ_{ML} the estimator's overlap is the location m^* of the good critical point; at λ_{ML} it jumps from 0 to $m^*(\lambda_{\mathrm{ML}})$. For $p = 3$, $\lambda_{\mathrm{ML}} = 1.20656$ ($q = 0.645007$) and $m^*(\lambda_{\mathrm{ML}}) = 0.80312$.[6] Both thresholds agree with the replica predictions of Ros et al. (2019): $\lambda_c(3) = 1.15470$ and $\lambda_{\mathrm{ML}}(3) = 1.20656$ against their $r_c = \sqrt{6}$ and $r_{\mathrm{1ST}} \approx 2.56$, that is 1.2068. Between λ_c and λ_{ML} the good minimum exists but is not the deepest; between the statistical and the algorithmic thresholds lies the equatorial band. For $\lambda = N^\alpha$ with $\alpha < \frac{p-2}{2}$, Langevin dynamics at positive temperature, started from the Gibbs measure of the noise alone restricted to $m > 0$, reaches overlap $2N^{-1/2+\varepsilon}$ within time e^{cN^ε} only with probability exponentially small in a power of N (Ben Arous et al. 2020, Theorem 2.10); their proof locates the obstruction in the weak curvature of the spike near the equator rather than in the count of minima.

[6] For $p = 2$ the estimator is the top eigenvector, whose overlap grows continuously from 0 at the BBP threshold (Chapter 5); Jagannath et al. (2020) stress that for $p \ge 3$ it jumps.

13.6 Measuring it

Five scripts test the formulas; the first, `13-kacrice-complexity.py`, computes every constant and curve of the chapter.

THE EXACT FORMULA AT SMALL N. `13-kacrice-count.py` finds every critical point of the 3-spin on S_N, $N = 3, \dots, 7$, by Newton's method on $\nabla H_N = \mu\sigma$, $\|\sigma\|^2 = N$, from the endpoints of a gradient-descent sweep and then from random points, until three batches in a row find no new critical value; every instance satisfies the Morse identity $\sum_k (-1)^k \mathrm{Crt}_{N,k}(\mathbb{R}) = 1 + (-1)^{N-1}$. The averages agree with (13.9), evaluated with 400000 GOE matrices (Figure 13.3(a)), also index by index: at $N = 7$ the formula gives 4.83, 10.69, 15.72, 17.71 critical points of indices 0 to 3, the count 4.84, 10.60, 15.58, 17.63. Without the sweep, random-start Newton missed about 6% of the minima at $N = 7$, which the Morse identity cannot detect when a minimum and a saddle of index 1 are missed together.

WHERE THE MINIMA ARE AT FINITE N. The exponent $\Theta_{0,3}(-E_\infty) = 0.0132$ is small: at $N = 128$ the formula gives only 455 minima on average. Integrated against $g(u) = u$, (13.9) gives the mean energy of a local minimum in the expected count, $\mathbb{E}\sum_{\text{minima}} H_N/N$ over $\mathbb{E}\mathrm{Crt}_{N,0}(\mathbb{R})$, as a ratio of expectations over the smallest GOE eigenvalue, which `13-kacrice-minima.py` samples with the tridiagonal model of Dumitriu and Edelman (2002) and Sturm bisection.[7] It is -1.319, -1.415, -1.482, -1.525 and -1.557 at $N = 16$, 32, 64, 128 and 256, far above $-E_\infty = -1.633$; the share of the expected count below $-E_\infty$ falls from 1.8% to 0.02%, and the gap to the threshold times $\sqrt{N}$ stays between 1.21 and 1.26. A heuristic explains the rate: for a minimum at $t = -2 + x$ the weight e^{-Nat^2} gains e^{4aNx}, while the probability that the smallest eigenvalue of A exceeds $-2 + x$ costs about $e^{-cN^2x^3}$; the balance is at $x \asymp N^{-1/2}$.

[7] The tridiagonal model has exactly the eigenvalue law of the GOE at every N, so a sample of the smallest eigenvalue costs $O(N)$ operations per bisection step instead of $O(N^3)$.

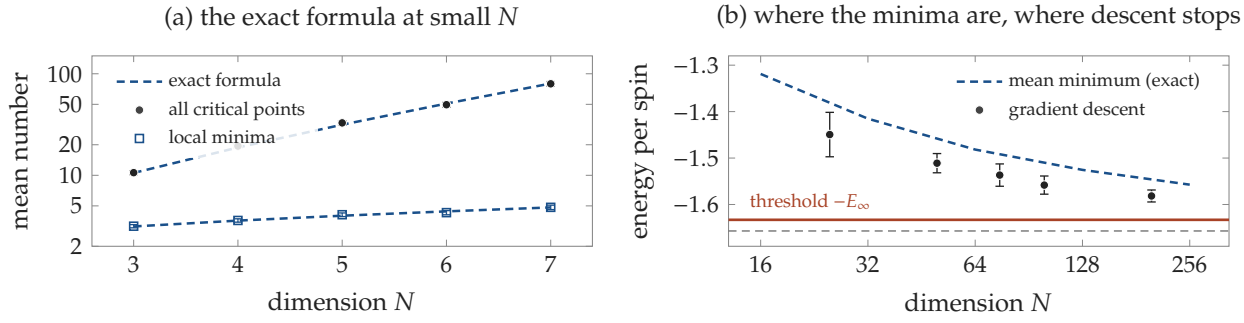


Figure 13.3. The pure spherical 3-spin at finite N. (a) Mean numbers of critical points and of local minima: Theorem 13.4 evaluated with 400000 GOE matrices (lapis dashed; Monte Carlo error below 0.02%) against exhaustive counts on 200, 150, 120, 100 and 80 random Hamiltonians at $N = 3, \dots, 7$ (seeds $1300 + N$; 95% Student-t intervals). (b) The mean energy of a local minimum in the expected count, from Theorem 13.4 with 40000 samples of the smallest GOE eigenvalue per N (lapis; Monte Carlo errors below 0.002; seed 1313), and where projected gradient descent from uniform random starts stops (95% Student-t intervals over 10, 8, 6, 5 and 4 Hamiltonians at $N = 25$, 50, 75, 100, 200, with 8 starts each, 4 at $N = 200$; seeds $130 + N$); threshold carnelian, ground state gray dashed. Scripts: `13-kacrice-count.py`, `-minima.py`, `-descent.py`.

WHERE DESCENT STOPS. `13-kacrice-descent.py` runs projected gradient descent with step 0.15 until the gradient falls below $10^{-5}\sqrt{N}$. The endpoints have mean energy −1.449, −1.511, −1.537, −1.558 and −1.582 at $N = 25$, 50, 75, 100 and 200: below the exact mean minimum by about 0.06 at $N = 25$ and 0.03 at $N = 200$, as befits minima with large basins, and far from $-E_\infty$, though the gap times $\sqrt{N}$ falls from 0.92 to 0.73. Single runs range from −1.735 ($N = 25$, where the ground state itself fluctuates below $-E_0$) to −1.181, and from −1.620 to −1.545 at $N = 200$. At the $N = 100$ endpoints the Hessian divided by $\sqrt{6}$ follows the semicircle of Lemma 13.1 centered at the energy (mean eigenvalue 1.932 against $-t(\bar{u}) = 1.908$ at the mean energy $\bar{u} = -1.558$; median largest eigenvalue 3.84 against the edge 3.91), but that semicircle reaches down to −0.09, while every endpoint has a positive smallest eigenvalue (median 0.079, range 0.021 to 0.166): descent stops at minima as close to marginal as the size allows.

THE SPIKED TENSOR. `13-kacrice-spiked.py` runs the same descent on $H_{N,\lambda}$ at $N = 80$ (step $0.3/\lambda$ for $\lambda > 2$; Figure 13.2(b)). From warm starts at overlap 0.5 descent lands on the good minimum: for $\lambda = 1.2$, 1.5, 2, 3 and 6 the final overlap is 0.810, 0.918, 0.959, 0.983 and 0.996 against $m^* = 0.798$, 0.905, 0.953, 0.981 and 0.995 (intervals below 0.01), an excess of at most 0.013, presumably a finite-size effect. Below λ_c warm starts stop at overlaps 0.22 to 0.60 ($\lambda = 0.9$) and 0.33 to 0.75 ($\lambda = 1.1$), beyond the band of positive annealed complexity of minima ($m \le 0.24$ and 0.34), where at $\lambda = 1.1$ that complexity is only about −0.001: $e^{80\Sigma} \approx 0.9$, so such minima are not rare at this size. At $\lambda = 0.6$ all ten end at $|m| \le 0.27$. From random starts (overlap about $N^{-1/2} = 0.11$) descent reaches the spike in no run at $\lambda \le 0.9$, in 1 to 4

of 10 for $1.1 \le \lambda \le 3$, and in 8 and 7 of 10 at $\lambda = 4$ and 6; here $N^{(p-2)/2} = 9$.

13.7 The status board

For the pure spherical p-spin, the landscape predictions are theorems for mean counts at every energy and fixed index, and for typical counts below the threshold; the ground state is known for every $p \ge 3$. What is open is dynamical: that gradient descent ends at the threshold is a conjecture, proved only for Langevin dynamics of the pure model on dimension-free time scales, in the zero-temperature limit (Chapter 14). For the spiked tensor, the typical landscape is a physical prediction and the algorithmic threshold is proved in one direction.

STATUS BOARD · TAP complexity ↔ Kac–Rice and the second moment

PREDICTION	STATUS	PROOF OR EVIDENCE
Mean counts below u: all, minima, fixed index; none of fixed index above $-E_\infty$	proved	Auffinger et al. (2013, Theorems 2.5, 2.8, 2.14, 2.15); Theorem 13.5
Typical number = mean number (pure model)	proved under	below $-E_\infty$, $p \ge 3$ (Subag 2017, Theorem 1); above, $p \ge 32$ (Subag and Zeitouni 2021)
Typical number = mean number (mixed models)	false	full mixtures (Auffinger and Ben Arous 2013, Corollary 4.1)
Ground state $-E_0$, the zero of the complexity of minima	proved	Auffinger et al. (2013) (even p), Subag (2017) (all $p \ge 3$)
Lowest critical point of index k at $-E_k$	conjectured	Auffinger et al. (2013, Remark 2.16); layers proved as probability bounds
Gradient descent from a random start ends at $-E_\infty$	conjectured	Cugliandolo and Kurchan (1993); Figure 13.3; Langevin, $\beta \to \infty$: Sellke (2024)
Spiked tensor: mean landscape, λ_c, equatorial band; estimator, λ_{ML}	proved	Ben Arous et al. (2019, Theorems 1–2, Propositions 1–2); Jagannath et al. (2020, Theorem 1.2)
Spiked tensor: typical landscape, λ_c, λ_{ML}	conjectured	replica-symmetric replicated Kac–Rice (Ros et al. 2019); its thresholds match proved ones
Descent from a random start needs $\lambda \gtrsim N^{(p-2)/2}$	proved under	sufficient (Ben Arous et al. 2020, Theorem 2.1); necessary for Langevin from a Gibbs start (Theorem 2.10)

13.8 Where the translation fails

WHAT WAS ASSUMED. The model is pure and the disorder Gaussian. The exact formula rests on Euler's identity, which makes the Hessian at energy u a GOE matrix with a deterministic shift, false for mixed models (Auffinger and Ben Arous 2013), and on the joint density of GOE eigenvalues. Two 2026 preprints treat other couplings: Chen et al. (2026) prove that the annealed complexity is unchanged when each coupling is a strictly subgaussian variable plus an independent nonzero Gaussian part (one step of their proof, they write, "was suggested by ChatGPT (OpenAI)"), and Kim (2026) constructs a symmetric bounded law, matching the first $2p$ Gaussian moments, whose mean count in a compact energy window grows at a strictly larger exponential rate. All limits fix λ, p and the index, so the algorithmic regime $\lambda = N^{\alpha}$ lies outside the landscape theorems, which moreover count in mean.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. The annealed count is wrong as a typical count for full mixtures (Auffinger and Ben Arous 2013). For mixed models part (iii) fails: integrating the dynamical equations, Folena et al. (2020) find gradient descent from equilibrated starts going below the threshold energy of the dominant marginal minima. For the spiked tensor, the thermodynamic replica method gives partly incorrect predictions for the landscape, which the replicated Kac–Rice computation corrects (Ros et al. 2019), and the proof of Ben Arous et al. (2020) locates the failure of descent for $\lambda \ll N^{(p-2)/2}$ in the weak curvature of the signal near the equator, not in a rugged landscape. At accessible sizes the asymptotic landscape is a poor guide: for $N \le 256$ the exact formula places almost all minima above $-E_\infty$.

OPEN PROBLEMS. As far as we know, these are open as of October 2026. (1) The typical complexity of the spiked tensor near the isolated minimum and in the band near λ_c, left open by Ben Arous et al. (2019) and expected by Ros et al. (2019) to be within reach. (2) Concentration of the count above the threshold for $3 \le p \le 31$; Subag and Zeitouni (2021) need $p \ge 32$. (3) The energy of the lowest critical point of index $k \ge 1$, conjectured to tend to $-E_k$ (Auffinger et al. 2013, Remark 2.16). (4) That gradient flow, not only low-temperature Langevin dynamics, ends at $-E_\infty$; Sellke (2024, Remark 1.5) explains why his method does not reach it. (5) Failure of descent below $N^{(p-2)/2}$ from the uniform start, conjectured for Langevin dynamics by Ben Arous et al. (2020, Sections 1 and 2.4).

Exercises

13.1 CONSOLIDATION. (The covariances.) Complete the proof of Lemma 13.1: differentiate $C(x, y) = N^{1-p}\langle x\rangle y^p$ three and four times and check that at $x = y = \sqrt{N}e_N$ only $\frac{p(p-1)}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ survives.

13.2 CONSOLIDATION. (Counting with signs.) (a) Show $\mathbb{E}\det(tI - A) = N^{-(N-1)/2}\mathrm{He}_{N-1}(t\sqrt{N})$, with He_n the probabilists' Hermite polynomial.

(b) Deduce $\sum_k (-1)^k \mathbb{E}\mathrm{Crt}_{N,k}(\mathbb{R}) = 1 + (-1)^{N-1}$ for every $p \geq 2$. *(Hint:* $\mathbb{E}\mathrm{He}_{2m}(cZ) = (2m-1)!!\,(c^2-1)^m$ *for* $Z \sim \mathcal{N}(0,1)$.*)*

13.3 BY HAND. (Selberg's constant.) Use Mehta's integral

$$\int_{\mathbb{R}^n} |\Delta(x)| \prod_{i=1}^n e^{-x_i^2/2}\,\mathrm{d}x = (2\pi)^{n/2} \prod_{j=1}^n \frac{\Gamma(1+j/2)}{\Gamma(3/2)}$$

to show $c_N = 2\sqrt{2}\,(2/N)^{N/2}\Gamma(1+\frac{N}{2})$ in Lemma 13.2, and recover $\gamma_N = 2\sqrt{2}$ without Lemma 13.3.

13.4 BY HAND. (The equator and the good point.) (a) For $p \geq 3$ show that the spike has no gradient and no Hessian along the sphere at $m = 0$, where minima have complexity $\ell_p - \frac{p-2}{p}$ for every λ. (b) Show that $m^{2p-4}(1-m^2)$ is maximal at $m^2 = \frac{p-2}{p-1}$, deduce (13.5), and check $\lambda_c(4) = \sqrt{27}/4$ and $m^*(\lambda_c) = m_c$. (c) For $p = 3$ show $m^*(\lambda)^2 = \frac{1}{2}(1+\sqrt{1-4/(3\lambda^2)})$.

13.5 BY HAND. (The estimator's threshold.) Show that λ_{ML} of Theorem 13.11 is $(pq^{p-2}(1-q))^{-1/2}$ at the root $q \in (0,1)$ of $\frac{q^2}{p(1-q)} + q + \log(1-q) = 0$, and compute $\lambda_{\mathrm{ML}}(3) = 1.20656$, $\lambda_{\mathrm{ML}}(4) = 1.40567$.

13.6 TRANSLATION. (No minima above the threshold, at speed N^2.) Using the large deviation principle of the empirical measure at speed N^2 (Remark A.17), prove $\mathbb{P}(\lambda_{(0)}(W) > -2+\delta) \leq e^{-c_\delta N^2}$, deduce that a local minimum above $-E_\infty + \varepsilon$ has probability at most e^{-cN^2}, and say which sentence of the physicist's argument this replaces.

13.7 TRANSLATION. (Quenched equals annealed.) Let random variables $X_N \geq 0$ satisfy $\mathbb{E}X_N^2 \sim (\mathbb{E}X_N)^2$. Show that $X_N/\mathbb{E}X_N \to 1$ in probability and that $\mathbb{P}(X_N > 0) \geq (\mathbb{E}X_N)^2/\mathbb{E}X_N^2$; with Theorem 13.8, prove that $\mathrm{GS}_N \to -E_0$ in probability. Which replica assumption does each step replace?

13.8 TRANSLATION. (TAP states.) Show that stationarity of (13.2) in q at energy $h = H_N(\sigma)/N$ reads $\beta z = \frac{-h \pm \sqrt{h^2 - E_\infty^2}}{\sqrt{2}(p-1)}$ with $z = (1-q)q^{p/2-1}$ (Auffinger et al. 2013, Section 6), so that TAP states exist only below the threshold.

13.9 MEASUREMENT. (Two scalings.) Modify the spiked-tensor script to find, for $N = 40$, 80 and 160, the λ at which half of the random starts reach $m > \frac{1}{2}$, and compare its growth with $N^{(p-2)/2}$; modify the script for the minima to test whether the gap of the mean minimum to the threshold, times $\sqrt{N}$, stays near 1.2 up to $N = 1024$.

13.10 RESEARCH. (The typical spiked landscape.) Compute the second moment of the number of critical points of $H_{N,\lambda}$ near the equator for $\lambda < \lambda_c$, and decide whether the band of minima there is typical, as Ros et al. (2019) predict.

Notes and References

The physics of the complexity grew from counting TAP states: Kurchan et al. (1993), Crisanti and Sommers (1995) and Cavagna et al. (1998) counted them for the spherical p-spin, Kurchan and Laloux (1996) found the layers of saddles, and Cugliandolo and Kurchan (1993) the threshold in the dynamics.

The rigorous theory is due to Auffinger et al. (2013), extended to mixed models by Auffinger and Ben Arous (2013); Subag's second moment reaches the ground state without the Parisi formula. The measurements are new.

TRANSLATING THE LITERATURE. Auffinger et al. (2013) and Subag (2017) use our H_N, $\mathrm{Crt}_{N,k}$, Θ_p, $\Theta_{k,p}$, E_k and E_∞, with a GOE of variances $(1+\delta_{ij})/(2N)$: their λ_k^N is our $\lambda_{(k)}/\sqrt{2}$. Their Remark 2.7 states the Crisanti–Sommers complexity as $\Theta_{0,p}(2^{-1/2}E)$; it should read $\Theta_{0,p}(\sqrt{2}E)$, as in their Lemma 6.3, eq. (6.19), to ten digits by our script (with $2^{-1/2}E$ it would be constant below the threshold). Physicists use energies $E = u/\sqrt{2}$ (the 3-spin threshold is -1.154701, its ground state -1.171675) and temperatures $\sqrt{2}$ times smaller. Ben Arous et al. (2019) maximize $f = -H_{N,\lambda}/(\sqrt{2}N)$ on the unit sphere: their λ is our $\lambda/\sqrt{2}$ and their energy x our $-u/\sqrt{2}$. Jagannath et al. (2020) and Ben Arous et al. (2020) use our λ; the former write λ_s for our λ_c and λ_c for our λ_{ML}. Ros et al. (2019) write $r = p\lambda/\sqrt{2}$; Huang and Sellke (2025) and Sellke maximize H_N.

References

Anderson, G. W., A. Guionnet, and O. Zeitouni (2010). *An Introduction to Random Matrices*. Vol. 118. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press.

Auffinger, A. and G. Ben Arous (2013). "Complexity of random smooth functions on the high-dimensional sphere". In: *The Annals of Probability* 41.6, pp. 4214–4247. arXiv: 1110.5872.

Auffinger, A., G. Ben Arous, and J. Černý (2013). "Random Matrices and Complexity of Spin Glasses". In: *Communications on Pure and Applied Mathematics* 66.2, pp. 165–201. arXiv: 1003.1129.

Ben Arous, G., R. Gheissari, and A. Jagannath (2020). "Algorithmic thresholds for tensor PCA". In: *The Annals of Probability* 48.4, pp. 2052–2087. arXiv: 1808.00921.

Ben Arous, G., S. Mei, A. Montanari, and M. Nica (2019). "The Landscape of the Spiked Tensor Model". In: *Communications on Pure and Applied Mathematics* 72.11, pp. 2282–2330. arXiv: 1711.05424.

Cavagna, A., I. Giardina, and G. Parisi (1998). "Stationary points of the Thouless–Anderson–Palmer free energy". In: *Physical Review B* 57.18, pp. 11251–11257. arXiv: cond-mat/9710272.

Chen, W.-K., T.-L. Lu, and A. Sen (2026). *Universality of Annealed Complexity in the Spherical Pure p-Spin Model*. arXiv preprint, v1 of 16 October 2026; unrefereed. arXiv: 2610.23701v1.

Crisanti, A. and H.-J. Sommers (1992). "The spherical p-spin interaction spin glass model: the statics". In: *Zeitschrift für Physik B Condensed Matter* 87.3, pp. 341–354.

Crisanti, A. and H.-J. Sommers (1995). "Thouless–Anderson–Palmer Approach to the Spherical p-Spin Spin Glass Model". In: *Journal de Physique I* 5.7, pp. 805–813.

Cugliandolo, L. F. and J. Kurchan (1993). "Analytical solution of the off-equilibrium dynamics of a long-range spin-glass model". In: *Physical Review Letters* 71.1, pp. 173–176. arXiv: cond-mat/9303036.

Dumitriu, I. and A. Edelman (2002). "Matrix models for beta ensembles". In: *Journal of Mathematical Physics* 43.11, pp. 5830–5847.

Fan, Z., S. Mei, and A. Montanari (2021). "TAP free energy, spin glasses and variational inference". In: *The Annals of Probability* 49.1, pp. 1–45.

Folena, G., S. Franz, and F. Ricci-Tersenghi (2020). "Rethinking Mean-Field Glassy Dynamics and Its Relation with the Energy Landscape: The Surprising Case of the Spherical Mixed p-Spin Model". In: *Physical Review X* 10.3, p. 031045.

Fyodorov, Y. V. (2004). "Complexity of Random Energy Landscapes, Glass Transition, and Absolute Value of the Spectral Determinant of Random Matrices". In: *Physical Review Letters* 92.24. Erratum: Phys. Rev. Lett. 93, 149901 (2004), p. 240601. arXiv: cond-mat/0401287.

Hopkins, S. B., J. Shi, and D. Steurer (2015). "Tensor principal component analysis via sum-of-square proofs". In: *Proceedings of the 28th Conference on Learning Theory (COLT)*, pp. 956–1006. arXiv: 1507.03269.

Huang, B. and M. Sellke (2025). "Tight Lipschitz hardness for optimizing mean field spin glasses". In: *Communications on Pure and Applied Mathematics* 78.1, pp. 60–119. arXiv: 2110.07847.

Jagannath, A., P. Lopatto, and L. Miolane (2020). "Statistical thresholds for tensor PCA". In: *The Annals of Applied Probability* 30.4, pp. 1910–1933. arXiv: 1812.03403.

Kim, T. (2026). *Regularized Bulk Universality versus Bounded-Disorder Nonuniversality for Annealed Complexity of Spherical p-Spin Landscapes*. arXiv preprint, v1 of 30 July 2026; unrefereed. arXiv: 2607.27613v1.

Kurchan, J., G. Parisi, and M. A. Virasoro (1993). "Barriers and metastable states as saddle points in the replica approach". In: *Journal de Physique I* 3.8, pp. 1819–1838.

Kurchan, J. and L. Laloux (1996). "Phase space geometry and slow dynamics". In: *Journal of Physics A: Mathematical and General* 29.9, pp. 1929–1948.

Maïda, M. (2007). "Large deviations for the largest eigenvalue of rank one deformations of Gaussian ensembles". In: *Electronic Journal of Probability* 12, pp. 1131–1150.

Montanari, A. and E. Richard (2014). "A statistical model for tensor PCA". In: *Advances in Neural Information Processing Systems 27 (NIPS 2014)*, pp. 2897–2905. arXiv: 1411.1076.

Ros, V., G. Ben Arous, G. Biroli, and C. Cammarota (2019). "Complex Energy Landscapes in Spiked-Tensor and Simple Glassy Models: Ruggedness, Arrangements of Local Minima, and Phase Transitions". In: *Physical Review X* 9.1, p. 011003. arXiv: 1804.02686.

Sellke, M. (2024). "The threshold energy of low temperature Langevin dynamics for pure spherical spin glasses". In: *Communications on Pure and Applied Mathematics* 77.11, pp. 4065–4099. arXiv: 2305.07956.

Subag, E. (2017). "The complexity of spherical p-spin models—A second moment approach". In: *The Annals of Probability* 45.5, pp. 3385–3450. arXiv: 1504.02251.

Subag, E. and O. Zeitouni (2021). "Concentration of the complexity of spherical pure p-spin models at arbitrary energies". In: *Journal of Mathematical Physics* 62.12, p. 123301. arXiv: 2109.03163.

Talagrand, M. (2006). "Free energy of the spherical mean field model". In: *Probability Theory and Related Fields* 134.3, pp. 339–382.

Dynamical Mean-Field Theory

14

Gradient flow in high dimension

THE DICTIONARY —— the generating functional ↔ Gaussian conditioning

IN THIS CHAPTER. Run gradient flow or Langevin dynamics in a random landscape and let the dimension tend to infinity at fixed times. Physicists predict, from a generating functional without replicas, that the correlation $C(t,s)$ and the response $R(t,s)$ obey closed equations, which for the spherical p-spin model predict a dynamical transition, aging, and an energy that stops at a threshold above the ground state. Rigorous versions hold on fixed time horizons, by stochastic calculus for spin glasses at positive temperature and by Gaussian conditioning for gradient methods with smooth losses on generalized linear models; we prove the theory completely for two linear flows, least squares and the spherical 2-spin, where the self-consistent kernel is a Marchenko–Pastur or semicircle transform. Integrations of the equations agree with simulations where the equations are a theorem; beyond times of order one results exist only in special cases, and aging for $p \geq 3$ remains a conjecture.

14.1 The question: where does gradient flow go?

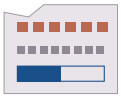
Most estimators in high dimension are computed by optimization, and the replica and comparison chapters describe the optimum. They do not say what a gradient method finds in a time that does not grow with the dimension, which in a nonconvex landscape can be something else. *Dynamical mean-field theory* is the physicists' answer: as the dimension grows, the trajectory of a high-dimensional flow is described by a few deterministic functions of two times, which solve closed equations.

A LANDSCAPE FROM TENSOR PCA. To estimate v from a noisy tensor $Y = \lambda v^{\otimes p} + G$ one maximizes the likelihood over the sphere; when λ is small, a gradient method sees the landscape of the noise, the *spherical p-spin model*. On $S_N = \{x \in \mathbb{R}^N : \|x\|^2 = N\}$,

$$H_N(x) = N^{-(p-1)/2} \sum_{i_1,\dots,i_p=1}^{N} G_{i_1\cdots i_p} x_{i_1} \cdots x_{i_p}, \qquad \mathbb{E} H_N(x) H_N(y) = N\xi\Big(\frac{\langle x \rangle y}{N}\Big),$$

with i.i.d. standard Gaussian entries and $\xi(q) = q^p$, the book's normalization (mixtures have $\xi(q) = \sum_p c_p^2 q^p$); we minimize. By Chapter 13 the ground state is $-E_0$, $E_0(3) = 1.65700$, and the threshold $-E_\infty$, $E_\infty = 2\sqrt{(p-1)/p}$.

PHYSICS
NUMERICS
PROVED IN PART



The dynamics is the Langevin diffusion at temperature $T \geq 0$ from x_0 uniform on S_N and independent of H_N,

$$\mathrm{d}x_t = -\nabla_{\mathrm{sp}} H_N(x_t)\,\mathrm{d}t + \sqrt{2T}\,P^{\perp}_{x_t}\,\mathrm{d}B_t - T\,\frac{N-1}{N}\,x_t\,\mathrm{d}t, \tag{14.1}$$

with $P^{\perp}_x = I - xx^{\top}/N$, $\nabla_{\mathrm{sp}} = P^{\perp}_x \nabla$, a Brownian motion B and an Itô correction that keeps $\|x_t\|^2 = N$; $T = 0$ is gradient flow. A trajectory is summarized by its energy $e_N(t) = H_N(x_t)/N$, the *correlation function* $C_N(t,s) = \frac{1}{N}\langle x_t, x_s\rangle$ and the *response function* $R_N(t,s) = \frac{1}{N}\sum_i \partial \mathbb{E}[x_{t,i}]/\partial h_i(s)$ to a small field h added to the drift at time $s < t$; in this chapter R is never an overlap.[1]

THE PHYSICAL PREDICTION · Crisanti, Horner and Sommers 1993; Cugliandolo and Kurchan 1993. As $N \to \infty$ at fixed times, $C_N \to C$ and $R_N \to R$, the solution of the closed equations, for $t > s$,

$$\begin{aligned} \partial_t C(t,s) &= -\mu(t)C(t,s) + \int_0^t \xi''(C(t,u))R(t,u)C(u,s)\,\mathrm{d}u \\ &\quad + \int_0^s \xi'(C(t,u))R(s,u)\,\mathrm{d}u, \\ \partial_t R(t,s) &= -\mu(t)R(t,s) + \int_s^t \xi''(C(t,u))R(t,u)R(u,s)\,\mathrm{d}u, \end{aligned} \tag{14.2}$$

with $C(t,t) = R(t,t) = 1$ and $\mu(t) = T + \int_0^t [\xi'(C(t,u)) + C(t,u)\xi''(C(t,u))]R(t,u)\,\mathrm{d}u$; the energy tends to $e(t) = -\int_0^t \xi'(C(t,u))R(t,u)\,\mathrm{d}u$. For the pure p-spin with $p \geq 3$ the dynamics equilibrates above a temperature T_d and ages below it, and the energy it reaches tends, as $T \to 0$, to the threshold $-E_\infty$ (-1.63299 for $p = 3$), strictly above the ground state $-E_0$ (-1.65700): gradient methods do not find the bottom of the landscape.

In learning, gradient flow $\dot\beta_t = -\nabla \hat L(\beta_t)$ on an empirical risk $\hat L(\beta) = \frac{1}{n}\sum_i \mathrm{loss}(\langle x_i, \beta\rangle, y_i)$, with $x_i \sim \mathcal{N}(0, I_d)$ and $n/d \to \alpha$, uses every sample at every time; unlike the online algorithm of Chapter 15, the trajectory and the data become dependent and the dynamics acquires memory. Dynamical mean-field theory predicts a closed effective process for one coordinate and one sample (Agoritsas et al. 2018; Mignacco et al. 2021a), and for least squares an explicit learning curve, (14.6) below.

How do physicists derive closed equations for an N-dimensional random dynamics, and what has been proved: for which models, on which time scales, and under which hypotheses?

14.2 The physics: a generating functional without replicas

A physicist computes averages over the noise and the disorder from a generating functional of trajectories, in the formalism of Martin et al. (1973), Janssen (1976) and De Dominicis (1976), first applied to spin glasses

[1] Cugliandolo and Kurchan (1993) and Crisanti et al. (1993) use the covariance $Nq^p/2$; rescaling time shows that their energies and temperatures are ours divided by $\sqrt{2}$ and their times ours multiplied by $\sqrt{2}$. Their threshold -1.154701, ground state -1.171675 and dynamical temperature $\sqrt{3/8}$ for $p = 3$ are our -1.632993, -1.656998 and $\sqrt{3}/2$. We write the equations for a general ξ, valid in both normalizations.

by Sompolinsky and Zippelius (1981). The functional is normalized by construction, so the disorder can be averaged directly.[2]

[2] De Dominicis (1978) titled his paper "Dynamics as a substitute for replicas in systems with quenched random impurities".

THE MARTIN–SIGGIA–ROSE INTEGRAL. *Paths.* Physicists write (14.1), with a field h, as $\dot{x}_i = h_i - \partial_i H - \mu x_i + \eta_i$, with white noise η of strength $2T$ and a Lagrange multiplier $\mu(t)$. Noise averages are then integrals over paths of a delta function of this equation, whose Jacobian is 1 in Itô's convention. Writing the delta function as an integral over a response field $\hat{x}$ and averaging the noise, $\mathbb{E} e^{i\int \hat{x}\eta} = e^{-T\int \hat{x}^2}$, gives

$$\mathbb{E}_\eta \mathcal{O}(x) = \int \mathcal{D}x\, \mathcal{D}\hat{x}\, \mathcal{O}(x)\, e^{S[x,\hat{x}]},$$
$$S = \sum_i \int \mathrm{d}t \left\{ i\hat{x}_i \big(h_i - \dot{x}_i - \mu x_i - \partial_i H(x)\big) - T\hat{x}_i^2 \right\}.$$

The total mass is 1 for every h and every disorder. Differentiating in h gives $R(t,s) = \frac{1}{N}\sum_i \langle x_i(t)\, i\hat{x}_i(s)\rangle$, and every moment of $\hat{x}$ alone vanishes.

Disorder. The Gaussian H enters linearly, through $-\sum_i \int i\hat{x}_i \partial_i H(x)$, so its average is the exponential of half the variance. Since $\mathbb{E}[\partial_i H(x)\partial_j H(y)] = \delta_{ij}\xi'(\langle x\rangle y/N) + y_i x_j \xi''(\langle x\rangle y/N)/N$, the averaged exponent is

$$\frac{N}{2}\int\!\!\int \mathrm{d}t\ \mathrm{d}s \left\{ \xi'(C(t,s))\, Q(t,s) + \xi''(C(t,s))\, K(t,s)K(s,t) \right\},$$

with $C(t,s) = \frac{1}{N}\sum_i x_i(t)x_i(s)$, $K(t,s) = \frac{1}{N}\sum_i x_i(t)\, i\hat{x}_i(s)$ and $Q(t,s) = \frac{1}{N}\sum_i i\hat{x}_i(t)\, i\hat{x}_i(s)$.

Saddle point and effective process. Fixing C, K, Q by delta functions factorizes the integral over coordinates and leaves an integral $\int e^{N\Sigma}$ over the order parameters, dominated as $N \to \infty$ by a saddle point at which C is the correlation, $K = R$ the response, and $Q = 0$. There the term in Q becomes a centered Gaussian noise ϕ with $\mathbb{E}\phi(t)\phi(s) = \xi'(C(t,s))$, and the term in KK a memory term: each coordinate follows the single-site process

$$\dot{x}(t) = -\mu(t)x(t) + \int_0^t \xi''(C(t,u))R(t,u)x(u)\,\mathrm{d}u + \phi(t) + \eta(t) + h(t), \quad (14.3)$$

with $C(t,s) = \mathbb{E}x(t)x(s)$ and $R(t,s) = \mathbb{E}\,\partial x(t)/\partial h(s)$ determined self-consistently. ◇

The uncontrolled steps are the path integral, which is not a measure, the exchange of the disorder average with it, and the saddle point in a space of functions of two times, with the assumption that the order parameters concentrate. The dynamical cavity method reaches (14.3) without a path integral, by adding one coordinate to a large system and treating its effect by linear response.

CLOSING THE EQUATIONS. For a spherical model (14.3) is linear. Multiply it by $x(s)$, $s < t$, and average. The noise $\eta(t)$ is independent of the past, and *Novikov's formula*, Gaussian integration by parts for ϕ, gives

$$\mathbb{E}[\phi(t)x(s)] = \int_0^s \mathbb{E}[\phi(t)\phi(u)]\, \mathbb{E}\frac{\partial x(s)}{\partial \phi(u)}\,\mathrm{d}u = \int_0^s \xi'(C(t,u))R(s,u)\,\mathrm{d}u,$$

since a change of $\phi(u)$ acts as a field: this is the C-equation of (14.2). Differentiating (14.3) in $h(s)$ gives the R-equation, and $C(t,t)=1$, with Itô's correction $2T$, fixes μ. Averaging $H(x(t))$ against the Gaussian term of the action gives $\mathbb{E}\langle H(x(t))\rangle = -\int_0^t \mathrm{d}u\, \sum_i \langle i\hat{x}_i(u)\, x_i(t)\rangle\, \xi'(C(t,u)) = -N\int_0^t \xi'(C(t,u))R(t,u)\,\mathrm{d}u$. ◇

For a pure model Euler's identity $\langle x, \nabla H\rangle = pH$ gives $\mu = T - pe$. The closure is special to spherical models: with a non-quadratic single-site potential, C and R do not determine the effective process.[3]

[3] This is the Sompolinsky–Zippelius theory of soft spins; Ben Arous and Guionnet (1997) proved that its effective process is the limit at high temperature.

14.2.1 Long times: equilibrium, aging, threshold

Physicists then let $t\to\infty$ after $N\to\infty$. Above a dynamical temperature the solution becomes invariant under time translations, obeys the *fluctuation–dissipation theorem* $R(\tau) = -\frac{1}{T}C'(\tau)$, and $C(\tau)\to 0$. Below it, Cugliandolo and Kurchan (1993) found, numerically and then analytically, *aging*: $C(t_w+\tau, t_w)$ decays to a plateau q and then to 0 on a time scale that grows with the age t_w, with a fluctuation–dissipation relation modified by a factor $x<1$.

COMPUTATION 14.1 · THE DYNAMICAL TEMPERATURE AND THE THRESHOLD

1. *Two scales.* For large s and $\tau = t-s$, split $C(t,s)$ into a fast part, decaying from 1 to q with $R = -\frac{1}{T}\partial_\tau C$, and a slow part, decaying from q to 0 with $R = \frac{x}{T}\partial_s C$.
2. *Energy.* On each part $-\int \xi'(C)R$ becomes an integral over the value c of C:
$$e_\infty = -\frac{1}{T}\int_q^1 \xi'(c)\,\mathrm{d}c - \frac{x}{T}\int_0^q \xi'(c)\,\mathrm{d}c = -\frac{1}{T}\big[\xi(1) - (1-x)\xi(q)\big];$$
with $q=0$ this is the equilibrium energy $-\xi(1)/T$.
3. *Plateau.* The same splitting in the C-equation for $1\ll\tau\ll s$, with μ computed likewise, gives $qT^2 = \xi'(q)(1-q)\big[(1-q)+xq\big]$.
4. *Dynamical temperature.* At the transition the slow decay still has $x=1$, so $T^2 = \xi'(q)(1-q)/q$, and a solution $q>0$ first appears at $T_d^2 = \max_q \xi'(q)(1-q)/q$. For $\xi = q^p$: $q_d = (p-2)/(p-1)$, $T_d^2 = p(p-2)^{p-2}/(p-1)^{p-1}$, and $T_d = \sqrt{3}/2 = 0.86603$ for $p=3$.
5. *Aging.* Below T_d, Cugliandolo and Kurchan impose marginality, $\xi''(q)(1-q)^2 = T^2$; with step 3 it gives $x = (p-2)(1-q)/q$. For $p=3$ and $T=0.7$: $q=0.64385$, $x=0.55315$, $e_\infty = -1.25819$.
6. *Threshold.* As $T\to 0$, $1-q\sim T/\sqrt{\xi''(1)}$ and $x\sim(p-2)(1-q)$, and step 2 gives $e_\infty \to -[\xi'(1)+(p-2)\xi(1)]/\sqrt{\xi''(1)} = -2(p-1)/\sqrt{p(p-1)} = -E_\infty$.

The separation of scales, the slow part and marginality are ansätze;[4] marginality says that the dynamics settles where the Hessian has a vanishing

[4] Marginality and the energy with this x are equations (7) and (8) of Cugliandolo and Kurchan (1993), in their normalization.

smallest eigenvalue, and in the landscape of Chapter 13 the Hessian at a critical point of energy E is a semicircle of radius $2\sqrt{p(p-1)}$ shifted by $-pE$, which touches zero exactly at the *threshold energy* $-E_\infty$.

14.2.2 The same theory for learning

In coordinates of order one, $w = \sqrt{d}\,\beta$ and $A = X/\sqrt{d}$, gradient flow on $\hat{L}$ reads $\dot{w}_t = -\frac{1}{\alpha}A^\top \ell(Aw_t; y)$, where $\ell(r; y) = \partial_r \mathrm{loss}(r, y)$ acts row by row.

THE DYNAMICAL CAVITY FOR LEARNING. Add a coordinate w_0, with a new column a_0 of A, so that $\dot{w}_0 = -\frac{1}{\alpha}a_0^\top \ell(r; y)$. On the old residuals this force is a Gaussian process of covariance C_ℓ/α, by the central limit theorem; the effect of w_0 on its own fields gives $-\Gamma_t w_0$; and the old trajectory's response feeds back with delay. Removing a sample gives the equation for its field, and closing the averages gives

$$\begin{aligned} \dot{w}_t &= -\Gamma_t w_t - \int_0^t R_\ell(t,s) w_s \,\mathrm{d}s + \zeta_t, \qquad \zeta \sim \mathrm{GP}(0, C_\ell/\alpha), \\ r_t &= -\frac{1}{\alpha}\int_0^t R_w(t,s)\,\ell(r_s; y)\,\mathrm{d}s + \omega_t, \quad \omega \sim \mathrm{GP}(0, C_w), \end{aligned} \tag{14.4}$$

with $\Gamma_t = \mathbb{E}\ell'(r_t; y)$, $C_w(t,s) = \mathbb{E}w_t w_s$, $C_\ell(t,s) = \mathbb{E}\ell(r_t; y)\ell(r_s; y)$, and responses R_w and $R_\ell = \mathbb{E}\rho$ defined, for $s < t$, by

$$\partial_t R_w(t,s) = -\Gamma_t R_w(t,s) - \int_s^t R_\ell(t,u) R_w(u,s)\,\mathrm{d}u, \qquad R_w(s,s) = 1,$$

$$\rho(t,s) = -\frac{\ell'(r_t; y)}{\alpha}\Big[\int_s^t R_w(t,u)\rho(u,s)\,\mathrm{d}u + R_w(t,s)\ell'(r_s; y)\Big]. \quad \diamond$$

This is the system that Celentano et al. (2021) later proved, a planted teacher entering as an extra fixed coordinate. For the square loss it is linear.

COMPUTATION 14.2 · THE MEMORY KERNEL OF LEAST SQUARES

1 *Square loss.* For $\ell(r; y) = r - y$, $\ell' = 1$ and $\Gamma_t = 1$; the response equations of (14.4) are deterministic, linear and invariant under time translations. With $\tau = t - s$ and $\gamma = 1/\alpha$,

$$R_w'(\tau) = -R_w(\tau) - \int_0^\tau R_\ell(\tau - u) R_w(u)\,\mathrm{d}u, \qquad R_w(0) = 1,$$

$$R_\ell(\tau) = -\gamma R_w(\tau) - \gamma\int_0^\tau R_w(\tau - u) R_\ell(u)\,\mathrm{d}u.$$

2 *Laplace transforms.* With $\hat{f}(p) = \int_0^\infty e^{-p\tau} f(\tau)\,\mathrm{d}\tau$: $p\hat{R}_w - 1 = -\hat{R}_w - \hat{R}_\ell\hat{R}_w$ and $\hat{R}_\ell = -\gamma\hat{R}_w - \gamma\hat{R}_w\hat{R}_\ell$.

3 *Eliminate.* The second gives $\hat{R}_\ell = -\gamma\hat{R}_w/(1 + \gamma\hat{R}_w)$, and the first becomes

$$\hat{R}_w(p)\Big(p + \frac{1}{1 + \gamma\hat{R}_w(p)}\Big) = 1, \tag{14.5}$$

that is, $\gamma p\hat{R}_w^2 + (p + 1 - \gamma)\hat{R}_w - 1 = 0$.

4 *Recognize.* At $z = -p$ the Marchenko–Pastur equation (A.9) is $-\gamma p m^2 - (p+1-\gamma)m + 1 = 0$, the same quadratic, whose roots have product $-1/(\gamma p) < 0$. So $\hat{R}_w(p) = m_\gamma(-p) = \int \mu_\gamma(\mathrm{d}\lambda)/(\lambda + p)$, and $R_w(\tau) = \int e^{-\tau\lambda}\mu_\gamma(\mathrm{d}\lambda)$.

The self-consistent memory kernel is the Laplace transform of the Marchenko–Pastur law, and it is the limit of the response $\frac{1}{d}\operatorname{tr} e^{-\tau S}$ of the flow (Theorem 14.2); time-stepping step 1 reproduces it to 10^{-5}.[5] Solving the system, or diagonalizing the flow and replacing the spectrum by its limit (Advani et al. 2020), gives for $\|\beta^*\| = r$ and noise variance σ^2 the excess risk and the training error

$$\begin{aligned} \mathcal{E}(t) &= r^2 \int e^{-2t\lambda}\mu_\gamma(\mathrm{d}\lambda) + \sigma^2\gamma \int \frac{(1-e^{-t\lambda})^2}{\lambda}\,\mu_\gamma(\mathrm{d}\lambda), \\ \mathcal{L}(t) &= \frac{r^2}{2}\int \lambda e^{-2t\lambda}\mu_\gamma(\mathrm{d}\lambda) + \frac{\sigma^2}{2}\Big(1-\gamma+\gamma\int e^{-2t\lambda}\mu_\gamma(\mathrm{d}\lambda)\Big). \end{aligned} \tag{14.6}$$

[5] Celentano et al. (2021, Section 4.3) note, in a heuristic sketch for strongly convex risks, a generalized Marchenko–Pastur equation; for the square loss it is exact.

14.3 The stone

PHYSICS · *the generating functional*

1 · Average and saddle point

The path integral has mass 1, so the disorder is averaged without replicas; the action is N times a functional of C, R, $\langle \hat{x}\hat{x}\rangle$, and the saddle point is exact.

2 · Effective process

Each coordinate follows (14.3), with memory kernel $\xi''(C)R$ and colored noise of covariance $\xi'(C)$.

3 · Closure

For spherical models the process is linear, and Novikov's formula closes it on (C, R): the equations (14.2).

PROOF · *fixed horizons, conditioning and concentration*

Fix a horizon. Itô calculus and concentration of C_N and the integrated response replace the saddle point (Ben Arous et al. 2006); for learning, gradient descent becomes an AMP iteration with memory, with deterministic state evolution.

Gaussian conditioning (Lemma A.13): given the past, the new field is a combination of past fields, the memory, plus a fresh Gaussian vector, the noise (Gerbelot et al. 2024).

Limit points satisfy closed equations because the covariance of the drift is a polynomial in overlaps; uniqueness by Grönwall. For flows, a contraction sends the step to zero (Celentano et al. 2021).

4 · The linear case

For least squares the kernel solves the Marchenko–Pastur equation (Computation 14.2); for $p = 2$ it is a semicircle transform.

Theorems 14.2 and 14.3: spectral law and Gaussian quadratic forms give every limit, uniformly in time, with no saddle point.

5 · Equilibrium and aging

Above T_d: time-translation invariance and FDT; below: aging, with factor x and marginality.

FDT solution for small β only (Dembo et al. 2007); up to T_d, no counterpart. Aging proved for $p = 2$ only (Ben Arous et al. 2001).

6 · Threshold and statics

As $T \to 0$ the energy tends to $-E_\infty$; in benign landscapes the long-time limit is the replica-symmetric optimum.

The threshold for the pure p-spin at low temperature (Sellke 2024); convex losses, given exponential convergence (Celentano et al. 2021, unpublished). The pure p-spin is a counterexample: $-E_\infty > -E_0$.

The rigorous route keeps the physicists' objects but not their method: concentration replaces the saddle point, and Gaussian conditioning, the dynamical cavity method made exact (Gerbelot et al. 2024), replaces the derivation of the effective process. Every rigorous version of the equations holds at fixed times as $N \to \infty$. The routes part at long times, where each prediction needs a separate theorem; these exist for the energy of the pure p-spin (Sellke 2024), for convex losses in a preprint (Celentano et al. 2021), and for the linear flows below, where the self-consistency reconstructs the spectral law of the matrix that drives the flow.

14.4 The proof: two linear flows

For a linear flow the dynamical mean-field limits reduce to spectral laws and to concentration of Gaussian quadratic forms.

Lemma 14.1 (Gaussian quadratic forms). *Let $g \sim \mathcal{N}(0, I_m)$, and let a symmetric $m \times m$ matrix A and a vector $u \in \mathbb{R}^m$ be random and independent of g. Then*

$$\mathbb{E}\left[(g^\top A g - \operatorname{tr} A)^4 \,\middle|\, A\right] \le 60\|A\|_F^4 \le 60\, m^2 \|A\|^4,$$
$$\mathbb{E}\left[(u^\top g)^4 \,\middle|\, u\right] = 3\|u\|^4.$$

Consequently, if along a sequence $m \to \infty$ each $g_m \sim \mathcal{N}(0, I_m)$ is independent of (A_m, u_m), and almost surely $\limsup_m \|A_m\| < \infty$ and $\limsup_m \|u_m\|/\sqrt{m} < \infty$, then almost surely $\frac{1}{m}(g_m^\top A_m g_m - \operatorname{tr} A_m) \to 0$ and $\frac{1}{m} u_m^\top g_m \to 0$.

Proof. The plan is to diagonalize A, compute a fourth moment, and apply

the Borel–Cantelli lemma. Given $A = \sum_k a_k v_k v_k^\top$, the $\zeta_k = \langle v_k \rangle g^2 - 1$ are independent copies of $Z^2 - 1$, which has mean 0, $\mathbb{E}(Z^2-1)^2 = 2$ and $\mathbb{E}(Z^2-1)^4 = 105 - 60 + 18 - 4 + 1 = 60$. Terms of $\mathbb{E}(\sum_k a_k \zeta_k)^4$ containing some ζ_k exactly once vanish, so

$$\mathbb{E}\Big(\sum_k a_k \zeta_k\Big)^4 = 60 \sum_k a_k^4 + 12 \sum_{k \neq l} a_k^2 a_l^2 \le 60 \Big(\sum_k a_k^2\Big)^2 = 60 \|A\|_F^4,$$

and $\|A\|_F^2 \le m \|A\|^2$; given u, $u^\top g \sim \mathcal{N}(0, \|u\|^2)$. For the consequence fix $\varepsilon > 0$ and $K < \infty$: by Markov's inequality $\mathbb{P}(|g_m^\top A_m g_m - \operatorname{tr} A_m| > \varepsilon m,\ \|A_m\| \le K) \le 60 K^4/(\varepsilon^4 m^2)$, which is summable, so by the Borel–Cantelli lemma, almost surely, for all large m either $\|A_m\| > K$ or the deviation is at most εm. Letting K run through the integers and ε through $1/j$ proves the first claim; the second uses $\mathbb{P}(|u_m^\top g_m| > \varepsilon m,\ \|u_m\| \le K\sqrt{m}) \le 3K^4/(\varepsilon^4 m^2)$. ∎

14.4.1 Least squares

Theorem 14.2 (gradient flow on least squares). *Let $X \in \mathbb{R}^{n \times d}$ have i.i.d. $\mathcal{N}(0,1)$ entries, $\varepsilon \sim \mathcal{N}(0, I_n)$, and β^* uniform on the sphere of radius r in $\mathbb{R}^d$, independent, and $y = X\beta^* + \sigma\varepsilon$ with $\sigma \ge 0$. Let β_t be the gradient flow of $\hat{L}(\beta) = \frac{1}{2n}\|y - X\beta\|^2$ from $\beta_0 = 0$, $S = X^\top X/n$, and $n, d \to \infty$ with $d/n \to \gamma \in (0, \infty)$; let μ_γ be the Marchenko–Pastur law (Theorem A.16) and m_γ its Stieltjes transform. Then, almost surely:*

(i) (Response.) The response $R_d(t,s) = \frac{1}{d} \operatorname{tr} e^{-(t-s)S}$ satisfies $\sup_{0 \le s \le t} |R_d(t,s) - R(t-s)| \to 0$, with $R(\tau) = \int e^{-\tau\lambda} \mu_\gamma(\mathrm{d}\lambda)$, whose Laplace transform at $p > 0$ is $m_\gamma(-p)$, the unique positive solution of (14.5).

(ii) (Correlation and risk.) For every $t_{\max} < \infty$, uniformly in $s, t \le t_{\max}$, $\hat{L}(\beta_t) \to \mathcal{L}(t)$ and $\langle \beta_t - \beta^*, \beta_s - \beta^* \rangle \to C(t,s)$, where

$$C(t,s) = r^2 \int e^{-(t+s)\lambda} \mu_\gamma(\mathrm{d}\lambda) + \sigma^2 \gamma \int \frac{(1 - e^{-t\lambda})(1 - e^{-s\lambda})}{\lambda} \mu_\gamma(\mathrm{d}\lambda),$$

the last integrand being 0 at $\lambda = 0$; in particular $\|\beta_t - \beta^*\|^2 \to \mathcal{E}(t) = C(t,t)$.

(iii) (Long times.) If $\gamma \neq 1$, the convergence of $\|\beta_t - \beta^*\|^2$ and $\hat{L}(\beta_t)$ is uniform in $t \ge 0$; as $t \to \infty$, $\mathcal{E}(t) \to \sigma^2\gamma/(1-\gamma)$ and $\mathcal{L}(t) \to \frac{\sigma^2}{2}(1-\gamma)$ if $\gamma < 1$, and $\mathcal{E}(t) \to r^2(1 - 1/\gamma) + \sigma^2/(\gamma - 1)$, $\mathcal{L}(t) \to 0$ if $\gamma > 1$. If $\gamma = 1$ and $\sigma > 0$, $\mathcal{E}(t) \to \infty$.

For a deterministic β^* with $\|\beta^*\| = r$ the same conclusions hold in probability.

With $\alpha = 1/\gamma$ the limits in (iii) are the ridgeless risks of Theorem 9.2: the flow converges to the minimum-norm least-squares solution, and away from the interpolation threshold the limits $d \to \infty$ and $t \to \infty$ can be taken in either order.[6]

[6] For the excess risk averaged over the teacher and the noise (the Bayes risk), at each fixed t, the limit $\mathcal{E}(t)$ is Theorem 5 of Ali et al. (2019), who allow i.i.d. entries with mean 0 and variance 1, not necessarily Gaussian, and a general covariance.

PROOF. The plan is to write the flow in closed form, to express each quantity through normalized traces, handled by the Marchenko–Pastur law, and Gaussian quadratic and linear forms, handled by Lemma 14.1, and to obtain uniformity from Lipschitz bounds in time and the case $t = \infty$ from the spectral gap. The cases $r = 0$ or $\sigma = 0$ are included.

Closed form. Let $b = X^\top y/n$ and $g_t(\lambda) = (1 - e^{-t\lambda})/\lambda$, with $g_t(0) = t$. Since $\nabla\hat{L}(\beta) = S\beta - b$ and $\frac{\mathrm{d}}{\mathrm{d}t} g_t(\lambda) = e^{-t\lambda} = 1 - \lambda g_t(\lambda)$, $\beta_t = g_t(S)b$, and a field h added to the drift changes β_t by $\int_0^t e^{-(t-s)S} h(s)\,\mathrm{d}s$. With $\underline{S} = XX^\top/n$, $Xf(S)X^\top/n = f(\underline{S})\underline{S}$ for every f, as for polynomials. Since $b = S\beta^* + \sigma X^\top\varepsilon/n$ and $g_t(S)S = I - e^{-tS}$,

$$\beta_t - \beta^* = -e^{-tS}\beta^* + \frac{\sigma}{n} g_t(S) X^\top\varepsilon, \qquad y - X\beta_t = Xe^{-tS}\beta^* + \sigma e^{-t\underline{S}}\varepsilon.$$

Hence, with $u_{ts} = X(g_s(S)e^{-tS} + g_t(S)e^{-sS})\beta^*$, $B_{ts} = Xg_t(S)g_s(S)X^\top$ and $v_t = e^{-t\underline{S}} X e^{-tS}\beta^*$,

$$\langle \beta_t - \beta^*, \beta_s - \beta^* \rangle = \beta^{*\top} e^{-(t+s)S}\beta^* - \frac{\sigma}{n} u_{ts}^\top\varepsilon + \frac{\sigma^2}{n}\,\varepsilon^\top \frac{B_{ts}}{n}\varepsilon,$$
$$\hat{L}(\beta_t) = \frac{1}{2}\beta^{*\top} S e^{-2tS}\beta^* + \frac{\sigma}{n} v_t^\top\varepsilon + \frac{\sigma^2}{2n}\varepsilon^\top e^{-2t\underline{S}}\varepsilon.$$

Spectrum. By Theorem A.16, almost surely μ_S, the spectral distribution of S, converges weakly to μ_γ and $\lambda_{\max}(S) \to (1 + \sqrt{\gamma})^2$; so for large d the spectrum lies in $[0, K_0]$, $K_0 = (1+\sqrt{\gamma})^2+1$, and $\frac{1}{d}\operatorname{tr} f(S) \to \int f\,\mathrm{d}\mu_\gamma$ for every continuous f on $[0, K_0]$. The spectral distribution of $\underline{S}$ is $\frac{d}{n}\mu_S + (1 - \frac{d}{n})\delta_0 \to \gamma\mu_\gamma + (1-\gamma)\delta_0$. And $\|b\| \le r\|S\| + \sigma\|S\|^{1/2}\|\varepsilon\|/\sqrt{n}$ is bounded for large d.

Fixed times. Write $\beta^* = r\,g/\|g\|$ with $g \sim \mathcal{N}(0, I_d)$ independent of (X, ε). Since $1 - e^{-x} \le \min(1, x)$, we have $\lambda g_t(\lambda)^2 \le t$ and $\lambda g_t(\lambda) g_s(\lambda) \le \min(t, s)$. Hence $\|B_{ts}\|/n \le \min(t, s)$, $\|u_{ts}\|^2 \le 2(t+s)nr^2$, and $\|v_t\|^2 = n\,\beta^{*\top} S e^{-4tS}\beta^*$ is at most $n\|S\|r^2$, while $e^{-(t+s)S}$, Se^{-2tS} and $e^{-2t\underline{S}}$ have norms at most 1, $\|S\|$ and 1. Lemma 14.1, applied to g with matrices depending on X and to ε with matrices and vectors depending on (X, g), replaces each quadratic form by its trace and each linear form by 0, up to $o(m)$. The traces converge, since $\frac{1}{n}\operatorname{tr}\frac{B_{ts}}{n} = \frac{d}{n}\cdot\frac{1}{d}\operatorname{tr}(Sg_t(S)g_s(S))$:

$$\tfrac{1}{d}\operatorname{tr} e^{-(t+s)S} \to \int e^{-(t+s)\lambda}\,\mathrm{d}\mu_\gamma, \qquad \tfrac{1}{n}\operatorname{tr}\tfrac{B_{ts}}{n} \to \gamma\int \lambda g_t g_s\,\mathrm{d}\mu_\gamma,$$
$$\tfrac{1}{d}\operatorname{tr}(Se^{-2tS}) \to \int \lambda e^{-2t\lambda}\,\mathrm{d}\mu_\gamma, \qquad \tfrac{1}{n}\operatorname{tr} e^{-2t\underline{S}} \to 1 - \gamma + \gamma\int e^{-2t\lambda}\,\mathrm{d}\mu_\gamma,$$

and $\|g\|^2/d \to 1$. So the limits hold almost surely at every (t, s) of a countable set.

Uniformity. Since $\dot\beta_t = e^{-tS}b$, $\|\dot\beta_t\| \le \|b\|$ and $\|\beta_t - \beta^*\| \le r + t_{\max}\|b\|$, so the correlation is Lipschitz in each variable on $[0, t_{\max}]^2$ with constant $\|b\|(r + t_{\max}\|b\|)$, and $|\frac{\mathrm{d}}{\mathrm{d}t}\hat{L}(\beta_t)| = \|\dot\beta_t\|^2 \le \|b\|^2$; the limits are Lipschitz too, their integrands having bounded t-derivatives on $[0, K_0]$. A grid of mesh η gives $\limsup_d \sup|\cdot| \le c\eta$, proving (ii). The response $\frac{1}{d}\operatorname{tr} e^{-\tau S} = \int e^{-\tau\lambda}\,\mathrm{d}\mu_S$ converges for every τ, uniformly on compact sets by the same argument. For $\tau \ge \tau_0$ and $\delta > 0$ it is within $\mu_S((0,\delta]) + e^{-\tau_0\delta}$ of $\mu_S(\{0\}) = (1 - n/d)_+$, which tends to $\mu_\gamma(\{0\})$, since X has full rank; so the portmanteau theorem

for $[0,\delta]$ gives $\limsup_d \mu_S((0,\delta]) \le \mu_\gamma((0,\delta])$, which is small for small δ. With the same bound for R, the convergence is uniform in $\tau \ge 0$. By Fubini's theorem the Laplace transform of R is $m_\gamma(-p)$, which solves (14.5) by Computation 14.2. This proves (i).

Long times. Let $\gamma \neq 1$ and $a = (1-\sqrt{\gamma})^2 > 0$. The smallest positive eigenvalue λ_+ of S tends to a: for $\gamma < 1$ it is $\lambda_{\min}(S)$, and for $\gamma > 1$ it is $\frac{d}{n}\lambda_{\min}(XX^\top/d) \to \gamma(1-\gamma^{-1/2})^2 = a$ by Theorem A.16 applied to $X^\top$. Since b lies in the range of S, $\beta_\infty = S^+ b$ minimizes $\hat{L}$, and on that range $g_t(\lambda) \le 1/\lambda$, so $\|\beta_t - \beta_\infty\| = \|e^{-tS}S^+ b\| \le e^{-ta/2}\,2\|b\|/a$ for large d. With $\|\beta_t - \beta^*\| \le r + 2\|b\|/a$ and $\hat{L}(\beta_t) - \hat{L}(\beta_\infty) = \frac{1}{2}(\beta_t-\beta_\infty)^\top S(\beta_t-\beta_\infty)$, the excess risk and the training error at every $t \ge \tau_0$ are within $Ke^{-\tau_0 a/2}$ of their values at β_∞, with K bounded for large d; the limits converge exponentially too, because μ_γ has no mass in $(0,a)$. It remains to take $d\to\infty$ at $t=\infty$. With P_0 the projection onto $\ker S$, $\beta_\infty - \beta^* = -P_0\beta^* + \frac{\sigma}{n}S^+X^\top\varepsilon$, and $P_0 S^+ = 0$, so

$$\|\beta_\infty - \beta^*\|^2 = \beta^{*\top}P_0\beta^* + \frac{\sigma^2}{n}\,\varepsilon^\top \frac{XS^{+2}X^\top}{n}\varepsilon.$$

The first term tends to $r^2(1-1/\gamma)_+$, as $\operatorname{tr} P_0 = (d-n)_+$. In the second $\|XS^{+2}X^\top/n\| = \|S^+\| \le 2/a$, and $\frac{1}{n}\operatorname{tr}(XS^{+2}X^\top/n) = \frac{d}{n}\cdot\frac{1}{d}\sum_{\lambda_k>0}\lambda_k^{-1} \to \gamma\int_{\lambda>0}\lambda^{-1}\,\mathrm{d}\mu_\gamma$, the positive eigenvalues lying in $[a/2, K_0]$. The Marchenko–Pastur equation at $z=0$ gives $\int \lambda^{-1}\,\mathrm{d}\mu_\gamma = 1/(1-\gamma)$ for $\gamma<1$, and applied to $XX^\top/d$ it gives $\int_{\lambda>0}\lambda^{-1}\,\mathrm{d}\mu_\gamma = 1/(\gamma(\gamma-1))$ for $\gamma>1$; these are the values in (iii). Next, $y - X\beta_\infty = \sigma(I-P_X)\varepsilon$, with P_X the projection onto the column space of X, of rank $\min(n,d)$, so $\hat{L}(\beta_\infty) \to \frac{\sigma^2}{2}(1-\gamma)_+$. If $\gamma=1$, the density of μ_1 is about $1/(\pi\sqrt{\lambda})$ near 0, so $\int\lambda^{-1}\,\mathrm{d}\mu_1 = \infty$ and $\mathcal{E}(t)\to\infty$ by monotone convergence.

A deterministic teacher. Each quantity is a function $F(X,\varepsilon,\beta^*)$ with $F(XO,\varepsilon,O^\top\beta^*) = F(X,\varepsilon,\beta^*)$ for orthogonal O, since this leaves y unchanged and replaces β_t by $O^\top\beta_t$. For Haar-distributed O independent of (X,ε), (XO,ε) has the law of (X,ε), so $F(X,\varepsilon,\beta^*)$ has, for each d, the law of $F(X,\varepsilon,O\beta^*)$, with a uniform teacher; convergence in probability depends only on these laws. ∎

14.4.2 The spherical 2-spin at zero temperature

Theorem 14.3 (gradient flow on the spherical 2-spin). *Let $W \sim \mathrm{GOE}(N)$ and $H_N(x) = -\frac{1}{\sqrt{2}}\langle x, Wx\rangle$ on S_N, so that $\mathbb{E}H_N(x)H_N(y) = N(\langle x\rangle y/N)^2$. Let x_t be the gradient flow $\dot{x}_t = -\nabla_{\mathrm{sp}}H_N(x_t)$ from x_0 uniform on S_N and independent of W, and let $\mathcal{M}(u) = \int e^{u\lambda}\sigma_{\mathrm{sc}}(\mathrm{d}\lambda) = I_1(2u)/u$, with σ_{sc} the semicircle law and I_1 a modified Bessel function.*

(i) For all $t\ge 0$, $x_t = \sqrt{N}\,e^{\sqrt{2}tW}x_0/\|e^{\sqrt{2}tW}x_0\|$.

(ii) Almost surely, uniformly on $[0,t_{\max}]^2$ for every $t_{\max}$, $C_N(t,s)\to C(t,s)$ and $e_N(t)\to e(t)$, where

$$C(t,s) = \frac{\mathcal{M}(\sqrt{2}(t+s))}{\sqrt{\mathcal{M}(2\sqrt{2}t)\,\mathcal{M}(2\sqrt{2}s)}}, \qquad e(t) = -\frac{\mathcal{M}'(2\sqrt{2}t)}{\sqrt{2}\,\mathcal{M}(2\sqrt{2}t)}.$$

(iii) The energy decreases strictly to the ground-state energy $-\sqrt{2}$, with $t\,(e(t)+\sqrt{2}) \to \tfrac{3}{8}$. For fixed $\tau \geq 0$, $C(t_w+\tau, t_w) \to 1$ as $t_w \to \infty$, while for fixed $c > 0$, $C((1+c)t_w, t_w) \to (4(1+c)/(2+c)^2)^{3/4}$.

Part (iii) is aging at zero temperature: after a long wait the correlation does not decay on times of order one, but it decays on times proportional to the age, along a universal curve. Cugliandolo and Dean (1995) found this curve, their equation (III.25) at $T = 0$, in their physical solution of the 2-spin dynamics, and Ben Arous et al. (2001) proved aging for the soft-constraint dynamics at positive temperature.

Proof. The plan is to verify the explicit solution, to reduce C_N and e_N to Gaussian quadratic forms, and to evaluate the limits by Laplace's method at the edge of the semicircle. The covariance follows from $\mathbb{E}\,\mathrm{tr}(AW)\,\mathrm{tr}(BW) = 2\,\mathrm{tr}(AB)/N$ for symmetric A, B (Definition A.12).

(i) Since $\nabla H_N(x) = -\sqrt{2}Wx$, the flow is $\dot{x} = \sqrt{2}(Wx - \frac{1}{N}\langle x, Wx\rangle x)$, a smooth field tangent to S_N with a unique solution. For $v_t = e^{\sqrt{2}tW}x_0$, differentiating $\sqrt{N}\,v_t/\|v_t\|$ gives $\sqrt{N}\big(\sqrt{2}Wv_t/\|v_t\| - v_t\langle v_t, \sqrt{2}Wv_t\rangle/\|v_t\|^3\big)$, which is the field at x_t.

(ii) Write $x_0 = \sqrt{N}\,g/\|g\|$ with $g \sim \mathcal{N}(0, I_N)$ independent of W. Then

$$C_N(t,s) = \frac{g^\top e^{\sqrt{2}(t+s)W} g}{\sqrt{g^\top e^{2\sqrt{2}tW} g \cdot g^\top e^{2\sqrt{2}sW} g}}, \qquad e_N(t) = -\frac{g^\top W e^{2\sqrt{2}tW} g}{\sqrt{2}\, g^\top e^{2\sqrt{2}tW} g}.$$

By Theorem A.15, almost surely $\|W\| \to 2$ and the spectral distribution of W converges weakly to σ_{sc}; so for large N, $\frac{1}{N}\,\mathrm{tr}\, f(W) \to \int f\,\mathrm{d}\sigma_{\mathrm{sc}}$ and $\|f(W)\| \leq \max_{[-3,3]}|f|$ for continuous f on $[-3,3]$, and Lemma 14.1 gives $\frac{1}{N} g^\top f(W) g - \frac{1}{N}\,\mathrm{tr}\, f(W) \to 0$. Taking $f(\lambda) = e^{u\lambda}$ and $\lambda e^{u\lambda}$ at countably many u, with denominators at least $e^{-3|u|}\|g\|^2/N$, gives convergence on a dense set of times; numerators and denominators are Lipschitz in $t, s \leq t_{\max}$ with constants bounded when $\|W\| \leq 3$, so the convergence is uniform.

(iii) The derivative of $\mathcal{M}'/\mathcal{M}$ is the variance of λ under the law proportional to $e^{u\lambda}\sigma_{\mathrm{sc}}(\mathrm{d}\lambda)$, so e decreases strictly, and the mean is below 2, so $e > -\sqrt{2}$. Substituting $\lambda = 2 - y/u$,

$$\mathcal{M}(u) = \frac{e^{2u}}{2\pi u^{3/2}} J_0(u), \qquad J_k(u) = \int_0^{4u} y^k e^{-y}\sqrt{y(4-y/u)}\,\mathrm{d}y \to 2\Gamma\big(k+\tfrac{3}{2}\big)$$

by dominated convergence. So $\mathcal{M}(u) \sim e^{2u}/(2\sqrt{\pi}\,u^{3/2})$, and the tilted mean of λ is $2 - J_1(u)/(uJ_0(u))$ with $J_1/J_0 \to \frac{3}{2}$; with $u = 2\sqrt{2}t$, $t(e(t)+\sqrt{2}) = J_1(u)/(4J_0(u)) \to \frac{3}{8}$. In $C(t,s)$ the exponentials cancel, as $2\sqrt{2}(t+s) = \frac{1}{2}(4\sqrt{2}t + 4\sqrt{2}s)$, and as $t, s \to \infty$ with t/s fixed

$$C(t,s) = \Big(\frac{(2\sqrt{2}t)(2\sqrt{2}s)}{(\sqrt{2}(t+s))^2}\Big)^{3/4}(1+o(1)) = \Big(\frac{4ts}{(t+s)^2}\Big)^{3/4}(1+o(1)),$$

which gives the aging limit for $t = (1+c)t_w$, $s = t_w$, and 1 for $t = t_w + \tau$. ∎

At $T = 0$ these limits, with $\mu = -2e$ and an explicit response, solve (14.2) for $\xi(q) = q^2$: the response equation becomes a convolution identity for $\mathcal{M}$ (Exercise 14.5), the counterpart of the Marchenko–Pastur equation of Computation 14.2, and our integrator reproduces $e(t)$ to 10^{-4}.

14.5 The proof: the general theorems

Theorem 14.4 (dynamical mean-field theory on fixed time horizons).

(i) (Spherical spin glasses; Ben Arous et al. (2006, Theorem 1.2), Sellke (2024, Theorem 4.3).) For a mixed model, $\xi(q) = \sum_{p\le m} c_p^2 q^p$, run the Langevin dynamics at $T > 0$ with a soft spherical constraint, a confining potential $N f(\|x\|^2/N)$ with f' locally Lipschitz, of polynomial growth and growing fast enough at infinity, from x_0 independent of the disorder and satisfying a tensorizing concentration hypothesis. For every horizon $t_{\max}$, $C_N(t,s)$ and the integrated response $\chi_N(t,s) = \frac{1}{N}\langle x_t, B_s\rangle$ converge uniformly on $[0,t_{\max}]^2$, almost surely, to the unique bounded continuous solution of the soft-constraint analogue of (14.2). For the pure p-spin with the hard constraint (14.1) and $T > 0$, they converge to the solution of (14.2).

(ii) (Gradient descent and SGD; Gerbelot et al. (2024, Theorem 3.2).) For first-order iterations driven by a matrix with i.i.d. $\mathcal{N}(0,1/d)$ entries, $n/d \to \alpha$, with pseudo-Lipschitz updates, a deterministic start, and conditions ensuring that certain limits exist, including gradient descent, SGD with batches of size proportional to n, momentum and Langevin noise: for every fixed number of iterations, pseudo-Lipschitz functions of the iterates converge in probability to their expectations under the time discretization of (14.4).

(iii) (Gradient flow; Celentano et al. (2021, Theorem 2, Corollary 4.1), unpublished.) Let $A_{ij} = \bar{X}_{ij}/\sqrt{d}$ with $\bar{X}_{ij}$ i.i.d., centered, of variance 1 and sub-Gaussian, let $\ell(r;z)$ be Lipschitz in r with Lipschitz Jacobian, uniformly in z, with $\mathbb{E}\,\ell(0;z)^2 < \infty$, and let (w_0, z) be independent of A with converging empirical laws and second moments. Then (14.4) has a solution, unique among those with locally bounded kernels, and the empirical laws of $(w_{t,i})_{t\le t_{\max}}$, $i \le d$, and of $(z_i, (r_{t,i})_{t\le t_{\max}})$, $i \le n$, converge in probability, weakly on path space, to the laws of the processes of (14.4).

Proof sketch. (i) After moment bounds, tightness and concentration of C_N, χ_N and auxiliary functions, Proposition 1.3 of Ben Arous et al. (2006) shows, by Itô's formula and Gaussian change of measure, that every limit point solves closed equations, because $\mathbb{E}[\nabla H(x)\nabla H(y)^\top]$ is a polynomial in overlaps; Grönwall's inequality gives uniqueness. Sellke couples the hard sphere with steeper and steeper soft constraints, whose limit equations Dembo et al. (2007) identified. (ii) Induction on the iterations by Gaussian conditioning (Lemma A.13, part (iv)): the known part of the matrix produces the memory kernels, the fresh part the Gaussian processes. (iii) Time is

discretized, gradient descent becomes message passing with memory (Celentano et al. 2020), state evolution applies, and a contraction controls the vanishing step. ■

In the linear flows of Section 14.4 independence handles the dependence between path and disorder, and the spectral law replaces the saddle point; with a nonlinear drift, Girsanov's theorem or the conditioning lemma must untangle it, and the closed equations must be derived. None of the three parts goes beyond fixed horizons or covers gradient flow on a spin glass with $p \geq 3$, since the spin-glass proofs need $T > 0$.[7]

Long times need different tools. For the pure p-spin the energy is settled at low temperature; Sellke maximizes H_N with drift $\beta \nabla_{\mathrm{sp}} H_N$ and unit noise: our dynamics (14.1) for $-H_N$ at $T = 1/(2\beta)$, our time being β times his.

Theorem 14.5 (the threshold energy; Sellke 2024, Theorems 1.1 and 1.7). *Let $p \geq 2$, let H_N be the pure p-spin Hamiltonian, and let x_s solve $\mathrm{d}x_s = \big(\beta \nabla_{\mathrm{sp}} H_N(x_s) - \frac{N-1}{2N} x_s\big)\,\mathrm{d}s + P^{\perp}_{x_s}\,\mathrm{d}B_s$ on S_N, with invariant law proportional to $e^{2\beta H_N}$. Let $c > 0$ denote a constant that may depend on all parameters but N.*

(i) For every $\eta > 0$ there are $s_0(p, \eta)$ and $\beta_0(p, \eta)$ such that for $\beta \geq \beta_0$, N large and every $x_0 \in S_N$, possibly depending on H_N, with probability at least $1 - e^{-cN}$, $H_N(x_s)/N \geq E_\infty - \eta$ for all $s \in [s_0, s_0 + e^{cN}]$.

(ii) For every $\beta_0 < \infty$ there is $\delta(p, \beta_0) > 0$ such that for $\beta \leq \beta_0$, every $s_{\max}$, N large and x_0 independent of H_N, with probability at least $1 - e^{-cN}$, $H_N(x_s)/N \leq E_\infty - \delta$ for all $s \leq s_{\max}$.

Proof sketch. (i) Near an approximate saddle point the low-temperature dynamics is close to an Ornstein–Uhlenbeck process that gains energy along directions of positive curvature, so it climbs to the lowest energy of an approximate local maximum, at least $E_\infty - o(1)$ because the bulk of the Hessian spectrum is a semicircle shifted by a multiple of the energy. (ii) On bounded times the dynamics is close to a Lipschitz function of the disorder, and no such algorithm exceeds $E_\infty + o(1)$ (Huang and Sellke 2025); a lower bound on the gradient at finite β improves this to $E_\infty - \delta$. ■

So for the pure p-spin, low-temperature Langevin dynamics reaches $-E_\infty$ in the limit $T \to 0$ after $N \to \infty$, after a time of order $1/T$ in our units. Since $E_\infty < E_0$ for $p \geq 3$, where $-E_0$ is the zero of the annealed complexity (Auffinger et al. 2013) and the typical ground state (Subag 2017), dynamics and statics differ. For convex learning problems they agree, under a hypothesis.[8]

Proposition 14.6 (dynamics and statics for convex losses; Celentano et al. 2021, Theorems 3–4, Proposition 4.3, unpublished). *In Theorem 14.4(iii) for a planted model, let $\ell = \partial_r \mathrm{loss}$ be time-independent and add a ridge $\lambda \geq 0$. If the solution of (14.4) converges exponentially fast (their Definition 4.2), its limit solves explicit fixed-point equations; if the loss is twice differentiable and convex and the teacher and the noise are nondegen-*

[7] Dembo and Subag (2020) extend part (i) to starts uniform in the spherical bands where the low-temperature Gibbs measure concentrates, for pure models and small perturbations, by conditioning on a critical point.

[8] In inference, Liang et al. (2023) prove the closed equations for Langevin dynamics in a spiked matrix model and, for semicircular noise, its overlap as $t \to \infty$. The unrefereed preprint of Fan et al. (2025) (arXiv:2504.15558v2) shows, for an adaptive Langevin sampler in Bayesian linear regression under approximate time-translation invariance, convergence to an equilibrium given by scalar fixed-point equations.

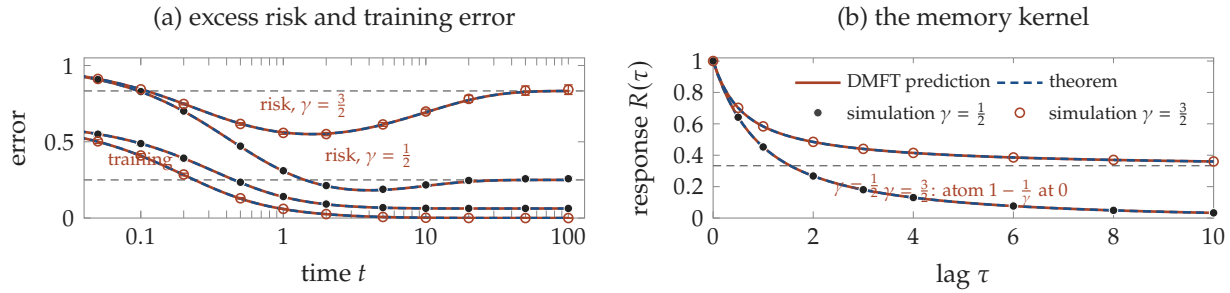


Figure 14.1. Gradient flow on least squares, $r = 1$, $\sigma = \frac{1}{2}$. (a) Excess risk and training error: the prediction (14.6) (carnelian) and Theorem 14.2 (lapis dashed), which coincide, and simulations with $n = 2000$, $d = 1000$ ($\gamma = \frac{1}{2}$, filled) and $n = 1000$, $d = 1500$ ($\gamma = \frac{3}{2}$, hollow): means over 10 independent data sets with Student-t 95% intervals (seeds $1400 + 100\lfloor 10\gamma \rfloor + k$). All 44 intervals contain the curves; dashed lines mark the limits 0.25 and 0.8333 at $t = \infty$. (b) The response: the kernel equations of Computation 14.2 integrated in time (carnelian), the Laplace transform of μ_γ (lapis dashed), and $\frac{1}{d}\operatorname{tr} e^{-\tau S}$ averaged over the same data sets (points; intervals below 10^{-3}). Script: `14-dmft-leastsquares.py`.

> *erate, these are equivalent to the equations characterizing the empirical risk minimizer, whose solution, if one exists, is unique when $\lambda > 0$ or the loss is strictly convex. Exponential convergence holds for strongly convex losses with $\alpha > 1$, and for convex losses with $\lambda > 0$.*

14.6 Measuring it

Four scripts, each under two minutes, measure the chapter's claims; `14-dmft-ck.py` integrates (14.2) by Heun's method with trapezoidal quadrature, a second-order scheme, and Richardson extrapolation from steps 0.05 and 0.025.[9]

Least squares. The curves of Figure 14.1(a) show early stopping: the risk is smallest at $t = 4.00$ (0.1822, against 0.25 at convergence) for $\gamma = \frac{1}{2}$, and at $t = 1.58$ (0.5502, against 0.8333) for $\gamma = \frac{3}{2}$ (Exercise 14.3). In (b), for $\gamma = \frac{3}{2}$, the response tends to the mass $\frac{1}{3}$ of the null space of S, which the flow never leaves.[10]

The 2-spin. In Figure 14.2(b) the scaled curves approach the limit of Theorem 14.3(iii) slowly, $C(2t_w, t_w)$ being 0.915968 at $t_w = 10$ and 0.915503 at $t_w = 100$, against 0.915452.[11] The intervals widen at large times (±0.13 at $t_w = 8$, $\tau = 80$): the top eigenvalue gap of W is about 0.019 at $N = 1500$, and once $2\sqrt{2}t$ exceeds its inverse the finite system begins to align with the top eigenvector.

The Cugliandolo–Kurchan equations. The integrator reproduces Theorem 14.3 for $p = 2$, the equilibrium energy at $T = 1.2$, and an independent integrator. At $T = 0.7$ the energy is -1.2228 at $t = 50$, descending toward the predicted aging value -1.2582; at $T = 0$ it is -1.5821 and -1.6054 at $t = 20$ and 50, approaching -1.6330 from above. In (b), $C(t_w + 20, t_w)$ at

[9] For $p = 2$, $T = 0$ the error in $e(t)$ on $[0, 10]$ is 4.7×10^{-3} and 1.1×10^{-3} at the two steps, and 9.1×10^{-5} after extrapolation. For $p = 3$, $T = 1.2$, $e(50) = -0.83328$, against $-1/T = -0.83333$. An Euler integrator in the normalization of Cugliandolo and Kurchan (steps 0.02 and 0.01, extrapolated, converted) gives $e(5) = -1.5012$ and $e(10) = -1.5518$ at $T = 0$, against our -1.5007 and -1.5513.

[10] At step 0.01 the kernel equations agree with the Laplace transform of μ_γ to 5.8×10^{-6} ($\gamma = \frac{1}{2}$) and 9.8×10^{-6} ($\gamma = \frac{3}{2}$), four times better than at step 0.02; the measured $\frac{1}{d}\operatorname{tr} e^{-2S}$ is 0.26774 and 0.48504, against 0.26759 and 0.48483.

[11] For the energy, $e(10) + \sqrt{2} = 0.037331$, against $3/(8t) = 0.0375$, and the simulated $e(10)$ is -1.3751 ± 0.0081, against -1.3769.

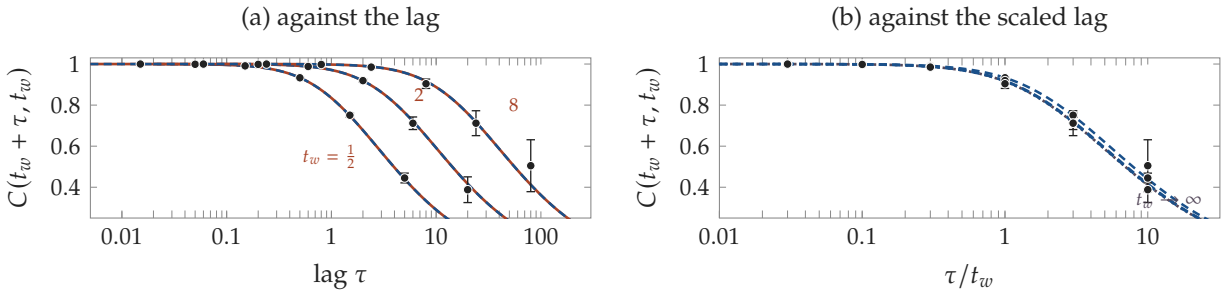


Figure 14.2. Aging of zero-temperature gradient flow on the spherical 2-spin. (a) $C(t_w + \tau, t_w)$ for $t_w = \frac{1}{2}$, 2, 8: Theorem 14.3(ii) (carnelian with lapis dashes, which coincide) and simulations at $N = 1500$ (means over 10 independent GOE matrices and starts, seeds $1420 + k$, with Student-t 95% intervals); the decay time grows with the age. (b) The same against τ/t_w, with the aging limit $(4(1 + c)/(2 + c)^2)^{3/4}$ of part (iii) dotted: the curves collapse. All 18 intervals contain the limits. Script: `14-dmft-twospin.py`.

$T = 0.7$ is 0.293, 0.367, 0.448, 0.525 for $t_w = 2.5$, 5, 10, 20, while at $T = 1.2$ it settles at 0.0107, 0.0122, 0.0131, 0.0133. That aging persists as $t_w \to \infty$ is the conjecture; these curves are evidence, not proof.

The simulations agree with the equations where these are a theorem: at $T = 1.2$ and 0.7, 15 of the 18 intervals at $N = 100$ contain the curves, and the others miss by less than 0.01. At $T = 0$ the finite systems lag from $t \approx 7$: at $t = 20$ the energy is -1.519 ± 0.028 at $N = 100$ and -1.495 ± 0.032 at $N = 50$, against -1.582, a gap that shrinks by a factor 1.4 from $N = 50$ to 100 (two sizes do not determine a rate). At fixed N gradient flow converges to a critical point, and at these sizes the energy has nearly stopped decreasing by $t = 20$; the equations describe $N \to \infty$ first.[12]

[12] Renormalizing a noisy step to the sphere lowers the effective temperature by a factor of about $1 - T\,\mathrm{d}t$; the script corrects for it.

14.7 The status board

On fixed time horizons the theory is a theorem for spherical spin glasses at positive temperature and for gradient methods with smooth updates; for gradient flow in learning it is, as of October 2026, proved only in an unrefereed preprint. Linear flows are settled at all times. Beyond times of order one, the threshold is a theorem for the pure p-spin at low temperature, dynamics agrees with statics for convex losses under a hypothesis (in the same preprint), and aging for $p \geq 3$ is a conjecture.

STATUS BOARD · the generating functional ↔ Gaussian conditioning

PREDICTION	STATUS	PROOF OR EVIDENCE
Equations (14.2), fixed horizons, start independent of the disorder	proved under	$T > 0$: pure (Sellke 2024, Thm. 4.3); mixed, soft constraint (Ben Arous et al. 2006, Thm. 1.2); open at $T = 0$, $p \geq 3$

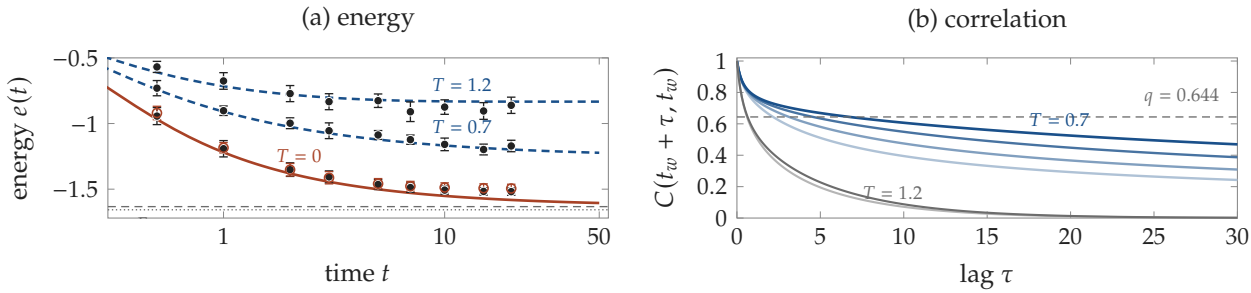


Figure 14.3. The spherical 3-spin, $T_d = 0.866$. (a) Energy at $T = 1.2$, 0.7 and 0: the equations (14.2) with Richardson extrapolation (lapis dashed for $T > 0$, where Theorem 14.4(i) makes them the limit; carnelian for $T = 0$, a prediction), and simulations of (14.1): means over 12 disorder samples at $N = 100$ (seeds $1430 + k$, filled) and 24 at $N = 50$ (seeds $2440 + k$, hollow, $T = 0$ only), with Student-t 95% intervals. Dashed: the threshold $-E_\infty = -1.633$; dotted: the ground state $-E_0 = -1.657$. (b) $C(t_w + \tau, t_w)$ from the equations for $t_w = 2.5$, 5, 10, 20, darker for larger t_w: aging at $T = 0.7$, with the predicted plateau $q = 0.644$ dashed, and time-translation invariance at $T = 1.2$ (curves for $t_w = 2.5$ and 20). Scripts: `14-dmft-ck.py` and `14-dmft-pspin.py`.

Linear flows: least squares, all times; 2-spin at $T = 0$, aging	proved	Theorems 14.2 and 14.3; Bayes risk, fixed t: Ali et al. (2019, Thm. 5); $T > 0$: Ben Arous et al. (2001)
Equilibrium (TTI, FDT) solution above T_d	proved under	small β only (Dembo et al. 2007, Thm. 1.3)
Dynamical transition at T_d, aging below it, $p \geq 3$	conjectured	numerics (Kim and Latz 2001), Figure 14.3; mixing dichotomy far from T_d (Gheissari and Jagannath 2019)
Low-temperature dynamics reaches $-E_\infty$ (pure p-spin)	proved	Sellke (2024, Thms. 1.1, 1.7), $T \to 0$ after $N \to \infty$
The same threshold picture for mixed models	open	bounds do not match; numerics depend on preparation (Folena et al. 2020)
DMFT for gradient descent and SGD, finitely many steps	proved under	Gaussian design, pseudo-Lipschitz updates (Gerbelot et al. 2024, Thm. 3.2)
DMFT for gradient flow, fixed horizons; its limit = empirical risk minimizer, convex losses	preprint	smooth losses, sub-Gaussian design; limit given exponential convergence: Celentano et al. (2021, Thms. 2–4), arXiv:2112.07572v3

Long-time dynamics = statics in general	false	pure p-spin: $-E_\infty > -E_0$ (Auffinger et al. 2013; Sellke 2024; Subag 2017)
DMFT for persistent SGD, non-smooth losses, phase retrieval	heuristic	Mignacco et al. (2021a,b); outside the theorems' hypotheses

14.8 Where the translation fails

WHAT WAS ASSUMED. Every theorem takes $N \to \infty$ at a fixed horizon, the order in which physicists derive the equations but not the one in which they use them. The spin-glass theorems need $T > 0$ and, except for Dembo and Subag (2020) and Sellke's lower bound, a start independent of the disorder; the hard constraint is covered only for pure models. The learning theorems need smooth losses and Gaussian or sub-Gaussian designs in the proportional regime. Our proofs add linearity: Theorem 14.2 uses the square loss and the rotation invariance of the Gaussian design, Theorem 14.3 uses $p = 2$ and $T = 0$.

WHERE THE PHYSICS IS WRONG, OR NOT YET RIGHT. Long-time dynamics is not statics: the pure p-spin stops at $-E_\infty$, above the ground state, and in the convex case agreement needs exponential convergence. For mixed spherical models the threshold picture fails: Folena et al. (2020) find, integrating the equations, that the asymptotic energy depends on how the system was prepared, and the rigorous bounds do not match (Sellke 2024). On times that grow with N the equations lose their meaning, as activated jumps and, at modest sizes, convergence to a critical point take over (Figures 14.2 and 14.3). The predictions most used in machine learning, for persistent mini-batch SGD, phase retrieval with the square loss, hinge losses and ReLU networks, rest on the same derivation outside every theorem's hypotheses (Mignacco et al. 2021a,b).

OPEN PROBLEMS. First, the equations for zero-temperature gradient flow with $p \geq 3$, which the limit $T \to 0$ of Sellke's results does not give (Sellke 2024, Remark 1.5). Second, the transition at T_d and the aging solution for $p \geq 3$, announced as future work by Ben Arous et al. (2006); only the solution at small β is proved (Dembo et al. 2007). Third, when the dynamical mean-field solution of a learning problem converges exponentially, which Celentano et al. (2021) call open, and its long-time limit for nonconvex losses. Fourth, horizons that grow with the dimension, such as the $\log d$ needed to escape a saddle. Fifth, the asymptotic energy of mixed models.

Exercises

14.1 CONSOLIDATION. (The explicit flow.) Verify Theorem 14.3(i) and the covariance of H_N, and show that $e(0) = 0$ and $e'(0) = -2 = -\xi'(1)$, the initial slope predicted by (14.2).

14.2 CONSOLIDATION. (The ridgeless limits.) Derive $\int \lambda^{-1}\mu_\gamma(\mathrm{d}\lambda) = 1/(1-\gamma)$ for $\gamma < 1$ from (A.9) at $z = 0$, and $\int_{\lambda>0} \lambda^{-1}\mu_\gamma(\mathrm{d}\lambda) = 1/(\gamma(\gamma-1))$ for $\gamma > 1$, and check that the limits in Theorem 14.2(iii) are those of Theorem 9.2 with $\alpha = 1/\gamma$.

14.3 CONSOLIDATION. (Early stopping.) Show that $\mathcal{E}'(0) = -2r^2$ and that for $\gamma \neq 1$ and $\sigma > 0$, $\mathcal{E}(t) < \mathcal{E}(\infty)$ for large t. For $\gamma = \frac{1}{2}$, $r = 1$, $\sigma = \frac{1}{2}$, find the best time numerically ($t = 4.00$).

14.4 CONSOLIDATION. (The dictionary.) For $p = 4$ compute T_d, q_d and E_∞ in our normalization and in that of Cugliandolo and Kurchan (answers: 0.769800 and 0.544331; $\frac{2}{3}$; 1.732051 and 1.224745). Show that for $p = 2$ the maximum in step 4 of Computation 14.1 is approached as $q \to 0$, giving $T_d = \sqrt{2}$, the static critical temperature of the spherical 2-spin in our normalization.

14.5 BY HAND. (The semicircle solves the response equation.) Show that $\mathcal{M}(u) = \sum_k C_k u^{2k}/(2k)!$ with the Catalan numbers C_k, and use $C_{k+1} = \sum_{i+j=k} C_i C_j$ to prove $\mathcal{M}'(u) = \int_0^u \mathcal{M}(v)\mathcal{M}(u-v)\,\mathrm{d}v$. Deduce that $R(t,s) = \mathcal{M}(\sqrt{2}(t-s))\sqrt{\mathcal{M}(2\sqrt{2}s)/\mathcal{M}(2\sqrt{2}t)}$ solves the response equation of (14.2) for $\xi(q) = q^2$, $T = 0$, $\mu = -2e$.

14.6 BY HAND. (The interpolation threshold.) For $\gamma = 1$ show that $R(\tau) \sim (\pi\tau)^{-1/2}$ and $\mathcal{E}(t) \sim \frac{2(2-\sqrt{2})}{\sqrt{\pi}}\sigma^2\sqrt{t}$. *(Hint: use* $\int_0^\infty (1-e^{-u})^2 u^{-3/2}\,\mathrm{d}u = 2\sqrt{\pi}(2-\sqrt{2})$.*)*

14.7 BY HAND. ⋆ (Edges and aging.) Let the spectral law of a symmetric J tend to ρ, with largest point λ_+ and $\rho([\lambda_+ - \varepsilon, \lambda_+]) \sim \kappa_0 \varepsilon^{a+1}$, $a > -1$, and run the spherical gradient flow of $-\frac{1}{2}\langle x, Jx\rangle$. Show that $C((1+c)t_w, t_w) \to (4(1+c)/(2+c)^2)^{(a+1)/2}$. What does the Marchenko–Pastur edge give? *(Hint: the Laplace transform of ρ behaves like $e^{u\lambda_+}u^{-(a+1)}$.)*

14.8 TRANSLATION. (The fluctuation–dissipation theorem as a theorem.) For $\gamma < 1$ let $\mathrm{d}w_t = -S(w_t - \hat{w})\,\mathrm{d}t + \sqrt{2T}\,\mathrm{d}B_t$, with $\hat{w}$ the least-squares solution, started from its stationary law. Show that the correlation of $w_t - \hat{w}$, $C(\tau) = \frac{T}{d}\operatorname{tr}(e^{-\tau S}S^{-1})$, and $R(\tau) = \frac{1}{d}\operatorname{tr} e^{-\tau S}$ satisfy $R = -C'/T$ exactly at every d, and find their limits. Which physical step does this replace?

14.9 TRANSLATION. ⋆ (What the saddle point needs.) Let $X = UDV^\top$ with V Haar-distributed and independent of (U, D), and let the spectral law of $X^\top X/n$ tend to ν, with bounded norm. Show that Theorem 14.2(i)–(ii) hold with ν in place of μ_γ, while (14.5) fails in general: the saddle point encodes the spectral law, and the closed equation the i.i.d. design. *(Hint: rotation invariance of V replaces the uniform teacher.)*

14.10 MEASUREMENT. (Universality and size.) In `14-dmft-leastsquares.py` replace the Gaussian design by Rademacher entries and the teacher by re_1: do the simulations still match (14.6), which step of the proof fails,

and how does a random teacher repair it? Then run `14-dmft-pspin.py` at $N = 150$ and check whether the lag at $T = 0$ shrinks further.

14.11 RESEARCH. ★★ (Gradient flow on the 3-spin.) Prove (14.2) for zero-temperature gradient flow on the pure spherical 3-spin on fixed horizons, or identify which step of the argument of Celentano et al. (2021), written for flows driven by a rectangular i.i.d. matrix, fails for a Gaussian 3-tensor with a hard constraint.

Notes and References

In 1993 Crisanti et al. (1993) wrote the equations for the spherical p-spin and Cugliandolo and Kurchan (1993) found the aging solution and the threshold; Cugliandolo and Dean (1995) solved the 2-spin exactly, and Cugliandolo (2024) surveys the field. The rigorous theory began with soft spins at high temperature (Ben Arous and Guionnet 1997) and continued with the spherical models (Ben Arous et al. 2001, 2006; Dembo et al. 2007; Sellke 2024). For learning, Agoritsas et al. (2018) derived the equations for the perceptron, by the dynamical cavity method and by a path integral, Mignacco et al. (2021a,b) applied them to SGD and phase retrieval, and Advani et al. (2020) computed the learning curves of linear models through the Marchenko–Pastur spectrum; Celentano et al. (2020) reduced first-order methods to message passing, the step on which Gerbelot et al. (2024) and Celentano et al. (2021) build.

Lemma 14.1, the proofs of Theorems 14.2 and 14.3, and Computations 14.1 and 14.2 were written for this book. For isotropic Gaussian designs, Theorem 14.2 strengthens Theorem 5 of Ali et al. (2019) from the Bayes risk at each fixed time to the risk itself, almost surely, uniformly in time and at $t = \infty$, and adds the correlation, the training error and the response. Theorem 14.3 is the zero-temperature, hard-constraint case of the dynamics that Cugliandolo and Dean (1995) solved by physical arguments.

TRANSLATING THE LITERATURE. Cugliandolo and Kurchan (1993) write $G(t, t')$ for the response and $z(t)$ for the multiplier, with noise strength $2T$; their normalization is converted in Section 14.1. Ben Arous et al. (2006) use unit noise, the drift $-\frac{\beta}{2}\nabla H_J$ and covariance $4N\nu(q)$, which is our system with $T = \frac{1}{2}$ and $\xi = \beta^2\nu$; they track $\chi(s,t) = \int_0^t R(s,u)\,\mathrm{d}u$ and put the later time first, as does Sellke (2024). Celentano et al. (2021) write δ for our α, X for our $A = X/\sqrt{d}$, θ for our $w = \sqrt{d}\,\beta$, u and w for our ζ and ω, and R_θ, C_θ for our R_w, C_w; Gerbelot et al. (2024) also take $X_{ij} \sim \mathcal{N}(0, 1/d)$, in discrete time.

References

Advani, M. S., A. M. Saxe, and H. Sompolinsky (2020). "High-dimensional dynamics of generalization error in neural networks". In: *Neural Networks* 132, pp. 428–446. arXiv: 1710.03667.

Agoritsas, E., G. Biroli, P. Urbani, and F. Zamponi (2018). "Out-of-equilibrium dynamical mean-field equations for the perceptron model". In: *Journal of Physics A: Mathematical and Theoretical* 51.8, p. 085002. arXiv: 1710.04894.

Ali, A., J. Z. Kolter, and R. J. Tibshirani (2019). "A Continuous-Time View of Early Stopping for Least Squares Regression". In: *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*. Vol. 89. Proceedings of Machine Learning Research, pp. 1370–1378. arXiv: 1810.10082.

Auffinger, A., G. Ben Arous, and J. Černý (2013). "Random Matrices and Complexity of Spin Glasses". In: *Communications on Pure and Applied Mathematics* 66.2, pp. 165–201. arXiv: 1003.1129.

Ben Arous, G., A. Dembo, and A. Guionnet (2001). "Aging of spherical spin glasses". In: *Probability Theory and Related Fields* 120.1, pp. 1–67.

Ben Arous, G., A. Dembo, and A. Guionnet (2006). "Cugliandolo-Kurchan equations for dynamics of spin-glasses". In: *Probability Theory and Related Fields* 136.4, pp. 619–660. arXiv: math/0409273.

Ben Arous, G. and A. Guionnet (1997). "Symmetric Langevin spin glass dynamics". In: *The Annals of Probability* 25.3, pp. 1367–1422.

Celentano, M., C. Cheng, and A. Montanari (2021). *The high-dimensional asymptotics of first order methods with random data*. Version 3, 18 April 2026; unpublished as of October 2026. arXiv: 2112.07572.

Celentano, M., A. Montanari, and Y. Wu (2020). "The estimation error of general first order methods". In: *Proceedings of the 33rd Conference on Learning Theory (COLT)*. Vol. 125. Proceedings of Machine Learning Research, pp. 1078–1141. arXiv: 2002.12903.

Crisanti, A., H. Horner, and H.-J. Sommers (1993). "The spherical p-spin interaction spin-glass model: the dynamics". In: *Zeitschrift für Physik B Condensed Matter* 92.2, pp. 257–271.

Cugliandolo, L. F. and D. S. Dean (1995). "Full dynamical solution for a spherical spin-glass model". In: *Journal of Physics A: Mathematical and General* 28.15, pp. 4213–4234. arXiv: cond-mat/9502075.

Cugliandolo, L. F. and J. Kurchan (1993). "Analytical solution of the off-equilibrium dynamics of a long-range spin-glass model". In: *Physical Review Letters* 71.1, pp. 173–176. arXiv: cond-mat/9303036.

Cugliandolo, L. F. (2024). "Recent Applications of Dynamical Mean-Field Methods". In: *Annual Review of Condensed Matter Physics* 15, pp. 177–213. arXiv: 2305.01229.

De Dominicis, C. (1976). "Techniques de renormalisation de la théorie des champs et dynamique des phénomènes critiques". In: *Le Journal de Physique Colloques* 37.C1, pp. C1-247–C1-253.

De Dominicis, C. (1978). "Dynamics as a substitute for replicas in systems with quenched random impurities". In: *Physical Review B* 18.9, pp. 4913–4919.

Dembo, A., A. Guionnet, and C. Mazza (2007). "Limiting Dynamics for Spherical Models of Spin Glasses at High Temperature". In: *Journal of Statistical Physics* 128.4, pp. 847–881. arXiv: math/0609546.

Dembo, A. and E. Subag (2020). "Dynamics for Spherical Spin Glasses: Disorder Dependent Initial Conditions". In: *Journal of Statistical Physics* 181.2, pp. 465–514. arXiv: 1908.01126.

Fan, Z., J. Ko, B. Loureiro, Y. M. Lu, and Y. Shen (2025). *Dynamical mean-field analysis of adaptive Langevin diffusions: Replica-symmetric fixed point and empirical Bayes*. arXiv: 2504.15558.

Folena, G., S. Franz, and F. Ricci-Tersenghi (2020). "Rethinking Mean-Field Glassy Dynamics and Its Relation with the Energy Landscape: The Surprising Case of the Spherical Mixed p-Spin Model". In: *Physical Review X* 10.3, p. 031045.

Gerbelot, C., E. Troiani, F. Mignacco, F. Krzakala, and L. Zdeborová (2024). "Rigorous Dynamical Mean-Field Theory for Stochastic Gradient Descent Methods". In: *SIAM Journal on Mathematics of Data Science* 6.2, pp. 400–427. arXiv: 2210.06591.

Gheissari, R. and A. Jagannath (2019). "On the spectral gap of spherical spin glass dynamics". In: *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 55.2, pp. 756–776. arXiv: 1608.06609.

Huang, B. and M. Sellke (2025). "Tight Lipschitz hardness for optimizing mean field spin glasses". In: *Communications on Pure and Applied Mathematics* 78.1, pp. 60–119. arXiv: 2110.07847.

Janssen, H.-K. (1976). "On a Lagrangean for classical field dynamics and renormalization group calculations of dynamical critical properties". In: *Zeitschrift für Physik B Condensed Matter and Quanta* 23.4, pp. 377–380.

Kim, B. and A. Latz (2001). "The dynamics of the spherical p-spin model: From microscopic to asymptotic". In: *Europhysics Letters* 53.5, pp. 660–666.

Liang, T., S. Sen, and P. Sur (2023). "High-dimensional asymptotics of Langevin dynamics in spiked matrix models". In: *Information and Inference: A Journal of the IMA* 12.4, pp. 2720–2752. arXiv: 2204.04476.

Martin, P. C., E. D. Siggia, and H. A. Rose (1973). "Statistical Dynamics of Classical Systems". In: *Physical Review A* 8.1, pp. 423–437.

Mignacco, F., F. Krzakala, P. Urbani, and L. Zdeborová (2021a). "Dynamical mean-field theory for stochastic gradient descent in Gaussian mixture classification". In: *Journal of Statistical Mechanics: Theory and Experiment* 2021.12, p. 124008. arXiv: 2006.06098.

Mignacco, F., P. Urbani, and L. Zdeborová (2021b). "Stochasticity helps to navigate rough landscapes: comparing gradient-descent-based algorithms in the phase retrieval problem". In: *Machine Learning: Science and Technology* 2.3, p. 035029. arXiv: 2103.04902.

Sellke, M. (2024). "The threshold energy of low temperature Langevin dynamics for pure spherical spin glasses". In: *Communications on Pure and Applied Mathematics* 77.11, pp. 4065–4099. arXiv: 2305.07956.

Sompolinsky, H. and A. Zippelius (1981). "Dynamic Theory of the Spin-Glass Phase". In: *Physical Review Letters* 47.5, pp. 359–362.

Subag, E. (2017). "The complexity of spherical p-spin models—A second moment approach". In: *The Annals of Probability* 45.5, pp. 3385–3450. arXiv: 1504.02251.

15 Online SGD

Teacher–student networks, plateaus and the information exponent

THE DICTIONARY — order-parameter equations ↔ concentration of summary statistics

IN THIS CHAPTER. A student network learns a teacher network by stochastic gradient descent on one fresh Gaussian sample per step. In 1995 Biehl and Schwarze, and Saad and Solla, reduced the dynamics in dimension $d \to \infty$, with time counted in units of d samples, to differential equations for a few overlaps, and Saad and Solla found a plateau on which the student units stay interchangeable. We derive the equations, the plateau and its escape rate, and prove by a martingale decomposition and Grönwall's lemma that the overlaps follow the equations within $O(d^{-1/2})$ on every fixed horizon, so a random start cannot specialize there; exact simulations place the escape at $c + (\ln d)/(2\varrho)$, with the spread the linearized dynamics predicts, a law not proved. For a single neuron the search phase is governed by the information exponent k: vanilla online SGD needs about d, $d \log d$ or d^{k-1} samples up to logarithms (Ben Arous, Gheissari and Jagannath), a cost that smoothing and sample reuse beat.

15.1 The question: learning from one sample at a time

In a *teacher–student* problem a teacher network with fixed, unknown weights labels random inputs, and a student network with the same architecture adjusts its weights to imitate it. In *online stochastic gradient descent*, or one-pass SGD, the student takes one gradient step on each example and never sees it again, so the number of steps equals the number of samples. How fast does the student's error fall, why does it stall, and how many samples does a random start cost?

The networks are *soft committee machines*: two-layer networks with K hidden units for the student, M for the teacher, and output weights fixed to one.[1] For an input $x \in \mathbb{R}^d$ the fields and outputs of the student, with weights $w_k \in \mathbb{R}^d$, and of the teacher, with weights $w_n^* \in \mathbb{R}^d$, are

$$\lambda_k = \frac{\langle w_k \rangle x}{\sqrt{d}}, \qquad \nu_n = \frac{\langle w_n^* \rangle x}{\sqrt{d}}, \qquad \hat{y} = \sum_{k=1}^{K} g(\lambda_k), \qquad y = \sum_{n=1}^{M} g(\nu_n),$$

with the sigmoid $g(z) = \operatorname{erf}(z/\sqrt{2}) = 2\Phi(z) - 1$, $g'(z) = \sqrt{2/\pi}\, e^{-z^2/2}$, and independent $\mathcal{N}(0, I_d)$ inputs $x^1, x^2, \dots$. The gradient of the loss $\frac{1}{2}(\hat{y} - y)^2$ of

[1] A committee machine adds the votes of its hidden units with equal weights; in the soft version the votes are smooth functions of the fields rather than signs.

one example in w_k is $(\hat{y}-y)g'(\lambda_k)x/\sqrt{d}$, so a step of size η on the $(\mu+1)$st example reads

$$w_k^{\mu+1} = w_k^{\mu} + \frac{\eta}{\sqrt{d}}\,\Delta_k\,x^{\mu+1}, \qquad \Delta_k = g'(\lambda_k)\,(y-\hat{y}), \tag{15.1}$$

with the fields, the outputs and Δ_k computed from w^μ and the new input $x^{\mu+1}$. As everywhere in the book, η is the learning rate and time is the number of samples divided by d, $t=\mu/d$.

The student enters the error only through its *overlaps*, the physicist's order parameters and the statistician's summary statistics,

$$Q_{ik} = \frac{\langle w_i, w_k\rangle}{d}, \qquad R_{in} = \frac{\langle w_i, w_n^*\rangle}{d}, \qquad T_{nm} = \frac{\langle w_n^*, w_m^*\rangle}{d}.$$

We collect Q_{ik}, $i\le k$, and R_{in} in $\Omega=(Q,R)\in\mathbb{R}^p$, $p=K(K+1)/2+KM$; the teacher's overlaps T are fixed. For an input independent of the weights, (λ,ν) is centered Gaussian with covariance

$$C(\Omega) = \begin{pmatrix} Q & R \\ R^\top & T \end{pmatrix}, \tag{15.2}$$

so the *generalization error*, the student's population risk, is a function of the overlaps:

$$\varepsilon_g(\Omega) = \tfrac{1}{2}\mathbb{E}\Big(\sum_k g(\lambda_k) - \sum_n g(\nu_n)\Big)^2, \qquad (\lambda,\nu)\sim\mathcal{N}\big(0, C(\Omega)\big).$$

THE PHYSICAL PREDICTION · Saad and Solla 1995; Biehl, Riegler and Wöhler 1996. Let $d\to\infty$ with $t=\mu/d$ fixed. The overlaps concentrate on the solution of

$$\frac{\mathrm{d}R_{in}}{\mathrm{d}t} = \eta\,\mathbb{E}[\Delta_i\nu_n], \qquad \frac{\mathrm{d}Q_{ik}}{\mathrm{d}t} = \eta\,\mathbb{E}[\Delta_i\lambda_k+\Delta_k\lambda_i] + \eta^2\,\mathbb{E}[\Delta_i\Delta_k], \tag{15.3}$$

where $(\lambda,\nu)\sim\mathcal{N}(0,C(\Omega))$ and $\Delta_i = g'(\lambda_i)\big(\sum_n g(\nu_n) - \sum_j g(\lambda_j)\big)$, and the learning curve is $\varepsilon_g(\Omega(t))$. If $K=M\ge 2$ and the teacher units are orthonormal, then a student started at random is trapped near a fixed point at which every student unit has the same overlap with every teacher unit: the *plateau*, with error

$$\varepsilon_\star = \frac{K}{\pi}\Big(\frac{\pi}{6} - K\arcsin\frac{1}{2K}\Big) \quad \text{as } \eta\to 0,$$

which is 0.011611 for $K=2$. It leaves the plateau, each student unit specializing to one teacher unit, after a time that grows like $\ln d$.

Biehl and Schwarze (1995) wrote such equations for on-line gradient descent in a single sigmoidal unit and in committee machines; Saad and Solla (1995b,c) solved them for general K and M and studied the unspecialized phase that produces the plateau, whose error $\varepsilon_\star$ and escape rate they gave in closed form (Saad and Solla 1995a); and Biehl et al. (1996) described

analytically how the length of the plateau depends on the initial conditions. A random start, with overlaps of order $d^{-1/2}$, gives the $\ln d$.[2]

The plateau is one face of a problem that every random start in high dimension faces: a random unit vector has overlap of order $d^{-1/2}$ with the vector to be learned, and where the population loss is flat near zero overlap, the student must first find the signal. For a single neuron the flatness is measured by the *information exponent* of Ben Arous et al. (2021), the order k of the first nonvanishing derivative of the population loss at zero overlap; for a single-index model $y = h(\langle\theta\rangle x)$ it is the index of the first nonzero Hermite coefficient of h. How many samples does online SGD need as a function of k? And for all three questions: how do physicists derive the answers, and what is proved?

[2] Biehl and Schwarze's paper appeared in February 1995; Saad and Solla's letter, received in October 1994, appeared in May 1995.

15.2 The physics: closing the equations on the overlaps

15.2.1 From weights to overlaps

Taking inner products of (15.1) with $w_k^{\mu+1}$ and w_n^*, and recalling that λ_k is the inner product of the new input with w_k^μ divided by $\sqrt{d}$, gives exact updates:

$$\begin{aligned} R_{in}^{\mu+1} &= R_{in}^{\mu} + \frac{\eta}{d}\,\Delta_i \nu_n, \\ Q_{ik}^{\mu+1} &= Q_{ik}^{\mu} + \frac{\eta}{d}\,(\Delta_i\lambda_k + \Delta_k\lambda_i) + \frac{\eta^2}{d}\,\frac{|x^{\mu+1}|^2}{d}\,\Delta_i\Delta_k. \end{aligned} \tag{15.4}$$

Each step moves the overlaps by $O(1/d)$, and one unit of time by $O(1)$. The last term, of second order in η, carries the fluctuation of the gradient: it is nonnegative on the diagonal on average, and makes the students' norms grow even where the average gradient vanishes.

Self-averaging (Saad and Solla). Sum (15.4) over a window of L steps, $1 \ll L \ll d$. The overlaps change by $O(L/d)$ in the window, so to leading order all L increments are taken at the same Ω. The new input is independent of the weights, so the fields are jointly Gaussian with covariance $C(\Omega)$, by the central limit theorem (exactly so for Gaussian inputs), and $|x|^2/d \approx 1$. The increments are then independent, with mean $f(\Omega)/d$, where f collects the right sides of (15.3), and size $O(1/d)$; their sum is $L f(\Omega)/d$ plus fluctuations of order $\sqrt{L}/d$. With $\delta t = L/d$,

$$\frac{\Omega(t+\delta t) - \Omega(t)}{\delta t} = f(\Omega) + O\big(L^{-1/2}\big) + O(\delta t),$$

and both errors are small: the overlaps are *self-averaging*, and $\Omega' = f(\Omega)$. For erf the averages are explicit (Computation 15.1). ◇

The derivation assumes what it needs: that the fluctuations of successive windows are negligible, which is the concentration of the overlaps; that $|x|^2/d$ may be replaced by one inside an average that also contains the fields, which depend on x; and that the solution stays where the equations make sense. It then uses the equations at all times, including the long times at

which plateaus are analyzed, although the window controls errors only for t of order one: the limits $d \to \infty$ and $t \to \infty$ are exchanged without comment. The activation $\operatorname{erf}(z/\sqrt{2})$ is chosen because its Gaussian averages are explicit; three of them close the equations.

computation 15.1 · Gaussian averages for $g(z) = \operatorname{erf}(z/\sqrt{2})$

1 *An average of signs.* Let (u, v, w, z) be centered Gaussian with covariance (c_{ab}). For $\zeta \sim \mathcal{N}(0,1)$ independent of u, $\mathbb{E}[\operatorname{sgn}(u-\zeta) \mid u] = 2\Phi(u) - 1 = g(u)$, so with ζ, ζ' independent standard normals independent of (u, v), $\mathbb{E}[g(u)g(v)] = \mathbb{E}[\operatorname{sgn}(u-\zeta)\operatorname{sgn}(v-\zeta')]$.

2 *Sheppard's formula.* A centered Gaussian pair with correlation ρ is $(\langle a\rangle G, \langle b\rangle G)$ with G standard Gaussian in $\mathbb{R}^2$ and unit vectors a, b at the angle $\theta = \arccos\rho$; the signs differ when the direction of G falls in two opposite sectors of total angle 2θ, so $\mathbb{E}[\operatorname{sgn}\langle a\rangle G \operatorname{sgn}\langle b\rangle G] = 1 - 2\theta/\pi = \frac{2}{\pi}\arcsin\rho$. With the variances $1 + c_{11}$, $1 + c_{22}$ and covariance c_{12} of $(u-\zeta, v-\zeta')$,

$$I_2(u, v) = \mathbb{E}[g(u)g(v)] = \frac{2}{\pi}\arcsin\frac{c_{12}}{\sqrt{(1+c_{11})(1+c_{22})}}.$$

3 *Derivatives in the covariance.* By Lemma A.2, the derivative of $\mathbb{E}F(u, w)$ in c_{13} is $\mathbb{E}[\partial_u\partial_w F]$ and in c_{11} it is $\frac{1}{2}\mathbb{E}[\partial_u^2 F]$. Differentiating $I_2(u, w)$, with $\Lambda = (1+c_{11})(1+c_{33}) - c_{13}^2$,

$$\mathbb{E}[g'(u)g'(w)] = \frac{2}{\pi\sqrt{\Lambda}}, \qquad \mathbb{E}[g''(u)g(w)] = -\frac{2c_{13}}{\pi(1+c_{11})\sqrt{\Lambda}}.$$

4 *Three points.* Gaussian integration by parts (Theorem A.1) in v gives $\mathbb{E}[g'(u)v\,g(w)] = c_{12}\mathbb{E}[g''(u)g(w)] + c_{23}\mathbb{E}[g'(u)g'(w)]$, that is,

$$I_3(u, v, w) = \mathbb{E}[g'(u)\,v\,g(w)] = \frac{2}{\pi\sqrt{\Lambda}}\Big(c_{23} - \frac{c_{12}c_{13}}{1+c_{11}}\Big).$$

5 *Four points.* The factor $g'(u)g'(v) = \frac{2}{\pi}e^{-(u^2+v^2)/2}$ tilts the Gaussian law. For nonsingular c, completing the square gives $\mathbb{E}[g'(u)g'(v)F(w, z)] = \frac{2}{\pi}\det(I + c_{uv})^{-1/2}\mathbb{E}'F(w, z)$, where c_{uv} is the covariance of (u, v) and, under $\mathbb{E}'$, (w, z) is centered Gaussian with covariance $c' = c_{wz} - c_{wz,uv}(I + c_{uv})^{-1}c_{uv,wz}$. By step 2, and by continuity for singular c,

$$\begin{aligned} I_4(u, v, w, z) &= \mathbb{E}[g'(u)g'(v)g(w)g(z)] \\ &= \frac{4}{\pi^2\sqrt{\det(I + c_{uv})}}\arcsin\frac{c'_{wz}}{\sqrt{(1+c'_{ww})(1+c'_{zz})}}. \end{aligned}$$

Each expectation in (15.3) is a sum of such terms over the units, for instance $\mathbb{E}[\Delta_i\nu_n] = \sum_m I_3(\lambda_i, \nu_n, \nu_m) - \sum_j I_3(\lambda_i, \nu_n, \lambda_j)$, and ε_g is a sum of

I_2 terms. The script `15-sgd-ode.py` checks I_2, I_3 and I_4, including terms with repeated units, against Gauss–Hermite quadrature: the largest error is 7×10^{-17}.[3]

[3] Sheppard (1899) computed the probability that a bivariate normal pair falls in a quadrant, which is step 2 of Computation 15.1.

15.2.2 The plateau

Take $K = M \geq 2$ and orthonormal teacher units, $T = I_K$. Relabeling the student units, or the teacher units, maps the equations to themselves. A random start $w_k^0 = \sqrt{q_0}\, g_k$, $g_k \sim \mathcal{N}(0, I_d)$, has $Q(0) = q_0 I + O(d^{-1/2})$ and $R(0) = O(d^{-1/2})$, which in the limit is invariant under all relabelings, and so is the solution. Every student unit then has the same overlap with every teacher unit, the same norm and the same overlap with the other students: the solution stays in a three-dimensional subspace $\mathcal{S}$ on which no student unit can align with one teacher unit more than with another, and settles near a fixed point, the *plateau*.

COMPUTATION 15.2 · THE PLATEAU AND ITS ESCAPE RATE AS $\eta \to 0$ (SAAD AND SOLLA)

1 *Gradient flow.* Since $\nabla_{w_k} \varepsilon_g = -\mathbb{E}[\Delta_k x]/\sqrt{d}$, the terms of (15.3) linear in η are the overlaps of the population gradient flow $\mathrm{d}w_k/\mathrm{d}t = -\eta d\, \nabla_{w_k} \varepsilon_g$, and the overlaps of a critical point of ε_g are a fixed point of these terms.

2 *The symmetric line.* Try all student vectors equal to $w = s \sum_n w_n^*$, $s > 0$. By Gaussian integration by parts $\mathbb{E}[\Delta_k x]$ is a combination of the w_j and w_n^*; it is the same for every k and unchanged when the teacher units are permuted, so it is a multiple of $\sum_n w_n^*$. The point is critical if and only if the error $\varepsilon(s)$ along the line has $\varepsilon'(s) = 0$.

3 *The error on the line.* There $\lambda_k = s \sum_n \nu_n$, so $Q_{ik} = Ks^2 =: q$ and $R_{kn} = s$, and since $\frac{2}{\pi} \arcsin \frac{1}{2} = \frac{1}{3}$,

$$\varepsilon(s) = \frac{1}{\pi}\left[K^2 \arcsin \frac{q}{1+q} - 2K^2 \arcsin \frac{s}{\sqrt{2(1+q)}} + \frac{K\pi}{6}\right].$$

4 *The critical point.* With $a = q/(1+q)$ and $b = s/\sqrt{2(1+q)}$, $\varepsilon'(s) = 0$ reads $a'/\sqrt{1-a^2} = 2b'/\sqrt{1-b^2}$, where $a' = 2Ks/(1+q)^2$ and $b' = 2/(2(1+q))^{3/2}$. At $s_\star^2 = 1/(K(2K-1))$, $1+q = 2K/(2K-1)$, so $a = b = 1/(2K)$, and $a' = 2b'$ because $4K^2 s_\star^2 = 2(1+q)$.

5 *The plateau.* Since $a = b$ at $s_\star$, $\varepsilon_\star = \varepsilon(s_\star) = \frac{K}{\pi}\left(\frac{\pi}{6} - K \arcsin \frac{1}{2K}\right)$, at $Q_{ik} = 1/(2K-1)$ and $R_{kn} = s_\star$. For $K = 2$: $Q_{ik} = \frac{1}{3}$, $R_{kn} = 1/\sqrt{6}$, $\varepsilon_\star = \frac{2}{\pi}(\frac{\pi}{6} - 2 \arcsin \frac{1}{4}) = 0.011611$.

6 *The escape rate.* Perturb $R = s_\star J + \epsilon E$, with J the all-ones matrix and $E = I - J/K$, keeping Q. With $\Lambda_\star = 2(1+q) - s_\star^2 = (2K+1)/K$, differentiating the I_3 terms changes the η-linear part of $\mathrm{d}R_{in}/\mathrm{d}t$

η	point	Q_{11}	Q_{12}	R_{in}	ε_g	ϱ	unstable directions
$\to 0$	A=B	0.333 333	0.333 333	0.408 248	0.011 611	$0.065750\,\eta$	1
0.1	A	0.334 297	0.334 297	0.408 543	0.012 085 8	0.006 621	3
0.1	B	0.338 824	0.332 031	0.409 235	0.012 085 2	0.006 532	1
0.5	A	0.339 579	0.339 579	0.410 158	0.014 677 0	0.034 341	3
0.5	B	0.367 720	0.325 386	0.414 394	0.014 644 8	0.031 618	1
1	A	0.352 963	0.352 963	0.414 234	0.021 156 5	0.074 807	3
1	B	0.435 085	0.310 213	0.426 138	0.020 656 5	0.059 140	1

Table 15.1. Symmetric fixed points of (15.3) for $K = M = 2$, $T = I$: the overlaps (all R_{in} are equal), the error, the largest eigenvalue ϱ of the linearization, and the number of positive eigenvalues among seven. A has identical students; B is the limit of the solution from the random start. The row $\eta \to 0$ is Computation 15.2. Script: `figures/scripts/15-sgd-ode.py`.

by

$$\frac{2\eta\epsilon}{\pi} E_{in}\left(s_\star \Lambda_\star^{-3/2} - \frac{K s_\star \Lambda_\star^{-1/2}}{1+q} + \frac{Kq}{(1+q)\sqrt{1+2q}}\right) + O(\epsilon^2),$$

whose last two terms cancel, while $\mathrm{d}Q/\mathrm{d}t$ does not change at first order because the rows of E sum to zero. So E is an unstable direction, with the rate

$$\varrho = \frac{2K\eta}{\pi\sqrt{2K-1}\,(2K+1)^{3/2}} + O(\eta^2),$$

which is $4\sqrt{15}\,\eta/(75\pi) = 0.06575\,\eta$ for $K = 2$ and $6\sqrt{35}\,\eta/(245\pi) = 0.04612\,\eta$ for $K = 3$.

At this plateau the student units are identical, each equal to $s_\star \sum_n w_n^*$: the network has learned the teacher's average direction and nothing more. Numerically, at small η the other eigenvalues of the linearization are negative or of order η^2. A finite learning rate adds one feature (Table 15.1), which Saad and Solla (1995a) saw in their solutions: the students' norms exceed their mutual overlaps. The identical-student point persists as a fixed point A, but the term in η^2 makes it unstable in two more directions, in which the students separate; the solution from a random start settles instead at a nearby fixed point B, where the students have the same overlaps with the teacher but different components orthogonal to it. A and B coincide as $\eta \to 0$; their errors differ by 6×10^{-7} at $\eta = 0.1$ and 5×10^{-4} at $\eta = 1$, where B has the single unstable direction E, with $\varrho = 0.05914$.

15.2.3 Leaving the plateau

THE LENGTH OF THE PLATEAU. Near the plateau, the coordinate $\epsilon = \frac{1}{2}(R_{11} + R_{22} - R_{12} - R_{21})$ along E (for $K = 2$) grows like $e^{\varrho t}$. A random start gives ϵ of order $d^{-1/2}$, and the approach to the plateau, linear in such small

perturbations, carries it to $\mathcal{A}d^{-1/2}$ at some time $t_{\rm p}$, where $\mathcal{A}$ collects the projections on E of the initial overlaps and of the fluctuations of SGD, both of order $d^{-1/2}$, and has a law that does not depend on d. The student leaves the plateau when ϵ is of order one, at

$$t_{\rm esc} \approx t_{\rm p} + \frac{1}{\varrho}\ln\frac{\sqrt{d}}{|\mathcal{A}|} = c + \frac{\ln d}{2\varrho} - \frac{\ln|\mathcal{A}|}{\varrho}. \tag{15.5}$$

If $\mathcal{A}$ is Gaussian, as a linear functional of Gaussian initial overlaps and nearly Gaussian noise, then $t_{\rm esc} - \mathbb{E}t_{\rm esc}$ has the law of $-(\ln|Z| - \mathbb{E}\ln|Z|)/\varrho$ with $Z \sim \mathcal{N}(0,1)$, whose standard deviation is $\pi/(2\sqrt{2}\,\varrho)$. ◇

The $\ln d$ is the classical prediction; the law of the *escape time* carries the same linearization one step further, and we have not found it in the sources cited here. The derivation linearizes over a time of order $\ln d$, lets the noise act only through $\mathcal{A}$, and ignores the nonlinear part of the escape. For $\eta = 1$ it predicts a slope $1/(2\varrho) = 8.45$ in $\ln d$ and a spread 18.8 (Section 15.5). The equations themselves, started at $R(0) = \delta_0 I$, escape at $t_{\rm esc} = \varrho^{-1}\ln(1/\delta_0) + c(\delta_0)$ with $c = 2.88, 3.84, 4.37$ for $\delta_0 = 10^{-2}, 10^{-3}, 3\times10^{-5}$.[4]

[4] Integrated in floating point at $\eta = 1$, the solution from the symmetric start leaves $\mathcal{S}$ near $t = 600$: rounding errors of order 10^{-16} grow like $e^{\varrho t}$ along E, a numerical shadow of the escape.

15.2.4 The search phase of a single neuron

For a single unit the same mechanism acts at the start. Ben Arous et al. (2021) run SGD on the sphere, $w \leftarrow (w - \frac{\delta}{d}\nabla\ell)/|w - \frac{\delta}{d}\nabla\ell|$, with ∇ the spherical gradient of the loss ℓ of one sample and a step-size parameter δ, for a population loss $\phi(m)$ of the overlap $m = \langle w\rangle\theta$ with the unit vector θ to be learned; the information exponent is the k with $\phi^{(j)}(0) = 0$ for $j < k$ and $\phi^{(k)}(0) < 0$, for ϕ with a bounded derivative of order $k+1$. For $y = \mathrm{He}_k(\langle\theta\rangle x)/\sqrt{k!}$ and the correlation loss $\ell = -y\,\mathrm{He}_k(\langle w\rangle x)/\sqrt{k!}$, with He_k the monic Hermite polynomial of degree k,[5] $\phi(m) = -m^k$.

[5] The Hermite polynomials $\mathrm{He}_1 = z$, $\mathrm{He}_2 = z^2 - 1$, $\mathrm{He}_3 = z^3 - 3z$ satisfy $\mathbb{E}\,\mathrm{He}_j(X)\mathrm{He}_k(Y) = k!\,m^k\delta_{jk}$ for a standard Gaussian pair with correlation m (Section A.8).

THE SEARCH PHASE, AFTER BEN AROUS, GHEISSARI AND JAGANNATH. With $u = \langle w\rangle x$ and $x_1 = \langle\theta\rangle x$, the normalized step expanded to second order in δ/d gives the drift per unit time

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \delta k\, m^{k-1}(1-m^2) - \frac{\delta^2}{2}\,\mathbb{E}[a^2]\,m, \qquad a = -\frac{\mathrm{He}_k(x_1)\mathrm{He}_k'(u)}{k!}, \tag{15.6}$$

whose last term is the retraction caused by the normalization, with $\mathbb{E}[a^2] = k$ at $m = 0$. From $m \approx d^{-1/2}$: for $k = 1$, m reaches $\frac{1}{2}$ in a time of order one, $n \asymp d$; for $k = 2$ the drift is linear with rate $r = \delta(2-\delta)$, and the time is $(\ln d)/(2r)$, $n \asymp d\ln d$; for $k \geq 3$ the drift beats the retraction only if $\delta \lesssim m^{k-2} \approx d^{-(k-2)/2}$, and then $t \approx m_0^{2-k}/\delta \approx d^{k-2}$, $n \asymp d^{k-1}$. ◇

Started at $m = 0$, (15.6) predicts no learning for $k \geq 2$: zero overlap is a fixed point, as $\mathcal{S}$ is invariant for the committee machine, and the escape is driven by the fluctuations that the equation discards. The erf committee machine has information exponent one in this sense, since $\mathbb{E}[Zg(Z)] = \mathbb{E}g'(Z) = 1/\sqrt{\pi} \neq 0$; its plateau is a second search with a linear instability, as for $k = 2$, hence the $\ln d$.

15.3 The stone

PHYSICS · *Saad and Solla's equations*

PROOF · *concentration of the overlaps*

1 · Reduce

The state of the network is the overlaps Q, R, T, and the error depends on them alone.

The same summary statistics; for Gaussian inputs, an exact Markov chain (Lemma 15.3): rotation invariance replaces the typical input.

2 · Average

The fields of a fresh input are Gaussian by the central limit theorem, with covariance $C(\Omega)$; average the update.

Given the past, the fields are exactly $\mathcal{N}(0, C(\Omega^\mu))$ (Lemma 15.2); the drift is Lipschitz on bounded sets by Gaussian interpolation (Lemma 15.4).

3 · Self-average

Fluctuations of the increments are $O(1/d)$ and average out over a window of $1 \ll L \ll d$ steps.

Drift plus martingale: Doob's inequality bounds the martingale by $O(d^{-1/2})$ uniformly on $[0, \bar{t}]$, and $|x|^2/d - 1$ costs a bias of order $d^{-1/2}$.

4 · Close

Integrate the averaged increments; the solution is assumed to exist and to stay physical.

A discrete Grönwall inequality up to a stopping time gives $e^{L_B \bar{t}} O(d^{-1/2})$; existence and uniqueness for the equations come from SGD itself (Theorem 15.5).

5 · Plateau

An unstable symmetric fixed point, with error $\varepsilon_\star$ and escape rate ϱ.

The symmetric subspace is invariant; on fixed horizons SGD from a random start follows the symmetric solution (Corollary 15.6).

6 · Escape

The asymmetry grows like $e^{\varrho t}$ from $d^{-1/2}$, so $t_{\text{esc}} \approx c + (\ln d)/(2\varrho)$.

No counterpart for the committee machine: the bound $e^{L_B \bar{t}}/\sqrt{d}$ is useless at $\bar{t} \asymp \ln d$, and we know of no proof as of 2026.

7 · Search

For a single neuron $\mathrm{d}m/\mathrm{d}t \propto m^{k-1}$ from $m \approx d^{-1/2}$: $n \asymp d$, $d \ln d$, d^{k-1}.

Sufficiency up to $\log^2 d$ and necessity, for spherical online SGD (Theorem 15.7); smoothing reaches $d^{k/2}$, and reuse beats both.

On fixed horizons the proof follows the derivation step by step: the typical input becomes a conditional Gaussian law, self-averaging becomes Doob's inequality, the window becomes a discrete Grönwall inequality. It adds what the physics does not ask for: control of the region the overlaps visit, and the well-posedness of the equations, which it takes from SGD itself. The routes part at long times, where the physicist puts the fluctuations of order $d^{-1/2}$ back by hand and the theorem stops. For single neurons the mathematicians went beyond the horizon, controlling the martingale at the scale $d^{-1/2}$ over times of order $\ln d$ and d^{k-2}; for the committee machine that step has not been taken.

15.4 The proof: concentration of the overlaps on fixed horizons

We prove the equations of the box on every fixed horizon, with the rate $d^{-1/2}$, for general K, M and T, noiseless labels, and activations that include erf.

Assumption 15.1.

(i) The activation g is three times continuously differentiable, g, g', g'', g''' are bounded, and $G_\star = \sup_z |z g'(z)| < \infty$.

(ii) For each d the teacher vectors $w_1^*, \dots, w_M^* \in \mathbb{R}^d$ have overlaps $T_{nm} = \langle w_n^*, w_m^* \rangle / d$ given by a fixed positive semidefinite $M \times M$ matrix T. The inputs $x^1, x^2, \dots$ are independent $\mathcal{N}(0, I_d)$ vectors, and SGD runs as in (15.1) with the labels $y = \sum_n g(v_n)$.

(iii) The initial student vectors are deterministic, and $\max_k Q_{kk}^0 \le B_0$.

For erf, $G_\star = \sqrt{2/\pi}\, e^{-1/2} = 0.484$; tanh and the logistic function also qualify, the rectified linear unit does not. Write $G_r = \sup_z |g^{(r)}(z)|$, $D = (K+M) G_0 G_1$, which bounds $|\Delta_k|$, $\tau_T = \max_n T_{nn}$, and $\mathcal{F}_\mu = \sigma(x^1, \dots, x^\mu)$. Let $\mathcal{K}$ be the set of Ω with $C(\Omega)$ positive semidefinite and $\mathcal{K}_B = \{\Omega \in \mathcal{K} : \max_k Q_{kk} \le B\}$, a closed, bounded and convex set, because $C(\Omega)$ is affine in Ω and $|Q_{ik}| \le B$, $R_{in}^2 \le B\tau_T$ on it. Every Ω^μ lies in $\mathcal{K}$, since $C(\Omega^\mu)$ is the Gram matrix of the vectors $w_k^\mu/\sqrt{d}$ and $w_n^*/\sqrt{d}$. For $z = (\lambda, \nu) \in \mathbb{R}^{K+M}$ let $h(z) \in \mathbb{R}^p$

have the components $h_{R,in} = \eta\Delta_i\nu_n$ and $h_{Q,ik} = \eta(\Delta_i\lambda_k + \Delta_k\lambda_i) + \eta^2\Delta_i\Delta_k$, with $\Delta_i = \Delta_i(z)$ as in the box, and let $f(\Omega) = \mathbb{E}h(Z)$, $Z \sim \mathcal{N}(0, C(\Omega))$: the right side of (15.3).

Lemma 15.2 (one step). *Under Assumption 15.1, for every $\mu \geq 0$:*

(i) given $\mathcal{F}_\mu$, the fields $Z^{\mu+1} = (\lambda, \nu)$ of the new input are $\mathcal{N}(0, C(\Omega^\mu))$, and $d(\Omega^{\mu+1} - \Omega^\mu) = h(Z^{\mu+1}) + \eta^2(|x^{\mu+1}|^2/d - 1)\, v(Z^{\mu+1})$, where $v_{Q,ik} = \Delta_i\Delta_k$ and $v_{R,in} = 0$;

(ii) $\mathbb{E}[d(\Omega^{\mu+1} - \Omega^\mu) \mid \mathcal{F}_\mu] = f(\Omega^\mu) + b^\mu$ with $|b^\mu| \leq \beta/\sqrt{d}$, $\beta = \eta^2 D^2\sqrt{K(K+1)}$;

(iii) $\mathbb{E}[|d(\Omega^{\mu+1} - \Omega^\mu)|^2 \mid \mathcal{F}_\mu] \leq c_2(1 + \max_k Q^\mu_{kk})$, with c_2 depending only on η, K, M, G_0, G_1 and T;

(iv) $|Q^{\mu+1}_{kk} - Q^\mu_{kk}| \leq \frac{c_1}{d}\left(1 + |x^{\mu+1}|^2/d\right)$ for every k, with $c_1 = 2\eta G_\star(K + M)G_0 + \eta^2 D^2$.

Proof. The plan is to read off the identities from (15.4) and to use that the new input is independent of the past. Writing $|x|^2/d = 1 + (|x|^2/d - 1)$ in (15.4) gives the identity of (i). Given $\mathcal{F}_\mu$ the weights w^μ are fixed and $x^{\mu+1} \sim \mathcal{N}(0, I_d)$ is independent of them, so $Z^{\mu+1}$ is a linear image of a standard Gaussian vector, centered, with $\mathbb{E}[\lambda_i\lambda_k \mid \mathcal{F}_\mu] = \langle w^\mu_i, w^\mu_k\rangle/d = Q^\mu_{ik}$ and likewise for the other entries. For (ii), $\mathbb{E}[h(Z^{\mu+1}) \mid \mathcal{F}_\mu] = f(\Omega^\mu)$ by (i), and $b^\mu = \eta^2\mathbb{E}[(|x^{\mu+1}|^2/d - 1)v(Z^{\mu+1}) \mid \mathcal{F}_\mu]$; each of the $K(K+1)/2$ nonzero components of v is at most D^2, and $\mathbb{E}||x|^2/d - 1| \leq (\mathrm{Var}|x|^2/d)^{1/2} = (2/d)^{1/2}$. For (iii), $(\eta\Delta_i\nu_n)^2 \leq \eta^2 D^2\nu_n^2$, and by $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$ the squared Q-components are at most $3\eta^2D^2(\lambda_k^2 + \lambda_i^2) + 3\eta^4D^4(|x|^2/d)^2$; take conditional expectations, with $\mathbb{E}[\nu_n^2 \mid \mathcal{F}_\mu] = T_{nn}$, $\mathbb{E}[\lambda_k^2 \mid \mathcal{F}_\mu] = Q^\mu_{kk}$ and $\mathbb{E}(|x|^2/d)^2 = 1 + 2/d \leq 3$. For (iv), (15.4) gives $d(Q^{\mu+1}_{kk} - Q^\mu_{kk}) = 2\eta g'(\lambda_k)\lambda_k(y - \hat{y}) + \eta^2(|x|^2/d)\Delta_k^2$, with $|g'(\lambda_k)\lambda_k| \leq G_\star$, $|y - \hat{y}| \leq (K+M)G_0$ and $\Delta_k^2 \leq D^2$. ∎

Part (iv) is the only use of $G_\star$: a student's norm changes by $O(1/d)$ in every step, whatever the input, because the first-order term contains $g'(\lambda_k)\lambda_k$. Part (i) says more than its statement uses, and the simulations of Section 15.5 rest on the stronger form.

Lemma 15.3 (exact reduction). *Under Assumption 15.1, let r be the rank of $C(\Omega^\mu)$ and C^+ its pseudoinverse. Given $\mathcal{F}_\mu$, $|x^{\mu+1}|^2 = (Z^{\mu+1})^\top C^+ Z^{\mu+1} + \xi$ with $\xi \sim \chi^2_{d-r}$ independent of $Z^{\mu+1}$. Consequently the conditional law of $\Omega^{\mu+1}$ given $\mathcal{F}_\mu$ depends only on Ω^μ: the overlaps form a time-homogeneous Markov chain, which depends on d only through the factor $1/d$ in (15.4) and the degrees of freedom of ξ.*

Proof. Let V be the $d \times (K+M)$ matrix with columns $w^\mu_k/\sqrt{d}$ and $w^*_n/\sqrt{d}$, so that $Z^{\mu+1} = V^\top x$ for $x = x^{\mu+1}$ and $V^\top V = C(\Omega^\mu)$. The orthogonal projection onto the range of V is $P = VC^+V^\top$. Given $\mathcal{F}_\mu$, Px and $x - Px$ are independent centered Gaussian vectors, $|x - Px|^2 \sim \chi^2_{d-r}$, and $|Px|^2 =$

$x^\top V C^+ V^\top x = (Z^{\mu+1})^\top C^+ Z^{\mu+1}$. By (15.4) the increment is a fixed function of Ω^μ, $Z^{\mu+1}$ and $|x^{\mu+1}|^2$, whose joint conditional law depends only on Ω^μ. ∎

Lemma 15.4 (the drift is Lipschitz). *Under (i) of Assumption 15.1, for every $B > 0$ there are constants F_B and L_B, depending only on B, η, K, M, $G_0, \dots, G_3$ and T, such that $|f(\Omega)| \le F_B$ and $|f(\Omega) - f(\Omega')| \le L_B|\Omega - \Omega'|$ for all $\Omega, \Omega' \in \mathcal{K}_B$. The same holds for ε_{g}.*

Proof. The plan is to compare two Gaussian expectations by the interpolation lemma, which needs second derivatives of h and no square root of the covariance. Each Δ_i is a sum of at most $K + M$ terms $\pm g'(\lambda_i) g(z_c)$, so Δ_i and its partial derivatives of first and second order are bounded by a constant A_0 depending on $G_0, \dots, G_3$ and $K + M$. Since $\partial_a \partial_b (\Delta_i z_e) = z_e \partial_a \partial_b \Delta_i + \delta_{ae} \partial_b \Delta_i + \delta_{be} \partial_a \Delta_i$, every component h_c is twice continuously differentiable with $|h_c(z)| + \max_{a,b} |\partial_a \partial_b h_c(z)| \le A(1 + |z|)$ for a constant A. For $\Omega \in \mathcal{K}_B$, $\mathbb{E}|Z| \le (\operatorname{tr} C(\Omega))^{1/2} \le (KB + \operatorname{tr} T)^{1/2} =: s_B$, so $|f(\Omega)| \le \sqrt{p} A(1 + s_B) =: F_B$. Now let $\Omega, \Omega' \in \mathcal{K}_B$, and let $X \sim \mathcal{N}(0, C(\Omega))$ and $Y \sim \mathcal{N}(0, C(\Omega'))$ be independent. The vector $Z_s = \sqrt{s} X + \sqrt{1 - s}\, Y$ has covariance $C(\Omega_s)$ with $\Omega_s = s\Omega + (1 - s)\Omega' \in \mathcal{K}_B$, by convexity, so $\mathbb{E}|\partial_a \partial_b h_c(Z_s)| \le A(1 + s_B)$. Each entry of Ω appears at most twice in $C(\Omega)$, so $\sum_{a,b} |C(\Omega)_{ab} - C(\Omega')_{ab}| \le 2\sqrt{p}\, |\Omega - \Omega'|$, and Lemma A.2 gives

$$|\mathbb{E} h_c(X) - \mathbb{E} h_c(Y)| \le \frac{A(1 + s_B)}{2} \sum_{a,b} |C(\Omega)_{ab} - C(\Omega')_{ab}| \le A(1 + s_B) \sqrt{p}\, |\Omega - \Omega'|.$$

Over the p components, $L_B = pA(1 + s_B)$. For ε_{g} the integrand $\frac{1}{2}(\sum_k g(\lambda_k) - \sum_n g(\nu_n))^2$ is bounded with bounded derivatives of order two, and the same argument applies. ∎

Theorem 15.5 (the ODE limit on fixed horizons). *Let Assumption 15.1 hold, fix $\bar{t} > 0$, and put $B = B_0 + 2c_1 \bar{t} + 1$. Then the equations (15.3) with $\Omega(0) = \Omega^0$ have a unique solution $\Omega \colon [0, \bar{t}] \to \mathcal{K}$; it stays in $\mathcal{K}_{B-1/2}$; and there is a constant C, depending only on g, K, M, η, T, B_0 and $\bar{t}$, such that for every $d \ge 1$*

$$\mathbb{E} \max_{0 \le \mu \le d\bar{t}} |\Omega^\mu - \Omega(\mu/d)| \le \frac{C}{\sqrt{d}}.$$

In particular $\sup_{t \le \bar{t}} |\varepsilon_{\mathrm{g}}(\Omega^{\lfloor dt \rfloor}) - \varepsilon_{\mathrm{g}}(\Omega(t))| \to 0$ in L^1 as $d \to \infty$.

Proof. The plan is to compare SGD with the solution of a globally Lipschitz extension of the equations, splitting each step into drift, bias and a martingale increment, bounding the martingale by Doob's inequality and the rest by a discrete Grönwall inequality, up to the first time the overlaps leave $\mathcal{K}_B$; a priori bounds place this time beyond the horizon with high probability, and the same bounds show that the extension solves the original equations. Write $n = \lfloor d\bar{t} \rfloor$.

Step 1: an extension. Let Π_B be the Euclidean projection onto the closed convex set $\mathcal{K}_B$, which is 1-Lipschitz, and $f_B = f \circ \Pi_B$. By Lemma 15.4,

f_B is bounded by F_B and L_B-Lipschitz on $\mathbb{R}^p$, so $\Omega' = f_B(\Omega)$ has a unique solution $\tilde\Omega$ on $[0,\infty)$ with $\tilde\Omega(0) = \Omega^0$, and $|\tilde\Omega(s) - \tilde\Omega(s')| \le F_B|s - s'|$.

Step 2: a priori bounds. Let $S = \sum_{\nu=1}^{n} |x^\nu|^2$, a χ^2 variable with $nd \le d^2\bar t$ degrees of freedom. Summing (iv) of Lemma 15.2, for every $\mu \le n$ and every k,

$$Q_{kk}^{\mu} \le B_0 + c_1\bar t + c_1 S/d^2. \tag{15.7}$$

On the event $G = \{S \le d^2\bar t + d^2/(2c_1)\}$, therefore, $\Omega^\mu \in \mathcal{K}_{B-1/2}$ for all $\mu \le n$. Since $\mathbb{E}S = nd$ and $\operatorname{Var} S = 2nd \le 2d^2\bar t$, Chebyshev's inequality gives $\mathbb{P}(G^c) \le 8c_1^2\bar t/d^2$.

Step 3: a decomposition up to a stopping time. Let $\tau = \min\{\mu : \Omega^\mu \notin \mathcal{K}_B\}$, with $\min\emptyset = \infty$, a stopping time for $(\mathcal{F}_\mu)$, and $\xi^{\nu+1} = d(\Omega^{\nu+1} - \Omega^\nu) - f(\Omega^\nu) - b^\nu$. By (ii) of Lemma 15.2, $\mathbb{E}[\xi^{\nu+1} \mid \mathcal{F}_\nu] = 0$, and by (iii), $\mathbb{E}[\mathbf{1}\{\nu < \tau\}|\xi^{\nu+1}|^2 \mid \mathcal{F}_\nu] \le c_2(1+B)$, a conditional variance being at most the conditional second moment. So $M_\mu = \sum_{\nu<\mu} \mathbf{1}\{\nu < \tau\}\xi^{\nu+1}$ is a square-integrable martingale. For $\mu \le \tau \wedge n$ every Ω^ν with $\nu < \mu$ lies in $\mathcal{K}_B$, where $f = f_B$, and summing the steps gives

$$\Omega^\mu = \Omega^0 + \frac{1}{d}\sum_{\nu<\mu}\big(f_B(\Omega^\nu) + b^\nu\big) + \frac{M_\mu}{d}.$$

Step 4: Grönwall. Subtract $\tilde\Omega(\mu/d) = \Omega^0 + \sum_{\nu<\mu}\int_{\nu/d}^{(\nu+1)/d} f_B(\tilde\Omega(s))\,\mathrm{d}s$ and let $e_\mu = |\Omega^\mu - \tilde\Omega(\mu/d)|$. The three differences are bounded in turn: $|f_B(\Omega^\nu) - f_B(\tilde\Omega(\nu/d))| \le L_B e_\nu$; $\int_{\nu/d}^{(\nu+1)/d}|f_B(\tilde\Omega(\nu/d)) - f_B(\tilde\Omega(s))|\,\mathrm{d}s \le L_B F_B/(2d^2)$; and $|b^\nu| \le \beta/\sqrt{d}$. With $M^* = \max_{\mu\le n}|M_\mu|$ and $a_d = \bar t L_B F_B/(2d) + \bar t\beta/\sqrt{d}$, and since $\mu \le n \le d\bar t$,

$$e_\mu \le \frac{L_B}{d}\sum_{\nu<\mu} e_\nu + a_d + \frac{M^*}{d} \qquad (\mu \le \tau\wedge n).$$

Since $e_0 = 0$, induction on μ gives $e_\mu \le (a_d + M^*/d)(1 + L_B/d)^\mu \le (a_d + M^*/d)\,e^{L_B\bar t}$ for $\mu \le \tau\wedge n$.

Step 5: the martingale. The norm of a vector martingale is a nonnegative submartingale, so Doob's maximal inequality and the orthogonality of martingale increments give $\mathbb{E}(M^*)^2 \le 4\mathbb{E}|M_n|^2 = 4\sum_{\nu<n}\mathbb{E}[\mathbf{1}\{\nu<\tau\}|\xi^{\nu+1}|^2] \le 4nc_2(1+B)$. Hence

$$\mathbb{E}\max_{\mu\le\tau\wedge n} e_\mu \le e^{L_B\bar t}\Big(a_d + 2\sqrt{\bar t c_2(1+B)/d}\Big).$$

Step 6: the bad event. On G, $\tau > n$ by Step 2, so $\max_{\mu\le n} e_\mu = \max_{\mu\le\tau\wedge n} e_\mu$ there. On $\mathcal{K}$, $|Q_{ik}| \le \max_j Q_{jj}$ and $|R_{in}| \le (Q_{ii} + T_{nn})/2$, so $|\Omega^\mu| \le c_3(1 + \max_k Q_{kk}^\mu)$ and $e_\mu \le |\Omega^\mu| + |\Omega^0| + F_B\bar t$. By (15.7) and $\mathbb{E}S^2 \le 2d^2\bar t + d^4\bar t^2$, $\mathbb{E}\max_{\mu\le n} e_\mu^2 \le c_4$ with c_4 independent of d, and the Cauchy–Schwarz inequality gives $\mathbb{E}[\mathbf{1}_{G^c}\max_{\mu\le n} e_\mu] \le (8c_4c_1^2\bar t)^{1/2}/d$. Adding Steps 5 and 6, and using $d \ge 1$, gives $\mathbb{E}\max_{\mu\le n}|\Omega^\mu - \tilde\Omega(\mu/d)| \le C/\sqrt{d}$.

Step 7: the extension solves the equations. Steps 1–6 use the initial vectors only through B_0, so their conclusion holds in every dimension $d' \ge K + M$

for initial vectors with overlaps exactly Ω^0, which exist because $C(\Omega^0)$ is positive semidefinite of size $K+M \le d'$. Fix $t \le \bar{t}$ and let $\mu' = \lfloor d't \rfloor$. In dimension d', on G we have $\Omega^{\mu'} \in \mathcal{K}_{B-1/2}$, so the distance from $\tilde{\Omega}(t)$ to $\mathcal{K}_{B-1/2}$ is at most $\mathbb{E}|\tilde{\Omega}(t) - \Omega^{\mu'}|/\mathbb{P}(G) \le (C/\sqrt{d'} + F_B/d')/\mathbb{P}(G)$, which tends to zero as $d' \to \infty$. Hence $\tilde{\Omega}(t) \in \mathcal{K}_{B-1/2}$ for $t \le \bar{t}$, so $f_B(\tilde{\Omega}) = f(\tilde{\Omega})$ and $\tilde{\Omega}$ solves (15.3). If $\hat{\Omega}$ is another solution in $\mathcal{K}$ with the same start, let t_1 be the supremum of the times up to which $\hat{\Omega} = \tilde{\Omega}$. If $t_1 < \bar{t}$, then by continuity $\hat{\Omega}$ stays in $\mathcal{K}_B$ for a while after t_1, where it solves $\Omega' = f_B(\Omega)$, and uniqueness for f_B contradicts the choice of t_1. So $\Omega = \tilde{\Omega}$, and the bound holds. Finally, ε_g is Lipschitz on $\mathcal{K}_B$ by Lemma 15.4 and bounded on $\mathcal{K}$, and $\mathbb{P}(G^c) \to 0$, which gives the last claim. ∎

The constant, $C = e^{L_B \bar{t}}(\bar{t} L_B F_B/2 + \bar{t}\beta + 2\sqrt{\bar{t} c_2(1+B)}) + (8c_4 c_1^2 \bar{t})^{1/2}$, grows exponentially with the horizon, which is why the theorem says nothing at times of order $\ln d$.[6]

[6] The norm of a vector martingale is a submartingale by Jensen's inequality, and Doob's inequality bounds $\mathbb{E}\max_{\mu \le n}|M_\mu|^2$ by $4\mathbb{E}|M_n|^2$ (Durrett 2019).

Corollary 15.6 (no specialization on fixed horizons). *Let $K = M \ge 2$, $T = I_K$, and $w_k^0 = \sqrt{q_0}\, g_k$ with $g_1, \dots, g_K$ independent $\mathcal{N}(0, I_d)$ vectors, independent of the inputs. Let Ω_{sym} solve (15.3) with $\Omega_{\mathrm{sym}}(0) = (q_0 I_K, 0)$. Then:*

(i) for every $t \ge 0$, $\Omega_{\mathrm{sym}}(t)$ lies in $\mathcal{S} = \{\Omega : Q_{ii} = Q_{11},\ Q_{ik} = Q_{12}\ (i \ne k),\ R_{in} = R_{11}$ for all $i, k, n\}$;

(ii) for every $\bar{t} > 0$ there is a constant C' with $\mathbb{E}\min\{1, \max_{\mu \le d\bar{t}}|\Omega^\mu - \Omega_{\mathrm{sym}}(\mu/d)|\} \le C'/\sqrt{d}$.

In particular, for every $\bar{t}$ and $\epsilon > 0$, the probability that two overlaps R_{in}^μ and R_{jm}^μ with $\mu \le d\bar{t}$ differ by more than ϵ tends to zero: on no fixed horizon do the student units specialize.

Proof. The plan is to deduce (i) from the symmetry of the equations and the uniqueness in Theorem 15.5, and (ii) from the theorem, applied given the initialization, and Grönwall's inequality in the initial condition. For (i), let P and P' be $K \times K$ permutation matrices and $\Gamma(Q, R) = (PQP^\top, PRP'^\top)$. If $(\lambda, \nu) \sim \mathcal{N}(0, C(\Omega))$, then $(P\lambda, P'\nu) \sim \mathcal{N}(0, C(\Gamma\Omega))$, because $P'TP'^\top = T$; and $h(P\lambda, P'\nu) = \Gamma h(\lambda, \nu)$, because $\hat{y}$ and y are symmetric functions of the fields, which are merely relabeled. Hence $f(\Gamma\Omega) = \Gamma f(\Omega)$, and $\Gamma\Omega_{\mathrm{sym}}$ solves the equations with the same start, since $\Gamma(q_0 I, 0) = (q_0 I, 0)$; it stays in $\mathcal{K}$, as $C(\Gamma\Omega)$ is $C(\Omega)$ conjugated by a permutation matrix. By uniqueness $\Gamma\Omega_{\mathrm{sym}} = \Omega_{\mathrm{sym}}$ for every Γ, and the points fixed by every Γ form $\mathcal{S}$. For (ii), let E_d be the event $\max_k Q_{kk}^0 \le 2q_0$; since $dQ_{kk}^0/q_0 \sim \chi^2_d$, Chebyshev's inequality gives $\mathbb{P}(E_d^c) \le 2K/d$. On E_d, Theorem 15.5 with $B_0 = 2q_0$, applied given $g_1, \dots, g_K$, bounds the conditional mean of $\max_{\mu \le d\bar{t}}|\Omega^\mu - \Omega(\mu/d; \Omega^0)|$ by $C/\sqrt{d}$, where $\Omega(\cdot\,; \Omega^0)$ is the solution from Ω^0. That solution and Ω_{sym} stay in $\mathcal{K}_B$ on $[0, \bar{t}]$, where f is L_B-Lipschitz, so Grönwall's inequality gives $|\Omega(t; \Omega^0) - \Omega_{\mathrm{sym}}(t)| \le e^{L_B \bar{t}}|\Omega^0 - (q_0 I, 0)|$, and the moments of Gaussian

vectors give

$$\mathbb{E}|\Omega^0 - (q_0 I, 0)|^2 = \frac{2Kq_0^2 + \frac{1}{2}K(K-1)q_0^2 + K^2 q_0}{d}.$$

The three bounds give (ii). Since all R_{in} are equal on $\mathcal{S}$, $|R_{in}^\mu - R_{jm}^\mu| \le 2|\Omega^\mu - \Omega_{\mathrm{sym}}(\mu/d)|$, and Markov's inequality applied to (ii) gives the last claim. ∎

Numerically $\Omega_{\mathrm{sym}}(t)$ converges to B, so $\lim_{t\to\infty}\lim_{d\to\infty}\varepsilon_{\mathrm{g}}(\Omega^{\lfloor dt\rfloor}) = \varepsilon_\star(\eta) > 0$, the error of B, with the inner limit in probability, while at each finite d the student escapes: the plateau is the statement that the two limits do not commute, and the corollary is its rigorous half. Veiga et al. (2022) remark that the equations, started uncorrelated with the teacher, never specialize.

What the general theorems add. Goldt et al. (2020, Theorem 1.1) state $\max_\mu \mathbb{E}|\Omega^\mu - \Omega(\mu/d)| \le C(\bar{t})/\sqrt{d}$ for activations bounded with two bounded derivatives, with label noise and trained second-layer weights. Veiga et al. (2022) write that their proof fixes "a number of holes" in that argument, in particular in bounding the students' norms, and prove (their Theorem 3.1) a bound for Lipschitz activations with a Lipschitz vector field, with constants that track the width and the scaling of the learning rate, the basis of their phase diagram. Our hypothesis $G_\star < \infty$ makes the norm bound pathwise.

Beyond the horizon: the information exponent. For single neurons the search phase was treated rigorously. Write $a_d \ll b_d$ for $a_d/b_d \to 0$.

Theorem 15.7 (Ben Arous, Gheissari and Jagannath 2021, Theorems 1.3 and 1.4; stated). *Let the population loss be $\phi(m)$, with $\phi' < 0$ on $(0,1)$ and information exponent k, let the sample-wise errors satisfy the moment bounds of their Assumption B, and run spherical online SGD with step-size parameter δ, from the uniform distribution on $\{m \ge 0\}$, for $n = td$ steps, t growing at most polynomially in d.*

(i) If $t^{-1} \ll \delta \ll t^{-1/2}$ and $t \gg 1$ ($k = 1$), $t \gg (\log d)^2$ ($k = 2$), or $t \gg d^{k-2}(\log d)^2$ ($k \ge 3$), then $m(w^n) \to 1$ in probability.

(ii) If $t \ll 1$, $t \ll \log d$ or $t \ll d^{k-2}$ for $k = 1, 2$ or $k \ge 3$, with $\delta = O(1)$ for $k \le 2$ and $\delta = O(t^{-1/2})$ for $k \ge 3$, then $\max_{\mu\le n}|m(w^\mu)| \to 0$ in probability.

So $n \gg d$, $d\log^2 d$, $d^{k-1}\log^2 d$ samples suffice and $n \ll d$, $d\log d$, d^{k-1} do not, each for its range of step sizes; for $k \ge 2$ almost all samples are spent in the search phase (their Theorem 1.5), and the authors expect the thresholds to be sharp without the logarithms. The bound d^{k-1} belongs to vanilla online SGD, not to gradient methods: online SGD on a smoothed correlation loss reaches population loss ϵ with $n \asymp d^{k/2} + d/\epsilon$ samples up to polylogarithmic factors, for $k \ge 3$ and an initial overlap of order $d^{-1/2}$ (Damian et al. 2023, Theorem 1), matching the lower bound $d^{k/2}$ for correlational statistical queries; a two-layer network trained by an SGD-based algorithm that reuses minibatches learns polynomial single-index

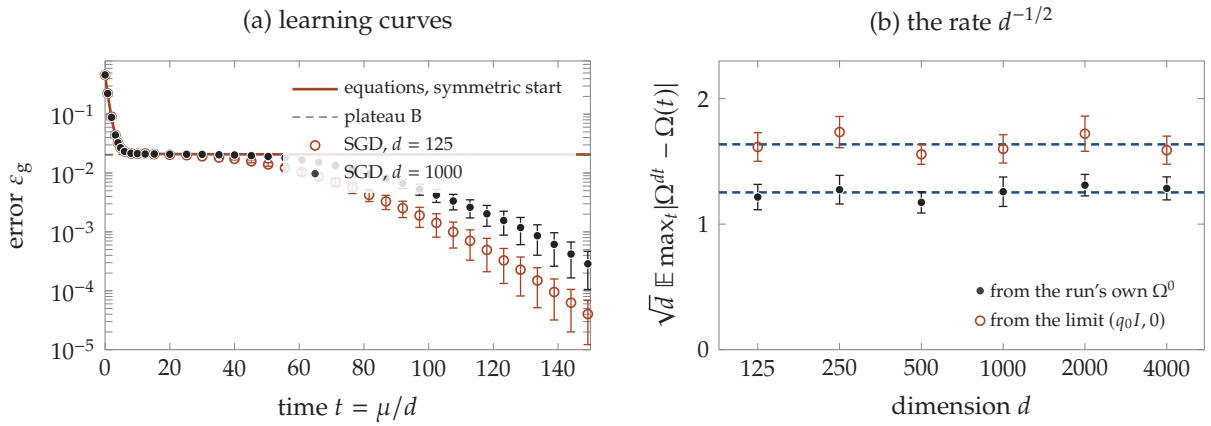


Figure 15.1. Online SGD for the soft committee machine ($K = M = 2$, erf, $T = I$, $\eta = 1$, $q_0 = \frac{1}{4}$), simulated exactly in law through Lemma 15.3. (a) Mean error over 100 runs at $d = 125$ (hollow) and $d = 1000$ (filled), with Student-t 95% intervals (seed 1515), against the solution of (15.3) from the random-start limit $(q_0 I, 0)$ (carnelian), which Corollary 15.6 proves to be the limit on every fixed horizon, and the error 0.020657 of the fixed point B (dashed). (b) $\sqrt{d}$ times the mean, over 40 runs per d (seed 2026), of $\max_t|\Omega^{dt} - \Omega(t)|$ over $t = 0.2, 0.4, \dots, 10$, with 95% intervals, for the solution from each run's own initial overlaps (filled; Theorem 15.5) and from $(q_0 I, 0)$ (hollow; Corollary 15.6); lapis dashed: the rate $C/\sqrt{d}$, with C the mean over d. Scripts: `15-sgd-plateau.py`, `15-sgd-rate.py`, `15-sgd-ode.py`.

targets with $n \simeq d$ polylog d (Lee et al. 2024); and two passes over a batch find directions that one pass cannot (Dandi et al. 2024).

15.5 Measuring it

The committee machine has $K = M = 2$, orthonormal teacher units, $\eta = 1$ and the random start $w_k^0 = g_k/2$, so $q_0 = \frac{1}{4}$. Direct simulation in $\mathbb{R}^d$ costs $O(d)$ per step; Lemma 15.3 reduces it to $O(1)$, exactly in law: given Ω^μ, draw independent $\nu, \zeta \sim \mathcal{N}(0, I_2)$, set $\lambda = R\nu + L\zeta$ with $LL^\top = Q - RR^\top$ and $|x|^2 = |\nu|^2 + |\zeta|^2 + \chi^2_{d-4}$, and apply (15.4). Against direct SGD in $\mathbb{R}^{200}$ (200 runs) the chain (400 runs) agrees on all seven overlaps at $t = 2, 4, 6, 8$ within 1.91 standard errors. The single neuron reduces to a chain for m in the same way. The four scripts have fixed seeds and run in under a minute each.

THE EQUATIONS ON FIXED HORIZONS. In Figure 15.1(a) both dimensions follow the solution from the random-start limit down to the plateau (at $t = 10$, 0.02126 and 0.02136 against 0.02130); then the simulations leave it, the smaller dimension first, while the equations, by Corollary 15.6 the correct limit at every fixed time, reach 0.02066, the error of B, and stay. In Figure 15.1(b) the largest deviation on $[0, 10]$ falls from 0.109 at $d = 125$ to 0.020 at $d = 4000$, and $\sqrt{d}$ times it stays between 1.17 and 1.31, every interval containing the average 1.25: the rate of Theorem 15.5. From the symmetric limit, which misses the $d^{-1/2}$ asymmetry of the start, the constant

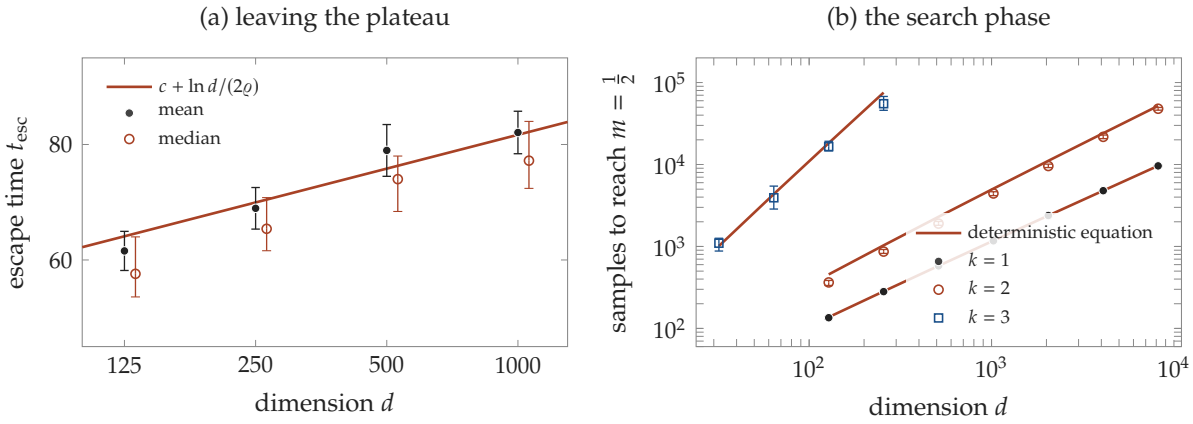


Figure 15.2. (a) Time at which online SGD for the committee machine of Figure 15.1 leaves the plateau, the first time on a grid of spacing 0.4 with $\varepsilon_{\rm g} < \varepsilon_\star/2$, $\varepsilon_\star = 0.020657$; 100 runs per d (seed 1515), all escaped before $t = 180$: means with Student-t 95% intervals (filled) and medians with distribution-free 95% intervals from binomial order statistics (hollow, drawn 6% to the right). Carnelian: the heuristic (15.5), $c + \ln d/(2\varrho)$ with $\varrho = 0.05914$ from B and c fitted to the means. (b) Median number of samples until spherical online SGD reaches overlap $\frac{1}{2}$ for $y = \mathrm{He}_k(\langle\theta\rangle x)/\sqrt{k!}$ with the correlation loss, information exponent $k = 1, 2, 3$, step sizes $\delta = \frac{1}{2}$ for $k \le 2$ and $\delta = 1/(2\sqrt{d})$ for $k = 3$; 200 runs per d (seeds 301–303), distribution-free 95% intervals. Carnelian: the deterministic equation (15.6) from the median initial overlap $0.674/\sqrt{d}$. Scripts: `15-sgd-plateau.py`, `15-sgd-exponent.py`.

is 1.56 to 1.73.

THE ESCAPE. In Figure 15.2(a) the mean escape times at $d = 125, 250, 500, 1000$ are 61.6, 68.9, 79.0, 82.1 (±3.4 to ±4.5); a line in $\ln d$, weighted by their standard errors, fits them with slope 10.2 ± 2.3 (95%), against $1/(2\varrho) = 8.45$, and the medians give 9.7. At $t = 40$, on the plateau, $\sqrt{d}\,\mathbb{E}|\epsilon|$ is 2.86, 3.02, 2.82, 3.14: the asymmetry is of order $d^{-1/2}$, as assumed. The pooled standard deviation of the escape times is 19.1, against $\pi/(2\sqrt{2}\varrho) = 18.8$, and the 10%, 25%, 50%, 75%, 90% quantiles of the centered times are −19.3, −13.7, −4.5, 7.9, 24.0, against −19.2, −13.1, −4.1, 8.6, 24.3 for $-(\ln|Z| - \mathbb{E}\ln|Z|)/\varrho$. A factor eight in d cannot separate $\ln d$ from a small power of d; the slope, the spread and the shape of the law together are the evidence.

THE SEARCH PHASE. In Figure 15.2(b) the fitted exponents of d are 1.03, 1.17, 1.90 for $k = 1, 2, 3$. For $k = 1$ the samples divided by d rise from 1.06 to 1.17 toward 1.192, the time the limiting equation takes from $m = 0$ to $\frac{1}{2}$; the equation from the median start is within 1.5%. For $k = 2$ they grow linearly in $\ln d$ with slope 0.71 ± 0.06, against $1/(2r) = 0.667$, and the deterministic equation overestimates them by 8 to 25%: at the scale $d^{-1/2}$ the noise is as large as the drift. For $k = 3$ the median divided by d^2 stays between 0.84 and 1.07; 7 to 14% of the runs are still short of $\frac{1}{2}$ after $4d^2$ steps. Our step sizes satisfy part (ii) of Theorem 15.7, not the window of part (i): the experiment illustrates the scaling, not the theorem.

15.6 The status board

On fixed horizons the equations are a theorem and a random start provably cannot specialize; the $\ln d$ length of the plateau is a heuristic that the measurements support. The thresholds of online SGD are proved up to logarithms; they belong to one algorithm, not to gradient methods.

STATUS BOARD · order-parameter equations ↔ concentration of summary statistics

PREDICTION	STATUS	PROOF OR EVIDENCE
Saad–Solla equations on fixed horizons, erf-type activations, rate $d^{-1/2}$	proved	Goldt et al. (2020, Thm. 1.1), with gaps fixed by Veiga et al. (2022, Thm. 3.1); Theorem 15.5 for $G_\star < \infty$
Plateau as $\eta \to 0$: $\varepsilon_\star = \frac{K}{\pi}(\frac{\pi}{6} - K \arcsin \frac{1}{2K})$, escape rate $\frac{2K\eta}{\pi\sqrt{2K-1}(2K+1)^{3/2}}$	proved	a fixed point and an eigenvalue of the equations: Saad and Solla (1995a); Computation 15.2
No specialization on a fixed horizon from a random start	proved	Corollary 15.6 (this book); noted for the equations by Veiga et al. (2022)
Plateau length $(\ln d)/(2\varrho) + O(1)$, spread $\pi/(2\sqrt{2}\varrho)$	heuristic	Biehl et al. (1996) for the dependence on the start; observed here ($d \le 1000$): slope 10.2 ± 2.3 against 8.45, spread 19.1 against 18.8
Spherical online SGD from a random start: $n \gg d$, $d \log^2 d$, $d^{k-1} \log^2 d$ samples suffice, $n \ll d$, $d \log d$, d^{k-1} do not ($k = 1, 2, \ge 3$)	proved	Ben Arous et al. (2021, Thms. 1.3–1.4), each in a range of step sizes (Theorem 15.7)
The same thresholds without logarithmic factors	conjectured	expected by Ben Arous et al. (2021); observed here: exponents 1.03, 1.17, 1.90
$d^{k/2}$ samples suffice for online SGD on a smoothed loss ($k \ge 3$)	proved	Damian et al. (2023, Thm. 1), up to polylogarithmic factors
d^{k-1} samples are necessary for every gradient method	false	smoothing (Damian et al. 2023); reuse of minibatches gives d polylog d for polynomial single-index targets (Lee et al. 2024)

15.7 Where the translation fails

What was assumed. The theorem uses exactly Gaussian inputs, through the conditional law of the fields, where the physics needs a central limit theorem; for other inputs the overlaps are not a Markov chain. It uses fresh samples; with reuse, memory appears (Chapter 14). It fixes K, M and η as $d \to \infty$, the classical scaling, outside which other limits hold (Arnaboldi et al. 2023b; Veiga et al. 2022). It needs a bounded activation with three bounded derivatives and $G_\star < \infty$, which excludes the rectified linear unit, whose vector field is not Lipschitz at degenerate overlaps (Veiga et al. 2022), and the Hermite targets of the search phase; noiseless labels (Exercise 15.7); a start independent of the data; and a fixed horizon. The teacher has the student's architecture, an idealization.

Where the physics is wrong, or not yet right. The equations are deterministic, and at the plateau the dynamics is governed by the fluctuations of order $d^{-1/2}$ that they discard: from a symmetric start they describe a student that never learns, the correct limit at fixed t (Corollary 15.6) and wrong at every finite d at times of order $\ln d$. The physicist puts in by hand the asymmetry and its size, $d^{-1/2}$ times a random factor that includes the noise of SGD. For the search phase the deterministic equation gets the exponent right and the constant wrong, by up to 25% for $k = 2$. The cost d^{k-1} belongs to vanilla online SGD, not to the problem: smoothing gives $d^{k/2}$, reuse beats the information exponent (Damian et al. 2023; Dandi et al. 2024; Lee et al. 2024), and the lower bounds for statistical-query and low-degree algorithms are $d^{k_\star/2}$, with the generative exponent $k_\star \le k$ (Damian et al. 2024), a COLT extended abstract whose proofs are in its arXiv version.

Open problems. A proof, for the soft committee machine from a random start, that $t_{\rm esc} - (\ln d)/(2\varrho)$ converges in law, to the law of $c - \varrho^{-1} \ln|\mathcal{A}|$ of (15.5); Veiga et al. (2022, Section 3) leave the extra $\log d$ time scale of two-layer networks open, and Arnaboldi et al. (2023a) give prefactors for $d \log d$ in a related problem in an unrefereed preprint. Whether the symmetric solution converges to B for every learning rate below the critical one, which we see only numerically. Theorem 15.5 without $G_\star < \infty$, for the rectified linear unit through degenerate overlaps, and for non-Gaussian inputs. The thresholds of Theorem 15.7 without logarithmic factors.

Exercises

15.1 consolidation. (The updates.) Derive (15.4) from (15.1), and show that two student units that are equal at some step stay equal, so that a student started with identical units never specializes, at every d.

15.2 consolidation. (Linear networks.) For $K = M = 1$, $T = 1$ and $g(z) = z$, show that the equations read $\mathrm{d}R/\mathrm{d}t = \eta(1 - R)$, $\mathrm{d}Q/\mathrm{d}t = 2\eta(R - Q) + \eta^2(1 - 2R + Q)$, and that $\varepsilon_{\rm g} = \frac{1}{2}(Q - 2R + 1)$ satisfies $\mathrm{d}\varepsilon_{\rm g}/\mathrm{d}t = -\eta(2 - \eta)\varepsilon_{\rm g}$: the error decays for $\eta < 2$, fastest at $\eta = 1$. Which hypothesis of Assumption 15.1 fails, and where does the proof use it?

15.3 CONSOLIDATION. (Larger committees.) Evaluate Computation 15.2 for $K = 3$ ($Q_{ik} = \frac{1}{5}$, $\varepsilon_\star = 0.020297$, $\varrho = 0.04612\,\eta$), and show that $\varepsilon_\star = (\frac{\pi}{6} - \frac{1}{2})\frac{K}{\pi} - \frac{1}{48\pi K} + O(K^{-3})$.

15.4 CONSOLIDATION. (Symmetry.) For $K = M$ and $T = I$, show that the overlaps invariant under relabeling student and teacher units by the same permutation form a four-dimensional invariant subspace of the equations, containing $\mathcal{S}$ and the point $Q = R = I$.

15.5 BY HAND. (The four-point average.) Carry out step 5 of Computation 15.1, and check it for $u = v = w = z \sim \mathcal{N}(0, c)$: $\mathbb{E}[g'(u)^2 g(u)^2] = 4\arcsin(c/(1+3c))/(\pi^2\sqrt{1+2c})$, which is 0.059125 at $c = 1$.

15.6 BY HAND. (The escape rate.) Carry out step 6 of Computation 15.2 for general K. *(Hint: The rows of E sum to zero; at the plateau $\Lambda_\star = (2K+1)/K$, $1 + q = 2K/(2K-1)$, $1 + 2q = (2K+1)/(2K-1)$.)*

15.7 BY HAND. (Label noise.) For labels $y = \sum_n g(\nu_n) + \sigma_\varepsilon \varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$ independent, show that (15.3) gains only the term $\eta^2\sigma_\varepsilon^2 \mathbb{E}[g'(\lambda_i)g'(\lambda_k)]$ in $\mathrm{d}Q_{ik}/\,\mathrm{d}t$, so perfect learning is not a fixed point. Which bounds of Lemma 15.2 fail when Δ_k is unbounded, and what replaces them?

15.8 BY HAND. (The search phase.) Derive (15.6), and show that $\mathbb{E}[a^2] = 1$ for $k = 1$ and $2 + 8m^2$ for $k = 2$. For $k = 1$ and $\delta = \frac{1}{2}$, show that the solution from $m = 0$ reaches $\frac{1}{2}$ at $t = 1.1919$; for $k = 2$, solve $\mathrm{d}m/\,\mathrm{d}t = m(r - cm^2)$, $r = \delta(2-\delta)$, $c = 2\delta + 4\delta^2$.

15.9 TRANSLATION. (Self-averaging as Doob's inequality.) Identify the steps of the proof of Theorem 15.5 that do the work of the two inequalities $1 \ll L$ and $L \ll d$ of the physical derivation, and explain why the proof needs no window.

15.10 TRANSLATION. ★ (Beyond Gaussian inputs.) For inputs with independent uniform ± 1 coordinates, show by an example with $K = 1$ that the conditional law of λ_1 given the past is not a function of Q_{11}. Which step of the proof of Theorem 15.5 fails, and what must a quantitative central limit theorem for the fields provide?

15.11 MEASUREMENT. (A smaller learning rate.) Run `15-sgd-plateau.py` with $\eta = \frac{1}{2}$, with the plateau error and escape rate of B from Table 15.1; compare the slope of the mean escape time with $1/(2\varrho) = 15.8$ and the spread with $\pi/(2\sqrt{2}\varrho) = 35.1$, and say why the horizon must grow.

15.12 RESEARCH. (The plateau length.) Prove that for $K = M = 2$, erf, $T = I$ and a random start, $t_{\mathrm{esc}} - (\ln d)/(2\varrho)$ converges in law, and identify the limit. *(Hint: Combine Theorem 15.5 up to a time at which the solution is near B with a linearization at B controlled at the scale $d^{-1/2}$, as Ben Arous et al. (2021) control the search phase for $k = 2$.)*

Notes and References

In 1995 Biehl and Schwarze (1995) found exponential convergence at small learning rates and a critical rate beyond which perfect learning is impossible; Saad and Solla (1995b,c) treated realizable, over-realizable and unrealizable cases and the role of the learning rate; Riegler and Biehl (1995) made the hidden-to-output weights adaptive; and Biehl et al. (1996) found several fixed points, cascades of plateaus, and fixed points that split as the learning rate changes. Engel and Van den Broeck (2001, Chapters 9 and 13) give the textbook account.

The ODE method of stochastic approximation (Ljung 1977) has a fixed dimension and a vanishing step; here the dimension grows, so it applies to the overlaps, not to the weights. Wang et al. (2017) proved scaling limits for online regression and principal component analysis, in an unrefereed preprint; Goldt et al. (2020) used the equations for over-parameterized students; Veiga et al. (2022) drew the phase diagram in the width and the learning rate; Arnaboldi et al. (2023b) unified the classical, high-dimensional and mean-field regimes; Ben Arous et al. (2024, Theorem 2.3) proved ballistic and diffusive limits for localizable and asymptotically closable summary statistics, with a critical step size; and Collins-Woodfin et al. (2024) gave deterministic equivalents of streaming SGD for generalized linear models. For several passes over the data, dynamical mean-field theory takes over (Chapter 14).

Theorem 15.5 arranges the argument of Goldt et al. (2020) and Veiga et al. (2022); to our knowledge the hypothesis $G_\star < \infty$, the derivation of well-posedness from SGD, Lemma 15.3 with its use for simulation, Corollary 15.6, and the law (15.5) of the escape time are new here. The plateau error and the escape rate of Computation 15.2 are equation (7) and the time constant of Saad and Solla (1995a), who also saw the students separate at finite learning rates; we computed B and its stability, which may be among the split fixed points of Biehl et al. (1996), a paper we have not read in full.

TRANSLATING THE LITERATURE. Goldt et al. (2020) write N for our d, use our λ, ν, Q, R, T, measure time by μ/N, and add second-layer weights and label noise. Veiga et al. (2022) write p for the number of hidden units, γ for the learning rate, and Ω for the overlap matrix that contains Q, R and T. Ben Arous et al. (2021) write N for the dimension, M for the number of samples, $\alpha = M/N$ for our time t, X_t for the iterate, θ_N for the vector to be learned, $m_N(x) = x \cdot \theta_N$ for our overlap m, and δ_N/N for our step δ/d; their thresholds $\alpha_c(N, k) = 1, \log N, N^{k-2}$ are thresholds in our t. Damian et al. (2023) write $k^\star$ for the information exponent.

References

Arnaboldi, L., F. Krzakala, B. Loureiro, and L. Stephan (2023a). *Escaping mediocrity: how two-layer networks learn hard generalized linear models with SGD*. arXiv: 2305.18502.

Arnaboldi, L., L. Stephan, F. Krzakala, and B. Loureiro (2023b). "From high-dimensional & mean-field dynamics to dimensionless ODEs: A unifying approach to SGD in two-layers networks". In: *Proceedings of Thirty Sixth Conference on Learning Theory*. Vol. 195. Proceedings of Machine Learning Research, pp. 1199–1227. arXiv: 2302.05882.

Ben Arous, G., R. Gheissari, and A. Jagannath (2021). "Online stochastic gradient descent on non-convex losses from high-dimensional inference". In: *Journal of Machine Learning Research* 22.106, pp. 1–51. arXiv: 2003.10409.

Ben Arous, G., R. Gheissari, and A. Jagannath (2024). "High-dimensional limit theorems for SGD: Effective dynamics and critical scaling". In: *Communications on Pure and Applied Mathematics* 77.3, pp. 2030–2080. arXiv: 2206.04030.

Biehl, M. and H. Schwarze (1995). "Learning by on-line gradient descent". In: *Journal of Physics A: Mathematical and General* 28.3, pp. 643–656.

Biehl, M., P. Riegler, and C. Wöhler (1996). "Transient dynamics of on-line learning in two-layered neural networks". In: *Journal of Physics A: Mathematical and General* 29.16, pp. 4769–4780.

Collins-Woodfin, E., C. Paquette, E. Paquette, and I. Seroussi (2024). "Hitting the High-dimensional notes: an ODE for SGD learning dynamics on GLMs and multi-index models". In: *Information and Inference: A Journal of the IMA* 13.4, iaae028. arXiv: 2308.08977.

Damian, A., E. Nichani, R. Ge, and J. D. Lee (2023). "Smoothing the Landscape Boosts the Signal for SGD: Optimal Sample Complexity for Learning Single Index Models". In: *Advances in Neural Information Processing Systems 36*, pp. 752–784. arXiv: 2305.10633.

Damian, A., L. Pillaud-Vivien, J. D. Lee, and J. Bruna (2024). "Computational-Statistical Gaps in Gaussian Single-Index Models". In: *Proceedings of Thirty Seventh Conference on Learning Theory*. Vol. 247. Proceedings of Machine Learning Research, p. 1262. arXiv: 2403.05529.

Dandi, Y., E. Troiani, L. Arnaboldi, L. Pesce, L. Zdeborová, and F. Krzakala (2024). "The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks: Breaking the Curse of Information and Leap Exponents". In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research, pp. 9991–10016. arXiv: 2402.03220.

Durrett, R. (2019). *Probability: Theory and Examples*. 5th ed. Cambridge University Press.

Engel, A. and C. Van den Broeck (2001). *Statistical Mechanics of Learning*. Cambridge University Press.

Goldt, S., M. S. Advani, A. M. Saxe, F. Krzakala, and L. Zdeborová (2020). "Dynamics of stochastic gradient descent for two-layer neural networks in the teacher–student setup". In: *Journal of Statistical Mechanics: Theory and Experiment* 2020.12, p. 124010. arXiv: 1906.08632.

Lee, J. D., K. Oko, T. Suzuki, and D. Wu (2024). "Neural network learns low-dimensional polynomials with SGD near the information-theoretic limit". In: *Advances in Neural Information Processing Systems 37*, pp. 58716–58756. arXiv: 2406.01581.

Ljung, L. (1977). "Analysis of recursive stochastic algorithms". In: *IEEE Transactions on Automatic Control* 22.4, pp. 551–575.

Riegler, P. and M. Biehl (1995). "On-line backpropagation in two-layered neural networks". In: *Journal of Physics A: Mathematical and General* 28.20, pp. L507–L513.

Saad, D. and S. A. Solla (1995a). "Dynamics of On-Line Gradient Descent Learning for Multilayer Neural Networks". In: *Advances in Neural Information Processing Systems 8*. MIT Press, pp. 302–308.

Saad, D. and S. A. Solla (1995b). "Exact Solution for On-Line Learning in Multilayer Neural Networks". In: *Physical Review Letters* 74.21, pp. 4337–4340.

Saad, D. and S. A. Solla (1995c). "On-line learning in soft committee machines". In: *Physical Review E* 52.4, pp. 4225–4243.

Sheppard, W. F. (1899). "On the application of the theory of error to cases of normal distribution and normal correlation". In: *Philosophical Transactions of the Royal Society of London. Series A* 192, pp. 101–167.

Veiga, R., L. Stephan, B. Loureiro, F. Krzakala, and L. Zdeborová (2022). "Phase Diagram of Stochastic Gradient Descent in High-Dimensional Two-Layer Neural Networks". In: *Advances in Neural Information Processing Systems 35*, pp. 23244–23255. arXiv: 2202.00293.

Wang, C., J. Mattingly, and Y. M. Lu (2017). *Scaling Limit: Exact and Tractable Analysis of Online Learning Algorithms with Applications to Regularized Regression and PCA*. arXiv: 1712.04332.

Epilogue

The stone, read across

The prologue derived one number, the capacity 2 of the perceptron, in four languages, and found that the physicist's replicas and the probabilist's comparison inequality, set side by side, do the same work in the same order. Fifteen chapters later the same exercise has been carried out on fifteen predictions, from the free entropy of the simplest disordered systems to the plateaus of stochastic gradient descent. Each chapter closed its account with a status board that says, in one word per prediction, how much of it is known. Read one at a time, the boards record theorems and conjectures. Read side by side, they show a pattern: which physical steps the proofs justify, which steps they replace, and which steps they have not been able to touch. This epilogue sets the boards side by side, reads the stone across the chapters, and collects what is false, what rests on unrefereed preprints, and what is open.

The fifteen predictions

Tables E.1 and E.2 list the main prediction of the prologue and of each chapter, the derivation that produced it, the route by which it was proved, and the status of the stone at the foot of the opening page. The precise statements, with their hypotheses, are in the chapters.

Of the sixteen stones of the prologue and the chapters, eleven read *proved* and five read *in part*. None reads *conjectured* or *false*: every main prediction of the book is a theorem in at least part of its range. This reflects the plan, which chose predictions whose proofs a chapter could present, not the state of the field. The boards are less selective. They list 149 predictions in all, the statements that the chapters derived or discussed, and Figure E.1 counts their statuses. Seventy-one are proved, twenty-four more are proved under extra hypotheses, nine are proved only in unrefereed preprints, fifteen are conjectured, eight are heuristic, nineteen are known to be false, and three are open.

Two features of the counts stand out. The first is how much is proved: almost two thirds of the rows, 95 of 149, are theorems, with or without extra hypotheses. Physicists derived most of these first, and in most cases a decade or more passed before the proof: Gardner's capacity of 1988 waited until 2003 for $\kappa > 0$ (for patterns with entries ± 1), Parisi's formula of 1979 until 2006, the Saad–Solla equations of 1995 until 2020, with gaps that Veiga et al. (2022) closed, and the threshold energy that Cugliandolo and Kurchan (1993) found in the dynamics until Sellke (2024), for Langevin dynamics of the pure spherical p-spin on time scales that do not grow with the dimension, in the limit of zero temperature. The exceptions are instructive. The transition of spiked covariance matrices was found by physicists (Hoyle and Rattray 2004) and by mathematicians (Baik et al. 2005) within three

	THE PREDICTION	DERIVED BY	PROVED BY	STONE
P	the capacity of the spherical perceptron: 2 at $\kappa = 0$, and $\alpha_c(\kappa)$ for $\kappa \geq 0$	Gardner's replicas (1988), after Cover's count at $\kappa = 0$	Cover's count at $\kappa = 0$; Gordon's comparison (Stojnic 2013); the volume (Shcherbina and Tirozzi 2003); conic geometry (Amelunxen et al. 2014)	proved
PART I. TWO LANGUAGES				
1	the free entropy self-averages and is a saddle point	Hubbard–Stratonovich and the replica trick	Gaussian concentration; Laplace's method on types	proved
2	the REM freezes at $\beta_c = \sqrt{2 \log 2}$, with one-step breaking below	Derrida (1980, 1981); Gross and Mézard (1984)	first and second moments; Poisson–Dirichlet weights (Bovier and Kurkova 2004; Ruelle 1987)	proved
3	the SK free entropy: replica symmetric above the de Almeida–Thouless line, Parisi's formula everywhere	Sherrington and Kirkpatrick (1975); de Almeida and Thouless (1978); Parisi (1979)	interpolation: Guerra and Toninelli (2002), Guerra (2003), Talagrand (2006); the line with a field in 2026 preprints	proved
PART II. INFERENCE ON THE NISHIMORI LINE				
4	the mutual information of $\mathbb{Z}_2$ synchronization is replica symmetric	the replica formula on the Nishimori line (Lesieur et al. 2017)	interpolation and overlap concentration (Barbier and Macris 2019; Deshpande et al. 2016; Lelarge and Miolane 2019)	proved
5	PCA sees the spike above $\lambda = 1$: eigenvalue $\lambda + 1/\lambda$, overlap $1 - 1/\lambda^2$	an eigenvalue can split off when the entries have a mean (Jones et al. 1978); replicas for covariances (Hoyle and Rattray 2004)	large means (Füredi and Komlós 1981); at the threshold (Baik et al. 2005; Benaych-Georges and Nadakuditi 2011; Féral and Péché 2007), here by the resolvent	proved
6	AMP's iterates are Gaussian with the variance of state evolution	the cavity and TAP (Donoho et al. 2009; Thouless et al. 1977)	Gaussian conditioning (Bayati and Montanari 2011; Bolthausen 2014), for a fixed number of iterations	proved
7	the Bayes error of the teacher–student perceptron; perfect learning of the Ising teacher at $\alpha_{\mathrm{IT}} = 1.2447$	Gardner and Derrida (1989); Györgyi (1990); Sompolinsky et al. (1990)	adaptive interpolation (Barbier et al. 2019), under hypotheses (h1)–(h5)	in part

Table E.1. The main prediction of the prologue and of each chapter of Parts I and II, who derived it and how, who proved it and how, and the status of the stone at the foot of the opening page ("in part": proved under extra hypotheses or in part of the range).

	THE PREDICTION	DERIVED BY	PROVED BY	STONE
PART III. COMPARISON AND CONVEXITY				
8	Gardner's capacity for every margin	Gardner's volume (1988)	Gordon's min–max and the CGMT, for $\kappa \geq 0$ (Gordon 1985; Stojnic 2013; Thrampoulidis et al. 2018)	 in part
9	the ridge risk and its peak at $\alpha = 1$	replicas at zero temperature; random matrices (Krogh and Hertz 1992)	the Marchenko–Pastur resolvent (Dobriban and Wager 2018; Hastie et al. 2022); the CGMT	 proved
10	the LASSO risk as a scalar fixed point; the Donoho–Tanner threshold	replicas (Kabashima et al. 2009; Rangan et al. 2012)	state evolution (Bayati and Montanari 2012); the CGMT (Thrampoulidis et al. 2018); conic geometry (Amelunxen et al. 2014)	 proved
11	the logistic MLE exists exactly above $\alpha_{\mathrm{MLE}}(r)$; the max-margin error	statisticians, without physics (Candès and Sur 2020); the error: Opper et al. (1990); Gardner's volume with a teacher in Chapter 11	the kinematic formula (Candès and Sur 2020); the CGMT (Montanari et al. 2025)	 proved
12	kernel learning curves through an effective ridge; Gaussian equivalence	replicas and cavities (Bordelon et al. 2020; Canatar et al. 2021; Gerace et al. 2020)	deterministic equivalents (Cheng and Montanari 2024); universality (Hu and Lu 2023; Mei and Montanari 2022)	 in part
PART IV. LANDSCAPES AND DYNAMICS				
13	the complexity of the spherical p-spin; threshold $-E_\infty$ above the ground state $-E_0$	TAP complexity (Cavagna et al. 1998; Crisanti and Sommers 1995)	Kac–Rice and the GOE (Auffinger et al. 2013); the second moment (Subag 2017)	 in part
14	closed equations for correlation and response; dynamics stops at the threshold	the generating functional (Crisanti et al. 1993; Cugliandolo and Kurchan 1993)	large deviations (Ben Arous et al. 2006); Sellke (2024); conditioning (Gerbelot et al. 2024)	 in part
15	the Saad–Solla equations and the plateau of online SGD	order-parameter equations (Biehl and Schwarze 1995; Saad and Solla 1995)	martingales on fixed horizons (Goldt et al. 2020; Veiga et al. 2022)	 proved

Table E.2. The main prediction of each chapter of Parts III and IV, as in Table E.1.

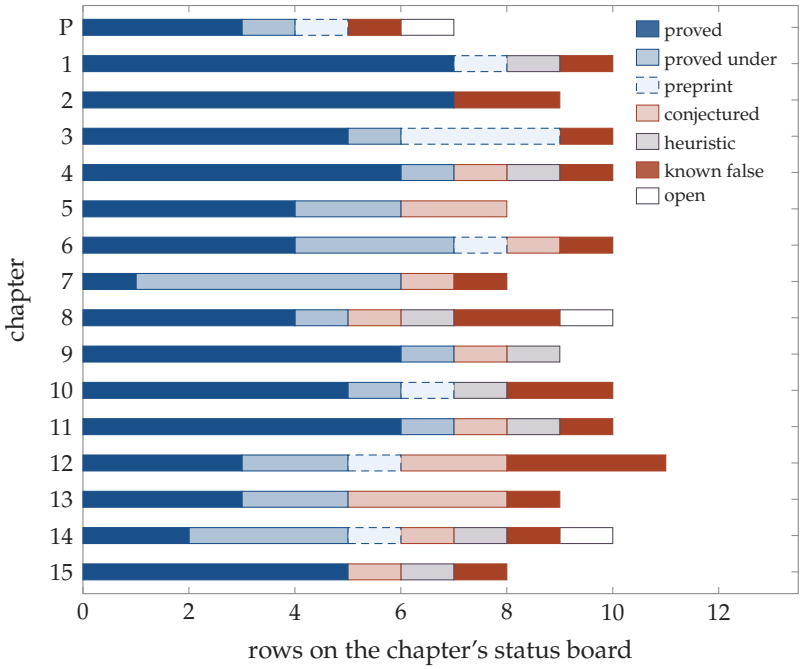


Figure E.1. The statuses on the boards of the prologue (P) and the fifteen chapters, counted by `figures/scripts/16-epilogue-boards.py`, which reads every `\statusrow` of the chapters' sources. Lapis: proved, in a refereed source or in the book (filled), under extra hypotheses (light), or only in an unrefereed preprint (dashed). Carnelian: conjectured (light) or known to be false (filled). Granite: heuristic (light) or open (white).

days of each other, at the turn of February and March 2004 (Chapter 5); the theory of the logistic MLE was derived and proved by statisticians, with conic geometry, message passing and the leave-one-out analysis of robust regression, and no replica computation preceded it (Chapter 11).

The second feature is where the rows that are not proved sit. They cluster in four places: below the de Almeida–Thouless line and its analogues, where replica symmetry breaks (Chapters 3 and 8); in the hard phases of inference, where the question is not what can be learned but what can be learned efficiently (Chapters 5 and 7); away from Gaussian data, where universality fails or is not proved (Chapters 10 and 12); and in Part IV, where the question is where an algorithm goes after a time that grows with the dimension (Chapters 13 to 15).

The stone, read across

Each chapter's stone paired the steps of one derivation with the steps of one proof. The same pairs recur from chapter to chapter, and the stone below collects them.

PHYSICS · *the physical derivation*	PROOF · *the rigorous route*
1 · Replicate	
Compute $\mathbb{E}Z^n$ for integer n: n copies of the system, coupled by the disorder, which interact only through their overlaps Q_{ab} (Chapters 1 to 3 and 8).	Interpolate: a path from the model to a decoupled system whose derivative is an overlap fluctuation (Chapters 3, 4 and 7); or compare: Gordon's inequality replaces the Gaussian matrix by two vectors (Chapters 8 to 11). At $n = 2$ the replica computation is the second moment method (Chapter 2).
2 · Assume replica symmetry	
All off-diagonal overlaps equal one number q, the overlap of two typical configurations.	A reason for the overlap to concentrate: high temperature (Chapter 3), the Nishimori identity with a small perturbation (Chapters 4 and 7), convexity (Chapters 8 to 11), or a fixed time horizon (Chapters 6, 14 and 15). Where there is none, the ansatz fails.
3 · Find the saddle point	
Extremize the replicated action in q; choose the extremum by physical arguments, often a minimum where Laplace's method would ask for a maximum.	Matching bounds: an inequality with a remainder of fixed sign for every q, such as Guerra's bound, and a converse (Aizenman–Sims–Starr, adaptive interpolation, the reverse direction of the CGMT). The theorem decides between minimum and maximum.
4 · Break the symmetry	
Parisi's hierarchy of overlaps; one step in the REM, with $m = \beta_c/\beta$; a continuum in SK; ultrametricity.	Poisson–Dirichlet weights and Ruelle's cascades (Chapter 2); Guerra's broken bound and Talagrand's converse (Chapter 3); the Ghirlanda–Guerra identities and Panchenko's theorem.
5 · Continue to $n \to 0$	
Take the result for integer n, continue it analytically, and divide by n.	No counterpart in this book: no proof continues in n. In the REM the exact moment function is not analytic in n (Chapter 2).
6 · Remove one variable	
The cavity: the field on a removed spin or sample is Gaussian once the Onsager reaction is subtracted (Chapters 6, 10 and 11).	Gaussian conditioning on the past of an iteration (Chapters 6 and 14); leave-one-out identities for resolvents (Chapters 9 and 12).

7 · Count the states	
The complexity of TAP solutions, or a replicated Kac–Rice computation (Chapter 13).	The Kac–Rice formula with the GOE determinant for the mean; Subag's second moment for the typical count (Chapter 13).
8 · Follow the dynamics	
A generating functional, or a dynamical cavity, closes equations for the correlation and the response (Chapter 14); for online learning, ordinary differential equations for the order parameters (Chapter 15).	Large deviations of the empirical process; conditioning along an iteration; martingales and Grönwall's lemma (Chapters 14 and 15). All on fixed time horizons.

Read down the right-hand column, and three things are visible. First, the proofs rarely follow a derivation line by line, but they often follow its architecture. Guerra's interpolation, Gordon's comparison and Bolthausen's conditioning each take one step of the physics, replication, the Gaussian volume integral and the cavity, and turn it into an identity or an inequality whose remainder has a sign. Second, the replica-symmetric ansatz never becomes a lemma. Every proof of a replica-symmetric formula in this book replaces the ansatz by a reason for the overlap to concentrate, and the book met four reasons: high temperature, Bayes optimality, convexity, and a fixed horizon. Where none of them holds, as below the de Almeida–Thouless line, for the perceptron with a negative margin, and in the long-time dynamics of rugged landscapes, either replica symmetry is false or nothing is proved. Third, the continuation $n \to 0$, the step that mathematicians distrust most, is the one step that no proof reproduces. The rigorous theory does not justify the replica trick; it makes the trick unnecessary, one model at a time.

What the proofs assume

The twenty-four rows marked *proved under* name the hypotheses that the theorems need and the predictions do not, and the same few hypotheses recur.

Gaussian data. Gordon's comparison is a statement about Gaussian processes, and the main theorems of Part III start from Gaussian inputs. Universality theorems extend some of them to independent entries (Chapters 9 and 10) and to random features (Chapter 12), but not to general isotropic designs, for which the Gaussian risk of least squares provably fails and that of the LASSO fails in simulations (Han and Shen 2023) (Chapter 10), and not to real data.

Bayes optimality. The proofs of the replica formulas of Part II use the Nishimori identity, which holds only when the statistician knows the prior and the channel. At the wrong temperature the replica-symmetric formula is false (Chapter 4), and the mismatched student of Chapter 7 is open.

Convexity. The two-sided comparison of Chapter 8 needs convex sets

and convex–concave objectives. Without convexity Gordon gives one side, which is why Gardner's formula for the negative perceptron is only an upper bound on its capacity.

Fixed horizons. State evolution holds for a fixed number of iterations as the dimension grows (Chapter 6), dynamical mean-field theory for fixed times, at positive temperature or with smooth losses (Chapter 14), and the Saad–Solla equations on fixed intervals of rescaled time (Chapter 15). The physically interesting questions, such as where gradient flow ends, how long a plateau lasts, or whether AMP converges, are about the opposite order of limits.

Numerical evaluation. Several thresholds are theorems up to the numerical solution of a scalar equation: the information-theoretic thresholds of sparse PCA (Chapter 5), $\alpha_{\mathrm{IT}} = 1.2447$ for the Ising teacher (Chapter 7), and the capacity 0.8331 of the Ising perceptron, whose proof rests on numerical conditions (Chapter 8). In each case the book's scripts recompute the number; in Chapter 7 the recomputation disagrees with the value printed in the source.

Where the physics was wrong

Nineteen rows are marked *known false*. They are of three kinds, and only the last is a failure of physics, or of anyone, in the ordinary sense.

Seven rows record replica-symmetric answers that physics itself rejects. Below the critical temperature of the REM the replica-symmetric entropy is negative (Chapter 2); below the de Almeida–Thouless line of SK the replica-symmetric saddle point is unstable (Chapter 3); for the perceptron with a negative margin, and for the fewest errors beyond capacity, the replica-symmetric solution is unstable (three rows: the prologue, Chapters 8 and 11); for binary weights the replica-symmetric capacity $4/\pi$ exceeds the counting bound 1 (Chapter 8); for ℓ_0 reconstruction the replica-symmetric success solution is unstable at every load (Chapter 10). In each case physicists found the failure themselves. For the REM, SK and binary weights they also found the broken solution; for the negative perceptron they predict full replica-symmetry breaking, and for ℓ_0 reconstruction the broken computation was announced and is still open. Theorems confirm the failure for the REM, for SK, for binary weights and for the negative perceptron, the last so far only below $\kappa = -0.622$ and as $\kappa \to -\infty$; for the fewest errors beyond capacity and for ℓ_0 reconstruction the evidence is the physicists' own stability computation. Where theorems reach the broken solution at all, they confirm it too.

Four record shortcuts whose failure the physics itself explains: the annealed average below the critical temperature (Chapter 1) and in learning curves (Chapter 7), naive mean field without the Onsager term (Chapter 6), and the naive continuation of the integer moments of the REM to $n \to 0$ (Chapter 2).

The remaining eight record extrapolations beyond the regime in which a derivation or a theorem holds, made by physicists and mathematicians alike: the Gaussian approximation to learning curves at zero ridge in

fixed dimension, which fails on the circle, where the interpolant is nearly piecewise linear and its noise term is the $2\sigma^2/3$ of linear interpolation (Belkin et al. 2018), not the formula's σ^2 (Chapter 12); linear Gaussian equivalence in polynomial scalings or after one large gradient step, and the Gaussian covariate model for the classification error of real feature maps on real images (Chapter 12); universality of the Gaussian formulas for every isotropic design, which fails for least squares and, in simulations, for the LASSO (Chapter 10); equality of the typical and mean numbers of critical points in mixed models (Chapter 13); equality of the long-time dynamics and the statics (Chapter 14); and the claim that d^{k-1} samples are necessary for every gradient method to learn a single-index target of information exponent k, which smoothing and the reuse of samples both beat (Chapter 15). The replica-symmetric formula at the wrong temperature (Chapter 4) belongs here too: the formula is right on the Nishimori line and was never claimed off it.

The preprints

Nine rows rest on unrefereed preprints. Four of them are preprints posted between 2013 and 2024 whose results, as the boards use them, have no refereed version: El Karoui's analysis of unregularized robust regression with non-Gaussian designs, whose published revision keeps only ridge-regularized estimators (Chapter 10), the dynamical mean-field theory of gradient flow of Celentano et al. (2021) (Chapter 14), the long-horizon AMP of Li and Wei (2022) (Chapter 6), and the deterministic equivalent for general kernels of Misiakiewicz and Saeed (2024) (Chapter 12). The other five appeared in 2026 and settle questions, most of them open for decades: replica symmetry of SK on the whole high-temperature side of the de Almeida–Thouless line with a field (Kusuoka and Nakajima 2026; Lopatto 2026c); full replica-symmetry breaking below it (Lopatto 2026a,b); the convergence of the overlap of SK itself to the Parisi measure (Aronow and Lopatto 2026a); large deviations of the SK free entropy at the order predicted by Parisi and Rizzo (2009) (Aronow and Lopatto 2026b; Pei 2026); and the capacity of the Ising perceptron without numerical conditions (Shmalo 2026). Table E.3 lists them with their arXiv versions as of 25 October 2026, together with one more 2026 preprint that Chapter 3 reports in its text but not on its board.

These results are reported as their authors state them, and the boards do not count them as proved. By their authors' accounts, AI systems also did a substantial part of the mathematical work in several of them. The book does not treat them as more or less likely to be right than other preprints on that account. What it changes is the weight of independent checking, and two of them invite such checking in forms that are unusual in the field: a formalization in Lean (Chapter 3) and a computer-assisted proof (Chapter 8).

RESULT	PREPRINT	WHAT THE AUTHORS SAY ABOUT AI ASSISTANCE
Replica symmetry of SK for $h > 0$ wherever $\beta^2 \mathbb{E}\,\mathrm{sech}^4 \le 1$ (Chapter 3)	Lopatto (2026c), arXiv:2604.11921v2	written with the assistance of large language models, which suggested arguments, helped draft and revise, and coded simulations
The same without the Parisi formula, with rate $1/N$ where $\beta^2 \mathbb{E}\,\mathrm{sech}^4 < 1$	Kusuoka and Nakajima (2026), arXiv:2608.23413v2	for their key step the authors "asked ChatGPT 5.6 Sol to construct a proof"; they revised the argument in detail, take full responsibility for it, and formalized the proof in Lean 4 with Codex
Full RSB at $h = 0$ for every $\beta > 1$	Lopatto (2026a), arXiv:2607.11756v4	written with the assistance of large language models
The Parisi measure for $h > 0$ has an interval as support, with a density inside	Lopatto (2026b), arXiv:2610.00127v1	written with the assistance of large language models; the observables of the proof were located with "a massive number of numerical experiments orchestrated by large language models"
The zero-temperature Parisi measure has support $[0, 1)$ (in the text of Chapter 3)	Chen (2026), arXiv:2607.18032v1	the arXiv record states that "ChatGPT 5.6 is the actual author"
The overlap law of SK is the Parisi measure	Aronow and Lopatto (2026a), arXiv:2609.17442v1	written with the assistance of large language models
Upper tail $e^{-Na\delta^{6/5}}$ of the SK free entropy (Chapter 1)	Aronow and Lopatto (2026b), arXiv:2609.09103v1; Pei (2026), arXiv:2610.06480v2	Aronow and Lopatto: written with the assistance of large language models; Pei: none stated
Capacity 0.8330786 of the Ising perceptron (Chapter 8)	Shmalo (2026), arXiv:2607.11010v1	"Most of the work reported here was done by two AI systems"; the proof has not yet been independently reproduced or peer reviewed

Table E.3. The 2026 preprints that the chapters report, with the arXiv versions current on 25 October 2026 and what each says about the use of AI systems, quoted or paraphrased from the papers as the chapters report them. None is refereed. The boards mark every result in the table *preprint* except the fifth, which Chapter 3 reports only in its text.

Open problems

Each chapter closed with its open problems. Ten of them, in the chapters' order, show the shape of what is not known as of October 2026.

(i) A mathematical framework for the replica method with broken symmetry, which would explain, among other things, why the extremum in the REM is a minimum over $[n, 1]$ (Mézard and Montanari 2009, notes to ch. 8); and, more modestly, a class of disordered systems in which the moment problem for Z_N is solved, so that the replica trick is justified (Tanaka 2007) (Chapters 1 and 2).

(ii) The order of the fluctuations of $\log Z_N$ for SK below the critical temperature, whose variance the preprint of Aronow and Lopatto (2026b) places between the orders $N^{4/15}$ and $N^{7/15}$ (Chapters 1 and 3).

(iii) Whether the local minimizer of the TAP free energy near the posterior mean of $\mathbb{Z}_2$ synchronization, which Celentano et al. (2023) construct for every $\lambda > 1$, is global, as they conjecture from numerics (Chapters 4 and 6).

(iv) Whether some polynomial-time algorithm beats message passing in the hard phases of sparse PCA (Lesieur et al. 2017) and of the Ising teacher–student perceptron (Barbier et al. 2019) (Chapters 5 and 7).

(v) The capacity of the spherical perceptron with a negative margin, for which physics predicts full replica-symmetry breaking (Franz et al. 2017) and rigorous bounds match only to leading order as $\kappa \to -\infty$ (Montanari et al. 2024) (Chapter 8).

(vi) Universality of the existence threshold of the logistic MLE for non-Gaussian inputs (Candès and Sur 2020, Sec. 5), and a characterization of the designs that share the Gaussian LASSO risk (Chapters 10 and 11).

(vii) A criterion, for a given kernel and data distribution, for the exactness of the Gaussian approximation to the learning curve, and the exchange of the limits $\lambda \to 0$ and $d \to \infty$ for random features (Mei and Montanari 2022, Remark 1) (Chapter 12).

(viii) The typical complexity of the spiked tensor model near its isolated minimum, and in its band of minima near the signal strength at which that minimum appears, which Ros et al. (2019) computed by a replicated Kac–Rice formula and Ben Arous et al. (2019) leave open (Chapter 13).

(ix) That gradient flow ends at the threshold energy of the pure spherical p-spin, which Sellke (2024) proves only for Langevin dynamics on time scales that do not grow with the dimension, in the limit of zero temperature, and not for the flow itself (his Remark 1.5); and the dynamical transition and aging for $p \geq 3$ (Chapters 13 and 14).

(x) The length of the plateau of online SGD on a committee machine, of order $\log d$ and with a limiting law, from a random start (Veiga et al. 2022, Section 3) (Chapter 15).

Many of these share one feature: they are about the regime in which the four reasons for concentration of the previous sections fail, or about times that grow with the dimension. The physics of these regimes is rich and, in most cases, believed; the mathematics is where the frontier is.

What to read next

On the physical side, Mézard et al. (1987) collect the founding papers of spin-glass theory with commentary, Nishimori (2001) develops the Nishimori line, Engel and Van den Broeck (2001) is the textbook of the statistical mechanics of learning, Mézard and Montanari (2009) treat the cavity method and message passing for inference and coding, and Zdeborová and Krzakala (2016) review the statistical physics of inference. On the mathematical side, Talagrand (2011a,b) and Panchenko (2013) develop the theory of mean-field spin glasses, and Bovier (2006) that of the random energy models. Between them, Montanari and Sen (2024) teach replica and cavity computations alongside interpolation bounds, on two models, for readers without physics,

the lecture notes of Krzakala and Zdeborová (2024) follow several replica computations with their rigorous counterparts, Bandeira et al. (2018) explain the physicists' heuristics for computational thresholds to mathematicians, Feng et al. (2022) unify the theory of approximate message passing, and Thrampoulidis et al. (2018) set out the convex Gaussian min–max theorem with its applications to estimation. For dynamics, Cugliandolo (2024) surveys dynamical mean-field methods and their recent applications.

References

Amelunxen, D., M. Lotz, M. B. McCoy, and J. A. Tropp (2014). "Living on the edge: phase transitions in convex programs with random data". In: *Information and Inference: A Journal of the IMA* 3.3, pp. 224–294. arXiv: 1303.6672.

Aronow, P. M. and P. Lopatto (2026a). *Overlap structure of mixed even p-spin models*. arXiv: 2609.17442v1.

Aronow, P. M. and P. Lopatto (2026b). *Quantitative Parisi formulas and fluctuations in the Sherrington-Kirkpatrick model*. arXiv preprint, version 1 (September 2026). arXiv: 2609.09103.

Auffinger, A., G. Ben Arous, and J. Černý (2013). "Random Matrices and Complexity of Spin Glasses". In: *Communications on Pure and Applied Mathematics* 66.2, pp. 165–201. arXiv: 1003.1129.

Baik, J., G. Ben Arous, and S. Péché (2005). "Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices". In: *The Annals of Probability* 33.5, pp. 1643–1697. arXiv: math/0403022.

Bandeira, A. S., A. Perry, and A. S. Wein (2018). "Notes on computational-to-statistical gaps: predictions using statistical physics". In: *Portugaliae Mathematica* 75.2, pp. 159–186. arXiv: 1803.11132.

Barbier, J., F. Krzakala, N. Macris, L. Miolane, and L. Zdeborová (2019). "Optimal errors and phase transitions in high-dimensional generalized linear models". In: *Proceedings of the National Academy of Sciences* 116.12, pp. 5451–5460. arXiv: 1708.03395.

Barbier, J. and N. Macris (2019). "The adaptive interpolation method: a simple scheme to prove replica formulas in Bayesian inference". In: *Probability Theory and Related Fields* 174.3-4, pp. 1133–1185. arXiv: 1705.02780.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: 1001.3448.

Bayati, M. and A. Montanari (2012). "The LASSO Risk for Gaussian Matrices". In: *IEEE Transactions on Information Theory* 58.4, pp. 1997–2017. arXiv: 1008.2581.

Belkin, M., D. J. Hsu, and P. P. Mitra (2018). "Overfitting or perfect fitting? Risk bounds for classification and regression rules that interpolate". In: *Advances in Neural Information Processing Systems 31*. arXiv: 1806.05161.

Ben Arous, G., A. Dembo, and A. Guionnet (2006). "Cugliandolo-Kurchan equations for dynamics of spin-glasses". In: *Probability Theory and Related Fields* 136.4, pp. 619–660. arXiv: math/0409273.

Ben Arous, G., S. Mei, A. Montanari, and M. Nica (2019). "The Landscape of the Spiked Tensor Model". In: *Communications on Pure and Applied Mathematics* 72.11, pp. 2282–2330. arXiv: 1711.05424.

Benaych-Georges, F. and R. R. Nadakuditi (2011). "The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices". In: *Advances in Mathematics* 227.1, pp. 494–521. arXiv: 0910.2120.

Biehl, M. and H. Schwarze (1995). "Learning by on-line gradient descent". In: *Journal of Physics A: Mathematical and General* 28.3, pp. 643–656.

Bolthausen, E. (2014). "An Iterative Construction of Solutions of the TAP Equations for the Sherrington–Kirkpatrick Model". In: *Communications in Mathematical Physics* 325.1, pp. 333–366. arXiv: 1201.2891.

Bordelon, B., A. Canatar, and C. Pehlevan (2020). "Spectrum Dependent Learning Curves in Kernel Regression and Wide Neural Networks". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research, pp. 1024–1034. arXiv: 2002.02561.

Bovier, A. (2006). *Statistical Mechanics of Disordered Systems: A Mathematical Perspective*. Vol. 18. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press.

Bovier, A. and I. Kurkova (2004). "Derrida's generalised random energy models 1: models with finitely many hierarchies". In: *Annales de l'Institut Henri Poincaré (B) Probabilités et Statistiques* 40.4, pp. 439–480.

Canatar, A., B. Bordelon, and C. Pehlevan (2021). "Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks". In: *Nature Communications* 12, p. 2914. arXiv: 2006.13198.

Candès, E. J. and P. Sur (2020). "The phase transition for the existence of the maximum likelihood estimate in high-dimensional logistic regression". In: *The Annals of Statistics* 48.1, pp. 27–42. arXiv: 1804.09753.

Cavagna, A., I. Giardina, and G. Parisi (1998). "Stationary points of the Thouless–Anderson–Palmer free energy". In: *Physical Review B* 57.18, pp. 11251–11257. arXiv: cond-mat/9710272.

Celentano, M., C. Cheng, and A. Montanari (2021). *The high-dimensional asymptotics of first order methods with random data*. Version 3, 18 April 2026; unpublished as of October 2026. arXiv: 2112.07572.

Celentano, M., Z. Fan, and S. Mei (2023). "Local convexity of the TAP free energy and AMP convergence for $\mathbb{Z}_2$-synchronization". In: *The Annals of Statistics* 51.2, pp. 519–546. arXiv: 2106.11428.

Chen, H.-B. (2026). *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. arXiv: 2607.18032v1.

Cheng, C. and A. Montanari (2024). "Dimension free ridge regression". In: *The Annals of Statistics* 52.6, pp. 2879–2912. arXiv: 2210.08571.

Crisanti, A., H. Horner, and H.-J. Sommers (1993). "The spherical p-spin interaction spin-glass model: the dynamics". In: *Zeitschrift für Physik B Condensed Matter* 92.2, pp. 257–271.

Crisanti, A. and H.-J. Sommers (1995). "Thouless–Anderson–Palmer Approach to the Spherical p-Spin Spin Glass Model". In: *Journal de Physique I* 5.7, pp. 805–813.

Cugliandolo, L. F. and J. Kurchan (1993). "Analytical solution of the off-equilibrium dynamics of a long-range spin-glass model". In: *Physical Review Letters* 71.1, pp. 173–176. arXiv: cond-mat/9303036.

Cugliandolo, L. F. (2024). "Recent Applications of Dynamical Mean-Field Methods". In: *Annual Review of Condensed Matter Physics* 15, pp. 177–213. arXiv: 2305.01229.

De Almeida, J. R. L. and D. J. Thouless (1978). "Stability of the Sherrington–Kirkpatrick solution of a spin glass model". In: *Journal of Physics A: Mathematical and General* 11.5, pp. 983–990.

Derrida, B. (1980). "Random-Energy Model: Limit of a Family of Disordered Models". In: *Physical Review Letters* 45.2, pp. 79–82.

Derrida, B. (1981). "Random-energy model: An exactly solvable model of disordered systems". In: *Physical Review B* 24.5, pp. 2613–2626.

Deshpande, Y., E. Abbe, and A. Montanari (2016). "Asymptotic mutual information for the balanced binary stochastic block model". In: *Information and Inference: A Journal of the IMA*. Published online 2016-12-26; in print as volume 6, issue 2 (2017), pages 125–170. arXiv: 1507.08685.

Dobriban, E. and S. Wager (2018). "High-dimensional asymptotics of prediction: Ridge regression and classification". In: *The Annals of Statistics* 46.1, pp. 247–279. arXiv: 1507.03003.

Donoho, D. L., A. Maleki, and A. Montanari (2009). "Message-passing algorithms for compressed sensing". In: *Proceedings of the National Academy of Sciences* 106.45, pp. 18914–18919. arXiv: 0907.3574.

Engel, A. and C. Van den Broeck (2001). *Statistical Mechanics of Learning*. Cambridge University Press.

Feng, O. Y., R. Venkataramanan, C. Rush, and R. J. Samworth (2022). "A Unifying Tutorial on Approximate Message Passing". In: *Foundations and Trends in Machine Learning* 15.4, pp. 335–536. arXiv: 2105.02180.

Féral, D. and S. Péché (2007). "The Largest Eigenvalue of Rank One Deformation of Large Wigner Matrices". In: *Communications in Mathematical Physics* 272.1, pp. 185–228. arXiv: math/0605624.

Franz, S., G. Parisi, M. Sevelev, P. Urbani, and F. Zamponi (2017). "Universality of the SAT-UNSAT (jamming) threshold in non-convex continuous constraint satisfaction problems". In: *SciPost Physics* 2.3, p. 019. arXiv: 1702.06919v2 [cond-mat.dis-nn].

Füredi, Z. and J. Komlós (1981). "The eigenvalues of random symmetric matrices". In: *Combinatorica* 1.3, pp. 233–241.

Gardner, E. (1988). "The space of interactions in neural network models". In: *Journal of Physics A: Mathematical and General* 21.1, pp. 257–270.

Gardner, E. and B. Derrida (1989). "Three unfinished works on the optimal storage capacity of networks". In: *Journal of Physics A: Mathematical and General* 22.12, pp. 1983–1994.

Gerace, F., B. Loureiro, F. Krzakala, M. Mézard, and L. Zdeborová (2020). "Generalisation error in learning with random features and the hidden manifold model". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research, pp. 3452–3462. arXiv: 2002.09339.

Gerbelot, C., E. Troiani, F. Mignacco, F. Krzakala, and L. Zdeborová (2024). "Rigorous Dynamical Mean-Field Theory for Stochastic Gradient Descent Methods". In: *SIAM Journal on Mathematics of Data Science* 6.2, pp. 400–427. arXiv: 2210.06591.

Goldt, S., M. S. Advani, A. M. Saxe, F. Krzakala, and L. Zdeborová (2020). "Dynamics of stochastic gradient descent for two-layer neural networks in the teacher–student setup". In: *Journal of Statistical Mechanics: Theory and Experiment* 2020.12, p. 124010. arXiv: 1906.08632.

Gordon, Y. (1985). "Some inequalities for Gaussian processes and applications". In: *Israel Journal of Mathematics* 50.4, pp. 265–289.

Gross, D. J. and M. Mézard (1984). "The simplest spin glass". In: *Nuclear Physics B* 240.4, pp. 431–452.

Guerra, F. (2003). "Broken Replica Symmetry Bounds in the Mean Field Spin Glass Model". In: *Communications in Mathematical Physics* 233.1, pp. 1–12.

Guerra, F. and F. L. Toninelli (2002). "The Thermodynamic Limit in Mean Field Spin Glass Models". In: *Communications in Mathematical Physics* 230.1, pp. 71–79.

Györgyi, G. (1990). "First-order transition to perfect generalization in a neural network with binary synapses". In: *Physical Review A* 41.12, pp. 7097–7100.

Han, Q. and Y. Shen (2023). "Universality of Regularized Regression Estimators in High Dimensions". In: *The Annals of Statistics* 51.4, pp. 1799–1823. arXiv: 2206.07936.

Hastie, T., A. Montanari, S. Rosset, and R. J. Tibshirani (2022). "Surprises in high-dimensional ridgeless least squares interpolation". In: *The Annals of Statistics* 50.2, pp. 949–986. arXiv: 1903.08560.

Hoyle, D. C. and M. Rattray (2004). "Principal-component-analysis eigenvalue spectra from data with symmetry-breaking structure". In: *Physical Review E* 69.2, p. 026124.

Hu, H. and Y. M. Lu (2023). "Universality Laws for High-Dimensional Learning With Random Features". In: *IEEE Transactions on Information Theory* 69.3, pp. 1932–1964. arXiv: 2009.07669.

Jones, R. C., J. M. Kosterlitz, and D. J. Thouless (1978). "The eigenvalue spectrum of a large symmetric random matrix with a finite mean". In: *Journal of Physics A: Mathematical and General* 11.3, pp. L45–L48.

Kabashima, Y., T. Wadayama, and T. Tanaka (2009). "A Typical Reconstruction Limit for Compressed Sensing Based on L_p-Norm Minimization". In: *Journal of Statistical Mechanics: Theory and Experiment* 2009.09, p. L09003. arXiv: 0907.0914.

Krogh, A. and J. A. Hertz (1992). "Generalization in a linear perceptron in the presence of noise". In: *Journal of Physics A: Mathematical and General* 25.5, pp. 1135–1147.

Krzakala, F. and L. Zdeborová (2024). *Statistical Physics Methods in Optimization and Machine Learning: An Introduction to Replica, Cavity & Message-Passing techniques*. Lecture notes, https://sphinxteam.github.io/EPFLDoctoralLecture2021/Notes.pdf. Unpublished draft, version of 9 June 2024.

Kusuoka, S. and S. Nakajima (2026). *A quantitative replica-symmetric bound for Sherrington–Kirkpatrick model in the entire de Almeida–Thouless region*. arXiv: 2608.23413v2.

Lelarge, M. and L. Miolane (2019). "Fundamental limits of symmetric low-rank matrix estimation". In: *Probability Theory and Related Fields* 173.3-4, pp. 859–929.

Lesieur, T., F. Krzakala, and L. Zdeborová (2017). "Constrained low-rank matrix estimation: phase transitions, approximate message passing and applications". In: *Journal of Statistical Mechanics: Theory and Experiment* 2017.7, p. 073403. arXiv: 1701.00858.

Li, G. and Y. Wei (2022). *A Non-Asymptotic Framework for Approximate Message Passing in Spiked Models*. Preprint, version 2 (2023). arXiv: 2208.03313.

Lopatto, P. (2026a). *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. arXiv: 2607.11756v4.

Lopatto, P. (2026b). *Parisi measure of the Sherrington–Kirkpatrick model with external field*. arXiv: 2610.00127v1.

Lopatto, P. (2026c). *Replica symmetry up to the de Almeida–Thouless line in the Sherrington–Kirkpatrick model*. arXiv: 2604.11921v2.

Mei, S. and A. Montanari (2022). "The Generalization Error of Random Features Regression: Precise Asymptotics and the Double Descent Curve". In: *Communications on Pure and Applied Mathematics* 75.4, pp. 667–766. arXiv: 1908.05355.

Mézard, M. and A. Montanari (2009). *Information, Physics, and Computation*. Oxford Graduate Texts. Oxford: Oxford University Press.

Mézard, M., G. Parisi, and M. A. Virasoro (1987). *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications*. Vol. 9. World Scientific Lecture Notes in Physics. Singapore: World Scientific.

Misiakiewicz, T. and B. Saeed (2024). *A non-asymptotic theory of Kernel Ridge Regression: deterministic equivalents, test error, and GCV estimator*. arXiv: 2403.08938.

Montanari, A., F. Ruan, Y. Sohn, and J. Yan (2025). "The generalization error of max-margin linear classifiers: Benign overfitting and high dimensional asymptotics in the overparametrized regime". In: *The Annals of Statistics* 53.2, pp. 822–853. arXiv: 1911.01544.

Montanari, A. and S. Sen (2024). "A Friendly Tutorial on Mean-Field Spin Glass Techniques for Non-Physicists". In: *Foundations and Trends in Machine Learning* 17.1, pp. 1–173. arXiv: 2204.02909.

Montanari, A., Y. Zhong, and K. Zhou (2024). "Tractability from overparametrization: the example of the negative perceptron". In: *Probability Theory and Related Fields* 188.3-4, pp. 805–910. arXiv: 2110.15824v3 [cs.LG].

Nishimori, H. (2001). *Statistical Physics of Spin Glasses and Information Processing: An Introduction*. Vol. 111. International Series of Monographs on Physics. Oxford: Oxford University Press.

Opper, M., W. Kinzel, J. Kleinz, and R. Nehl (1990). "On the ability of the optimal perceptron to generalise". In: *Journal of Physics A: Mathematical and General* 23.11, pp. L581–L586.

Panchenko, D. (2013). *The Sherrington-Kirkpatrick Model*. Springer Monographs in Mathematics. New York: Springer New York.

Parisi, G. (1979). "Infinite Number of Order Parameters for Spin-Glasses". In: *Physical Review Letters* 43.23, pp. 1754–1756.

Parisi, G. and T. Rizzo (2009). "Phase diagram and large deviations in the free energy of mean-field spin glasses". In: *Physical Review B* 79.13, p. 134205.

Pei, Y. R. (2026). *Upper deviations of the Sherrington-Kirkpatrick free energy: the fifth-order law and the onset of constrained Parisi measures*. arXiv preprint, version 2 (October 2026). arXiv: 2610.06480.

Rangan, S., A. K. Fletcher, and V. K. Goyal (2012). "Asymptotic Analysis of MAP Estimation via the Replica Method and Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 58.3, pp. 1902–1923. arXiv: 0906.3234.

Ros, V., G. Ben Arous, G. Biroli, and C. Cammarota (2019). "Complex Energy Landscapes in Spiked-Tensor and Simple Glassy Models: Ruggedness, Arrangements of Local Minima, and Phase Transitions". In: *Physical Review X* 9.1, p. 011003. arXiv: 1804.02686.

Ruelle, D. (1987). "A mathematical reformulation of Derrida's REM and GREM". In: *Communications in Mathematical Physics* 108.2, pp. 225–239.

Saad, D. and S. A. Solla (1995). "Exact Solution for On-Line Learning in Multilayer Neural Networks". In: *Physical Review Letters* 74.21, pp. 4337–4340.

Sellke, M. (2024). "The threshold energy of low temperature Langevin dynamics for pure spherical spin glasses". In: *Communications on Pure and Applied Mathematics* 77.11, pp. 4065–4099. arXiv: 2305.07956.

Shcherbina, M. and B. Tirozzi (2003). "Rigorous Solution of the Gardner Problem". In: *Communications in Mathematical Physics* 234.3, pp. 383–422. arXiv: math-ph/0112003v1 [math-ph].

Sherrington, D. and S. Kirkpatrick (1975). “Solvable Model of a Spin-Glass”. In: *Physical Review Letters* 35.26, pp. 1792–1796.

Shmalo, Y. (2026). *The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions*. arXiv preprint, v1 of 13 July 2026; unrefereed. arXiv: 2607.11010v1 [math.PR].

Sompolinsky, H., N. Tishby, and H. S. Seung (1990). “Learning from examples in large neural networks”. In: *Physical Review Letters* 65.13, pp. 1683–1686.

Stojnic, M. (2013). *Another look at the Gardner problem*. arXiv preprint. arXiv: 1306.3979v1 [math.OC].

Subag, E. (2017). “The complexity of spherical p-spin models—A second moment approach”. In: *The Annals of Probability* 45.5, pp. 3385–3450. arXiv: 1504.02251.

Talagrand, M. (2006). “The Parisi formula”. In: *Annals of Mathematics* 163.1, pp. 221–263.

Talagrand, M. (2011a). *Mean Field Models for Spin Glasses: Volume I: Basic Examples*. Vol. 54. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Talagrand, M. (2011b). *Mean Field Models for Spin Glasses: Volume II: Advanced Replica-Symmetry and Low Temperature*. Vol. 55. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Berlin: Springer Berlin Heidelberg.

Tanaka, T. (2007). “Moment Problem in Replica Method”. In: *Interdisciplinary Information Sciences* 13.1, pp. 17–23.

Thouless, D. J., P. W. Anderson, and R. G. Palmer (1977). “Solution of ‘Solvable model of a spin glass’”. In: *Philosophical Magazine* 35.3, pp. 593–601.

Thrampoulidis, C., E. Abbasi, and B. Hassibi (2018). “Precise Error Analysis of Regularized M-Estimators in High Dimensions”. In: *IEEE Transactions on Information Theory* 64.8, pp. 5592–5628. arXiv: 1601.06233.

Veiga, R., L. Stephan, B. Loureiro, F. Krzakala, and L. Zdeborová (2022). “Phase Diagram of Stochastic Gradient Descent in High-Dimensional Two-Layer Neural Networks”. In: *Advances in Neural Information Processing Systems 35*, pp. 23244–23255. arXiv: 2202.00293.

Zdeborová, L. and F. Krzakala (2016). “Statistical physics of inference: thresholds and algorithms”. In: *Advances in Physics* 65.5, pp. 453–552. arXiv: 1511.02476.

A Toolkit

A

Gaussian calculus, concentration and random matrices

This appendix collects the tools that the book's proofs share: Gaussian integration by parts and interpolation, the comparison and concentration inequalities they yield, Laplace's method and Varadhan's lemma, the conditional law of a Gaussian matrix given linear constraints, the semicircle and Marchenko–Pastur laws, the Kac–Rice formula, and the numerical methods of the scripts. Results with short proofs that use only what precedes them are proved, including the semicircle law and the norm of a GOE matrix; the others are stated with their sources. Each item names the chapters that use it and, where there is one, the physical step that it replaces.

Vectors are columns; $|x|$ is the Euclidean norm, $\langle x \rangle y$ the inner product, $\|A\|$ the operator norm and $\|A\|_F$ the Frobenius norm of a matrix; $Z \sim \mathcal{N}(0, I_n)$ is a standard Gaussian vector, and ∂_i, ∂_{ij} are partial derivatives.

A.1 Gaussian integration by parts and interpolation

A centered Gaussian vector X in $\mathbb{R}^n$ with covariance $\Sigma = \mathbb{E}XX^\top$ has the law of $\Sigma^{1/2}Z$, where $\Sigma^{1/2}$ is the positive semidefinite square root of Σ, which may be singular. A function $F\colon \mathbb{R}^n \to \mathbb{R}$ has *moderate growth* if for every $a > 0$ some C_a satisfies $|F(x)| \le C_a e^{a|x|^2}$ for all x. Polynomials, bounded functions, $e^{\langle b \rangle x}$ and $\log\cosh$ have moderate growth, and so do sums and products of such functions and their compositions with linear maps. If F has moderate growth, then $F(X)$ has moments of all orders, because $\mathbb{E}e^{a|X|^2} \le \mathbb{E}e^{a\|\Sigma\||Z|^2} < \infty$ when $2a\|\Sigma\| < 1$.

Theorem A.1 (Gaussian integration by parts). *Let X be a centered Gaussian vector in $\mathbb{R}^n$ with covariance Σ, and let $F\colon \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable, with F and its first partial derivatives of moderate growth. Then, for each i,*

$$\mathbb{E}\big[X_i F(X)\big] = \sum_{j=1}^{n} \Sigma_{ij}\, \mathbb{E}\big[\partial_j F(X)\big]. \tag{A.1}$$

More generally, if (Y, X) is a centered Gaussian vector in $\mathbb{R} \times \mathbb{R}^n$, then $\mathbb{E}[YF(X)] = \sum_j \mathbb{E}[YX_j]\,\mathbb{E}[\partial_j F(X)]$. For $n = 1$ and $X = Z \sim \mathcal{N}(0,1)$ this is Stein's identity *$\mathbb{E}[Z f(Z)] = \mathbb{E}f'(Z)$.*

Proof. The plan is to integrate by parts in one variable, to pass to a standard Gaussian vector by Fubini's theorem, and to reach a general covariance by a linear change of variables. Let $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$, so that $\phi'(z) = -z\phi(z)$.

If $g\colon \mathbb{R} \to \mathbb{R}$ is continuously differentiable and $|g(z)| + |g'(z)| \le Ce^{az^2}$ with $a < \frac{1}{2}$, then integration by parts on $[-L, L]$ gives

$$\int_{-L}^{L} g'(z)\phi(z)\,\mathrm{d}z = g(L)\phi(L) - g(-L)\phi(-L) + \int_{-L}^{L} z g(z)\phi(z)\,\mathrm{d}z.$$

The boundary terms are at most $Ce^{(a-1/2)L^2}$, and both integrals converge absolutely as $L \to \infty$; so $\mathbb{E}g'(Z) = \mathbb{E}[Zg(Z)]$ for $Z \sim \mathcal{N}(0,1)$.

Now let $Z \sim \mathcal{N}(0, I_n)$, let $G\colon \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable with G and $\partial_k G$ of moderate growth, and fix $a < \frac{1}{2}$. With the coordinates z' other than z_k held fixed, the bound $|G(z)| + |\partial_k G(z)| \le C_a e^{a|z'|^2} e^{az_k^2}$ allows the one-variable identity in z_k. Since $\mathbb{E}[(1 + |Z_k|)e^{a|Z|^2}] < \infty$, Fubini's theorem integrates the identity over z' and gives $\mathbb{E}[Z_k G(Z)] = \mathbb{E}[\partial_k G(Z)]$.

Finally let $A = \Sigma^{1/2}$, so that X has the law of AZ, and let $G(z) = F(Az)$. Then $\partial_k G(z) = \sum_j A_{jk}\,\partial_j F(Az)$, and G and $\partial_k G$ have moderate growth because $|Az| \le \|A\||z|$. The previous step gives

$$\mathbb{E}[X_i F(X)] = \sum_k A_{ik}\mathbb{E}[Z_k G(Z)] = \sum_k A_{ik}\mathbb{E}[\partial_k G(Z)],$$

which is $\sum_j (AA^\top)_{ij}\mathbb{E}[\partial_j F(X)]$, and $AA^\top = \Sigma$. For the general form, apply (A.1) to the vector (Y, X) and to the function $(y, x) \mapsto F(x)$, whose derivative in y vanishes. ∎

Multiplying by a Gaussian variable is, in expectation, differentiating along its covariance. With $F(x) = x_j x_k x_l$ the identity gives Wick's formula $\mathbb{E}X_iX_jX_kX_l = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$, the form physicists usually meet. It extends to complex-valued F by linearity, and it gives the interpolation lemma below, the I-MMSE relation (Theorem 1.12), and, in the proof of state evolution (Theorem 6.4), the average derivative that becomes the Onsager coefficient.[1]

[1] Statisticians name the identity after Charles Stein, whose paper on estimating the mean of a multivariate normal distribution uses it to compute the risk of an estimator (Stein 1981).

Lemma A.2 (Gaussian interpolation). *Let X and Y be independent centered Gaussian vectors in $\mathbb{R}^n$ with covariances Σ^X and Σ^Y, and let $F\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable, with F and its partial derivatives of first and second order of moderate growth. For $t \in [0,1]$ let $Z_t = \sqrt{t}\,X + \sqrt{1-t}\,Y$ and $\varphi(t) = \mathbb{E}F(Z_t)$. Then φ is continuously differentiable on $[0,1]$, with*

$$\varphi'(t) = \frac{1}{2}\sum_{i,j=1}^{n} \left(\Sigma^X_{ij} - \Sigma^Y_{ij}\right) \mathbb{E}\left[\partial_{ij} F(Z_t)\right], \tag{A.2}$$

and therefore $\mathbb{E}F(X) - \mathbb{E}F(Y) = \frac{1}{2}\int_0^1 \sum_{i,j}(\Sigma^X_{ij} - \Sigma^Y_{ij})\,\mathbb{E}[\partial_{ij}F(Z_t)]\,\mathrm{d}t$.

Proof. The plan is to differentiate under the expectation and to remove the factors $X_i/\sqrt{t}$ and $Y_i/\sqrt{1-t}$ with Theorem A.1. By the Cauchy–Schwarz inequality $|Z_t|^2 \le |X|^2 + |Y|^2$, so for every $a > 0$ the function F and its

first and second derivatives, evaluated at Z_t, are bounded by $C_a e^{a(|X|^2+|Y|^2)}$, uniformly in t. For $0 < t < 1$,

$$\frac{\mathrm{d}}{\mathrm{d}t}F(Z_t) = \sum_i \partial_i F(Z_t)\Big(\frac{X_i}{2\sqrt{t}} - \frac{Y_i}{2\sqrt{1-t}}\Big),$$

which on $[\delta, 1-\delta]$ is bounded by $\sqrt{n}\, C_a(|X| + |Y|)e^{a(|X|^2+|Y|^2)}/(2\sqrt{\delta})$, an integrable function for small a; so we may differentiate under the expectation. Apply the general form of Theorem A.1 to the Gaussian vector (X, Y) and to $G_i(x, y) = \partial_i F(\sqrt{t}\, x + \sqrt{1-t}\, y)$, whose derivatives are $\partial_{x_j} G_i = \sqrt{t}\, \partial_{ij} F$ and $\partial_{y_j} G_i = \sqrt{1-t}\, \partial_{ij} F$; since X and Y are independent, this gives

$$\mathbb{E}[X_i\, \partial_i F(Z_t)] = \sqrt{t} \sum_j \Sigma^X_{ij}\, \mathbb{E}[\partial_{ij} F(Z_t)],$$
$$\mathbb{E}[Y_i\, \partial_i F(Z_t)] = \sqrt{1-t} \sum_j \Sigma^Y_{ij}\, \mathbb{E}[\partial_{ij} F(Z_t)].$$

Substituting gives (A.2) on $(0, 1)$. By dominated convergence, φ and the right side of (A.2) are continuous on $[0, 1]$, so by the mean value theorem φ is differentiable at the end points too, and the integral formula is the fundamental theorem of calculus. ∎

The lemma uses X and Y only through the law of Z_t, which is centered Gaussian with covariance $t\Sigma^X + (1-t)\Sigma^Y$. Its message is a sign rule: if $(\Sigma^X_{ij} - \Sigma^Y_{ij})\, \partial_{ij} F \geq 0$ for all i, j, then $\mathbb{E}F(X) \geq \mathbb{E}F(Y)$. For a free entropy the second derivatives are Gibbs covariances, and the derivative takes the form that every interpolation of the book uses.

Corollary A.3 (interpolation of a free entropy). *Let S be a finite set, $w : S \to (0, \infty)$, and let $X = (X(\sigma))_{\sigma\in S}$ and $Y = (Y(\sigma))_{\sigma\in S}$ be independent centered Gaussian processes with covariances $C^X(\sigma, \tau) = \mathbb{E}X(\sigma)X(\tau)$ and $C^Y(\sigma, \tau)$. Let $Z_t = \sqrt{t}\, X + \sqrt{1-t}\, Y$, let $\langle\cdot\rangle_t$ be the average under the Gibbs weights proportional to $w(\sigma)e^{Z_t(\sigma)}$, and let σ^1, σ^2 be two independent samples (replicas) from them. Then, with $\Delta = C^X - C^Y$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}\log\sum_{\sigma\in S} w(\sigma)e^{Z_t(\sigma)} = \frac{1}{2}\,\mathbb{E}\big[\big\langle\Delta(\sigma^1,\sigma^1)\big\rangle_t - \big\langle\Delta(\sigma^1,\sigma^2)\big\rangle_t\big]. \qquad \text{(A.3)}$$

Proof. The function $F(x) = \log\sum_\sigma w(\sigma)e^{x_\sigma}$ on $\mathbb{R}^S$ is infinitely differentiable, with $|F(x)| \leq |\log\sum_\sigma w(\sigma)| + \max_\sigma|x_\sigma|$, $\partial_\sigma F = p_\sigma$ and $\partial_{\sigma\tau}F = \delta_{\sigma\tau}p_\sigma - p_\sigma p_\tau$, where $p_\sigma(x) = w(\sigma)e^{x_\sigma}/\sum_\tau w(\tau)e^{x_\tau}$; all of these have moderate growth. At $x = Z_t$, $\sum_\sigma \Delta(\sigma,\sigma)p_\sigma = \big\langle\Delta(\sigma^1,\sigma^1)\big\rangle_t$ and $\sum_{\sigma,\tau}\Delta(\sigma,\tau)p_\sigma p_\tau = \big\langle\Delta(\sigma^1,\sigma^2)\big\rangle_t$, and Lemma A.2 gives (A.3). ∎

In mean-field models the covariances are functions of the overlap, so (A.3) expresses the derivative as a Gibbs average of a function of R_{12}. This is the computation behind Guerra and Toninelli's superadditivity and Guerra's replica-symmetric bound (Theorems 3.2 and 3.3) and behind the interpolation bounds of Chapters 4 and 7. Where the replica method

computes $\mathbb{E}Z^n$ with n replicas and continues to $n \to 0$, the interpolation needs two replicas and a sign.

A.2 Comparison inequalities

For the maximum of a Gaussian vector the sign rule gives two classical inequalities. For a family $(X_t)_{t\in T}$ indexed by an infinite set we write $\mathbb{E}\sup_t X_t$ for the supremum of $\mathbb{E}\max_{t\in T_0} X_t$ over finite subsets $T_0 \subset T$; for a process with continuous paths on a separable metric space this is the expectation of the supremum, by monotone convergence.

Theorem A.4 (Slepian's lemma and the Sudakov–Fernique inequality). *Let $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ be centered Gaussian vectors.*

(i) (Slepian.) If $\mathbb{E}X_i^2 = \mathbb{E}Y_i^2$ for all i and $\mathbb{E}X_iX_j \geq \mathbb{E}Y_iY_j$ for all $i \neq j$, then $\mathbb{P}(\max_i X_i > u) \leq \mathbb{P}(\max_i Y_i > u)$ for every $u \in \mathbb{R}$; in particular $\mathbb{E}\max_i X_i \leq \mathbb{E}\max_i Y_i$.

(ii) (Sudakov–Fernique.) If $\mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2$ for all i, j, then $\mathbb{E}\max_i X_i \leq \mathbb{E}\max_i Y_i$.

The comparisons of expectations extend to infinite index sets, with $\mathbb{E}\sup$ as above.

Proof. The plan is to apply Lemma A.2 to smooth functions that approximate $\mathbf{1}\{\max_i x_i \leq u\}$ and $\max_i x_i$. Only the laws of X and Y enter, so we may take them independent; write $a_{ij} = \mathbb{E}X_iX_j - \mathbb{E}Y_iY_j$.

(i) Let $h\colon \mathbb{R} \to [0,1]$ be infinitely differentiable and nonincreasing, equal to 1 on $(-\infty, 0]$ and to 0 on $[1, \infty)$, and let $F_k(x) = \prod_i h(k(x_i - u))$ for $k \geq 1$. For $i \neq j$, $\partial_{ij}F_k(x) = k^2 h'(k(x_i-u))\,h'(k(x_j-u))\prod_{l\neq i,j} h(k(x_l-u)) \geq 0$, because $h' \leq 0$; and $a_{ii} = 0$, $a_{ij} \geq 0$ by hypothesis. Lemma A.2 gives $\mathbb{E}F_k(X) \geq \mathbb{E}F_k(Y)$. As $k \to \infty$, F_k decreases to the indicator of $\{\max_i x_i \leq u\}$, and monotone convergence gives $\mathbb{P}(\max_i X_i \leq u) \geq \mathbb{P}(\max_i Y_i \leq u)$. Integrating, $\mathbb{E}\max_i X_i = \int_0^\infty \mathbb{P}(\max_i X_i > u)\,\mathrm{d}u - \int_{-\infty}^0 \mathbb{P}(\max_i X_i \leq u)\,\mathrm{d}u \leq \mathbb{E}\max_i Y_i$.

(ii) For $\beta > 0$ let $F_\beta(x) = \beta^{-1}\log\sum_i e^{\beta x_i}$, which satisfies $\max_i x_i \leq F_\beta(x) \leq \max_i x_i + \beta^{-1}\log n$. With $p_i = e^{\beta x_i}/\sum_k e^{\beta x_k}$, we have $\partial_i F_\beta = p_i$ and $\partial_{ij}F_\beta = \beta(\delta_{ij}p_i - p_ip_j)$. Since $\sum_i p_i = 1$,

$$\sum_{i,j} a_{ij}\,\partial_{ij}F_\beta = \beta\Big(\sum_i a_{ii}p_i - \sum_{i,j} a_{ij}p_ip_j\Big) = \frac{\beta}{2}\sum_{i,j} p_ip_j\big(a_{ii} + a_{jj} - 2a_{ij}\big),$$

and $a_{ii} + a_{jj} - 2a_{ij} = \mathbb{E}(X_i - X_j)^2 - \mathbb{E}(Y_i - Y_j)^2 \leq 0$. Lemma A.2 gives $\mathbb{E}F_\beta(X) \leq \mathbb{E}F_\beta(Y)$, hence $\mathbb{E}\max_i X_i \leq \mathbb{E}\max_i Y_i + \beta^{-1}\log n$ for every $\beta > 0$. For an infinite index set, apply (i) or (ii) to each finite subset. ∎

Part (ii) needs no equal variances; applied to a GOE matrix it gives $\mathbb{E}\lambda_{\max}(W) \leq 2$ at every size (Theorem A.15). Gordon's inequality, with a minimum over one index and a maximum over another, is proved in Chapter 8 (Theorem 8.1) from Lemma A.2 in the same way.[2]

[2] Inequality (i) goes back to the paper of Slepian (1962) on the one-sided barrier problem for Gaussian noise; (ii) is due to Sudakov (1971) and Fernique (1975), and the proof by F_β follows Chatterjee (2005).

A.3 Concentration

Physicists call the free entropy of a disordered system self-averaging; the rigorous counterpart is concentration for smooth or Lipschitz functions of many independent Gaussian variables. The variance bound rests on a case of Lemma A.2.

Lemma A.5 (a covariance identity). *Let Z and Z' be independent standard Gaussian vectors in $\mathbb{R}^n$, and let $f, g\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable, with f, g and their first and second partial derivatives of moderate growth. Then*

$$\operatorname{Cov}\big(f(Z), g(Z)\big) = \int_0^1 \mathbb{E}\Big[\big\langle \nabla f(Z), \nabla g(tZ + \sqrt{1-t^2}\,Z')\big\rangle\Big]\,\mathrm{d}t.$$

Proof. Let Z_1, Z_2, Z_3 be independent standard Gaussian vectors in $\mathbb{R}^n$, and apply Lemma A.2 in $\mathbb{R}^{2n}$ to $X = (Z_1, Z_1)$, $Y = (Z_2, Z_3)$ and $F(x, y) = f(x)g(y)$. The two covariance matrices differ only in their off-diagonal blocks, which are I_n for X and 0 for Y, and $\partial_{x_i}\partial_{y_i} F = \partial_i f(x)\,\partial_i g(y)$; so $\varphi'(t) = \sum_i \mathbb{E}[\partial_i f(U_t)\,\partial_i g(V_t)]$ with $(U_t, V_t) = \sqrt{t}\,(Z_1, Z_1) + \sqrt{1-t}\,(Z_2, Z_3)$. This pair is centered Gaussian with $\mathbb{E}U_tU_t^{\top} = \mathbb{E}V_tV_t^{\top} = I_n$ and $\mathbb{E}U_tV_t^{\top} = tI_n$, so it has the law of $(Z, tZ + \sqrt{1-t^2}\,Z')$. Finally $\varphi(1) = \mathbb{E}f(Z)g(Z)$ and $\varphi(0) = \mathbb{E}f(Z)\,\mathbb{E}g(Z)$. ∎

Theorem A.6 (Gaussian Poincaré inequality). *Let $Z \sim \mathcal{N}(0, I_n)$.*

(i) If $f\colon \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and f and its partial derivatives have moderate growth, then $\operatorname{Var} f(Z) \le \mathbb{E}|\nabla f(Z)|^2$.

(ii) If f is L-Lipschitz, that is $|f(x) - f(y)| \le L|x - y|$ for all x, y, then $\operatorname{Var} f(Z) \le L^2$.

(iii) If $X \sim \mathcal{N}(0, \Sigma)$ and f is as in (i), then $\operatorname{Var} f(X) \le \mathbb{E}\langle \nabla f(X), \Sigma \nabla f(X)\rangle \le \|\Sigma\|\,\mathbb{E}|\nabla f(X)|^2$.

For complex-valued f, (i) holds with $\operatorname{Var} f = \mathbb{E}|f - \mathbb{E}f|^2$ *and* $|\nabla f|^2 = \sum_k |\partial_k f|^2$.

Proof. The plan is to prove (i) for smooth f from Lemma A.5 and to reach the general case by Gaussian smoothing. If f is twice continuously differentiable with derivatives of first and second order of moderate growth, then Lemma A.5 with $g = f$ and the Cauchy–Schwarz inequality give

$$\operatorname{Var} f(Z) \le \int_0^1 \big(\mathbb{E}|\nabla f(Z)|^2\big)^{1/2}\big(\mathbb{E}|\nabla f(tZ + \sqrt{1-t^2}\,Z')|^2\big)^{1/2}\,\mathrm{d}t = \mathbb{E}|\nabla f(Z)|^2,$$

because $tZ + \sqrt{1-t^2}\,Z'$ is again standard Gaussian. For f as in (i), let $f_\varepsilon(x) = \mathbb{E}f(x + \varepsilon Z'')$ for $0 < \varepsilon \le 1$, with Z'' standard Gaussian. Writing f_ε as the convolution of f with the density of $\mathcal{N}(0, \varepsilon^2 I_n)$ and differentiating the density shows that f_ε is infinitely differentiable with derivatives of moderate growth, and dominated convergence gives $\nabla f_\varepsilon(x) = \mathbb{E}\nabla f(x + \varepsilon Z'')$. As $\varepsilon \to 0$, $f_\varepsilon \to f$ and $\nabla f_\varepsilon \to \nabla f$ pointwise, and since $|x + \varepsilon z|^2 \le$

$2|x|^2 + 2|z|^2$, both are bounded by $C_a' e^{2a|x|^2}$ uniformly in ε; for $a < \frac{1}{8}$ dominated convergence carries the inequality for f_ε to the limit. If f is L-Lipschitz, then f_ε is smooth and L-Lipschitz, so $|\nabla f_\varepsilon| \le L$ and $\operatorname{Var} f_\varepsilon(Z) \le L^2$; and $|f_\varepsilon - f| \le \varepsilon L\, \mathbb{E}|Z''| \le \varepsilon L\sqrt{n}$, which gives (ii) as $\varepsilon \to 0$. Part (iii) is (i) for $g(z) = f(\Sigma^{1/2} z)$, for which $|\nabla g|^2 = \langle \nabla f, \Sigma \nabla f\rangle$. For complex f, add the inequalities for the real and imaginary parts. ∎

For the Sherrington–Kirkpatrick model, as a function of the $N(N-1)/2$ couplings, $|\nabla \log Z_N|^2 = \frac{\beta^2}{N} \sum_{i<j} \langle \sigma_i \sigma_j \rangle^2 \le \beta^2 (N-1)/2$, so the free entropy per spin has variance at most $\beta^2/(2N)$ at every N (Chapter 1).[3]

[3] Chernoff (1981) proved the inequality in dimension one by expanding f in Hermite polynomials. Houdré and Pérez-Abreu (1995) develop covariance identities like Lemma A.5; Boucheron et al. (2013, Chapter 3) derive the inequality from the Efron–Stein inequality and the central limit theorem.

Theorem A.7 (concentration of Lipschitz functions). *Let $Z \sim \mathcal{N}(0, I_n)$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz.*

(i) For every $\lambda \in \mathbb{R}$, $\mathbb{E} e^{\lambda (f(Z) - \mathbb{E} f(Z))} \le e^{\pi^2 \lambda^2 L^2/8}$; hence, for every $u \ge 0$,

$$\mathbb{P}\big(f(Z) - \mathbb{E} f(Z) \ge u\big) \le e^{-2u^2/(\pi^2 L^2)},$$

and $\mathbb{P}(|f(Z) - \mathbb{E} f(Z)| \ge u) \le 2e^{-2u^2/(\pi^2 L^2)}$.

(ii) (The sharp constant.) For every $\lambda \in \mathbb{R}$, $\mathbb{E} e^{\lambda (f(Z) - \mathbb{E} f(Z))} \le e^{\lambda^2 L^2/2}$; hence $\mathbb{P}(f(Z) - \mathbb{E} f(Z) \ge u) \le e^{-u^2/(2L^2)}$ and $\mathbb{P}(|f(Z) - \mathbb{E} f(Z)| \ge u) \le 2e^{-u^2/(2L^2)}$.

If $X \sim \mathcal{N}(0, \Sigma)$, the same bounds hold for $f(X)$ with L^2 replaced by $\|\Sigma\| L^2$.

We prove (i), which loses the factor $\pi^2/4$ in the exponent and suffices for every limit theorem of the book. Part (ii) is due to Cirel'son et al. (1976); with the median in place of the mean it also follows from the Gaussian isoperimetric inequality of Borell (1975) and Sudakov and Tsirel'son (1978). Boucheron et al. (2013, Chapter 5) prove (ii) through the Gaussian logarithmic Sobolev inequality; Vershynin (2018, Theorem 5.2.2) states the bound with an unspecified constant.

Proof of (i). The plan, known as the Maurey–Pisier argument (Pisier 1986), is to connect Z to an independent copy Z' along a quarter circle and to average the exponential along the arc by Jensen's inequality. Assume first that f is continuously differentiable with $|\nabla f| \le L$. For $\theta \in [0, \pi/2]$ let $Z_\theta = Z \sin\theta + Z' \cos\theta$ and $W_\theta = Z \cos\theta - Z' \sin\theta$, so that $\frac{\mathrm{d}}{\mathrm{d}\theta} Z_\theta = W_\theta$, $Z_0 = Z'$ and $Z_{\pi/2} = Z$. The pair (Z_θ, W_θ) is the image of (Z, Z') under a rotation of $\mathbb{R}^{2n}$, so it has the law of (Z, Z'). With $A_\theta = \frac{\lambda\pi}{2} \langle \nabla f(Z_\theta), W_\theta \rangle$, Jensen's inequality for the uniform probability measure on $[0, \pi/2]$ gives

$$e^{\lambda (f(Z) - f(Z'))} = \exp\Big(\frac{2}{\pi} \int_0^{\pi/2} A_\theta \, \mathrm{d}\theta \Big) \le \frac{2}{\pi} \int_0^{\pi/2} e^{A_\theta} \, \mathrm{d}\theta.$$

Given Z_θ, A_θ is centered Gaussian with variance $\frac{\lambda^2\pi^2}{4} |\nabla f(Z_\theta)|^2 \le \frac{\lambda^2 \pi^2 L^2}{4}$, so $\mathbb{E} e^{A_\theta} \le e^{\pi^2 \lambda^2 L^2/8}$, and by Tonelli's theorem $\mathbb{E} e^{\lambda (f(Z) - f(Z'))} \le e^{\pi^2 \lambda^2 L^2/8}$. Jensen's inequality in Z' gives $\mathbb{E} e^{-\lambda f(Z')} \ge e^{-\lambda \mathbb{E} f(Z)}$, so by independence $\mathbb{E} e^{\lambda (f(Z) - \mathbb{E} f(Z))} \le \mathbb{E} e^{\lambda (f(Z) - f(Z'))}$. For a general L-Lipschitz f, the smoothed f_ε of the proof of Theorem A.6 satisfies $|\nabla f_\varepsilon| \le L$ and $|f - f_\varepsilon| \le \varepsilon L \sqrt{n}$,

so $\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \le e^{2|\lambda|\varepsilon L\sqrt{n}}\,\mathbb{E}e^{\lambda(f_\varepsilon-\mathbb{E}f_\varepsilon)}$; let $\varepsilon \to 0$. Markov's inequality gives $\mathbb{P}(f-\mathbb{E}f \ge u) \le e^{-\lambda u+\pi^2\lambda^2L^2/8}$, which equals $e^{-2u^2/(\pi^2L^2)}$ at $\lambda = 4u/(\pi^2L^2)$; the bound for $-f$ gives the two-sided form. For $X = \Sigma^{1/2}Z$, the function $z \mapsto f(\Sigma^{1/2}z)$ is $\|\Sigma\|^{1/2}L$-Lipschitz. ∎

Corollary A.8 (concentration of a free entropy). *Let S be a finite set, $w\colon S \to (0,\infty)$, and $(H(\sigma))_{\sigma\in S}$ a centered Gaussian process with $v = \max_\sigma \operatorname{Var} H(\sigma)$. Both $F = \log\sum_\sigma w(\sigma)e^{H(\sigma)}$ and $F = \max_\sigma H(\sigma)$ satisfy $\operatorname{Var} F \le v$ and, for every $u \ge 0$, $\mathbb{P}(|F-\mathbb{E}F| \ge u) \le 2e^{-2u^2/(\pi^2 v)}$, and $\mathbb{P}(|F-\mathbb{E}F| \ge u) \le 2e^{-u^2/(2v)}$ with the sharp constant.*

Proof. Write $H = AZ$ with Z standard Gaussian and rows a_σ of A, so that $|a_\sigma|^2 = \operatorname{Var} H(\sigma) \le v$. The gradient of $z \mapsto \log\sum_\sigma w(\sigma)e^{\langle a_\sigma\rangle z}$ is $\sum_\sigma p_\sigma a_\sigma$ for Gibbs weights p_σ, of norm at most $\max_\sigma|a_\sigma| \le \sqrt{v}$; and $|\max_\sigma\langle a_\sigma\rangle z - \max_\sigma\langle a_\sigma, z'\rangle| \le \max_\sigma|\langle a_\sigma, z-z'\rangle| \le \sqrt{v}\,|z-z'|$. Both functions are $\sqrt{v}$-Lipschitz, and Theorems A.6 and A.7 apply. ∎

A deterministic part of the Hamiltonian, such as an external field, goes into the weights w. For the Sherrington–Kirkpatrick model $\operatorname{Var} H_N(\sigma) = \beta^2(N-1)/2$ for every σ, so the corollary with the sharp constant gives

$$\mathbb{P}\Big(\Big|\frac{1}{N}\log Z_N - \mathbb{E}\frac{1}{N}\log Z_N\Big| \ge u\Big) \le 2e^{-Nu^2/\beta^2},$$

the bound of Theorem 1.8; the constant of part (i) gives the exponent $4Nu^2/(\pi^2\beta^2)$. The bounds are summable in N, so the Borel–Cantelli lemma turns the convergence of $\mathbb{E}\frac{1}{N}\log Z_N$ into almost sure convergence. Chapters 2 to 4 use the corollary in this way, and Chapter 8 uses Theorem A.7 for the values of Gaussian min–max problems.[4]

[4] The sharp form of Corollary A.8 for $\max_\sigma H(\sigma)$ is often called the Borell–TIS inequality, after Borell (1975) and Cirel'son et al. (1976).

A.4 Laplace's method and large deviations

A physicist evaluates an integral $\int e^{Nf}$ at the maximum of f, the saddle point, after a Hubbard–Stratonovich transformation has turned a sum over configurations into such an integral. Laplace's method justifies the first step; large deviations and Varadhan's lemma evaluate the sum over configurations directly.

Definition A.9 (large deviation principle). Let $(\mathcal{X}, d)$ be a metric space. A *rate function* is a lower semicontinuous function $I\colon \mathcal{X} \to [0,\infty]$; it is *good* if its level sets $\{I \le \alpha\}$ are compact. Probability measures μ_N on $\mathcal{X}$ satisfy the *large deviation principle* with speed N and rate function I if, for every closed set C and every open set G,

$$\limsup_{N\to\infty}\frac{1}{N}\log\mu_N(C) \le -\inf_C I, \qquad \liminf_{N\to\infty}\frac{1}{N}\log\mu_N(G) \ge -\inf_G I.$$

By Cramér's theorem the empirical mean of N i.i.d. copies of X, with $\mathbb{E}e^{\langle\lambda\rangle X} < \infty$ for every λ, satisfies the principle with the Legendre transform

of $\lambda \mapsto \log \mathbb{E}e^{\langle\lambda\rangle X}$ as rate (Dembo and Zeitouni 2010, Chapter 2). The case that spin systems need has a short proof.

Lemma A.10 (the entropy of a magnetization). *Let $\sigma_1, \dots, \sigma_N$ be independent uniform signs, $m_N = \frac{1}{N}\sum_i \sigma_i$, and $I(m) = \frac{1+m}{2}\log(1+m) + \frac{1-m}{2}\log(1-m)$, with $0\log 0 = 0$. For every $m \in \{-1, -1+\frac{2}{N}, \dots, 1\}$,*

$$\frac{e^{-NI(m)}}{N+1} \le \mathbb{P}(m_N = m) \le e^{-NI(m)}.$$

Consequently the laws of m_N satisfy the large deviation principle on $[-1,1]$ with the good rate function I.

Proof. Let $k = N(1+m)/2$, $p = k/N$ and $b_j = \binom{N}{j}p^j(1-p)^{N-j}$. Since $p^k(1-p)^{N-k} = e^{-N(\log 2 - I(m))}$, $\mathbb{P}(m_N = m) = \binom{N}{k}2^{-N} = b_k e^{-NI(m)}$. The $N+1$ numbers b_j sum to 1, so $b_k \le 1$; and b_k is the largest of them, because $b_{j+1}/b_j = (N-j)p/((j+1)(1-p))$ is at least 1 for $j < k$ and at most 1 for $j \ge k$, so $b_k \ge 1/(N+1)$. The function I is continuous on the compact set $[-1,1]$, hence a good rate function. For a closed set C, summing the upper bound over the at most $N+1$ points of C gives $\mathbb{P}(m_N \in C) \le (N+1)e^{-N\inf_C I}$. For an open set G and $m \in G$, the lattice points m'_N nearest to m lie in G for large N and tend to m, and the lower bound gives $\liminf \frac{1}{N}\log \mathbb{P}(m_N \in G) \ge -\lim I(m'_N) = -I(m)$. ∎

Theorem A.11 (Laplace's method and Varadhan's lemma).

(i) (Laplace, exponential order.) Let ν be a Borel measure on a Borel set $A \subset \mathbb{R}^d$ with $\nu(A) > 0$, and let $f : A \to \mathbb{R}$ be measurable and bounded above, with $\int_A e^{N_0 f}\,\mathrm{d}\nu < \infty$ for some $N_0 > 0$. Let M be the essential supremum of f with respect to ν; it equals $\sup_A f$ when f is continuous and every nonempty relatively open subset of A has positive measure. Then $\frac{1}{N}\log\int_A e^{Nf}\,\mathrm{d}\nu \to M$ as $N \to \infty$.

(ii) (Laplace, with the prefactor.) Let $A \subset \mathbb{R}^d$ be a Borel set and $f, g : A \to \mathbb{R}$ measurable, with $\int_A |g|e^{N_0 f}\,\mathrm{d}x < \infty$ for some N_0. Suppose that x^* is an interior point of A, that f is twice continuously differentiable near x^* with $\nabla f(x^*) = 0$ and $H = -\nabla^2 f(x^*)$ positive definite, that g is continuous at x^*, and that $\sup\{f(x) : x \in A,\ |x - x^*| \ge \delta\} < f(x^*)$ for every $\delta > 0$. Then, as $N \to \infty$,

$$\int_A g(x)e^{Nf(x)}\,\mathrm{d}x = e^{Nf(x^*)}\Big(\frac{2\pi}{N}\Big)^{d/2}(\det H)^{-1/2}\big(g(x^*) + o(1)\big). \tag{A.4}$$

(iii) (Varadhan.) If probability measures μ_N on a metric space $\mathcal{X}$ satisfy the large deviation principle with a good rate function I, and $F : \mathcal{X} \to \mathbb{R}$ is continuous and bounded above, then

$$\lim_{N\to\infty}\frac{1}{N}\log\int_{\mathcal{X}} e^{NF}\,\mathrm{d}\mu_N = \sup_{x\in\mathcal{X}}\big(F(x) - I(x)\big). \tag{A.5}$$

Proof. (i) Let $J_N = \int e^{Nf}\,\mathrm{d}\nu$. For $N \ge N_0$, $J_N \le e^{(N-N_0)M} J_{N_0}$, so the upper limit of $\frac{1}{N}\log J_N$ is at most M. For $\varepsilon > 0$ the set $\{f > M-\varepsilon\}$ has positive measure, by the definition of the essential supremum, and finite measure, because $e^{N_0 f} \ge e^{N_0(M-\varepsilon)}$ on it. So $J_N \ge e^{N(M-\varepsilon)}\nu(\{f > M-\varepsilon\})$, and the lower limit of $\frac{1}{N}\log J_N$ is at least $M-\varepsilon$.

(ii) The plan is to compare f with its quadratic approximation on a small ball and to show that the rest of A contributes an exponentially smaller amount. We may assume $x^* = 0$ and $f(0) = 0$. Let $\mu > 0$ be the smallest eigenvalue of H, fix $\varepsilon \in (0, \frac{1}{2})$, and choose $\delta > 0$ such that the ball $B = \{|x| < \delta\}$ lies in A and $\|\nabla^2 f(x) + H\| \le \varepsilon\mu$ and $|g(x) - g(0)| \le \varepsilon$ on B. Taylor's formula $f(x) = \int_0^1 (1-s)\, x^\top \nabla^2 f(sx)\, x\,\mathrm{d}s$ gives $|f(x) + \frac{1}{2}x^\top H x| \le \frac{1}{2}\varepsilon\mu|x|^2 \le \frac{\varepsilon}{2} x^\top H x$ on B, hence

$$-\tfrac{1+\varepsilon}{2}\, x^\top H x \le f(x) \le -\tfrac{1-\varepsilon}{2}\, x^\top H x \qquad (x \in B).$$

Let $c_N = (2\pi/N)^{d/2}(\det H)^{-1/2} = \int_{\mathbb{R}^d} e^{-Nx^\top Hx/2}\,\mathrm{d}x$. The upper bound gives $\int_B e^{Nf} \le (1-\varepsilon)^{-d/2} c_N$. The lower bound gives $\int_B e^{Nf} \ge (1+\varepsilon)^{-d/2} c_N - \int_{|x|\ge\delta} e^{-N\mu|x|^2/2}\,\mathrm{d}x$, and the last integral is at most $e^{-N\mu\delta^2/4}\int e^{-N\mu|x|^2/4}\,\mathrm{d}x = o(c_N)$. Also $|\int_B (g-g(0))e^{Nf}| \le \varepsilon(1-\varepsilon)^{-d/2} c_N$. Outside B, $\eta = -\sup\{f(x) : x \in A \setminus B\}$ is positive, and for $N \ge N_0$ the integral of $|g|e^{Nf}$ over $A \setminus B$ is at most $e^{-(N-N_0)\eta}\int_A |g| e^{N_0 f} = o(c_N)$. Collecting the three estimates,

$$\limsup_{N\to\infty}\Big|\frac{1}{c_N}\int_A g e^{Nf} - g(0)\Big| \le |g(0)|\big((1-\varepsilon)^{-d/2} - (1+\varepsilon)^{-d/2}\big) + \frac{\varepsilon}{(1-\varepsilon)^{d/2}},$$

which tends to 0 with ε.

(iii) The plan is to get the lower bound from small open sets and the upper bound by covering a compact level set of I with finitely many small balls. Write $J_N = \int e^{NF}\,\mathrm{d}\mu_N$. Lower bound: for x with $I(x) < \infty$ and $\varepsilon > 0$, the set $G = \{F > F(x) - \varepsilon\}$ is open and contains x, so $J_N \ge e^{N(F(x)-\varepsilon)}\mu_N(G)$ and $\liminf \frac{1}{N}\log J_N \ge F(x) - \varepsilon - I(x)$; take the supremum over x and let $\varepsilon \to 0$. Upper bound: let $M = \sup F$, fix $\alpha < \infty$ and $\varepsilon > 0$, and let $K = \{I \le \alpha\}$, which is compact. By the lower semicontinuity of I and the continuity of F, each $x \in K$ is the center of an open ball B_x whose closure $\bar B_x$ satisfies $\inf_{\bar B_x} I \ge I(x) - \varepsilon$ and $\sup_{\bar B_x} F \le F(x) + \varepsilon$. Finitely many of these balls, centered at $x_1, \dots, x_k$, cover K, and the closed set $C = \mathcal{X} \setminus \bigcup_j B_{x_j}$ satisfies $\inf_C I \ge \alpha$. Then

$$J_N \le \sum_{j=1}^{k} e^{N(F(x_j)+\varepsilon)}\mu_N(\bar B_{x_j}) + e^{NM}\mu_N(C).$$

The upper limit of $\frac{1}{N}\log$ of a sum of $k+1$ terms is the largest of the upper limits for the terms, so the principle gives $\limsup \frac{1}{N}\log J_N \le \max\{\sup(F - I) + 2\varepsilon,\ M - \alpha\}$. Let $\alpha \to \infty$ and $\varepsilon \to 0$. ∎

For the Curie–Weiss model, $Z_N = \sum_\sigma e^{NF(m(\sigma))} = 2^N \mathbb{E} e^{NF(m_N)}$ with $F(m) = \beta m^2/2 + hm$, and Lemma A.10 with part (iii) gives the limit $\log 2 + \sup_m(F(m) - I(m))$ of $\frac{1}{N}\log Z_N$. The physicist instead linearizes

the square by the Hubbard–Stratonovich identity, sums over σ, and applies part (ii) to $\int \exp(N[-\beta y^2/2 + \log 2\cosh(\beta y + h)])\,\mathrm{d}y$. The two variational formulas agree because I is the Legendre transform of log cosh, the logarithmic moment generating function of a uniform sign; for $\beta = 1.5$ and $h = 0.1$ both give 0.896082602766, and exact summation exceeds it by 1.80×10^{-4} at $N = 1000$ (Chapter 1 carries out both computations). Part (iii) also makes the integer moments $\mathbb{E}Z_N^n$ rigorous at fixed n as $N \to \infty$, as integrals over the large deviations of the overlap matrix of n replicas; it says nothing about the continuation to $n \to 0$.[5]

[5] The lemma is due to Varadhan (1966). Dembo and Zeitouni (2010, Chapter 4) prove it under a weaker tail condition on F, and Anderson et al. (2010, Theorem D.8) state it for bounded continuous F.

A.5 Gaussian matrices and conditioning

Definition A.12 (the Gaussian orthogonal ensemble). A random symmetric $N \times N$ matrix W is a GOE(N) matrix if its entries W_{ij}, $i \le j$, are independent and centered Gaussian, with variance $1/N$ off the diagonal and $2/N$ on it. Equivalently, $W = (G+G^\top)/\sqrt{2N}$ for a matrix G with i.i.d. $\mathcal{N}(0,1)$ entries, or W has density proportional to $e^{-N\,\mathrm{tr}(W^2)/4}$ with respect to Lebesgue measure on the entries $(W_{ij})_{i\le j}$. Its covariance is $\mathbb{E}W_{ab}W_{cd} = (\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})/N$, so that $\mathrm{tr}(AW) \sim \mathcal{N}(0, 2\,\mathrm{tr}(A^2)/N)$ for every symmetric A, and $OWO^\top$ is again GOE(N) for every orthogonal O, because it is centered Gaussian with the same covariance.

The cavity method removes a site and asserts that the field acting on it is Gaussian. The analysis of approximate message passing removes the past instead: conditioned on what earlier iterates revealed, the matrix is a known part plus a fresh Gaussian matrix projected away from the directions already explored. For $M \in \mathbb{R}^{N\times t}$ of rank t, let $P_M = M(M^\top M)^{-1}M^\top$ be the orthogonal projection onto its column space, $P_M^\perp = I - P_M$, $M^+ = (M^\top M)^{-1}M^\top$, and, for $Y \in \mathbb{R}^{N\times t}$,

$$E_M(Y) = YM^+ + (M^+)^\top Y^\top - (M^+)^\top M^\top Y M^+. \tag{A.6}$$

Lemma A.13 (Gaussian conditioning). *Let $W \sim$ GOE(N), let $\widetilde{W}$ be an independent* GOE(N) *matrix, and let $1 \le t < N$.*

(i) (Fixed directions.) If $M \in \mathbb{R}^{N\times t}$ is deterministic of rank t, then $W = E_M(WM) + P_M^\perp W P_M^\perp$, and $P_M^\perp W P_M^\perp$ is independent of WM. Hence, given $WM = Y$, the matrix W is distributed as $E_M(Y) + P_M^\perp \widetilde{W} P_M^\perp$: for every bounded measurable h, $\mathbb{E}[h(W) \mid WM] = \hat{h}_M(WM)$ almost surely, with $\hat{h}_M(Y) = \mathbb{E}\,h\big(E_M(Y) + P_M^\perp \widetilde{W} P_M^\perp\big)$. In particular $\mathbb{E}[W \mid WM] = E_M(WM)$.

(ii) (Directions chosen from the past.) Let $m^0 \in \mathbb{R}^N$ be deterministic and $m^s = F_s(Wm^0, \dots, Wm^{s-1})$ for $1 \le s < t$, with measurable maps F_s. Let $M_t = [m^0|\cdots|m^{t-1}]$, $Y_t = WM_t$ and $\mathcal{S}_t = \sigma(Y_t)$. If M_t has rank t almost surely, then given $\mathcal{S}_t$ the matrix W is distributed as $E_{M_t}(Y_t) + P_{M_t}^\perp \widetilde{W} P_{M_t}^\perp$: for every bounded measurable h, $\mathbb{E}[h(W) \mid$

$\mathcal{S}_t] = \hat{h}_{M_t}(Y_t)$ *almost surely.*

(iii) (The fresh field.) If P is an orthogonal projection and e a unit vector with $Pe = e$, then $P\widetilde{W}e \sim \mathcal{N}\big(0, (P + ee^\top)/N\big)$, the law of $N^{-1/2}(Pz + \zeta e)$ for independent $z \sim \mathcal{N}(0, I_N)$ and $\zeta \sim \mathcal{N}(0,1)$.

(iv) (Rectangular matrices.) Let $A \in \mathbb{R}^{n\times d}$ have i.i.d. $\mathcal{N}(0, 1/n)$ entries, let $\widetilde{A}$ be an independent copy, and let $Q \in \mathbb{R}^{d\times t_1}$ and $M \in \mathbb{R}^{n\times t_2}$ be deterministic of full column rank. Then

$$A = M(M^\top M)^{-1}(A^\top M)^\top + P_M^\perp (AQ)(Q^\top Q)^{-1}Q^\top + P_M^\perp A P_Q^\perp,$$

and the last term is independent of $(AQ, A^\top M)$. Given $AQ = Y$ and $A^\top M = X$, the matrix A is therefore distributed as $M(M^\top M)^{-1}X^\top + P_M^\perp Y(Q^\top Q)^{-1}Q^\top + P_M^\perp \widetilde{A} P_Q^\perp$. The analogue of (ii) holds, with the same proof.

Proof. (i) The plan is to split W into the part determined by WM and a part uncorrelated with it. Write $P = P_M$ and $Y = WM$. Expanding $W = (P + P^\perp)W(P + P^\perp)$ gives $W = WP + PW - PWP + P^\perp W P^\perp$. Since $P = MM^+$, we have $WP = YM^+$, then $PW = (WP)^\top = (M^+)^\top Y^\top$ by symmetry, and $PWP = (M^+)^\top M^\top Y M^+$; so $W = E_M(WM) + P^\perp W P^\perp$. The entries of WM and of $P^\perp W P^\perp$ are jointly Gaussian, and by the covariance of Definition A.12

$$\mathbb{E}\big[(P^\perp W P^\perp)_{ij}(WM)_{kl}\big] = \frac{1}{N}\big(P^\perp_{ik}(P^\perp M)_{jl} + (P^\perp M)_{il}P^\perp_{jk}\big) = 0,$$

because $P^\perp M = 0$; so they are independent. Since $P^\perp W P^\perp$ has the law of $P^\perp \widetilde{W} P^\perp$, integrating it out (Fubini's theorem) gives the formula for $\mathbb{E}[h(W) \mid WM]$.

(ii) The plan is an induction in which each new direction is split into its component along the old directions, which the past determines, and a fresh component, to which (i) applies. For M of rank s and Y with $M^\top Y$ symmetric, let $L_{M,Y}$ be the law of $E_M(Y) + P_M^\perp \widetilde{W} P_M^\perp$. We prove by induction on $s \le t$ that, for every bounded measurable H,

$$(\mathrm{C}_s) \qquad \mathbb{E}\, H(W, Y_s) = \mathbb{E} \int H(w, Y_s)\, L_{M_s,Y_s}(\,\mathrm{d}w),$$

which for $s = t$ and $H(w, y) = h(w)k(y)$ is the claim; recall that M_s is a function of Y_s. For $s = 1$, $M_1 = m^0$ is deterministic and (C_1) is (i). Assume (C_s) for some $s < t$, and let H be bounded and measurable in (w, y), $y \in \mathbb{R}^{N\times(s+1)}$. Since $m^s = F_s(Y_s)$ and $Y_{s+1} = [Y_s | W m^s]$, applying (C_s) to $(w, y) \mapsto H(w, [y|wF_s(y)])$ gives

$$\mathbb{E}\, H(W, Y_{s+1}) = \mathbb{E} \int H(w, [Y_s|w m^s])\, L_{M_s,Y_s}(\,\mathrm{d}w).$$

Fix $M = M_s$, $Y = Y_s$ and $m = m^s$ inside the integral, and write $P^\perp = P_M^\perp$. The vector $u = P^\perp m$ is not zero, because $M_{s+1} = [M|m]$ has rank $s + 1$;

let $e = u/|u|$, and let $Q = P^\perp - ee^\top$, the projection onto the orthogonal complement of the columns of M_{s+1}. Under $L_{M,Y}$ write $w = E_M(Y) + V$ with $V = P^\perp \widetilde{W} P^\perp$. Then $Vm = |u|\,Ve$, so $wM_{s+1} = [Y|E_M(Y)m + |u|\,Ve]$ is a function of Ve. By the decomposition of (i), applied to w and M_{s+1}, $w = E_{M_{s+1}}(wM_{s+1}) + QwQ$, and $QwQ = Q\widetilde{W}Q$ because $M^+Q = 0$ and $QP^\perp = Q$. The Gaussian pair $(Ve, Q\widetilde{W}Q)$ is uncorrelated, since Definition A.12 gives

$$\mathbb{E}\big[(Q\widetilde{W}Q)_{ij}(P^\perp\widetilde{W}e)_k\big] = \tfrac{1}{N}\big((QP^\perp)_{ik}(Qe)_j + (Qe)_i(QP^\perp)_{jk}\big) = 0,$$

because $Qe = 0$. So $Q\widetilde{W}Q$ may be replaced by $Q\widetilde{W}'Q$, with $\widetilde{W}'$ an independent copy, without changing the joint law. Hence, with $\bar{H}$ defined by the first line,

$$\bar{H}(\hat{y}) = \int H(w', \hat{y})\, L_{M_{s+1},\hat{y}}(\,\mathrm{d}w'),$$
$$\int H(w, [Y|wm])\, L_{M,Y}(\,\mathrm{d}w) = \int \bar{H}(wM_{s+1})\, L_{M,Y}(\,\mathrm{d}w).$$

Applying (C_s) to $(w, y) \mapsto \bar{H}([y|wF_s(y)])$, in which M_{s+1} is the function $[M_s|F_s(y)]$ of y, we obtain $\mathbb{E}\,H(W, Y_{s+1}) = \mathbb{E}\,\bar{H}(Y_{s+1})$, which is (C_{s+1}).

(iii) By Definition A.12, $\mathbb{E}(\widetilde{W}e)_a(\widetilde{W}e)_c = \sum_{b,d} e_b e_d\, \mathbb{E}\widetilde{W}_{ab}\widetilde{W}_{cd} = (\delta_{ac} + e_a e_c)/N$, so $P\widetilde{W}e$ has covariance $P(I + ee^\top)P/N = (P + ee^\top)/N$, which is also the covariance of $N^{-1/2}(Pz + \zeta e)$.

(iv) The decomposition is $A = P_M A + P_M^\perp A P_Q + P_M^\perp A P_Q^\perp$, rewritten with the identities $P_M A = M(M^\top M)^{-1}(A^\top M)^\top$ and $AP_Q = AQ(Q^\top Q)^{-1}Q^\top$. Since $\mathbb{E}A_{ab}A_{kc} = \delta_{ak}\delta_{bc}/n$, the covariance of $(P_M^\perp A P_Q^\perp)_{ij}$ with $(AQ)_{kl}$ is $\frac{1}{n}(P_M^\perp)_{ik}(P_Q^\perp Q)_{jl} = 0$, and with $(A^\top M)_{kl}$ it is $\frac{1}{n}(P_M^\perp M)_{il}(P_Q^\perp)_{kj} = 0$; the rest is as in (i) and (ii). ∎

In symmetric AMP (Chapter 6) the iterates are $h^{s+1} = Wm^s - b_s m^{s-1}$ with $m^s = f_s(h^s)$, so the past is a function of $Y_t = WM_t$ with $M_t = [m^0|\cdots|m^{t-1}]$, and (ii) applies. Given the past, the next field Wm^t has the law of $E_{M_t}(Y_t)m^t + |P^\perp m^t|\,P^\perp\widetilde{W}e$, with $P^\perp = P_{M_t}^\perp$ and e the direction of $P^\perp m^t$. The first term, $Y_t M_t^+ m^t + (M_t^+)^\top Y_t^\top P^\perp m^t$, combines past fields and past iterates, and Chapter 6 extracts the Onsager correction from it; by (iii) the second is a fresh Gaussian vector, the rigorous form of the Gaussianity of the cavity field. Bolthausen (2014) introduced the conditioning to construct solutions of the TAP equations, Bayati and Montanari (2011) used it to prove state evolution, and Feng et al. (2022) give a tutorial account. Part (iv) serves the rectangular AMP of Chapters 7 and 10, and Chapter 14 uses the same idea for gradient flow.

A.6 Random matrices

Physicists study a random spectrum through the Green's function $(z - A)^{-1}$; mathematicians use the resolvent, its negative, and its normalized trace, the Stieltjes transform.

Definition A.14 (Stieltjes transform and resolvent). The *Stieltjes transform* of a probability measure μ on $\mathbb{R}$ is $s_\mu(z) = \int \mu(\mathrm{d}x)/(x-z)$ for $z \in \mathbb{C} \setminus \operatorname{supp}\mu$. The *resolvent* of a symmetric matrix $A \in \mathbb{R}^{N\times N}$, with eigenvalues λ_k and orthonormal eigenvectors u_k, is $R_A(z) = (A - zI)^{-1} = \sum_k u_k u_k^\top/(\lambda_k - z)$ for $z \notin \{\lambda_k\}$. With the empirical spectral distribution $\mu_A = \frac{1}{N}\sum_k \delta_{\lambda_k}$ and, for $v \in \mathbb{R}^N$, the spectral measure $\nu_v = \sum_k \langle u_k\rangle v^2 \delta_{\lambda_k}$,

$$\tfrac{1}{N}\operatorname{tr} R_A(z) = s_{\mu_A}(z), \qquad v^\top R_A(z) v = \int \frac{\nu_v(\mathrm{d}x)}{x-z}.$$

These objects have the following properties; z, w lie outside $\operatorname{supp}\mu$, or outside the spectrum of A.

(i) The function s_μ is analytic, $s_\mu(\bar z) = \overline{s_\mu(z)}$, $\Im s_\mu(z) > 0$ when $\Im z > 0$, and $|s_\mu(z)| \le 1/\operatorname{dist}(z, \operatorname{supp}\mu)$. For $z, w \notin \mathbb{R}$, $|s_\mu(z) - s_\mu(w)| \le |z-w|/(|\Im z||\Im w|)$. On each interval of $\mathbb{R}\setminus\operatorname{supp}\mu$, s_μ is real and increasing.

(ii) (Inversion.) If $a < b$ are not atoms of μ, then $\mu((a,b)) = \lim_{\eta\downarrow 0}\frac{1}{\pi}\int_a^b \Im s_\mu(x+i\eta)\,\mathrm{d}x$.

(iii) (Continuity.) If μ_N and μ are probability measures and $s_{\mu_N}(z) \to s_\mu(z)$ for every $z \in \mathbb{C}\setminus\mathbb{R}$, then $\mu_N \to \mu$ weakly; the converse also holds.

(iv) (Resolvent identities.) $AR_A(z) = I + zR_A(z)$; $R_A(z) - R_A(w) = (z-w)R_A(z)R_A(w)$, so that $\frac{\mathrm{d}}{\mathrm{d}z}R_A(z) = R_A(z)^2$; $R_{A+B}(z) = R_A(z) - R_{A+B}(z)BR_A(z)$; $R_A(z)^\top = R_A(z)$; and $\|R_A(z)\| = 1/\operatorname{dist}(z, \{\lambda_k\}) \le 1/|\Im z|$.

(v) (Rank-one perturbations.) For $\theta \in \mathbb{R}$ and $v \in \mathbb{R}^N$, $\det(A + \theta vv^\top - zI) = \det(A - zI)\,(1 + \theta\, v^\top R_A(z) v)$. Hence a real z outside the spectrum of A is an eigenvalue of $A + \theta vv^\top$ if and only if $1 + \theta\, v^\top R_A(z) v = 0$, and then every unit eigenvector $\hat u$ for z satisfies

$$\langle \hat u\rangle v^2 = \frac{1}{\theta^2\, v^\top R_A(z)^2 v} = \frac{1}{\theta^2 \frac{\mathrm{d}}{\mathrm{d}z}\big(v^\top R_A(z) v\big)}. \tag{A.7}$$

(vi) (Schur complement.) For $z \notin \mathbb{R}$, $R_A(z)_{ii} = \big(A_{ii} - z - a_i^\top R_{A^{(i)}}(z) a_i\big)^{-1}$, where $A^{(i)}$ is A with row and column i removed and a_i is column i of A without its ith entry.

Proofs. The plan is to read (i) and (iv) off the spectral decomposition, to take (ii) and (iii) from the literature, and to reduce (v) and (vi) to determinant and block-inverse formulas. (i) Differentiate under the integral sign; use $\Im\frac{1}{x-z} = \frac{\Im z}{|x-z|^2}$, $|x-z| \ge \operatorname{dist}(z, \operatorname{supp}\mu)$ and $\big|\frac{1}{x-z} - \frac{1}{x-w}\big| = \frac{|z-w|}{|x-z||x-w|}$; on $\mathbb{R}\setminus\operatorname{supp}\mu$ the derivative $\int \mu(\mathrm{d}x)/(x-z)^2$ is positive. (ii) and (iii) are Theorems 2.4.3 and 2.4.4 of Anderson et al. (2010); their part (b) gives vague convergence

to the measure whose transform is s_μ, which is μ, and vague convergence to a probability measure is weak convergence. (iv) The first identity is $(A-z)R_A(z) = I$. Next, $R_A(z)-R_A(w) = R_A(z)\big[(A-w)-(A-z)\big]R_A(w)$ and $R_{A+B}-R_A = R_{A+B}\big[(A-z)-(A+B-z)\big]R_A$; the rest follows from the spectral decomposition. (v) Factor $A+\theta vv^\top - z = (A-z)(I+\theta R_A(z)vv^\top)$ and use $\det(I+xy^\top) = 1+y^\top x$. If $(A+\theta vv^\top)\hat u = z\hat u$, then $(A-z)\hat u = -\theta\langle v,\hat u\rangle v$, so $\hat u = -\theta\langle v,\hat u\rangle R_A(z)v$, and $|\hat u| = 1$ gives (A.7), with $v^\top R_A(z)^2 v = \frac{\mathrm{d}}{\mathrm{d}z} v^\top R_A(z)v$ by (iv). (vi) is the formula for the first diagonal entry of the inverse of a block matrix, after a permutation that moves i to the first position. ∎

Chapter 5 proves the BBP transition (Theorem 5.1) from (v) with $A = W$ and $\theta vv^\top = \frac{\lambda}{N}xx^\top$: the outlier solves $1+\lambda\, v^\top R_W(z)v = 0$ with $v = x/\sqrt{N}$, and (A.7) gives the overlap of its eigenvector with x. Chapter 9 expresses the ridge risk through the Stieltjes transform of the Marchenko–Pastur law.[6]

[6] Wigner introduced random symmetric matrices as models of the spectra of heavy nuclei (Wigner 1955); the resolvent method was pioneered by Pastur and coworkers, as Anderson et al. (2010) recount.

Theorem A.15 (the semicircle law and the norm of a GOE matrix). *Let $W \sim \mathrm{GOE}(N)$, let μ_N be its empirical spectral distribution and $s_N = s_{\mu_N}$, and let $\sigma_{\mathrm{sc}}(\mathrm{d}x) = \frac{1}{2\pi}\sqrt{4-x^2}\,\mathbf{1}_{|x|\le 2}\,\mathrm{d}x$ be the semicircle law. Its Stieltjes transform is*

$$s(z) = \tfrac{1}{2}\big(-z+\sqrt{z-2}\,\sqrt{z+2}\big), \tag{A.8}$$

with principal square roots; for $\Im z > 0$ it is the solution of $s^2+zs+1=0$ with $\Im s > 0$.

(i) Almost surely $\mu_N \to \sigma_{\mathrm{sc}}$ weakly. Quantitatively, for $\Im z = \eta > 0$,

$$|\mathbb{E}s_N(z) - s(z)| \le \frac{1}{\eta}\Big(\frac{1}{N\eta^2}+\frac{2}{N^2\eta^4}\Big),$$
$$\mathbb{E}|s_N(z)-\mathbb{E}s_N(z)|^2 \le \frac{2}{N^2\eta^4}.$$

(ii) For every N, $\mathbb{E}\lambda_{\max}(W) \le 2$ and, for $u \ge 0$, $\mathbb{P}(\lambda_{\max}(W) \ge 2+u) \le e^{-Nu^2/\pi^2}$; with the sharp constant of Theorem A.7, $e^{-Nu^2/4}$.

(iii) Almost surely $\lambda_{\max}(W) \to 2$, $\lambda_{\min}(W) \to -2$ and $\|W\| \to 2$.

(iv) (Wigner matrices; stated.) Let $W_{ij} = W_{ji} = X_{ij}/\sqrt{N}$, where the X_{ij}, $i<j$, are i.i.d. with mean 0 and variance 1, independent of the i.i.d. diagonal entries X_{ii}, which have mean 0 and finite variance. Then $\mu_N \to \sigma_{\mathrm{sc}}$ weakly in probability. If moreover $\mathbb{E}X_{12}^4 < \infty$, then $\lambda_{\max}(W) \to 2$ and $\|W\| \to 2$ almost surely.

Proof of (i)–(iii). The plan is to derive the quadratic equation for $\mathbb{E}s_N$ by Gaussian integration by parts, to control the fluctuations of s_N by the Poincaré inequality, to bound the top eigenvalue by the Sudakov–Fernique inequality, and to conclude with concentration. Fix z with $\eta = \Im z > 0$ and write $R = R_W(z)$, a smooth function of the independent entries $(W_{ab})_{a\le b}$ with $|R_{ab}| \le \|R\| \le 1/\eta$. Differentiating $(W-z)R = I$ in the entry W_{ab}, which for $a<b$ moves both W_{ab} and W_{ba}, gives $\partial R/\partial W_{ab} = -R(e_ae_b^\top + e_be_a^\top)R$ for $a<b$ and $\partial R/\partial W_{aa} = -Re_ae_a^\top R$. Stein's identity in the single variable W_{ab},

of variance $(1+\delta_{ab})/N$, gives for all a, b

$$\mathbb{E}[W_{ab}R_{ba}] = -\frac{1}{N}\,\mathbb{E}\big[R_{aa}R_{bb} + R_{ab}^2\big],$$

because R is symmetric. Summing over a, b and using $\operatorname{tr}(WR) = N + z\operatorname{tr} R$ from Definition A.14 (iv),

$$1 + z\,\mathbb{E}s_N = -\mathbb{E}[s_N^2] - \frac{1}{N^2}\mathbb{E}\operatorname{tr}(R^2), \qquad \left|\frac{1}{N^2}\operatorname{tr}(R^2)\right| \le \frac{1}{N\eta^2}.$$

For the variance write $W_{ab} = \sqrt{(1+\delta_{ab})/N}\,g_{ab}$ with independent standard normals g_{ab}, $a \le b$. Since $\partial s_N/\partial W_{ab} = -\frac{2}{N}(R^2)_{ab}$ for $a < b$ and $-\frac{1}{N}(R^2)_{aa}$ for $a = b$,

$$\sum_{a\le b}\left|\frac{\partial s_N}{\partial g_{ab}}\right|^2 = \frac{2}{N^3}\sum_{a,b}|(R^2)_{ab}|^2 = \frac{2}{N^3}\|R^2\|_F^2 \le \frac{2}{N^2}\|R\|^4 \le \frac{2}{N^2\eta^4},$$

and Theorem A.6 gives $\mathbb{E}|s_N - \mathbb{E}s_N|^2 \le 2/(N^2\eta^4)$. Let $m_N = \mathbb{E}s_N(z)$. Since $\mathbb{E}[s_N^2] = m_N^2 + \mathbb{E}[(s_N - m_N)^2]$, the two displays give $m_N^2 + zm_N + 1 = \varepsilon_N$ with $|\varepsilon_N| \le 1/(N\eta^2) + 2/(N^2\eta^4)$. The roots of $x^2 + zx + 1$ are $s = s(z)$ and $s' = -z - s$, and $\Im s' = -\eta - \Im s < -\eta$; since $\Im m_N > 0$, $|m_N - s'| > \eta$, and $|m_N - s| = |\varepsilon_N|/|m_N - s'| \le |\varepsilon_N|/\eta$. This is the first bound of (i).

For almost sure convergence, Chebyshev's inequality and the variance bound give $\mathbb{P}(|s_N(z) - m_N(z)| > N^{-1/4}) \le 2N^{-3/2}\eta^{-4}$, which is summable, so by the Borel–Cantelli lemma $s_N(z) \to s(z)$ almost surely, for each z; hence, almost surely, for all z in a countable dense subset of $\{\Im z > 0\}$. The equicontinuity in Definition A.14 (i) extends this to every z with $\Im z > 0$, and conjugation to every $z \notin \mathbb{R}$. That s is the Stieltjes transform of σ_{sc} is a classical computation (Anderson et al. 2010, Section 2.4.1), so Definition A.14 (iii) gives $\mu_N \to \sigma_{\mathrm{sc}}$ weakly, almost surely.

(ii) For unit vectors u, v, Definition A.12 gives $\mathbb{E}(u^\top Wu)(v^\top Wv) = 2\langle u\rangle v^2/N$, so $\mathbb{E}(u^\top Wu - v^\top Wv)^2 = \frac{4}{N}(1 - \langle u\rangle v^2) \le \frac{8}{N}(1 - \langle u\rangle v)$, which is $\mathbb{E}(Y_u - Y_v)^2$ for $Y_u = \frac{2}{\sqrt{N}}\langle g\rangle u$ with $g \sim \mathcal{N}(0, I_N)$. By Theorem A.4 (ii) over the unit sphere, $\mathbb{E}\lambda_{\max}(W) = \mathbb{E}\sup_u u^\top Wu \le \frac{2}{\sqrt{N}}\mathbb{E}|g| \le 2$. With $W = (G + G^\top)/\sqrt{2N}$ as in Definition A.12, $\lambda_{\max}(W) = \sqrt{2/N}\,\sup_u u^\top Gu$ is a $\sqrt{2/N}$-Lipschitz function of the N^2 standard normal entries of G, because $|u^\top(G - G')u| \le \|G - G'\| \le \|G - G'\|_F$; Theorem A.7 gives the tail bounds.

(iii) By (ii) and the Borel–Cantelli lemma, $\limsup_N \lambda_{\max}(W) \le 2 + u$ almost surely for every $u > 0$. For $\varepsilon \in (0, 2)$, a continuous function $0 \le g \le 1$ supported in $[2-\varepsilon, 2+\varepsilon]$ with $\int g\,\mathrm{d}\sigma_{\mathrm{sc}} > 0$ has $\int g\,\mathrm{d}\mu_N \to \int g\,\mathrm{d}\sigma_{\mathrm{sc}} > 0$ almost surely by (i), so μ_N eventually charges $[2-\varepsilon, 2+\varepsilon]$ and $\lambda_{\max}(W) \ge 2 - \varepsilon$. Since $-W$ is again GOE(N), $\lambda_{\min}(W) \to -2$, and $\|W\| = \max(\lambda_{\max}(W), -\lambda_{\min}(W)) \to 2$. ∎

Part (iv) is Wigner's theorem in the form of Anderson et al. (2010, Theorem 2.1.21); Bai and Silverstein (2010, Chapter 2) treat the almost sure versions. For the norm, Bai and Yin (1988) gave necessary and sufficient conditions for the almost sure convergence of the largest eigenvalue, among

them the finite fourth moment. The proof of (i) is that of Anderson et al. (2010, Section 2.4.1), with the Poincaré inequality for the concentration step and explicit errors.

Theorem A.16 (the Marchenko–Pastur law; stated). *Let $X \in \mathbb{R}^{n\times d}$ have i.i.d. entries of mean 0 and variance 1, and let $n, d \to \infty$ with $d/n \to \gamma \in (0,\infty)$; in the book's notation $\gamma = 1/\alpha$. Almost surely, the empirical spectral distribution of the $d \times d$ matrix $X^\top X/n$ converges weakly to the Marchenko–Pastur law μ_γ. It has density $\rho_\gamma(x) = \sqrt{(b-x)(x-a)}/(2\pi\gamma x)$ on $[a,b]$, with $a = (1-\sqrt{\gamma})^2$ and $b = (1+\sqrt{\gamma})^2$, and, when $\gamma > 1$, an atom of mass $1 - 1/\gamma$ at 0. Its Stieltjes transform $m = s_{\mu_\gamma}$ solves*

$$\gamma z\, m(z)^2 + (z - 1 + \gamma)\, m(z) + 1 = 0, \tag{A.9}$$

with $\Im m(z) > 0$ for $\Im z > 0$; for real $z < 0$, $m(z)$ is the positive root, $m(z) = \big(1 - \gamma - z - \sqrt{(1-\gamma-z)^2 - 4\gamma z}\big)/(2\gamma z)$. The $n \times n$ matrix $XX^\top/n$ has the same nonzero eigenvalues, and the Stieltjes transform of its limiting spectral distribution is $\underline{m}(z) = \gamma m(z) - (1-\gamma)/z$. If the entries also have a finite fourth moment, then $\lambda_{\max}(X^\top X/n) \to b$ almost surely and, for $\gamma < 1$, $\lambda_{\min}(X^\top X/n) \to a$ almost surely.

The law is due to Marčenko and Pastur (1967); Bai and Silverstein (2010, Chapters 3 and 5) prove it and the limits of the extreme eigenvalues, due to Yin et al. (1988) for the largest and to Bai and Yin (1993) for the smallest. For Gaussian entries the proof of Theorem A.15 (i) adapts, with (A.9) in place of $s^2 + zs + 1 = 0$. The kit's `mp_stieltjes` returns the positive root of (A.9) at $z < 0$. Chapters 9, 12 and 14 use the theorem.

Remark A.17 (large deviations of the GOE spectrum). The landscape computations of Chapter 13 also use large deviations of a GOE matrix. Its empirical spectral distribution satisfies a large deviation principle at speed N^2, with a good rate function built from Voiculescu's noncommutative entropy (Ben Arous and Guionnet 1997), and $\lambda_{\max}(W)$ satisfies one at speed N (Ben Arous et al. 2001), with the good rate function $J(x) = \infty$ for $x < 2$ and

$$J(x) = \frac{1}{2}\int_2^x \sqrt{y^2-4}\,\mathrm{d}y = \frac{x}{4}\sqrt{x^2-4} - \log\frac{x+\sqrt{x^2-4}}{2} \qquad (x \ge 2).$$

These are Theorems 2.6.1 and 2.6.6 of Anderson et al. (2010), with potential $V(x) = x^2/4$ and $\beta = 1$ for our normalization; we computed the closed form of J from their rate function.

A.7 The Kac–Rice formula

Physicists count TAP solutions and critical points of random landscapes with a delta function; the Kac–Rice formula is the theorem behind the computation.

Counting with a delta function. If the critical points of a smooth function f on $\mathbb{R}^n$ are nondegenerate, then near each of them the map $x \mapsto$

$\nabla f(x)$ is a diffeomorphism, and the change of variables $y = \nabla f(x)$ gives $\int_U \delta(\nabla f(x))|\det \nabla^2 f(x)|\,\mathrm{d}x = 1$ on a small neighborhood U of the point. Summing over critical points,

$$\#\{x \in D : \nabla f(x) = 0,\ f(x) \in B\} = \int_D \delta(\nabla f(x))\,|\det \nabla^2 f(x)|\,\mathbf{1}\{f(x) \in B\}\,\mathrm{d}x.$$

For a random f, take expectations, exchange them with the integral, and write $\mathbb{E}[\delta(\nabla f(x))\Phi(x)] = \mathbb{E}[\Phi(x) \mid \nabla f(x) = 0]\, p_{\nabla f(x)}(0)$, where $p_{\nabla f(x)}$ is the density of $\nabla f(x)$. The result is (A.10). ◇

A proof replaces δ by an approximate identity and must show that almost surely every critical point is nondegenerate and none lies on the boundary of D, and that the limit can be exchanged with the expectation; the regularity hypotheses below buy this. We state the formula as Auffinger et al. (2013, Lemma 3.1) and Ben Arous et al. (2019, Lemma 1) use it for Gaussian fields on spheres.

Theorem A.18 (the Kac–Rice formula; stated). *Let f be a centered Gaussian field on the unit sphere $S^{N-1} \subset \mathbb{R}^N$ with twice continuously differentiable paths, and let (U_α, ψ_α) be a finite atlas of S^{N-1}. Write $f^\alpha = f \circ \psi_\alpha^{-1}$, with partial derivatives f_i^α and f_{ij}^α. Assume that*

(i) for every α and every $x \in \psi_\alpha(U_\alpha)$, the Gaussian vector $(f_i^\alpha(x), f_{ij}^\alpha(x))_{1 \le i \le j < N}$ is nondegenerate;

(ii) for some $\beta > 0$ and constants K_α, $\max_{i,j} \mathbb{E}\big(f_{ij}^\alpha(x) - f_{ij}^\alpha(y)\big)^2 \le K_\alpha |\log|x - y||^{-1-\beta}$ for all $x, y \in \psi_\alpha(U_\alpha)$.

Let ∇f and $\nabla^2 f$ be the gradient and the covariant Hessian in an orthonormal frame field, p_σ the density of $\nabla f(\sigma)$, and $i(\cdot)$ the number of negative eigenvalues. For a Borel set $B \subset \mathbb{R}$ and $0 \le k < N$, let $\mathcal{A}_k(\sigma)$ be the event $\{f(\sigma) \in B,\ i(\nabla^2 f(\sigma)) = k\}$ and $\mathrm{Crt}_k(B)$ the number of critical points σ at which $\mathcal{A}_k(\sigma)$ holds. Then, with $\mathrm{d}\sigma$ the surface measure,

$$\mathbb{E}\,\mathrm{Crt}_k(B) = \int_{S^{N-1}} \mathbb{E}\big[|\det \nabla^2 f(\sigma)|\,\mathbf{1}_{\mathcal{A}_k(\sigma)} \,\big|\, \nabla f(\sigma) = 0\big]\, p_\sigma(0)\,\mathrm{d}\sigma. \quad \text{(A.10)}$$

Auffinger et al. (2013) take hypotheses (i) and (ii) from Corollaries 11.3.2 and 11.3.5 of Adler and Taylor (2007), which make f almost surely a Morse function with mean-square derivatives, and prove the formula as Adler and Taylor prove their Theorem 12.4.1. The general statement for random fields on manifolds is Theorem 12.1.1 of Adler and Taylor (2007), whose hypotheses (a)–(g) concern the continuity, moments and moduli of continuity of the fields and the continuity of certain conditional densities; Chapter 11 of that book treats subsets of $\mathbb{R}^n$, and Azaïs and Wschebor (2009) give another account. We know these conditions through Auffinger et al. (2013) and Subag (2017, Appendix III), not from the book itself. The formula scales to the sphere of radius $\sqrt{N}$.

The formula gives a mean, which can exceed the typical number of critical points by an exponential factor; a second-moment computation shows when

it does not (Subag 2017). The conditional law of the Hessian given $\nabla f(\sigma) = 0$ and $f(\sigma) = x$ is a Gaussian regression; for the pure spherical p-spin model it is a multiple of a GOE matrix minus a multiple of the identity (Auffinger et al. 2013, Lemma 3.2), which turns (A.10) into an expectation over the GOE (Theorem 13.5). In dimension one the same computation, with $\delta(X(t))|X'(t)|$, gives Rice's formula: if X is a stationary centered Gaussian process with continuously differentiable paths, $\lambda_0 = \mathbb{E}X(t)^2$ and $\lambda_2 = \mathbb{E}X'(t)^2$, then $X(t) = u$ has on average $\frac{T}{\pi}\sqrt{\lambda_2/\lambda_0}\, e^{-u^2/(2\lambda_0)}$ solutions in $[0, T]$, because $X(t)$ and $X'(t)$ are independent. For $X(t) = \sum_{k=1}^{K}(a_k \cos kt + b_k \sin kt)$ with independent standard normal coefficients the mean number of zeros in $[0, 2\pi)$ is $2\sqrt{(K+1)(2K+1)/6}$, which is 12.410 for $K = 10$; 20000 draws (seed 7), with the zeros counted as the roots on the unit circle of a polynomial of degree $2K$, give 12.400 ± 0.033 (a 95% Student-t interval).[7]

[7] Kac (1943) found that a random algebraic polynomial of degree n with independent standard normal coefficients has about $(2/\pi)\log n$ real roots on average, and Rice (1944) derived the formula for the crossings of random noise.

A.8 Numerics

This section says what the numerical routines of the kit `figures/scripts/rosettakit.py` compute, how accurate they are, and how they fail. All numbers below were computed with the kit by the script `figures/scripts/17-toolkit-numerics.py`, which also prints the numbers quoted for the Curie–Weiss model in Section A.4 and for Rice's formula in Section A.7.

Gaussian expectations. The function `gauss(f, n)` computes $\mathbb{E}f(Z) = \int f(z)\,\mathrm{D}z$ by the n-point Gauss–Hermite rule $\sum_k w_k f(x_k)$, by default with $n = 121$. The nodes x_k are the zeros of the monic Hermite polynomial He_n, these polynomials being orthogonal with $\mathbb{E}\,\mathrm{He}_j(Z)\mathrm{He}_k(Z) = k!\,\delta_{jk}$, and the weights are $w_k = n!/(n\,\mathrm{He}_{n-1}(x_k))^2$. The rule is exact for polynomials of degree at most $2n-1$, and for f with $2n$ continuous derivatives its error is $f^{(2n)}(\xi)\,n!/(2n)!$ for some real ξ. Table A.1 shows what this means in practice: entire integrands such as cos are exact to rounding at $n = 21$; integrands analytic only in a strip around the real axis, such as tanh and log cosh, converge roughly like $e^{-c\sqrt{n}}$; and an integrand with a kink converges like $1/n$. For such integrands `gauss_split(f, a)` puts the kink at a and applies 200-point Gauss–Legendre rules to f times the normal density on $(-12, a)$ and $(a, 12)$; its error for $\mathbb{E}|Z|$ is 5×10^{-15}. Numpy documents its routine for these nodes and weights as tested up to 100 nodes; at 121, 161 and 201 nodes the weights are symmetric and sum to 1 within 2.2×10^{-16}. The kit's tail $H(x) = \mathbb{P}(Z > x)$ returns 0 at $x = 38$, where the true value falls below the smallest normal double; `logH` returns $\log H(x)$ stably.

Fixed points. Saddle-point and state-evolution equations have the form $x = F(x)$. The function `fixed_point(F, x0, damping, tol)` iterates the damped map $T_\delta(x) = (1-\delta)x + \delta F(x)$, $\delta \in (0, 1]$, until the largest change of a coordinate is below `tol`, and returns the point, the number of iterations, and whether it converged. For scalar F, continuously differentiable near a fixed point x^* with $\lambda = F'(x^*)$, the slope $T'_\delta(x^*) = 1 - \delta(1-\lambda)$ shows, by the mean value theorem, that the iteration converges to x^* from nearby starting points, geometrically with ratio $r = |1-\delta(1-\lambda)|$, when $0 < \delta(1-\lambda) < 2$, and that x^* repels for every δ when $\lambda > 1$. Damping therefore stabilizes fixed

n	$\mathbb{E}\cos Z$	$\mathbb{E}\tanh(2+\sqrt{2}Z)$	$\mathbb{E}\log\cosh(1.5Z+0.3)$	$\mathbb{E}\lvert Z\rvert$
11	2.8×10^{-14}	1.4×10^{-3}	5.1×10^{-4}	6.0×10^{-2}
21	$<10^{-15}$	7.4×10^{-5}	4.0×10^{-6}	3.1×10^{-2}
41	$<10^{-15}$	1.3×10^{-6}	7.0×10^{-7}	1.6×10^{-2}
61	$<10^{-15}$	6.8×10^{-8}	4.1×10^{-8}	1.1×10^{-2}
121	$<10^{-15}$	5.7×10^{-11}	1.9×10^{-11}	5.4×10^{-3}
201	$<10^{-15}$	3.9×10^{-14}	2.6×10^{-14}	3.3×10^{-3}

Table A.1. Absolute errors of the n-point Gauss–Hermite rule of the kit (`gauss`) against reference values: closed forms for cos and $|z|$, mpmath quadrature at 30 digits for tanh and log cosh.

points with $\lambda<-1$, which the plain iteration misses because it oscillates; it slows the convergence when $0<\lambda<1$; and it never finds a fixed point with $\lambda>1$. For a vector equation the same holds with the eigenvalues of the Jacobian. When the change between iterates falls below `tol`, the error is about $\texttt{tol}\cdot r/(1-r)$, large near a continuous transition, where $\lambda\to1$. For $\mathbb{Z}_2$ synchronization, $q=1-\mathrm{mmse}(\lambda^2 q)$ with the Rademacher prior, the undamped iteration from $q=1$ with $\texttt{tol}=10^{-13}$ takes 18 iterations at $\lambda=2$, where $F'(q^*)=0.197$, and 608 at $\lambda=1.02$, where $F'(q^*)=0.962$; it stops with errors 1.5×10^{-14} and 2.5×10^{-12}, against the estimates 2.5×10^{-14} and 2.5×10^{-12}, and with $\delta=\frac12$ it needs two to three times as many iterations. Near a transition a script should check the residual $|F(x)-x|$, or solve $F(x)=x$ by bracketing and Brent's method (`root`), whose convergence does not depend on F'.

Several fixed points. When F has several fixed points, a script must find all of them and say which one it reports. The recipe is to tabulate $F(q)-q$ on a grid, refine each sign change with `root`, classify each root by its slope, and run the damped iteration from $q_0=1$ (the informative branch) and from q_0 near 0 (the uninformative one). A Bayes-optimal error is given by the global extremum of the potential whose stationarity condition is $q=F(q)$; an algorithm such as AMP follows its state evolution from its own starting point. For spiked Wigner matrices the replica-symmetric formula (Theorem 4.1) gives the mutual information per coordinate as the minimum over $\gamma\ge0$ of $\Psi(\gamma)=\frac{\lambda^2}{4}+\frac{\gamma^2}{4\lambda^2}-\frac{\gamma}{2}+I(\gamma)$, with I the mutual information of the scalar channel; with $\gamma=\lambda^2 q$, the I-MMSE relation $I'(\gamma)=\frac12\mathrm{mmse}(\gamma)$ (Theorem 1.12) gives $\frac{\mathrm{d}\Psi}{\mathrm{d}\gamma}=\frac12(q-F(q))$. The potential difference of two fixed points is therefore an integral of the residual,

$$\Psi(q_2)-\Psi(q_1)=\frac{\lambda^2}{2}\int_{q_1}^{q_2}\big(q-F(q)\big)\,\mathrm{d}q,$$

an equal-area rule. For the sparse Rademacher prior with $\rho=0.05$ at $\lambda=0.92$ there are three fixed points: $q=0$, stable, with $F'(0)=\lambda^2=0.846$; $q=0.12934$, unstable, with $F'=1.213$; and $q=0.75471$, stable, with $F'=0.550$. The damped iteration ($\delta=\frac12$) reaches 0.75471 from $q_0=1$ and 0.3, and 0 from $q_0=0.1$ and 10^{-3}. Since $\Psi(0.75471)-\Psi(0)=-0.0116$, the

informative fixed point is the Bayes-optimal one, while AMP started without information stays at 0: the hard phase of Chapter 5. Solving $\Psi(q_+) = \Psi(0)$ for λ gives the information-theoretic threshold $\lambda_{\mathrm{IT}} = 0.86447$.

INTERVALS. The figures show 95% intervals of two kinds; each caption says which, over how many independent replicates. For R replicates with mean $\bar{v}$ and sample variance s^2 (divisor $R-1$), `mean_ci` returns $\bar{v} \pm t_{R-1}s/\sqrt{R}$, with t_{R-1} the 0.975 quantile of Student's distribution with $R-1$ degrees of freedom. The coverage is exactly 95% for normal replicates and tends to 95% for replicates of finite variance; for skewed replicates and small R it falls short, and for exponential replicates we measured 87.7%, 90.0% and 92.6% at $R = 5$, 10 and 30 (20000 repetitions each, seed 2026). For J successes in n independent trials, `clopper_pearson` returns $[\theta_-(J), \theta_+(J)]$, where $\theta_-(k)$ is the $\alpha/2$ quantile of the Beta distribution with parameters $(k, n-k+1)$, or 0 if $k = 0$, and $\theta_+(k)$ is the $1-\alpha/2$ quantile of the Beta distribution with parameters $(k+1, n-k)$, or 1 if $k = n$, with $\alpha = 0.05$ (Clopper and Pearson 1934).

Proposition A.19. *For every $\theta \in [0,1]$ and $J \sim \mathrm{Bin}(n,\theta)$, $\mathbb{P}\big(\theta_-(J) \le \theta \le \theta_+(J)\big) \ge 1-\alpha$.*

PROOF. For $1 \le k \le n$, $\mathbb{P}(J \ge k) = \mathbb{P}(B_k \le \theta)$, where B_k has the Beta distribution with parameters $(k, n-k+1)$: for independent uniform $U_1, \dots, U_n$ on $[0,1]$, the number of $U_i \le \theta$ has the law of J, and it is at least k exactly when the kth smallest U_i, which has the law of B_k, is at most θ. The end points $\theta_\pm(k)$ increase with k. If $\theta_+(J) < \theta$ has positive probability, let k^* be the largest k with $\theta_+(k) < \theta$; then $k^* < n$ and $\mathbb{P}(\theta_+(J) < \theta) = \mathbb{P}(J \le k^*) = \mathbb{P}(B_{k^*+1} > \theta) \le \mathbb{P}(B_{k^*+1} > \theta_+(k^*)) = \alpha/2$. In the same way, with k_* the smallest $k \ge 1$ with $\theta_-(k) > \theta$, $\mathbb{P}(\theta_-(J) > \theta) = \mathbb{P}(B_{k_*} \le \theta) \le \mathbb{P}(B_{k_*} \le \theta_-(k_*)) = \alpha/2$; the two failures have total probability at most α. ∎

The interval is conservative: the infimum of its coverage over θ is 96.10% for $n = 10$, 95.08% for $n = 50$, and 95.002% for $n = 200$ and $n = 1000$. Every script uses fixed seeds and prints the numbers that its chapter quotes.

Notes and References

Vershynin (2018, Chapters 5 and 7) treats the Gaussian tools, including the inequalities of Slepian and Sudakov–Fernique (its Section 7.2). Boucheron et al. (2013) and Ledoux (2001) treat concentration, Dembo and Zeitouni (2010) large deviations, Anderson et al. (2010) and Bai and Silverstein (2010) random matrices, and Adler and Taylor (2007) and Azaïs and Wschebor (2009) the Kac–Rice formula. The induction in part (ii) of Lemma A.13 was written for this book.

TRANSLATING THE LITERATURE. Anderson et al. (2010) and Bai and Silverstein (2010) define the Stieltjes transform as we do; Ben Arous et al. (2019) and much of the free-probability literature use the Cauchy transform $\int \mu(\mathrm{d}x)/(z-x) = -s_\mu(z)$, in which the semicircle's transform reads

$(z - \sqrt{z^2 - 4})/2$; and the physicist's Green's function $(z - A)^{-1}$ is $-R_A(z)$. The GOE of Anderson et al. (2010) is ours; Auffinger et al. (2013) normalize it with $\mathbb{E}M_{ij}^2 = (1 + \delta_{ij})/(2N)$, so that their semicircle lives on $[-\sqrt{2}, \sqrt{2}]$; Statisticians write the Marchenko–Pastur ratio as $\gamma = p/n$, our $d/n = 1/\alpha$ (Chapter 9).

References

Adler, R. J. and J. E. Taylor (2007). *Random Fields and Geometry*. Springer Monographs in Mathematics. New York: Springer.

Anderson, G. W., A. Guionnet, and O. Zeitouni (2010). *An Introduction to Random Matrices*. Vol. 118. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press.

Auffinger, A., G. Ben Arous, and J. Černý (2013). "Random Matrices and Complexity of Spin Glasses". In: *Communications on Pure and Applied Mathematics* 66.2, pp. 165–201. arXiv: 1003.1129.

Azaïs, J.-M. and M. Wschebor (2009). *Level Sets and Extrema of Random Processes and Fields*. Hoboken, NJ: Wiley.

Bai, Z. D. and Y. Q. Yin (1988). "Necessary and Sufficient Conditions for Almost Sure Convergence of the Largest Eigenvalue of a Wigner Matrix". In: *The Annals of Probability* 16.4, pp. 1729–1741.

Bai, Z. D. and Y. Q. Yin (1993). "Limit of the Smallest Eigenvalue of a Large Dimensional Sample Covariance Matrix". In: *The Annals of Probability* 21.3, pp. 1275–1294.

Bai, Z. and J. W. Silverstein (2010). *Spectral Analysis of Large Dimensional Random Matrices*. 2nd ed. Springer Series in Statistics. New York: Springer.

Bayati, M. and A. Montanari (2011). "The Dynamics of Message Passing on Dense Graphs, with Applications to Compressed Sensing". In: *IEEE Transactions on Information Theory* 57.2, pp. 764–785. arXiv: 1001.3448.

Ben Arous, G., A. Dembo, and A. Guionnet (2001). "Aging of spherical spin glasses". In: *Probability Theory and Related Fields* 120.1, pp. 1–67.

Ben Arous, G. and A. Guionnet (1997). "Large deviations for Wigner's law and Voiculescu's non-commutative entropy". In: *Probability Theory and Related Fields* 108.4, pp. 517–542.

Ben Arous, G., S. Mei, A. Montanari, and M. Nica (2019). "The Landscape of the Spiked Tensor Model". In: *Communications on Pure and Applied Mathematics* 72.11, pp. 2282–2330. arXiv: 1711.05424.

Bolthausen, E. (2014). "An Iterative Construction of Solutions of the TAP Equations for the Sherrington–Kirkpatrick Model". In: *Communications in Mathematical Physics* 325.1, pp. 333–366. arXiv: 1201.2891.

Borell, C. (1975). "The Brunn-Minkowski inequality in Gauss space". In: *Inventiones Mathematicae* 30.2, pp. 207–216.

Boucheron, S., G. Lugosi, and P. Massart (2013). *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford: Oxford University Press.

Chatterjee, S. (2005). *An error bound in the Sudakov–Fernique inequality*. arXiv: math/0510424.

Chernoff, H. (1981). "A Note on an Inequality Involving the Normal Distribution". In: *The Annals of Probability* 9.3, pp. 533–535.

Cirel'son, B. S., I. A. Ibragimov, and V. N. Sudakov (1976). "Norms of Gaussian sample functions". In: *Proceedings of the Third Japan–USSR Symposium on Probabil-*

ity Theory. Vol. 550. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, pp. 20–41.

Clopper, C. J. and E. S. Pearson (1934). "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial". In: *Biometrika* 26.4, pp. 404–413.

Dembo, A. and O. Zeitouni (2010). *Large Deviations Techniques and Applications*. Vol. 38. Stochastic Modelling and Applied Probability. Corrected reprint of the second edition (1998). Berlin: Springer.

Feng, O. Y., R. Venkataramanan, C. Rush, and R. J. Samworth (2022). "A Unifying Tutorial on Approximate Message Passing". In: *Foundations and Trends in Machine Learning* 15.4, pp. 335–536. arXiv: 2105.02180.

Fernique, X. (1975). "Régularité des trajectoires des fonctions aléatoires gaussiennes". In: *École d'Été de Probabilités de Saint-Flour IV—1974*. Vol. 480. Lecture Notes in Mathematics. Springer, pp. 1–96.

Houdré, C. and V. Pérez-Abreu (1995). "Covariance Identities and Inequalities for Functionals on Wiener and Poisson Spaces". In: *The Annals of Probability* 23.1, pp. 400–419.

Kac, M. (1943). "On the average number of real roots of a random algebraic equation". In: *Bulletin of the American Mathematical Society* 49.4, pp. 314–320.

Ledoux, M. (2001). *The Concentration of Measure Phenomenon*. Vol. 89. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society.

Marčenko, V. A. and L. A. Pastur (1967). "Distribution of eigenvalues for some sets of random matrices". In: *Mathematics of the USSR-Sbornik* 1.4, pp. 457–483.

Pisier, G. (1986). "Probabilistic methods in the geometry of Banach spaces". In: *Probability and Analysis (Varenna, 1985)*. Vol. 1206. Lecture Notes in Mathematics. Springer, pp. 167–241.

Rice, S. O. (1944). "Mathematical Analysis of Random Noise". In: *Bell System Technical Journal* 23.3, pp. 282–332.

Slepian, D. (1962). "The One-Sided Barrier Problem for Gaussian Noise". In: *Bell System Technical Journal* 41.2, pp. 463–501.

Stein, C. M. (1981). "Estimation of the Mean of a Multivariate Normal Distribution". In: *The Annals of Statistics* 9.6, pp. 1135–1151.

Subag, E. (2017). "The complexity of spherical p-spin models—A second moment approach". In: *The Annals of Probability* 45.5, pp. 3385–3450. arXiv: 1504.02251.

Sudakov, V. N. (1971). "Gaussian random processes and measures of solid angles in Hilbert space". In: *Soviet Mathematics Doklady* 12, pp. 412–415.

Sudakov, V. N. and B. S. Tsirel'son (1978). "Extremal properties of half-spaces for spherically invariant measures". In: *Journal of Soviet Mathematics* 9.1, pp. 9–18.

Varadhan, S. R. S. (1966). "Asymptotic probabilities and differential equations". In: *Communications on Pure and Applied Mathematics* 19.3, pp. 261–286.

Vershynin, R. (2018). *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press.

Wigner, E. P. (1955). "Characteristic Vectors of Bordered Matrices With Infinite Dimensions". In: *Annals of Mathematics* 62.3, pp. 548–564.

Yin, Y. Q., Z. D. Bai, and P. R. Krishnaiah (1988). "On the limit of the largest eigenvalue of the large dimensional sample covariance matrix". In: *Probability Theory and Related Fields* 78.4, pp. 509–521.

Index

aging, 291
annealed average, 13
annealed bound, 55
approximate message passing, 119
auxiliary optimization, 211

BBP transition, 104
belief propagation, 120
bias factor, 228

capacity
  of the perceptron, 161
Carleman's condition, 22
cavity field, 184
cavity method, 119
Clopper–Pearson interval, 366
complete monotonicity, 88
complexity
  annealed, 268
  quenched, 268
concentration of measure, 351
convex Gaussian min–max theorem, 169
  one-sided, 231
convex Gaussian min-max theorem, 193
correlation function, 289
critical point, 268
  index, 268
Curie–Weiss model, 14

de Almeida–Thouless condition, 206
de Almeida–Thouless line, 58
decoupling principle, 204
descent cone, 214
deterministic equivalent, 251
Donoho–Tanner phase transition, 204
Doob's inequality, 321
double descent, 182
dynamical cavity method, 290
dynamical mean-field theory, 288
dynamical transition, 292

early stopping, 301
effective ridge, 188, 246
entropy
  of a level, 35
evidence, 77

fixed-point iteration, 364
fluctuation–dissipation theorem, 291
fractional moment, 38
free entropy, 13, 77
  of a generalized linear model, 140
  of SK, 55
  of the solution set, 162

Gauss–Hermite quadrature, 364
Gaussian conditioning, 356
Gaussian covariate model, 248
Gaussian equivalence, 246
Gaussian integration by parts, 348
Gaussian interpolation, 349
Gaussian Poincaré inequality, 352
generalization error, 310
generalized cross-validation, 195
generalized linear model, 139
generalized random energy model, 45
Ghirlanda–Guerra identities, 66
Gibbs learning, 140
Gibbs measure, 13
GOE, 356
Gordon's comparison inequality, 167
gradient flow, 289
ground state, 98
  spherical p-spin, 278
Grönwall's inequality, 321
Guerra's bound, 63

Hamiltonian, 13
hard phase, 102, 143
Hermite expansion, 248
Hermite polynomials, 315

Hubbard–Stratonovich
transformation, 14

implicit bias, 225
information exponent, 311
information-theoretic threshold,
101
interpolation
adaptive, 87
interpolation threshold, 183
isostaticity, 165, 227

Kac–Rice formula, 362
kernel ridge regression, 245
kinematic formula, 214

Langevin dynamics, 289
Laplace's method, 353
large deviation principle, 354
LASSO, 203
learning curve, 245
leave-one-out, 228, 249
level count, 38
leverage, 188
lifted Gordon bound, 173
likelihood-ratio test
rescaled, 236
logistic regression, 224

M-estimator, 204
magnetization, 119
Marchenko–Pastur law, 188, 362
Laplace transform, 293
margin, 225
marginal stability, 268, 292
Martin–Siggia–Rose integral, 290
max-margin classifier, 225
maximal stability, 164
maximum likelihood estimator
tensor PCA, 267
Maxwell construction, 101
mean-field bound, 19
Mercer decomposition, 245
method of types, 18
minimum mean-square error, 77
minimum-norm interpolant, 245
minimum-norm interpolator, 182
mutual information, 77

naive mean field, 120
Nishimori line, 77
Novikov's formula, 290

online SGD, 309
Onsager reaction term, 120
optimal ridge, 193
order parameter, 13
order parameters
overlaps, 310
orthogonal code, 42
overlap, 13, 55, 98
concentration, 85
in the REM, 35
matrix, 162
teacher–student, 140
overlap map, 100

Parisi formula
equation, 59
Parisi measure, 66
Parisi parameter, 36
participation ratio, 34
partition function, 13
perceptron
Ising, 166
Ising teacher, 139
spherical, 161
spherical teacher, 139
plateau, 313
escape time, 315
Plefka's condition, 122
Poisson–Dirichlet distribution, 45
proximal map, 228
pseudo-Lipschitz function, 123
pure states, 59

quenched average, 13

random energy model, 34
random features, 246
replica, 77
replica method
uncontrolled steps, 141
replica moments, 44
replica symmetry
in the REM, 36
zero temperature, 206
replica symmetry breaking

one step, 36
replica trick, 56
replica-symmetric potential, 100
replica-symmetry breaking
full, 59
replicas, 13, 55
replicon, 58, 164
resolvent, 187, 358
response function, 289
ridge regression, 182

saddle point, 14
scalar channel, 77
search phase, 315
second moment method, 64, 278
secular equation, 105
self-averaging, 15, 311
semicircle law, 104, 362
separability, 225
Shannon limit, 42
Sheppard's formula, 313
Sherman–Morrison formula, 248
Sherrington–Kirkpatrick model,
54
signed teacher field, 225
Slepian's lemma, 168, 350
slice capacity, 227
soft committee machine, 309
soft thresholding, 204
sparse PCA, 98
specialization, 322
spectral initialization, 129
spherical p-spin model, 288
spherical model, 98
spherical p-spin model, 267
spiked tensor model, 280
spiked Wigner model, 76, 97
spinodal, 101
state evolution, 121
GAMP, 142
statistical dimension, 214
Stein's identity, 348
Stieltjes transform, 104, 358
Sudakov–Fernique inequality, 350
sum rule, 63
susceptibility, 164, 184, 228

TAP equations, 120
TAP free energy, 121, 269
teacher–student scenario, 139
teacher–student setup, 309
tempered overfitting, 255
tensor PCA, 267
threshold energy, 268, 292

ultrametricity, 59

Varadhan's lemma, 356
variance inflation, 214
version space, 140, 226

weak recovery, 98
weak threshold, 215

$\mathbb{Z}_2$ synchronization, 76
$\mathbb{Z}_2$ synchronization, 97, 118

How This Book Was Made

Colophon

Roles

The Replica Rosetta Stone was written by Claude Opus 5.5, an AI model made by Anthropic, in October 2026. Xiaoyu Li conceived the series *The Missing Textbooks* and proposed this volume: its subject, the exact predictions of high-dimensional statistics and learning theory that came from statistical physics, and their proofs; its gap, a parallel text for readers fluent in one of the two languages; its form, the physical derivation and the rigorous route set side by side with a status board in every chapter; the rule that every example come from inference and learning; and the one-language works it should be measured against, among them the introduction of Barbier and Macris to adaptive interpolation and the reviews and lecture notes of Zdeborová and Krzakala. Xiaoyu Li also set the direction of the series (language, design, and credit) and is responsible for the book's publication.

Process

A coordinating instance of the model first surveyed the literature with four research assistants, separate instances that verified, with web search and the papers themselves, the replica method and interpolation with the Nishimori line and the perceptron; the cavity method, message passing and random landscapes; Gaussian comparison and regression; and dynamics, together with the history of the field and the books and notes that already exist. They found that two works already teach replica computations next to rigorous arguments for selected models, the tutorial of Montanari and Sen and the lecture notes of Krzakala and Zdeborová, and the plan was narrowed to claim, as far as the authors know, a systematic parallel text of this kind with a status board for every prediction. Their digests corrected the plan in several places before a chapter was written. The threshold of the Ising teacher–student perceptron that Barbier and coauthors print, 1.249, does not agree with their own formula, which gives 1.2447. The often-quoted critical density 0.09 of sparse PCA mixes two different densities, 0.0964 and 0.0856. A margin in a proposition of Altschuler and Tikhomirov is about five times smaller than stated, without affecting their theorem. The arXiv version of a paper on the TAP free energy misprints the formula for the spectral start of message passing. Two papers of Ben Arous, Dembo and Guionnet had been conflated, the volume of the 2022 Les Houches school that the plan would have cited has not, as far as could be verified, appeared, and a cited tutorial had changed its title. And a cluster of preprints from 2026, several of them written, by their authors' accounts, with the help of AI systems, had settled questions about the Sherrington–Kirkpatrick model and the binary perceptron that were open when the plan was conceived;

the status word *preprint* was added for them.

The coordinator then wrote the plan (the thesis, the global normalizations, which the two literatures disagree on, fixed labels for results that chapters cite across the book, and a specification for each chapter), a style guide (the anatomy of a chapter, the voice, a proof style that follows Halmos, and the conventions of the scripts), the volume's typography with its stone and status boards, a shared kit for the scripts, and the prologue.

Separate instances of the same model drafted the fifteen chapters and the appendix, up to thirteen at a time. Each verified its references and the claims it attributes to papers, wrote and ran the scripts behind its figures, checked its theorems and constants numerically, and built its chapter with the volume's checks for errors, references, layout, and style. Some authors found that their specifications were wrong, and said so. For generalized linear models the straight interpolation path of the plan has a remainder without a sign, so Chapter 7 replaced the planned one-sided bound by an exact sum rule, a bound that is conditional on the concentration of the overlap, and counting theorems that bracket the threshold of the Ising teacher. Chapter 4 proved the whole replica-symmetric formula for $\mathbb{Z}_2$ synchronization, where the plan had asked for one direction. The reports of six authors were lost when the work was paused, as described below, and their referees checked those chapters against their specifications instead. Some proofs and observations are, as far as the authors and referees could find, the book's own, and the chapters mark them. Among them are the bracket $1 \le \alpha_{\mathrm{IT}} \le 1.4480$ for the Ising teacher, with exact recovery above the threshold (Chapter 7); the closed form $(1 + \log 2)/2$ of an upper bound on the capacity of the binary perceptron, and the crossover $\kappa = -0.6219$ below which Stojnic's lifted bound beats Gardner's formula (Chapter 8); a proof of the ridgeless limits of ridge regression by a monotone comparison and the mean of the inverse Wishart matrix (Chapter 9); the slicing of Gardner's volume by the teacher (Chapter 11); and the comparison on the circle in which the Gaussian approximation to a kernel learning curve fails at zero ridge (Chapter 12).

Each chapter, and the prologue and the appendix together, then received one independent referee pass from a separate instance, which re-derived the results line by line and checked them numerically, reran the scripts and compared every printed number with the text, worked the exercises, and checked the claims about papers against the papers; it corrected what it found directly. No referee found a false main theorem, except in one degenerate case: at zero signal, where its maximizer is not unique (Chapter 4). The referees found real errors elsewhere. Some cited theorems had been stated with the wrong hypotheses: a step size of order $d^{-1/2}$ where the source requires order $t^{-1/2}$ (Chapter 15), a state-evolution theorem applied to a channel that is not Lipschitz (Chapter 7), a failure of universality for the LASSO that its authors prove only for least squares (Chapter 10), and a theorem on gradient flow that covers only the Bayes risk (Chapter 14). Some board rows claimed more than their sources, and three in Chapter 7 became *proved under*; one row of Chapter 4 went the other way, from

conjectured to *proved under*, when its referee found a theorem of Barbier and Panchenko. Credit was wrong in several places: the exact replica moments of the random energy model had been derived by Ogure and Kabashima (Chapter 2), the plateau formulas of online SGD are in a paper of Saad and Solla from 1995, who also saw the student units separate at a finite learning rate (Chapter 15), the ingredients of the circle example were in print (Chapter 12), and the uniqueness of the maximizer in $\mathbb{Z}_2$ synchronization is due to Deshpande, Abbe and Montanari (Chapter 4). The margin below which Stojnic's lifted bound proves Gardner's formula wrong, quoted as -0.63 from a paper that states no threshold, is -0.622 (Chapter 8), and the jump of a sparse prior at density 0.05, printed as 0.585 in one chapter, is 0.583 (Chapters 4 and 5). Many theorem numbers had been taken from arXiv versions that differ from the published papers; the referees corrected those they could check and say which version the others come from. Figures had points off the plot (Chapter 9), a curve drawn against the wrong axis and a legend that hid data (Chapter 11); two confidence intervals were corrected (Chapter 15); a side note blamed undercovering intervals for an outlier that is a rare fluctuation (Chapter 4); a cut that the pause had interrupted was repaired (Chapter 14); and a compressed revision had dropped a qualification of Sellke's theorem (Chapter 13). Each of these errors is corrected in the text. The book also reports errors in the sources, found by the research assistants, the authors, or the referees, and confirmed by the referees: a threshold printed as 1.249 that the paper's own formula puts at 1.2447 (Chapter 7), a proposition whose margin is about five times smaller than stated, and a figure caption that contradicts its paper's own definition (Chapter 8), misprints in arXiv versions of papers on message passing (Chapters 5 and 6), a reversed inequality (Chapter 11), a parameter defined in two inconsistent ways (Chapter 12), and a misprinted remark on the complexity of the spherical spin glass (Chapter 13). The chapters state each of them where the source is used.

A usage limit stopped the running agents once, and each was resumed with its context. Later the work was paused, at the editor's request, while several authors and referees were still at work, and their contexts did not survive the pause. When the work resumed in a new session, the state of each chapter was reconstructed from its files, and every referee pass that had not finished was run again, from the start or, where the earlier referee had already edited the chapter, as a pass that checked those edits and completed the review, with the earlier referee's scratch computations available to it.

While the chapters were being refereed, the coordinator wrote the preface, the notation, and the epilogue, whose figure is computed from the chapters' status boards by a script, and revised them as the referee reports came in. It then integrated the book: it reconciled numbers and notation between chapters, gave every work cited in several chapters a single bibliographic entry, and checked the whole book for layout, undefined references, and overflowing lines. A last referee pass checked the epilogue, the preface, the notation, this colophon, and the volume's README against the chapters.

Limits

No human has yet checked every proof or every attribution. Web search and the papers themselves were available while the book was written, but some full texts could not be opened: several classic physics papers are behind paywalls and were checked against their abstracts, against later papers that restate them, or by recomputing their formulas, and the chapters say so where it matters. The preprints of 2026 are reported as their authors state them; the book checked their statements and versions, not their proofs. One referee pass per chapter finds many errors but not all of them, and a referee that is an instance of the same model may share the author's blind spots. Results that the book presents as its own have been checked by the same model only. The scripts are small: exact enumeration reaches sixteen spins, and simulations reach dimensions of a few thousand, which confirms the arithmetic of the theorems and shows finite-size effects, not the behavior of large systems.

Typeset with pdfLaTeX in `missingbook`, the series class built on KOMA-Script, with the volume style `rosetta`; Palatino (newpx) for text and mathematics, Source Sans for labels, biblatex and biber for references, and Ti*k*Z and pgfplots for figures, whose data were computed with Python, NumPy, and SciPy, and checked with mpmath and SymPy. Physics is set in carnelian, proof in lapis, and the structure of the book in granite. The title page carries one fact in three registers: Gardner's replica formula for the capacity of the spherical perceptron, its curve, and the theorem that proves it for nonnegative margins; the curve is drawn by `figures/scripts/00-emblem.py`. The foot of each chapter's opening page, and of each part page, carries a status stone for the main prediction of the chapter or the part.